JEE MAIN JANUARY 2020

OFFICIAL QUESTION PAPERS WITH DETAILED SOLUTIONS

6 SETS HAVING 450 QUESTIONS WITH DETAILED SOLUTIONS OF PHYSICS, CHEMISTRY AND MATHEMATICS

CAREERS360

7th January Shift-1

Physics

Q. 1 A 60 HP electric motor lifts an elevator having a maximum total load capacity of 2000 kg. If the frictional force on the elevator is 4000N , then the speed of the elevator at full load is close to: (1 HP=746 W, g=10 ms-2).

Correct Answer:

$1.9\ ms^{-1}$

Solution:

Power -

Instantaneous power-

$$P = \frac{dw}{dt} = P = \vec{F} \cdot \vec{v}$$

Where, $\vec{F} \rightarrow force$

$\vec{v} \rightarrow velocity$

i.e. power is equal to the scalar product of force with velocity

so, net force act on elevator =mg+F

F_{net}=(2000 x 10+4000)=24000 N

using power =F_{net} x speed

So speed=$v = \frac{60 * 746}{24000}$=1.9 m/s

So, Option (4) is correct.

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Q. 2 Consider a coil of wire carrying current I, forming a magnetic dipole. The magnetic flux through an infinite plane that contains the circular coil and excluding the circular coil area is given by ϕi. The magnetic flux through the area given by ϕo. Which of the following is correct?

Correct Answer:
ϕi = $-\phi_o$

Solution:

Magnetic flux - - wherein

Magnetic flux-

The total number of magnetic lines of force passing normally through an area placed in a magnetic field is equal to
the magnetic flux linked with that area.

I.e for the below figure

Net magnetic flux through the surface is given by

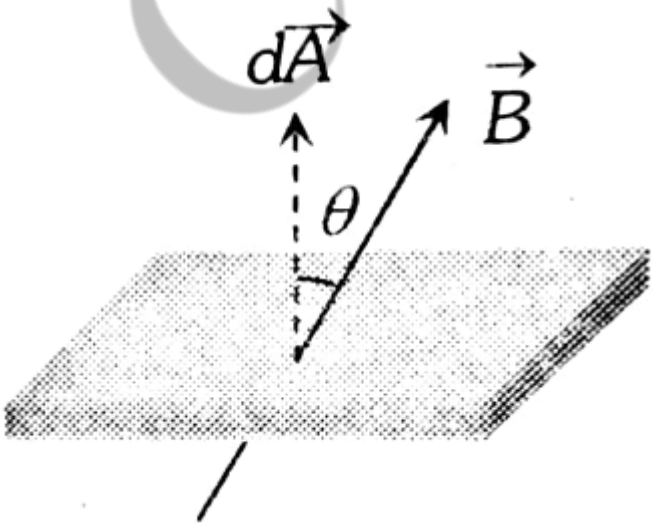
$$\phi_B = \oint \vec{B} \cdot d\vec{A} = BA\cos\Theta$$

where

$\phi_B =$ Magnetic Flux

$B =$ Magnetic field

$\theta =$ The angle between area vector and magnetic field vector



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Flux going right comes back to the left (forms closed loop)

$$\int B.dA = 0$$

Flux inside the coil come bach through outside

$\phi_{coil} = -\phi_{out\ side}$
$\phi_0 = -\phi_i$

So option (4) is correct.

Q. 3 A polarizer-analyzer set is adjusted such that the intensity of light coming out of the analyzer is just 10% of the original intensity. Assuming that the polarizer-analyzer set does not absorb any light, the angle by which the analyzer needs to be rotated further to reduce the output intensity to be zero is:

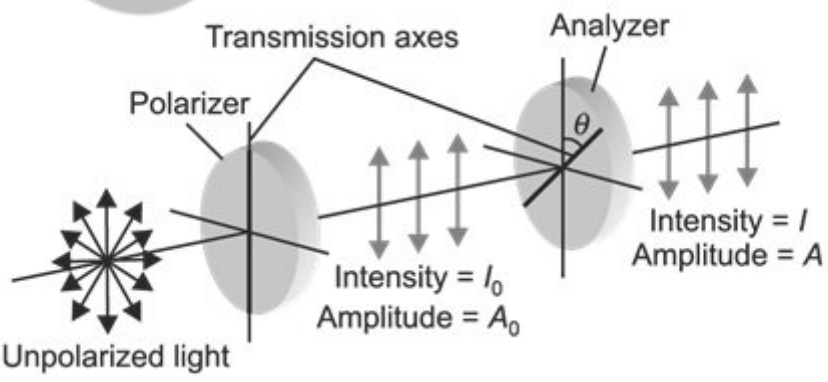
Correct Answer:
18.4°

Solution:

Malus's Law -

<u>Malus's Law-</u>

This law states that the intensity of the polarized light transmitted through the analyzer varies as the square of the cosine of the angle between the plane of transmission of the analyzer and the plane of the polarizer.



As,

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$$I = I_0 \cos^2 \theta$$

output intensity is given by $I = I_0 Cos^2(\theta)$

Initial output intensity=10% of I_0

I.e $\frac{10I_0}{100} = I_0 Cos^2(\theta) \Rightarrow \theta = 71.57$

Final output intensity=O

means new angle is 90^0

the angle by which the analyser needs to be rotated further is $90^0 - \theta = 18.4^0$

So Option (3) is correct.

Q. 4 A long solenoid of radius R carries a time(t) - dependent current $I(t) = I_o t(1-t)$. A ring of radius 2R is placed coaxially near its middle. During the time interval $0 \leq t \leq 1$, the induced current (IR) and the induced EMF (VR) in the ring changes as:-

Correct Answer:
At $t = 0.5$ direction of I_R reverses and v_r is zero

Solution:

Faraday's law of induction -

Faraday's First Law-

Whenever the number of magnetic lines of force (Magnetic Flux) passing through a circuit changes an emf called induced emf is produced in the circuit. The induced emf persists only as long as there is a change of flux.

Faraday's Second Law-

The induced emf is given by the rate of change of magnetic flux linked with the circuit.

i.e Rate of change of magnetic Flux= $\varepsilon = \frac{-d\phi}{dt}$

where $d\phi \rightarrow \phi_2 - \phi_1$= change in flux

So,

$B = \mu_0 n I_0 t(1-t)$

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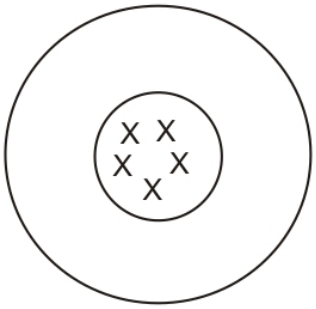
using $B_0 = \mu_0 n I_0$

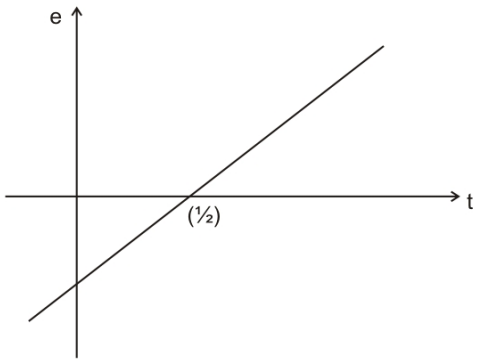
SO $B = B_0 t(1-t)$

Now considering solenoid as ideal solenoid extended up to infinite and ring as its centre



$$\phi = BA_s = B_0 t(1-t)A_s$$

$$e = -\frac{d\phi}{dt} = -B_0 A_s (1-2t)$$



Since induced emf is changing so current will also be changing

because $i = \frac{E}{r}$

So, since direction of emf is changing so direction of current is also changing.

And V_R will be zero at t=0.5 sec

So the option (4) is correct.

Q. 5 The radius of gyration of a uniform rod of length l, about an axis passing through a point $\frac{L}{4}$ away from the centre of the rod and perpendicular to it is:-

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Option 1:

$\frac{1}{8}l$

Option 2:

$\sqrt{\frac{7}{48}}l$

Option 3:

$\sqrt{\frac{3}{8}}l$

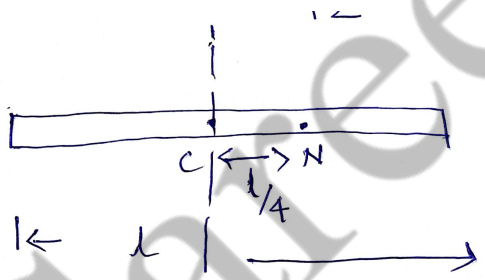
Option 4:

$\frac{1}{4}l$

Correct Answer:

$\sqrt{\frac{7}{48}}l$

Solution:



Moment of inertia of rod about an axis perpendicular through COM

$$I_{COM} = \frac{Ml^2}{12}$$

And $I_N = I_{COM} + M(\frac{l}{4})^2 = \frac{Ml^2}{12} + M(\frac{l^2}{16}) = \frac{7Ml^2}{48}$

Radius of Gyration

$$K = \sqrt{\frac{I_N}{M}} = \sqrt{\frac{7Ml^2}{48M}} = \frac{l}{4} * \sqrt{\frac{7}{3}}$$

$$= l\sqrt{\frac{7}{48}}$$

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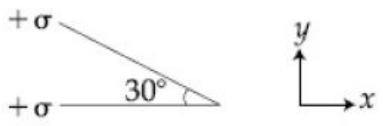
So the option (2) is correct.

Q. 6 Two infinite planes each with uniform surface charge density +σ are kept in each such a way that the angle between them is 300. The electric field in the region shown between them is given by:



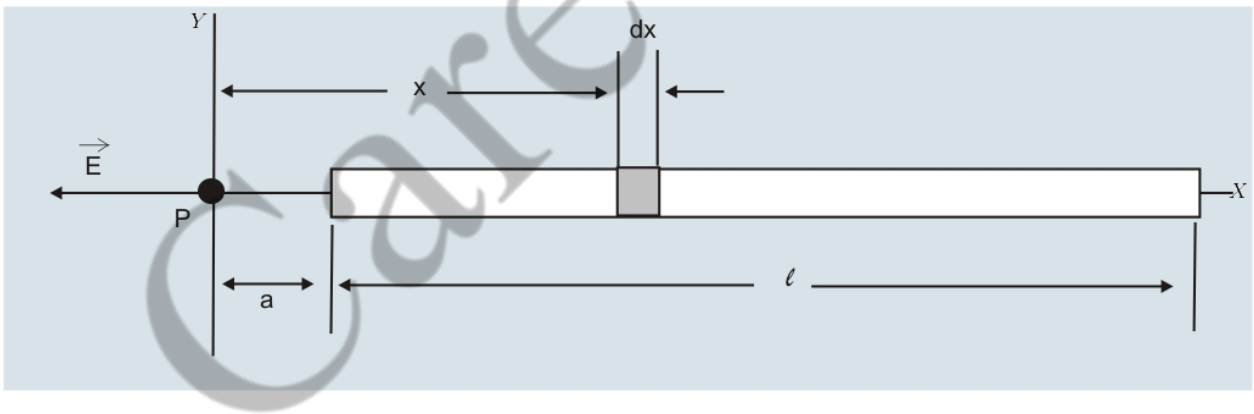
Correct Answer:

$$\frac{\sigma}{2\epsilon_0}\left[\left(1-\frac{\sqrt{3}}{2}\right)\hat{y}-\frac{\hat{x}}{2}\right]$$

Solution:

Electric field due to continuous charge distribution -

So, let us consider a rod of length l which has uniformly positive charge per unit length lying on x-axis, dx is the length of one small section. This rod is having a total charge Q and dq is the charge on dx segment. The charge per unit length of the rod is λ. We have to calculate the electric field at a point P which is located along the axis of the rod at a distance of 'a' from the nearest end of Rod as shown in the figure -



The field $d\vec{E}$ at P due to each segment of charge on the rod is in the negative "x" direction because every segment of the rod carry positive charge. In this every segment of the rod is producing electric field in the negative "x" direction, so the sum of electric field can be added directly and can be integrated because all electric field lies in the same direction.

Now, here - $dq = \lambda.dx$

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$$dE = k_e \frac{dq}{x^2} = k_e \frac{\lambda dx}{x^2}$$

The total lield at P is

$$E = \int_a^{l+a} k_e \lambda \frac{dx}{x^2}$$

If k_e and $\lambda = Q/l$ are constants and can be removed from the integral, then

$$E = k_e \lambda \int_a^{l+a} \frac{dx}{x^2} = k_e \lambda \left[-\frac{1}{x} \right]_a^{l+a}$$

$$\Rightarrow k_e \frac{Q}{l} \left(\frac{1}{a} - \frac{1}{l+a} \right) = \frac{k_e Q}{a(l+a)}$$

Now if we slide the rod toward the origin and the $a \to 0$, then due to that end, the electric field is infinite.

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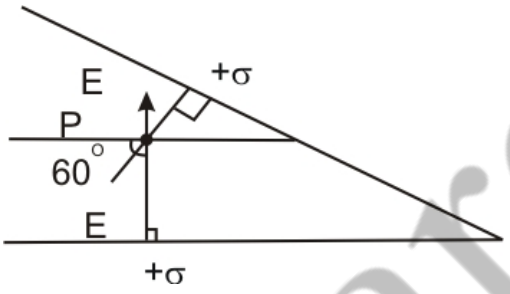
Assuming sheet to be non conducting



$$E_{net} = E((1 - \frac{\sqrt{3}}{2})\hat{j} - \frac{1}{2}\hat{i})$$

So,

$$E = \frac{\sigma}{2\epsilon_0}$$

So,

$$E_{net} = \frac{\sigma}{2\epsilon_0}((1 - \frac{\sqrt{3}}{2})\hat{y} - \frac{1}{2}\hat{x})$$

Where y and x are corresponding unit vectors

So option (4) is correct.

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Q. 7 Two moles of an ideal gas with $\frac{C_p}{C_v}=\frac{5}{3}$ are mixed 3 moles of another ideal gas with $\frac{C_p}{C_v}=\frac{4}{3}$. The value of $\frac{C_p}{C_v}$ for the mixture is:-

Option 1:
1.50

Option 2:
1.45

Option 3:
1.47

Option 4:
1.42

Correct Answer:
1.42

Solution:
For ideal gas:- $C_p-C_v=R$

For first case:-

$$\frac{C_{p1}}{C_{v1}}=\frac{5}{3}\ and\ C_{p1}-C_{v1}=R$$

$$C_{p1}=\frac{5}{3}C_{v1}\ and\ \frac{5}{3}C_{v1}-C_{v1}=R\Rightarrow\frac{2}{3}C_{v1}=R\Rightarrow C_{v1}=\frac{3}{2}R$$

So, $C_{p1}=\frac{5}{2}R$

For second case:-

$$\frac{C_{p2}}{C_{v2}}=\frac{4}{3}\ and\ C_{p2}-C_{v2}=R$$

$$C_{p2}=\frac{4}{3}C_{v2}\ and\ \frac{4}{3}C_{v2}-C_{v2}=R\Rightarrow C_{v2}=3R \text{ and } C_{p2}=4R$$

Now, $Y_{mix}=\frac{n_1C_{p1}+n_2C_{p2}}{n_1C_{v1}+n_2C_{v2}}=\frac{2\times\frac{5}{2}R+3\times4R}{2\times\frac{3}{2}R+3\times3R}=1.417=1.42$

So option (4) is correct.

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Q. 8 Speed of transverse wave of a straight wire (mass=6.0 g, length=60 cm and area of cross-section=1.0 mm2) is 90 ms-1. If the young's modulus of wire is $16 \times 10^{11} Nm^{-2}$, the extension of wire over its natural length is:

Correct Answer:

$0.03\ mm$

Solution:

Hooke's law -

Hooke's law states that if the deformation is small, the stress in a body is proportional to the corresponding strain, i.e.,

$$Stress\ \alpha\ Strain$$

$$\Rightarrow Stress = E(Strain)$$

$$\Rightarrow E = \frac{Stress}{Strain}$$

Where E is called as Modulus of elasticity and it depends on the nature of the material and temperature of the body and is independent of the dimensions of the body.

Unit of Modulus of elasticity= N/m^2

Young's Modulus(Y):- It is defined as the ratio of longitudinal stress to longitudinal strain.

$$Y = \frac{longitudinal\ \ stress}{longitudinal\ \ strain} = \frac{F/A}{\Delta L/L} = \frac{Fl}{A\Delta L}$$

The linear mass density

$\mu = \frac{m}{l} = \frac{6 \times 10^{-3}}{60 \times 10^{-2}}$
$= 10^{-2} Kgm^{-1}$
$given\ v = 90ms^{-1}$
$v = \sqrt{\frac{T}{\mu}}$
$T = \mu v^2 = 81N$
$Youngs\ modulus$
$Y = \frac{\frac{F}{A}}{\frac{\Delta l}{l}}$

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$$\Delta l = \frac{FL}{AY} = \frac{81 \times 0.6}{10^{-6} \times 16 \times 10^{11}} = 0.03mm$$

So option (1) is correct.

Q. 9 The time period of revolution of electron in its ground state orbit in a hydrogen atom is $1.6 \times 10^{-16} s$. The frequency of revolution of the electron in its first excited state (in s^{-1}) is:-

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Correct Answer:
7.8×10^{14}

Solution:
The velocity of the n_{th} orbit

$$V_n \ \alpha \ \frac{z}{n}$$

the radius of the n_{th} orbit

$$r_n \ \alpha \ \frac{n^2}{z}$$

$$T = \frac{2\pi r}{V}$$

$$\Rightarrow T \ \propto \ \frac{n^2 \times n}{z \times z} = \frac{n^3}{z^2}$$

$$\Rightarrow T \propto \frac{1}{f}$$

$$\Rightarrow f \ \propto \ \frac{z^2}{n^3}$$

$$f_1 = \frac{1}{1.6 \times 10^{-16}} \ s^{-1}$$

and

$$\frac{f_1}{f_2} = (\frac{n_2}{n_1})^3 = 8$$

$$\Rightarrow f_2 = \frac{f_1}{8} = 7.8 \times 10^{14} \ s^{-1}$$

So, option (3) is correct.

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Q. 10 A satellite of mass m is launched vertically upwards with an initial speed u from the surface of the earth. After it reaches a height of R(R=radius of r=earth), it ejects a rocket of mass $\frac{m}{10}$ so that subsequently the satellite moves in a circular orbit. The kinetic energy of the rocket is (G is the gravitational constant; M is the mass of the earth):

Correct Answer:

$5m\left(u^2-\frac{119}{200}\frac{GM}{R}\right)$

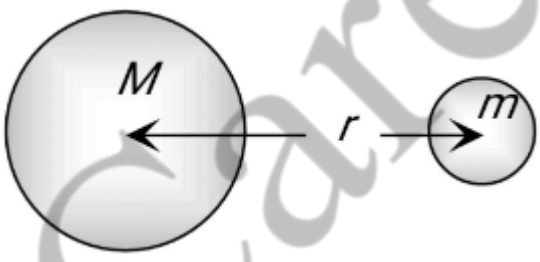
Solution:

Gravitational Potential Energy (U) -

It is the amount of work done in bringing a body from ∞ to that point against gravitational force.

- It is Scalar quantity
- SI Unit: Joule
- Dimension : $\left[ML^2T^{-2}\right]$
- **Gravitational Potential energy at a point**

If the point mass M is producing the field



Then gravitational force on test mass m at a distance r from M is given by $F=\frac{GMm}{R^2}$

And the amount of work done in bringing a body from ∞ to r

$$= W=\int_{\infty}^{r}\frac{GMm}{x^2}dx=-\frac{GMm}{r}$$

And this is equal to gravitational potential energy

So $U=-\frac{GMm}{r}$

$U\rightarrow$ gravitational potential energy

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$M \rightarrow$ Mass of source-body

$M \rightarrow$ mass of test body

$R \rightarrow$ distance between two

Note- U is always negative in the gravitational field because Force is attractive in nature.

Means As the distance r increases U becomes less negative

I.e U will increase as r increases

And for $r = \infty$, U=o which is maximum

- **Gravitational Potential energy of discrete distribution of masses**

$$U = -G\left[\frac{m_1 m_2}{r_{12}} + \frac{m_2 m_3}{r_{23}} + \cdots\right]$$

$U \rightarrow$ Net Gravitational Potential Energy

$r_{12}, r_{23} \rightarrow$ The distance of masses from each other

- **Change of potential energy**

if a body of mass m is moved from r_1 to r_2

Then Change of potential energy is given as

$$\Delta U = GMm\left[\frac{1}{r_1} - \frac{1}{r_2}\right]$$

$\Delta U \rightarrow$ change of energy

$r_1, r_2 \rightarrow$ distances

If $r_1 > r_2$ then the change in potential energy of the body will be negative.

I.e To decrease potential energy of a body we have to bring that body closer to the earth.

- **The relation between Potential and Potential energy**

As $U = \frac{-GMm}{r} = m\left[\frac{-GM}{r}\right]$

But $V = -\frac{GM}{r}$

So $U = mV$

Where $V \rightarrow$ Potential

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$u \rightarrow$ Potential energy

$r \rightarrow$ distance

- Gravitational Potential Energy at the center of the earth relative to infinity

$$U_{centre} = mV_{centre}$$

$V_{centre} \rightarrow Potential\ at\ centre$

$$U = m\left(-\frac{3}{2}\frac{GM}{R}\right)$$

$M \rightarrow$ mass of body

$M \rightarrow$ Mass of earth

- **The gravitational potential energy at height 'h' from the earth's surface**

$$U_h = -\frac{GMm}{R+h}$$

Using $GM = gR^2$

$$U_h = -\frac{gR^2m}{R+h}$$

$$U_h = -\frac{mgR}{1+\frac{h}{R}}$$

$U_h \rightarrow$ The potential energy at the height h

$r \rightarrow$ Radius of earth

-

Before the rocket rejection

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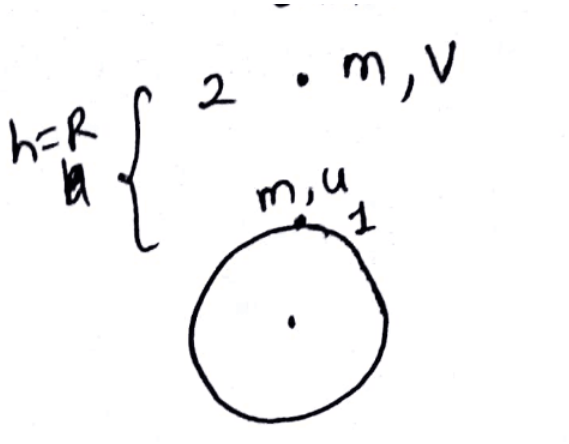
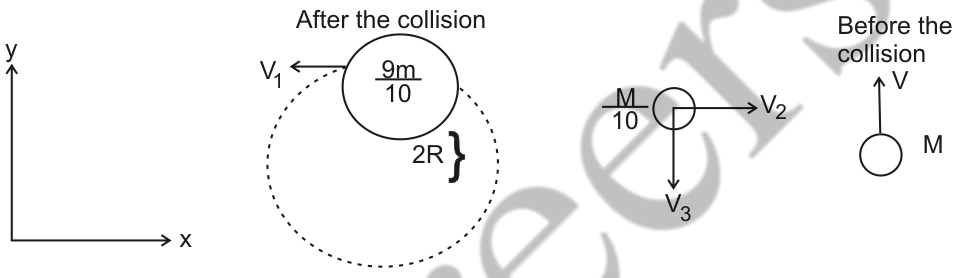


Apply energy conservation

$$\frac{1}{2}mu^2 - \frac{GMm}{R} = \frac{1}{2}mV^2 - \frac{GMm}{2R}$$
$$\Rightarrow \frac{1}{2}mV^2 = \frac{1}{2}mu^2 - -\frac{GMm}{2R}$$
$$\Rightarrow V = \sqrt{u^2 - \frac{GM}{R}}$$

After the rocket rejection



Apply momentum conservation

Along y-axis

$$mV = \frac{m}{10}V_3 \Rightarrow V_3 = 10V$$

Along x-axis

$$\frac{9m}{10}V_1 = \frac{m}{10}V_2 \Rightarrow V_2 = 9V_1$$

And since $V_1 = V_{orbital} = \sqrt{\frac{GM}{2R}}$

So Kinetic energy of the rocket

$$K = \frac{1}{2} * \frac{m}{10} * (V_2^2 + V_3^2) = \frac{1}{2} * \frac{m}{10} * (\frac{81GM}{2R} + 100(u^2 - \frac{GM}{R}))$$
$$K = 5m(u^2 - \frac{119}{200} * \frac{GM}{R})$$

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So option (3) is correct.

Q. 11 If we used a magnification of 375 from a compound microscope of tube length 150mm and an objective of focal length 5mm, the focal length of the eyepiece should be close to :

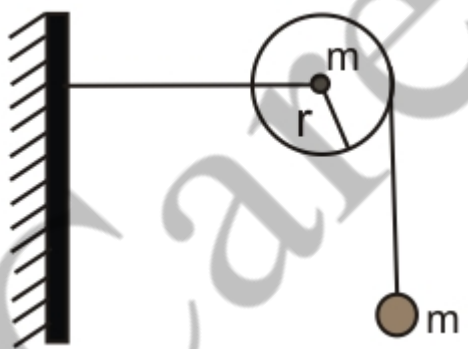
Correct Answer:
22mm

Solution:
The magnification is given by

$$M = \frac{L}{f_0}(1 + \frac{D}{f_e})$$
$$\Rightarrow 375 = \frac{150}{5}(1 + \frac{25}{f_e})$$
$$\Rightarrow f_e = 22\ mm$$

So option (3) is correct.

Q. 12 As shown in the figure, a bob of mass m is tied by a massless string whose other end portion is wound on a flywheel (disc) of radius r and mass m. When released from rest the bob starts falling vertically. When it has covered a distance h, the angular speed of the wheel will be:-



Option 1:
$r\sqrt{\frac{3}{2gh}}$

Option 2:
$r\sqrt{\frac{3}{4gh}}$

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Option 3:

$\frac{1}{r}\sqrt{\frac{4gh}{3}}$

Option 4:

$\frac{1}{r}\sqrt{\frac{2gh}{3}}$

Correct Answer:

$\frac{1}{r}\sqrt{\frac{4gh}{3}}$

Solution:

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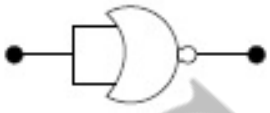
Conservation Of angular momentum -

$$mg - T = ma$$
$$T \times r = I\alpha$$
$$T = \frac{mr^2}{2} \times \frac{a}{r} \times \frac{1}{r}$$
$$T = \frac{ma}{2}$$
$$mg = \frac{3ma}{2}$$
$$a = \frac{2g}{3}$$
$$Also,\ v = \sqrt{2as} = \sqrt{\frac{4gh}{3}}$$
$$also, v = \omega r$$
$$\omega = \frac{v}{r}$$
$$\Rightarrow \sqrt{\frac{4gh}{3}} \times \frac{1}{r} = \frac{1}{r}\sqrt{\frac{4gh}{3}}$$

Hence option (3) is correct.

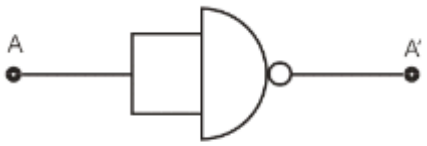
Q. 13 Which of the following gives a reversible operation?

Correct Answer:



Solution:

For reversible operation, NOT gate is used. If an input is Athen output= $\bar{A}$. The following circuit represents the NOT gate



So Option (2) is correct.

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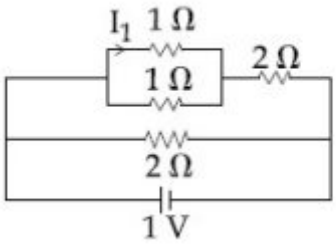
Q. 14 The current I1 (in A) flowing through 1Ω resistor in the following circuit is:-



Correct Answer:
0.25

Solution:

Ohms Law -

1. Ohm's law

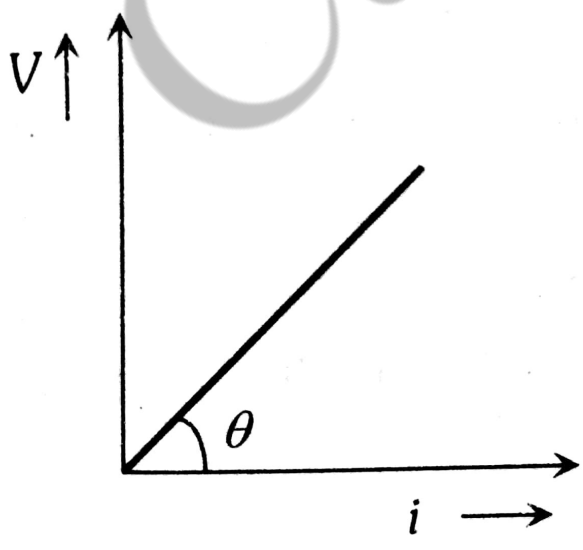
In a conductor, if all external physical conditions like temperature and pressure are kept constant the Current flowing through a conductor is directly proportional to the Potential difference across two ends.

$V \propto I$

$V = IR$

$R-$ Electric Resistance

- The graph between V and I



The slope gives the resistance

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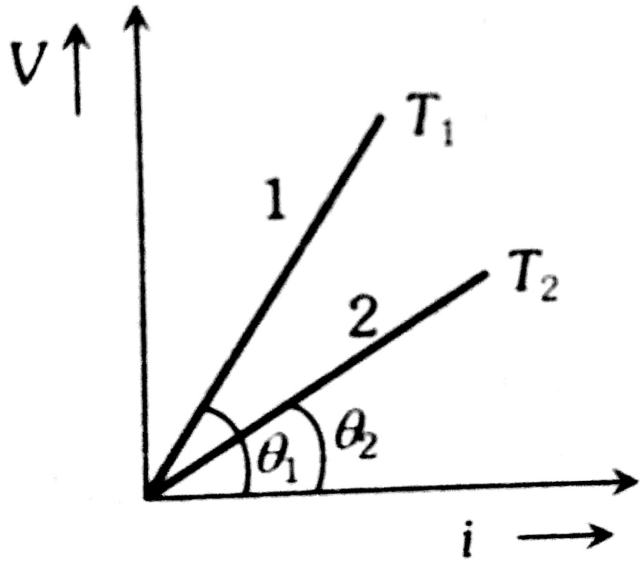
- The graph between V and I at different temperatures



Here T1>T2. The resistance of a conductor increases with increase in temperature

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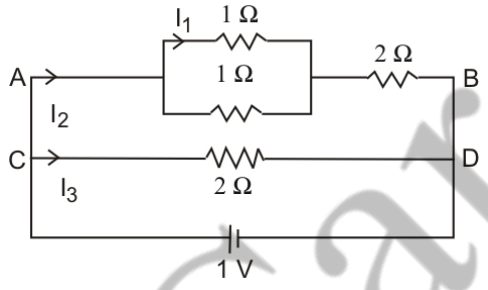

$$R_{AB} = \frac{1}{2} + 2 = \frac{5}{2}$$

$$R_{CD} = 2$$

$$R_{eq} = \frac{2.5 * 2}{2.5 + 2} = \frac{10}{9}$$

$$I = \frac{V}{R_{eq}} = \frac{10}{9}$$

As

$$\Delta V_{AB} = \Delta V_{CD}$$
$$\Rightarrow 2.5 * I_2 = 2 * I_3 ...(1)$$

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and $I_2 + I_3 = I = \frac{10}{9} ...(2)$

From equation (1) and (2)

$I_3 = \frac{50}{81}, I_2 = \frac{40}{81},$

and $I_1 = \frac{I_2}{2} = \frac{20}{81} = 0.2469$

$I_1 = 0.25$ (when considered up to two decimal point)

Or,

$I_1 = 0.2$ (when considered up to one decimal point)

Answer given by the NTA is $I_1 = 0.2$ But most appropriate answer is $I_1 = 0.25$

Q. 15 A LCR circuit behaves like a damped harmonic oscillator. Comparing it with a physical spring-mass damped oscillator having dampling constant 'b', the correct equivalence would be:-

---

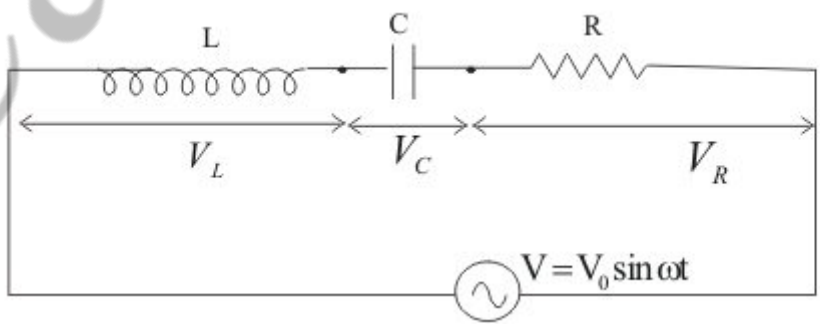
Correct Answer:

$L \leftrightarrow m, C \leftrightarrow \frac{1}{k}, R \leftrightarrow b$

Solution:

Series LCR circuit -

Series LCR circuit-



The Figure given above shows a circuit containing a capacitor ,resistor and inductor connected in series through an alternating / sinosoidal voltage source.

As they are in series so the same amount of current will flow in all the three circuit components and for the voltage, vector sum of potential drop across each component would be equal to the applied voltage.

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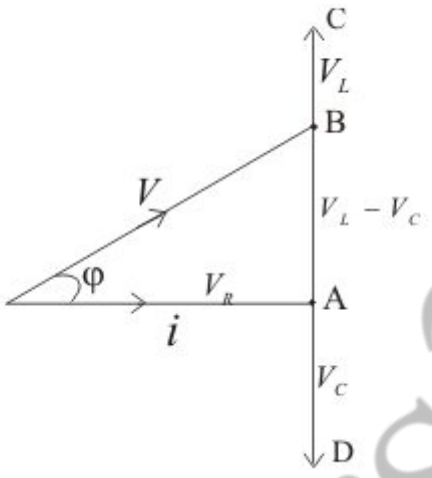
Let 'i' be the amount of current in the circuit at any time and V_L,V_C and V_R the potential drop across L,C and R respectively then

$$v_R = iR \rightarrow \text{ Voltage is in phase with i}$$

$$v_{L=i\omega L} \rightarrow \text{ voltage is leading i by } 90^\circ \qquad \varepsilon$$

$$v_c = i/\omega c \rightarrow \text{ voltage is lagging behind i by } 90^\circ$$

By all these we can draw phasor diagram as shown below -



One thing should be noticed that we have assumed that V_L is greater than V_C which makes i lags behind V. If $V_C > V_L$ then i lead V. So as per our assumption, there resultant will be (V_L -V_C). So, from the above phasor diagram V will represent resultant of vectors V_R and (V_L -V_C). So the equation become -

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$= i\sqrt{R^2 + (X_L - X_C)^2}$$

$$= i\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$= iZ$$

where,

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

Here, Z is called Impedence of this circuit.

Now come to the phase angle. The phase angle for this case is given as -

$$\tan\varphi = \frac{V_L - V_C}{V_R} = \frac{X_L - X_C}{R} = \frac{\omega L - \frac{1}{\omega C}}{R}$$

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Now from the equation of the phase angle three cases will arise. These three cases are -

(i) When, $\omega L > \frac{1}{\omega C}$

then, tanφ is positive i.e. φ is positive and voltage leads the current i.

(ii) When $\omega L < \frac{1}{\omega C}$

then, tanφ is negative i.e. φ is negative and voltage lags behind the current i.

(iii) When $\omega L = \frac{1}{\omega C}$,

then tanφ is zero i.e. φ is zero and voltage and current are in phase. **This is called electrical resonance.**

-

For the spring-mass damping system, the governing equation is given by

$$m\frac{dx^2}{dt^2} + b\frac{dx}{dt} + kx = 0.....(1)$$

For an LCR damped oscillator, the equation is

$$L\frac{dQ^2}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = 0.....(2)$$

Comparing 1 and 2

$L \leftrightarrow m$
$C \leftrightarrow \frac{1}{k}$
$R \leftrightarrow b$

So the answer is option (3)

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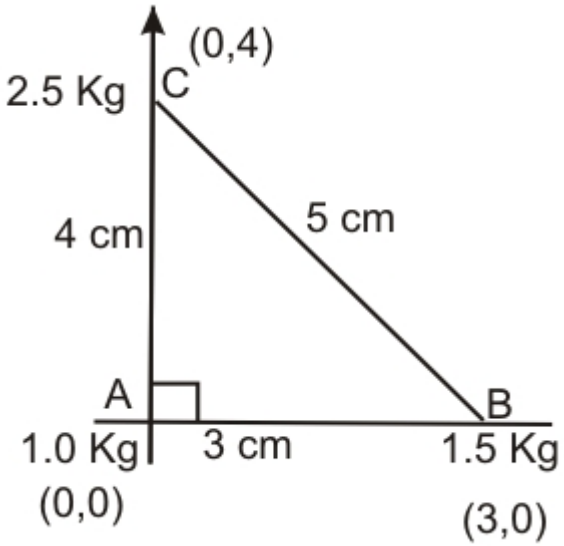
Q. 16 Three point particles of masses 1.0 kg, 1.5 kg and 2.5 kg are placed at three corners of a right triangle of sides 4.0 cm, 3.0 cm, and 5.0 cm as shown in the figure. The centre of mass of the system is at a point:



Option 1:
$2.0\ cm$ right and $0.9\ cm$ above $1\ kg$ mass

Option 2:
$0.9\ cm$ right and $2.0\ cm$ above $1\ kg$ mass

Option 3:
$0.6\ cm$ right and $2.0\ cm$ above $1\ kg$ mass

Option 4:
$1.5\ cm$ right and $1.2\ cm$ above $1\ kg$ mass

Correct Answer:
$0.9\ cm$ right and $2.0\ cm$ above $1\ kg$ mass

Solution:

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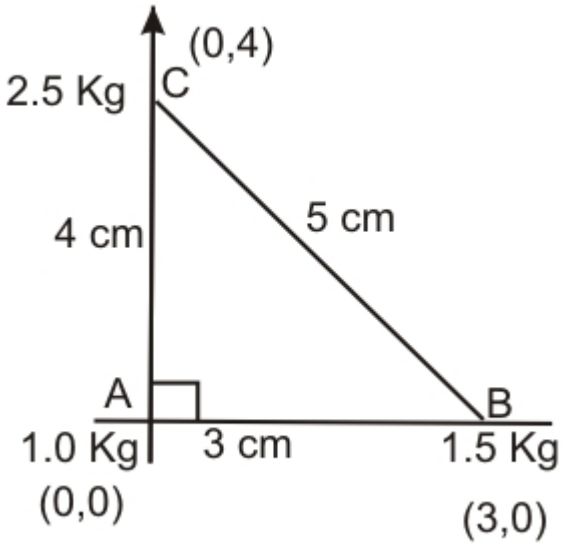

let m1=1 kg, m2=1.5 kg and m3=2.5 kg

x1=0, x2=3, x3=0 and y1=0, y2=0, y3=4

$x_{com} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$ and $y_{com} = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3}$

Let point A be origin and mass m_1=1.0 kg be at origin.

So, $x_{com} = \frac{1 \times 0 + 1.5 \times 3 + 2.5 \times 0}{1 + 1.5 + 2.5} = \frac{4.5}{5} = 0.9$

and $y_{com} = \frac{1 \times 0 + 1.5 \times 0 + 2.5 \times 4}{1 + 1.5 + 2.5} = \frac{10}{5} = 2$

so centre of mass of the system is at (0.9,2).

So from the figure we can say that the 0.9 cm right and 2 cm above the 1 kg mass.

So option (2) is correct.

Q. 17 If the magnetic field in a plane electromagnetic wave is given by $\vec{B} = 3 \times 10^{-8} sin(1.6 \times 10^3 x + 48 \times 10^{10} t)\ \vec{j}\ T$; then what will be the expression for the electric field?

Correct Answer:

$\vec{E} = \left(9 \sin\left(1.6 \times 10^3 x + 48 \times 10^{10} t\right) \hat{k}\ V/m\right)$

Solution:

Nature of Electromagnetic Waves -

It is also seen from Maxwell's equations that the magnitude of the electric and the magnetic fields in an electromagnetic wave are related as - $B_0 = \frac{E_o}{c}$

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given, $\vec{B} = 3 \times 10^{-8} sin(1.6 \times 10^3 x + 48 \times 10^{10} t)T$

$$\left|\vec{E}\right| = BC = 3 \times 10^{-8} sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \times (3 \times 10^8)$$
$$= 9sin(1.6 \times 10^3 x + 48 \times 10^{10} t)T$$

wave is propagating in -x direction, i.e., in - i direction.

the direction of the EMW wave is in the direction of $\vec{E} \times \vec{B}$.

Since B is in j direction and EMW is in -i direction. Therefore E is in (k) direction.

So Option (1) is correct.

Q. 18 Visible height of wavelength 6000×10^{-8} cm falls normally on a single slit and produces a diffraction pattern. It is found that the second diffraction min. is at 60 from the central max. If the first minimum is produced at θ_1, the θ_1 is close to,

---

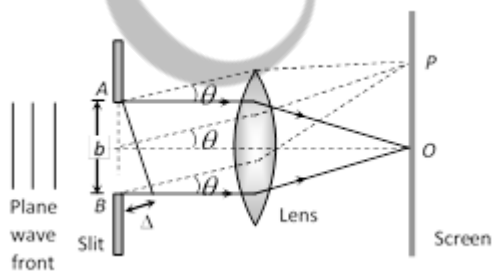
Correct Answer:
25^o

Solution:

Fraunhofer diffraction by a single slit -

Fraunhofer diffraction by a single slit

let's assume a plane wave front is incident on a slit AB (of width b). Each and every part of the exposed part of the plane wave front (i.e. every part of the slit) acts as a source of secondary wavelets spreading in all directions. The diffraction is obtained on a screen placed at a large distance. (In practice, this condition is achieved by placing the screen at the focal plane of a converging lens placed just after the slit).



- The diffraction pattern consists of a central bright fringe (central maxima) surrounded by dark and bright lines (called secondary minima and maxima).
- At point O on the screen, the central maxima is obtained. The wavelets originating from points A and B meets in the same phase at this point, hence at O, intensity is maximum

Secondary minima : For obtaining nth secondary minima at P on the screen, path difference between the diffracted waves

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$$\Delta x = b \sin\theta = n\lambda$$

1. Angular position of nth secondary minima:

$$\sin\theta \approx \theta = \frac{n\lambda}{b}$$

2. Distance of nth secondary minima from central maxima:

$x_n = D \cdot \theta = \frac{n\lambda D}{b}$ where D = Distance between slit and screen.

$f \approx D$ = Focal length of converging lens.

Secondary maxima : For nth secondary maxima at P on the screen.

Path difference $\Delta x = b \sin\theta = (2n+1)\frac{\lambda}{2}$; where n = 1, 2, 3 .....

(i) Angular position of nth secondary maxima

$$\sin\theta \approx \theta \approx \frac{(2n+1)\lambda}{2b}$$

(ii) Distance of nth secondary maxima from central maxima:

$$x_n = D \cdot \theta = \frac{(2n+1)\lambda D}{2b}$$

Central maxima : The central maxima lies between the first minima on both sides.

(i) The Angular width d central maxima = $2\theta = \frac{2\lambda}{b}$

(ii) Linear width of central maxima $= 2x = 2D\theta = 2f\theta = \frac{2\lambda f}{b}$

Intensity distribution: if the intensity of the central maxima is I_0 then the intensity of the first and secondary maxima are

found to be $\frac{I_0}{22}$ and $\frac{I_0}{61}$. Thus diffraction fringes are of unequal width and unequal intenstities.

(i) The mathematical expression for in intensity distribution on the screen is given by:

$I = I_o \left(\frac{\sin\alpha}{\alpha}\right)^2$ where α is just a convenient connection between the angle θ that locates a point on the viewing screening and light intensity I.

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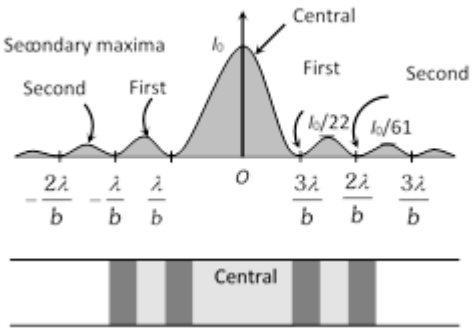

$\phi =$ Phase difference between the top and bottom ray from the slit width b.

Also $\alpha = \frac{1}{2}\phi = \frac{\pi b}{\lambda}\sin\theta.$

(ii) As the slit width increases relative to wavelength the width of the control diffraction maxima decreases; that is, the light undergoes less flaring by the slit. The secondary maxima also decreases in width and becomes weaker.

(iii) If b>>λ, the secondary maxima due to the slit disappear; we then no longer have single slit diffraction.

-

$d\ sin\theta = path\ difference = 2\lambda$

So,

$d\ sin\ 60 = 2\lambda$

$\Rightarrow \frac{\lambda}{d} = \frac{\sqrt{3}}{4}$

For first minima,

$d\ sin\theta_2 = \lambda$

$sin\theta_2 = \frac{\lambda}{d} = \frac{\sqrt{3}}{4}$

$\Rightarrow \theta_2 = sin^{-1}(\frac{\sqrt{3}}{4}) = 25.64^\circ = 25^\circ$

So option (3) is correct.

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Q. 19 A parallel plate capacitor has plates of area A separated by distance'd' between them. It is filled with a dielectric which has a
dielectric constant that varies as $\mathrm{k}(x) = \mathrm{K}(1 + \alpha x)$ where $'x'$ is the distance measured from one of the plates. If $(\alpha \mathrm{d}) << 1$, the total capacitance of the system is best given by the expression:

Correct Answer:
$\frac{AK\epsilon_0}{d}\left(1+\frac{\alpha d}{2}\right)$

Solution:

Dielectrics -

Dielectric: A dielectric is an insulating material in which all the electrons are tightly bounded to the nuclei of the atoms and no free electrons are available for the conduction of current. They are non-conducting materials. They do not have free charged particles like conductors have. They are of two types.

1. Polar : The centre of +ve and –ve charges do not coincide. Example HCl, H_2O, They have their own dipole moment

2. Non-Polar : The centers of +ve and –ve charges coincide. Example CO_2, C_6H_6. They do not have their own dipole moment.

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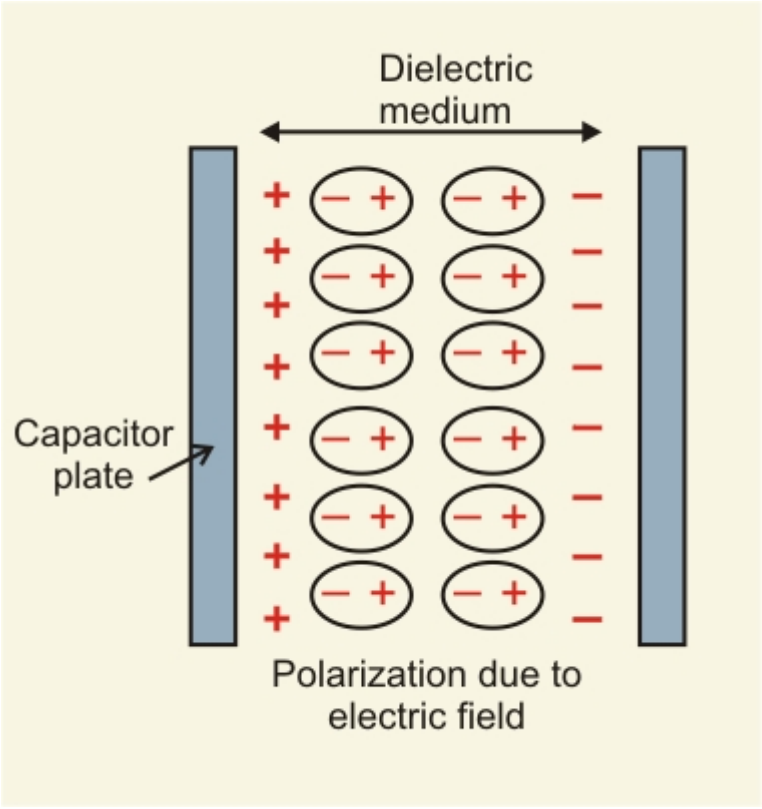

When a dielectric slab is exposed to an electric field, the two charges experience force in opposite directions. The molecules get elongated and develops a surface charge density σ_p . This leads to development of an induced electric field Ep , which is in opposition direction of external electric field Eo . Then net electric field E is given by $E = E_o - E_i$.

This indicates that net electric field is decreased when dielectric is introduced.

The ratio $\frac{E_0}{E} = K$ is called dielectric constant of the dielectric. Hence, Electric field inside a dielectric is $E_i = \frac{E_0}{K}$.

$$E = E_0 - E_i \text{ and } E = \frac{E_0}{k}$$

$$So,\ E_0 - E_i = \frac{E_0}{K}$$

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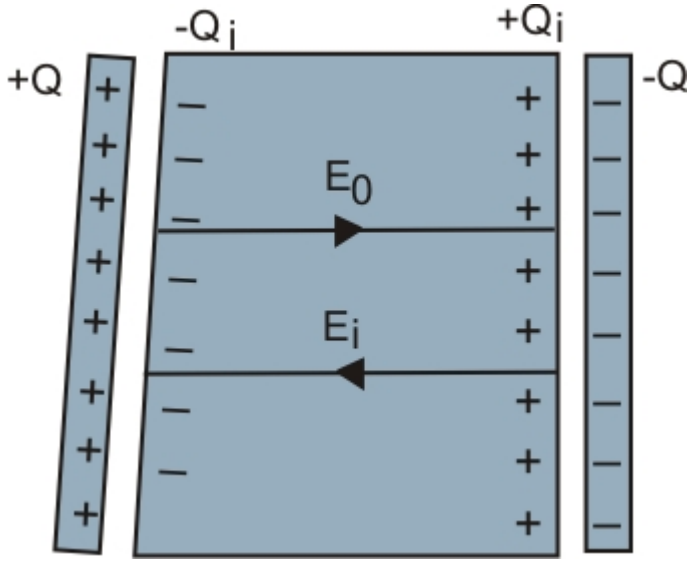
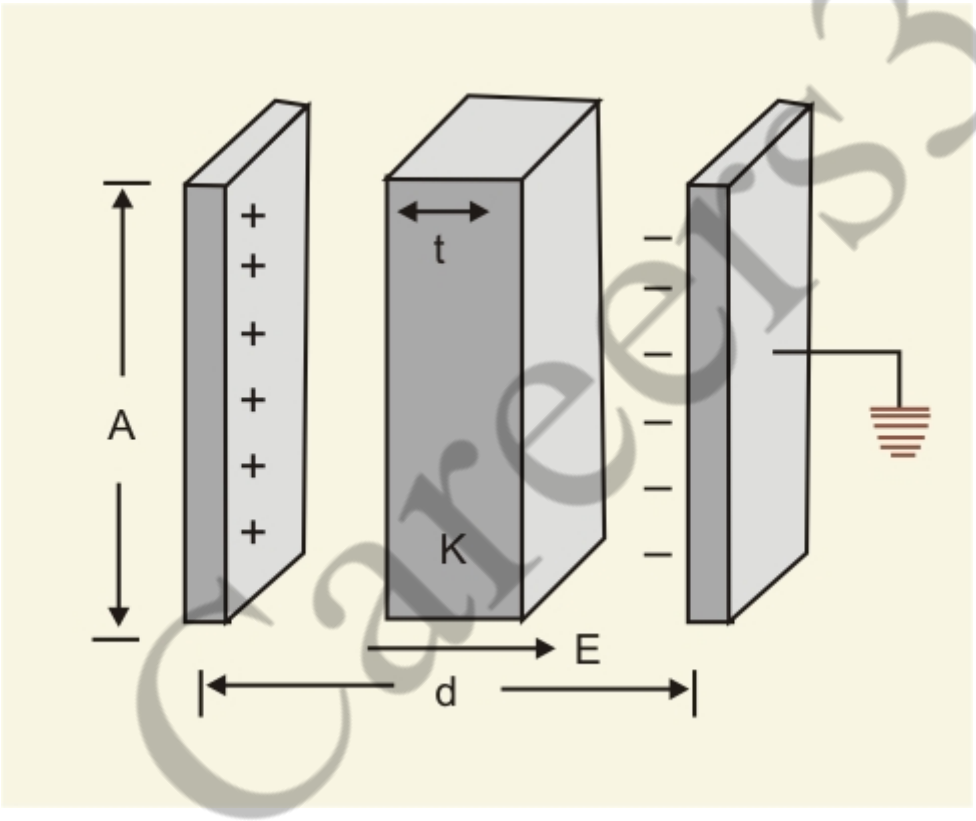


This is irrespective of the thickness of the dielectric slab,i.e., whether it fills up the entire space between the charged plates or any part of it.



$$C' = \frac{\epsilon_0 A}{d - t + \frac{t}{k}}$$

-

If K filled between the plates -

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$$C' = K\frac{\epsilon_0 A}{d} = C' = Ck$$

Given $K = K_0 + K_0\alpha x$

$$V = -\int_0^d Edr = V = \int_0^d \frac{\sigma}{K\epsilon_0}dx$$

$$V = \frac{\sigma}{\varepsilon_0}\int_0^d \frac{1}{K + K\alpha x}dx = \frac{\sigma}{K\alpha\varepsilon_0}[ln(K + K\alpha d) - lnK]$$

$$V = \frac{\sigma}{K\alpha\varepsilon_0}ln(1 + \alpha d)$$

For

$$ln(1 + \alpha d) = \alpha d - \frac{(\alpha d)^2}{2} + \frac{(\alpha d)^3}{3}.....$$

$$V = \frac{\sigma}{K\alpha\varepsilon_0}ln(1 + \alpha d)$$

$$C = \frac{Q}{V} = \frac{\sigma A}{V} = \frac{\sigma A}{\frac{\sigma}{K\varepsilon_0\alpha}ln(1 + \alpha d)}$$

here, $C_0 = \frac{\varepsilon_0 A}{d}$

$$C = \frac{Kd\alpha}{ln\,(1 + \alpha d)}C_0 = \frac{KdC_0\alpha}{\alpha d - \frac{(\alpha d)^2}{2} + \frac{(\alpha d)^3}{3}.....} = \frac{KdC_0\alpha}{(\alpha d) * (1 - \frac{\alpha d}{2} + \frac{(\alpha d)^2}{3}...)}$$

since $\alpha d << 1$

So

$$C = \frac{KdC_0\alpha}{(\alpha d) * (1 - \frac{\alpha d}{2})} = \frac{KdC_0\alpha}{(\alpha d)} * (1 - \frac{\alpha d}{2})^{-1} = \frac{KdC_0\alpha}{(\alpha d)} * (1 + \frac{\alpha d}{2})$$

$$= \frac{AK\varepsilon_0}{d}(1 + \frac{\alpha d}{2})$$

So, option (2) is correct.

Q. 20 A liter of dry air at STP expands adiabatically to a volume of 3l. If $\gamma = 1.40$, then work done (in J) by air is $(3^{1.4} = 4.6555)$ [take air to be an ideal gas]:-

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Correct Answer:
90.5

Solution:

Adiabatic process -

Work done in adiabatic process -

$$W = \int_{V_i}^{V_f} PdV = \int_{V_i}^{V_f} \frac{K}{V^\gamma} dV \Rightarrow W = \frac{[P_i V_i - P_f V_f]}{(\gamma - 1)} = \frac{\mu R (T_i - T_f)}{(\gamma - 1)}$$

$$W_{adiabatic} = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

At STP, Pressure = P_1 = 100000 Pascal

Volume = V_1 = 1 liter (As given in question)

So,

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$\Rightarrow \ 1 \times (1)^\gamma = P_2 \times (3)^{1.4}$$

$$P_2 = \frac{1}{4.65} \ atm$$

$$W_{adiabatic} = (\frac{1 \ \times 1 - \frac{1}{4.65} \times 3}{0.4}) \times 101.33$$

$$= 90.5 \ Joule$$

So, option (3) is correct.

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Q. 21 A non isotropic solid metal cube has coefficients of linear expansion as $5 \times 10^{-5}/^oC$ along the x-axis and $5 \times 10^{-6}/^oC$ along the y-axis and z-axis. If the coefficient of volume expansion of the solid is $C \times 10^{-6}/^oC$, then the value of C is:-

Correct Answer:
60

Solution:

Thermal Expansion and its types -

Let the length of cube be L.

Given,$\alpha_x = 5 \times 10^{-5}/^oC$ and $\alpha_y = 5 \times 10^{-6}/^oC = \alpha_z$

Let the coefficient of volume expansion be γ

After increase in temperature:-

Increase in volume of cube=$L^3 \times \gamma - - - - - - - - - - - - (1)$

and we can also write it as:- $L(1+\alpha_x)(1+\alpha_y)(1+\alpha_z) - - - - - - - - - - - - - - (2)$\

from equation 1 and 2 we get:-

$\gamma L^3 = L^3(1+\alpha_x)(1+\alpha_y)(1+\alpha_z)$

$\Rightarrow \gamma = (\alpha_x + \alpha_y + \alpha_z)\ by\ ignoring\ \alpha_x\alpha_y,\ \alpha_x\alpha_z,\ \alpha_y\alpha_z, and\ \alpha_x\alpha_y\alpha_z$

$\Rightarrow \gamma = (5 \times 10^{-5} + 5 \times 10^{-6} + 5 \times 10^{-6})$

$\Rightarrow \gamma = 6 \times 10^{-5}/^oC$

From question:- $\gamma = C \times 10^{-6}/^oC$

So, C=60

SO the correct answer is option 1.

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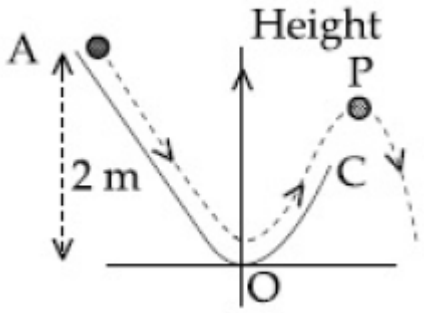
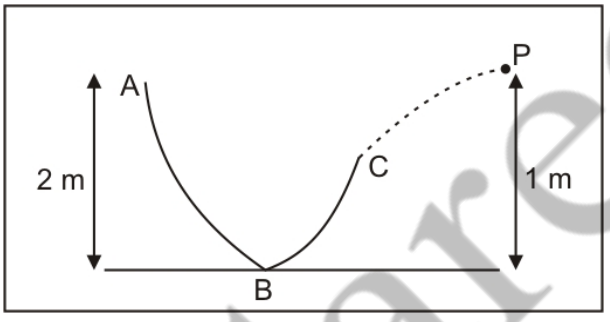
Q. 22 A particle (m=1 kg) slides down a frictionless block(AOC) starting from rest at a point A (height 2 m). After reaching C, the particle continues to move freely in air as a projectile. When it reaches its highest point P (height 1 m), the kinetic energy of the particle (in J) is (take g=10 ms^{-2}):-



Correct Answer: 10

Solution:

Law of Conservation of Energy -



Loss of potential energy= Gain in kinetic energy

$$1 \times 10(2-1) = K_f - K_i = K_f - 0$$
$$K_f = 10J$$

So, Correct Answer is option 4.

Q. 23 A carnot engine separates between two reservoirs of temperature 900 K and 300 K. The engine performs 1200 J of work per cycle. The heat energy (in J) delivered by the engine to the low temperature reservoir in a cycle is:-

Correct Answer:
600

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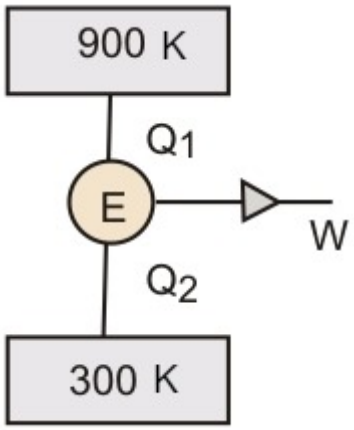
Solution:
Carnot Engine -

So $\eta = \frac{W}{Q_1} = 1 - \frac{T_2}{T_1}$

where

$T_1 =$ Source temperature, $T_2 =$ Sink Temperature and $(T_1 > T_2)$

and $T_1 \, and \, T_2$ are in kelvin



given, W=1200 J

$\eta_{carnot} = 1 - \frac{T_L}{T_H} = 1 - \frac{300}{900} = \frac{6}{9}$

$\eta_{carnot} = \frac{W}{Q_1}$

So, $\frac{W}{Q_1} = \frac{6}{9} \Rightarrow Q_1 = 1800J$

So, $W = Q_1 - Q_2 \Rightarrow Q_2 = Q_1 - W = 1800 - 1200 = 600J$

So, the Correct answer is option 1.

Q. 24 A beam of electromagnetic radiation of intensity $6.4 \times 10^{-5} W/cm^2$ is comprised of wavelength $\lambda = 310nm$. It falls normally on a metal (work function ϕ=2 eV) of surface area 1 cm^2. If one in 10^3 photons ejects an electron, total number of electrons ejected in 1s is 10^x (hc=1240 eVnm, 1 eV=1.6 x 10^{-19}J), then x is :-

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Correct Answer:
11

Solution:
Energy in 1 second = $6.4 \times 10^{-5} \times \ 1 \ \times 1 = 6.4 \times 10^{-5} Joule$

$$Energy(eV) = \frac{6.4 \times 10^{-5}}{1.6 \times 10^{-19}} = 4 \times 10^{14} eV$$

$$Energy\ of\ one\ photon = \frac{12400}{3100} = 4\ eV$$

$$Number\ of\ photon\ = \frac{4 \times 10^{14}}{4} = 10^{14}$$

$$Number\ of\ electrons\ = \frac{10^{14}}{10^{3}} = 10^{11}$$

$$So,\ x = 11$$

So the correct answer is option 1.

Q. 25 A loop ABCDEFA of straight edges has six corner points A(0,0,0), B(5,0,0),C(5,5,0), D(0,5,0), E(0,5,5) and F(0,0,5). The magnetic field in this region is $\vec{B} = (3\hat{i} + 4\hat{j})T$.The quantity of flux through the loop ABCDEFA (in Wb) is:-

Correct Answer:
175

Solution:

Magnetic flux - - wherein

Magnetic flux-

The total number of magnetic lines of force passing normally through an area placed in a magnetic field is equal to
the magnetic flux linked with that area.

I.e for the below figure

Net magnetic flux through the surface is given by

$$\phi_B = \oint \vec{B} \cdot d\vec{A} = BA\cos\Theta$$

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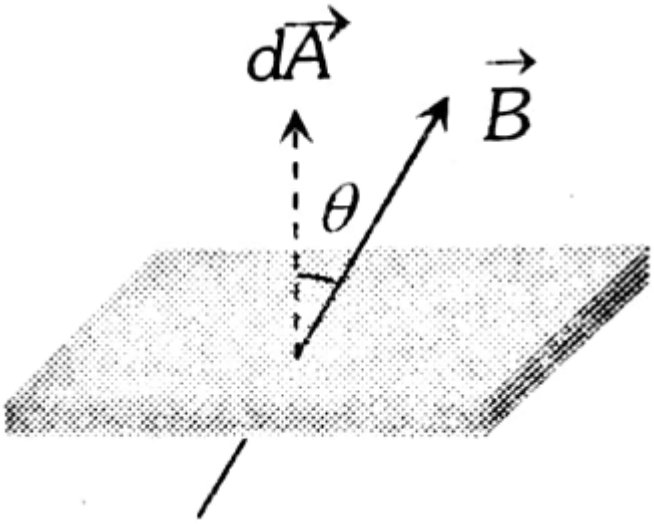
where

$\phi_B =$ Magnetic Flux

$B =$ Magnetic field

$\Theta =$ The angle between area vector and magnetic field vector



- Magnetic flux is a scalar quantity.
- Unit of magnetic flux -

  It's S.I. unit is Weber (wb) or $Tesla \times m^2$ and its C.G.S. unit is maxwell(Mx).

  and $1\ wb = 1\ Tm^2$ and $1\ Mx = 10^{-8}\ wb$

- The dimension of magnetic flux is $ML^2T^{-2}A^{-1}$
- if θ = 0 then $\phi = BA$ and Flux will be positive.
- If $\theta = \frac{\pi}{2}$ then Flux will be zero (i.e $\phi = 0$)

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$\vec{B} = (3\hat{i} + 4\hat{j})T$

Total area vectot=area of ABCD+area of DEFA=$5^2\hat{k} + 5^2\hat{i} = 25(\hat{i} + \hat{k})$

Total Magnetic flux=$\vec{B}.\vec{A} = (3\hat{i} + 4\hat{j}).(25(\hat{i} + \hat{k})) = (75 + 100)wb = 175wb$

So option 3 will be correct answer.

Chemistry

Q. 1 At 35^{o}C, the vapour pressure of CS_2 is 512mm Hg and that of acetone is 344mm Hg. A solution of CS_2 in acetone has a total vapour pressure of 600mm Hg. The false statement among the following is:

Option 1:
Heat must be absorbed in order to produce the solution at 35^{o} C

Option 2:
a mixture of 100 mL CS_2 and 100 mL acetone has a volume <200 mL

Option 3:
CS_2 and acetone are less attracted to each other than to themselves

Option 4:
Raoult's law is not obeyed by this system

Correct Answer:
a mixture of 100 mL CS_2 and 100 mL acetone has a volume <200 mL

Solution:
We have:

$P_{observed}$ = 600Hg

We know:

$P_{calc} = P^{o}_{A}x_A + P^{o}_{B}x_B$
= 512x + 344 - 344x
= 168x + 344
= 512 (if x =1 for maximum value)

Thus, $P_{calc} < P_{observed}$
Thus, the process is deviated from Raoult's law, showing positive deviation from Raoult's law.

$\Delta H > 0$, i.e, heat absorbed.

This means solution will show +ve deviation Hence

$V_s > V_A + V_B$

200 < 100 + 100

it is wrong.

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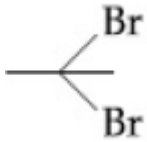
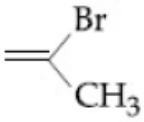
Therefore, **Option(2) is correct.**

Q. 2 1-methyl ethylene oxide when treated with an excess of HBr produces:

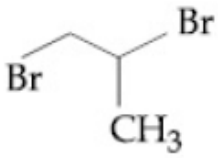
Option 1:



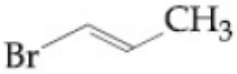
Option 2:



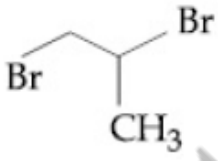
Option 3:



Option 4:



Correct Answer:



Solution:

Markonikov and Anti-markonikov Reaction -

Markovnikov's rule

This rule states that the acid hydrogen of the protic acid gets attached to the carbon with more hydrogen substituents and negative part adds to the atom with less number of hydrogen atoms.

Mechanism

The addition of halogens and halogen acids takes place by electrophilic addition(EA) reaction. +E mechanism is that when electrons of the π-bonds are transferred to that atomm of the multiple bond to which the reagent finally gets attached. First the electrophile(H^+) adds to the positive C atom and hence this step is slow and the rate-determining step. Afterwards the negative part of the reagent (Br^-) adds to the positive C atom. Thus, it is known as (+E) reaction.

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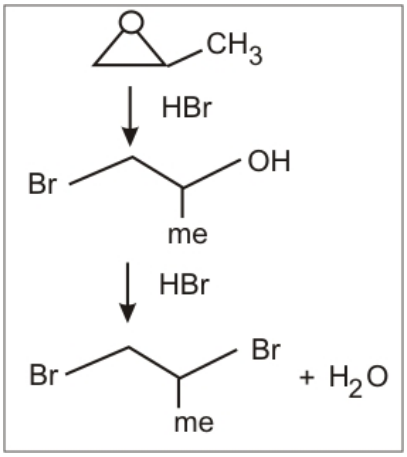
Anti-Markovnikov's rule

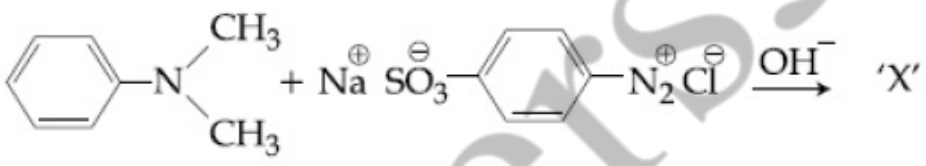
In the presence of peroxide, such as benzoyl peroxide and light, the addition of HBr(not HCl and HI) to unsymmetrical alkenes occurs contrary to Markovnikov's rule.

-



Therefore, **Option(3) is correct.**

Q. 3 Consider the following reaction:



The product 'X' is used :

Option 1:

In protein estimation as an alternative to ninhydrin

Option 2:

in laboratory test for phenols

Option 3:

as food grade colourant

Option 4:

In acid base titration as an indicator

Correct Answer:

In acid base titration as an indicator

Solution:

The reaction will be -

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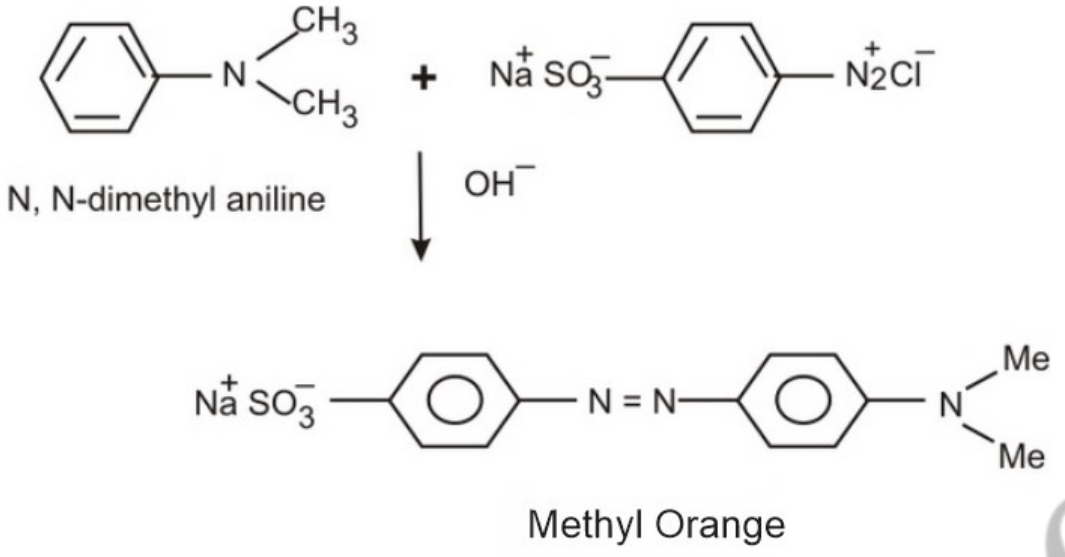

Methyl orange is used in acid-base titration as an indicator.

Therefore, **Option(4) is correct.**

Q. 4 A solution of m-chloroaniline, m-chlorophenol and m-chlorobenzoic acid in ethyl acetate was extracted initially with a saturated solution of $NaHCO_3$ to give fraction A. The leftover organic phase was extracted with dilute NaOH solution to give fraction B. The final organic layer was labelled as fraction C. Fractions A, B and C contain respectively:

Option 1:
m-chlorobenzoic acid, m-chloroaniline and m-chlorophenol

Option 2:
m-chlorobenzoic acid, m-chlorophenol and m-chloroaniline

Option 3:
m-chlorophenol,m-chlorobenzoic acid and m-chloroaniline

Option 4:
m-chloroaniline ,m-chlorobenzoic acid and m-chlorophenol

Correct Answer:
m-chlorobenzoic acid, m-chlorophenol and m-chloroaniline

Solution:
As we have learnt,

Carboxylic acids react with $\mathrm{NaHCO_3}$ while Phenol and Aniline do not.

Out of phenol and aniline, only Phenol reacts with NaOH.

Therefore, the given reactions occur as follows:

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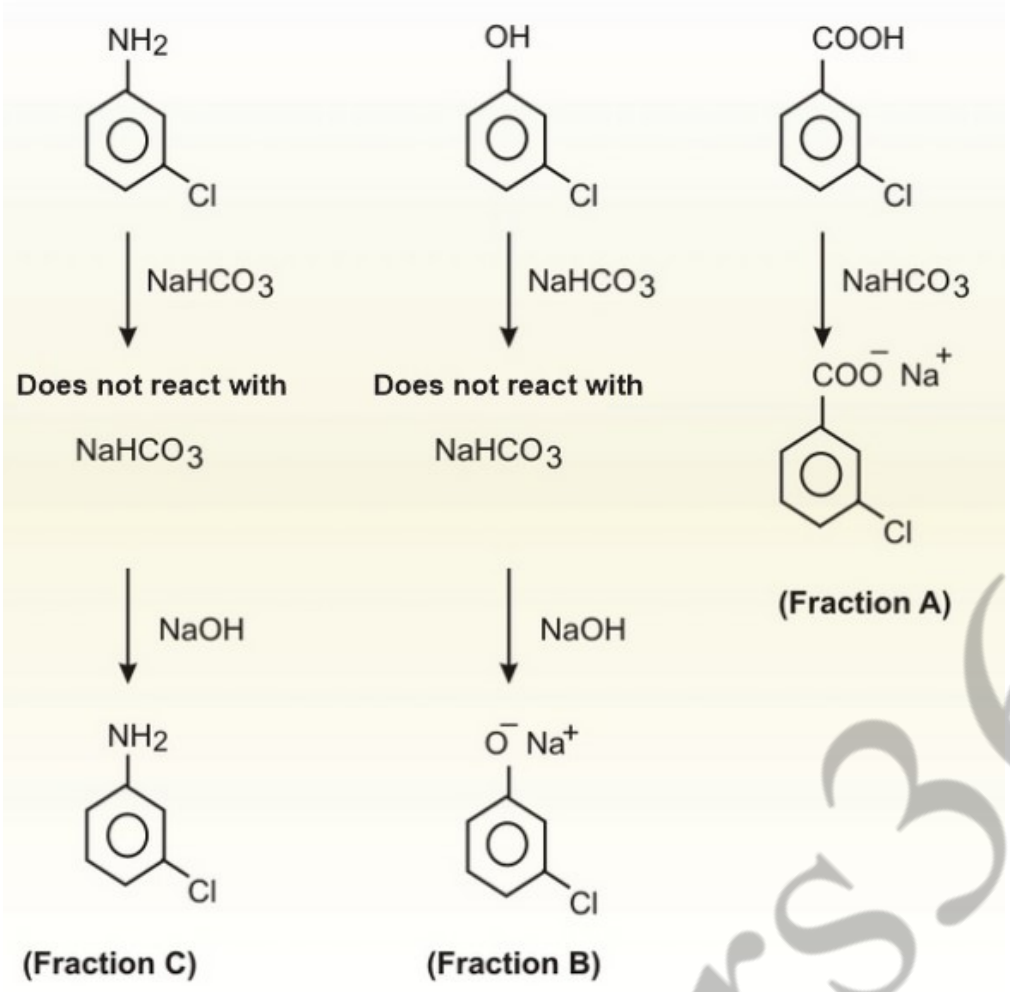

Therefore, **Option(2) is correct.**

Q. 5 The purest form of commercial iron is:

Option 1:
wrought iron

Option 2:
pig iron

Option 3:
scrap iron and pig iron

Option 4:
cast iron

Correct Answer:
wrought iron

Solution:
The purest form of iron is wrought iron.
Therefore, **Option(1) is correct.**

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Q. 6 Amongst the following statements, that which was not proposed by Dalton was:

Option 1:
All the atom of a given element has identical properties including identical mass. An atom of a different elements differs in mass.

Option 2:
matter consists of indivisible atoms.

Option 3:
when gases combine or reproduced in a chemical reaction they do so in a simple ratio by volume provided all gases are at the same T & P.

Option 4:
Chemical reactions involve reorganization of atom. These are neither created nor destroyed in a chemical reaction.

Correct Answer:
when gases combine or reproduced in a chemical reaction they do so in a simple ratio by volume provided all gases are at the same T & P.

Solution:
Gay Lussac's Law of Gaseous Volumes :

This law was given by Gay Lussac in 1808. He observed that when gases combine or are produced in a chemical reaction they do so in a simple ratio by volume provided all gases are at the same temperature and pressure.

-

Option(3) is correct as it was proposed by Gay Lussac's law of gaseous volume, because the theory given by the Dalton was for atoms and its properties don't depend on volume.

Therefore, **Option(3) is correct.**

Q. 7 The atomic radius of Ag is closest to:

Option 1:
Au

Option 2:
Ni

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Option 3:
Hg

Option 4:
Cu

Correct Answer:
Au

Solution:
Atomic Size/Radii -

Because of the lanthanide contraction, the size of the elements of the 4d series is similar to their corresponding elements of 5d series.

Thus, the atomic radius of Ag is closest to Au.

Therefore, **Option(1) is correct.**

Q. 8 The IUPAC name of the complex $[Pt(NH_3)_2Cl(NH_2CH_3)]Cl$ is:

Option 1:
Diammine(methanamine)chlorido platinum(II)chloride

Option 2:
Diamminechlorido(methanamine)platinum(II)chloride

Option 3:
Diamminechlorido(aminomethane)platinum(II)chloride

Option 4:
Bisammine(methanamine)chlorido platinum(II)chloride

Correct Answer:
Diamminechlorido(methanamine)platinum(II)chloride

Solution:
IUPAC Nomenclature of Coordination or Complex Compound -

The names of coordination compounds are derived by following the principles of additive nomenclature. Thus, the groups that surround the central atom must be identified in the name. They are listed as prefixes to the name of the central atom along with any appropriate multipliers. The following rules are used when naming coordination compounds:
(i) The cation is named first in both positively and negatively charged coordination entities.
(ii) The ligands are named in alphabetical order before the name of the central atom/ion. (This procedure is reversed from writing formula).
(iii) Names of the anionic ligands end in –o, those of neutral and cationic ligands are the same except aqua for H2O, ammine for NH3, carbonyl for CO and nitrosyl for NO. While writing the formula of

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coordination entity, these are enclosed in brackets ( ).
(iv) Prefixes mono, di, tri, etc., are used to indicate the number of individual ligands in the coordination entity. When the names of the ligands include a numerical prefix, then the terms, bis, tris, tetrakis are used, the ligand to which they refer being placed in parentheses. For example, $[NiCl_2(PPh_3)_2]$ is named as dichloridobis(triphenylphosphine)nickel(II).
(v) Oxidation state of the metal in cation, anion or neutral coordination entity is indicated by Roman numeral in parenthesis.
(vi) If the complex ion is a cation, the metal is named same as the element. For example, Co in a complex cation is called cobalt and Pt is called platinum. If the complex ion is an anion, the name of the metal ends with the suffix – ate. For example, Co in a complex anion, $[Co(SCN)_4]^{2-}$ is called cobaltate. For some metals, the Latin names are used in the complex anions, e.g., ferrate for Fe.
(vii) The neutral complex molecule is named similar to that of the complex cation.

-

The IUPAC name of the given compound is:
Diamminechlorido(methanamine)platinum(II)chloride.

Therefore, **Option(2) is correct.**

Q. 9 The theory that can completely /properly explain the nature of bonding in $[Ni(CO)_4]$

Option 1:
Molecular orbital theory

Option 2:
Crystal field theory

Option 3:
Wemer's theory

Option 4:
Valence bond theory

Correct Answer:
Molecular orbital theory

Solution:
As we have learnt,

Molecular Orbital Theory -

Molecular orbital theory (MO theory) provides an explanation of chemical bonding that accounts for the paramagnetism of the oxygen molecule. It also explains the bonding in a number of other molecules, such as violations of the octet rule and more molecules with more complicated bonding that are difficult

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to describe with Lewis structures. Additionally, it provides a model for describing the energies of electrons in a molecule and the probable location of these electrons. Unlike valence bond theory, which uses hybrid orbitals that are assigned to one specific atom, MO theory uses the combination of atomic orbitals to yield molecular orbitals that are delocalized over the entire molecule rather than being localized on its constituent atoms. MO theory also helps us understand why some substances are electrical conductors, others are semiconductors, and still others are insulators. The table given below explains the major differences between the valence bond theory and molecular orbital theory.

Comparison of Bonding Theories

Valence Bond Theory	Molecular Orbital Theory
considers bonds as localized between one pair of atoms	considers electrons delocalized throughout the entire molecule
creates bonds from overlap of atomic orbitals (s, p, d...) and hybrid orbitals (sp, sp^2, sp^3...)	combines atomic orbitals to form molecular orbitals (σ, σ*, π, π*)
forms σ or π bonds	creates bonding and antibonding interactions based on which orbitals are filled
predicts molecular shape based on the number of regions of electron density	predicts the arrangement of electrons in molecules
needs multiple structures to describe resonance	

Molecular orbital theory describes the distribution of electrons in molecules in much the same way that the distribution of electrons in atoms is described using atomic orbitals. Using quantum mechanics, the behavior of an electron in a molecule is still described by a wave function, Ψ, analogous to the behavior in an atom. Just like electrons around isolated atoms, electrons around atoms in molecules are limited to discrete (quantized) energies. The region of space in which a valence electron in a molecule is likely to be found is called a molecular orbital (Ψ2). Like an atomic orbital, a molecular orbital is full when it contains two electrons with opposite spin.

We will consider the molecular orbitals in molecules composed of two identical atoms (H_2 or Cl_2, for example). Such molecules are called homonuclear diatomic molecules. In these diatomic molecules, several types of molecular orbitals occur.

The mathematical process of combining atomic orbitals to generate molecular orbitals is called the linear combination of atomic orbitals (LCAO). The wave function describes the wavelike properties of an electron. Molecular orbitals are combinations of atomic orbital wave functions. Combining waves can lead to constructive interference, in which peaks line up with peaks, or destructive interference, in which peaks line up with troughs as shown in the figure below. In orbitals, the waves are three dimensional, and they combine with in-phase waves producing regions with a higher probability of electron density and out-of-phase waves producing nodes, or regions of no electron density.

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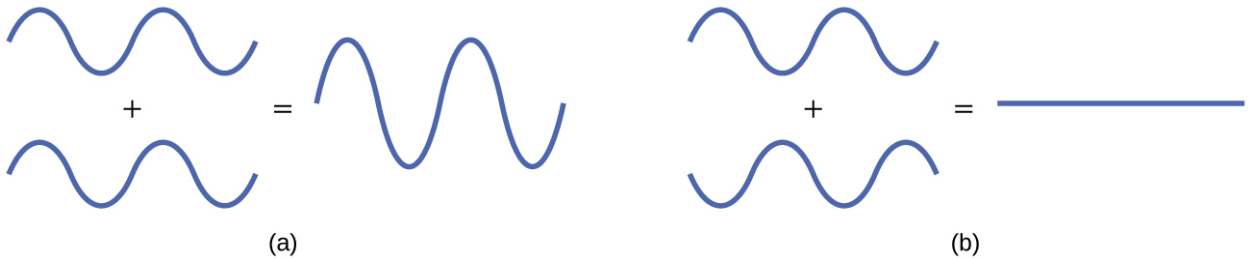

(a) (b)

(a) When in-phase waves combine, constructive interference produces a wave with greater amplitude. (b) When out-of-phase waves combine, destructive interference produces a wave with less (or no) amplitude.

-

The metal-carbonyl π is formed by the donation of a pair of electrons from a filled *d* orbital of metal into the vacant antibonding π^* orbital of carbon monoxide.

Therefore, **Option(1) is correct.**

Q. 10 **Match the following:**

Species	Deficiency disease
(i) Riboflavin	(a) Beriberi
(ii) Thiamine	(b) Scurvy
(iii) Pyridoxine	(c) Cheilosis
(iv) Ascorbic acid	(d) Convulsions

Option 1:

$(i) - (a),\ (ii) - (b),\ (iii) - (c),\ (iv) - (d)$

Option 2:

$(i) - (c),\ (ii) - (a),\ (iii) - (d),\ (iv) - (b)$

Option 3:

$(i) - (c),\ (ii) - (d),\ (iii) - (a),\ (iv) - (b)$

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Option 4:
$(i)-(d),\ (ii)-(b),\ (iii)-(a),\ (iv)-(c)$

Correct Answer:
$(i)-(c),\ (ii)-(a),\ (iii)-(d),\ (iv)-(b)$

Solution:
Vitamin A - Night blindness , Xeropthalmia
Vitamin (Thiamine) B_1 - Beriberi

Vitamin (Riboflavin) B_2 - Cheilosis

Vitamin (Niacin) B_3 - Pellagra

Vitamin (Pyridoxine) B_6 - Convulsions , Anaemia

Vitamin B_{12} - Pernicious anaemia

Vitamin C (Ascorbic acid) - Scurvy

Vitamin D - Rickets (in chilldrens)

Osteomalacia (in adults)

Vitamin E - Increased RBCs fragility , muscular weakness

Vitamin K - Poor blood clotting

-

Hence, the correct match is

(i) Riboflavin	-	(c) Cheilosis
(ii) Thiamine	-	(a) Beriberi
(iii) Pyridoxine	-	(d) Convulsions
(iv) Ascorbic acid	-	(b) Scurvy

Therefore, **Option(2) is correct.**

Q. 11 The dipole moments of **CCl_4, $CHCl_3$** and **CH_4** are in the order:

Option 1:
$CH_4 = CCl_4 < CHCl_3$

Option 2:
$CHCl_3 < CH_4 = CCl_4$

Option 3:
$CH_4 < CCl_4 < CHCl_3$

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Option 4:

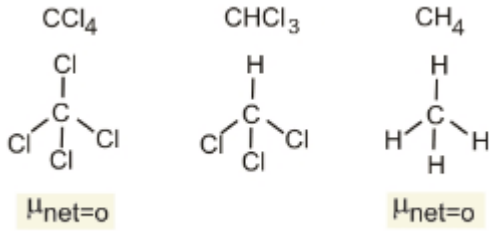
$CCl_4 < CH_4 < CHCl_3$

Correct Answer:

$CH_4 = CCl_4 < CHCl_3$

Solution:

Let us first look at the structures of the given compounds



$\text{Clearly } \mu \neq 0 \text{ of } \text{CHCl}_3$

Therefore, **Option(1) is correct.**

Q. 12 Given that the standard potentials(**E^o**) of **$Cu^{2+}|Cu$ and $Cu^{+}|Cu$** are **0.34V** and **0.522V** respectively, the **E^o** of **$Cu^{2+}|Cu^{+}$** is**:**

Option 1:

$-0.182V$

Option 2:

$+0.158V$

Option 3:

$-0.158V$

Option 4:

$0.182V$

Correct Answer:

$+0.158V$

Solution:

As we have learnt,

Feasibility and Gibbs Free Energy of Reaction -

Let n faraday charge be taken out of a cell of emf E, then work done by the cell will be calculated as:
Work = Charge × Potential
Work done by the cell is equal to the decrease in the free energy.

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$$-\Delta G = nFE$$

Similarly, maximum obtainable work from the cell at standard condition will be:

$$W_{max} = nFE^0_{cell} \quad \text{where } E^0_{cell} = \text{ standard emf of standard cell potential}$$

$$-\Delta G^\circ = nFE^0_{cell}$$

Variation of E.M.F. (E_{cell}) with Temperature

The temperature coefficient of e.m.f. of the cell is written as:

$$\Delta G = \Delta H + T\left(\frac{d(\Delta G)}{dT}\right)_{\text{Const. Pressure}}$$

$$\Rightarrow \quad -nFE_{cell} = \Delta H + T\left(\frac{d\left(-nFE_{cell}\right)}{dT}\right)_P \qquad [\because \Delta G = -nFE_{cell}]$$

$$\Rightarrow \quad \left(\frac{dE_{cell}}{dT}\right)_P = \frac{\Delta H}{nFT} + \frac{E_{cell}}{T}$$

Enthalpy Change: From above equation, we have:

$$\Delta H = \Delta G - T\left(\frac{d}{dT}(\Delta G)\right)_P$$

Thus, enthalpy(ΔH) is given as:

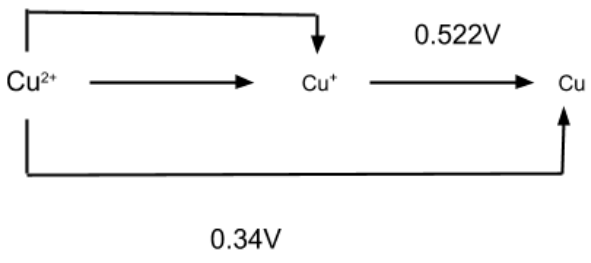
$$\Delta H = -nF\left[E_{cell} - T\left(\frac{dE_{cell}}{dT}\right)_P\right]$$

Entropy change: For entropy change we have:

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta S = -\left(\frac{d}{dT}(\Delta G)\right)_P = nF\left(\frac{dE_{cell}}{dT}\right)_P$$

-



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We have:

$\Delta G = -nFE^o$

$-2xFx0.34V = -1xFxE^o - 1xFx0.522V$

$-E^o = -2x0.34V + 0.522V$

Thus, $E^o = 0.158V$

Therefore, **Option(2) is correct.**

Q. 13 The electron gain enthalpy(in kJ/mol) of fluorine, chlorine, bromine and iodine respectively are:

Option 1:
$-333,\ -349,\ -325\ and\ -296$

Option 2:
$-349,\ -333,\ -325\ and\ -296$

Option 3:
$-296,\ -325\ -333,\ and\ -349$

Option 4:
$-333,\ -325\ -349\ and\ -296$

Correct Answer:
$-333,\ -349,\ -325\ and\ -296$

Solution:
Electron Gain Enthalpy or Electron Affinity -

Electron Affinity

Electron gain enthalpy is the energy change that occurs when an electron is added to a neutral gaseous atom to form a negative ion. It is also known as electron affinity.

$A(g) + e \rightarrow A^-(g) + \Delta_{eg}H$

Variation of Electron Affinity

- The electron gain enthalpy becomes less negative in going from top to bottom in a group.
- In moving from top to bottom in a group, both the atomic size and the nuclear charge increase. But the effect of the increase in atomic size is more prevalent than the nuclear charge.
- Halogens have the most negative electron gain enthalpies. In moving down from chlorine to iodine, the electron gain enthalpies become less negative due to the increase in their atomic radii.

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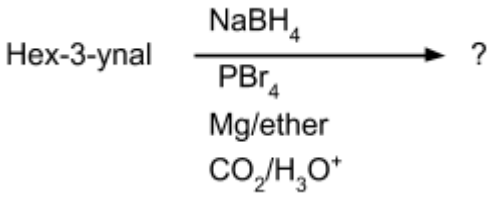
- Chlorine has the most negative electron gain enthalpy value than fluorine. Because fluorine is very small in size due to which there is a very strong inter-electronic repulsion for the incoming electron, thus its electron gain enthalpy is less than chlorine.

The electron gain enthalpy values are given below:

Fluorine = -333kJ/mol
Chlorine = -349kJ/mol
Bromine = -325kJ/mol
Iodine = -296kJ/mlol

Therefore, **Option(1) is correct.**

Q. 14 What is the product of following reaction:?



Option 1:

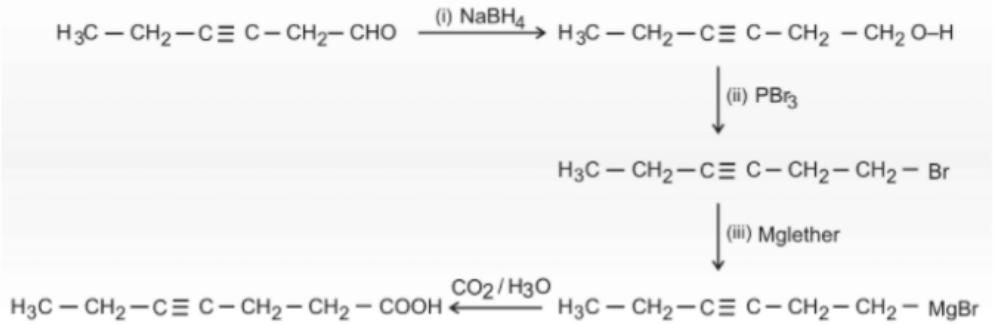
Option 2:

Option 3:

Option 4:

Correct Answer:

Solution:
The reactions occur as follows:



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Therefore, **Option(2) is correct.**

Q. 15 Consider the following reactions:

(a) $(CH_3)_3CCH(OH)CH_3 \xrightarrow{\text{conc. } H_2SO_4}$

(b) $(CH_3)_2CHCH(Br)CH_3 \xrightarrow{\text{alc. KOH}}$

(c) $(CH_3)_2CHCH(Br)CH_3 \xrightarrow{(CH_3)_3O^-K^+}$

(d) $(CH_3)_2COHCH_2CHO \xrightarrow{\Delta}$

Which of these reaction(s) will not produce Saytzeff product?

Option 1:
(c) only

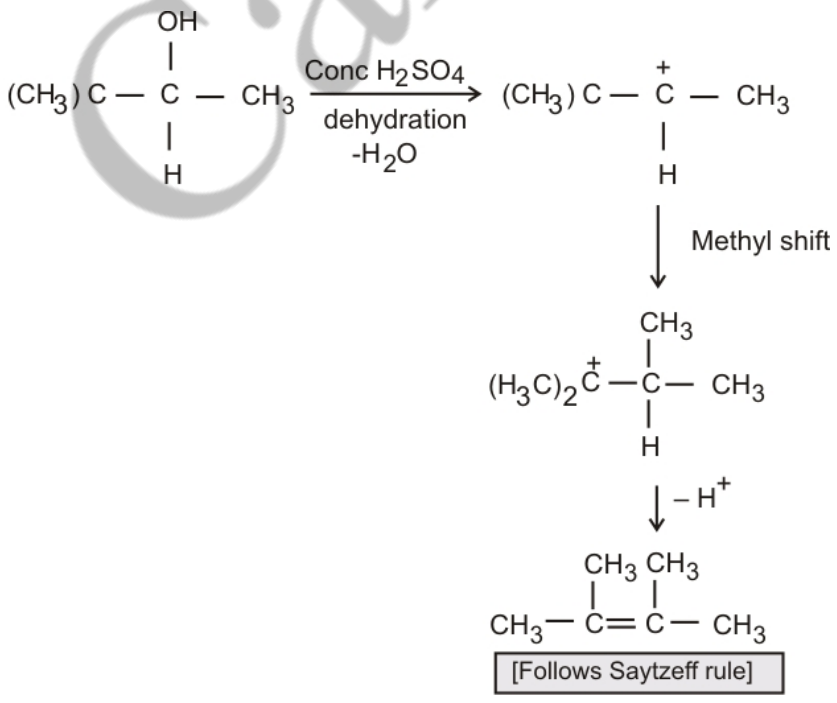
Option 2:
(b) and (d)

Option 3:
(a), (c) and (d)

Option 4:
(d) only

Correct Answer:
(c) only

Solution:
The reactions occur as follows:

(A) 

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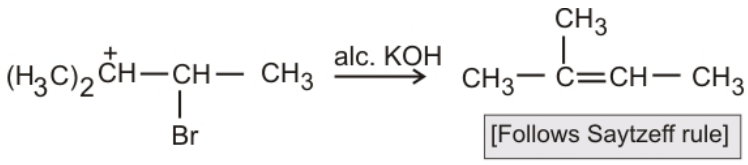
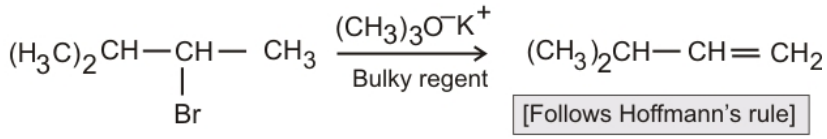
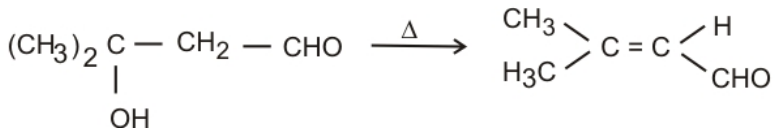
(B) 

(C) 

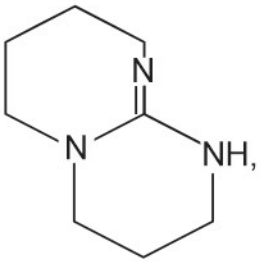
Because of Bulky reagent

(D) 

Therefore, **Option(1) is correct.**

Q. 16 The increasing order of pK_b for the following compounds will be:

$NH_2-CH=NH$,
(A)


(B)

CH_3NHCH_3
(C)

Option 1:
(b)<(c)<(a)

Option 2:
(c)<(a)<(b)

Option 3:
(a)<(b)<(c)

Option 4:
(c) > (a) > (b)

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Correct Answer:
(c) > (a) > (b)

Solution:
As we have learnt,

$pK_b = -\log k_b$
So, as K_b increases, pK_b decreases.

The conjugate acid of guanidine is most resonance stabilized followed by imidine.

Thus, the increasing order of basicity is:
(b) > (a) > (c)

pK_b order is given as:

(c) > (a) > (b)

Therefore, **Option(4) is correct.**

Q. 17 The number of orbitals associated with quantum numbers $n = 5, m_s = +1/2$ is:

Option 1:
25

Option 2:
11

Option 3:
15

Option 4:
50

Correct Answer:
25

Solution:
We have:

n = 5, m_s = +1/2

Thus, the values of l are from 0 to (n-1)

l = 0 to 4

Thus, values of l are 5s, 5p, 5d, 5f, and 5g

Now, the total number of orbitals $= n^2 = 5^2 = 25$

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Therefore, **Option(1) is correct.**

Q. 18 The relative strength of interionic/intermolecular forces in decreasing order is:

Option 1:
ion - dipole > dipole - dipole > ion -ion

Option 2:
ion - ion > ion - dipole > dipole-dipole

Option 3:
ion - dipole > ion -ion > dipole-dipole

Option 4:
dipole-dipole > ion - dipole > ion -ion

Correct Answer:
ion - ion > ion - dipole > dipole-dipole

Solution:
The correct order of intermolecular forces is:

ion-ion > ion-dipole > dipole-dipole

Therefore, **Option(2) is correct.**

Q. 19 In comparison to the zeolite process for the removal of permanent hardness, the synthetic resins method is:

Option 1:
less efficient as the resins cannot be regenerated

Option 2:
less efficient as it exchange only anions

Option 3:
More efficient as it can exchange only cations

Option 4:
More efficient as it exchange both cations as well as anions

Correct Answer:
More efficient as it exchange both cations as well as anions

Solution:
As we have learnt,

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The synthetic resins method is more efficient as it can exchange both cations and anions.
Therefore, **Option(4) is correct.**

Q. 20 The oxidation number of potassium in $K_2O, K_2O_2 \; and \; KO_2,$ respectively is :

Option 1:
$+1, \; +1 \; and \; +1$

Option 2:
$+2, \; +1 \; and \; +\frac{1}{2}$

Option 3:
$+1, \; +2 \; and \; +4$

Option 4:
$+1, \; +4 \; and \; +2$

Correct Answer:
$+1, \; +1 \; and \; +1$

Solution:

$$For \; K_2O : 2x - 2 = 0$$

$$x = +1$$

$$For K_2O_2 : 2x - 2 = 0$$

(In peroxide, the oxidation state of oxygen is -1)

$$x = +1$$

$$For KO_2 : x - 2(1/2) = 0$$

(In superoxide, the oxidation state of oxygen is -1/2)

$$x = +1$$

Therefore, **Option(1) is correct.**

Q. 21 Two solutions A and B each of 100L was made by dissolving 4g of NaOH and 9.8g of H_2SO_4 in water, respectively. The pH of the resultant solution obtained from mixing 40L of solution A and 10L of solution B is_____:

Option 1:
pH = 10.6

Option 2:
pH = 12.7

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Option 3:
pH = 9.8

Option 4:
pH = 11

Correct Answer:
pH = 10.6

Solution:
For given solutions, we have:

Moles of NaOH 4/40 = 0.1 moles

Moles of H_2SO_4 = 9.8/98 = 0.1 moles

Molarity of NaOH = 0.1/100L
And molarity of H_2SO_4 = 0.1/100

Now, 40L of NaOH solution and 10L of H_2SO_4 solution are added, thus we get:

Total volume = 50L

Milliequivalents of NaOH = 40x(0.1/100)x1 = 0.04

Milliequivalents of H_2SO_4 = 10x(0.2/100)x2 = 0.02

Thus, Meq of NaOH left = 0.04 - 0.02 = 0.02

$[OH^-] = 4x10^{-4}$

$pOH = -log[4x10^{-4}]$

$pOH = -log4 - log10^{-4}$

pOH = -0.60 + 4 = 3.4

Further, we know:

pH = 14 - 3.4

pH = 10.6

Hence, **the option number (1) is correct.**

Q. 22 The number of chiral carbons in chloramphenicol is _____

Option 1:
2

Option 2:

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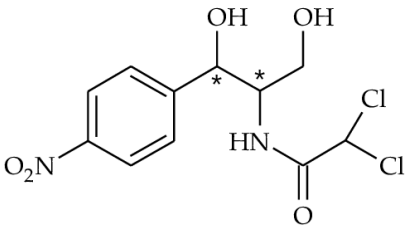
Option 3:

Option 4:

Solution:

The structure of chloramphenicol is given below:



Thus, the number of chiral carbons in chloramphenicol is 2.

Q. 23 For the reaction

$$A(l) \rightarrow 2B(g)$$
$$\Delta U = 2.1\ kcal, \Delta S = 20\ cal\ K^{-1}\ at\ 300\ K$$
$$Hence\ \Delta G\ in\ kcl\ is - - - - --$$

Option 1:
-2.7 Kcal

Option 2:
3.3 Kcal

Option 3:
- 3.3 Kcal

Option 4:
2.7 Kcal

Correct Answer:
-2.7 Kcal

Solution:
We know:

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta H = \Delta U + 2RT$$

Thus, we have:

$$\Delta G = \Delta U + 2RT - T\Delta S$$

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On putting the given values we get:

$$\Delta G = 2.1 + 2 \times 2 \times 300 \times 10^{-3} - 300 \times 20 \times 10^{-3}$$

$$\Delta G = 2.1 + 4 \times 300 \times 10^{-3} - 300 \times 20 \times 10^{-3}$$

$$\Delta G = 1200 \times 10^{-3} - 6000 \times 10^{-3}$$

$$\Delta G = 2.1 + 1.2 - 6$$

$$\Delta G = 3.3 - 6$$

$$\Delta G = -2.7 \text{ Kcal}$$

Q. 24 During the nuclear explosion, one of the products is ^{90}Sr with a half-life of 6.93 years. If $1\ \mu g \text{ of } ^{90}\text{Sr}$ was absorbed in the bones of a newly born baby in place of Ca, how much time, in the year, is required to reduce it by 90% if it is not lost metabolically----------

---

Correct Answer:
23.03

Solution:
We know this formula:-

$$t = \frac{2.303}{k} \log \frac{[R_o]}{[R]}$$

For 90% completion, the required time will be -

$$t = \frac{2.303}{k} \log \frac{[R_o]}{[0.1R_o]}$$

$$t = \frac{2.303}{k} \log 10$$

$$t = \frac{2.303}{k}$$

Half-life is given as 6.93 years. So,

$$k = \frac{0.693}{t_{\frac{1}{2}}}$$

$$k = \frac{0.693}{6.93} = 0.1$$

Thus, t is given as fro 90% completion:

$$t = \frac{2.303}{k}$$

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$$t = \frac{2.303}{0.1}$$

Thus, t = 23.03 years

Q. 25 Chlorine reacts with hot and concentrated NaOH and produces compounds (X) and (Y). Compound (X) gives white precipitate with silver nitrate solution. The average bond order between Cl and O atoms in (Y) is:

Correct Answer:
1.66

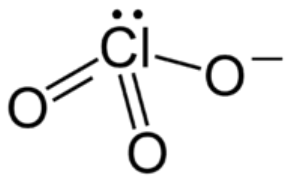
Solution:
As we have learnt,

Chlorine disproportionates in the presence of hot and concentrated NaOH to form chloride ions and chlorate ions

$$Cl_2 + 6NaOH(\text{hot and conc.}) \rightarrow \underset{Y}{NaClO_3} + \underset{X}{5NaCl} + 3H_2O$$

$$NaCl + AgNO_3 \rightarrow \underset{\text{white(ppt)}}{AgCl(s)} + NaNO_3(aq)$$

The structure of the Chlorate ion (ClO_3^-) is given below:



The average bond order is given as:

$$B.O = \frac{5}{3} = 1.66$$

Hence, **the option number (1) is correct.**

Maths

Q. 1 The area of the region, enclosed by the circle $x^2 + y^2 = 2$ which is not common to the region bounded by the parabola $y^2 = x$ and the straight line $y = x$, is :

Option 1:
$\frac{1}{3}(12\pi - 1)$

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Option 2:

$\frac{1}{6}(12\pi - 1)$

Option 3:

$\frac{1}{3}(6\pi - 1)$

Option 4:

$\frac{1}{6}(24\pi - 1)$

Correct Answer:

$\frac{1}{6}(12\pi - 1)$

Solution:

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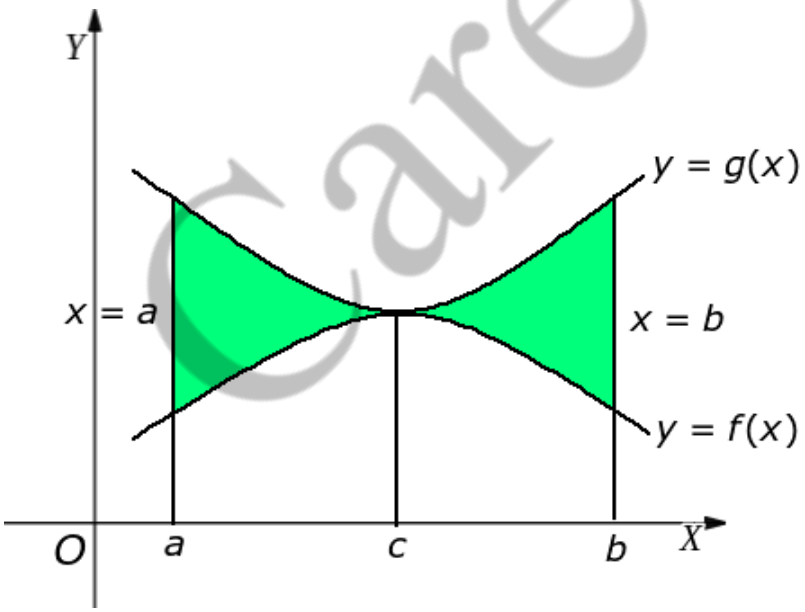
Area Bounded by Curves When Intersects at More Than One Point -

Area bounded by the curves $y = f(x)$, $y = g(x)$ and intersect each other in the interval [a, b]

First find the point of intersection of these curves $y = f(x)$ and $y = g(x)$, let the point of intersection be x = c



Area of the shaded region

$$= \int_a^c \{f(x) - g(x)\}dx + \int_c^b \{g(x) - f(x)\}dx$$

When two curves intersects more than one point

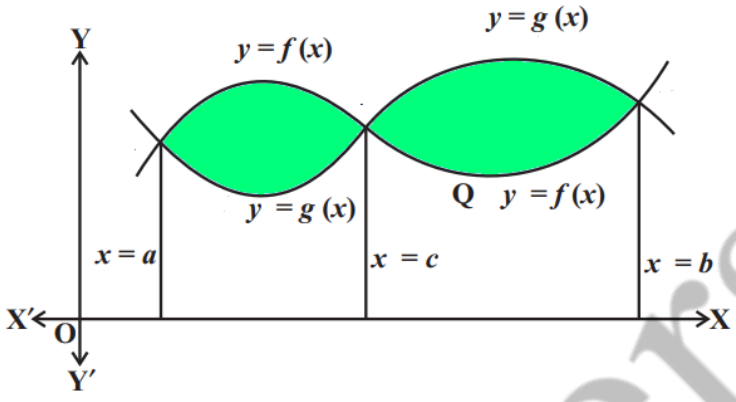
rea bounded by the curves *y=f(x), y=g(x)* and intersect each other at three points at *x = a, x = b amd x = c.*

To find the point of intersection, solve *f(x) = g(x).*

For x ∈ (a, c), f(x) > g(x) and for x ∈ (c, b), g(x) > f(x).

Area bounded by curves,

$$\mathrm{A} = \int_a^b |f(x) - g(x)|\, dx$$
$$= \int_a^c (f(x) - g(x))\, dx + \int_c^b (g(x) - f(x))\, dx$$



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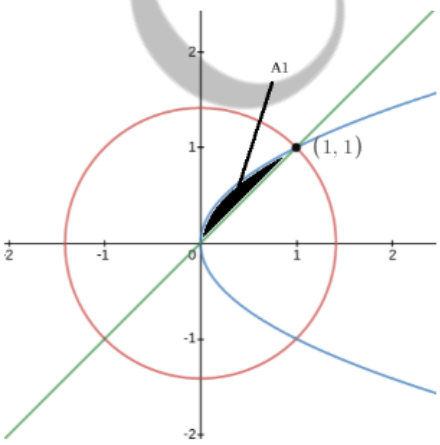
$$x^2 + y^2 = 2 \Rightarrow r = \sqrt{2}$$
$$y^2 = x$$
$$y = x$$



Area between parabola and line is A1

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$$A1 = \int_0^1 (y^2 - y)dx$$
$$A1 = \int_0^1 (\sqrt{x} - x)dx = 1/6$$
$$\text{Required area } = 2\pi - 1/6$$

Correct Option (2)

Q. 2 Total number of 6-digit numbers in which only the digits 1,3,5,7 and 9 appear and each of these digit appears at least once, is :

Option 1:
5^6

Option 2:
$\frac{1}{2}(6!)$

Option 3:
$6!$

Option 4:
$\frac{5}{2}(6!)$

Correct Answer:
$\frac{5}{2}(6!)$

Solution:
PERMUTATION OF OBJECTS WHEN FEW ARE IDENTICAL -

If there n objects of which p objects are of one type, q objects of another type, r objects of yet another type and all others are distinct, the total number of ways of arranging all the objects is $\frac{n!}{p!q!r!}$

Now,

6 places, 5 digit so one digit will repeat

First select the digit to be repeated: 5 options for this

Then arrange these 6 digits in 6 places : 6! / 2!

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So, total: $5 \times \frac{6!}{2!}$

Correct Option (4)

Q. 3 An unbiased coin is tossed 5 times.Suppose that a variable X is assigned the value k when consecutive heads are obtained for k = 3,4,5, otherwise X takes the value -1. Then the expected value of X, is :

Option 1:
$\frac{1}{8}$

Option 2:
$\frac{3}{16}$

Option 3:
$-\frac{1}{8}$

Option 4:
$-\frac{3}{16}$

Correct Answer:
$\frac{1}{8}$

Solution:
An unbiased coin is tossed 5 times, so total number of outcome is $2^5 = 32$.

Now, the probability for getting number of heads occurring simultaneously

$$\text{for k} = 0$$
$$\text{E(k)} = \{\text{TTTTT}\} = 1$$
$$\text{P(k)} = \frac{1}{32}$$

$$\text{for k} = 1$$
$$\text{E(k)} = \{\text{HTTTT}, \text{THTTT}, \text{TTHTT}, \text{TTTHT}, \text{TTTTH}, \text{HTHTT}, \text{HTTHT}, \text{HTTTH}, \text{THTHT}, \text{THTTH}, \text{TTHTH}, \text{HTHTH}\} = 12$$
$$\text{P(k)} = \frac{12}{32}$$

$$\text{for k} = 2$$
$$\text{E(k)} = \{\text{HHTTT}, \text{THHTT}, \text{TTHHT}, \text{TTTHH}, \text{HHTHT}, \text{HHTTH}, \text{THHTH}, \text{HTHHT}, \text{HTTHH}, \text{THTHH}, \text{HHTHH}\} = 11$$
$$\text{P(k)} = \frac{11}{32}$$

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$\text{for k} = 3$
$\text{E(k)} = \{\text{HHHTT, THHHT, TTHHH, HHHTH, HTHHH}\} = 5$
$\text{P(k)} = \dfrac{5}{32}$

$\text{for k} = 4$
$\text{E(k)} = \{\text{HHHHT, THHHH}\} = 2$
$\text{P(k)} = \dfrac{2}{32}$

$\text{for k} = 5$
$\text{E(k)} = \{\text{HHHHH}\} = 1$
$\text{P(k)} = \dfrac{1}{32}$

K	0	1	2	3	4	5
P(k)	$\frac{1}{32}$	$\frac{12}{32}$	$\frac{11}{32}$	$\frac{5}{32}$	$\frac{2}{32}$	$\frac{1}{32}$

$\text{For k} = 0, 1, 2; \text{X} = -1 \text{ and for k} = 3, 4, 5; \text{X} = \text{k}$.

Now expected value is

$$\sum X \times P(k) = (-1) \times \frac{1}{32} + (-1) \times \frac{12}{32} + (-1) \times \frac{11}{32} + 3 \times \frac{5}{32} + 4 \times \frac{2}{32} + 5 \times \frac{1}{32}$$
$$= \frac{1}{8}$$

Q. 4 If $\text{Re}\left[\dfrac{z-1}{2z+i}\right] = 1$, where $z = x + iy$, then the point (x, y) lies on a :

Option 1:

circle whose centre is at $\left[-\dfrac{1}{2}, -\dfrac{3}{2}\right]$.

Option 2:

straight line whose slope is $\dfrac{3}{2}$.

Option 3:

circle whose diameter is $\dfrac{\sqrt{5}}{2}$.

Option 4:

straight line whose slope is $-\dfrac{2}{3}$.

Correct Answer:

circle whose diameter is $\dfrac{\sqrt{5}}{2}$.

Solution:

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Conjugate of complex numbers and their properties -

The complex conjugate of a complex number a + ib (a, b are real numbers and b ≠ 0) is a − ib.

It is denoted as $\bar{z}$.

i.e. if z = a + ib, then its conjugate is $\bar{z}$ = a - ib.

Conjugate of complex numbers is obtained by changing the sign of the imaginary part of the complex number. The real part of the number is left unchanged.

Note:

- When a complex number is added to its complex conjugate, the result is a real number. i.e. z = a + ib, $\bar{z}$ = a - ib

  Then the sum, z + $\bar{z}$= a + ib + a - ib = 2a (which is real)

- When a complex number is multiplied by its complex conjugate, the result is a real number i.e. z = a + ib, $\bar{z}$ = a - ib

  Then the product, z • $\bar{z}$ = (a + ib) • (a - ib) = a^2 - $(ib)^2$

  = a^2 + b^2 (which is real)

-

Circle(Definition) -

General Form:

The equation of a circle with centre at (h,k) and radius r is

$$\Rightarrow (x-h)^2+(y-k)^2=r^2$$
$$\Rightarrow x^2+y^2-2hx-2ky+h^2+k^2-r^2=0 \quad \ldots(i)$$
Which is of the form :
$$\mathbf{x^2+y^2+2gx+2fy+c=0} \quad \ldots(ii)$$

This is known as the general equation of the circle.

Compare eq (i) and eq (ii)

h = -g, k = -h and $c=h^2+k^2-r^2$

Coordinates of the centre (-g,-f)

Radius $=g^2+f^2-c$

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$$Re\left(\frac{z-1}{2z+i}\right)=1$$

put $z=x+iy$

$$\text{Re}\left(\frac{(x+iy)-1}{2(x+iy)+i}\right)=1$$

$$\text{Re}\left(\left(\frac{(x-1)+iy}{2x+i(2y+1)}\right)\left(\frac{2x-i(2y+1}{2x-i(2y+1)}\right)\right)=1$$

$$\Rightarrow 2x^2+2y^2+2x+3y+1=0$$

$$x^2+y^2+x+\frac{3}{2}y+\frac{1}{2}=0$$

$\Rightarrow$ locus is a circle whose

$$\text{Centre is } \left(-\frac{1}{2},-\frac{3}{4}\right) \text{ and radius } \frac{\sqrt{5}}{4}.$$

$$\Rightarrow \text{ diameter } =\frac{\sqrt{5}}{2}$$

Correct Option (3)

Q. 5 If f(a+b+1-x)=f(x), for all x, where a and b are fixed positive real numbers,

then $\frac{1}{a+b}\int_a^b x(f(x)+f(x+1))dx$ is equal to :

Option 1:

$$\int_{a-1}^{b-1} f(x)dx$$

Option 2:

$$\int_{a+1}^{b+1} f(x+1)dx$$

Option 3:

$$\int_{a-1}^{b-1} f(x+1)dx$$

Option 4:

$$\int_{a+1}^{b+1} f(x)dx$$

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Correct Answer:

$$\int_{a+1}^{b+1} f(x)dx$$

Solution:

Properties of the Definite Integral (Part 2) - King's Property -

Property 4 (King's Property)

This is one of the most important properties of definite integration.

$$\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{f(x)}\ \mathbf{dx} = \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{f(a+b-x)}\ \mathbf{dx}$$

-

$$f(a+b+1-x) = f(x)$$

$$\text{put x} = 1+\text{x}$$

$$f(a+b-x) = f(x+1)$$

$$I = \frac{1}{a+b}\int_a^b [x(f(x)+f(x+1))]\,dx \text{.....1}$$

$$I = \frac{1}{a+b}\int_a^b [(a+b-x)(f(a+b-x)+f(a+b-x+1))]\,dx$$

$$I = \frac{1}{a+b}\int_a^b [(a+b-x)(f(x+1)+f(x))]\,dx \text{.....2}$$

Adding 1 and 2

$$I = \frac{1}{a+b}\int_a^b [(a+b)(f(x+1)+f(x))]\,dx$$

$$I = \int_a^b [(f(x+1)+f(x))]\,dx$$

$$2\text{I} = \int_{\text{a}}^{\text{b}} \text{f(x+1)dx} + \int_{\text{a}}^{\text{b}} \text{f(x)dx}$$
$$2I = \int_{\text{a}}^{\text{b}} \text{f(a+b+1-x)dx} + \int_{\text{a}}^{\text{b}} \text{f(x)dx}$$
$$2\text{I} = 2\int_{\text{a}}^{\text{b}} \text{f(x)dx}$$

Correct Option (4)

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Q. 6 If the distance between the foci of an ellipse is 6 and the distance between its directrices is 12, then the length of its latus rectum is :

Option 1:
$2\sqrt{3}$

Option 2:
$\sqrt{3}$

Option 3:
$\frac{3}{\sqrt{2}}$

Option 4:
$3\sqrt{2}$

Correct Answer:
$3\sqrt{2}$

Solution:

Length of Latusrectum -

Length of Latusrectum

$$\text{Let Latusrectum } LL' = 2\alpha$$
$$S(ae, 0) \text{ is focus, then } LS = SL' = \alpha$$
$$\text{coordinates of L and L' become } (ae, \alpha) \text{ and } (ae, -\alpha) \text{ respectively}$$
$$\text{Eq of ellipse, } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$
$$\therefore \frac{(ae)^2}{a^2} + \frac{\alpha^2}{a^2} = 1 \Rightarrow \alpha^2 = b^2\left(1 - e^2\right)$$
$$\alpha^2 = b^2\left(\frac{b^2}{a^2}\right) \quad \left[b^2 = a^2\left(1 - e^2\right)\right]$$
$$\alpha = \frac{b^2}{a}$$
$$\Rightarrow 2\alpha = LL' = \frac{2b^2}{a}$$

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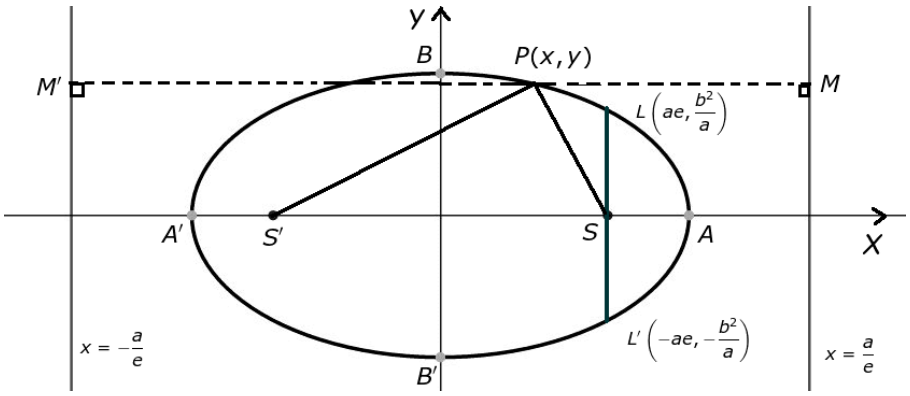


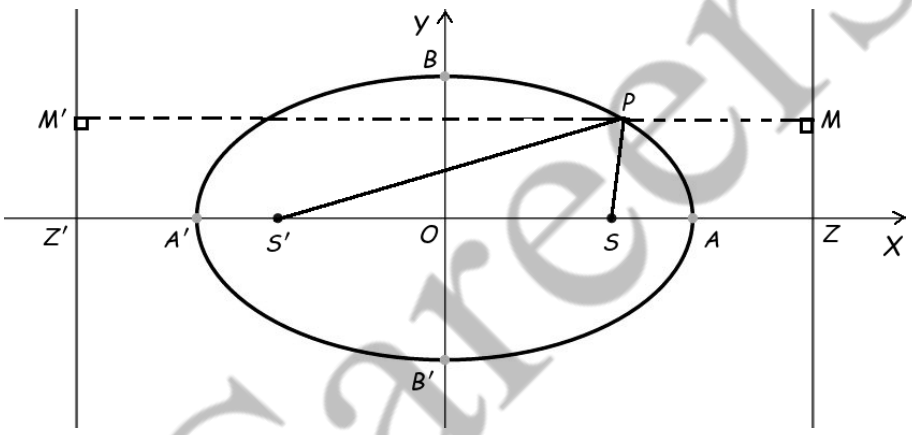
End Point of Latus rectum:

$$\mathrm{L} = \left(\mathrm{ae}, \frac{\mathrm{b}^2}{\mathrm{a}}\right) \text{ and } \mathrm{L}' = \left(\mathrm{ae}, -\frac{\mathrm{b}^2}{\mathrm{a}}\right)$$

Focal Distance of a Point:

The sum of the focal distance of any point on the ellipse is equal to the major axis.

Let P(x, y) be any point on the ellipse.



Here, $$SP = ePM = e\left(\frac{a}{e} - x\right) = a - ex$$

$$S'P = ePM' = e\left(\frac{a}{e} + x\right) = a + ex$$

Now, SP + S'P = a – ex + a + ex = 2a =AA' = constant.

Thus the sum of the focal distances of a point on the ellipse is constant.

-

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given

$$2ae = 6$$
$$\therefore ae = 3 \quad \text{.....(1)}$$

Also,

$$\frac{2a}{e} = 126$$
$$\therefore \frac{a}{e} = 63 \quad \text{.....(2)}$$

from (1) and (2)

$$e = \frac{1}{\sqrt{2}}, a = 3\sqrt{2}$$

since , $b^2 = a^2(1 - e^2)$

Substitute the values of 'e' and 'a' in above equation .

$$\Rightarrow b^2 = 9$$
$$\therefore b = \pm 3$$

length of latus rectum = $\frac{2b^2}{a} = \frac{2 \times 9}{3\sqrt{2}} = 3\sqrt{2}$

Correct Option (4)

Q. 7 The logical statement

$(p \Rightarrow q) \wedge (q \Rightarrow \sim p)$ is equivalent to :

Option 1:
$\sim p$

Option 2:
P

Option 3:
q

Option 4:
$\sim q$

Correct Answer:
$\sim p$

Solution:

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Practise Session - 2 -

Q1. Write the truth table for the following statement pattern:

1. $\sim(p \vee((\sim q \Rightarrow q) \wedge q))$
2. $(p \wedge q) \vee(\sim r)$

-

p	q	$((p \rightarrow q) \quad \wedge(q \rightarrow \neg p))$
F	F	T
F	T	T
T	F	F
T	T	F

which is equal to $\sim p$

Correct Option (1)

Q. 8 The greatest positive integer k, for which 49^k+1 is a factor of the sum $49^{125}+49^{124}+....+49^2+49+1$, is :

Option 1:
32

Option 2:
60

Option 3:
65

Option 4:
63

Correct Answer:
63

Solution:
Sum of n-term of a GP

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Let S_n be the sum of n terms of the G.P. with the first term 'a' and common ratio 'r'. Then

$$S_n = a\left(\frac{r^n - 1}{r - 1}\right)$$

The above formula does not hold for r = 1, For r = 1 the sum of n terms of the G.P. is $S_n = na$.

Now,

$$1 + 49 + 49^2 + ...49^{125}$$

$$\Rightarrow \frac{1(49^{126} - 1)}{48}$$
$$\Rightarrow \frac{(49^{63} - 1)(49^{63} + 1)}{48}$$

As $x^n - y^n$ is always divisible by (x - y), so $\frac{(49^{63} - 1)}{48}$ is an integrer

Hence, K=63

Correct Option (4)

Q. 9 A vector $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}\ (\alpha, \beta \epsilon R)$ lies in the plane of the vectors, $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{i} - \hat{j} + 4\hat{k}$. If $\vec{A}$ bisects the angle between $\vec{B}$ and $\vec{c}$, then:

Option 1:
$\vec{a} \cdot \hat{i} + 3 = 0$

Option 2:
$\vec{a} \cdot \hat{k} + 4 = 0$

Option 3:
$\vec{a} \cdot \hat{i} + 1 = 0$

Option 4:
$\vec{a} \cdot \hat{k} + 2 = 0$

Correct Answer:
$\vec{a} \cdot \hat{k} + 2 = 0$

Solution:

Equation of The Plane Bisecting the Angle Between Two Planes -

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-

angle bisector can be $\vec{a} = \vec{a} = \lambda(\hat{b} + \hat{c})$ or $\vec{a} = \mu(\hat{b} - \hat{c})$

$$\vec{a} = \lambda\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}} + \frac{\hat{i}-\hat{j}+4\hat{k}}{3\sqrt{2}}\right) = \frac{\lambda}{3\sqrt{2}}[3\hat{i}+3\hat{j}+\hat{i}-\hat{j}+4\hat{k}]$$
$$= \frac{\lambda}{3\sqrt{2}}[4\hat{i}+2\hat{j}+4\hat{k}]$$

Compare it with $\vec{a} = \alpha\hat{i}+2\hat{j}+\beta\hat{k}$

$$\frac{2\lambda}{3\sqrt{2}} = 2 \Rightarrow \lambda = 3\sqrt{2}$$
$$\vec{a} = 4\hat{i}+2\hat{j}+4\hat{k}$$

Not in option so now consider $\vec{a} = \mu\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}} - \frac{\hat{i}-\hat{j}+4\hat{k}}{3\sqrt{2}}\right)$

$$\vec{a} = \frac{\mu}{3\sqrt{2}}(3\hat{i}+3\hat{j}-\hat{i}+\hat{j}-4\hat{k})$$
$$= \frac{\mu}{3\sqrt{2}}(2\hat{i}+4\hat{j}-4\hat{k})$$

Compare it with $\vec{a} = \alpha\hat{i}+2\hat{j}+\beta\hat{k}$

$$\frac{4\mu}{3\sqrt{2}} = 2 \Rightarrow \mu = \frac{3\sqrt{2}}{2}$$
$$\vec{a} = \hat{i}+2\hat{j}-2\hat{k}$$
$$\vec{a}\cdot\hat{k}+2 = 0$$
$$-2+2 = 0$$

Correct Option (4)

Q. 10 If $y(\alpha) = \sqrt{2\left[\frac{\tan\alpha+\cot\alpha}{1+\tan^2\alpha}\right] + \frac{1}{\sin^2\alpha}}, \alpha\epsilon\left[\frac{3\pi}{4}, \pi\right]$, then, $\frac{dy}{d\alpha}$ at $\alpha = \frac{5\pi}{6}$ is :

Option 1:

$-\frac{1}{4}$

Option 2:

$\frac{4}{3}$

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Option 3:

4

Option 4:

-4

Correct Answer:

4

Solution:

Derivative of the Trigonometric Function (csc/sec/cot) -

Derivative of the Standard Function (Part 4)

11. $\dfrac{d}{dx}(\cot(\mathbf{x})) = -\csc^{\mathbf{2}}(\mathbf{x})$

12. $\dfrac{d}{dx}(\sec(\mathbf{x})) = \sec(\mathbf{x})\tan(\mathbf{x})$

13. $\dfrac{d}{dx}(\csc(\mathbf{x})) = -\csc(\mathbf{x})\cot(\mathbf{x})$

-

$$y(x) = \sqrt{2\frac{\tan x + \cot x}{1+\tan^2 x} + \frac{1}{\sin^2 x}}$$

$$= \sqrt{2\frac{\frac{1}{\sin x \cos x}}{1+\frac{\sin^2 x}{\cos^2 x}} + \frac{1}{\sin^2 x}}$$

$$= \sqrt{2\frac{\cos x}{\sin x} + \frac{1}{\sin^2 x}}$$

$$= \sqrt{\frac{2\cos x \sin x + 1}{\sin^2 x}}$$

$$= \sqrt{\frac{(\cos x + \sin x)^2}{\sin^2 x}}$$

$$= \frac{(\cos x + \sin x)}{\sin x}$$

$$|1 + \cot x| = \frac{\mathrm{d}}{\mathrm{d}x}(-1 - \cot x) = \csc^2 x$$

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put $x = \frac{5\pi}{6}$

we get 4

Correct Option (3)

Q. 11 If y=mx+4 is a tangent to both the parabolas, $y^2 = 4x$ and $x^2 = 2by,$ then b is equal to :

Option 1:
-64

Option 2:
128

Option 3:
-128

Option 4:
-32

Correct Answer:
-128

Solution:

Tangents of Parabola in Slope Form -

Tangents of Parabola in Slope Form

$$\text{Equation of the tangent to the parabola } y^2 = 4ax \text{ at the point } P(x_1, y_1) \text{ is } yy_1 = 2a(x + x_1)$$

m is the slope of the tangent, then

$$m = \frac{2a}{y_1} \Rightarrow y_1 = \frac{2a}{m}$$

(x_1, y_1) lies on the paarabola $y^2 = 4ax$

$$y_1^2 = 4ax_1 \Rightarrow \frac{4a^2}{m^2} = 4ax_1$$

$$\therefore \quad x_1 = \frac{a}{m^2}$$

put the value of x_1 and y_1 in the equation $yy_1 = 2a(x + x_1)$
we get

$$\Rightarrow \quad y = mx + \frac{a}{m}$$

which is equation of tangent of the parabola in slope form.

The coordinates of point of contact are

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$\left(\frac{a}{m^2}, \frac{2a}{m}\right)$

TIP

Equation of Parabola	Point of Contact	Tangent Equation
$y^2 = 4ax$	$\left(\frac{a}{m^2}, \frac{2a}{m}\right)$	$y = mx + \frac{a}{m}$
$y^2 = -4ax$	$\left(-\frac{a}{m^2}, \frac{2a}{m}\right)$	$y = mx - \frac{a}{m}$
$x^2 = 4ay$	$(2am, am^2)$	$y = mx - am^2$
$x^2 = -4ay$	$(2am, -am^2)$	$y = mx + am^2$

-

$$y = mx - am^2$$
$$y = \frac{x}{4} + 4$$
$$y = 4x, y = mx + \frac{1}{m}$$
$$\Rightarrow m = \frac{1}{4}$$
$$y = \frac{x}{4} + 4$$

also ,

$$x^2 = 2by$$
$$y = mx - \frac{bm^2}{2}$$
$$y = \frac{x}{4} + 4$$
$$\Longrightarrow \frac{bm^2}{2} = -4$$

since , $m = \frac{1}{4}$

Therefore , $b = \mathbf{-128}$

Correct Option (3)

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Q. 12

Let α be a root of the equation $x^2 + x + 1 = 0$ and the matrix $A = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha^4 \end{bmatrix}$,

Then the matrix A^{31} is equal to :

Option 1:

A

Option 2:

A^2

Option 3:

A^3

Option 4:

i_3

Correct Answer:

A^3

Solution:

Cube roots of unity -

z is a complex number

Let $z^3 = 1$

$\Rightarrow z^3 - 1 = 0$

$\Rightarrow (z - 1)(z^2 + z + 1) = 0$

$\Rightarrow z - 1 = 0$ or $z^2 + z + 1 = 0$

$$\therefore \ z = 1 \ \text{ or } \ z = \frac{-1 \pm \sqrt{(1-4)}}{2} = \frac{-1 \pm i\sqrt{3}}{2}$$

$$\text{Therefore, } \mathbf{z = 1,\ z = \frac{-1 + i\sqrt{3}}{2}} \ \text{ and } \ \mathbf{z = \frac{-1 - i\sqrt{3}}{2}}$$

If the second root is represented by **ω**, then the third root will be represented by **ω²**.

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$$\omega = \frac{-1+i\sqrt{3}}{2}, \quad \omega^2 = \frac{-1-i\sqrt{3}}{2}$$

Properties of cube roots:

i) $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$

ii) to find ω^n , first we write ω in multiple of 3 with remainder belonging to 0,1,2 like n=3q + r

Where r is from 0,1,2. Now $\omega^n = \omega^{3q+r} = (\omega^3)^q \cdot \omega^r = \omega^r$.

-

Multiplication of two matrices -

Matrix multiplication:

Two matrices A and B are conformable for the product AB if the number of columns in A and the number of rows in B is equal. Otherwise, these two matrices will be non-conformable for matrix multiplication. So on that basis,

i) AB is defined only if col(A) = row(B)

ii) BA is defined only if col(B) = row(A)

If

$$A = [a_{ij}]_{m \times n}$$
$$B = [b_{ij}]_{n \times p}$$
$$C = AB = [c_{ij}]_{m \times p}$$
$$\text{Where } c_{ij} = \sum_{j=1}^{n} a_{ij} b_{jk}, 1 \le i \le m, 1 \le k \le p$$
$$= a_{i1}b_{1k} + a_{i2}b_{2k} + a_{i3}b_{3k} + \ldots + a_{in}b_{nk}$$

For examples

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Suppose, two matrices are given

$$\mathrm{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{33} \end{bmatrix}_{2\times 3} \quad \text{and} \quad \mathrm{B} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}_{3\times 3}$$

To obtain the entries in row i of AB, we multiply the entries in row i of A by column j in B and add.
given matrices A and B, where the order of A are 2×3 and the order of B are 3×3, the product of AB will be a 2×3 matrix.

To obtain the entry in row 1, column 1 of AB, multiply the first row in A by the first column in B, and add.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} = \mathrm{a}_{11} \cdot \mathrm{b}_{11} + \mathrm{a}_{12} \cdot \mathrm{b}_{21} + \mathrm{a}_{13} \cdot \mathrm{b}_{31}$$

To obtain the entry in row 1, column 2 of AB, multiply the first row in A by the second column in B, and add.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \end{bmatrix} \begin{bmatrix} b_{12} \\ b_{22} \\ b_{32} \end{bmatrix} = \mathrm{a}_{11} \cdot \mathrm{b}_{12} + \mathrm{a}_{12} \cdot \mathrm{b}_{22} + \mathrm{a}_{13} \cdot \mathrm{b}_{32}$$

To obtain the entry in row 1, column 3 of AB, multiply the first row in A by the thired column in B, and add.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \end{bmatrix} \begin{bmatrix} b_{13} \\ b_{23} \\ b_{33} \end{bmatrix} = \mathrm{a}_{11} \cdot \mathrm{b}_{13} + \mathrm{a}_{12} \cdot \mathrm{b}_{23} + \mathrm{a}_{13} \cdot \mathrm{b}_{33}$$

We proceed the same way to obtain the second row of AB. In other words,
row 2 of A times column 1 of B;
row 2 of A times column 2 of B;
row 2 of A times column 3 of B.

When complete, the product matrix will be

$$\mathrm{AB} = \begin{bmatrix} a_{11} \cdot b_{11} + a_{12} \cdot b_{21} + a_{13} \cdot b_{31} & a_{11} \cdot b_{12} + a_{12} \cdot b_{22} + a_{13} \cdot b_{32} & a_{11} \cdot b_{13} + a_{12} \cdot b_{23} + a_{13} \cdot b_{33} \\ a_{21} \cdot b_{11} + a_{22} \cdot b_{21} + a_{23} \cdot b_{31} & a_{21} \cdot b_{12} + a_{22} \cdot b_{22} + a_{23} \cdot b_{32} & a_{21} \cdot b_{13} + a_{22} \cdot b_{23} + a_{23} \cdot b_{33} \end{bmatrix}$$

-

Given

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$x^2 + x + 1 = 0$
$\Rightarrow \alpha \, is \, the \, root \, of \, the \, eq$
$\alpha = \omega, \;\; \omega^2$

$$A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix}$$

$$A^2 = \frac{1}{3} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 3 \\ 0 & 3 & 0 \end{bmatrix}$$

$$A^4 = \frac{1}{3} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 3 \\ 0 & 3 & 0 \end{bmatrix} \times \frac{1}{3} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 3 \\ 0 & 3 & 0 \end{bmatrix}$$

$A^4 = I$
$\therefore A^{31} = A^{28} \cdot A^3 = A^3$

Correct option (3)

Q. 13 If $g(x) = x^2 + x - 1$ and $(gof)(x) = 4x^2 - 10x + 5$, then $f\left(\frac{5}{4}\right)$ is equal to :

Option 1:
$-\frac{3}{2}$

Option 2:
$-\frac{1}{2}$

Option 3:
$\frac{1}{2}$

Option 4:
$\frac{3}{2}$

Correct Answer:
$-\frac{1}{2}$

Solution:

Composition of function, Condition for Composite Function, Property of Composite Function -

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Here Domain of gof = Domain of f and

Co-domain of gof = Co-Domain of g

Properties of composition:

If f: A → B and g: B → C are one-one, then g o f: A → C is also one-one

If f: A → B and g: B → C are onto, then g o f: A → C is also onto

If go(f) is one-one, Then f is one-one.

If go(f) is onto, Then g is onto.

$$g(x) = x^2 + x - 1$$
$$gof(x) = 4x^2 - 10x + 5$$
$$f(x) = ax + b$$
$$gof(x) = (ax+b)^2 + ax + b - 1$$
$$gof(x) = a^2x^2 + b^2 + 2axb + ax + b - 1 = 4x^2 - 10x + 5$$
$$\implies a^2x^2 + x(2ab + a) + b^2 + b - 1 = 4x^2 - 10x + 5$$

Therefore , $a^2 = 4,\ 2ab + a = -10,\ b^2 + b - 1 = 5$

Solving the above equations,

$a = 2,\ b = -3,$

Therefore,

$$f(x) = 2x - 3$$
$$f\left(\frac{5}{4}\right) = 2 \times \frac{5}{4} - 3$$
$$= \frac{-1}{2}$$

Correct option (2)

Q. 14 Let α and β be two real roots of the equation $(k+1)\tan^2 x - \sqrt{2} \cdot \lambda \tan x = (1-k)$, where, $k(\neq -1)$ and λ are real numbers. If $\tan^2(\alpha + \beta) = 50,$ then a value of λ is :

Option 1:

$5\sqrt{2}$

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Option 2:
$10\sqrt{2}$

Option 3:
10

Option 4:
5

Correct Answer:
10

Solution:
As we have learnt,

Sum of roots:

$$\alpha + \beta = \frac{-\mathrm{b}}{\mathrm{a}}$$

Product of roots:

$$\alpha \cdot \beta = \frac{c}{a}$$

Trigonometric Ratio for Compound Angles (Part 2)

$$\tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$$
$$\tan(\alpha - \beta) = \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}$$

Now,

$$\tan\alpha + \tan\beta = \frac{\sqrt{2}\lambda}{1+k}$$
$$\tan\alpha \times \tan\beta = \frac{k-1}{1+k}$$

Since $\tan\alpha \& \tan\beta$ are the roots of the given equation

$$\tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta} = \frac{\frac{\sqrt{2}\lambda}{1+k}}{1 - \frac{k-1}{k+1}} = \frac{\lambda}{\sqrt{2}}$$

Now,

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$$\tan^2(\alpha+\beta)=\frac{\lambda^2}{2}=50$$
$$\lambda=10$$

Q. 15 Let P be a plane passing through the points $(2,1,0),(4,1,1)$ and $(5,0,1)$and r be any point $(2,1,6)$.Then the image of r in the plane P is :

Option 1:
$(6,5,2)$

Option 2:
$(6,5,-2)$

Option 3:
$(4,3,2)$

Option 4:
$(3,4,-2)$

Correct Answer:
$(6,5,-2)$

Solution:

Equation of a plane passing through three non collinear point -

Cartesian Form

Let (x_1, y_1, z_1), (x_2, y_2, z_2) and (x_3, y_3, z_3) be the coordinates of points A, B and C respectively.

Let P(x, y z) be any point on the plane.

Then, the vectors $\overrightarrow{PA}$, $\overrightarrow{BA}$ and $\overrightarrow{CA}$ are coplanar.

$$\left[\overrightarrow{\mathbf{PA}}\ \overrightarrow{\mathbf{BA}}\ \overrightarrow{\mathbf{CA}}\right]=0$$

$$\begin{vmatrix}\mathbf{x-x_1} & \mathbf{y-y_1} & \mathbf{z-z_1}\\ \mathbf{x_2-x_1} & \mathbf{y_2-y_1} & \mathbf{z_2-z_1}\\ \mathbf{x_3-x_1} & \mathbf{y_3-y_1} & \mathbf{z_3-z_1}\end{vmatrix}=0$$

which is required equation of the plane .

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Image of a Point in the Plane -

The image of the point $P(x_1, y_1, z_1)$ in the plane $ax + by + cz + d = 0$ is $Q(x_3, y_3, z_3)$ then coordinates of point Q is given by

$$\frac{\mathbf{x_3 - x_1}}{\mathbf{a}} = \frac{\mathbf{y_3 - y_1}}{\mathbf{b}} = \frac{\mathbf{z_3 - z_1}}{\mathbf{c}} = -\frac{\mathbf{2\left(ax_1 + by_1 + cz_1 + d\right)}}{\mathbf{a^2 + b^2 + c^2}}$$

-

Points of plane P (2,1,0), (4,1,1) and (5,0,1) and point R (2,1,6) .

Then the image of R in the plane P is:

$$\begin{vmatrix} x-2 & y-1 & z \\ 2 & 0 & 1 \\ 3 & -1 & 1 \end{vmatrix} = (x-2)[0+1] - (y-1)[2-3] + z[-2]$$

$\Rightarrow x - 2 + y - 1 - 2z = 0$

$\text{plane}, x + y - 2z = 3$

$$\Rightarrow \frac{x-2}{1} = \frac{y-1}{1} = \frac{z-6}{-2} = \frac{-2(2+1-12-3)}{6}$$

$\Rightarrow (x, y, z) = (6, 5, -2)$

Correct option (2)

Q. 16 Let $x^k + y^k = a^k, (a, k > 0)$ and $\frac{dy}{dx} + \left[\frac{y}{x}\right]^{\frac{1}{3}} = 0,$ then k is :

Option 1:

$\frac{1}{3}$

Option 2:

$\frac{3}{2}$

Option 3:

$\frac{2}{3}$

Option 4:

$\frac{4}{3}$

Correct Answer:

$\frac{2}{3}$

Solution:

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Derivative of the Polynomial Function -

Derivative of the Polynomial Function

Finding derivatives of functions by using the definition of the derivative can be a lengthy and, for certain functions, a rather challenging process.

In this section, we will learn rules for finding derivatives that allow us to bypass this process. Let's begin with the basics.

1. $\frac{d}{dx}(\mathbf{constant}) = \mathbf{0}$

2. $\frac{d}{dx}(\mathbf{x^n}) = \mathbf{nx^{n-1}}$

-

$x^k + y^k = a^k \qquad (a, k > 0)$
$\frac{dy}{dx} + \left(\frac{y}{x}\right)^{1/3} = 0$
$\Rightarrow \; kx^{k-1} + ky^{k-1} \cdot \frac{dy}{dx} = 0$
$\frac{dy}{dx} + \left(\frac{x}{y}\right)^{k-1} = 0$
$k - 1 = -1/3$
$k = 2/3$

Correct Option (3)

Q. 17 Let the function, $f : [-7, 0] \to R$ be continuous on $[-7, 0]$ and differentiable on $[-7, 0]$. If $f(-7) = -3$ and $f'(x) \leq 2$, for all $x\epsilon(-7, 0)$, then for all such functions f, $f(-1) + f(0)$ lies in the interval :

Option 1:
$[-6, 20]$

Option 2:
$[-\infty, 20]$

Option 3:
$[-\infty, 11]$

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Option 4:

$[-3, 11]$

Correct Answer:

$[-\infty, 20]$

Solution:

Lagrange's Mean Value Theorem -

Lagrange's Mean Value Theorem

Rolle's theorem is a special case of the **Mean Value Theorem**. In Rolle's theorem, we consider differentiable functions f defined on a closed interval [a, b] with f(a) = f(b) . The Mean Value Theorem generalized Rolle's theorem by considering functions that do not necessarily have equal value at the endpoints.

Statement

Let f (x) be a function defined on [a, b] such that

1. it is continuous on [a, b],
2. it is differentiable on (a, b).

Then there exists a real number c (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

-

Directly use

$$\frac{f(-1) - f(-7)}{-1 + 7} = f'(c) \quad \& \quad \frac{f(0) - f(-7)}{0 + 7} = f'(c)$$

$$\frac{f(-1) - f(-7)}{-1 + 7} = f'(2) \quad \& \quad \frac{f(0) - f(-7)}{0 + 7} = f'(2)$$

we get

$$f(-1) \leq 9 \quad \& \quad f(0) \leq 11$$

$$f(-1) + f(0) \leq 20$$

Correct Option (2)

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Q. 18 If $y = y(x)$ is the solution of the differential equation, $e^y\left[\frac{dy}{dx} - 1\right] = e^x$ such that $y(0) = 0$, then is equal to :

Option 1:
$log_e 2$

Option 2:
$2e$

Option 3:
$2 + log_e 2$

Option 4:
$1 + log_e 2$

Correct Answer:
$1 + log_e 2$

Solution:

Rules of Differentiation (Divide or Quotient Rule) -

Rules of Differentiation (Divide or Quotient Rule)

Quotient Rule

Let f(x) and g(x) be differentiable functions. Then

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)\cdot\frac{d}{dx}(f(x)) - f(x)\cdot\frac{d}{dx}(g(x))}{(g(x))^2}$$

OR

$$\text{if } h(x) = \frac{f(x)}{g(x)}, \text{ then } h'(x) = \frac{f'(x)g(x) - g'(x)f(x)}{(g(x))^2}$$

-

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{e^y}{e^x}\right) = \frac{e^y\cdot e^x\frac{\mathrm{d}y}{\mathrm{d}x} - e^x\cdot e^y}{(e^x)^2}$$

$$\int \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{e^y}{e^x}\right) = \int 1\cdot dx$$

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$$\left(\frac{e^y}{e^x}\right) = x$$

$$(e^y) = xe^x + c$$

at y(0)=0

c=1

y(1)=1+ln2

Correct Option (4)

Q. 19 Five numbers are in A.P., whose sum is 25 and product is 2520. If one of these five numbers is $-\frac{1}{2}$, then the greatest number amongst them is :

Option 1:
16

Option 2:
27

Option 3:
7

Option 4:
$\frac{21}{2}$

Correct Answer:
16

Solution:
Five terms are in AP

Let terms are a-2d, a-d, a, a+d, a+2d

Sum of the terms = 5a = 25

So, a=5

Product of the terms

(a-2d) (a-d) (a+d) (a+2d) a = 2520

Putting a = 5

$(25 - 4d^2) (25 - d^2) = 504$

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Putting $d^2 = t$

(25 - t) (25 - 4t) = 504

$4t^2$ - 125t + 121 = 0

t = 1 or t = 121/4

d = 1 or d = -1 or d= 11/2 or d = -11/2

But -1/2 can be a term of AP only for d= 11/2 or d = -11/2

Using d = 11/2

Greatest term = a+ 2d = 16

Correct Option (1)

Q. 20 If the system of linear equations

2x+2ay+az=0

2x+3by+bz=0,

2x+4cy+cz=0,

where $a, b, c \epsilon R$ are non-zero and distinct ; has a non-zero solution, then :

Option 1:
$a + b + c = 0$

Option 2:
a, b, c are in AP

Option 3:
$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.

Option 4:
a, b, c are in G.P.

Correct Answer:
$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.

Solution:

Cramer's law -

Cramer's law for the system of equations in two variables :

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Let $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$, where

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

on solving this equation by cross multiplication, we get

$$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{a_1c_2 - a_2c_1} = \frac{1}{a_1b_2 - a_2b_1}$$

$$\text{or } \frac{x}{\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}} = \frac{y}{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}} = \frac{1}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

$$\text{or } x = \frac{\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}, y = \frac{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

We can observe that first row in the numerator of x is of constants and 2nd row in the numerator is of constants, and the denominator is of the coefficient of variables.

We can follow this analogy for the system of equations of 3 variable where third in the numerator of the value of z will be of constant and denominator will be formed by the value of coefficients of the variables.

so let us consider the system of equations

$a_1x + b_1y + c_1z = d_1$ ...(i)

$a_2x + b_2y + c_2z = d_2$ ...(ii)

$a_3x + b_3y + c_3z = d_3$ ...(iii)

then Δ, which will be determinant of coefficient of variables, will be

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, \Delta_1 \text{ numerator of x is :}$$

$$\Delta_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}, \text{similarly } \Delta_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix} \text{ and}$$

$$\Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}, \text{if} \Delta \neq 0 \text{ then } x = \frac{\Delta_1}{\Delta}$$

$$y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}$$

i) If ? ≠ 0, then the system of equations has a unique finite solution and so equations are consistent, and solutions are $x = \frac{\Delta_1}{\Delta}, y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}$

ii) If ? = 0, and any of $\Delta_1 \neq 0 \; or \; \Delta_2 \neq 0 \; or \; \Delta_3 \neq 0$

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Then the system of equations is inconsistent and hence no solution exists.

iii) If all $\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$ then

System of equations is consistent and dependent and it has an infinite number of solution.

-

For non zero solutions D = 0

$$D = \begin{vmatrix} 2 & 2a & a \\ 2 & 3b & b \\ 2 & 4c & c \end{vmatrix} = 0$$

$$C_2 \to C_2 - 2C_3$$

$$D = \begin{vmatrix} 1 & 0 & a \\ 1 & b & b \\ 1 & 2c & c \end{vmatrix} = 0$$

on solving

$$-bc + 2ac - ab = 0$$

$$\frac{1}{a} + \frac{1}{c} = \frac{2}{b}$$

Correct Option (3)

Q. 21 $\lim\limits_{x \to 2} \dfrac{3^x + 3^{3-x} - 12}{3^{-\frac{x}{2}} - 3^{1-x}}$ is equal to _____.

Correct Answer: 36

Solution:

Limit of Algebraic function -

(c) Rationalization Method

This method is used when either numerator or denominator or both have fractional powers (like ½, ⅓ etc). After rationalization, the terms are factorized which on cancellation gives the final answer.

Let's go through some illustration to understand better

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Illustration 2:

$$\text{Evaluate } : \lim_{x\to a}\frac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}}$$

Rationalizing numerator and denominator we get,

$$= \lim_{x\to a}\frac{a+2x-3x}{3a+x-4x}\left(\frac{\sqrt{3a+x}+2\sqrt{x}}{\sqrt{a+2x}+\sqrt{3x}}\right) \qquad \left(\frac{0}{0}\ form\right)$$

$$= \lim_{x\to a}\frac{a-x}{3(a-x)}\left(\frac{\sqrt{3a+x}+2\sqrt{x}}{\sqrt{a+2x}+\sqrt{3x}}\right) \qquad \left(\frac{0}{0}\ form\right)$$

$$= \frac{1}{3}\left(\frac{2\sqrt{a}+2\sqrt{a}}{\sqrt{3a}+\sqrt{3a}}\right)$$

$$= \frac{2}{3\sqrt{3}}$$

-

$$\lim_{x\to 2}\frac{3^x+3^{3-x}-12}{3^{\frac{-x}{2}}-3^{1-x}}$$

$$\Rightarrow \lim_{x\to 2}\frac{3^x+\frac{3^3}{3^x}-12}{\frac{1}{3^{\frac{x}{2}}}-\frac{3}{3^x}}$$

$$\Rightarrow \lim_{x\to 2}\frac{3^{2x}+3^3-12\cdot 3^x}{3^{\frac{x}{2}}-3}$$

$$\text{Multiply by the conjugate of } 3^{\frac{x}{2}}-3 : \frac{3^x-9}{3^{\frac{x}{2}}+3}$$

$$= \lim_{x\to 2}\left(\frac{3^{2x}+3^3-12\cdot\ 3^x}{\frac{3^x-9}{3^{\frac{x}{2}}+3}}\right)$$

$$= \lim_{x\to 2}\left((3^x-3)\left(3^{\frac{x}{2}}+3\right)\right) = 36$$

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Q. 22 If the variance of the first n natural numbers is 10 and the variance of the first m even natural numbers is 16, then m+n is equal to _____.

Correct Answer:
18

Solution:

Dispersion (Variance and Standard Deviation) -

Variance and Standard Deviation

The mean of the squares of the deviations from the mean is called the variance and is denoted by σ^2 (read as sigma square).

Variance is a quantity which leads to a proper measure of dispersion.

The variance of n observations x_1 , x_2 ,..., x_n is given by

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Standard Deviation

The standard deviation is a number that measures how far data values are from their mean.

The positive square-root of the variance is called standard deviation. The standard deviation, usually denoted by σ and it is given by

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

-

$$\sigma^2 = \left(\frac{\sum x_i^2}{n}\right) - \left(\frac{\sum x_i}{n}\right)^2$$

variance of the first n natural numbers is 10

n = 1,2,3,4.....n

$$\sum x_i = 1 + 2 + 3.....n = \frac{n(n+1)}{2}$$

$$\sum x_i^2 = 1^2 + 2^2 + 3^2.....n^2 = \frac{n(n+1)(2n+1)}{6}$$

Using variance formula

$$\sigma^2 = \left(\frac{\sum x_i^2}{n}\right) - \left(\frac{\sum x_i}{n}\right)^2$$

$$\sigma^2 = \left(\frac{n(n+1)(2n+1)}{6n}\right) - \left(\frac{n(n+1)}{2n}\right)^2$$

$$\sigma^2 = \left(\frac{(n+1)(2n+1)}{6}\right) - \left(\frac{(n+1)}{2}\right)^2$$

$$\sigma^2 = \frac{n+1}{2}\left[\left(\frac{2n+1}{3}\right) - \left(\frac{n+1}{2}\right)\right]$$

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$$\sigma^2 = \frac{n+1}{2}\left(\frac{4n+2-3n-3}{6}\right)$$

$$\sigma_n = \frac{n^2-1}{12} = 10$$

$$n^2 = 121 \quad \Rightarrow \quad n = 11$$

variance of the first m even natural numbers is 16

N = 2,4,6,8.....2m

N = 2(1,2,3,4.....m)

$$\sum x_j = 2(1+2+3.....m) = 2 \times \frac{m(m+1)}{2} = m(m+1)$$

$$\sum x_j^2 = 2^2(1^2+2^2+3^2.....m^2) = 4 \times \frac{m(m+1)(2m+1)}{6}$$

Using variance formula

$$\sigma^2 = \left(\frac{\sum x_i^2}{n}\right) - \left(\frac{\sum x_i}{n}\right)^2$$

$$\sigma^2 = \left(\frac{2m(m+1)(2m+1)}{3m}\right) - \left(\frac{m(m+1)}{m}\right)^2$$

$$\sigma^2 = \left(\frac{2(m+1)(2m+1)}{3}\right) - (m+1)^2$$

$$\sigma^2 = \frac{m+1}{3}(4m+2-3m-3)$$

$$\sigma_{m(even)} = \frac{m^2-1}{3} = 16$$

$$m^2 = 49 \quad \Rightarrow \quad m = 7$$

Hence

$$n+m = 18$$

Q. 23 If the sum of the coefficients of all even powers of x in the product $(1+x+x^2+....+x^{2n})(1-x+x^2-x^3+....+x^{2n})$ is 61, then n is equal to _____.

Correct Answer:
30

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Solution:

Let $(1+x+x^2+....+x^{2n})(1-x+x^2-x^3+....+x^{2n}) = a_0+a_1x+a_2x^2+.....$

$Put\ x = 1$
$(2n+1).1 = a_0 + a_1 + a_2 + ...$
$Put\, x = -1$
$1.(2n+1) = a_0 - a_1 + a_2 + ...$
$From(i) + (ii)$
$4n + 2 = 2(a_0 + a_2 + ...)$
$So,\ 2n + 1 = 61$
$\Rightarrow n = 30$

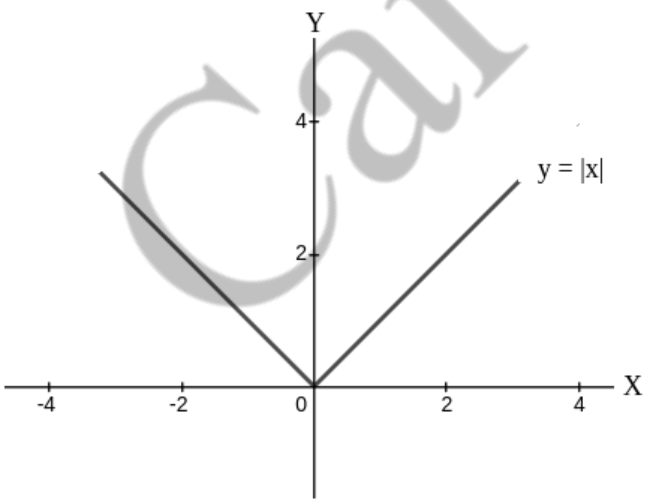
Q. 24 Let S be the set of points where the function, $f(x) = |2 - |x - 3||\ , x\epsilon R,$ is not differentiable. Then $\sum_{x\epsilon S} f(f(x))$ is equal to _____.

Correct Answer:
33

Solution:

Differentiability and Continuity -

Consider the function, f(x) = |x|, the modulus function is continuous at x = 0 but it is not differentiable at x= 0. As we see in the graph, at x = 0, it has a sharp edge, hence not differentiable.



Function fail to Differentiable if

a. both $Rf'(a)$ and $Lf'(a)$ exist but are not equal,
b. either or both $Rf'(a)$ and $Lf'(a)$ are not finite, and
c. either or both $Rf'(a)$ and $Lf'(a)$ do not exist.

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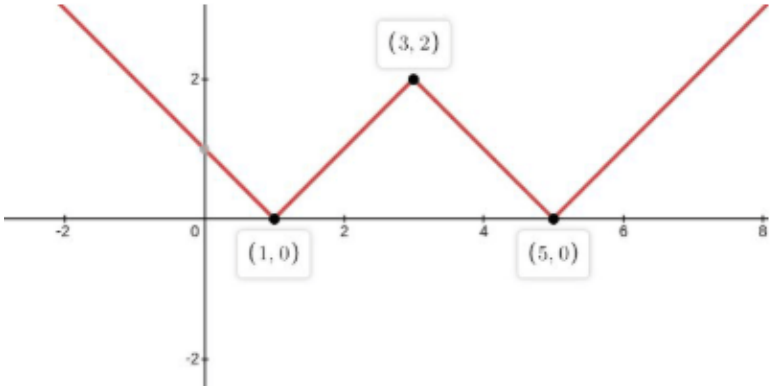
$f(x) = |2 - |x - 3||$
$f(f(x)) = |2 - ||2 - |x - 3|| - 3||$



$S = \{(1, 0), (3, 2), (5, 0)\}$
$\sum_{x \in S} f(f(x)) = \sum_{x \in S} |2 - ||2 - |x - 3|| - 3||$
$= 3$

Q. 25 Let $A(1, 0),\ B(6, 2)$ and $C\left(\frac{3}{2}, 6\right)$ be the vertices of a triangle ABC. If P is a point inside the triangle ABC such that the triangles APC, APB and BPC have equal areas, then the length of the line segment PQ, where Q is the point $\left(-\frac{7}{6}, -\frac{1}{3}\right)$,is ____.

Correct Answer:
5

Solution:

Centroid -

Centroid

Centroid of a triangle is the point of intersection of the medians of the triangle. A centroid divides the median in the ratio 2:1.

Whereas, the median is the line joining the mid-points of the sides and the opposite vertices.

The coordinates of the centroid of a triangle (G) whose vertices are A (x_1, y_1), B (x_2, y_2) and C(x_3, y_3), is given by

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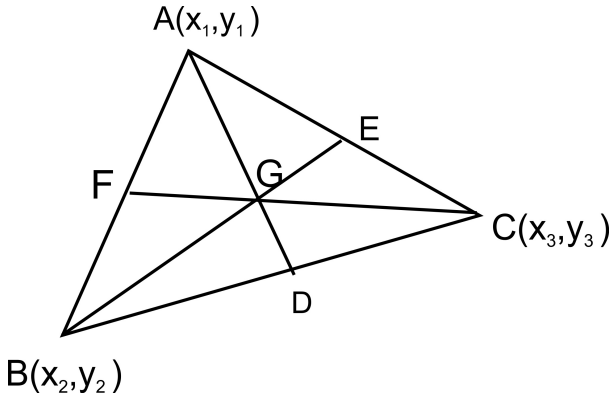

$$\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$$

If D (a_1, b_1), E (a_2, b_2) and F (a_3, b_3) are the mid point of ΔABC, then its centroid is given by

$$\left(\frac{a_1+a_2+a_3}{3}, \frac{b_1+b_2+b_3}{3}\right)$$

-

Distance between two points -

Distance between two points

Point A (x_1, y_1) and B (x_2, y_2) is two point on the plane then distance between them is given by

$$|AB| = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

The distance of a point A (x, y) from the origin O (0, 0) is given by

$$|OA| = \sqrt{x^2+y^2}$$

-

A(1,0) B(6,2) C(3/2,6)

Point P is the centroid of triangle ABC

P(17/6,8/3)

Distance between PQ is 5

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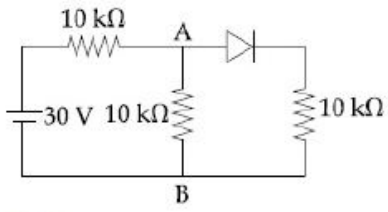
7th January Shift-2

Physics

Q. 1 In the figure, potential difference (in V) between A and B is :

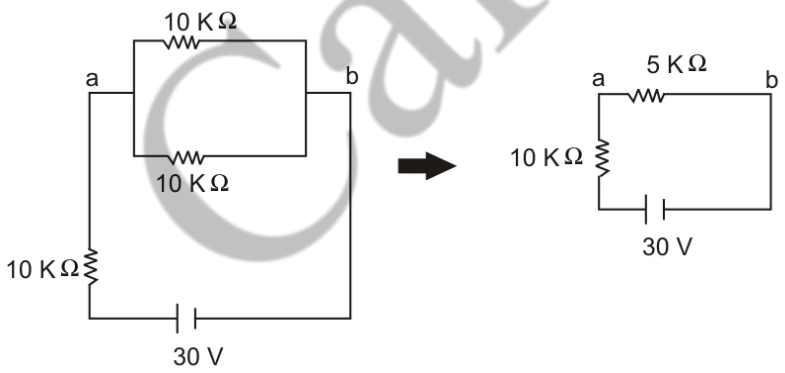


Correct Answer:
10

Solution:

A diode is forward biased

So the circuit will be as follows



The potential difference between a and b is

$$V_{ab} = \frac{30}{15 \times 10^3} \times 5 \times 10^3 V = 10V$$

Hence the correct option is (1).

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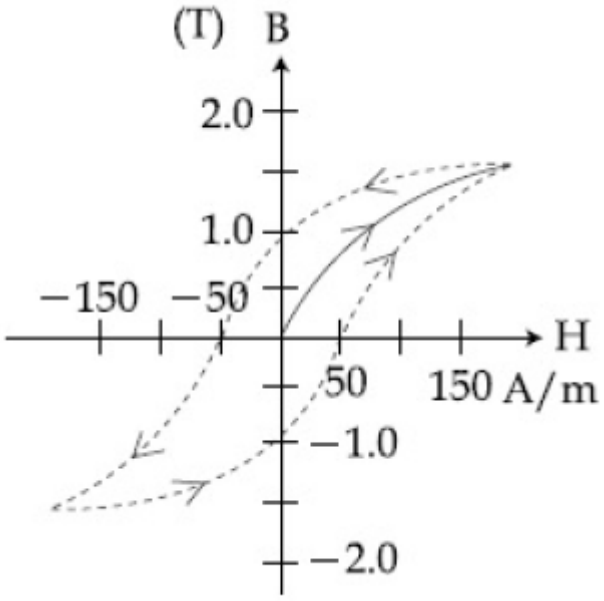
Q. 2 The figure gives experimentally measured B vs. H variation in a ferromagnetic material. The retentivity, co-ercivity and saturation, respectively, of the material are :



Option 1:
$1.5\ T,\ 50 A/m\ and\ 1.0\ T$

Option 2:
$1.0\ T,\ 50\ A/m\ and\ 1.5\ T$

Option 3:
$1.5\ T,\ 50 A/m\ and\ 1.0\ T$

Option 4:
$150\ A/m\ , 1.0\ T\ and\ 1.5\ T$

Correct Answer:
$1.0\ T,\ 50\ A/m\ and\ 1.5\ T$

Solution:

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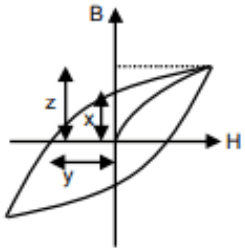

So, from the figure, we can see that the-

x = retentivity, y = coercivity, z = saturation magnetization

So, by matching with the diagram from the question and solution, option (2) is correct.

Q. 3 Two ideal Carnot engines operate in cascade (all heat given up by one engine is used by the other engine to produce work) between temperatures, T_1 and T_2. The temperature of the hot reservoir of the first engine is T_1 and the temperature of the cold reservoir of the second engine is T_2. T is temperature of the sink of first engine which is also the source for the second engine. How is T related to T_1 and T_2 if both the engines perform equal amount of work?

Option 1:

$t = \sqrt{t_1 t_2}$

Option 2:

$t = 0$

Option 3:

$t = \frac{t_1 + t_2}{2}$

Option 4:

$T = \frac{2T_1 T_2}{T_1 + T_2}$

Correct Answer:

$t = \frac{t_1 + t_2}{2}$

Solution:

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Let,
Q_H : Heat input to 1st engine
Q_L : Heat rejected from 1st engine
Q'_H : Heat input to 2nd engine
Q'_L : Heat rejected from 2nd engine

Since the heat sink of one engine is a source for other

So $Q_L = Q'_H$

And Work done by 1st engine = work done by 2nd engine

$$\mathrm{Q_H - Q_L = Q_L - Q'_L}$$
$$\mathrm{2Q_L = Q_H + Q'_L}$$

Let the lower temperature of the first engine=T

And in Carnot engine

$$\frac{Q_H}{Q_L} = \frac{T_1}{T},\ and\ \frac{Q'_H}{Q'_L} = \frac{Q_L}{Q'_L} = \frac{T}{T_2}$$

So

$$\mathrm{2Q_L = Q_H + Q'_L}$$

$$\mathrm{2 = \frac{T_1}{T} + \frac{T_2}{T}}$$

$$\mathrm{T = \frac{T_1+T_2}{2}}$$

Hence the correct option is (3).

Q. 4 A particle of mass 'm' and charge 'q' has an initial velocity $\vec{v} = v_0\hat{j}$ If an electric field $\overrightarrow{\mathrm{E}} = \mathrm{E_0}\hat{i}$ and magnetic field $\overrightarrow{\mathrm{B}} = \mathrm{B_0}\hat{i}$ act on the particle, its speed will double after a time:

Option 1:

$$\frac{\sqrt{3}mv_0}{qE_0}$$

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Option 2:

$\frac{3mv_0}{qE_0}$

Option 3:

$\frac{\sqrt{2}mv_0}{qE_0}$

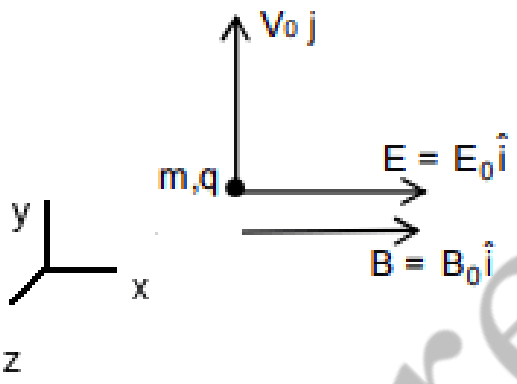
Option 4:

$\frac{2mv_0}{qE_0}$

Correct Answer:

$\frac{\sqrt{3}mv_0}{qE_0}$

Solution:



Initially $\vec{v} = v_0\hat{j}$

As both electric and magnetic field is in the x-direction

And initial velocity is perpendicular to both electric and magnetic fields.

So particle will move in a **helical motion**

and magnitude of velocity does not change in the y–z plane

and Speed will be increased due to electric field only i.e in the x-direction

and finally, the speed of the particle becomes double the initial speed.

i.e $(2v_0)^2 = v_0^2 + v_x^2 \quad ; \quad v_x = \sqrt{3}v_0$

And in x-direction $F_x = qE \Rightarrow a_x = \frac{qE}{m}$

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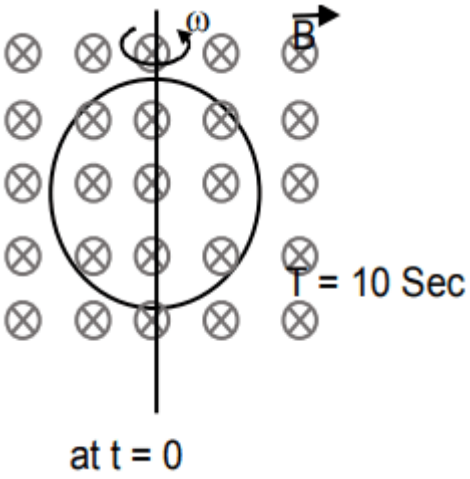
So $v_x = \sqrt{3}v_0 = 0 + \frac{qE}{m}t; \quad t = \frac{mv_0\sqrt{3}}{qE}$

Hence the correct option is (1).

Q. 5 A planer loop of wire rotates in a uniform magnetic field. Initially, at t = 0, the plane of the loop is perpendicular to the magnetic field. If it rotates with a period of 10 s about an axis in its plane then the magnitude of induced emf will be maximum and minimum, respectively at :



Option 1:
$2.5\ s\ and\ 7.5\ s$

Option 2:
$5.0\ s\ and\ 7.5\ s$

Option 3:
$5.0\ s\ and\ 10.0\ s$

Option 4:
$2.5\ s\ and\ 5.0\ s$

Correct Answer:
$2.5\ s\ and\ 5.0\ s$

Solution:

$\therefore \quad \omega = \frac{2\pi}{T} = \frac{\pi}{5}$

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We know that $\phi = BACos\theta = BASin\omega t$

And $\varepsilon = -\frac{d\phi}{dt} = BASin\omega t$

When $\omega t = \frac{\pi}{2}$

$\because$ ϕ will be minimum.

$\therefore$ ε will be maximum

$t = \frac{\frac{\pi}{2}}{\frac{\pi}{5}} = 2.5\text{sec}$

When $\omega t = \pi$

$\because$ ϕ will have maximum.

$\because$ ε will be minimum.

$t = \frac{\pi}{\pi/5} = 5$ sec.

Hence the correct option is (4).

Q. 6 In a Young's double slit experiment, the separation between the slits is 0.15 mm. In the experiment, a source of light of wavelength 589 nm is used and the interference pattern is observed on a screen kept 1.5 m away. The separation between the successive bright fringes on the screen is:

Option 1:

$4.9\ mm$

Option 2:

$6.9\ mm$

Option 3:

$5.9\ mm$

Option 4:

$3.9\ mm$

Correct Answer:

$5.9\ mm$

Solution:

As Fringe width is given by

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$$\beta = \frac{\lambda D}{d}$$

Given d=0.15 mm

D=1.5 m

$\lambda = 589mm$

Fringe width

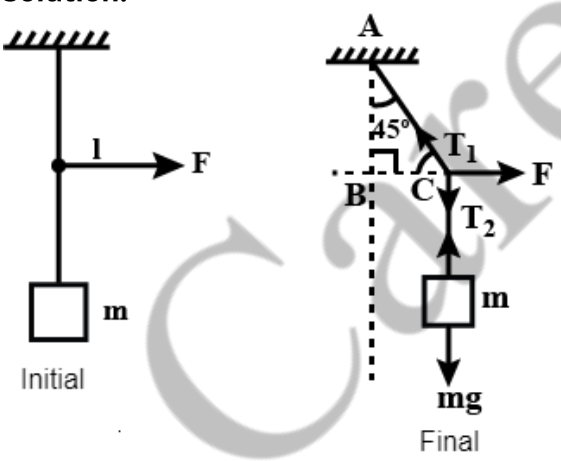
The fringe width is 5.9 mm

Hence the correct option is (3).

Q. 7 A mass of 10 kg is suspended by a rope of length 4m, from the ceiling. A force F is applied horizontally at the mid-point of the rope such that the top half of the rope makes an angle of 45° with the vertical. Then F (in newtons) equals: (Take $\mathrm{g} = 10\mathrm{ms}^{-2}$ and the rope to be massless):

Correct Answer:
100

Solution:



The system is in equilibrium

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$$\Rightarrow \sum \mathrm{F_y} = \mathrm{ma_y} = 0$$
$$\Rightarrow \mathrm{T_1} \sin 45^\circ - \mathrm{mg} = 0$$
$$\Rightarrow \frac{\mathrm{T_1}}{\sqrt{2}} = \mathrm{mg} \Rightarrow \mathrm{T_1} = \mathrm{mg}\sqrt{2} \ldots (\mathrm{I})$$
$$\Rightarrow \sum F_x = ma_x = 0$$
$$\Rightarrow \mathrm{F} - \mathrm{T_1} \cos 45^\circ = 0$$
$$\Rightarrow \mathrm{F} = \frac{\mathrm{T_1}}{\sqrt{2}} = \mathrm{mg} = 10 \times 10 = 100 \text{ N} \ldots (\text{using eq.} (I))$$

Q. 8 An elevator in a building can carry a maximum of 10 persons, with the average mass of each person being 68 kg. The mass of the elevator itself is 920 kg and it moves with a constant speed of $3 \text{ m } / \text{ s}$. The frictional force opposing the motion is 6000N. If the elevator is moving up with its full capacity, the power delivered by the motor to the elevator $\left(g = 10 \mathrm{m/s^2}\right)$ must be at least:

Option 1:
$48000\ W$

Option 2:
$56300\ W$

Option 3:
$66000\ W$

Option 4:
$62360\ W$

Correct Answer:
$66000\ W$

Solution:

The elevator moves upward hence frictional force (F) opposes their motion downward.

so, net force act on elevator =mg+F

F_{net}=((10*68+920) x 10+6000)=22000 N

using power =F_{net} x speed=22000*3=66000 Watt

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Hence the correct option is (3).

Q. 9 A stationary observer receives sound from two identical tuning forks, one of which approaches and the other one recedes with the same speed (much less than the speed of sound). The observer hears 2 beats/ sec. The oscillation frequency of each tuning fork is $v_0 = 1400\text{Hz}$ and the velocity of sound in air is $350\ \text{m}\ /\ \text{s}$. The speed of each tuning fork is close to:

Option 1:
$1\ m/s$

Option 2:
$\frac{1}{2}\ m/s$

Option 3:
$\frac{1}{8}\ m/s$

Option 4:
$\frac{1}{4}\ m/s$

Correct Answer:
$\frac{1}{4}\ m/s$

Solution:

given f_0=1400Hz

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$$f_1 = f_0\left[\frac{v_{sound}}{v_{sound} - v_{source}}\right]$$

$$f_2 = f_0\left[\frac{v_{sound}}{v_{sound} + v_{source}}\right]$$

Now, $beat frequency = f_1 - f_2$

$$\Rightarrow 2 = f_1 - f_2 \Rightarrow v_{source} = \frac{1}{4} m/s$$

Hence the correct option is (4).

Q. 10 A thin lens made of glass (refractive index = 1.5) of focal length $f = 16\ cm$ is immersed in a liquid of refractive index 1.42. If its focal length in liquid is f_l then the ratio f_l/f is closets to the integer:

Option 1:
9

Option 2:
1

Option 3:
5

Option 4:
17

Correct Answer:
9

Solution:

$$\frac{1}{f_a} = \left(\frac{\mu_g}{\mu_a} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f_l} = \left(\frac{\mu_g}{\mu_l} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

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$$\Rightarrow \frac{f_a}{f_l} = \frac{\left(\frac{\mu_g}{\mu_l} - 1\right)}{\left(\frac{\mu_g}{\mu_a} - 1\right)} = \frac{\left(\frac{1.50}{1.42} - 1\right)}{\left[\frac{1.50}{1} - 1\right]} = \frac{0.08}{(1.42)(0.5)}$$

$$\frac{\mathrm{f_l}}{\mathrm{f_a}} = \frac{(1.42)(0.5)}{0.08} = 8.875 \approx 9$$

Hence the correct option is (1).

Q. 11 A box weighs 196 N on a spring balance at the north pole. Its weight (in N) recorded on the same balance if it is shifted to the equator is close to (Take g=10 m/s^2 at the north pole and the radius of the earth =6400 km):

Correct Answer:
195.32

Solution:
Given weight = 196N, therefore mass = 19.6Kg

At pole centripetal acceleration is zero, there for w=mg

Due to the rotation of the earth, the effective weight is

$$w = mg - m\omega^2 R = mg - m\left(\frac{2\pi}{T}\right)^2 R$$

$$w = 196 - 19.6\left(\frac{2\pi}{24 \times 3600}\right)^2 \times 6400 \times 10^3$$

$$w = 195.32N$$

Hence the correct option is (1).

Q. 12 In a building there are 15 bulbs of 45 W, 15 bulbs of 100 W, 15 small fans of 10 W and 2 heaters of 1 kW. The voltage of the electric main is 220V. The minimum fuse capacity (rated value) of the building will be: (answer in amperes)

Correct Answer:
20

Solution:

$$Total\ power = 15 \times 45 + 15 \times 100 + 15 \times 10 + 2 \times 1000 = 4325W$$

Power =VI

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So current

$$I = \frac{P}{V} = \frac{4325}{220} = 19.65A$$

Minimum fuse current should be 20A.

Hence the correct option is (1).

Q. 13 An electron (of mass m) & a photon have the same energy E. Find the ratio of de Broglie wavelength of an electron and the wavelength of the photon (c = speed of light in vacuum).

Option 1:

$$\left(\frac{E}{2m}\right)^{1/2}$$

Option 2:

$$\frac{1}{c}\left(\frac{2E}{m}\right)^{1/2}$$

Option 3:

$$c\left(2mE\right)^{1/2}$$

Option 4:

$$\frac{1}{c}\left(\frac{E}{2m}\right)^{1/2}$$

Correct Answer:

$$\frac{1}{c}\left(\frac{E}{2m}\right)^{1/2}$$

Solution:

$$\lambda_d \text{ for electron} = \frac{h}{\sqrt{2mE}}$$

$$\lambda \text{ for photon} = \frac{hC}{E}$$

$$\text{Ratio} = \frac{h}{\sqrt{2mE}} \times \frac{E}{hC} = \frac{1}{C}\sqrt{\frac{E}{2m}}$$

Hence the correct option is (4).

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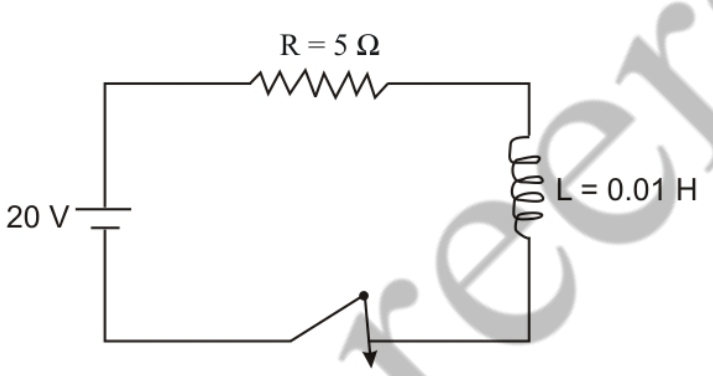
Q. 14 An emf of 20 V is applied at time t =0 to a circuit containing in series 10mH inductor and $5\ \Omega$ resistor. The ratio of the currents at time $t = \infty$ and at $t = 40\ s$ is close to:
$(Take\ e^2 = 7.389)$

Correct Answer:
1.15

Solution:

As we know that the -

$$i = i_o(1 - e^{\frac{-t}{L/R}})$$



So, from the question, we can write that -

$$i = \frac{20}{5}(1 - e^{\frac{-t}{0.01/5}})$$

$$= i = 4(1 - e^{-500t})$$

By putting t = ∞

We get, $i_o = 4$

Again at t = 40 sec

$$i_{40} = 4(1 - \frac{1}{(e^2)^{10000}})$$

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$$= 4(1 - \frac{1}{(7.389)^{10000}})$$

So,

$$\frac{i_\infty}{i_{40}} \approx 1.15$$

So option 2 is much closer. So option 1 will be correct.

Q. 15 Mass per unit area of a circular disc of radius a depends on the distance r from its centre as $\sigma(r) = A + Bx$. The moment of inertia of the disc about the axis, perpendicular to the plane and passing through its centre is :

Option 1:

$\pi a^4 \left(\frac{A}{4} + \frac{aB}{5}\right)$

Option 2:

$2\pi a^4 \left(\frac{A}{4} + \frac{aB}{5}\right)$

Option 3:

$2\pi a^4 \left(\frac{aA}{4} + \frac{B}{5}\right)$

Option 4:

$2\pi a^4 \left(\frac{A}{4} + \frac{B}{5}\right)$

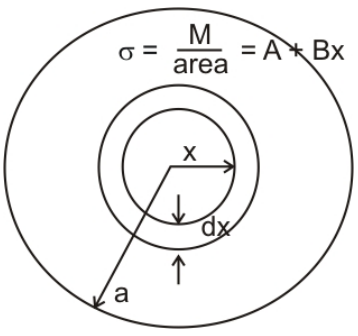
Correct Answer:

$2\pi a^4 \left(\frac{A}{4} + \frac{aB}{5}\right)$

Solution:

$\sigma = A + Bx$

We take an elemental ring of mass dm at a distance x from the centre.



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So Area of an elemental ring= $2\pi x dx$

As the moment of inertia for an elemental ring = $dI = dmx^2$

Total moment of inertia= $$I = \int dI = \int_0^a dmx^2 = \int_0^a \sigma dA x^2 = \int_0^a (A + Bx)x^2(2\pi x dx)$$
$$I = 2\pi(\int_0^a Ax^3 dx + \int_0^a Bx^4 dx) = 2\pi(A * \frac{a^4}{4} + B * \frac{a^5}{5})$$

Hence, the correct option is (2).

Q. 16 Under an adiabatic process, the volume of an ideal gas gets doubled. Consequently the mean collision time between the gas molecule changes from τ_1 to τ_2. If $\frac{C_p}{C_v} = \gamma$ for this gas then a good estimate for $\frac{\tau_2}{\tau_1}$ is

Option 1:
$\left(\frac{1}{2}\right)^{(\gamma+1)/2}$

Option 2:
$\frac{1}{2}$

Option 3:
$\left(\frac{1}{2}\right)^{\gamma}$

Option 4:
$(2)^{(1+\gamma)/2}$

Correct Answer:
$(2)^{(1+\gamma)/2}$

Solution:

As, relaxation time= $\tau \quad \alpha \quad \frac{V}{\sqrt{T}}$

and using $PV^{\gamma} = Const$ gives $\mathrm{T} \propto \frac{1}{\mathrm{V}^{\gamma-1}}$

$\tau \propto V^{1+(\gamma-1)/2}$
$\tau \propto V^{(1+\gamma)/2}$

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$$\frac{\tau_2}{\tau_1} = \left(\frac{2\ \mathrm{V}}{\mathrm{V}}\right)^{(1+\gamma)/2}$$
$$\frac{\tau_2}{\tau_1} = (2)^{(1+\gamma)/2}$$

Q. 17 The dimension of $\left[\frac{B^2}{2\mu_0}\right]$, where B is the magnetic field and μ_0 is permeability of vaccum, is :

Option 1:
ML^2T^{-2}

Option 2:
$ML^{-1}T^{-2}$

Option 3:
MLT^{-2}

Option 4:
ML^2T^{-1}

Correct Answer:
$ML^{-1}T^{-2}$

Solution:

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The energy density of Electromagnetic wave= $\frac{1}{2}\frac{B^2}{\mu_o}$

So, $\left[\frac{B^2}{2\mu_0}\right] = \left[\frac{Energy}{Volume}\right] = \left[\frac{ML^2T^{-2}}{L^3}\right] = ML^{-1}T^{-2}$

Hence, the correct option is (2).

Q. 18 The activity of a radioactive sample falls from 700s^{-1} to 500s^{-1} in 30 minutes. Its half life (in minutes) is close to :

Correct Answer:
62

Solution:
Activity at any time is given as

$$A = A_0 e^{-\lambda t}$$
$$\Rightarrow \frac{A}{A_0} = e^{-\lambda t}$$
$$\Rightarrow ln(\frac{A_0}{A}) = \lambda t$$

After half-life,

$$t = T_{\frac{1}{2}}$$
$$A = \frac{A_0}{2}$$

so $\ln 2 = \lambda T_{1/2}$

Now for $t = 30\ min$, $A_0 = 700\ s^{-1}$, $A = 500\ s^{-1}$

$$\ln\left[\frac{700}{500}\right] = \lambda(30)$$

from((i) and (ii)

$$\frac{\ln 2}{\ln(7/5)} = \frac{T_{1/2}}{(30)}$$

$$t_{1/2} = 61.81 \cong 62\ min$$

Hence the correct option is (1).

Q. 19 An ideal fluid flows (laminar flow) through a pipe of non-uniform diameter. The maximum and minimum diameters of the pipes are 6.4 cm and 4.8 cm, respectively. The ratio of the minimum and the maximum velocities of fluid in this pipe.

Option 1:
$\frac{\sqrt{3}}{2}$

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Option 2:

$\frac{3}{4}$

Option 3:

$\frac{81}{256}$

Option 4:

$\frac{9}{16}$

Correct Answer:

$\frac{9}{16}$

Solution:

By equation of continuity

$$A_1 v_1 = A_2 v_2$$

If the area is large velocity is minimum

Therefore

$$\frac{v_{min}}{v_{max}} = \left(\frac{4.8}{6.4}\right)^2 = \frac{9}{16}$$

Hence the correct option is (4).

Q. 20 The electric field of a plane electromagnetic wave is given by $\overrightarrow{\mathrm{E}}(\mathrm{t}) = \mathrm{E}_0 \frac{(\hat{\mathrm{i}} + \hat{\mathrm{j}})}{\sqrt{2}} \cos(\omega \mathrm{t} + \mathrm{Kz})$. A positively charged particle at the point $(0, 0, \pi/\mathrm{K})$ is given velocity $\mathrm{v}_0 \hat{\mathrm{k}}$ at $\mathrm{t} = 0$. The direction of force acting on a particle is:-

Option 1:

Paralle to $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$

Option 2:

Zero

Option 3:

antiparallel $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$

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Option 4:
parallel to $\hat{k}$

Correct Answer:
antiparallel $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

Solution:

Force due to electric field is in direction $-\frac{(\hat{i}+\hat{j})}{\sqrt{2}}$

because at t=0, $E=-\frac{(\hat{i}+\hat{j})}{\sqrt{2}}E_0$

Force due to the magnetic field is in direction $q(\vec{v}\times\vec{B})$ and $\vec{v}||\hat{k}$

therefore, it is parallel to $\vec{E}$

therefore the net force is antiparallel to $\frac{(\hat{i}+\hat{j})}{\sqrt{2}}$

Hence the correct option is (3).

Q. 21 The balancing length for a cell is 560 cm in a potentiometer experiment. When an external resistance of 10Ω is connected in parallel to the cell, the balancing length changes by 60 cm. If the internal resistance of the cell is $\frac{N}{10}\Omega$, where N is an integer then value of N is

---

Correct Answer: 12

Solution:

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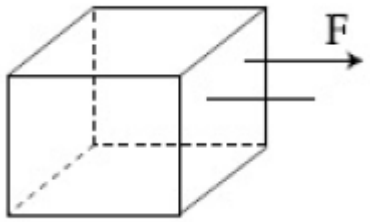
$$E \propto 560$$
$$\frac{E \times 10}{10+r} \propto 500$$
$$\frac{10+r}{10} = \frac{56}{50}$$
$$\Rightarrow r = 1.2 = \frac{n}{10}$$
$$\Rightarrow n = 12$$

Q. 22 Consider a uniform cubical box of side a on a rough floor that is to be moved by applying minimum possible force F at a point b above its centre of mass (see figure). If the coefficient of friction is $\mu = 0.4$, the maximum possible value of $100 \times \frac{b}{a}$ for box npt to topple before moving is



Correct Answer: 50

Solution:

For no toppling

$$F\left(\frac{a}{2}+b\right) \leq mg\frac{a}{2}$$

Using $F = \mu mg$

$$\mu\frac{a}{2} + \mu b \leq \frac{a}{2}$$

using $\mu = 0.4$

$$0.2a + 0.4b \leq 0.5a$$
$$0.4b \leq 0.3a$$
$$b \leq \tfrac{3a}{4}$$

i.e $b \leq 0.75a$ but this is not possible as b can maximum be equal to 0.5a

So $\therefore \quad \left(100\frac{\mathrm{b}}{\mathrm{a}}\right)_{\max} = 50.00$

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Q. 23 A 60 pF of capacitor is fully charged by a 20 V supply. It is then disconnected from the supply and is connected to another uncharged 60 pF capacitor in parallel. The electrostatic energy that is lost in this process by the time the charge is redistributed between them is (in nJ)

Correct Answer: 6

Solution:

$$Heat\ loss = \frac{1}{2}cv^2 - c\frac{v^2}{4} = c\frac{v^2}{4}$$
$$= 60 \times 10^{-12} \times 100 = 6nJ$$

Therefor n =6

Q. 24 The sum of two forces $\vec{P}$ and $\vec{Q}$ is $\vec{r}$ such that $\left|\vec{R}\right| = \left|\vec{P}\right|$. The angle θ (in degrees) that the resultant of $2\vec{P}$ and $\vec{Q}$ will make with $\vec{Q}$ is ,

Correct Answer:
90

Solution:
As

$$|\vec{P} + \vec{Q}| = |\vec{P}|$$
$$\because |\vec{R}| = |\vec{P}|$$
$$\Rightarrow P^2 + Q^2 + 2PQcos\theta = P^2$$
$$\Rightarrow Q + 2Pcos\theta = 0...(1)$$

Now Vector P is doubled but vector Q is the same so the angle between $2\vec{P}\ and\ \vec{Q}$ will be also θ

And If resultant of $2\vec{P}\ and\ \vec{Q}$ is $\vec{R'}$

Then from the figure angle between $\vec{R'}\ and\ \vec{Q}$ is α

and
$$tan\alpha = \frac{2Psin\theta}{Q + 2Pcos\theta}$$
$$From\ equation\ (1)$$
$$\text{we get } \tan\alpha = \infty \Rightarrow \alpha = 90^0$$

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Q. 25 M gram of steam at 100^oC is mixed with 200 g of ice at its melting point in a thermally insulated container. If it produces liquid water at 40^oC [heat of vaporization of water is 540 cal/g and heat of fusion of ice is 80 cal/g] the value of M is

Correct Answer:
40

Solution:

$$\mathrm{M_{ice}L_f + M_{ice}(40-0)C_w = m_{steam}L_v + m_{steam}(100-40)C_w}$$
$$\Rightarrow 200[80+40(1)] = \mathrm{M}[540+60(1)]$$
$$\Rightarrow 200(120) = \mathrm{M}(600)$$
$$\mathrm{M} = 40\mathrm{gm}$$

Chemistry

Q. 1 Within each pair elements F and Cl, S and Se and Li and Na, respectively, the elements that release more energy upon an electron gain are

Option 1:
Cl, Se and Na

Option 2:
Cl, S and Li

Option 3:
F, S and Li

Option 4:
F, Se and Na

Correct Answer:
Cl, S and Li

Solution:
From the given pairs, the more energy on gaining of an electron will be released by the following elements:
Cl, S, and Li

Therefore, **Option(2) is correct.**

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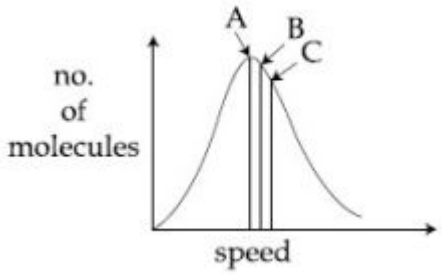
Q. 2 Identify the correct labels A. B and C in the following graph from the options given below:



Root mean square speed (V_{rms}) ; most probable speed (V_{mp}), Average speed (V_{av})

Option 1:
A - (V_{mp}), B- (V_{av}), C- (V_{rms})

Option 2:
A - (V_{mp}), B- (V_{rms}) , C- (V_{rms})

Option 3:
A - (V_{av}), B- (V_{rms}) ,C- (V_{av})

Option 4:
A - (V_{rms}) ,B- (V_{mp}), C- (V_{av})

Correct Answer:
A - (V_{mp}), B- (V_{av}), C- (V_{rms})

Solution:
Relation between u_{av}, u_{mp} and u_{rms}
The three types of molecular speeds, namely, most probable speed(v_{mp}), average speed (v_{av}) and root mean square speed(v_{rms}) of a gas at a given temperature are related to each other as follows:

$$v_{mp} : v_{av} : v_{ms} = \sqrt{\frac{2RT}{M}} : \sqrt{\frac{8RT}{\pi M}} : \sqrt{\frac{3RT}{M}}$$
$$v_{mp} : v_{av} : v_{rms} = 1.414 : 1.596 : 1.732$$
$$v_{mp} : v_{av} : v_{rms} = 1 : 1.128 : 1.224$$

For a particular gas, at a particular temperature:
$v_{mp} < v_{av} < v_{rms}$

It follows from the above relationship that:
Average speed(v_{av}) =0.921 x Root mean square speed(v_{rms})
Most probable speed(v_{mp}) = 0.817 x Root mean square speed(v_{rms})

$C_{RMS} > C_{Average} > C_{MPS.}$

A - $V_{MPS,}$ B - $V_{Average}$, C - V_{RMS}

Therefore, **Option(1) is correct.**

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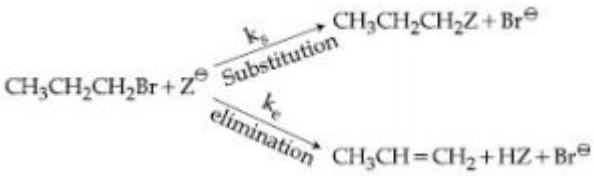
Q. 3 For the following reactions:



where

$Z^{\ominus} = CH_3CH_2O^{\ominus}$ (A) or $H_3C-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-O^{\ominus}$ (B),

k_s and k_e are respectively, the rate constants for substitution and elimination, and $\mu = \frac{k_s}{k_e}$, the correct option is _____.

Option 1:

$\mu_A > \mu_B and k_e(A) > k_e(B)$

Option 2:

$\mu_B > \mu_A and k_e(A) > k_e(B)$

Option 3:

$\mu_A > \mu_B and k_e(B) > k_e(A)$

Option 4:

$\mu_B > \mu_A and k_e(B) > k_e(A)$

Correct Answer:

$\mu_A > \mu_B and k_e(B) > k_e(A)$

Solution:

This is williamson synthesis.

Substitution is better for prrimary Z means A and Elimination tertiary or secondary Z means B.

$k_e(B) > k_e(A)$ will come.

Now, $\mu = \frac{k_s}{k_e}$

A favours substitution and B favour elimination.
So, for A, k_s will be greater than k_e.

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then $\mu_A > \mu_B$ will right.

Therefore, **Option(3) is correct.**

Q. 4 The ammonia (NH_3) released on quantitative reaction of 0.6g, urea (NH_2CONH_2) with sodium hydroxide $(NaOH)$ can be neutralised by:

Option 1:
200 ml of 0.2 N HCl

Option 2:
200 ml of 0.4 N HCl

Option 3:
100 ml of 0.1N HCl

Option 4:
100 ml of 0.2N HCl

Correct Answer:
100 ml of 0.2N HCl

Solution:
2 × mole of Urea = mole of NH_3 .....(1)
mole of NH_3 = mole of HCl .....(2)
$\therefore$ mole of HCl = 2 × mole of Urea

mole of HCl = $2 \times \frac{0.6}{60} = 0.02mol$ ...(i)

[We know , mole = M X V = N X n X V]

$100 \text{ ml} \times 0.2\text{N} \times 1 = 0.02 \text{ mol}$ ...as (i).

Therefore, **Option(4) is correct.**

Q. 5 The redox reaction among the following is:

Option 1:
Formation of ozone from atmospheric oxygen in presence of sunlight

Option 2:
Redox of H_2SO_4 with $NaOH$

Option 3:
Combination of dinitrogen with dioxygen of 2000K.

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Option 4:
Reaction of $[Co(H_2O)_6]Cl_3$ with $AgNo_3$

Correct Answer:
Combination of dinitrogen with dioxygen of 2000K.

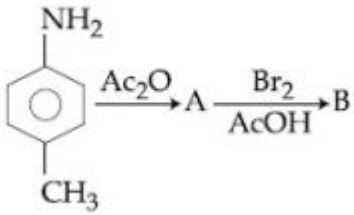
Solution:

$$N_2 + O_2 \rightarrow 2NO$$
$$3O_2 \rightarrow 2O_3$$
$$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$$
$$AgNO_3 + NaCl \rightarrow NaNO_3 + AgCl$$

showing redox reaction, both oxidation and reduction.

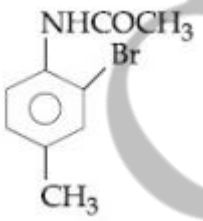
Therefore, **Option(3) is correct.**

Q. 6 In the following reaction sequence,

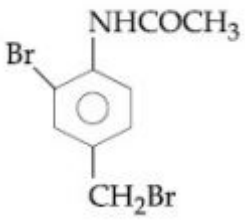


the major product B is:

Option 1:



Option 2:



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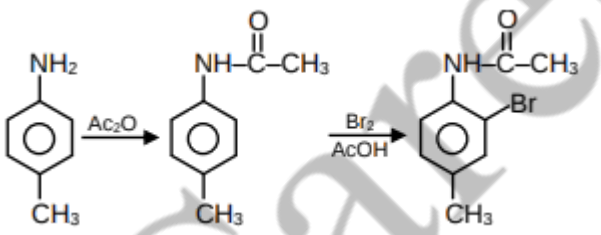
Option 3:

Option 4:

Correct Answer:

Solution:
The reaction will be -



Therefore, **Option(1) is correct.**

Q. 7 A chromatography column, packed with silica gel as stationary phase, was used to separate a mixture of compounds consisting of (A) benzanilide (B) aniline and (C) acetophenone. When the column is eluted with a mixture of solvents, hexane: ethyl acetate (20:80), the sequence of obtained compounds is:

Option 1:
(B), (A), and (C)

Option 2:
(C), (A) and (B)

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Option 3:
(B), (C) and (A)

Option 4:
(A) , (B) and (C)

Correct Answer:
(C), (A) and (B)

Solution:
Since, the solvent mixture is 20:80 in hexane : ethylacetate, the compound having a larger polarity will be obtained at the top.

The dipole moments of the compounds are in the order of

Acetophenone (C) > Benzanilide (A) > Aniline (B)

So, the order in which the compounds will be obtained is

(C) > (A) > (B)

Therefore, Option(2) is correct.

Q. 8 Which of the following statements is correct?

Option 1:
Gluconic acid can form cyclic (acetal/hemiacetal) structure

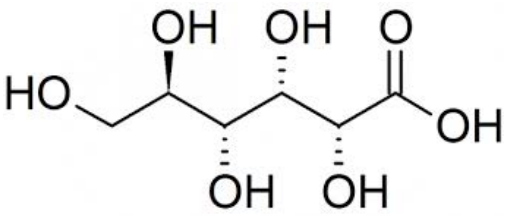
Option 2:
Gluconic acid is a dicarboxylic acid

Option 3:
Gluconic acid is obtained by oxidation of glucose with HNO_3

Option 4:
Gluconic acid is a partial oxidation product of glucose

Correct Answer:
Gluconic acid is a partial oxidation product of glucose

Solution:
Gluconic acid is obtained by partial oxidation of glucose by mild oxidising agents like Tollen's reagent or Fehling solution or Bromine water.



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Gluconic acid can not form cyclic hemiacetal or acetal.

It is to be noted that Glucose reacts with strong oxidising agents to form Saccharic acid.

Therefore, **Option(4) is correct.**

Q. 9 The refining method used when the metal and the impurities have low and high melting temperatures, respectively is:

Option 1:
vapour phase refining

Option 2:
distillation

Option 3:
liquation

Option 4:
zone refining

Correct Answer:
liquation

Solution:
As we have learnt,

Liquation -

In this method, a low melting metal like a tin can be made to flow on a sloping surface. In this way, it is separated from higher melting impurities.

-

Liquation, a technique for separating constituents of ore, a metal or an alloy by partial melting. When the material is heated to a temperature where one of the constituents melts and the other remains solid, the liquid constituent can be drained off.

Therefore, **Option(3) is correct.**

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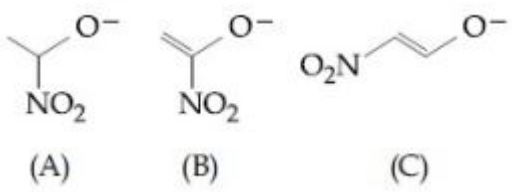
Q. 10 The correct order of statbility for the following alkoxides is:



(A) (B) (C)

Option 1:
C > A > B

Option 2:
C > B > A

Option 3:
B > A > C

Option 4:
B > C > A

Correct Answer:
C > B > A

Solution:
When a negative charge is delocalised with an electron-withdrawing group like (NO_2) then stability increases.

(A) The negative charge is localised

(B) The negative charge is **de**localised with the carbon of the alkene

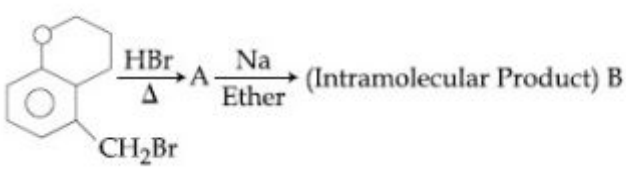
(C) Negative charge is **de**localised with NO_2 group

So, the order will be

C > B > A

Therefore, **Option(2) is correct.**

Q. 11 In the following reaction sequence, structure A and B, respectively will be:



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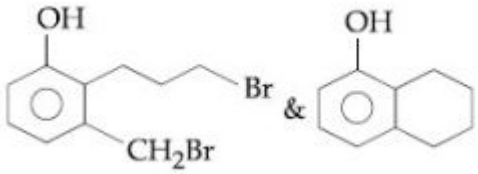
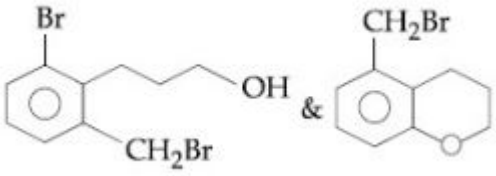
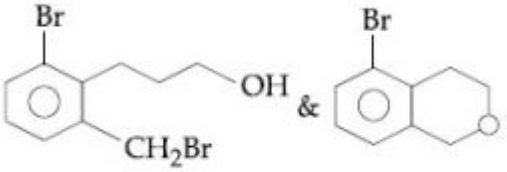
Option 1:



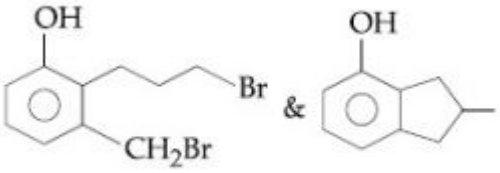
Option 2:



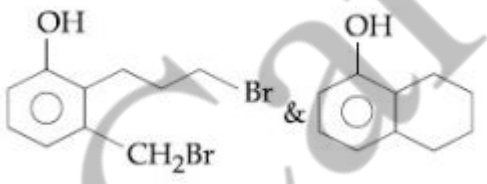
Option 3:



Option 4:

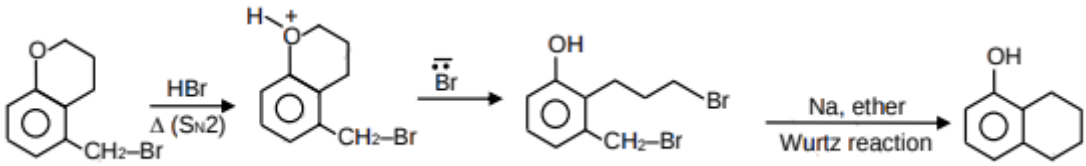


Correct Answer:



Solution:

The reaction will be -



Therefore, **Option(1) is correct.**

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Q. 12 The number of possible optical isomers for complexes MA_2B_2 with sp^3 and dsp^2 hybridized metal atom, respectively is:

Note : A and B are unidentate netural and unidentate monoanionic ligands, respectively.

Option 1:
0 and 1

Option 2:
0 and 0

Option 3:
2 and 2

Option 4:
0 and 2

Correct Answer:
0 and 0

Solution:
In tetrahedral structure of Ma_2b_2 which is sp^3 hybridized, the central metal is not chiral and hence it does not show optical isomerism.

In the case of a square planar Ma_2b_2 that is dsp^2 hybridized, there can exist two geometrical cis-trans isomers but no optical isomers.

Hence, there are no optical isomers possible for both the square planar as well as the tetrahedral form of complex type Ma_2b_2

Therefore, **Option(2) is correct.**

Q. 13 Among the statements (a) - (d), the correct ones are:

(a) Decomposition of hydrogen peroxide gives dioxygen.

(b) Like hydrogen peroxide, compounds, such as $KClO_3$, $Pb(NO_3)_2$ and $NaNO_3$ when heated, also liberate dioxygen.

(c) 2-Ethylanthraquinol is useful fo the industrial prepartion of hydrogen peroxide.

(d) Hydrogen peroxide is used for the manufacture of sodium perborate.

Option 1:
a, b, c and d

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Option 2:
a , b and c only

Option 3:
a, c and d only

Option 4:
a and c only

Correct Answer:
a, b, c and d

Solution:
1) $H_2O_2 \longrightarrow H_2O + O_2$

2) $2KClO_3 \longrightarrow 2KCl + 3O_2$

$2Pb(NO_3)_2 \longrightarrow 2PbO + 4NO_2 + O_2$

$2Na(NO_3) \longrightarrow 2NaNO_2 + O_2$

3) 2-Ethylanthaquinol is used in the industrial preparation of H_2O_2

4) Generally, peroxy compounds are formed using H_2O_2

Therefore, **Option(1) is correct.**

Q. 14 Among the statements (a) - (d), the incorrect ones are:

(a) Octahedral Co(III) complexes with strong field ligands have very high magnetic moments.

(b) When $\Delta_0 < P$, the d-electron configuration of Co(III) is an octahedral is $t_{eg}^4 e_g^2$.

(c) Wavelength of light absorbed by $[Co(en)_3]^{3+}$ is lower than that of $[CoF_6]^{3+}$

(d) If the Δ_0 for an octahedral complex of Co(III) is 18,000 cm^{-1}, the Δ_t for tetrahedral complex with the same ligand will be 16,000cm^{-1}

Option 1:
b and c only

Option 2:
a and b only

Option 3:
a and d only

Option 4:
c and d only

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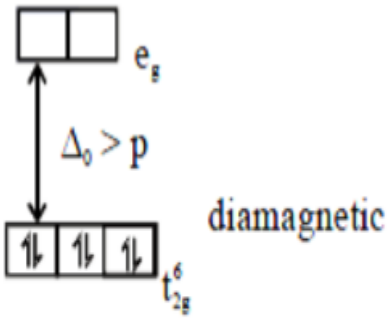
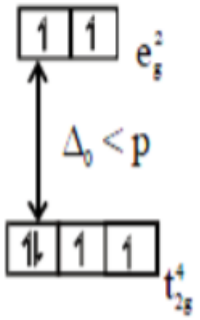
Correct Answer:
a and d only

Solution:

(a) Co^{+3} (with strong field ligands)



(b) If $\Delta_0 < p$;



(c) Splitting power of ethylenediamine (en) is greater than fluoride (F^-) ligand therefore more energy absorbed by $[Co(en)_3]^{3+}$ as compared to $[CoF_6]^{3-}$.

So wave length of light absorbed by $[Co(en)_3]^{3+}$ is lower than that of $[CoF_6]^{3-}$

(d) $\Delta_t = \frac{4}{9}\Delta_0$

so if $\Delta_0 = 18{,}000\ cm^{-1}$

$$\Delta_t = \frac{4}{9} \times 18000 = 8000\ cm^{-1}$$

Statement (a) and (d) are incorrect.

Therefore, **Option(3) is correct.**

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Q. 15 In the following reactions, products (A) and (B) respectively are:

$$NaOH + Cl_2 \rightarrow A + NaCl + side\ product$$

(hot and conc.)

$$Ca(OH)_2 + Cl_2 \rightarrow B + CaCl_2 + side\ product$$

(dry)

Option 1:
$NaClO_3\ and\ Ca(ClO_3)_2$

Option 2:
$NaOCl\ and\ Ca(ClO_3)_2$

Option 3:
$NaOCl\ and\ Ca(OCl)_2$

Option 4:
$NaClO_3\ and\ Ca(OCl)_2$

Correct Answer:
$NaClO_3\ and\ Ca(OCl)_2$

Solution:
As we have learnt,

Chlorine reacts with $NaOH$ and $Ca(OH)_2$ and undergoes disproportionation. The products depend upon the concentration of the base and also the temperature.

The reactions are given as

$$2Na(OH) + 3Cl_2 \longrightarrow NaClO_3 + 5NaCl + 3H_2O$$

(Hot and Conc.)

$$2Ca(OH)_2 + 2Cl_2 \longrightarrow Ca(OCl)_2 + CaCl_2 + 2H_2O$$

(Dry)

Therefore, **Option(4) is correct.**

Q. 16 Two open beakers one containing a solvent and the other containing a mixture of that solvent with non volatile solute are together sealed in a container. Over time:

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Option 1:
the volume of the solution and the solvent does not change

Option 2:
the volume of the solution increases and the volume of the solvent decreases

Option 3:
the volume of the solution decreases and the volume of the solvent increases

Option 4:
the volume of the solution does not change and the volume of the solvent decreases.

Correct Answer:
the volume of the solution increases and the volume of the solvent decreases

Solution:
There will be lowering in vapour pressure in solution due to which the transfer of solvent molecules will take place from the beaker containing pure solvent to that containing the solution.

Therefore, **Option(2) is correct.**

Q. 17 For the reaction

$2H_2(g) + 2NO(g) \rightarrow N_2(g) + 2H_2O(g)$ the observed rate experssion is, rate $= k_f\,[NO]^2\,[H_2]$. The rate expression for reverse reaction is:

Option 1:
$k_b[N_2][H_2O]^2$

Option 2:
$k_b[N_2][H_2O]$

Option 3:
$k_b[N_2][H_2O]^2/[H_2]$

Option 4:
$k_b[N_2][H_2O]^2/[NO]$

Correct Answer:
$k_b[N_2][H_2O]^2/[H_2]$

Solution:

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$$K_{eq} = \frac{k_f}{k_b} = \frac{[N_2][H_2O]^2}{[H_2]^2[NO]^2}$$

At equilibrium $r_f = r_b$

$$\underset{[Given]}{k_f[H_2][NO]^2} = k_b \frac{[N_2][H_2O]^2}{[H_2]}$$

Hence, rate expression for reverse reaction.

$$= k_b \frac{[N_2][H_2O]^2}{[H_2]}$$

Therefore, **Option(3) is correct.**

Q. 18 The bond order and the magnetic charateristics of CN^- are:

Option 1:
3, paramagnetic

Option 2:
$2\frac{1}{2}$, diamagnetic

Option 3:
3, diamagnetic

Option 4:
$2\frac{1}{2}$, paramagnetic

Correct Answer:
3, diamagnetic

Solution:
The bond order of CN^- is 3 and all the electrons are paired in this ion, thus it is diamagnetic in nature.

Therefore, **Option(3) is correct.**

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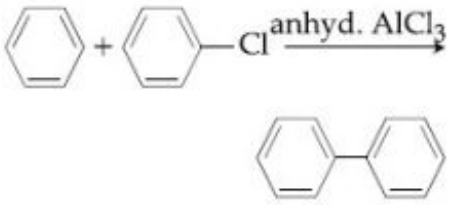
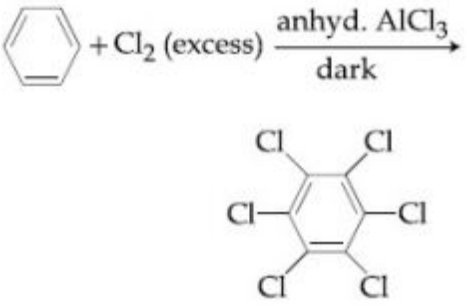
Q. 19 Consider the following reactions:

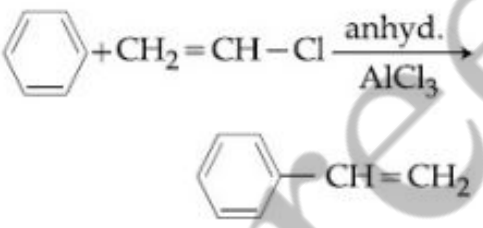
(a)



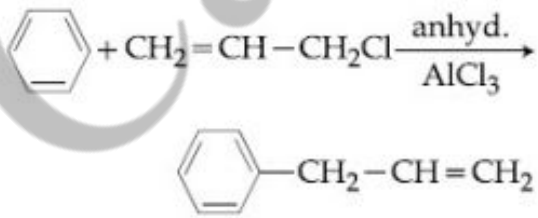
(b)



(c)



(d)



Which of these reactions are possible?

Option 1:
a and d

Option 2:
b and d

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Option 3:
b, a and d

Option 4:
a and d

Correct Answer:
b and d

Solution:
As we have learnt,

1. Aryl halides (a) do not give Friedel Craft's reaction with Benzene because the unstable Phenyl carbocation is very difficult to be obtained.

2. Vinyl halides (c) do not give Friedel Craft's reaction because the unstable vinyl carbocation is very difficult to be obtained.

Reactions given in (b) and (d) are correct.

Therefore, **Option(2) is correct.**

Q. 20 The equation that is incorrect is:

Option 1:
$(\Lambda_m^\circ)_{NaBr} - (\Lambda_m^\circ)_{NaI} = (\Lambda_m^\circ)_{KBr} - (\Lambda_m^\circ)_{NaBr}$

Option 2:
$(\Lambda_m^\circ)_{NaBr} - (\Lambda_m^\circ)_{NaCl} = (\Lambda_m^\circ)_{KBr} - (\Lambda_m^\circ)_{KCl}$

Option 3:
$(\Lambda_m^\circ)_{KCl} - (\Lambda_m^\circ)_{NaCl} = (\Lambda_m^\circ)_{KBr} - (\Lambda_m^\circ)_{NaBr}$

Option 4:
$(\Lambda_m^\circ)_{H_2O} = (\Lambda_m^\circ)_{HCl} + (\Lambda_m^\circ)_{NaOH} - (\Lambda_m^\circ)_{NaCl}$

Correct Answer:
$(\Lambda_m^\circ)_{NaBr} - (\Lambda_m^\circ)_{NaI} = (\Lambda_m^\circ)_{KBr} - (\Lambda_m^\circ)_{NaBr}$

Solution:
$(\Lambda_{\mathrm{m}}^0)_{\mathrm{NaBr}} - (\Lambda_{\mathrm{m}}^0)_{\mathrm{NaI}} = (\Lambda_{\mathrm{m}}^0)_{\mathrm{KBr}} - (\Lambda_{\mathrm{m}}^0)_{\mathrm{NaBr}}$

$(\Lambda_{\mathrm{m}}^0)_{\mathrm{Na}} + (\Lambda_{\mathrm{m}}^0)_{\mathrm{Br}} - (\Lambda_{\mathrm{m}}^0)_{\mathrm{Na}} - (\Lambda_{\mathrm{m}}^0)_{\mathrm{I}} = (\Lambda_{\mathrm{m}}^0)_{\mathrm{K}} + (\Lambda_{\mathrm{m}}^0)_{\mathrm{Br}} - (\Lambda_{\mathrm{m}}^0)_{\mathrm{Na}} + (\Lambda_{\mathrm{m}}^0)_{\mathrm{Br}}$

Both sides are not equal.

Therefore, **Option(1) is correct.**

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Q. 21 Consider the following reactions:

$$NaCl + K_2Cr_2O_7 + H_2SO_4(conc.) \rightarrow A + side\ products$$

$$A + NaOH \rightarrow B + side\ products$$

$$B + H_2SO_4(dil.) + H_2O_2 \rightarrow C + side\ products$$

The sum of the total number of atoms in one molecule each of A, B and C is:

Option 1:
18

Option 2:
16

Option 3:
14

Option 4:
20

Correct Answer:
18

Solution:
(A) = CrO_2Cl_2

(B) = Na_2CrO_4

(C) = CrO_5

Total atoms = 5 (A) + 7 (B) + 6 (C) = 18 atoms

Thus, the correct answer is 18.

Therefore, option(1) is correct

Q. 22 The number of sp^2 hybridized carbons present in "Aspartame" is_____.

Correct Answer:
9

Solution:
The aspartame structure is given below:

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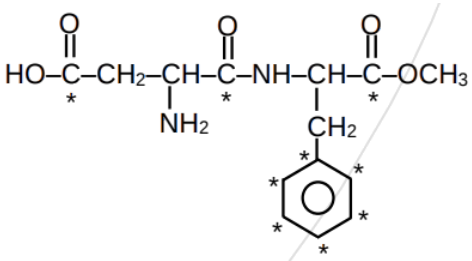

The number of sp^2 hybridised carbon atoms in aspartame is 9.

Q. 23 The flocculation value of HCl for arsenic sulphide solution is 30m molL^{-1}. If H_2SO_4 is used for the flocculation of arsenic sulphide, the amount, in grams of H_2SO_4 in 250 ml required for above purpose is_____

(molecular mass of H_2SO_4 = 98 g/mol)

Option 1:
0.37

Option 2:
23

Option 3:
3.7

Option 4:
0.45

Correct Answer:
0.37

Solution:
For 1L sol 30 mmol of HCl is required.

Thus, for 1L sol 15 mmol H_2SO_4 is required

For 250 mL of sol:
(15 x 250)/1000 mmol H_2SO_4 = 3.75 mmol

Thus, amount = 3.75 x 10^{-3} x 98 = 0.367

Therefore, **Option(1) is correct**

Q. 24 The standard heat of formation $(\Delta_f H^\circ_{298})$ of ethane (in kJ/mol) , if the heat of combustion of ethane, hydrogen and graphite are -1560, -393.5 and -286 kJ/mol, respectively is _____

Option 1:
-192.5

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Option 2:
192.5

Option 3:
- 292.5

Option 4:
292.5

Correct Answer:
-192.5

Solution:
The reactions we have:

$$C_2H_6 + \frac{7}{2}O_2 \rightarrow 2CO_2 + 3H_2O \qquad [\Delta H_c = -1560 kJ/mol] \quad \text{.....(i)}$$

$$2C + O_2 \rightarrow CO_2 \qquad [\Delta H_c = --286 kJ/mol] \quad \text{.....(ii)}$$

$$3H_2 + \frac{1}{2}O_2 \rightarrow H_2O \qquad [\Delta H_c = -393.5 kJ/mol] \quad \text{.....(iii)}$$

Multiply eq(ii) by 2 and eq(iii) by 3.

Now, ΔH^o_f = 2 x eq(ii) + 3eq(iii) - 1

Thus, the equaton is given as:

$$2C + 3H_2 \rightarrow C_2H_6$$

$$\Delta H^o_f = 2\,x\,(-286) + 3\,x\,(-393.5) - (-1560)$$

$$\Delta H^o_f = -572 - 1180.5 + 1560$$

$$\Delta H^o_f = -1752.5 + 1560 = -192.5 \text{ kJ/mol}$$

Hence, the option number (1) is correct.

Q. 25 3g of acetic acid is added to 250 mL of 0.1M HCl and the solution made up to 500 mL. To 20 mL of this solution $\frac{1}{2}$ mL. of 5 M NaOH is added. The pH of the solution is_____.

[Given: pK_a of acetic acid = 4.75, molar mass of acertic acid = 60 g/mol, log3 = 0.4771]

Neglect any chnages in volume.

Option 1:
5.22

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Option 2:
4.22

Option 3:
6.22

Option 4:
7.65

Correct Answer:
5.22

Solution:
Given,

m mole of acetic acid in 20 mL = 2
m mole of HCl in 20 mL = 1
m mole of NaOH = 2.5

$$\underset{\substack{2\\0.5}}{\mathrm{CH_3COOH}} + \underset{\substack{3/2\\0}}{\mathrm{NaOH(remaining)}} \rightarrow \underset{\substack{0\\3/2}}{\mathrm{CH_3COONa}} + \underset{0}{\mathrm{water}}$$

$$\mathrm{pH} = \mathrm{pK_a} + \log\frac{\frac{3}{2}}{2}$$

pH = 4.74 + log 3

pH = 4.74 + 0.48 = 5.22

Thus, the correct answer is 5.22

Hence, the option number (1) is correct.

Maths

Q. 1 If $3x + 4y = 12\sqrt{2}$ is a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{9} = 1$ for some $a \in R$, then the distance between the foci of the ellipse is :

Option 1:
$2\sqrt{5}$

Option 2:
$2\sqrt{7}$

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Option 3:

$2\sqrt{2}$

Option 4:

4

Correct Answer:

$2\sqrt{7}$

Solution:

What is Ellipse? -

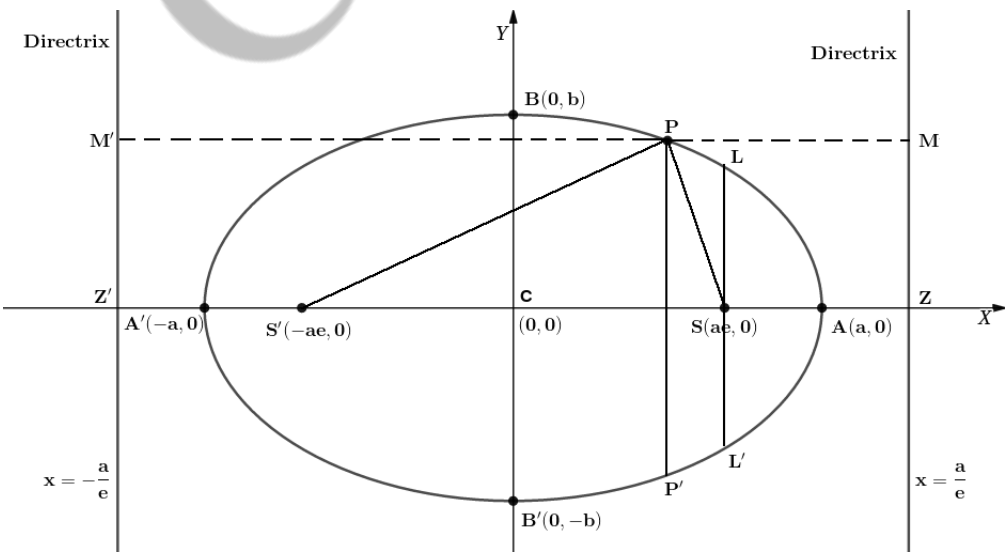
Standard Equation of Ellipse:

The standard form of the equation of an ellipse with center (0, 0) and major axis on the x-axis is

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \quad \text{where }, b^2=a^2\left(1-e^2\right)_{(a>b)}$$

1. a > b
2. the length of the major axis is 2a
3. the coordinates of the vertices are (±a, 0)
4. the length of the minor axis is 2b
5. the coordinates of the co-vertices are (0, ±b)

Important Terms related to ellipse:



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1. Centre: All chord passing through point C is bisected at point C. Here C is the origin, i.e. (0, 0).

2. Foci: Point S and S' is foci of the ellipse where, S is (ae, 0) and S' is (-ae, 0).

3. Directrices: The straight-line ZM and Z'M' are two directrices of the ellipse and their equations are x=ae and x=-ae.

4. Axis: In figure AA' is called the major axis and BB' is called the minor axis. 2a is called the length of the major Double Ordinateaxis and 2b is called the length of the minor axis.

5. Double Ordinate: If a line perpendicular to the major axis meets the curve at P and P', then PP' is called double ordinate.

6. Latusrectum: Double ordinate passing through focus is called latus rectum. Here LL' is a latus rectum.

-

Equation of Tangent of Ellipse in Parametric Form and Slope Form -

$$\text{The equation of tangent of slope m to the ellipse, } \frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \text{ are}$$
$$y=mx\pm\sqrt{a^2m^2+b^2} \text{ and coordinate of point of contact is}$$
$$\left(\mp\frac{a^2m}{\sqrt{a^2m^2+b^2}},\pm\frac{b^2}{\sqrt{a^2m^2+b^2}}\right)$$

-

$$3x+4y=12\sqrt{2}$$
$$\Rightarrow 4y=-3x+12\sqrt{2}$$
$$\Rightarrow y=-\tfrac{3}{4}x+3\sqrt{2}$$

condition of tangency $c^2=a^2m^2+b^2$

$$18=a^2\cdot\tfrac{9}{16}+9$$
$$a^2\cdot\tfrac{9}{16}=9$$
$$a^2=16$$
$$1a=4$$
$$e=\sqrt{1-\tfrac{b^2}{a^2}}=\sqrt{1-\tfrac{9}{16}}=\tfrac{\sqrt{7}}{4}$$

$$\therefore \text{ae}=\tfrac{\sqrt{7}}{4}\cdot 4=\sqrt{7}$$
$$\therefore \text{ focus are } (\pm\sqrt{7},0)$$

distance between foci = $2\sqrt{7}$

Correct Option (2)

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Q. 2 Let A,B,C and D be four non-empty sets. The contrapositive statement of '' If $A \subseteq B$ and $B \subseteq D$, then $A \subseteq C$ '' is :

Option 1:
If $A \subseteq C$, then $B \subset A$ or $D \subset B$

Option 2:
If $A \nsubseteq C$, then $A \subseteq B$ and $B \subseteq D$

Option 3:
If $A \nsubseteq C$, then $A \nsubseteq B$ and $B \subseteq D$

Option 4:
If $A \nsubseteq C$, then $A \nsubseteq B$ and $B \nsubseteq D$

Correct Answer:
If $A \nsubseteq C$, then $A \nsubseteq B$ and $B \nsubseteq D$

Solution:

Converse, Inverse, and Contrapositive -

Given an if-then statement "if p , then q ," we can create three related statements:

A **conditional statement** consists of two parts, a hypothesis in the “if” clause and a conclusion in the “then” clause. For instance, “If you are born in some country, then you are a citizen of that country”

"you are born in some country" is the hypothesis.

"you are a citizen of that country" is the conclusion.

To form the **contrapositive** of the conditional statement, interchange the hypothesis and the conclusion of the inverse statement.

The Contrapositive of “If you are born in some country, then you are a citizen of that country”

is “If you are not a citizen of that country, then you are not born in some country.”

Statement	If p, then q	$p \rightarrow q$
Converse	If q, then p	$q \rightarrow p$
Inverse	If not p, then not q	$(\sim p) \rightarrow (\sim q)$
Contrapositive	If not q, then not p	$(\sim q) \rightarrow (\sim p)$

-

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$\text{Let } P = A \subseteq B, Q = B \subseteq D, R = A \subseteq C$
contrapositive of $(P \wedge Q) \rightarrow R$ is
$\sim R \rightarrow \sim (P \wedge Q)$
i.e. $\sim R \rightarrow (\sim P \vee \sim Q)$

Hence, $\text{If } A \nsubseteq C, \text{ then } A \nsubseteq B \text{ or } B \neq D$

Correct Option (4)

Q. 3 The coefficient of x^7 in the expression $(1+x)^{10} + x(1+x)^9 + x^2(1+x)^8 + ... + x^{10}$ is :

Option 1:
420

Option 2:
330

Option 3:
210

Option 4:
120

Correct Answer:
330

Solution:
Binomial Theorem

$$(x+y)^n = {}^nC_0x^n + {}^nC_1x^{n-1}y + {}^nC_2x^{n-2}y^2 + \cdots + {}^nC_ny^n \quad where,\ n \in \mathbb{N}$$

Now,

Given series S is a GP, with a = $(1+x)^{10}$, r = x/(1+x), n = 11

So, S = $\frac{(1+x)^{10}\left(\frac{x}{1+x}^{11} - 1\right)}{\left(\frac{x}{1+x}\right) - 1}$

= $(1+x)^{11} - x^{11}$

Hence coefficient of x^7 is ${}^{11}C_7 = 330$

Correct option (2)

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Q. 4 In a workshop, there are five machines and the probability of any one of them to be out of service on a day is $\frac{1}{4}$. If the probability that at most two machines will be out of service on the same day is $\left(\frac{3}{4}\right)^3 k$, then k is equal to :

Option 1:
$\frac{17}{2}$

Option 2:
4

Option 3:
$\frac{17}{4}$

Option 4:
$\frac{17}{8}$

Correct Answer:
$\frac{17}{8}$

Solution:

Required probability = when no. the machine has fault + when only one machine has fault + when only two machines have the fault.

$$\mathrm{P(R)} = {}^5\mathrm{C}_0\left(\frac{3}{4}\right)^5 + {}^5\mathrm{C}_1\left(\frac{1}{4}\right)\left(\frac{3}{4}\right)^4 + {}^5\mathrm{C}_2\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^3$$

$$\mathrm{P(R)} = \frac{243}{1024} + \frac{405}{1024} + \frac{270}{1024} = \frac{918}{1024}$$

$$\mathrm{P(R)} = \frac{459}{512}$$

Correct Option (4)

$$\mathrm{P(R)} = \frac{27 \times 17}{64 \times 8} = \left(\frac{3}{4}\right)^3 \times \mathrm{k} = \left(\frac{3}{4}\right)^3 \times \frac{17}{8}$$

$$\therefore \mathrm{k} = \frac{17}{8}$$

Q. 5 The locus of the mid-points of the perpendiculars drawn from points on the line,x=2y to the line x=y is :

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Option 1:
$2x - 3y = 0$

Option 2:
$3x - 2y = 0$

Option 3:
$5x - 7y = 0$

Option 4:
$7x - 5y = 0$

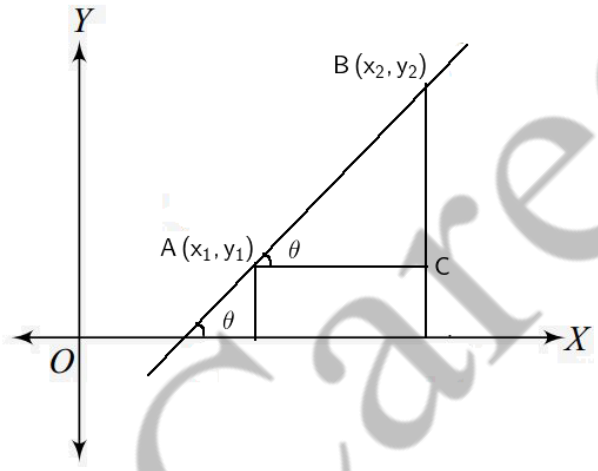
Correct Answer:
$5x - 7y = 0$

Solution:

Straight Line -

Straight Line

The slope of the line joining two Points



If$\mathrm{A}\left(\mathrm{x}_1, \mathrm{y}_1\right)$ and $\mathrm{B}\left(\mathrm{x}_2, \mathrm{y}_2\right)$ are two points on a straight line then the slope of the line is

$$.\tan\theta = \frac{BC}{AC} = \frac{y_2 - y_1}{x_2 - x_1}$$

-

Line parallel and perpendicular to a given line -

Line parallel and perpendicular to a given line

The equation of the line parallel to ax + by + c = 0 is given as ax + by + λ = 0, where λ is some constant.

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Equation of the given line is ax + by + c = 0

Its slope is (-a/b)

So, any equation of line parallel to ax + by + c = 0 is

$$y = \left(-\frac{a}{b}\right)x + c_1$$
$$ax + by - bc_1 = 0$$
$$ax + by + \lambda = 0$$

The equation of the line perpendicular to ax + by + c = 0 is given as bx - ay + λ = 0, where λ is some constant.

Equation of the given line is ax + by + c = 0

Its slope is (-a/b)

Slope of perpendicular line will be (b/a)

So, any equation of line perpendicular to ax + by + c = 0 is

$$y = \left(\frac{b}{a}\right)x + c_1$$
$$bx - by - ac_1 = 0$$
$$bx - ay + \lambda = 0$$

-

$$\text{slope of } PQ = \frac{x-a}{y-2a} = -1$$

$$\Rightarrow x - a = -y + 2a$$

$$\Rightarrow a = \frac{x+y}{3}$$

Using midpoint

$$2x = 2a + b$$
$$2y = a + b$$

$$a = 2x - 2y$$

$$\frac{x+y}{3} = 2(x-y)$$

so locus is $6x - 6y = x + y \quad \Rightarrow \quad 5x = 7y$

Correct Option (3)

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Q. 6 The value of α for which $4\alpha \int_{-1}^{2} e^{-\alpha|x|}dx = 5$, is :

Option 1:
$log_e 2$

Option 2:
$log_e \sqrt{2}$

Option 3:
$log_e \left[\frac{4}{3}\right]$

Option 4:
$log_e \left[\frac{3}{2}\right]$

Correct Answer:
$log_e 2$

Solution:

Integration as Reverse Process of Differentiation -

1. $\frac{\mathrm{d}}{\mathrm{dx}}(\mathrm{kx}) = \mathrm{k} \Rightarrow \int \mathrm{kdx} = \mathrm{kx} + \mathrm{C},$ where k is a constant
2. $\frac{\mathrm{d}}{\mathrm{dx}}\left(\frac{\mathrm{x}^{\mathrm{n}+1}}{\mathrm{n}+1}\right) = \mathrm{x}^{\mathrm{n}}, \mathrm{n} \neq -1 \Rightarrow \int \mathrm{x}^{\mathrm{n}}\mathrm{dx} = \frac{\mathrm{x}^{\mathrm{n}+1}}{\mathrm{n}+1} + \mathrm{C}, \mathrm{n} \neq -1$
3. $\frac{\mathrm{d}}{\mathrm{dx}}(\log|\mathrm{x}|) = \frac{1}{\mathrm{x}} \Rightarrow \int \frac{1}{\mathrm{x}}\mathrm{dx} = \log|\mathrm{x}| + \mathrm{C},$ when $\mathrm{x} \neq 0$
4. $\frac{\mathrm{d}}{\mathrm{dx}}(\mathrm{e}^{\mathrm{x}}) = \mathrm{e}^{\mathrm{x}} \Rightarrow \int \mathrm{e}^{\mathrm{x}}\mathrm{dx} = \mathrm{e}^{\mathrm{x}} + \mathrm{C}$
5. $\frac{\mathrm{d}}{\mathrm{dx}}\left(\frac{\mathrm{a}^{\mathrm{x}}}{\log_{\mathrm{e}}\mathrm{a}}\right) = \mathrm{a}^{\mathrm{x}}, \mathrm{a} > 0, \mathrm{a} \neq 1 \Rightarrow \int \mathrm{a}^{\mathrm{x}}\mathrm{dx} = \frac{\mathrm{a}^{\mathrm{x}}}{\log_{\mathrm{e}}\mathrm{a}} + \mathrm{C}$

-

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Integration Using Substitution -

Integration of Function f(ax + b)

If $\int f(x)dx = F(x)$ and a, b are constants, then

$$\int f(ax+b)dx = \frac{1}{a}F(ax+b)$$

we have, $\mathrm{I} = \int \mathrm{f(ax+b)dx}$

let $\mathrm{ax+b=t}$, then $\mathrm{a\,dx = dt}$

$$\begin{aligned}\therefore \quad \mathrm{I} &= \int \mathrm{f(ax+b)\,dx} \\ &= \frac{1}{\mathrm{a}}\int \mathrm{f(t)\,dt} \\ &= \frac{1}{\mathrm{a}}\mathrm{F(t)+c} \\ &= \frac{1}{\mathrm{a}}\mathrm{F(ax+b)+c}\end{aligned}$$

For example:

1. $\int \cos 2\mathrm{x}\,\mathrm{dx} = \frac{1}{2}\sin 2\mathrm{x} + \mathrm{c}$
2. $\int \frac{1}{\mathrm{x}+1}\,\mathrm{dx} = \log_{\mathrm{e}}|\mathrm{x}+1| + \mathrm{c}$
3. $\int \mathrm{e}^{2\mathrm{x}-3}\,\mathrm{dx} = \frac{1}{2}\mathrm{e}^{2\mathrm{x}-3} + \mathrm{c}$

-

Piecewise Definite integration -

Property 5

$$\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{f(x)dx} = \int_{\mathbf{a}}^{\mathbf{c}} \mathbf{f(x)dx} + \int_{\mathbf{c}}^{\mathbf{b}} \mathbf{f(x)dx} \text{ where } c \in \mathbb{R}$$

-

$$4\alpha\left\{\int_{-1}^{0} e^{\alpha x}dx + \int_{0}^{2} e^{-\alpha x}dx\right\} = 5$$

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$$\Rightarrow 4\alpha\left\{\left(\frac{e^{\omega x}}{\alpha}\right)_{-1}^{0}+\frac{\mathrm{e}^{-\alpha \mathrm{x}}}{-\alpha}\Big|_{0}^{2}\right\}$$

$$\Rightarrow 4\alpha\left\{\left(\frac{1-e^{-\alpha}}{\alpha}\right)-\left(\frac{e^{-2\alpha}-1}{\alpha}\right)\right\}=5$$

$$\Rightarrow 4\left(2-\mathrm{e}^{-\alpha}-\mathrm{e}^{-2\alpha}\right)=5; \quad \text{Put } \mathrm{e}^{-\alpha}=\mathrm{t}$$

$$\Rightarrow 4t^2+4t-3=0$$

$$\Rightarrow e^{-\alpha}=\frac{1}{2}\Rightarrow \alpha=\ln 2$$

Correct Option (1)

Q. 7 If sum of the first 40 terms of the series,3+4+8+9+13+14+18+19+.....is (102)m, then m is equal to :

Option 1:

10

Option 2:

25

Option 3:

5

Option 4:

20

Correct Answer:

20

Solution:

Sum of n terms of an AP -

$S_n=\frac{n}{2}[2a+(n-1)d]$

OR

$S_n=\frac{n}{2}[a+l]$

$a \rightarrow$ first term

$d \rightarrow$ common difference

$n \rightarrow$ number of terms

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$Given\ series\ \ 3+4+8+9+13+14+\ldots$
$Above\ series\ can\ be\ divided\ into\ two\ series\ having\ 20\ terms\ each$

$First\ series$
$S_1 = 3+8+13+\ldots$
$S_1 = \frac{20}{2}(2\times 3+(20-1)5) = 1010$

$Second\ series$
$S_2 = 4+9+14+\ldots$
$S_2 = \frac{20}{2}(2\times 4+(20-1)5) = 1030$

$So,\ S = S_1+S_2 = 2040 = 102m$
$m = 20$

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Correct Option (4)

Q. 8 If $\frac{3+i\sin\theta}{4-i\cos\theta}, \theta\epsilon\,[0,2\pi]$, is real number, then an argument of $\sin\Theta+i\cos\Theta$ is :

Option 1:
$\pi-\tan^{-1}\left[\frac{4}{3}\right]$

Option 2:
$-\tan^{-1}\left[\frac{3}{4}\right]$

Option 3:
$\pi-\tan^{-1}\left[\frac{3}{4}\right]$

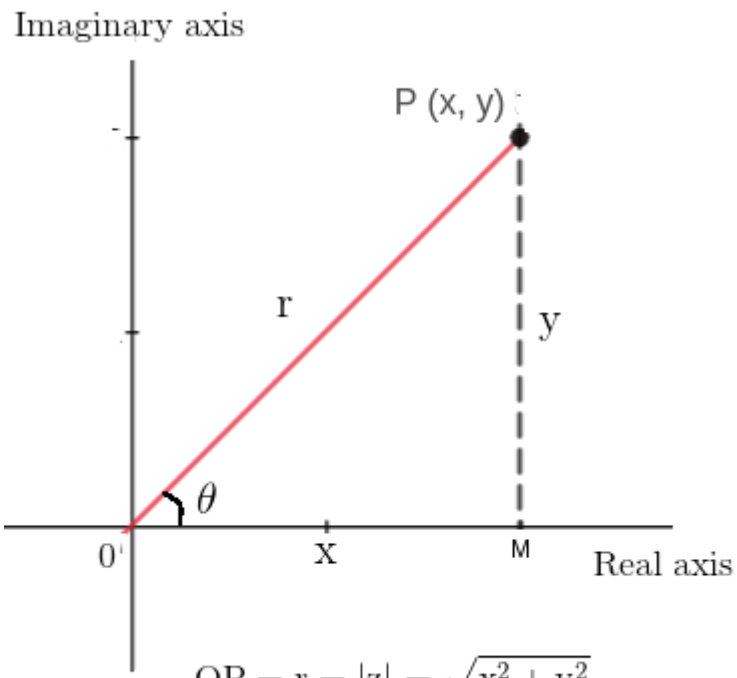
Option 4:
$\tan^{-1}\left[\frac{4}{3}\right]$

Correct Answer:
$\pi-\tan^{-1}\left[\frac{4}{3}\right]$

Solution:

Argument of complex number -

If OP forms some angle with the X-axis, let's denote it with ?, if (x,y) ≠ (0,0), then ? is called the argument of z. Where, $z = x + iy$



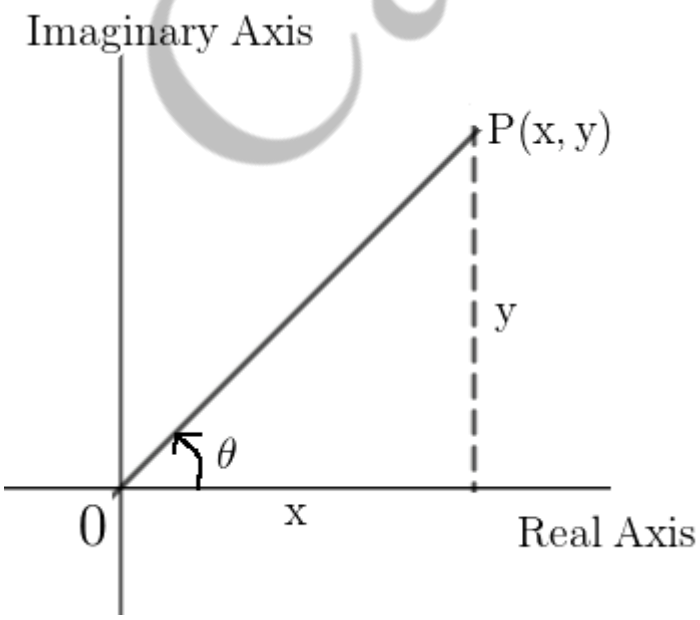
$$OP = r = |z| = \sqrt{x^2 + y^2}$$

$$\tan\theta = \frac{PM}{OM}$$

$$\tan\theta = \frac{y}{x} = \frac{Im(z)}{Re(z)} \Rightarrow \theta = \tan^{-1}\frac{y}{x}$$

$$\arg(z) = \theta = \tan^{-1}\frac{y}{x}$$

If ? lies between -? < ? ≤ ?, then ? is called principal argument. Value of ? differs depending on which quadrant point (x,y) lies. If it lies in 1st quadrant then it is ? (acute angle)



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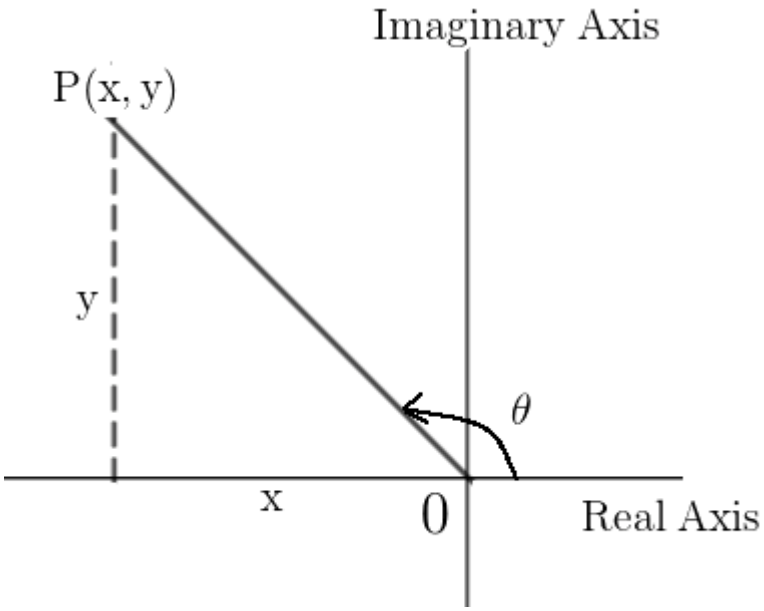
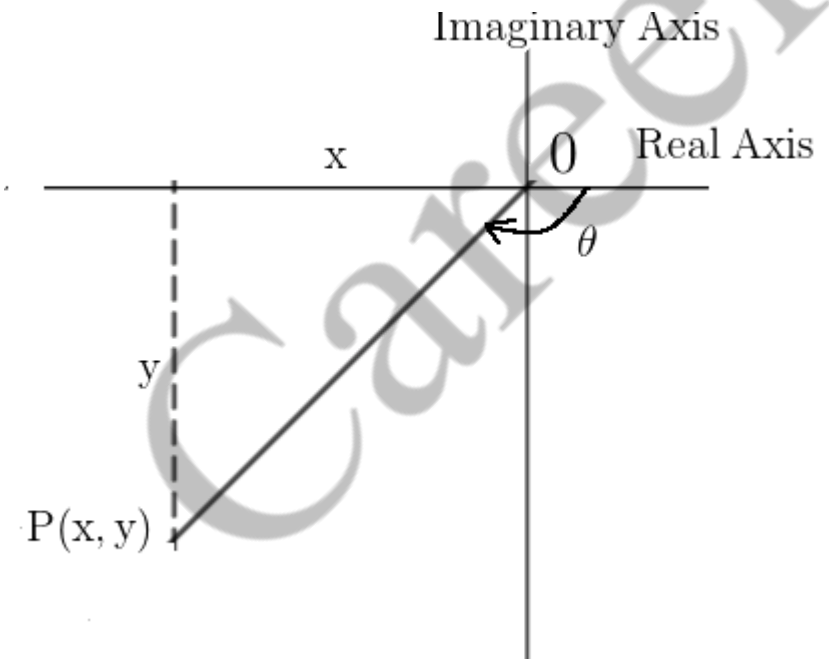
If the point lies in 2nd quadrant, then $arg(z) = \theta = \pi - \tan^{-1}\frac{y}{|x|}$

So it will be an obtuse +ve angle



If the point lies in lies in 3rd quadrant then $arg(z) = \theta = -\pi + \tan^{-1}\frac{y}{x}$

It will be an obtuse -ve angle



If the point lies in 4th quadrant then $arg(z) = \theta = -\tan^{-1}\frac{|y|}{x}$

It will be -ve acute angle

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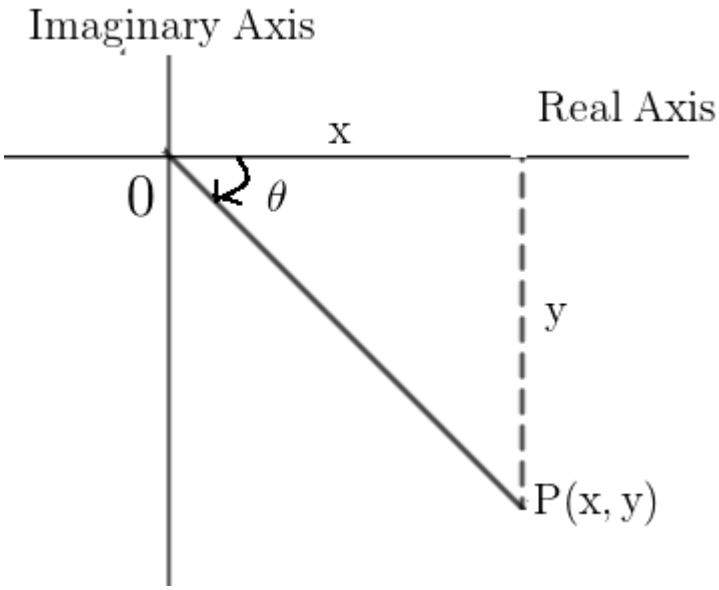

Note:

If $\arg(z) = \frac{\pi}{2}$ or $-\frac{\pi}{2}$, z is purely imaginary.
If $\arg(z) = 0$ or π, z is purely real.

-

$$z = \frac{(3 + i\sin\theta)}{(4 - i\cos\theta)} \times \frac{(4 + i\cos\theta)}{(4 + i\cos\theta)}$$

as z is purely real $\Rightarrow 3\cos\theta + 4\sin\theta = 0 \Rightarrow \tan\theta = -\frac{3}{4}$

$$\arg(\sin\theta + i\cos\theta) = \tan^{-1}\left(\frac{\cos\theta}{\sin\theta}\right) = \tan^{-1}\left(-\frac{4}{3}\right)$$

$$= \pi - \tan^{-1}\left(\frac{4}{3}\right)$$

Correct Option (1)

Q. 9 Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two 3×3 real matrices such that $b_{ij} = (3)^{(i+j-2)} a_{ji}$, where, i, j=1,2,3. if the determinant of B is 81, then the determinant of A is :

Option 1:
$1/9$

Option 2:
$1/81$

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Option 3:
$1/3$

Option 4:
3

Correct Answer:
$1/9$

Solution:

Matrices, order of matrices, row and column matrix -

A set of $\mathrm{m} \times \mathrm{n}$ numbers (real or complex) or objects or symbols arranged in form of a rectangular array having m rows and n columns and bounded by brackets [•] is called matrix of order m × n, read as m by n matrix.

E.g for m = 2 and n =3, we have $\begin{bmatrix} 2 & 4 & -3 \\ 5 & 4 & 6 \end{bmatrix}$ order of this matrix is 2×3

The m by n matrix is represented as :

$$\begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{bmatrix}$$

This representation can be represented in a more compact form as

$$[a_{ij}]_{m\times n}$$

Where a_{ij} represents element of i^{th} row and j^{th} column and i = 1,2,...,m; j = 1,2,...,n.

For example, to locate the entry in matrix A identified as a_{ij}, we look for the entry in row i, column j. In matrix A, shown below, the entry in row 2, column 3 is a_{23}.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Matrix is only a representation of the symbol, number or object. It does not have any value. Usually, a matrix denoted by capital letters.

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Properties of Determinants - Part 2 -

Property 5

If each element of a row (or a column) of a determinant is multiplied by a constant k, then the value of the determinant is multiplied by k.

For example

$$\text{Let, } \Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

and Δ' be the determinant obtained by multiplying the elements of the first row by k.

$$\Delta' = \begin{vmatrix} ka_1 & ka_2 & ka_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Expanding along first row, we get

$$\Delta' = \mathrm{ka_1(b_2c_3 - b_3c_2) - ka_2(b_1c_3 - b_3c_1) + ka_3(b_1c_2 - b_2c_1)}$$
$$= \mathrm{k[a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1)]}$$
$$\Delta' = \mathrm{k\Delta}$$

Hence,

$$\begin{vmatrix} ka_1 & ka_2 & ka_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \mathrm{k} \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Note:

1. By this property, we can take out any common factor from any one row or any one column of a given determinant.
2. If corresponding elements of any two rows (or columns) of a determinant are proportional (in the same ratio), then the determinant value is zero.

-

$$|B| = \begin{vmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{vmatrix}$$

$$|B| = \begin{vmatrix} 3^0a_{11} & 3^1a_{12} & 3^2a_{13} \\ 3^1a_{21} & 3^2a_{22} & 3^3a_{23} \\ 3^2a_{31} & 3^3a_{32} & 3^4a_{33} \end{vmatrix}$$

Taking Common 3^2 from R_3 and 3 from R_2

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$$|B| = 3^3 \begin{vmatrix} 3^0 a_{11} & 3^1 a_{12} & 3^2 a_{13} \\ 3^0 a_{21} & 3^1 a_{22} & 3^2 a_{23} \\ 3^0 a_{31} & 3^1 a_{32} & 3^2 a_{33} \end{vmatrix}$$

Taking Common 3^2 from C_3 and 3 from C_2

$$\Rightarrow 81 = 3^3 \cdot 3 \cdot 3^2 |\mathrm{A}| \Rightarrow 3^4 = 3^6 |\mathrm{A}| \Rightarrow |\mathrm{A}| = \frac{1}{9}$$

Correct Option (A)

Q. 10 Let f(x) be a polynomial of degree 5 such that $x = \pm 1$ are its critical points. If $\lim_{x \to 0} \left(2 + \frac{f(x)}{x^3}\right) = 4$, then which one of the following is not true ?

Option 1:
$f(1) - 4(-1) = 4$.

Option 2:
$x = 1$ is a point of maxima and $x = -1$ is a point of maxima and $x = -1$

Option 3:
f is an odd function.

Option 4:
$x = 1$ is a point of minima and $x = -1$ is a point of maxima of f.

Correct Answer:
$x = 1$ is a point of minima and $x = -1$ is a point of maxima of f.

Solution:

Algebraic Function of type 'infinity (∞) -

Algebraic Function of type 'infinity (∞)'

To find the limit of the type $x \to \infty$, write the given expression in the form of $\frac{N}{D}$ ($D \neq 0$), (N is numerator, D is denominator). Then divide both N and D by highest power of x occurring in both N and D to get a meaningful form.

An important result:

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If m, n are positive integers and $a_0, b_0 \neq 0$ are non-zero real numbers, then

$$\lim_{x\to\infty} \frac{a_0x^m + a_1x^{m-1} + \ldots + a_{m-1}x + a_m}{b_0x^n + b_1x^{n-1} + \ldots + b_{n-1}x + b_n} = \begin{cases} 0 & \text{if} \quad m < n \\ \frac{a_0}{b_0} & \text{if} \quad m = n \\ \infty & \text{if} \quad m > n \quad when \quad a_0b_0 > 0 \\ -\infty & \text{if} \quad m > n \quad when \quad a_0b_0 < 0 \end{cases}$$

-

$$f(x) = ax^5 + bx^4 + cx^3$$
$$\lim_{x\to 0}\left(2 + \frac{ax^5 + bx^4 + cx^3}{x^3}\right) = 4$$
$$\Rightarrow 2 + c = 4 \Rightarrow c = 2$$
$$f'(x) = 5ax^4 + 4bx^3 + 6x^2$$
$$= x^2\left(5ax^2 + 4bx + 6\right)$$

$$f'(1) = 0 \quad \Rightarrow \quad 5a + 4b + 6 = 0$$
$$f'(-1) = 0 \quad \Rightarrow \quad 5a - 4b + 6 = 0$$

On solving $b = 0$ and $a = -\frac{6}{5}$

$$f(x) = \tfrac{-6}{5}x^5 + 2x^3$$
$$f'(x) = -6x^4 + 6x^2$$
$$= 6x^2\left(-x^2 + 1\right)$$
$$= -6x^2(x + 1)(x - 1)$$

f(x) is maxima at x = 1 and minima at x = –1

Correct Option (4)

Q. 11 The number of ordered pairs (r,k) for which $6 \cdot {}^{35}C_r = (k^2 - 3) \cdot {}^{36}C_{r+1},$ where k is an integer, is :

Option 1:

4

Option 2:

6

Option 3:

2

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Option 4:

3

Correct Answer:

4

Solution:

As we have learnt

$${}^{n}C_{r} = \frac{n}{r} \cdot {}^{n-1}C_{r-1}$$

Now,

$${}^{36}C_{r+1}(k^2 - 3) = {}^{35}C_{r} \times 6$$

$$\frac{36}{r+1} \cdot {}^{35}C_{r}(k^2 - 3) = {}^{35}C_{r} \times 6$$

$$k^2 - 3 = \frac{r+1}{6} \Rightarrow k^2 = 3 + \frac{r+1}{6}$$

r should be less than or equal to 35

Hence for k to be an integer, r can be 5 and 35

For r=5 we get k = -2, 2

For r=35 we get k=-3,3

We get 4 ordered pair (5,-2), (5,2), (35,-3), (35, 3)

Correct Option (1)

Q. 12 Let $a_1, a_2, a_3, \cdots$ be a G.P. such that $a_1 < 0, a_1 + a_2 = 4$ and $a_3 + a_4 = 16$. If $\sum_{i=1}^{9} a_i = 4\lambda$, then λ is equal to :

Option 1:

171

Option 2:

$\frac{511}{3}$

Option 3:

-171

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Option 4:

-513

Correct Answer:

-171

Solution:

Given,

a1 + a2 = 4 $\Rightarrow a_1 + a_1 r = 4$ .....(i)

a3 + a4 = 16 $\Rightarrow a_1 r^2 + a_1 r^3 = 16$ .....(ii)

On dividing these

$\frac{1}{r^2} = \frac{1}{4} \Rightarrow r^2 = 4$

$\Rightarrow r = \pm 2$

$If\ r = 2,\ then\ a_1(1+2) = 4 \Rightarrow a_1 = \frac{4}{3}$

$If\ r = -2, \quad a_1(1-2) = 4 \Rightarrow a_1 = -4$

$As\ first\ term\ is\ negative,\ so,\ r = -2$

$\sum_{i=1}^{9} a_i = \frac{a_1\left(r^9-1\right)}{r-1} = \frac{(-4)\left((-2)^9-1\right)}{-2-1} = \frac{4}{3}(-513) = 4\lambda$
$\lambda = -171$

Correct Option (3)

Q. 13 Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$.If $\lambda = \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ and $\vec{d} = \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$, then the ordered pair, $\left(\lambda, \vec{d}\right)$ is equal to :

Option 1:

$\left[\frac{3}{2}, 3\vec{a} \times \vec{c}\right]$

Option 2:

$\left[-\frac{3}{2}, 3\vec{c} \times \vec{b}\right]$

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Option 3:

$\left[-\frac{3}{2}, 3\vec{a} \times \vec{b}\right]$

Option 4:

$\left[\frac{3}{2}, 3\vec{b} \times \vec{c}\right]$

Correct Answer:

$\left[-\frac{3}{2}, 3\vec{a} \times \vec{b}\right]$

Solution:

Addition and subtraction of Vectors -

Properties of vector addition

The sum of two vectors is always a vector.

1. $\vec{a}+\vec{b}=\vec{b}+\vec{a}$ (Commutative property)
2. $(\vec{a}+\vec{b})+\vec{c}=\vec{a}+(\vec{b}+\vec{c})$ (Associative property)
3. $\vec{a}+\overrightarrow{0}=\overrightarrow{0}+\vec{a}=\vec{a}$ (additive identity)
4. $\vec{a}+(-\vec{a})=(-\vec{a})+\vec{a}=\vec{0}$ (additive inverse)

Properties of vector Subtraction

$$|\vec{a}+\vec{b}+\vec{c}|^2=0$$
$$3+2(\vec{a}\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})=0$$
$$(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})=\frac{-3}{2}$$
$$\Rightarrow \lambda=\frac{-3}{2}$$

$$\overrightarrow{\mathrm{a}}+\overrightarrow{\mathrm{b}}+\overrightarrow{c}=\overrightarrow{0}$$

$$\vec{d}=\vec{a}\times\vec{b}+\vec{b}\times(-\vec{a}-\vec{b})+(-\vec{a}-\vec{b})\times\vec{a}$$
$$=\vec{a}\times\vec{b}+\vec{a}\times\vec{b}+\vec{a}\times\vec{b}$$
$$\vec{d}=3(\vec{a}\times\vec{b})$$

Correct Option (3)

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Q. 14 Let y=y(x) be the solution curve of the differentiable equation, $\left(y^2-x\right)\frac{dy}{dx}=1,$ satisfying y(0)=1. This curve intersects the x-axis at a point whose abscissa is :

Option 1:
$2+e$

Option 2:
2

Option 3:
$2-e$

Option 4:
$-e$

Correct Answer:
$2-e$

Solution:

Linear Differential Equation -

The linear differential equations are those in which the variable and its derivative occur only in the first degree.

An equation of the form

$$\frac{dy}{dx}+P(x)\cdot y=Q(x) \qquad \ldots(\mathrm{i})$$

Where P(x) and Q(x) are functions of *x* only or constant is called a linear equation of the first order.

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To solve the differential equation (i)

multiply both sides of Eq (i) by $\int e^{P(x)\,dx}$, we get

$$e^{\int P(x)dx}\left(\frac{dy}{dx}+P(x)y\right)=e^{\int P(x)dx}\cdot Q(x)$$

i.e. $$e^{\int P(x)dx}\cdot\frac{dy}{dx}+y\frac{d}{dx}\left(e^{\int P(x)dx}\right)=Qe^{\int P(x)dx}$$

or $$\frac{d}{dx}\left(ye^{\int P(x)dx}\right)=e^{\int P(x)dx}\cdot Q(x)$$

Integrating both sides, we get

or $$\int d\left(ye^{\int P(x)dx}\right)=\int\left(e^{\int P(x)dx}\cdot Q(x)\right)dx$$

$\Rightarrow$ $$ye^{\int P(x)dx}=\int Q(x)e^{\int P(x)dx}dx+C$$

Which is the required solution of the given differential equation.

The term $\int e^{P(x)\,dx}$ which convert the left hand expression of the equation into a perfect differential is called an Integrating factor (IF).

Thus, we remember the solution of the above equation as

$$y(\text{IF})=\int Q(\text{IF})dx+C$$

-

$$\frac{dx}{dy}+x=y^2$$

$$\text{I.F.}=e^{\int 1\cdot dx}=e^y$$

$$x\cdot e^y=\int y^2\cdot e^y\cdot dy$$

$$x\cdot e^y=y^2\cdot e^y-\int 2y\cdot e^y\cdot dy$$

$$x\cdot e^y=y^2e^y-2\left(y\cdot e^y-e^y\right)+c$$
$$x\cdot e^y=y^2e^y-2ye^y+2e^y+C$$
$$x=y^2-2y+2+c\cdot e^{-y}$$

$$x=0,\quad y=1$$
$$0=1-2+2+\frac{c}{e}\Rightarrow c=-e$$

Hence

At x-axis $y=0$,
$$x=0-0+2+(-e)\left(e^{-0}\right)$$
$$x=2-e$$

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Correct Option (3)

Q. 15 If Θ_1 and Θ_2 be respectively the smallest and the largest values of θ in $(0, 2\pi) - \{\pi\}$ which satisfy the equation $2\cot^2\theta - \frac{5}{\sin\theta} + 4 = 0$, then

, $\int_{\theta_1}^{\theta_2} \cos^2 3\theta \; d\theta$ is equal to :

Option 1:
$\frac{2\pi}{3}$

Option 2:
$\frac{\pi}{3}$

Option 3:
$\frac{\pi}{3} + \frac{1}{6}$

Option 4:
$\frac{\pi}{9}$

Correct Answer:
$\frac{\pi}{3}$

Solution:

Fundamental Formulae of Indefinite Integration (Trigonometric Functions) -

Trigonometric Functions

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1. $\frac{d}{dx}(-\cos x) = \sin x \Rightarrow \int \sin x \, dx = -\cos x + C$
2. $\frac{d}{dx}(\sin x) = \cos x \Rightarrow \int \cos x \, dx = \sin x + C$
3. $\frac{d}{dx}(\tan x) = \sec^2 x \Rightarrow \int \sec^2 x \, dx = \tan x + C$
4. $\frac{d}{dx}(-\cot x) = \csc^2 x \Rightarrow \int \csc^2 x \, dx = -\cot x + C$
5. $\frac{d}{dx}(\sec x) = \sec x \tan x \Rightarrow \int \sec x \tan x \, dx = \sec x + C$
6. $\frac{d}{dx}(-\csc x) = \csc x \cot x \Rightarrow \int \csc x \cot x \, dx = -\csc x + C$

-

Integration Using Substitution -

we have, $I = \int f(ax+b)dx$

let $ax + b = t$, then $a\,dx = dt$

$$\therefore \quad I = \int f(ax+b)\,dx$$
$$= \frac{1}{a}\int f(t)\,dt$$
$$= \frac{1}{a}F(t) + c$$
$$= \frac{1}{a}F(ax+b) + c$$

For example:

1. $\int \cos 2x \, dx = \frac{1}{2}\sin 2x + c$
2. $\int \frac{1}{x+1} \, dx = \log_e |x+1| + c$
3. $\int e^{2x-3} \, dx = \frac{1}{2}e^{2x-3} + c$

-

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Definite Integration -

Let f be a function of x defined on the closed interval [a, b] and F be another function such that $\frac{d}{dx}(F(x)) = f(x)$ for all x in the domain of f, then

$$\int_a^b f(x)dx = [F(x) + c]_a^b = F(b) - F(a)$$

Working Rule to evaluate definite Integral $\int_a^b f(x)dx$

1. First find the indefinite integration $\int f(x)\,dx$ and suppose the result be F(x)
2. Next find the F(b) and F(a)
3. And, finally value of definite integral is obtained by subtracting F(a) from F(b).
   Thus, $\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a).$

-

$$2\cot^2\theta - \frac{5}{\sin\theta} + 4 = 0$$
$$\frac{2\cos^2\theta}{\sin^2\theta} - \frac{5}{\sin\theta} + 4 = 0$$
$$2\cos^2\theta - 5\sin\theta + 4\sin^2\theta = 0, \sin\theta \neq 0$$
$$2\sin^2\theta - 5\sin\theta + 2 = 0$$
$$(2\sin\theta - 1)(\sin\theta - 2) = 0$$

$$\sin\theta = \frac{1}{2} \text{ and } \sin\theta \neq 2$$
$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$
$$\therefore \int_{\pi/6}^{5\pi/6} \cos^2 3\theta d\theta = \int_{\pi/6}^{5\pi/6} \frac{1+\cos 6\theta}{2} d\theta$$

$$= \frac{1}{2}\left[\theta + \frac{\sin 6\theta}{6}\right]_{\pi/6}^{5\pi/6} = \frac{1}{2}\left[\frac{5\pi}{6} - \frac{\pi}{6} + \frac{1}{6}(0-0)\right] = \frac{1}{2}\cdot\frac{4\pi}{6} = \frac{\pi}{3}$$

Correct Option (2)

Q. 16 Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $P_k = (\alpha)^k + (\beta)^k, k \geq 1$, then which one of the following statements is not true ?

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Option 1:
$P_1 + P_2 + P_3 + P_4 + P_5 = 26$

Option 2:
$P_5 = 11$

Option 3:
$P_5 = P_2 \cdot P_3$

Option 4:
$P_3 = P_5 - P_4$

Correct Answer:
$P_5 = P_2 \cdot P_3$

Solution:

Polynomial Equation of Higher Degree, Remainder theorem -

An equation of the form $a_0x^n + a_1x^{n-1} + ... + a_{n-1}x + a_n = 0$,
where $a_0, a_1, ..., a_n$ are constant and $a_0 \neq 0$

Is known as the polynomial equation of degree n which have n and only n roots.

$$\text{sum of all roots} = \sum \alpha_1 = \alpha_1 + \alpha_2 + ... + \alpha_n - 1 + \alpha_n = (-1)\frac{a_1}{a_0}$$

$$\text{sum of products taken two at a time } \sum \alpha_1\alpha_2 = \alpha_1\alpha_2 + \alpha_1\alpha_3 + ... + \alpha_1\alpha_n + \alpha_2\alpha_3 + ... + \alpha_2\alpha_n + ... + \alpha_{n-1}\alpha_n = (-1)^2\frac{a_2}{a_0}$$

$$\text{sum of products taken three at a time } \sum \alpha_1\alpha_2\alpha_3 = (-1)^3\frac{a_3}{a_0}$$

$$\text{product of all roots} = \alpha_1\alpha_2...\alpha_n = (-1)^n\frac{a_n}{a_0}$$

For example, suppose n = 3 and $ax^3 + bx^2 + cx + d = 0$ is polynomial equation with $a \neq 0$ and ?, ? and ? are the roots of the equation then :

$$\alpha + \beta + \gamma = -\frac{b}{a}$$
$$\sum \alpha\beta = \alpha\beta + \beta\gamma + \gamma\alpha = (-1)^2\frac{c}{a} = \frac{c}{a}$$
$$\alpha\beta\gamma = (-1)^3\frac{d}{a} = -\frac{d}{a}$$

-

$$\alpha + \beta = 1 \text{ and } \alpha\beta = -1$$

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$P_1 = \alpha + \beta = 1$
$P_2 = \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 1 - 2(-1) = 3$
$P_3 = \alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 + \beta^2 - \alpha\beta) = (1 \times (3 + 1)) = 4$
$P_4 = \alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2 = (P_2)^2 - 2(-1)^2 = 7$

$x^2 = x + 1$

$x^2 = x + 1 \Rightarrow x^5 = x^4 + x^3$
$\alpha^5 = \alpha^4 + \alpha^3$ and $\beta^5 = \alpha^4 + \alpha^3$

$\alpha^5 + \beta^5 = P_4 + P_3 = 11$

$\mathrm{P}_5 \neq \mathrm{P}_2 \times \mathrm{P}_3$

Correct option (3)

Q. 17 The area (in sq.units) of the region $\left\{(x, y)\, \epsilon R^2 \,\middle|\, 4x^2 \leq y \leq 8x + 12\right\}$ is :

Option 1:
$\frac{125}{3}$

Option 2:
$\frac{128}{3}$

Option 3:
$\frac{124}{3}$

Option 4:
$\frac{127}{3}$

Correct Answer:
$\frac{128}{3}$

Solution:

Area Bounded by Curves When Intersects at More Than One Point -

Area bounded by the curves $y = f(x)$, $y = g(x)$ and intersect each other in the interval [a, b]

First find the point of intersection of these curves $y = f(x)$ and $y = g(x)$, let the point of intersection be x = c

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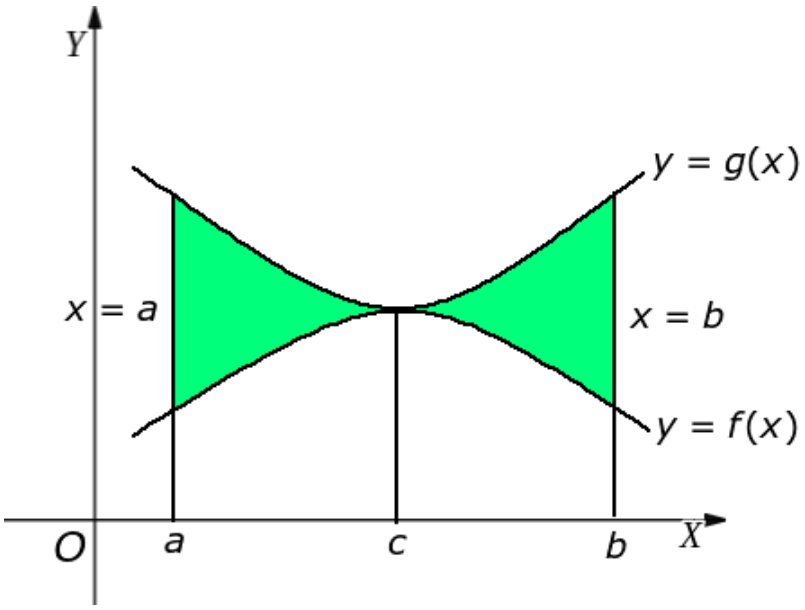


Area of the shaded region

$$= \int_a^c \{f(x) - g(x)\}dx + \int_c^b \{g(x) - f(x)\}dx$$

When two curves intersects more than one point

rea bounded by the curves *y=f(x), y=g(x)* and intersect each other at three points at *x = a, x = b amd x = c.*

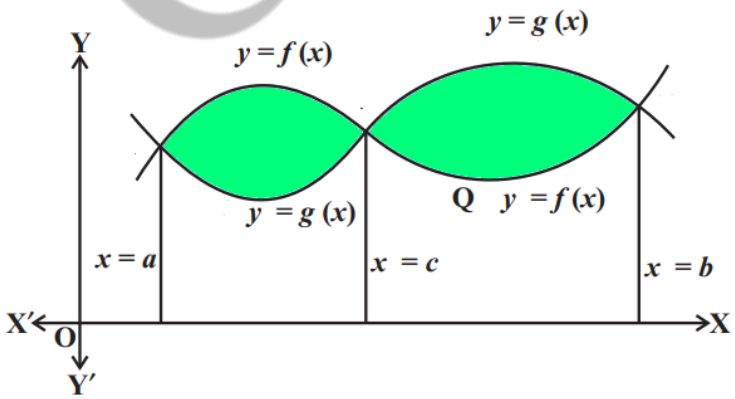
To find the point of intersection, solve *f(x) = g(x).*

For x ∈ (a, c), f(x) > g(x) and for x ∈ (c, b), g(x) > f(x).

Area bounded by curves,

$$\mathrm{A} = \int_a^b |f(x) - g(x)|\,dx$$

$$= \int_a^c (f(x) - g(x))\,dx + \int_c^b (g(x) - f(x))\,dx$$



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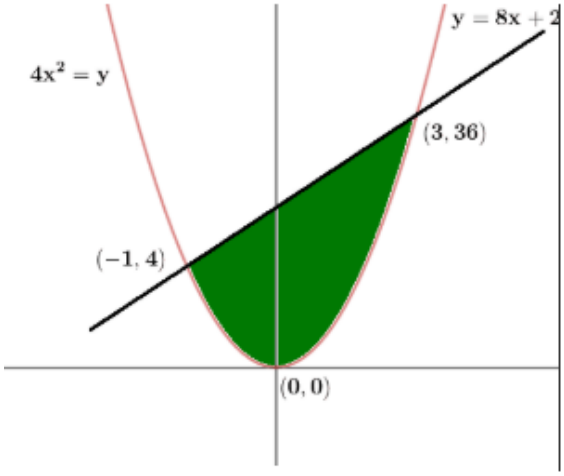
$$\int (8x + 12 - 4x^2)dx$$

Point of intersection (-1,4),(3,36)



$$\int_{-1}^{3} (8x + 12 - 4x^2)dx = 4x^2 + 12x - \frac{4x^3}{3}|_{-1}^{3}$$

$$\frac{128}{3}$$

Correct Option (2)

Q. 18 The value of c in the Lagrange's mean value theorem for the function $f(x) = x^3 - 4x^2 + 8x + 11$, when $x\epsilon\,[0, 1]$ is :

Option 1:

$$\frac{4 - \sqrt{7}}{3}$$

Option 2:

$$\frac{2}{3}$$

Option 3:

$$\frac{\sqrt{7} - 2}{3}$$

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Option 4:

$\frac{4-\sqrt{5}}{3}$

Correct Answer:

$\frac{4-\sqrt{7}}{3}$

Solution:

Lagrange's Mean Value Theorem -

Lagrange's Mean Value Theorem

Rolle's theorem is a special case of the **Mean Value Theorem**. In Rolle's theorem, we consider differentiable functions f defined on a closed interval [a, b] with f(a) = f(b) . The Mean Value Theorem generalized Rolle's theorem by considering functions that do not necessarily have equal value at the endpoints.

Statement

Let f (x) be a function defined on [a, b] such that

1. it is continuous on [a, b],
2. it is differentiable on (a, b).

Then there exists a real number c (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

-

f(x) is a polynomial function
Therefore, it is continuous and differentiable in [0, 1]
Here f(0) = 11, f(1) = 1 – 4 + 8 + 11 = 16

$$f'(x) = 3x^2 - 8x + 8$$
$$\therefore \quad f'(c) = \frac{f(1) - f(0)}{1 - 0} = \frac{16 - 11}{1} = 3c^2 - 8c + 8$$
$$\Rightarrow \quad 3c^2 - 8c + 3 = 0$$
$$C = \frac{8 \pm 2\sqrt{7}}{6} = \frac{4 \pm \sqrt{7}}{3}$$
$$\therefore \quad c = \frac{4 - \sqrt{7}}{3} \quad \in (0, 1)$$

Correct Option (1)

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Q. 19 Let y=y(x) be a function of x satisfying $y\sqrt{1-x^2}=k-x\sqrt{1-y^2}$ where k is a constant and $y\left(\frac{1}{2}\right)=-\frac{1}{4}$. Then $\frac{dy}{dx}$ at $x=\frac{1}{2}$, is equal to :

Option 1:

$-\frac{\sqrt{5}}{2}$

Option 2:

$\frac{\sqrt{5}}{2}$

Option 3:

$-\frac{\sqrt{5}}{4}$

Option 4:

$\frac{2}{\sqrt{5}}$

Correct Answer:

$-\frac{\sqrt{5}}{2}$

Solution:

Derivative of the Polynomial Function -

Derivative of the Polynomial Function

Finding derivatives of functions by using the definition of the derivative can be a lengthy and, for certain functions, a rather challenging process.

In this section, we will learn rules for finding derivatives that allow us to bypass this process. Let's begin with the basics.

1. $\frac{d}{dx}(\textbf{constant})=\mathbf{0}$

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The rule for differentiating constant functions is called the constant rule. It states that the derivative of a constant function is zero; that is, since a constant function is a horizontal line, the slope, or the rate of change, of a constant function, is 0.

2. $\frac{d}{dx}(\mathbf{x^n}) = \mathbf{nx^{n-1}}$

-

Rules of Differentiation (Chain Rule) -

Rules of Differentiation (Chain Rule)

Chain Rule or Derivation of Composite Function:

The chain rule, which states that the derivative of a composite function is the derivative of the outer function evaluated at the inner function times the derivative of the inner function.

Let f and g be functions. For all x in the domain of g for which g is differentiable at x and f is differentiable at g(x), the derivative of the composite function

h(x) = (f∘g)(x) = f (g(x)) Is given by

h′(x) = f′(g(x)) • g′(x)

-

$$x = 1/2,\ \ y = -1/4 \ \ \Rightarrow xy = -1/8$$

$$y \cdot \frac{1 \cdot (-2x)}{2\sqrt{1-x^2}} + y'\sqrt{1-x^2} = -\left[1 \cdot \sqrt{(1-y^2)} + \frac{x \cdot (-2y)}{2 \cdot \sqrt{1-y^2}}y'\right]$$

$$-\frac{xy}{\sqrt{1-x^2}} + y'\sqrt{1-x^2} = -\sqrt{1-y^2} + \frac{xy \cdot y'}{\sqrt{1-y^2}}$$

$$y'\left(\sqrt{1-x^2} - \frac{xy}{\sqrt{1-y^2}}\right) = \frac{xy}{\sqrt{1-x^2}} - \sqrt{1-y^2}$$

$$\text{put } x = 1/2 \ \ and\ y = -1/4$$

$$we\ get\ y' = -\frac{\sqrt{5}}{2}$$

Correct Option (1)

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Q. 20 Let the tangents drawn from the origin to the circle, $x^2 + y^2 - 8x - 4y + 16 = 0$ touch it at the points A and B. The $(AB)^2$ is equal to:

Option 1:
$\frac{32}{5}$

Option 2:
$\frac{64}{5}$

Option 3:
$\frac{52}{5}$

Option 4:
$\frac{56}{5}$

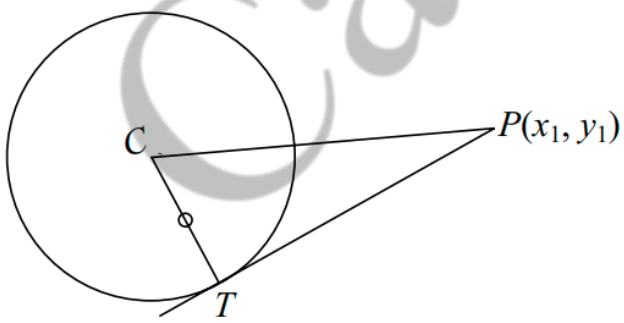
Correct Answer:
$\frac{64}{5}$

Solution:

Tangent from a Point to the Circle (NEW) -

Length of tangent (PT) from a point to a circle

The length of the tangent from a point P (x_1, y_1) to the circle $x^2 + y^2 = a^2$ is $\sqrt{x_1^2 + y_1^2 - a^2}$



In ΔPTC, $PT^2 = PC^2 - CT^2$

Here for the circle $x^2 + y^2 = a^2$ coordinate of C is $(0, 0)$

Hence, $PT^2 = \left(\sqrt{(x_1 - 0)^2 + (y_1 - 0)^2}\right)^2 - \left(\sqrt{a^2}\right)^2$

$\Rightarrow \quad PT = \sqrt{x_1^2 + y_1^2 - a^2} \qquad [\because CT = \text{radius} = a]$

The length of the tangent from a point P (x_1, y_1) to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $\sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c}$

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-

Length of the tangent from the origin (0,0) is $L = \sqrt{S_1} = \sqrt{16} = 4$

The radius of the circle $R = \sqrt{16 + 4 - 16} = 2$

Length of Chord of contact $\frac{2LR}{\sqrt{L^2 + R^2}} = \frac{2 \times 4 \times 2}{\sqrt{16 + 4}} = \frac{16}{\sqrt{20}}$

Correct Option (2)

Q. 21 If the system of linear equations,

x+y+z=6

x+2y+3z=10

$3x + 2y + \lambda z = \mu$

has more than two solutions, then $\mu - \lambda^2$ is equal to _____.

Correct Answer: 13

Solution:

Cramer's law -

Cramer's law for the system of equations in two variables :

$\text{Let } a_1x + b_1y = c_1 \text{ and } a_2x + b_2y = c_2, \text{where}$
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$
$\text{on solving this equation by cross multiplication, we get}$
$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{a_1c_2 - a_2c_1} = \frac{1}{a_1b_2 - a_2b_1}$
$\text{or } \frac{x}{\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}} = \frac{y}{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}} = \frac{1}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$
$\text{or } x = \frac{\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}, y = \frac{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$

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We can observe that first row in the numerator of x is of constants and 2nd row in the numerator is of constants, and the denominator is of the coefficient of variables.

We can follow this analogy for the system of equations of 3 variable where third in the numerator of the value of z will be of constant and denominator will be formed by the value of coefficients of the variables.

so let us consider the system of equations

$a_1x + b_1y + c_1z = d_1 \quad ...(i)$

$a_2x + b_2y + c_2z = d_2 \quad ...(ii)$

$a_3x + b_3y + c_3z = d_3 \quad ...(iii)$

then Δ, which will be determinant of coefficient of variables, will be

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, \Delta_1 \text{ numerator of x is :}$$

$$\Delta_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}, \text{similarly } \Delta_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix} \text{ and}$$

$$\Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}, \text{if} \Delta \neq 0 \text{ then } x = \frac{\Delta_1}{\Delta}$$

$$y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}$$

i) If ? ≠ 0, then the system of equations has a unique finite solution and so equations are consistent, and solutions are $x = \frac{\Delta_1}{\Delta}, y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}$

ii) If ? = 0, and any of $\Delta_1 \neq 0 \; or \; \Delta_2 \neq 0 \; or \; \Delta_3 \neq 0$

Then the system of equations is inconsistent and hence no solution exists.

iii) If all $\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$ then

System of equations is consistent and dependent and it has an infinite number of solution.

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x + y + z = 6 ..... (1)

x + 2y + 3z = 10 ..... (2)

3x + 2y + λz = μ..... (3)

from (1) and (2)

if z = 0 $\rightarrow$ x + y = 6 and x + 2y = 10

$\rightarrow$ y = 4, x = 2

(2, 4, 0)

if y = 0 $\rightarrow$ x + z = 6 and x + 3z = 10

$\rightarrow$ z = 2 and x = 4

(4, 0, 2)

so, 3x + 2y + λz = μ must pass through (2, 4, 0) and (4, 0, 2)

μ = 14

and 12 + 2λ = μ

λ = 1

$(\mu - \lambda^2)$ = 13

Q. 22 If the foot of the perpendicular drawn from the point (1,0,3) on line passing through $(\alpha, 7, 1)$ is $\left(\frac{5}{3}, \frac{7}{3}, \frac{17}{3}\right)$, then $\alpha$is equal to _____.

Correct Answer: 4

Solution:

Angle Between Two Lines -

Vector Form
Condition for Perpendicularity

The lines are perpendicular then cosΘ = 90°

$$i.e. \qquad \vec{b} \cdot \vec{b'} = 0 \qquad [\because \quad \cos 90^\circ = 0]$$

$$\Rightarrow \quad a_1a_2 + b_1b_2 + c_1c_2 = 0$$

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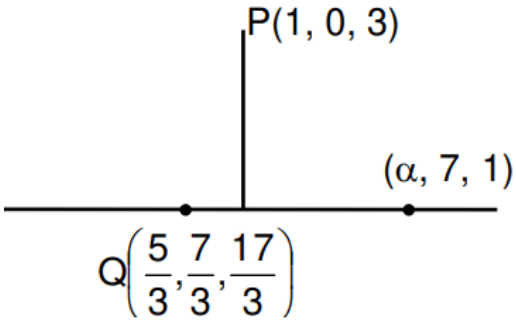
Condition for parallelism

The lines are parallel then $\vec{b} = \lambda\vec{b'}$ for some scalar λ.

$$\Rightarrow \qquad \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

-

Since PQ is perpendicular to L, therefore



$$\left(1-\frac{5}{3}\right)\left(\alpha-\frac{5}{3}\right)+\left(-\frac{7}{3}\right)\left(7-\frac{7}{3}\right)+\left(3-\frac{17}{3}\right)\left(1-\frac{17}{3}\right)=0$$

$$\alpha = 4$$

Q. 23 If the function f defined on $\left(-\frac{1}{3}, \frac{1}{3}\right)$ by

$$f(x) = \begin{cases} \frac{1}{x}\, log_e \left(\frac{1+3x}{1-2x}\right), & when x \neq 0 \\ k & when\ x = 0 \end{cases}$$

is continuous, then k is equal to _____.

Correct Answer: 5

Solution:

Logarithmic Limits -

Logarithmic Limits

To evaluate the Logarithmic limit we use the following results:

$$\lim_{x\to 0} \frac{\log_e(1+x)}{x} = 1$$

Proof:

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$$\lim_{x\to 0}\frac{\log_e(1+x)}{x}=\lim_{x\to 0}\frac{x-\frac{x^2}{2}+\frac{x^3}{3}-\cdots}{x}$$

[using Taylor series expansion of $\log_e(1+x)$]

$$=\lim_{x\to 0}\left(1-\frac{x}{2}+\frac{x^2}{3}-\cdots\right)$$

$$=1$$

-

$$\lim_{x\to 0} f(x)=\lim_{x\to 0}\left(\frac{1}{x}\log_e\left(\frac{1+3x}{1-2x}\right)\right)$$

$$\Rightarrow \lim_{x\to 0}\left(\frac{\ln(1+3x)}{x}-\frac{\ln(1-2x)}{x}\right)$$

$$\Rightarrow \lim_{x\to 0}\left(\frac{3\ln(1+3x)}{3x}+\frac{2\ln(1-2x)}{-2x}\right)=3+2=5$$

Q. 24 If the mean and variance of eight numbers 3,7,9,12,13,20,x and y be 10 and 25 respectively, then $x\cdot y$ is equal to _____.

Correct Answer: 54

Solution:

Mean -

Mean of the Ungrouped Data

If n observations in data are x_1, x_2, x_3, ..., x_n, then arithmetic mean $\bar{x}$ is given by

$$\bar{x}=\frac{x_1+x_2+x_3+\ldots\cdots+x_n}{n}=\frac{1}{n}\sum_{i=1}^{n}x_i$$

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Dispersion (Variance and Standard Deviation) -

Variance and Standard Deviation

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The mean of the squares of the deviations from the mean is called the variance and is denoted by σ^2 (read as sigma square).

Variance is a quantity which leads to a proper measure of dispersion.

The variance of n observations x_1 , x_2 ,..., x_n is given by

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

-

$$\text{Mean} = \bar{x} = \frac{3+7+9+12+13+20+x+y}{8} = 10$$
$$\Rightarrow x + y = 16$$
$$\text{variance } \sigma^2 = \frac{\sum(x_i)^2}{8} - (\bar{x})^2$$
$$\Rightarrow \frac{9+49+81+144+169+400+x^2+y^2}{8} - 100 = 25$$
$$\Rightarrow x^2 + y^2 = 148$$
$$(x+y)^2 = x^2 + y^2 + 2xy$$
$$xy = 54$$

Q. 25 Let $X = \{n \epsilon N : 1 \leq n \leq 50\}$. If $A = \{n \epsilon X : n \; is \; multiple \; of \; 2\}$ and $B = \{n \epsilon X : n \; is \; multiple \; of \; 7\}$, then the number of smallest elements in the smallest subset of X containing both A and B is ____.

Correct Answer:
29

Solution:

Cardinal number of some sets -

The number of distinct elements in a finite set A is called Cardinal number and it is denoted by n(A).

For example, if set A = {1, 3, 7, 11, 13} then n(A) = 5

Given A and B be any finite sets. then Number of Elements in the union of set A & B is given by

n (A ∪ B) = n (A) + n (B) – n (A ∩ B)

If (A ∩ B) = φ,then n(A ∪ B) = n (A) + n (B)

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Given A, B and C be any finite sets, then Number of Elements in the union of sets A, B & C is given by

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

-

A = {2, 4, 6, 8, 10, ..., 50}

B = {7, 14, 21, 28, 35, 42, 49}

n(A ∩ B) = {14, 28, 42}

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$
$$\Rightarrow 25 + 7 - 3 = 29$$

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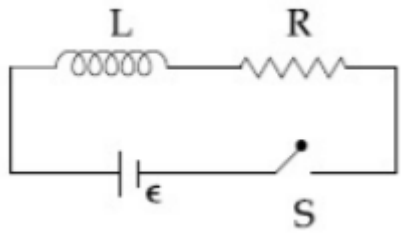
8th January shift-2

Physics

Q. 1



As shown in the figure, a battery of emf ϵ is connected to an inductor L and resistance R in series. The switch is closed at t = 0. The total charge that flows from the battery, between t = 0 and t = t_c (t_c is the time constant of the circuit) is :

Option 1:
$\frac{\epsilon L}{R^2}\left(1-\frac{1}{e}\right)$

Option 2:
$\frac{\epsilon L}{R^2}$

Option 3:
$\frac{\epsilon R}{eL^2}$

Option 4:
$\frac{\epsilon L}{eR^2}$

Correct Answer:
$\frac{\epsilon L}{eR^2}$

Solution:

As current at any time t is given as

$$I = I_0\left(1-e^{\frac{-t}{\tau}}\right) = \frac{\epsilon}{R}\left(1-e^{\frac{-t}{\tau}}\right)$$

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So $q = \int_0^{T_C} idt$

So integrating this will give total charge

$$q = \frac{\varepsilon}{R}\left[t - \frac{e^{-t/\tau}}{\frac{-1}{\tau}}\right]_0^{\tau}; \quad = \frac{\varepsilon}{R}\left[\tau + \tau e^{-1} - \tau\right]$$

$$q = \frac{\varepsilon}{R} \times \frac{1}{e} \times \frac{L}{R} \quad ; \quad q = \frac{\varepsilon L}{R^2 e}$$

Hence the correct option is (4).

Q. 2 A simple pendulum is being used to determine the value of gravitational acceleration g at a certain place. The length of the pendulum is 25.0 cm and a stop watch with 1 s resolution measures the time taken for 40 oscillations to be 50 s. The accuracy in g is :

Option 1:
2.40%

Option 2:
5.40%

Option 3:
4.40%

Option 4:
3.40%

Correct Answer:
4.40%

Solution:

As

- **The time period of oscillation of simple pendulum (T)-**

$$T = 2\pi\sqrt{\frac{l}{g}}$$

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where

m=mass of the bob

l = length of pendulum

g = acceleration due to gravity.

So

$$\frac{\Delta T}{T}=\frac{1}{2}\left(\frac{\Delta g}{g}+\frac{\Delta L}{L}\right)$$
$$\frac{\Delta g}{g}=\frac{2\Delta T}{T}+\frac{\Delta L}{L};\quad =2\left(\frac{1}{50}\right)+\frac{0.1}{25.0}$$
$$=4.4\%$$

Note - We are adding the error just for calculating maximum possible error.

Hence the correct option is (3).

Q. 3 A particle of mass m is dropped from a height h above the ground. At the same time another particle of the same mass is thrown vertically upwards from the ground with a speed of $\sqrt{2gh}$. If they collide head-on completely inelastically, the time taken for the combined mass to reach the ground, in units of $\sqrt{\frac{H}{g}}$ is :

Option 1:
$\sqrt{\frac{1}{2}}$

Option 2:
$\frac{1}{2}$

Option 3:
$\sqrt{\frac{3}{2}}$

Option 4:
$\sqrt{\frac{3}{4}}$

Correct Answer:
$\sqrt{\frac{3}{2}}$

Solution:

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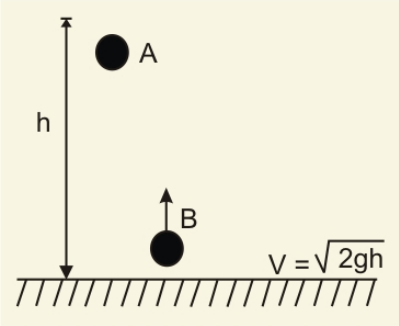

time for collision -

$$t_1 = \frac{h}{\sqrt{2gh}}$$

After t1 -

$$V_A = 0 - gt_1 = -\sqrt{\frac{gh}{2}}$$

and $V_B = \sqrt{2gh} - gt_1 = \sqrt{gh}\left[\sqrt{2} - \frac{1}{\sqrt{2}}\right]$

at the time of collision

$$\vec{P}_i = \vec{P}_f$$
$$\Rightarrow m\vec{V}_A + m\vec{V}_B = 2m\vec{V}_f$$
$$\Rightarrow -\sqrt{\tfrac{gh}{2}} + \sqrt{gh}\left[\sqrt{2} - \tfrac{1}{\sqrt{2}}\right] = 2\vec{V}_f$$
$$V_f = 0$$

and height from ground

$$= h - \frac{1}{2}gt_1^2 = h - \frac{h}{4} = \frac{3h}{4}$$

so time

$$= \sqrt{2 \times \frac{\left(\frac{3h}{4}\right)}{g}} = \sqrt{\frac{3h}{2g}}$$

Hence the correct option is (3).

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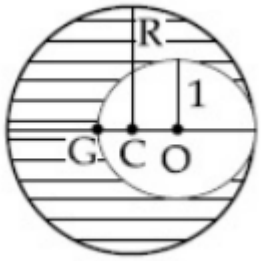
Q. 4



As shown in fig. when a spherical cavity (centred at O) of radius 1 is cut out of a uniform sphere of radius R (centred at C), the centre of mass of remaining (shaded) part of sphere is at G, i.e on the surface of the cavity. R can be determined by the equation :

Option 1:

$\left(R^2+R+1\right)(2-R)=1$

Option 2:

$\left(R^2+R-1\right)(2-R)=1$

Option 3:

$\left(R^2-R-1\right)(2-R)=1$

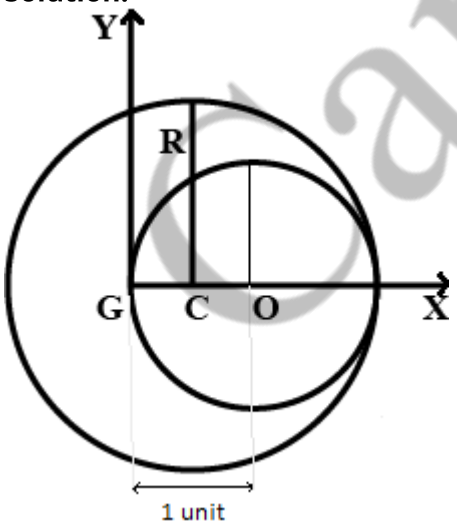
Option 4:

$\left(R^2-R+1\right)(2-R)=1$

Correct Answer:

$\left(R^2+R+1\right)(2-R)=1$

Solution:



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Taking centre at O for the cavity and the Center of the solid sphere at C.
Let origin be at G
Mass of the sphere is taken to be $\mathrm{m_s}$, which can be given as:

$$\mathrm{m_s} = \left(\frac{4}{3}\pi \mathrm{R}^3 \cdot \rho\right) \mathrm{m_s} = \left(\frac{4}{3}\pi \mathrm{R}^3 \rho\right)$$

Mass of cavity is taken to be $\mathrm{m_c}$, which can be given as:

$$\mathrm{m_c} = -\left(\frac{4}{3}\pi \times 1^2 . \rho\right)$$

where ρ is the density of the material.

Center of mass of sphere:

$$\mathrm{x_8} = \mathrm{GC} \quad [\mathrm{x_s \text{ can be given as } x_c - x_G = GC}]$$
$$\mathrm{x_s} = 2 - \mathrm{R}$$

Center of mass of cavity:

$$x_c = 2 \times 1 - 1$$
$$x_c = 1$$

$$\mathrm{x_{com}} = \frac{\mathrm{m_s x_s + m_c x_c}}{\mathrm{m_x + m_c}} = \frac{\left(\frac{4}{3}\pi \mathrm{R}^3 \rho\right) \times (2 - \mathrm{R}) + \left(-\frac{4}{3}\pi\rho\right) \times 1}{\frac{4\pi}{3}\pi \mathrm{R}^3 \rho + \left(-\frac{4}{3}\pi\rho\right)}$$

Therefore the centre of mass is at G, $\mathrm{x_{com}} = 0$

$$\Rightarrow 0 = \left(\frac{4}{3}\pi R^3 \rho\right)(2 - R) - \frac{4}{3}\pi\rho$$
$$0 = R^3(2 - R) - 1 \Rightarrow 2R^3 - R^4 - 1 = 0$$
$$R^3 + R^3 - R^4 - 1 = 0$$
$$\Rightarrow R^3 - R^3(R - 1) = 1$$
$$\Rightarrow (R - 1)\left(R^2 + 1 + R\right) - R^3(R - 1) = 0$$
$$\Rightarrow (R - 1)\left(R^2 + 1 + R - R^3\right) = 0$$
$$\Rightarrow R^2 + 1 + R - R^3 = 0$$
$$\Rightarrow \mathrm{R^2 + R - R^3 + 2 = 1}$$
$$\Rightarrow \mathrm{2R^2 - R^2 + 2R - R - R^3 + 2 = 1}$$
$$\Rightarrow \mathrm{2R^2 - R^3 - R^2 + 2R - R + 2 = 1}$$
$$\Rightarrow \mathrm{R^2(2 - R) + R(2 - R) + 1(2 - R) = 1}$$
$$\Rightarrow \mathrm{(2 - R)\left(R^2 + R + 1\right) = 1}$$

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Q. 5 A plane electromagnetic wave of frequency 25 GHz is propagating in vacuum along the z-direction. At a particular point in space and time, the magnetic field is given by $\vec{B} = 5 \times 10^{-8}\hat{j}T$. The corresponding electric field $\vec{E}$ is (speed of light c = 3 x 10^8 ms^{-1})

Option 1:
$-1.66 \times 10^{-16}\hat{i}V/m$

Option 2:
$1.66 \times 10^{-16}\hat{i}V/m$

Option 3:
$-15\hat{i}V/m$

Option 4:
$15\hat{i}V/m$

Correct Answer:
$15\hat{i}V/m$

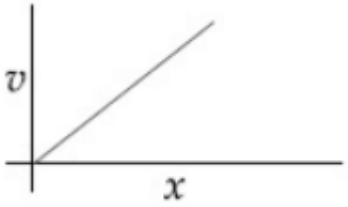
Solution:

$\frac{E}{B} = c$
$E = B \times c$
$= 15\hat{i}\ \mathrm{V/m}$

Hence the correct option is (4).

Q. 6 A particle of mass m and charge q is released from rest in uniform electric field. If there is no other force on the particle, the dependence of its speed v on the distance x travelled by it is correctly given by (graphs are schematic and not drawn to scale)

Option 1:

1. 

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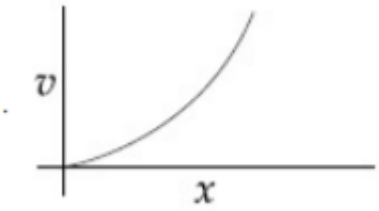
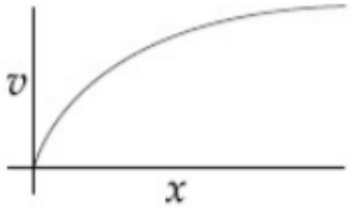
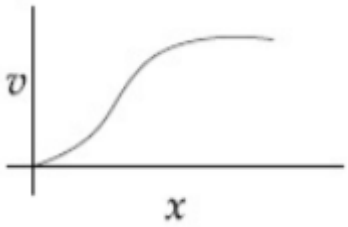
Option 2:



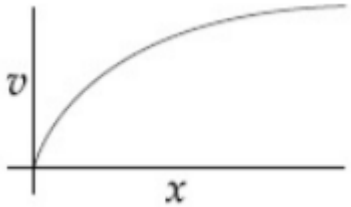
Option 3:



Option 4:



Correct Answer:



Solution:

$$u = 0, a = \frac{qE}{m}, distance = x$$

$$\Rightarrow using \;\; v^2 = u^2 + 2ax$$

$$\Rightarrow v^2 = \frac{2qE}{m} * x$$

So graph would be option 3.

Hence the correct option is (3).

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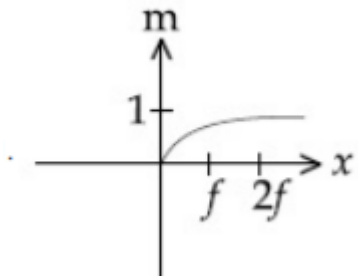
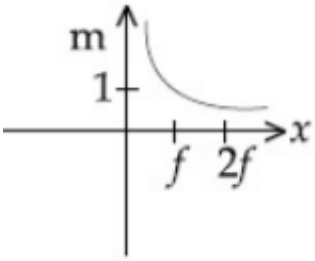
Q. 7 An object is gradually moving away from the focal point of a concave mirror along the axis of the mirror. The graphical representation of the magnitude of linear magnification (m) versus distance of the object from the mirror (x) is correctly given by

(Graphs are drawn spherically and are not to scale)

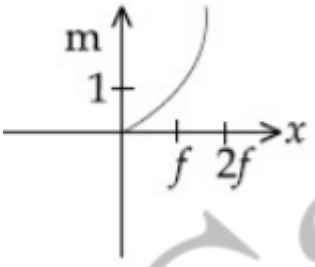
Option 1:



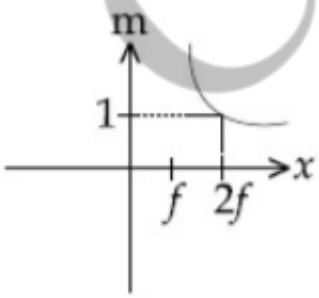
Option 2:



Option 3:



Option 4:



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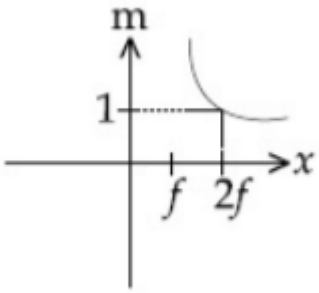
Correct Answer:



Solution:
In this question, we can use the elimination method by looking at the graph.

As we know that the

$$m = \frac{-v}{u}$$

And by using the mirror formula. we can write that the -

$$m = \frac{-f}{u-f} = \frac{-1}{1-\frac{u}{f}}$$

Now by using this equation, we can start eliminating the option. As we can see that the at u=f, the m become infinite. So the option (1) and (3) is incorrect. If we u = 2f, then the m =1.

So option (2) is also wrong.

Hence the correct option is (4).

Q. 8 A uniform sphere of mass 500 g rolls without slipping on a plane horizontal surface with its centre moving at a speed of 5.00 cm/s. Its kinetic energy is :

Option 1:
$6.25 \times 10^{-4} J$

Option 2:
$1.13 \times 10^{-3} J$

Option 3:
$8.75 \times 10^{-4} J$

Option 4:
$8.75 \times 10^{-3} J$

Correct Answer:
$8.75 \times 10^{-4} J$

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Solution:

The kinetic energy of sphere is kinetic energy of translation+ kinetic energy of rotation

$$K = \frac{1}{2}mv^2(1 + \frac{k^2}{R^2})$$

Where K is the radius of gyration

$$K = \frac{1}{2} \times 0.5(\frac{5}{100})^2(1 + \frac{2}{5})$$
$$K = \frac{35}{4} \times 10^{-4}J = 8.75 \times 10^{-4}J$$

Hence the correct option is (3).

Q. 9 A particle moves such that its position vector $\hat{r}\,(t) = cos\omega t\hat{i} + sin\omega t\hat{j}$ where ω is a constant and t is time. Then which of the following statements is true for the velocity $\vec{v}(t)$ and acceleration $\vec{a}(t)$ of the particle :

Option 1:
$\vec{V}$ and $\vec{A}$ both are parallel to $\vec{r}$

Option 2:
$\vec{V}$ is perpendicular to $\vec{r}$ and $\vec{A}$ is directed away from the origin

Option 3:
$\vec{V}$ and $\vec{A}$ both are perpendicular to $\vec{r}$

Option 4:
$\vec{V}$ is perpendicular to $\vec{r}$ and $\vec{A}$ is directed towards the origin

Correct Answer:
$\vec{V}$ is perpendicular to $\vec{r}$ and $\vec{A}$ is directed towards the origin

Solution:
$$\vec{r} = cos\omega t\ i + sin\omega t\ j$$
$$\vec{v} = \frac{d\vec{r}}{dt} = -\omega sin\omega t i + \omega cos\omega t j$$

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$$\vec{a} = \frac{d\vec{v}}{dt} = -\omega^2(cos\omega t\ i + sin\omega t\ j) = -\omega^2\vec{r}$$

implies a is anti parallel to r

$$\vec{v}.\vec{r} = 0$$

Implies v is perpendicular to r

hence the correct option is (4).

Q. 10 A transverse wave travels on a taut steel wire with a velocity of v when tension in it is 2.06 x 10^4 N . When the tension is changed to t, the velocity changed to v/2. The value of T (in N) is close to :

Correct Answer:
5.15 x 10^3

Solution:

Velocity

$$v \propto \sqrt{T}$$

$$\frac{T_1}{T_2} = (\frac{v_1}{v_2})^2$$

$$\frac{T_1}{T_2} = (\frac{v}{v/2})^2 = 4$$

$$\Rightarrow T_2 = \frac{T_1}{4} = 0.515 \times 10^4$$

$$T_2 = 5.15 \times 10^3 N$$

Hence the correct option is (1)

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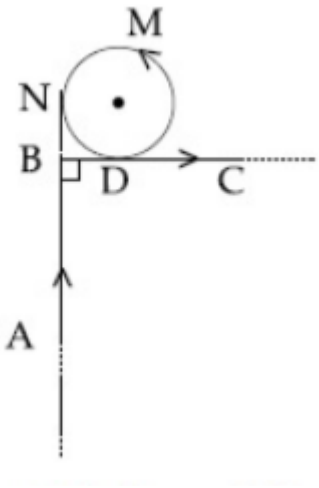
Q. 11 A very long wire ABDMNDC is shown in figure carrying current I. AB and BC parts are straight , long and at right angle. At D wire forms circular turn DMND of radius R . AB, BC parts are tangential to circular turn at N and D . Magnetic field at the centre of circle is :



Option 1:
$\frac{\mu_0 I}{2\pi R}\left(\pi+\frac{1}{\sqrt{2}}\right)$

Option 2:
$\frac{\mu_0 I}{2R}$

Option 3:
$\frac{\mu_0 I}{2\pi R}(\pi+1)$

Option 4:
$\frac{\mu_0 I}{2\pi R}\left(\pi-\frac{1}{\sqrt{2}}\right)$

Correct Answer:
$\frac{\mu_0 I}{2\pi R}\left(\pi+\frac{1}{\sqrt{2}}\right)$

Solution:

As

Magnetic Field due to current in straight wire -

$$B=\frac{\mu_0}{4\pi}\cdot\frac{i}{r}\left(\sin\phi_1+\sin\phi_2\right)$$

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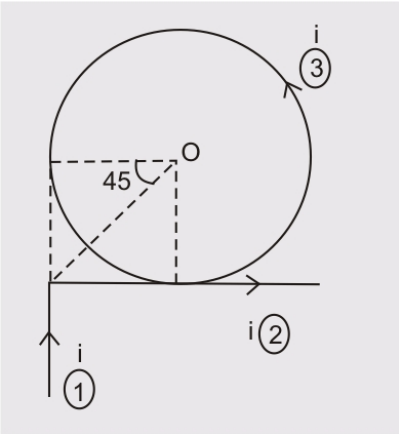
So



Magnitude of net magnetic field at O is

$$B_O = B_1 + B_2 + B_3$$
$$= -\frac{\mu_0 i}{4\pi R}(sin90 - sin45) + \frac{\mu_0 i}{4\pi R}(sin90 + sin45) + \frac{\mu_0 i}{2R}$$
$$= \frac{\mu_0 i}{4\pi R}(\sqrt{2} + 2\pi)$$

The direction of magnetic field due to first conductor (in to the plane) is opposite to that of other two conductors (out of the plane).

Hence the correct option is (1)

Q. 12 In a double-slit experiment, at a certain point on the screen the path difference betweeen the two interfering waves is $\frac{1}{8}$th of a wavelength. The ratio of the intensity of light at that point to that at the centre of a bright fringe is :

Option 1:
0.568

Option 2:
0.853

Option 3:
0.760

Option 4:
0.672

Correct Answer:
0.853

Solution:

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Intensity is given by $I = I_0 \cos^2\left(\frac{\Delta\phi}{2}\right)$

Where I_0=maximum intensity=intensity of light at the center of a bright fringe

and using the Relation $\Delta x = \frac{2\pi}{\lambda}\Delta\phi$

we get $\frac{I}{\mathrm{I}_0} = \cos^2\left[\frac{\frac{2\pi}{\lambda}\times\Delta x}{2}\right] = \cos^2\left(\frac{\pi}{8}\right); \quad \frac{I}{\mathrm{I}_0} = 0.853$

Hence the correct option is (2).

Q. 13 A galvanometer having a coil resistance 100 Ω gives a full scale deflection when a current of 1 mA is passed through it. What is the value of the resistance (in range of $k\Omega$)which can convert this galvanometer into a voltmeter giving full scale deflection for a potential difference of 10 V ?

Correct Answer:
8.9

Solution:

$$V_g = i_g R_g = 0.1\mathrm{V}$$
$$V = 10\mathrm{V}$$
$$\mathrm{R} = \mathrm{R}_g\left(\frac{\mathrm{V}}{\mathrm{V}_g} - 1\right)$$
$$= 100 \times 99 = 9.9\mathrm{K}\Omega$$

Hence the correct option is (1).

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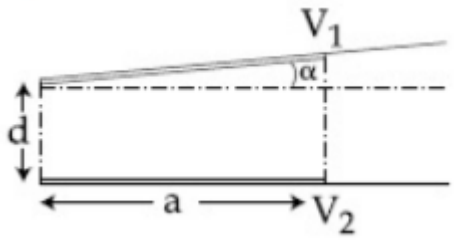
Q. 14 A capacitor is made of two square plates each of side 'a' making a very small angle α between them, as shown in figure. The capacitance will be close to :



Option 1:

$\frac{\epsilon_0 a^2}{d}\left(1-\frac{\alpha a}{4d}\right)$

Option 2:

$\frac{\epsilon_0 a^2}{d}\left(1+\frac{\alpha a}{4d}\right)$

Option 3:

$\frac{\epsilon_0 a^2}{d}\left(1-\frac{\alpha a}{2d}\right)$

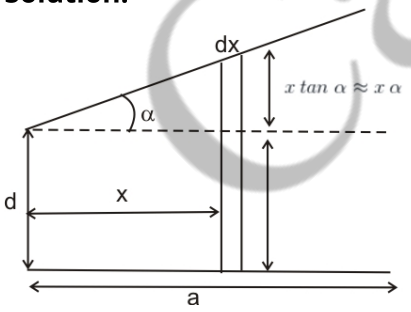
Option 4:

$\frac{\epsilon_0 a^2}{d}\left(1-\frac{3\alpha a}{2d}\right)$

Correct Answer:

$\frac{\epsilon_0 a^2}{d}\left(1-\frac{\alpha a}{2d}\right)$

Solution:



Take an elemental strip of length dx at a distance x from one end

So the capacitance of an elemental strip is $\mathrm{dc}=\frac{\varepsilon_0 \mathrm{adx}}{\mathrm{d}+\alpha \mathrm{x}}$

So integrating this will give total capacitance= $C=\int \mathrm{dc}=\int_0^a \frac{\varepsilon_0 \mathrm{adx}}{\mathrm{d}+\alpha \mathrm{x}}$

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$\Rightarrow \quad c = \frac{\varepsilon_0 a}{\alpha}[\ln(d+\alpha x)]_0^a$

$C = \frac{\varepsilon_0 a}{\alpha}\ln\left(1+\frac{\alpha a}{d}\right) \approx \frac{\varepsilon_0 a^2}{d}\left(1-\frac{\alpha a}{2d}\right)$ .....(Using $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \ldots$)

Hence the correct option is (3).

Q. 15 An electron (mass m) with initial velocity $\vec{v} = v_0\hat{i} + v_0\hat{j}$ is in an electric field $\vec{E} = -E_0\hat{k}$. If λ_0 is initial de-Brogile wavelength of electron , its de-Brogile wavelength at time t is given by :

Option 1:

$$\frac{\lambda_0}{\sqrt{1+\frac{e^2E_0^2t^2}{m^2v_0^2}}}$$

Option 2:

$$\frac{\lambda_0}{\sqrt{2+\frac{e^2E_0^2t^2}{m^2v_0^2}}}$$

Option 3:

$$\frac{\lambda_0}{\sqrt{1+\frac{e^2E_0^2t^2}{2m^2v_0^2}}}$$

Option 4:

$$\frac{\lambda_0\sqrt{2}}{\sqrt{1+\frac{e^2E_0^2t^2}{m^2v_0^2}}}$$

Correct Answer:

$$\frac{\lambda_0}{\sqrt{1+\frac{e^2E_0^2t^2}{2m^2v_0^2}}}$$

Solution:

$$\text{Initially } m\left(\sqrt{2}v_0\right) = \frac{h}{\lambda_0}$$

We know v=u+at and here $a = \frac{qE}{m} = \frac{eE_0}{m}$

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$$\text{Velocity as a function of time } = v_0\hat{i} + v_0\hat{j} + \frac{eE_0}{m}t\hat{k}$$

So, magnitude of velocity:- $\sqrt{2v_0^2 + \frac{e^2E_0^2}{m^2}t^2}$

$$\text{so wavelength } \lambda = \frac{\mathrm{h}}{\mathrm{m}\sqrt{2\mathrm{v}_0^2 + \frac{\mathrm{e}^2\mathrm{E}_0^2}{\mathrm{m}^2}\mathrm{t}^2}}$$

$$\Rightarrow \lambda = \frac{\lambda_0}{\sqrt{1 + \frac{\mathrm{e}^2\mathrm{E}_0^2}{2\mathrm{m}^2\mathrm{v}_0^2}\mathrm{t}^2}}$$

Hence the correct option is (3).

Q. 16 A carnot engine having an efficiency of $\frac{1}{10}$ is being used as a refrigerator. If the work done on the system is 10 J , then the amount of heat absorbed (in Joules) from the reservoir at lower temperature is :

Correct Answer:
90

Solution:

$$(COP)_{refrigerator} = \frac{1}{\eta_{carnot}} - 1 = 10 - 1 = 9$$
$$So,\ (COP)_{refrigerator} = 9 = \frac{R.E}{W_{I/P}} = \frac{Q_2}{W_{I/P}}$$
$$\Rightarrow R.E = Q_2 = 9 \times 10 = 90J$$

Hence the correct option is (1).

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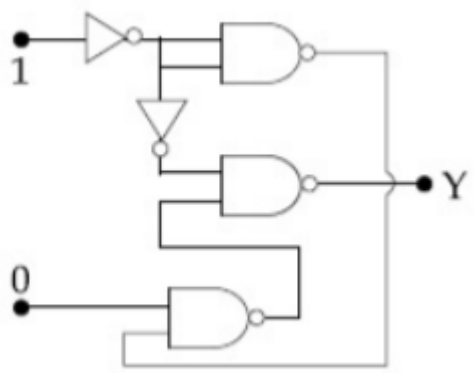
Q. 17 In the given circuit, value of Y is :



Option 1:
toggles between 0 and 1

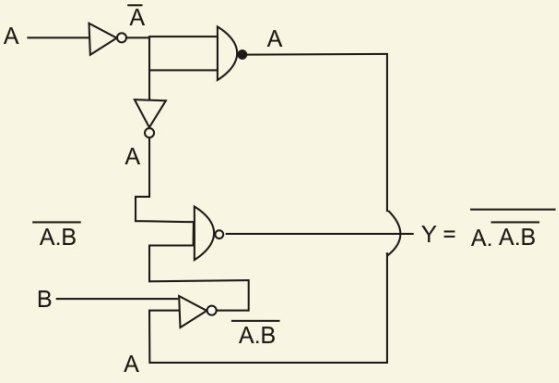
Option 2:
1

Option 3:
0

Option 4:
will not execute

Correct Answer:
0

Solution:



In NAND gate, when

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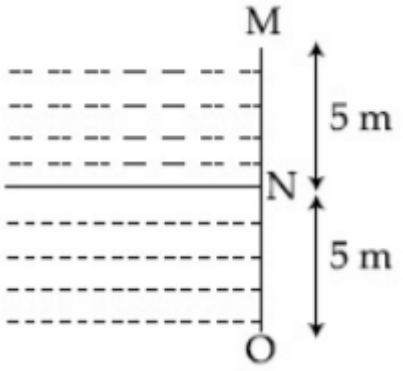
$y = \overline{A.\overline{A.B}}$
$Given$
$A = 1,\ B = 0$
$\Rightarrow A.B = 0$
$\overline{A.B} = 1\ and\ A.\overline{A.B} = 1$
$therefore\ \overline{A.\overline{A.B}} = 0$

Hence the correct option is (3)

Q. 18



Two liquids of densities ρ_1 and ρ_2 ($\rho_2 = 2\rho_1$) are filled up behind a square wall of side 10 m as shown in figure. Each liquid has a height of 5 m. The ratio of the forces due to these liquids exerted on upper part MN to that at the lower part NO is (Assume that the liquids are not mixing) :

Option 1:
1/2

Option 2:
2/3

Option 3:
1/4

Option 4:
1/3

Correct Answer:
1/4

Solution:

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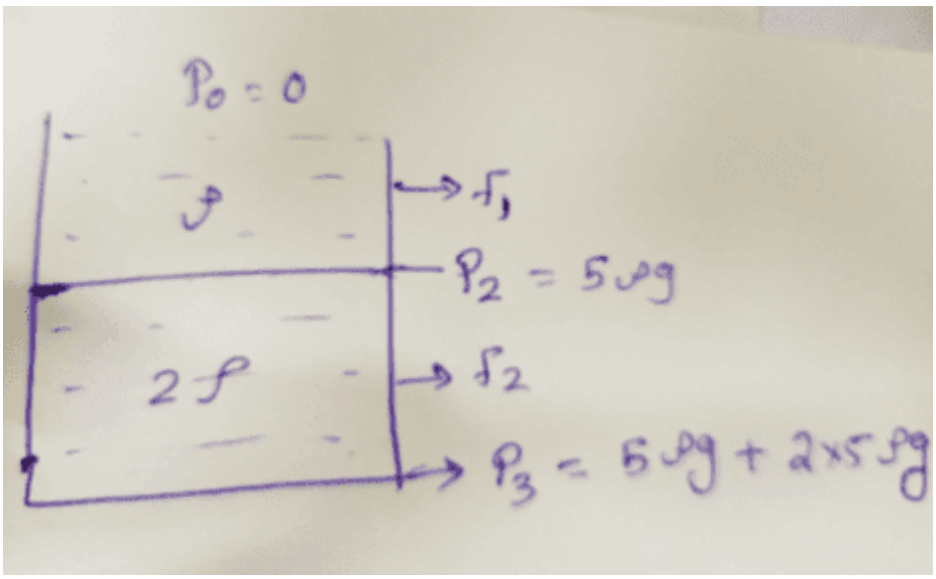

$$F_1 = \frac{P_0 + P_2}{2}A$$

$$F_2 = \frac{P_2 + P_3}{2}$$

$$\frac{F_1}{F_2} = \frac{P_O + P_2}{P_2 + P_3} = \frac{5\rho g}{20\rho g} = \frac{1}{4}$$

Hence the correct option is (3).

Q. 19 Consider a mixture of n moles of helium gas andd 2n moles of oxygen gas (molecules taken to be rigid) as an ideal gas. Its C_p/C_v value will be :

Option 1:
40/27

Option 2:
23/15

Option 3:
19/13

Option 4:
67/45

Correct Answer:
19/13

Solution:
For monoatomic gas (He)

$$c_v = \frac{3R}{2}, c_p = \frac{5R}{2}$$

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For diatomic gas (O_2)

$$c_v = \frac{5R}{2}, c_p = \frac{7R}{2}$$

$$\gamma_{\text{mixure}} = \frac{C_{p_{mix}}}{C_{v_{mx}}} = \frac{(n_1 C_{p_1} + n_2 C_{p_2})}{(n_1 C_{v_1} + n_2 C_{v_2})}$$

$$= \frac{n\frac{5}{2}R + 2n\frac{7}{2}R}{n\frac{3}{2}R + 2n\frac{5}{2}R} = \frac{19}{13}$$

Hence the correct option is (3).

Q. 20 Consider two charged metallic spheres S_1 and S_2 of radii R_1 and R_2 , respectively . The electric fields $E_1/E_2 = R_1/R_2$. Then the ratio V_1(on S_1)/V_2(on S_2) of the electrostatic potentials on each sphere is :

Option 1:
$\left(\frac{R_1}{R_2}\right)^3$

Option 2:
$\left(\frac{R_2}{R_1}\right)$

Option 3:
$\frac{R_1}{R_2}$

Option 4:
$\left(\frac{R_1}{R_2}\right)^2$

Correct Answer:
$\left(\frac{R_1}{R_2}\right)^2$

Solution:
Given

$$\frac{E_1}{E_2} = \frac{R_1}{R_2}$$
$$V = -E.R$$
$$\frac{V_1}{V_2} = \frac{E_1 R_1}{E_2 R_2} = (\frac{R_1}{R_2})^2$$

hence the correct option is (4)

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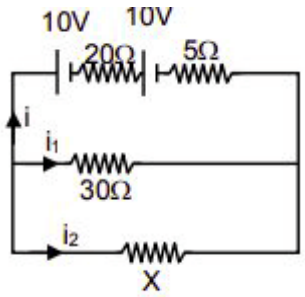
Q. 21 The series combination of two batteries, both of the same emf 10 V , but different internal resistance of 20 ω and 5 ω , is connected to the parallel combination of two resistors 30 ω and R ω . The voltage diffrence across the battery of internal resistance 20 ω is zero, the value of R (in ω) is _____ .

Correct Answer: 30

Solution:



$$V_1 = \varepsilon_1 - i.r_1$$
$$0 = 10 - i \times 20$$
$$i = 0.5A$$
$$V_2 = \varepsilon_2 - ir_2$$
$$= 10 - 0.5 \times 5$$
$$V_2 = 7.5V$$
$$O \cdot 5 = \frac{7.5}{30} + \frac{7.5}{x}$$
$$\frac{7.5}{x} = 0.25$$
$$x = \frac{7.5}{0.25} = 30$$

Hence option (1) is correct.

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Q. 22 Three containers C_1, C_2 and C_3 have water at different temperatures. The table below shows the final temperature T when different amounts of water is (given in litres) are taken from each container and mixed (assume no loss of heat during the process)

C_1	C_2	C_3	T
1l	2l	--	60°C
--	1l	2l	30°C
2l	--	1l	60°C
1l	1l	1l	θ

The value of θ (in °C to the nearest integer) is _____ .

Correct Answer:
50

Solution:

We know that:- $\theta_{mix} = \dfrac{m_1 s_1 \theta_1 + m_2 s_2 \theta_2}{m_1 s_1 + m_2 s_2}$ **and here s is same**

$$1\theta_1 + 2\theta_2 = (1+2)60$$
$$\theta_1 + 2\theta_2 = 180 - - - - - - - - - - (1)$$

$$0 \times \theta_1 + 1 \times \theta_2 + 2 \times \theta_3 = (1+2)30$$
$$\theta_2 + 2\theta_3 = 90 - - - - - - - - - - (2)$$

$$2 \times \theta_1 + 0 \times \theta_2 + 1 \times \theta_3 = (2+1)60$$
$$2\theta_1 + \theta_3 = 180 - - - - - - - - - - (3)$$

$$\text{and } \theta_1 + \theta_2 + \theta_3 = (1+1+1)\theta - - - - - - - - - - (4)$$

$$\text{from } (1) + (2) + (3)$$

$$3\theta_1 + 3\theta_2 + 3\theta_3 = 450 \Rightarrow \theta_1 + \theta_2 + \theta_3 = 150$$

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from (4) equation $150 = 3\theta \Rightarrow \theta = 50°C$

Q. 23 A ball is dropped from the top of a 100 m high tower on a planet . In the last $\frac{1}{2}$ s before hitting the ground, it covers a distance of 19 m. Acceleration due to gravity (in ms^{-2}) near the surface on that planet is _____ .

Correct Answer: 8

Solution:

Let total time taken by particle is t

So distance covered by particle in t sec= $S_t = ut + \frac{1}{2}gt^2 = \frac{1}{2}gt^2$

similarly distance covered by partical in $(t - \frac{1}{2})$ **sec=** $S_{t-\frac{1}{2}} = \frac{1}{2}g(t - \frac{1}{2})^2$

So distance covered by partical in last (1/2) sec of its journey= $S_t - S_{t-\frac{1}{2}}$

So $19 = S_t - S_{t-\frac{1}{2}} = \frac{1}{2}gt^2 - \frac{1}{2}g(t - \frac{1}{2})^2 = \frac{1}{2}g(t - \frac{1}{4}) ....(1)$

and $S_t = \frac{1}{2}gt^2 = 100..(2)$

from equation (1) and (2)

we get $19t^2 - 100t + 25 = 0 \Rightarrow t = 5, or \;\; t = 0.26$

putting t=5 we get g = 8 m/s2

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Q. 24 An asteroid is moving directly towards the centre of the earth. When at a distance of 10 R (R is the radius of the earth) from the earths centre , it has a speed of 12 km/s . Neglecting the effect of earths atmosphere, what will be the speed of the asteroid when it hits the surface of the earth (escape velocity from the earth is 11.2 km/s) ? Give your answer to the nearest integer in kilometer/s _____ .

Correct Answer:
16

Solution:

We will apply energy conservation here:-

$$\mathrm{KE_1 + PE_1 = KE_f + PE_f}$$

$$\frac{1}{2}\mathrm{mu_0^2} + \left(-\frac{\mathrm{GMm}}{10\mathrm{R}}\right) = \frac{1}{2}\mathrm{mv^2} + \left(-\frac{\mathrm{GMm}}{\mathrm{R}}\right)$$

$$\mathrm{v^2} = \mathrm{u_0^2} + \frac{2\mathrm{GM}}{\mathrm{R}}\left[1 - \frac{1}{10}\right]$$

$$\mathrm{v} = \sqrt{\mathrm{u_0^2} + \frac{9}{5}\frac{\mathrm{GM}}{\mathrm{R}}}$$

$$= \sqrt{12^2 + 0.9(11.2)^2}$$

$$= 16.028\mathrm{km/s}$$
$$= 16km/s$$

Q. 25 The first member of the Balmer series of hydrogen atom has a wavelength of 6561 $\mathring{A}$. The wavelength of the second member of the Balmer series (in nm) is _____ .

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Correct Answer:
486

Solution:

$$\frac{1}{\lambda} = \mathrm{R}Z^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\frac{1}{\lambda_1} = R(1)^2\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = \frac{5\mathrm{R}}{36}$$

$$\frac{1}{\lambda_2} = \mathrm{R}(1)^2\left(\frac{1}{2^2} - \frac{1}{4^2}\right) = \frac{3\mathrm{R}}{16}$$

$$\frac{\lambda_2}{\lambda_1} = \frac{20}{27}$$

$$\lambda_2 = \frac{20}{27} \times 6561\mathrm{A} = 4860\mathrm{A}$$

$$= 486\mathrm{nm}$$

Chemistry

Q. 1 Arrange the following bonds according to their average bond energies in descending order :

$C - Cl,\ C - Br,\ C - F,\ C - I$

Option 1:
$C - F > C - Cl > C - Br > C - I$

Option 2:
$C - Br > C - I > C - Cl > C - F$

Option 3:
$C - I > C - Br > C - Cl > C - F$

Option 4:
$C - Cl > C - Br > C - I > C - F$

Correct Answer:
$C - F > C - Cl > C - Br > C - I$

Solution:
The values of bond energies are given below:

$C - F \quad - \quad 485kJ/mol$

$C - Cl \quad - \quad 328kJ/mol$

$C - Br \quad - \quad 276kJ/mol$

$C - I \quad - \quad 240kJ/mol$

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Thus, the correct order of bond energies is:

$$C-F>C-Cl>C-Br>C-I$$

Therefore, **Option(1) is correct.**

Q. 2 Two monomers in maltose are :

Option 1:
$\alpha-D-glucose\ and\ \alpha-D-glucose$

Option 2:
$\alpha-D-glucose\ \ and\ \ \alpha-D-galactose$

Option 3:
$\alpha-D-glucose\ \ and\ \ \beta-D-glucose$

Option 4:
$\alpha-D-glucose\ \ and\ \ \alpha-D-fructose$

Correct Answer:
$\alpha-D-glucose\ and\ \alpha-D-glucose$

Solution:
As we have learnt,

Maltose is a disaccharaide of $\alpha-D-Glucose$. Two units of $\alpha-D-Glucose$ are connected by the C_1-C_4 linkage to form a molecule of Maltose.
Therefore, **Option(1) is correct.**

Q. 3 The correct order of the calculated spin-only magnetic moments of complexes

(A) to (D) is :

(A) $Ni(CO)_4$

(B) $\left[Ni(H_2O)_6\right]Cl_2$

(C) $Na_2\left[Ni(CN)_4\right]$

(D) $PdCl_2(PPh_3)_2$

Option 1:
$(A)\approx(C)<(B)\approx(D)$

Option 2:
$(C)\approx(D)<(B)<(A)$

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Option 3:
$(C) < (D) < (B) < (A)$

Option 4:
$(A) \approx (C) \approx (D) < (B)$

Correct Answer:
$(A) \approx (C) \approx (D) < (B)$

Solution:
As we have learnt,

$[Ni(CO)_4]$ contains $Ni(0)$ which has a d^{10} configuration having $n = 0$ and $\mu = 0$

$[Ni(H_2O)_6]^{2+}$ contains Ni^{2+} which has a d^8 configuration. In a weak ligand field, it has $n = 2$ and hence has $\mu = \sqrt{8}\ BM$

$[Ni(CN)]_4$ contains Ni^{2+} which has a d^8 configuration. In a strong ligand field, these electrons get paired up and hence $n = 0$ and $\mu = 0$

$[Pd(PPh_3)_2Cl_2]$ contains Pd^{2+} which has a d^8 configuration. These electrons get paired in the presence of strong ligand like PPh_3 and hence, $n = 0$ and correspondingly $\mu = 0$.

Therefore, the correct order of spin only magnetic moment of the complexes is

$(B) > (A) = (C) = (D)$

Therefore, **Option(4) is correct.**

Q. 4 Among the reactions (a) - (d), the reaction(s) that does / do not occur in the blast furnace during the extraction of iron is / are :

(a) $CaO + SiO_2 \rightarrow CaSiO_3$

(b) $3Fe_2O_3 + CO \rightarrow 2Fe_3O_4 + CO_2$

(c) $FeO + SiO_2 \rightarrow FeSiO_3$

(d) $FeO \rightarrow Fe + \frac{1}{2}O_2$

Option 1:
(C) and (D)

Option 2:
(A)

Option 3:
(A) and (D)

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Option 4:
(D)

Correct Answer:
(C) and (D)

Solution:
The following reactions do not occur in the blast furnace:

C) $\mathrm{FeO} + \mathrm{SiO_2} \rightarrow \mathrm{FeSiO_3}$

D) $\mathrm{FeO} \rightarrow \mathrm{Fe} + \frac{1}{2}\mathrm{O_2}$

Therefore, **Option(1) is correct.**

Q. 5 For the following Assertion and Reason, the correct option is :

Assertion : The pH of water increases with increase in temperature.

Reason : The dissociation of water into H^+ and OH^- is an exothermic reaction.

Option 1:
Assertion is not true, but reason is true.

Option 2:
Both assertion and reason are true, and the reason is the correct explanation for the assertion.

Option 3:
Both assertion and reason are false.

Option 4:
Both assertion and reason are true, but the reason is not the correct explanation for the assertion.

Correct Answer:
Both assertion and reason are false.

Solution:

The pH of water does not increase with the increase in temperature.
The dissociation of water molecules into ions is bond breaking and is, therefore, an endothermic process.

Therefore, **Option(3) is correct.**

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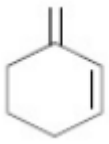
Q. 6 An unsaturated hydrocarbon X absorbs two hydrogen molecules on catalytic hydrogenation, and also gives following reaction:

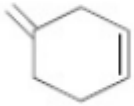
$$X \xrightarrow[H_2O_2]{O_3} A\ (3-\text{oxohexanedioic acid})$$

The compound X is correctly represented in

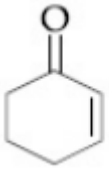
Option 1:



Option 2:

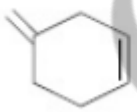


Option 3:



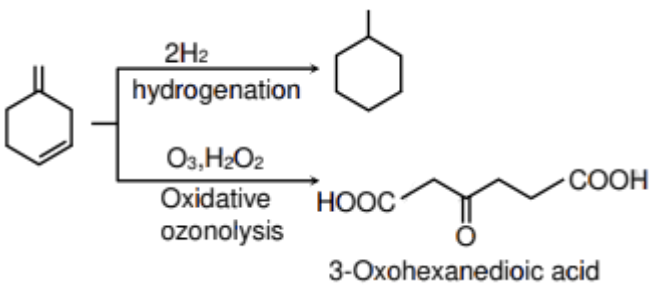
Option 4:

Correct Answer:



Solution:

The reaction occurs as:



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Therefore, **Option(2) is correct.**

Q. 7 Which of the following compounds is likely to show both Frenkel abnd Schottky defects in its crystalline form ?

Option 1:
AgBr

Option 2:
CsCl

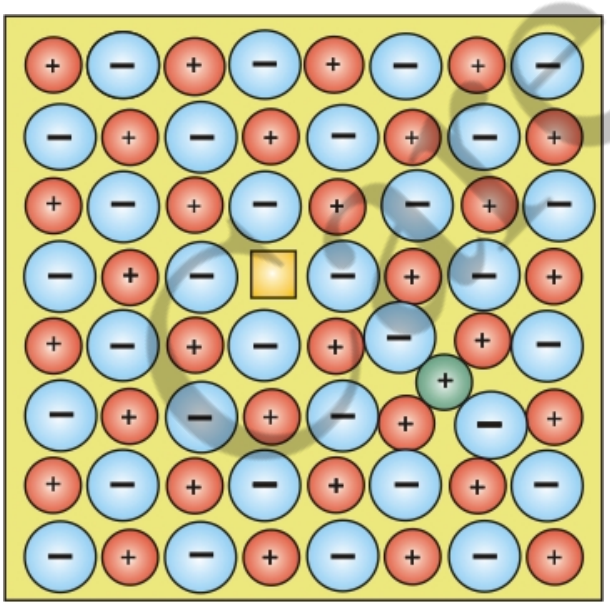
Option 3:
KBr

Option 4:
ZnS

Correct Answer:
AgBr

Solution:

Frenkel Defect: This defect is shown by ionic solids. The smaller ion (usually cation) is dislocated from its normal site to an interstitial site.



Frenkel defects

It creates a vacancy defect at its Original site and an interstitial defect at its new location. Frenkel defect is also called dislocation defect. It does not change the density of the solid. Frenkel defect is shown by ionic substance in which there is a large difference in the size of ions, for example, ZnS, AgCl, AgBr and AgI due to small size of Zn^{2+} and Ag^{+} ions.

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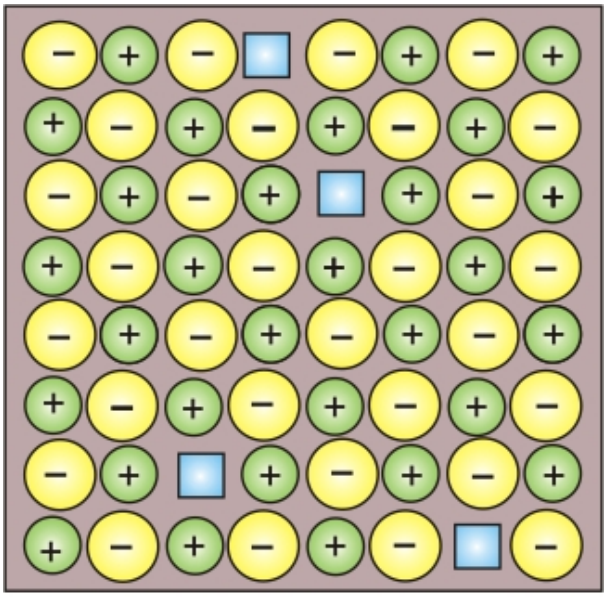
Schottky Defect: It is basically a vacancy defect in ionic solids. In order to maintain electrical neutrality, the number of missing cations and anions are equal.



Schottky defects

Like a simple vacancy defect, the Schottky defect also decreases the density of the substance. The number of such defects in ionic solids is quite significant. For example, in NaCl, there are approximately 10^6 Schottky pairs per cm^3 at room temperature. In 1 cm^3 there are about 10^{22} ions. Thus, there is one Schottky defect per 10^{16} ions. The schottky defect is shown by ionic substances in which the cation and anion are of almost similar sizes. For example, NaCl, KCl, CsCl and AgBr. It may be noted that AgBr shows both, Frenkel as well as Schottky defects.

Only AgBr can exhibit both Schottky and Frenkel defect.

Therefore, **Option(1) is correct.**

Q. 8 The radius of the second Bohr orbit, in terms of the Bohr radius, a_0, in Li^{2+} is :

Option 1:
$\frac{2a_0}{3}$

Option 2:
$\frac{4a_0}{3}$

Option 3:
$\frac{4a_0}{9}$

Option 4:
$\frac{2a_0}{9}$

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Correct Answer:

$\frac{4a_0}{3}$

Solution:
As we have learnt,

Radius, velocity and the energy of nth Bohr orbital -

$$r = \frac{n^2 a_o}{Z}.$$

$$\text{For } Li^{2+} \text{ r} = \frac{2^2 a_o}{3} = \frac{4a_o}{3}$$

Therefore, **Option(2) is correct.**

Q. 9 For the following Assertion and Reason, the correct option is :

Assertion: For hydrogenation reactions, the catalytic activity increases from Group 5 to Group 11 metals with maximum activity shown by Group 7-9 elements.

Reason: The reactants are most strongly adsorbed on group 7-9 elements.

Option 1:
The assertion is true, but the reason is false.

Option 2:
Both assertion and reason are true but the reason is not the correct explanation for the assertion.

Option 3:
Both assertion and reason are true but the reason is the correct explanation for the assertion.

Option 4:
Both assertion and reason are false.

Correct Answer:
Both assertion and reason are false.

Solution:
The metals of groups 7, 8 and 9 do not form hydride. Even from group 6, only chromium forms CrH. These hydrides conduct heat and electricity though not as efficiently as their parent metals do.

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"For hydrogenation reactions, the catalytic activity increases from Group 5 to Group 11 metals with maximum activity shown by Group 7-9 elements" this statement is wrong due to the metals of groups 7, 8 and 9 do not form hydride.

"The reactants do not strongly adsorb on group 7-9 elements" this statement is wrong due to the metals of groups 7, 8 and 9 do not form hydride.

Both assertion and reason are false.

Therefore, **Option(4) is correct.**

Q. 10 Preparation of Bakelite proceeds via reactions:

Option 1:
Electrophilic addition and dehydration

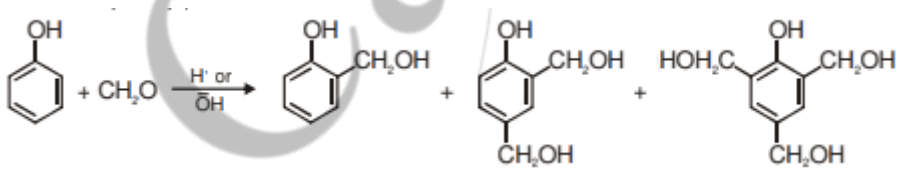
Option 2:
Condensation and elimination

Option 3:
Nucleophilic addition and dehydration

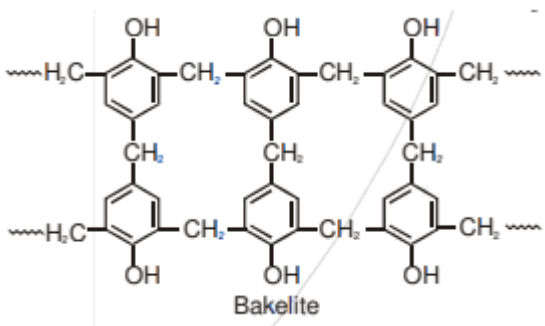
Option 4:
Electrophilic substitution and dehydration

Correct Answer:
Electrophilic substitution and dehydration

Solution:
Preparation of Bakelit follows Electrophilic substitution and dehydration reaction:



Dehydration
→



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Therefore, **Option(4) is correct**

Q. 11 White phosphorous on reaction with concentrated $NaOH$ solution in an inert atmosphere of CO_2 gives phosphine and compound (X). (X) on acidification with HCl gives compound (Y). The basicity of compound (Y) is :

Option 1:
4

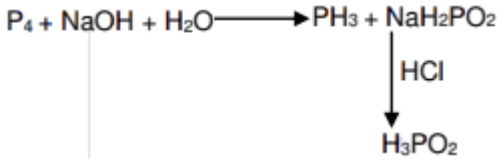
Option 2:
3

Option 3:
2

Option 4:
1

Correct Answer:
1

Solution:



Thus, basicity is 1.

Therefore, **Option(4) is correct.**

Q. 12 Among (a) - (d), the complexes that can display geometrical isomerism are :

(a) $[Pt(NH_3)_3Cl]^+$

(b) $[Pt(NH_3)Cl_5]^-$

(c) $[Pt(NH_3)_2Cl(NO_2)]$

(d) $[Pt(NH_3)_4ClBr]^{2+}$

Option 1:
$(d)\ and\ (a)$

Option 2:
$(c)\ and\ (d)$

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Option 3:
$(a)\ and\ (b)$

Option 4:
$(b)\ and\ (c)$

Correct Answer:
$(c)\ and\ (d)$

Solution:
Out of the given complexes,

Octahedral Ma_4bc and Square planar Ma_2bc can show geometrical isomerism

Hence, only the complexes given in (C) and (D) are capable of showing Geometrical isomerism.

Therefore, **Option(2) is correct.**

Q. 13 A metal (A) on heating in nitrogen gas gives compound B. B on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a dark blue-violet coloured solution. A and B respectively, are :

Option 1:
Na and Na_3N

Option 2:
Mg and Mg_3N_2

Option 3:
Mg and $Mg(NO_3)_2$

Option 4:
Na and $NaNO_3$

Correct Answer:
Mg and Mg_3N_2

Solution:
Magnesium reacts with air to form its nitride. The nitride upon hydrolysis yields ammonia which gives a deep blue color solution with $CuSO_4$

$$3Mg + N_2 \rightarrow \underset{(B)}{Mg_3N_2} \overset{H_2O}{\rightarrow} Mg(OH)_2 + NH_3$$
$$CuSO_4 + 2NH_3 \longrightarrow [Cu(NH_3)_2]SO_4$$

Therefore, **Option(2) is correct.**

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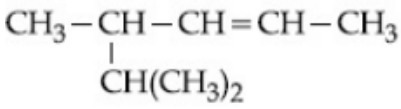
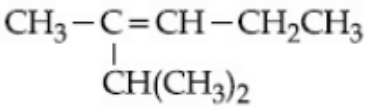
Q. 14 The major product $[B]$ in the following sequence of reaction is :

$$CH_3-\underset{CH(CH_3)_2}{\overset{}{C}}=CH-CH_2CH_3 \xrightarrow[(i)H_2SO_2,OH^{\ominus})]{(i)B_2H_6} [A] \xrightarrow[\Delta]{dil.H_2SO_4} [B]$$

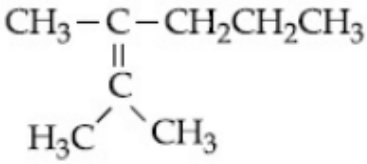
Option 1:



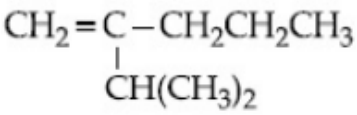
Option 2:



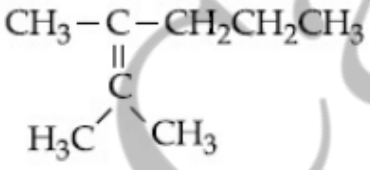
Option 3:



Option 4:



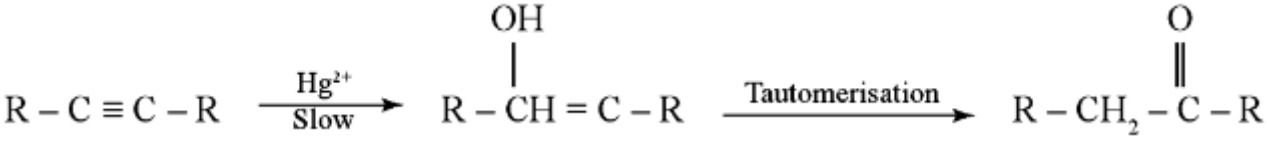
Correct Answer:



Solution:
As we have learnt,

Hydration, Hydroboration and Oxidation of Alkynes -

Hydration
Alkynes cannot be hydrated more easily than alkenes because of their low reactivity towards electrophilic addition reactions. The reaction occurs as follows:



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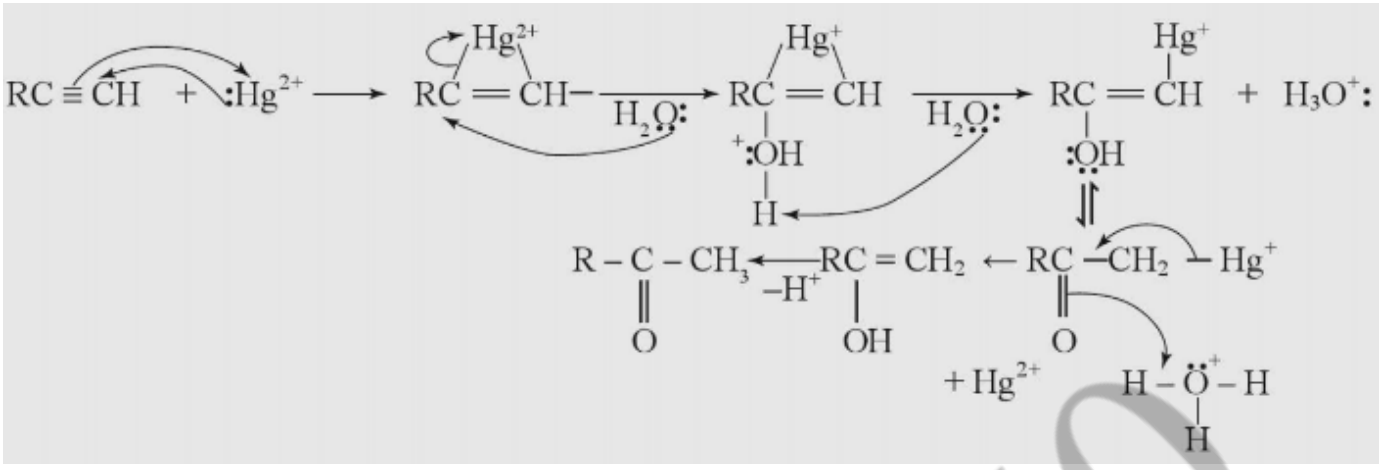
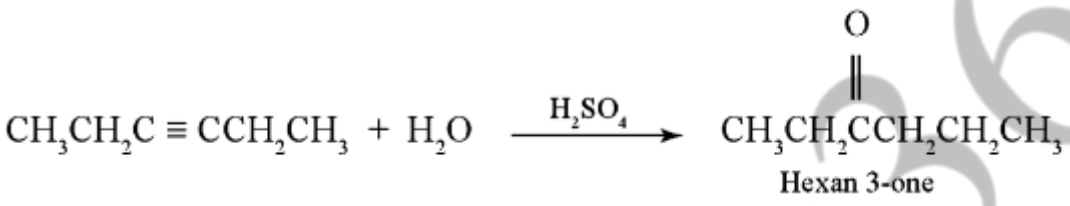
Mechanism

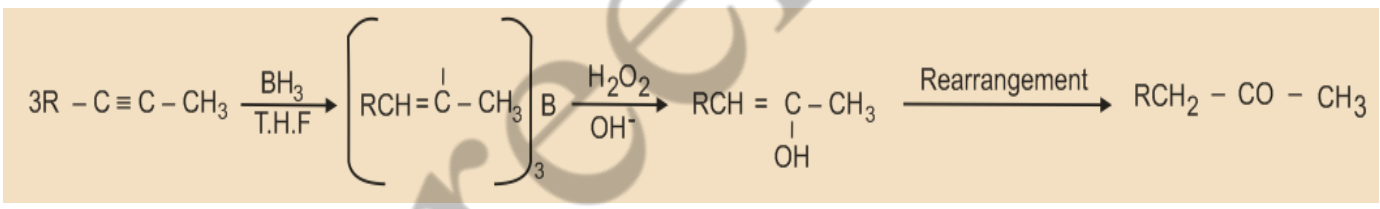


For example:

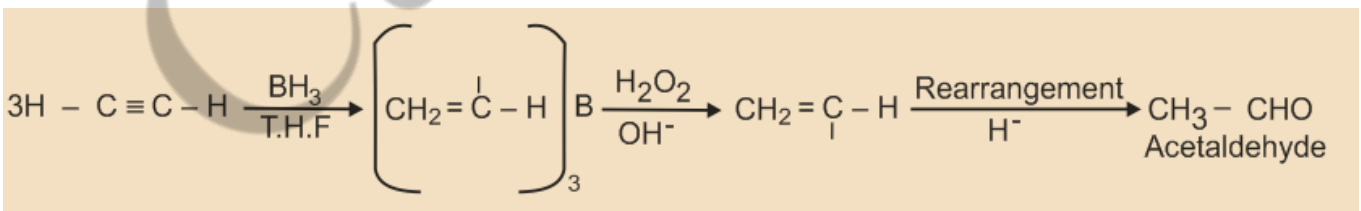


Hydroboration-Oxidation reaction

Alkynes react with BH_3 (in THF) and finally converted into carbonyl compounds. This method is useful for preparing aldehyde from terminal alkyne, which is otherwise not possible by hydration. The reaction occurs as follows:



For example:



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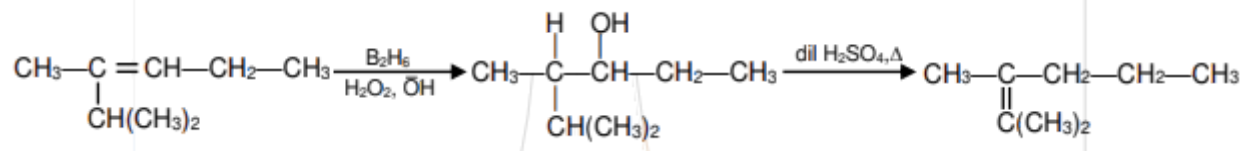

Therefore, **Option(3) is correct.**

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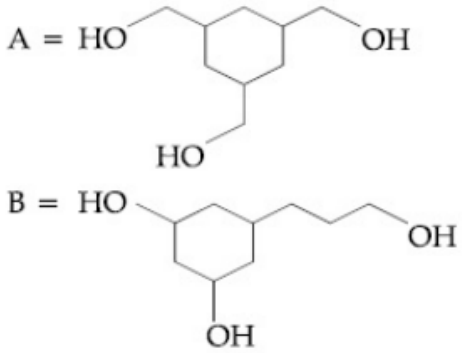
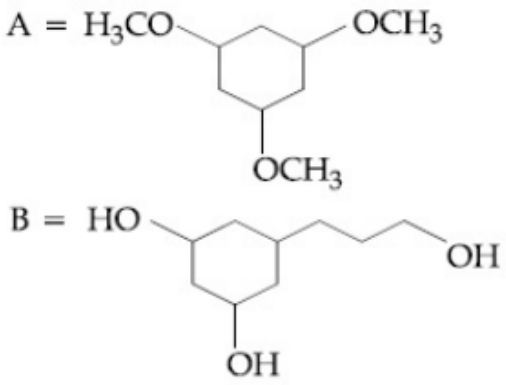
Q. 15 Among the compounds, A and B with molecular formula $C_9H_{18}O_3$, A is having higher boiling point than B. The possible structures of A and B are :

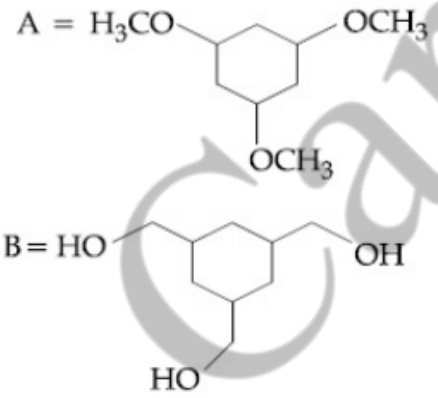
Option 1:



Option 2:



Option 3:



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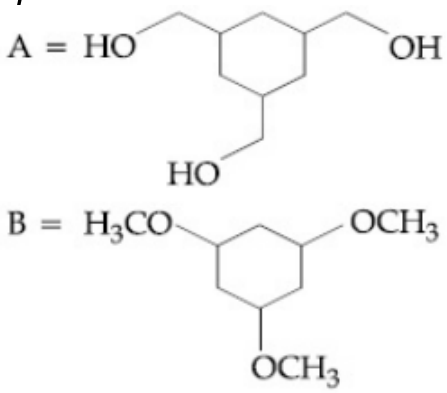
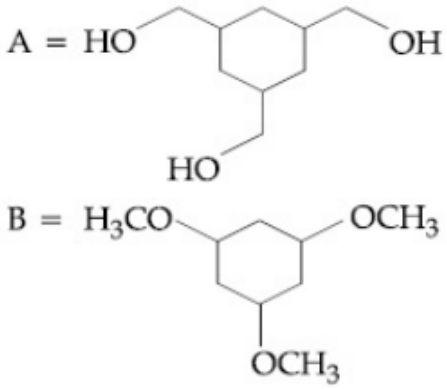
Option 4:



Correct Answer:

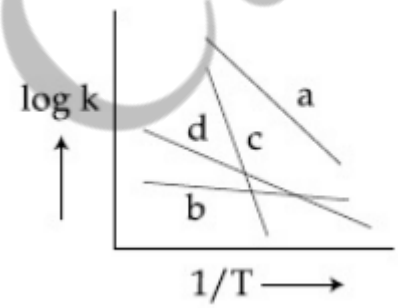


Solution:
In (A), extensive inter-molecular H-bonding is possible while in (B) there is no Inter-molecular H-bonding.

Option 4 has this type of arrangement that follows the above conditions.

Therefore, **Option(4) is correct.**

Q. 16 Consider the following plots of rate constant versus $\frac{1}{T}$ for four different reactions. Which of the following orders is correct for the activation energies of these reactions ?



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Option 1:
$E_b > E_a > E_d > E_c$

Option 2:
$E_a > E_c > E_d > E_b$

Option 3:
$E_b > E_d > E_c > E_a$

Option 4:
$E_c > E_a > E_d > E_b$

Correct Answer:
$E_c > E_a > E_d > E_b$

Solution:
The formula of rate constant -

$$\log k = \log A - \frac{E_a}{2.303RT}$$

$$\text{Slope} = -\frac{E_a}{2.303R}$$

From graph :

Negative Slope : c > a > d > b

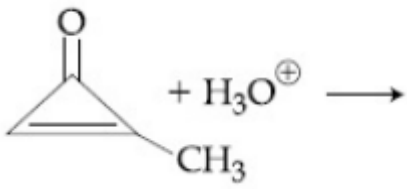
More the negative slope more will be the value of activation energy.

So,

$E_c > E_a > E_d > E_b$

Therefore, **Option(4) is correct.**

Q. 17 The major product in the following reaction is :



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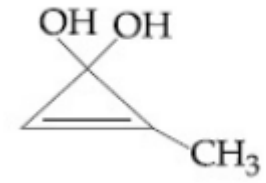
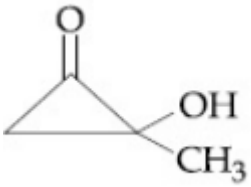
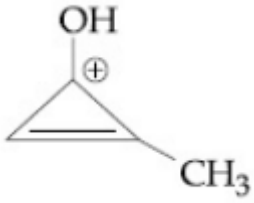
Option 1:



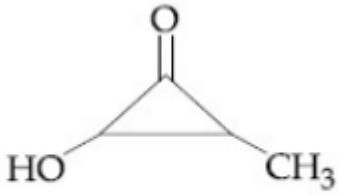
Option 2:



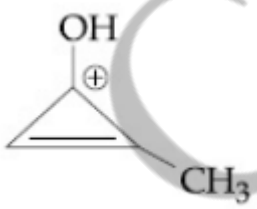
Option 3:



Option 4:

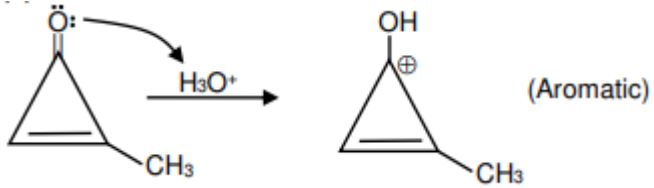


Correct Answer:



Solution:

In the given reaction, there is the formation of an aromatic species upon protonation of the carbonyl group and subsequent polarisation.



Therefore, **Option(3) is correct.**

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Q. 18 The increasing order of the atomic radii of the following elements is :

(a) C (b) O (c) F (d) Cl (e) Br

Option 1:
$(a) < (b) < (c) < (d) < (e)$

Option 2:
$(c) < (b) < (a) < (d) < (e)$

Option 3:
$(d) < (c) < (b) < (a) < (e)$

Option 4:
$(b) < (c) < (d) < (a) < (e)$

Correct Answer:
$(c) < (b) < (a) < (d) < (e)$

Solution:
The correct increasing order of the atomic radii of elements is given below:

$$F < O < C < Cl < Br$$

Atomic radii increase down a group and it decreases from left to right in a period.

Therefore, **Option(2) is correct.**

Q. 19 Hydrogen has three isotopes (A),(B), and (C). If the number of neutron(s) in (A), (B) and (C) respectively, are (x), (y) and (z), the sum of (x),(y) and (z) is :

Option 1:
4

Option 2:
3

Option 3:
2

Option 4:
1

Correct Answer:
3

Solution:

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As we have learnt,

Hydrogen has 3 isotopes. And these isotopes differ from each other because of presence of different number of neutrons.

1. Protium, ${}_1^1H$
2. Deuterium, ${}_1^2H$ or D
3. Tritium, ${}_1^3H$ or T

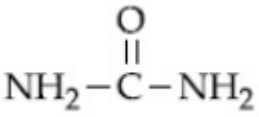
Number of neutrons in Protium is 0.
Number of neutrons in Deuterium is 1.
Number of neutrons in Tritium is 2.

Thus, the correct answer is $0+1+2=3$

Therefore, **Option(2) is correct.**

Q. 20 Kjeldahl's method cannot be used to estimate nitrogen for which of the following compounds?

Option 1:
$C_6H_5NO_2$

Option 2:


Option 3:
$CH_3CH_2-C\equiv N$

Option 4:
$C_6H_5NH_2$

Correct Answer:
$C_6H_5NO_2$

Solution:
As we have learnt,

Kjeldahl's method is not applicable to compounds containing nitrogen in nitro and azo compounds and nitrogen present in rings (pyridine) as nitrogen of these compounds does not convert to ammonium sulphate.
Therefore, **Option(1) is correct.**

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Q. 21 For an electrochemical cell

$Sn(s)\left|Sn^{2+}(aq,1M)\right|\left|Pb^{2+}(aq,1M)\right|Pb(s)$ the ratio $\frac{[Sn^{2+}]}{[Pb^{2+}]}$ When this cell attains equilibrium is ____.

$\left(Given: E^0_{Sn^{2+}|Sn} = -0.14V,\right.$

$\left. E^0_{Pb^{2+}|Pb} = -0.13V, \frac{2.303RT}{F} = 0.06\right)$

Option 1:
2.1544

Option 2:
1.11

Option 3:
7.15

Option 4:
3.14

Correct Answer:
2.1544

Solution:
As we have learnt,

Nernst Equation -

Thisequation gives the relationship between electrode potential and concentration of ions in the solution. In other words, it shows the dependency of electrode potential on the concentration of the ions with which the electrode is reversible.

For a single electrode involving the reduction process,

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The Nernst equation is

$$E = E^\circ + \frac{2.303RT}{nF} \log_{10} \frac{[M^{n+}]}{[M]}$$

Or

$$E = E^\circ - \frac{2.303RT}{nF} \log_{10} \frac{[M]}{[M^{n+}]}$$

At 25°C or 298K

$$E = E^\circ + \frac{2.303 \times 8.314 \times 298}{n \times 96500} \log_{10} \frac{[M^{n+}]}{[M]}$$

$$E = E^\circ + \frac{0.059}{n} \log_{10} \frac{[M^{n+}]}{[M]}$$

$$E = E^\circ - \frac{0.059}{n} \log_{10} \frac{[M]}{[M^{n+}]}$$

Here R = Gas constant

T = Absolute temperature

E° = emf of the cell

E = Electrode potential of cell

F = Faraday number

n = number of electrons transferred

- If electrode is solid its active mass is taken as one.
- For an electrochemical cell having net reaction:

$xA + yB \xrightarrow{ne^-} mC + nD$

The emf can be calculated as

$$E_{cell} = E^\circ_{cell} + \frac{0.059}{n} \log \frac{[A]^x[B]^y}{[C]^m[D]^n}$$

In using the above equation, the following facts should be kept in mind.

- Concentration or activity of gases is expressed in terms of their partial pressures.
- n, the number of electrons transferred should be calculated from the balanced net cell reaction.
- Concentration or activity of solids is taken to be unity.

-

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$$Sn + Pb^{2+} \longrightarrow Sn^{2+} + Pb$$

$$0 = 0.01 - \frac{0.06}{2}\log\left\{\frac{[Sn^{2+}]}{[Pb^{2+}]}\right\}$$

$$0.01 = \frac{0.06}{2}\log\left\{\frac{[Sn^{2+}]}{[Pb^{2+}]}\right\}$$

$$\frac{1}{3} = \log\left\{\frac{[Sn^{2+}]}{[Pb^{2+}]}\right\} \Rightarrow \frac{[Sb^{2+}]}{[Pb^{2+}]} = 10^{1/3} = 2.1544$$

Hence, the option number (1) is correct.

Q. 22 At constant volume, $4\ mol$ of an ideal gas when heated from $300\ K$ to $500\ K$ changes its internal energy by $5000\ J$. The molar heat capacity (**J**/mol · K) at constant volume is _____.

Correct Answer:
6.25

Solution:
As we have learnt,

$$\Delta U = nC_V\Delta T$$

$$5000 = 4 \times C_V(500 - 300)$$

$$C_V = 6.25\ JK^{-1}mol^{-1}$$

Thus, the correct answer is $6.25.$

Hence, **the option number (1) is correct**

Q. 23 In the following sequence of reactions the maximum number of atoms present in molecule 'C' in one plane is ____.

$$A \xrightarrow[Cu\ tu\ be]{Red\ hot} B \xrightarrow[Anhydrous\ AlCl_3]{CH_3Cl(I.eq.)} C$$

(A is a lowest molecular weight alkyne)

Option 1:
13

Option 2:

Option 3:

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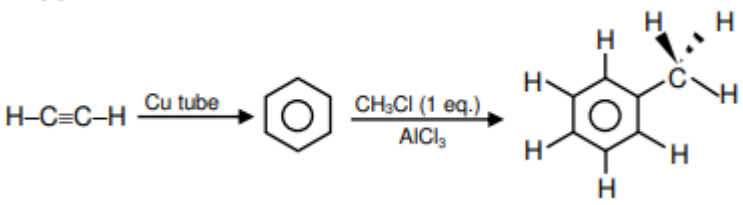
Option 4:

Correct Answer:
13

Solution:



Number of atoms in one plane = 13.

Q. 24 $NaClO_3$ is used, even in spacecrafts, to produce O_2. The daily consumption of pure O_2 by a person is $492L$ at $1\ atm, 300\ K$. How much amount of $NaClO_3$, in grams, is required to produce O_2 for the daily consumption of a person at $1\ atm, 300\ K$? _____.

$$NaClO_3(s) + Fe(s) \rightarrow O_2(g) + NaCl(s) + FeO(s)$$

$$R = 0.082\ L\ atm\ mol^{-1}K^{-1}$$

Correct Answer:
2130

Solution:
moles of $NaClO_3$ = moles of O_2

$$\text{moles of } O_2 = \frac{PV}{RT} = \frac{1 \times 492}{0.082 \times 300} = 20\text{ml}$$

Mass of $NaClO_3 = 20 \times 106.5 = 2130g$

Q. 25 Complexes (ML_5) of metals Ni and Fe have ideal square pyramidal and trigonal bipyramidal geometries, respectively.The sum of the $90°, 120°$ and $180° L - M - L$ angles in the two complexes is _____.

Option 1:
20

Option 2:

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Option 3:

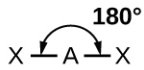
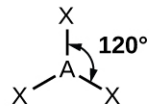
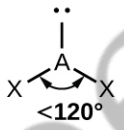
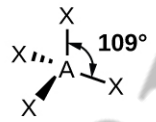
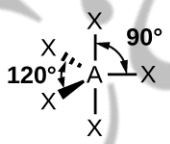
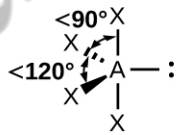
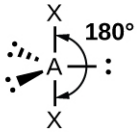
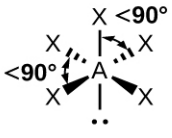
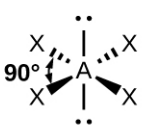
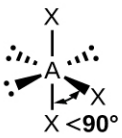
Option 4:

Correct Answer:
20

Solution:
As we have learnt,

Shapes of Molecules -

The ideal shapes of molecules, which are predicted on the basis of electron pairs and lone pairs of electrons are mentioned in the table below:

Number of electron pairs	Electron pair geometries: 0 lone pair	1 lone pair	2 lone pairs	3 lone pairs	4 lone pairs
2	 Linear				
3	 Trigonal planar	 Bent or angular			
4	 Tetrahedral	Trigonal pyramid	Bent or angular		
5	 Trigonal bipyramid	 Sawhorse or seesaw	T-shape	 Linear	
6	Octahedral	 Square pyramid	 Square planar	 T-shape	Linear

Predicting the geometry of molecules

The following procedure uses VSEPR theory to determine the geometry of the molecules:

1. Write the Lewis structure of the molecule or polyatomic ion.

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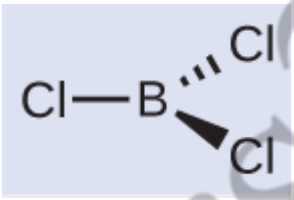
2. Count the number of regions of electron density (lone pairs and bonds) around the central atom. A single, double, or triple bond counts as one region of electron density.

3. Identify the electron-pair geometry based on the number of regions of electron density: linear, trigonal planar, tetrahedral, trigonal bipyramidal, or octahedral

4. Use the number of lone pairs to determine the molecular structure. If more than one arrangement of lone pairs and chemical bonds is possible, choose the one that will minimize repulsions, remembering that lone pairs occupy more space than multiple bonds, which occupy more space than single bonds. In trigonal bipyramidal arrangements, repulsion is minimized when every lone pair is in an equatorial position. In an octahedral arrangement with two lone pairs, repulsion is minimized when the lone pairs are on opposite sides of the central atom.

   For example, BCl3 has three electron pairs and no lone pairs of electrons. Thus these three electron pairs will arrange themselves in a trigonal planar geometry as shown below. The bond angles between each B-Cl bonds is 1200.



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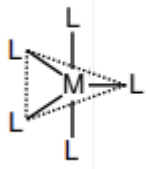
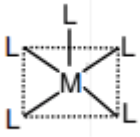


$\angle 120^\circ = 3; \angle 90^\circ = 6; \angle 180^\circ = 1 \Rightarrow Total = 10$



$\angle 90^\circ = 8; \angle 180^\circ = 2 \Rightarrow Total = 10$

Thus, **the correct answer is 20.**

Maths

Q. 1 Let A and B be two events such that the probability that exactly one of them occurs is $\frac{2}{5}$ and the probability that A or B occurs is $\frac{1}{2}$, then the probability pf both of them occur together is :

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Option 1:
0.10

Option 2:
0.20

Option 3:
0.01

Option 4:
0.02

Correct Answer:
0.10

Solution:

$$P(\text{exactly one}) = \frac{2}{5} = P(A) + P(B) - 2P(A \cap B)$$
$$P(A \cup B) = \frac{1}{2} = P(A) + P(B) - P(A \cap B)$$
$$\Rightarrow P(A \cap B) = \frac{1}{2} - \frac{2}{5} = \frac{1}{10}$$

Correct Option (1)

Q. 2 Let S be the set of all real roots of the equation, $3^x(3^x - 1) + 2 = |3^x - 1| + |3^x - 2|$. Then S :

Option 1:
is a singleton.

Option 2:
is an empty set.

Option 3:
contains at least four elements.

Option 4:
contains exactly two elements.

Correct Answer:
is a singleton.

Solution:

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Transcendental function -

Transcendental functions: the functions which are not algebraic are called transcendental functions. Exponential, logarithmic, trigonometric and inverse trigonometric functions are transcendental functions.

Exponential Function: function f(x) such that $\mathrm{f(x)} = \mathrm{a^x}$ is known as an exponential function.

$$\begin{aligned}&\text{base}: \quad \mathrm{a} > 0, \mathrm{a} \neq 1\\&\text{domain}: \mathrm{x} \in \mathbb{R}\\&\text{range}: \mathrm{f(x)} > 0\end{aligned}$$

Modulus Function, Properties of Modulus Function -

Modulus Function:

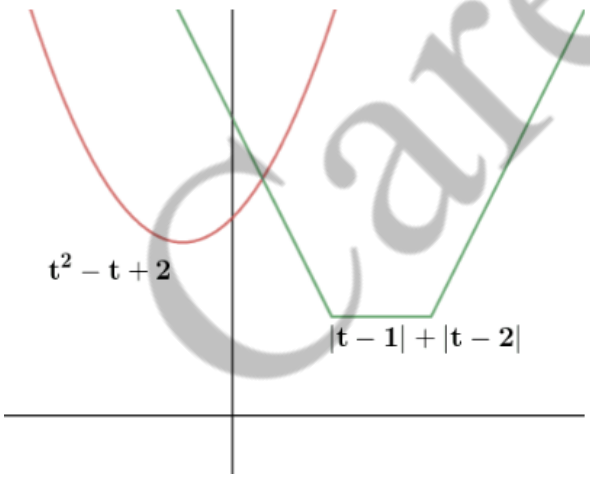
$$|\mathrm{x}|, \ \mathrm{x} \in \mathbb{R} = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$\text{Range} \in [0, \infty)$$

-

$$\text{put } 3^x = t$$

$$\begin{aligned}&t\,(t-1)+2 = |t-1| + |t-2|\\&t^2 - t + 2 = |t-1| + |t-2|\end{aligned}$$



singolton set

Correct Option (1)

Q. 3 The mean and variance of 20 observations are found to be 10 and 4, respectively. On rechecking, it was found that an observation 9 was incorrect and the correct observation was 11. Then the correct variance is :

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Option 1:
4.01

Option 2:
3.99

Option 3:
3.98

Option 4:
4.02

Correct Answer:
3.99

Solution:

Dispersion (Variance and Standard Deviation) -

Variance and Standard Deviation

The mean of the squares of the deviations from the mean is called the variance and is denoted by σ^2 (read as sigma square).

Variance is a quantity which leads to a proper measure of dispersion.

The variance of n observations x_1 , x_2 ,..., x_n is given by

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Standard Deviation

The standard deviation is a number that measures how far data values are from their mean.

The positive square-root of the variance is called standard deviation. The standard deviation, usually denoted by σ and it is given by

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Variance and Standard Deviation of a Discrete Frequency Distribution

The given discrete frequency distribution be

$$x \quad : \quad x_1, \quad x_2, \quad x_3, \quad \ldots \quad x_n$$

$$f \quad : \quad f_1, \quad f_2, \quad f_3, \quad \ldots \quad f_n$$

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$$\text{In this case, Variance } (\sigma^2) \quad = \frac{1}{N}\sum_{i=1}^{n} f_i\,(x_i - \bar{x})^2$$

$$\text{and, Standard Deviation}(\sigma) \; = \sqrt{\frac{1}{N}\sum_{i=1}^{n} f_i\,(x_i - \bar{x})^2}$$

$$\text{where, N} = \sum_{i=1}^{n} f_i$$

Variance and Standard deviation of a continuous frequency distribution

The formula for variance and standard deviation are the same as in the case of discrete frequency distribution. Here, x_i is the mid point of each class.

Another formula for Standard Deviation

$$\begin{aligned}
\text{Variance } (\sigma^2) &= \frac{1}{\text{N}}\sum_{i=1}^{n} f_i\,(x_i - \bar{x})^2 = \frac{1}{\text{N}}\sum_{i=1}^{n} f_i\left(x_i^2 + \bar{x}^2 - 2\bar{x}x_i\right) \\
&= \frac{1}{N}\left[\sum_{i=1}^{n} f_i x_i^2 + \sum_{i=1}^{n} \bar{x}^2 f_i - \sum_{i=1}^{n} 2\bar{x} f_i x_i\right] \\
&= \frac{1}{N}\left[\sum_{i=1}^{n} f_i x_i^2 + \bar{x}^2 \sum_{i=1}^{n} f_i - 2\bar{x}\sum_{i=1}^{n} x_i f_i\right] \\
&= \frac{1}{N}\left[\sum_{i=1}^{n} f_i x_i^2 + \bar{x}^2 N - 2\bar{x}\cdot N\bar{x}\right] \\
&= \frac{1}{N}\left[\sum_{i=1}^{n} f_i x_i^2 + \bar{x}^2 N - 2\bar{x}\cdot N\bar{x}\right] \\
&\left[\because \frac{1}{N}\sum_{i=1}^{n} x_i f_i = \bar{x} \text{ or } \sum_{i=1}^{n} x_i f_i = \text{N}\bar{x}\right] \\
&= \frac{1}{\text{N}}\sum_{i=1}^{n} f_i x_i^2 + \bar{x}^2 - 2\bar{x}^2 = \frac{1}{\text{N}}\sum_{i=1}^{n} f_i x_i^2 - \bar{x}^2
\end{aligned}$$

$$\text{or } \; \sigma^2 = \frac{1}{N}\sum_{i=1}^{n} f_i x_i^2 - \left(\frac{\sum_{i=1}^{n} f_i x_i}{N}\right)^2 = \frac{1}{N^2}\left[N\sum_{i=1}^{n} f_i x_i^2 - \left(\sum_{i=1}^{n} f_i x_i\right)^2\right]$$

$$\text{Thus, standard deviation } (\sigma) = \frac{1}{N}\sqrt{N\sum_{i=1}^{n} f_i x_i^2 - \left(\sum_{i=1}^{n} f_i x_i\right)^2}$$

Shortcut method to find variance and standard deviation

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The values of x_i in a discrete distribution or the mid points x_i of different classes in a continuous distribution are large and so the calculation of mean and variance becomes tedious and time consuming.

Here is the shortcut method to find variance and standard deviatio

Let the assumed mean be 'A' and the scale be reduced to 1/h times (h being the width of class-intervals).

Let the step-deviations or the new values be y_i .

i.e. $$y_i = \frac{x_i - \mathrm{A}}{h} \text{ or } x_i = \mathrm{A} + hy_i| \qquad \ldots(1)$$

We know that $$\bar{x} = \frac{\sum_{i=1}^{n} f_i x_i}{N} \qquad \ldots(2)$$

Replacing x_i from (1) in (2),

$$\begin{aligned}
\bar{x} &= \frac{\sum_{i=1}^{n} f_i\,(\mathrm{A} + hy_i)}{\mathrm{N}} \\
&= \frac{1}{\mathrm{N}}\left(\sum_{i=1}^{n} f_i \mathrm{A} + \sum_{i=1}^{n} h f_i y_i\right) = \frac{1}{\mathrm{N}}\left(\mathrm{A}\sum_{i=1}^{n} f_i + h\sum_{i=1}^{n} f_i y_i\right) \\
&= \mathrm{A}\cdot\frac{\mathrm{N}}{\mathrm{N}} + h\frac{\sum_{i=1}^{n} f_i y_i}{\mathrm{N}} \quad \left(\text{because } \sum_{i=1}^{n} f_i = \mathrm{N}\right)
\end{aligned}$$

Thus $$\bar{x} = \mathrm{A} + h\bar{y} \qquad \ldots(3)$$

Now Variance of the variable x

$$\begin{aligned}
\sigma_x^2 &= \frac{1}{\mathrm{N}}\sum_{i=1}^{n} f_i\,(x_i - \bar{x})^2 \\
&= \frac{1}{\mathrm{N}}\sum_{i=1}^{n} f_i\,(\mathrm{A} + hy_i - \mathrm{A} - h\bar{y})^2 \qquad (\text{Using (1) and (3)}) \\
&= \frac{1}{N}\sum_{i=1}^{n} f_i h^2\,(y_i - \bar{y})^2 \\
&= \frac{h^2}{\mathrm{N}}\sum_{i=1}^{n} f_i\,(y_i - \bar{y})^2 = h^2 \times \text{ variance of the variable } y_i
\end{aligned}$$

i.e. $$\sigma_x^2 = h^2\sigma_y^2$$

or $$\sigma_x = h\sigma_y \qquad \ldots(4)$$

From (3) and (4),

$$\sigma_x = \frac{h}{\mathrm{N}}\sqrt{\mathrm{N}\sum_{i=1}^{n} f_i y_i^2 - \left(\sum_{i=1}^{n} f_i y_i\right)^2}$$

-

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$$\sigma^2_{new} = 4 - \left[\frac{(11-9)}{20}\right]^2 = 4 - \frac{4}{200} = 3.99$$

Correct Option (2)

Q. 4 Let $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ be two vectors. If $\vec{c}$ is a vector such that $\vec{b} \times \vec{c} = \vec{b} \times \vec{a}$ and $\vec{c} \cdot \vec{a} = 0$, then $\vec{c} \cdot \vec{b}$ is equal to :

Option 1:
$\frac{1}{2}$

Option 2:
$-\frac{3}{2}$

Option 3:
$-\frac{1}{2}$

Option 4:
-1

Correct Answer:
$-\frac{1}{2}$

Solution:

Vector Triple Product -

For three vectors $\vec{\mathbf{a}}$, $\vec{\mathbf{b}}$ and $\vec{\mathbf{c}}$ vector triple product is defined as $\vec{\mathbf{a}} \times \left(\vec{\mathbf{b}} \times \vec{\mathbf{c}}\right)$.

$$\vec{\mathbf{a}} \times \left(\vec{\mathbf{b}} \times \vec{\mathbf{c}}\right) = (\vec{\mathbf{a}} \cdot \vec{\mathbf{c}}) \cdot \vec{\mathbf{b}} - \left(\vec{\mathbf{a}} \cdot \vec{\mathbf{b}}\right) \cdot \vec{\mathbf{c}}$$

-

$$\begin{aligned}
\vec{a} \times (\vec{b} \times \vec{c}) &= \vec{a} \times (\vec{b} \times \vec{a}) \\
-(\vec{a} \cdot \vec{b})\vec{c} &= (\vec{a} \cdot \vec{a})\vec{b} - (\vec{a} \cdot \vec{b})\vec{a} \\
-4\vec{c} &= 6(\hat{i} - \hat{j} + \hat{k}) - 4(\hat{i} - 2\hat{j} + \hat{k}) \\
-4\vec{c} &= 2\hat{i} + 2\hat{j} + 2\hat{k} \\
\vec{c} &= -\frac{1}{2}(\hat{i} + \hat{j} + \hat{k}) \\
\vec{b} \cdot \vec{c} &= -\frac{1}{2}
\end{aligned}$$

Correct Option (3)

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Q. 5 Let $f:(1,3)\to R$ be a function defined by $f(x)=\frac{x\,[x]}{1+x^2}$, where $[x]$ denotes the greatest integer $\leq x$. Then the range of f is :

Option 1:
$\left(\frac{2}{3},\frac{3}{5}\right]\cup\left(\frac{3}{4},\frac{4}{5}\right)$

Option 2:
$\left(\frac{2}{5},\frac{4}{5}\right]$

Option 3:
$\left(\frac{3}{5},\frac{4}{5}\right)$

Option 4:
$\left(\frac{2}{5},\frac{1}{2}\right)\cup\left(\frac{3}{5},\frac{4}{5}\right]$

Correct Answer:
$\left(\frac{2}{5},\frac{1}{2}\right)\cup\left(\frac{3}{5},\frac{4}{5}\right]$

Solution:

Piecewise function -

Greatest integer function

[1.75] = 1

[2.34] = 2

[-0.9] = -1

[-4.8] = -5

From the definition of [x], we

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can see that

Properties of greatest integer function:

i) $[x] \leq x < [x] + 1$

ii) $x - 1 < [x] < x$

iii) $I \leq x < I+1 \Rightarrow [x] = I$ where I belongs to integer.

iv) [[x]]=[x]v)

v) $[x] + [-x] = \begin{cases} 0, & if \quad x \quad belongs\ to \quad integer \\ -1, & if \quad x \quad doesn't\ belongs\ to \quad integer \end{cases}$

vi) [x] + [-x] = 2x if x belongs to integer

2[x] + 1 if x doesn't belongs to integer

-

Domain of function, Co-domain, Range of function -

All possible values of x for f(x) to be defined is known as a domain. If a function is defined from A to B i.e. f: A→B, then all the elements of set A is called Domain of the function.

If a function is defined from A to B i.e. f: A→B, then all the elements of set B are called Co-domain of the function.

The set of all possible values of f(x) for every x belongs to the domain is known as Range.

For example, let A = {1, 2, 3, 4, 5} and B = {1, 4, 8, 16, 25, 64, 125}. The function f : A -> B is defined by $f(x) = x^3$. So here,

Domain : Set A

Co-Domain : Set B

Range : {1, 8, 27, 64, 125}

The range can be equal to or less than codomain but cannot be greater than that.

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$$f(x)=\begin{cases}\frac{x}{x^2+1}; & x\in(1,2)\\ \frac{2x}{x^2+1}; & x\in[2,3)\end{cases}$$

$\therefore$ f(x) is a decreasing function

$$\therefore \quad y\in\left(\frac{2}{5},\frac{1}{2}\right)\cup\left(\frac{6}{10},\frac{4}{5}\right]$$

$$\Rightarrow \quad y\in\left(\frac{2}{5},\frac{1}{2}\right)\cup\left(\frac{3}{5},\frac{4}{5}\right]$$

Correct Option (4)

Q. 6 If α and β be the coefficients of x^4 and x^2 respectively in the expansion of

$\left(x+\sqrt{x^2-1}\right)^6+\left(x-\sqrt{x^2-1}\right)^6$, then :

Option 1:

$\alpha+\beta=30$

Option 2:

$\alpha-\beta=-132$

Option 3:

$\alpha+\beta=60$

Option 4:

$\alpha-\beta=60$

Correct Answer:

$\alpha-\beta=-132$

Solution:

We know

$$(x+y)^n+(x-y)^n=2\left[{}^nC_0\,x^n\,y^0+{}^n\,C_2\,x^{n-2}\,y^2+{}^n\,C_4\,x^{n-4}\,y^4+\ldots.\right]$$

Now,

$$\left(x+\sqrt{x^2-1}\right)^6+\left(x-\sqrt{x^2-1}\right)^6$$

$$=2\left[{}^6C_0x^6+{}^6\,C_2x^4\left(x^2-1\right)+{}^6\,C_4x^2\left(x^2-1\right)^2+{}^6\,C_6\left(x^2-1\right)^3\right]$$

$$=2\left[x^6+15\left(x^6-x^4\right)+15x^2\left(x^4-2x^2+1\right)+\left(x^6-1-3x^4+3x^2\right)\right]$$

$$=2\left(32x^6-48x^4+18x^2-1\right)$$

$\alpha=-96$ and $\beta=36$

$\therefore \quad \alpha-\beta=-132$

Correct Option (2)

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Q. 7 If a hyperbola passes through the point P(10,16) and it has vertices at $(\pm 6, 0)$, then the equation of the normal to it at P is :

Option 1:
$3x + 4y = 94$

Option 2:
$x + 2y = 42$

Option 3:
$2x + 5y = 100$

Option 4:
$x + 3y = 58$

Correct Answer:
$2x + 5y = 100$

Solution:

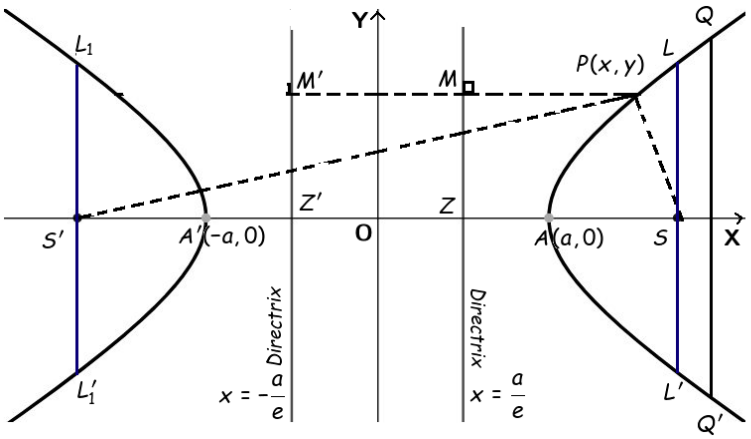
What is Hyperbola? -

Hyperbola:

The standard form of the equation of a hyperbola with center (0, 0) and major axis on the x-axis is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \text{where, } b^2 = a^2\left(e^2 - 1\right)$$

Important Terms related to Hyperbola:



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- **Centre:** All chord passing through point O is bisected at point O. Here O is the origin, i.e. (0, 0).
- **Foci:** Point S and S' is foci of the hyperbola where, S is (ae, 0) and S' is (-ae, 0).
- **Directrices:** The straight line ZM and Z'M' are two directrices of the hyperbola and their equations are x=ae and x=-ae.
- **Axis:** In figure AA' is called the transverse axis and the line perpendicular it through centre of hyperbola is called conjugate axis. 2a is length of transverse axis and 2b is length of conjugate axis.
- **Double Ordinate:** If a line perpendicular to axis of hyperbola meets the curve at Q and Q', then QQ' is called double ordinate.
- **Latusrectum:** Double ordinate passing through focus is called latus rectum. Here LL' is latusrectum.

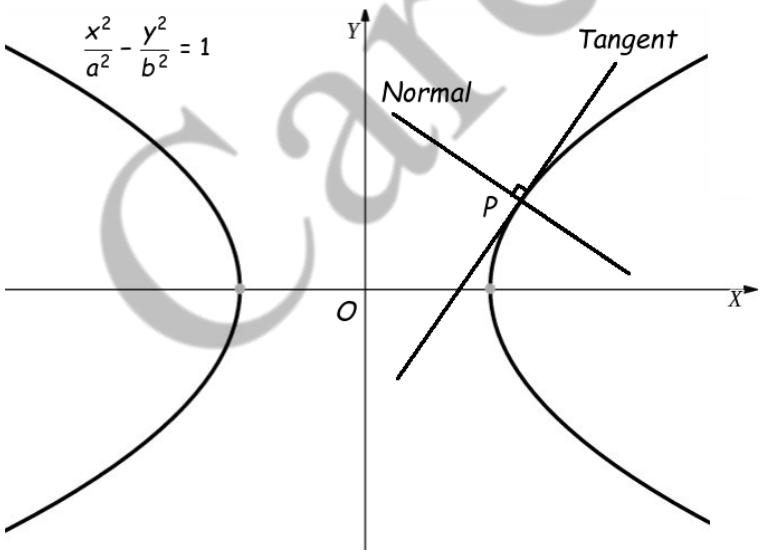
-

Equation of Normal of Hyperbola in Point Form -

Equation of Normal of Hyperbola in Point Form

Point form:

The equation of normal at (x_1, y_1) to the hyperbola, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$.



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Vertex is at (±6, 0)

∴a = 6

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Let the hyperbola is $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

Putting point P(10, 16) on the hyperbola

$\frac{100}{36}-\frac{256}{b^2}=1\Rightarrow b^2=144$

$\therefore$ hyperbola is $\frac{x^2}{36}-\frac{y^2}{144}=1$

$\therefore$ equation of normal is $\frac{a^2x}{x_1}+\frac{b^2y}{y_1}=a^2+b^2$

$\therefore$ putting we get $2x+5y=100$

Correct Option (3)

Q. 8 $\lim\limits_{x\to 0}\frac{\int_0^x t\sin(10t)dt}{x}$ is equal to :

Option 1:

0

Option 2:

$\frac{1}{10}$

Option 3:

$-\frac{1}{10}$

Option 4:

$-\frac{1}{5}$

Correct Answer:

0

Solution:

L' Hospital's Rule -

L'Hospital's Rule

L'Hospital's Rule states that, if $\lim\limits_{x\to 0}\ \frac{f(x)}{g(x)}\ is\ reduced\ to\ \frac{0}{0} or\ \frac{\infty}{\infty}$,

Then, differentiate numerator and denominator till this intermediate form is removed.

i.e. $\lim\limits_{x\to a}\ \frac{f(x)}{g(x)}=\lim\limits_{x\to a}\ \frac{f'(x)}{g'(x)}$,

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Provided that the limit $\lim_{x\to a}\ \frac{f'(x)}{g'(x)}$ exists. But, if again this take indeterminate form $\frac{0}{0} or \frac{\infty}{\infty}$Then,

$\lim_{x\to a}\ \frac{f(x)}{g(x)} = \lim_{x\to a}\ \frac{f'(x)}{g'(x)} = \lim_{x\to a}\ \frac{f''(x)}{g''(x)}$

And this process is continued till the indeterminate form $\frac{0}{0} or \frac{\infty}{\infty}$ is removed.

In some questions, we may have to apply the above process again and again till we get the non-indeterminate form

-

Using L'Hospital

$\lim_{x\to 0}\ \frac{x\sin(10x) - 0\sin(10\cdot 0)}{1} = 0$

$\lim_{x\to 0}\ \frac{x\sin(10x)}{1} = 0$

Correct option (1)

Q. 9 If a line, y=mx+c is a tangent to the circle, $(x-3)^2+y^2=1$ and it is perpendicular to a line L_1, where, l_1 is the tangent to the circle, $x^2+y^2=1$ at the point $\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)$;; then :

Option 1:
$c^2+7c+6=0$

Option 2:
$c^2-6c+7=0$

Option 3:
$c^2-7c+6=0$

Option 4:
$c^2+6c+7=0$

Correct Answer:
$c^2+6c+7=0$

Solution:

Slope and Equation of Tangent -

Tangent and Normal

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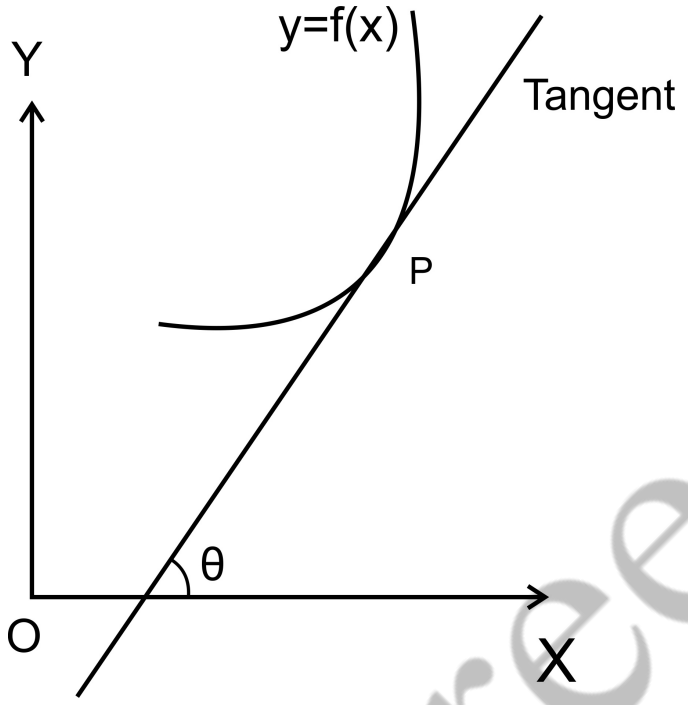
Slope and Equation of Tangent:

Let P(x_0, y_0) be a point on the continuous curve y = f(x), then the slope of the tangent to the curve at point P is

$$\left(\frac{dy}{dx}\right)_{(x_0,y_0)}.$$

$$\Rightarrow \left(\frac{dy}{dx}\right)_{(x_0,y_0)} = \tan\theta = \text{ slope of tangent at } P$$

Where Θ is the angle which the tangent at P (x makes with the positive direction of the x-axis as shown in the figure.



- If the tangent is parallel to x-axis then Θ = 0^0.

$$\Rightarrow \quad \tan\theta = 0$$

$$\therefore \quad \left(\frac{dy}{dx}\right)_{(x_0,y_0)} = 0$$

- If the tangent is perpendicular to x-axis then Θ = 90^0

$$\Rightarrow \quad \tan\theta \to \infty \quad \text{or} \quad \cot\theta = 0$$

$$\therefore \quad \left(\frac{dx}{dy}\right)_{(x_0,y_0)} = 0$$

Equation of Tangent:

Let the equation of curve y = f (x) and a point P (x_0, y_0) lies on this curve.

The slope of the tangent to the curve at a point P is

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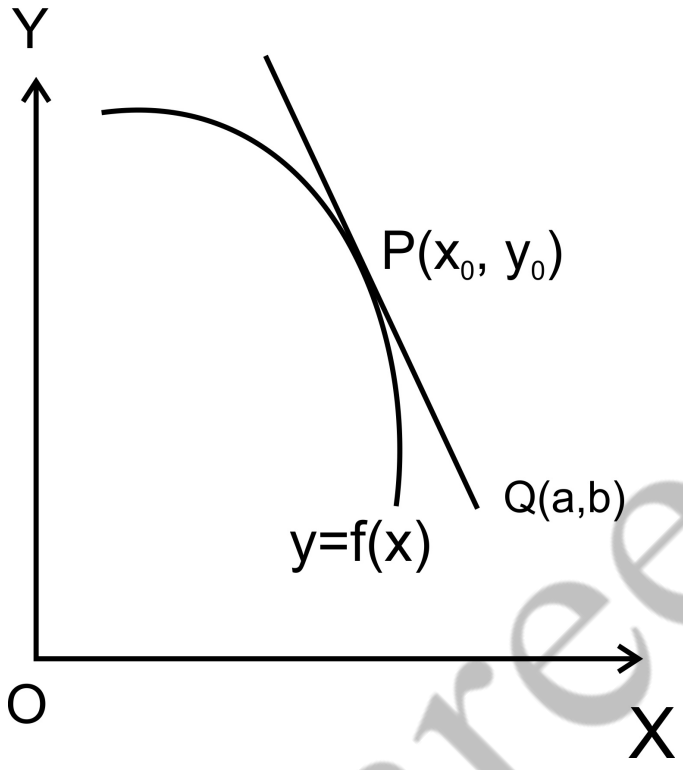
$\left(\frac{dy}{dx}\right)_{(x_0,y_0)}$.

Hence, the equation of the tangent at point P is

$$(y - y_0) = \left(\frac{dy}{dx}\right)_{(x_0,y_0)} \cdot (x - x_0)$$

Tangent from External Point:

If a point Q(a, b) does not lie on the curve y = f(x), then the equation of possible tangent to the curve y = f(x) (tangent passing through point Q (a, b)) can be found by solving point of contact $P(x_0, y_0)$ on the curve.



$P\ (x_0, y_0)$ lies on the curve y = f(x), then

$$y_0 = f(x_0)$$

Also, slope of PQ is

$$\frac{y_0 - b}{x_0 - a} = \left(\frac{dy}{dx}\right)_{(x_0,y_0)}$$

By solving the above two equations we get point of contact point P and equation of tangent PQ.

-

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Distance of a Point From a Line -

Distance of a point from a line

Perpendicular length from a point (x1,y1) to the line L : Ax + By + C = 0 is

$$\frac{|\mathbf{Ax_1} + +\mathbf{By_1} + \mathbf{C}|}{\sqrt{\mathbf{A^2} + \mathbf{B^2}}}$$

-

The slope of the tangent to $x^2 + y^2 = 1$ at $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

$x^2 + y^2 = 1$
$2x + 2yy' = 0$
$y' = -\frac{x}{y} = -1$
$y = mx + c$ is tangent of $x^2 + y^2 = 1$

so m = 1

y = x + c

now distance of (3, 0) from y = x + c is

$\left|\frac{c+3}{\sqrt{2}}\right| = 1$
$c^2 + 6c + 9 = 2$
$c^2 + 6c + 7 = 0$

Correct Option (4)

Q. 10 Let $\alpha = \frac{-1 + i\sqrt{3}}{2}$. If $a = (1 + \alpha)\sum_{k=0}^{100} \alpha^{2k}$ and $b = \sum_{k=0}^{100} \alpha^{3k}$, then a and b are the roots of the quadratic equation :

Option 1:
$x^2 + 101x + 100 = 0$

Option 2:
$x^2 + 102x + 101 = 0$

Option 3:
$x^2 - 102x + 101 = 0$

Option 4:
$x^2 - 101x + 100 = 0$

Correct Answer:
$x^2 - 102x + 101 = 0$

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Solution:
Cube roots of unity

$$\omega = \frac{-1+i\sqrt{3}}{2}, \quad \omega^2 = \frac{-1-i\sqrt{3}}{2}$$

Sum of n-term of a GP

Let S_n be the sum of n terms of the G.P. with the first term 'a' and common ratio 'r'. Then

$$S_n = a(\frac{r^n - 1}{r - 1})$$

Now,

$\alpha = \omega$,

b = $1 + \omega^3 + \omega^6 + ...\omega^{300} = 101$

a= $(1+\omega)(1+\omega^2+\omega^4+\omega^6.....+\omega^{200})$

$$a = (1+\omega)\frac{(\omega^{2(101)} - 1)}{\omega^2 - 1} = \frac{1-\omega^2}{1-\omega^2} = 1$$

Now, equation with roots 1 and 101 is

$x^2 - (1+101)x + 101*1 = 0$

$x^2 - 102x + 101 = 0$

Correct Option (3)

Q. 11 The mirror image of the point (1,2,3) in a plane is $\left(\frac{-7}{3}, -\frac{4}{3}, -\frac{1}{3}\right)$. which of the following points lies on this plane ?

Option 1:
$(1, -1, 1)$

Option 2:
$(-1, -1, 1)$

Option 3:
$(1, 1, 1)$

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Option 4:

$(-1, -1, -1)$

Correct Answer:

$(1, -1, 1)$

Solution:

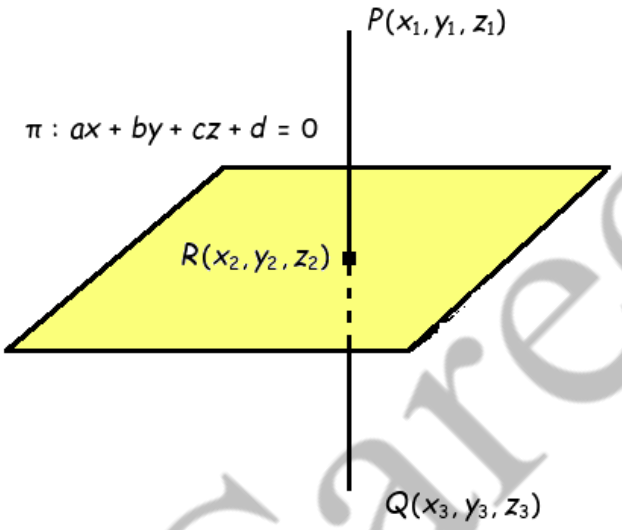
Image of a Point in the Plane -

The image of the point $P(x_1, y_1, z_1)$ in the plane $ax + by + cz + d = 0$ is $Q(x_3, y_3, z_3)$ then coordinates of point Q is given by

$$\frac{x_3 - x_1}{a} = \frac{y_3 - y_1}{b} = \frac{z_3 - z_1}{c} = -\frac{2(ax_1 + by_1 + cz_1 + d)}{a^2 + b^2 + c^2}$$

Proof:

Let point $Q(x_3, y_3, z_3)$ is the image of the point $P(x_1, y_1, z_1)$ in the plane $\pi : ax + by + cz + d = 0$.



Let line PQ meets the plane ax+by+cz+d=0 at point R, direction ratio of normal to plane π are (a, b, c), since, PQ perpendicular to plane π.

So direction ratio of PQ are a, b, c

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$\Rightarrow$ equation of line PQ is

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}=r \quad \text{(say)}$$

any point on line PQ may be taken as

$$(ar+x_1, br+y_1, cr+z_1)$$

Let $Q \equiv (ar+x_1, br+y_1, cr+z_1)$

since, R is the middle point of PQ

$$R \equiv \left(\frac{x_1+ar+x_1}{2}, \frac{y_1+br+y_1}{2}, \frac{z_1+cr+z_1}{2}\right)$$

$$\Rightarrow \quad R \equiv \left(x_1+\frac{ar}{2}, y_1+\frac{br}{2}, z_1+\frac{cr}{2}\right)$$

Since, point R lies in the plane π, we get

$$\Rightarrow \quad R \equiv \left(x_1+\frac{ar}{2}, y_1+\frac{br}{2}, z_1+\frac{cr}{2}\right)$$

$$\Rightarrow \quad a\left(x_1+\frac{ar}{2}\right)+b\left(y_1+\frac{br}{2}\right)+c\left(z_1+\frac{cr}{2}\right)+d=0$$

$$\Rightarrow \quad \left(a^2+b^2+c^2\right)\frac{r}{2}=-\left(ax_1+by_1+cz_1+d\right)$$

$$\Rightarrow \quad r=\frac{-2\left(ax_1+by_1+cz_1+d\right)}{a^2+b^2+c^2}$$

In the similar method we can also find the coordinates of point R.

-

P(1,2,3) and Q$\left(\frac{-7}{3}, \frac{-4}{3}, \frac{-1}{3}\right)$

Direction ratio of normal to the plane $\left(1-\frac{-7}{3}, 2-\frac{-4}{3}, 3-\frac{-1}{3}\right)$

we get $\frac{10}{3}, \frac{10}{3}, \frac{10}{3}$

and Midpoint of PQ is $\left(\frac{-2}{3}, \frac{1}{3}, \frac{4}{3}\right)$

equation of plane x + y + z = 1

Correct Option (1)

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Q. 12 The length of the perpendicular from the origin, on the normal to the curve, $x^2 + 2xy - 3y^2 = 0$ at the point (2,2) is ;

Option 1:
2

Option 2:
$2\sqrt{2}$

Option 3:
$4\sqrt{2}$

Option 4:
$\sqrt{2}$

Correct Answer:
$2\sqrt{2}$

Solution:

Line parallel and perpendicular to a given line -

Line parallel and perpendicular to a given line

The equation of the line parallel to ax + by + c = 0 is given as ax + by + λ = 0, where λ is some constant.

Equation of the given line is ax + by + c = 0

Its slope is (-a/b)

So, any equation of line parallel to ax + by + c = 0 is

$$y = \left(-\frac{a}{b}\right)x + c_1$$
$$ax + by - bc_1 = 0$$
$$ax + by + \lambda = 0$$

The equation of the line perpendicular to ax + by + c = 0 is given as bx - ay + λ = 0, where λ is some constant.

Equation of the given line is ax + by + c = 0

Its slope is (-a/b)

Slope of perpendicular line will be (b/a)

So, any equation of line perpendicular to ax + by + c = 0 is

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$$y = \left(\frac{b}{a}\right)x + c_1$$
$$bx - by - ac_1 = 0$$
$$bx - ay + \lambda = 0$$

-

Homogeneous Equations in Two Variables -

Homogeneous Equations in Two Variables

Homogeneous equations are those equations where each term has the same degree.

The equation $ax^2 + 2hxy + by^2 = 0$ is a homogeneous equation of second degree, it represents two straight lines through the origin.

The given equation is :
$$ax^2 + 2hxy + by^2 = 0 \qquad \ldots(i)$$
Divide the equation by x^2
$$\Rightarrow \qquad a + 2h\left(\frac{y}{x}\right) + b\left(\frac{y^2}{x^2}\right) = 0$$
now put $\frac{y}{x} = m$
$$\Rightarrow \qquad a + 2h(m) + b(m^2) = 0 \qquad \ldots(ii)$$
If m_1 and m_2 are roots, then
$$m_1 + m_2 = -\frac{2h}{b}$$
amd, $\qquad m_1 m_2 = \frac{b}{a}$
Thus, $y = m_1 x$ and $y = m_2 x$ are two straight lines given by Eq. (i)
Also, from the eq (iii)
$$m = \frac{-2h \pm \sqrt{(h^2 - ab)}}{2b} = \frac{y}{x}$$
$$\therefore \qquad by = \left\{-h + \sqrt{(h^2 - ab)}\right\} x$$
and $\qquad by = \left\{-h - \sqrt{(h^2 - ab)}\right\} x$
are two lines represented by Eq. (i)

-

Distance of a Point From a Line -

Distance of a point from a line

Perpendicular length from a point (x1,y1) to the line L : Ax + By + C = 0 is

$$\frac{|\mathbf{Ax_1} + +\mathbf{By_1} + \mathbf{C}|}{\sqrt{\mathbf{A^2} + \mathbf{B^2}}}$$

-

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$x^2 + 2xy - 3y^2 = 0$
$x^2 + 3xy - xy - 3y^2 = 0$
$(x - y)(x + 3y) = 0$
$x - y = 0 \quad x + 3y = 0$

(2, 2) satisfy x – y = 0

Normal

x + y = λ= 4 Hence

the perpendicular distance from the origin $= \left|\frac{0 + 0 - 4}{\sqrt{2}}\right| = 2\sqrt{2}$

Correct Option (2)

Q. 13 Which of the following statements is a tautology ?

Option 1:
$\sim (p \wedge \sim q) \rightarrow p \vee q$

Option 2:
$\sim (p \vee \sim q) \rightarrow p \wedge q$

Option 3:
$p \vee (\sim q) \rightarrow p \wedge q$

Option 4:
$\sim (p \vee \sim q) \rightarrow p \vee q$

Correct Answer:
$\sim (p \vee \sim q) \rightarrow p \vee q$

Solution:

Tautology And Contradiction -

Tautology

A compound statement is called **tautology** if it is always true for all possible truth values of its component statement.

For example, (p ⇒ q) ∨ (q ⇒ p)

Contradiction (fallacy)

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A compound statement is called a **contradiction** if it is always false for all possible truth values of its component statement.

For example, ~(p ⇒ q) ∨ (q ⇒ p)

Truth Table

p	q	$p \to q$	$q \to p$	$(p \to q) \vee (q \to p)$	$\sim ((p \to q) \vee (q \to p))$
T	T	T	T	T	F
T	F	F	T	T	F
F	T	T	F	T	F
F	F	T	T	T	F

Quantifiers

Quantifiers are phrases like 'These exist' and "for every". We come across many mathematical statements containing these phrases.

For example –

p : For every prime number x, √x is an irrational number.

q : There exists a triangle whose all sides are equal.

-

Practise Session - 1 -

Q1. Write the negations of the following:

1. Ram is a doctor or peon.
2. Room is clean and big.
3. $\sqrt{5}$ is a rational number.
4. If Ram is a doctor then he is smart

-

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$\sim (p \vee \sim q) \rightarrow p \vee q$

$(\sim p \wedge q) \rightarrow (p \vee q)$
$\sim \{(\sim p \wedge q) \wedge (\sim p \wedge \sim q)\}$
$\sim \{\sim p \wedge f\}$

Correct Option (4)

Q. 14 If $I = \int_1^2 \frac{dx}{\sqrt{2x^2 - 9x^2 + 12x + 4}}$, then :

Option 1:
$\frac{1}{6} < I^2 < \frac{1}{2}$

Option 2:
$\frac{1}{8} < I^2 < \frac{1}{4}$

Option 3:
$\frac{1}{9} < I^2 < \frac{1}{8}$

Option 4:
$\frac{1}{16} < I^2 < \frac{1}{9}$

Correct Answer:
$\frac{1}{9} < I^2 < \frac{1}{8}$

Solution:

Application of Inequality in Definite Integration -

Property 2

If m is the least value (global minimum) and M is the greatest value (global maximum) of the function f(x) on the interval [a, b], then

$$\mathbf{m(b-a) \leq \int_a^b f(x)\,dx \leq M(b-a)}$$

-

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$$f(x)=\frac{1}{\sqrt{2x^3-9x^2+12x+4}}$$

$$f'(x)=\frac{-1}{2}\frac{(6x^2-18x+12)}{(2x^3-9x^2+12x+4)^{\frac{3}{2}}}$$

$$=\frac{-6(x-1)(x-2)}{2\left(2x^3-9x^2+12x+4\right)^{\frac{3}{2}}}$$

$f(1)=\frac{1}{3}, \quad f(2)=\frac{1}{\sqrt{8}}$

$\frac{1}{3}<I<\frac{1}{\sqrt{8}}$

Correct Option (3)

Q. 15 If $A=\begin{pmatrix}2 & 2\\ 9 & 4\end{pmatrix}$ and $I=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$, then $10A^{-1}$ is equal to :

Option 1:

$6I-A$

Option 2:

$A-6I$

Option 3:

$4I-A$

Option 4:

$A-4I$

Correct Answer:

$A-6I$

Solution:

Inverse of a Matrix -

A non-singular square matrix “A” is said to be invertible if there exists a non-singular square matrix B such that AB = I = BA, (all matrix are of the same order, they must be for this), then B is called inverse of matrix A.

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$A^{-1} = B \Leftrightarrow AB = \mathbb{I}_n = BA$
we have,

$A(adjA) = A\mathbb{I}_n$
$\Rightarrow A^{-1}A(adjA) = A^{-1}\mathbb{I}_n|A|$
$\mathbb{I}_n(adjA) = A^{-1}|A|\mathbb{I}_n$
$A^{-1} = \frac{adjA}{|A|}$

Inverse of 2 x 2 matrix

Let A is a square matrix of order 2
$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$
Then,
$A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

-

Multiplication of two matrices -

Matrix multiplication:

Two matrices A and B are conformable for the product AB if the number of columns in A and the number of rows in B is equal. Otherwise, these two matrices will be non-conformable for matrix multiplication. So on that basis,

i) AB is defined only if col(A) = row(B)

ii) BA is defined only if col(B) = row(A)

If

$A = [a_{ij}]_{m \times n}$
$B = [b_{ij}]_{n \times p}$

$C = AB = [c_{ij}]_{m \times p}$
$\text{Where } c_{ij} = \sum_{j=1}^{n} a_{ij}b_{jk}, 1 \leq i \leq m, 1 \leq k \leq p$
$= a_{i1}b_{1k} + a_{i2}b_{2k} + a_{i3}b_{3k} + ... + a_{in}b_{nk}$

For examples

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Suppose, two matrices are given

$$\mathrm{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{33} \end{bmatrix}_{2\times 3} \quad \text{and} \quad \mathrm{B} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}_{3\times 3}$$

To obtain the entries in row i of AB, we multiply the entries in row i of A by column j in B and add.
given matrices A and B, where the order of A are 2×3 and the order of B are 3×3, the product of AB will be a 2×3 matrix.

To obtain the entry in row 1, column 1 of AB, multiply the first row in A by the first column in B, and add.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} = \mathrm{a}_{11} \cdot \mathrm{b}_{11} + \mathrm{a}_{12} \cdot \mathrm{b}_{21} + \mathrm{a}_{13} \cdot \mathrm{b}_{31}$$

To obtain the entry in row 1, column 2 of AB, multiply the first row in A by the second column in B, and add.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \end{bmatrix} \begin{bmatrix} b_{12} \\ b_{22} \\ b_{32} \end{bmatrix} = \mathrm{a}_{11} \cdot \mathrm{b}_{12} + \mathrm{a}_{12} \cdot \mathrm{b}_{22} + \mathrm{a}_{13} \cdot \mathrm{b}_{32}$$

To obtain the entry in row 1, column 3 of AB, multiply the first row in A by the thired column in B, and add.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \end{bmatrix} \begin{bmatrix} b_{13} \\ b_{23} \\ b_{33} \end{bmatrix} = \mathrm{a}_{11} \cdot \mathrm{b}_{13} + \mathrm{a}_{12} \cdot \mathrm{b}_{23} + \mathrm{a}_{13} \cdot \mathrm{b}_{33}$$

We proceed the same way to obtain the second row of AB. In other words,
row 2 of A times column 1 of B;
row 2 of A times column 2 of B;
row 2 of A times column 3 of B.

When complete, the product matrix will be

$$\mathrm{AB} = \begin{bmatrix} a_{11} \cdot b_{11} + a_{12} \cdot b_{21} + a_{13} \cdot b_{31} & a_{11} \cdot b_{12} + a_{12} \cdot b_{22} + a_{13} \cdot b_{32} & a_{11} \cdot b_{13} + a_{12} \cdot b_{23} + a_{13} \cdot b_{33} \\ a_{21} \cdot b_{11} + a_{22} \cdot b_{21} + a_{23} \cdot b_{31} & a_{21} \cdot b_{12} + a_{22} \cdot b_{22} + a_{23} \cdot b_{32} & a_{21} \cdot b_{13} + a_{22} \cdot b_{23} + a_{23} \cdot b_{33} \end{bmatrix}$$

-

$$A = \begin{bmatrix} 2 & 2 \\ 9 & 4 \end{bmatrix} \text{ and } A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$AA^{-1} = \begin{bmatrix} 2a+2c & 2b+2d \\ 9a+4c & 9b+4d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

On comparing we get

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$a = -\frac{2}{5}, \quad b = \frac{1}{5}, \quad c = \frac{9}{10} \ \& \ d = -\frac{1}{5}$

$\Rightarrow \quad 10A^{-1} = A - 61$

Correct Option (2)

Q. 16 The area (in sq.units) of the region $\left\{(x, y) \epsilon R^2 : x^2 \leq y \leq 3 - 2x\right\}$, is :

Option 1:
$\frac{31}{3}$

Option 2:
$\frac{32}{3}$

Option 3:
$\frac{29}{3}$

Option 4:
$\frac{34}{3}$

Correct Answer:
$\frac{32}{3}$

Solution:

Area Bounded by Curves When Intersects at More Than One Point -

Area bounded by the curves $y = f(x)$, $y = g(x)$ and intersect each other in the interval [a, b]

First find the point of intersection of these curves $y = f(x)$ and $y = g(x)$, let the point of intersection be x = c

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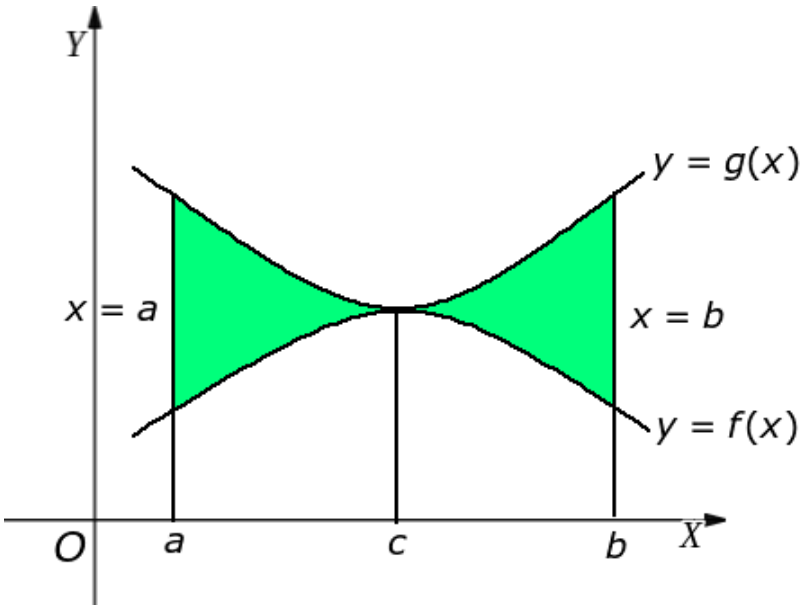


Area of the shaded region

$$= \int_a^c \{f(x) - g(x)\}dx + \int_c^b \{g(x) - f(x)\}dx$$

When two curves intersects more than one point

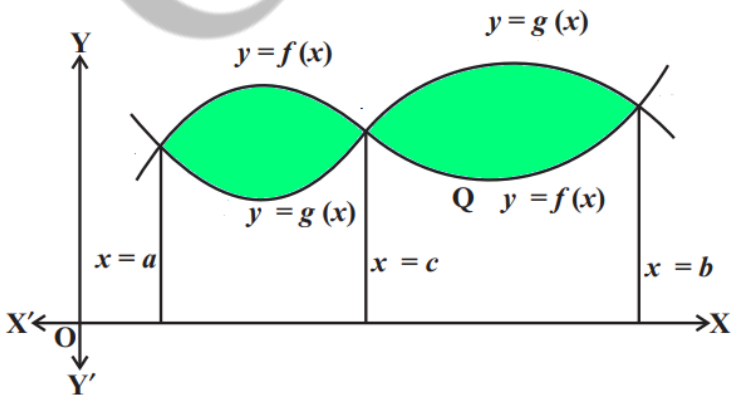
rea bounded by the curves *y=f(x), y=g(x)* and intersect each other at three points at *x = a, x = b amd x = c.*

To find the point of intersection, solve *f(x) = g(x).*

For x ∈ (a, c), f(x) > g(x) and for x ∈ (c, b), g(x) > f(x).

Area bounded by curves,

$$\mathrm{A} = \int_a^b |f(x) - g(x)|\,dx$$
$$= \int_a^c (f(x) - g(x))\,dx + \int_c^b (g(x) - f(x))\,dx$$



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Point of intersection of $y = x^2 \ \ \& \ \ y = -2x + 3$ is – 3, 1

Now, $\text{Area } 3 := \int_{-3}^{1} \left(-2x + 3 - x^2\right) dx$

$$\left[-x^2 + 3x - \frac{x^3}{3}\right]_{-3}^{1} = -2\left(\frac{1^2 - 3^2}{2}\right) + 3(1 - (-3)) - \left(\frac{1^3 + 3^3}{3}\right)$$

$$= 12 + 8 - \frac{28}{3} = \frac{32}{3}$$

Correct Option (2)

Q. 17 Let S be the set of all functions $f : [0, 1] \to R,$ which are continuous on [0,1] and differentiable on (0,1). Then for every f in S, there exists a $c\epsilon\,(0, 1)$, depending on f, such that :

Option 1:

$$\frac{f(1) - f(c)}{1 - c} = f'(c)$$

Option 2:

$$|f(c) - f(1)| < |f'(c)|$$

Option 3:

$$|f(c) + f(1)| < (1 + c)\,|f'(c)|$$

Option 4:

$$|f(c) - f(1)| < (1 - c)\,|f'(c)|$$

Correct Answer:

$$|f(c) - f(1)| < |f'(c)|$$

Solution:

Differentiability and Continuity -

Differentiability and Continuity

Theorem:

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A function f(x) is differentiable at every point of its domain, then it must be continuous in that domain. But the converse may or may not be true.

As we have studied in the previous concept that a function is differentiable at a point then it should be continuous at that point also and a discontinuous function is not differentiable. We will use this fact to prove the given theorem.

Proof:

Let a function f(x) is differentiable at x = a

$$\text{then,} \quad \lim_{x\to a} \frac{f(x)-f(a)}{x-a} \text{ exists finitely.}$$

$$\text{Let,} \quad f'(a) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a} \qquad \text{(i)}$$

$$\text{In order to prove that } f(x) \text{ is continuous at } x = a \text{ it is enough to}$$

$$\text{show that } \lim_{x\to a} f(x) = f(a)$$

$$\text{or wde have to show that } \lim_{x\to a} [f(x)-f(a)] = 0$$

$$\begin{aligned}\text{Now,} \quad \lim_{x\to a} [f(x)-f(a)] &= \lim_{x\to a} \frac{f(x)-f(a)}{x-a}(x-a) \\ &= \lim_{x\to a} \frac{f(x)-f(a)}{x-a} \lim_{x\to a}(x-a) \\ &= f'(a)\times 0 \\ &= 0\end{aligned}$$

Therefore, f(x) is continuous at a.

-

Let f(x)=mx+k m>0 for increasing function

f(0)=k and f(1)=m+k

$$c \in (0,1)$$

let c=0.5

a)

$$\begin{aligned}&|f(c)-f(1)| < (1-c)\,|f'(c)| \\ &|f(0.5)-f(1)| < (1-0.5)\,|f'(0.5)| \\ &|0.5m+k-m-k| < 0.5\times 1 \\ &0.5m < 0.5 \\ &m < 1\end{aligned}$$

but m can be any real positive number.

b)

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$|f(c) - f(1)| < |f'(c)|$
$|0.5m + k - m - k| < |m|$
$|-0.5m| < |m|$
$1 < 2$

Always true

c)

$$\frac{f(1) - f(c)}{1 - c} = f'(c)$$
$$\frac{m + k - 0.5m - k}{1 - 0.5} = 0.5m$$
$$m = 0.5m$$

Not Possible

d)

$$|f(c) + f(1)| < (1 + c)\,|f'(c)|$$
$|0.5m + k + m + k| < 1.5 \times 1$
$|1.5m + k| < 1.5$

Here nothing can be concluded

Correct Option (2)

Q. 18 The differential equation of the family of curves, $x^2 = 4b(y + b), b \in R, is:$

Option 1:
$xy'' = y'$

Option 2:
$x(y')^2 = x + 2yy'$

Option 3:
$x(y')^2 = x - 2yy'$

Option 4:
$x(y')^2 = 2yy' - x$

Correct Answer:
$x(y')^2 = x + 2yy'$

Solution:

Formation of Differential Equation and Solutions of a Differential Equation -

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Differential equations are used to represent a family of curves. Suppose that a family of curves is represented using a set of n arbitrary constants, then the differential equation corresponding to that family is a relation between x, y and derivatives of y upto order n, not containing any of the n arbitrary constants.

Given a relation between the variables x, y and n arbitrary constants C_1, C_2, ... C_n, if we differentiate n times in succession with respect to x, we have altogether n + 1 equations between which the n arbitrary constants can be eliminated. The result is a differential equation of the nth order.

Consider the equation of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad \ldots(4)$$

differentiate Eq (4) w.r.t. 'x'

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \left(\frac{dy}{dx}\right) = 0 \quad \text{or} \quad \frac{x}{a^2} = -\frac{yy'}{b^2} \qquad \ldots(5)$$

differentiate above equation w.r.t. 'x'

$$\frac{1}{a^2} = -\frac{yy'' + (y')^2}{b^2} = 0 \qquad \ldots(6)$$

eliminating the constant 'a' and 'b' from (5) and (6), we get

$$yy' = xyy'' + x(y')^2$$

This is the differential equation of the family of the equation of ellipse (4).

From the above examples, it is clear that the order of the differential equation of the family of curve is equal to the number of arbitrary constant that presents in family of curve.

NOTE:

If arbitrary constants appear in addition, subtraction, multiplication or division, then we can club them to reduce into one new arbitrary constant. Hence, the differential equation corresponding to a family of curves will have order exactly same as number of essential arbitrary constants in the equation of the curve.

-

$$x^2 = 4b(y + b)$$
$$2x = 4by'$$
$$\Rightarrow \quad b = \frac{x}{2y'}$$

So the Differential equation is

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$$x^2 = \frac{2x}{y'} \cdot y + \left(\frac{x}{y'}\right)^2$$

$$\Rightarrow x\left(\frac{dy}{dx}\right)^2 = 2y\frac{dy}{dx} + x$$

Correct Option (2)

Q. 19 The system of linear equations

$\lambda x + 2y + 2z = 5$

$2\lambda x + 3y + 5z = 8$

$4x + \lambda y + 6z = 10$ has :

Option 1:
no solution when $\lambda = 2$

Option 2:
infinitely many solutions when $\lambda = 2$

Option 3:
no solution when $\lambda = 8$

Option 4:
a unique solution when $\lambda = -8$

Correct Answer:
no solution when $\lambda = 2$

Solution:

Cramer's law -

Cramer's law for the system of equations in two variables :

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Let $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$, where

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

on solving this equation by cross multiplication, we get

$$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{a_1c_2 - a_2c_1} = \frac{1}{a_1b_2 - a_2b_1}$$

$$\text{or } \frac{x}{\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}} = \frac{y}{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}} = \frac{1}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

$$\text{or } x = \frac{\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}, y = \frac{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

We can observe that first row in the numerator of x is of constants and 2nd row in the numerator is of constants, and the denominator is of the coefficient of variables.

We can follow this analogy for the system of equations of 3 variable where third in the numerator of the value of z will be of constant and denominator will be formed by the value of coefficients of the variables.

so let us consider the system of equations

$a_1x + b_1y + c_1z = d_1$ ...(i)

$a_2x + b_2y + c_2z = d_2$ ...(ii)

$a_3x + b_3y + c_3z = d_3$ ...(iii)

then Δ, which will be determinant of coefficient of variables, will be

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, \Delta_1 \text{ numerator of x is :}$$

$$\Delta_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}, \text{similarly } \Delta_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix} \text{ and}$$

$$\Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}, \text{if} \Delta \neq 0 \text{ then } x = \frac{\Delta_1}{\Delta}$$

$$y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}$$

i) If ? ≠ 0, then the system of equations has a unique finite solution and so equations are consistent, and solutions are $x = \frac{\Delta_1}{\Delta}, y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}$

ii) If ? = 0, and any of $\Delta_1 \neq 0 \; or \; \Delta_2 \neq 0 \; or \; \Delta_3 \neq 0$

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Then the system of equations is inconsistent and hence no solution exists.

iii) If all $\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$ then

System of equations is consistent and dependent and it has an infinite number of solution.

-

$$D = \begin{vmatrix} \lambda & 2 & 2 \\ 2\lambda & 3 & 5 \\ 4 & \lambda & 6 \end{vmatrix}$$

D = (λ + 8) (2 – λ) for λ = 2

$$D_1 = \begin{vmatrix} 5 & 2 & 2 \\ 8 & 3 & 5 \\ 10 & 2 & 6 \end{vmatrix}$$

= 5[18 – 10] – 2 [48 – 50] + 2 (16 – 30]

= 40 + 4 – 28 $\neq$ 0 No solutions for λ= 2

Correct Option (1)

Q. 20 If the 10th term of an A.P. is $\frac{1}{20}$ and its 20th term is $\frac{1}{10}$, then the sum of its first 200 term is :

Option 1:
$50\frac{1}{4}$

Option 2:
100

Option 3:
50

Option 4:
$100\frac{1}{2}$

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Correct Answer:

$100\frac{1}{2}$

Solution:
General Term of an AP

n^{th} term (general term) of the A.P. is $a_n = a + (n - 1)d$.

Sum of n terms of an AP

The sum, S_n of n terms of an AP with the first term 'a' and common difference 'd' is given by

$S_n = \frac{n}{2}[2a + (n-1)d]$
OR
$S_n = \frac{n}{2}[a + l]$
$a \rightarrow$ first term
$d \rightarrow$ common difference
$n \rightarrow$ number of terms

Now,

$a_{10} = a + 9d = 1/20$
$a_{20} = a + 19d = 1/10$
$On\ subtracting\ them\ ,we\ get\ d = 1/200$
$a + 9/10 = 1/20 \Rightarrow a = 1/200$
$S_{200} = \frac{200}{2}\left[2/200 + (199) \times 1/200\right] = \frac{201}{2}$

Correct Option (4)

Q. 21 Let a line $y = mx(m > 0)$ intersect the parabola, $y^2 = x$ at a point P, oyher than the origin. Let the tangent to it at P meet the x-axis at the point Q. If area $(\Delta OPQ) = 4\ sq.units$, then m is equal to ____.

Option 1:
2

Option 2:
1

Option 3:
0.5

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Option 4:

4

Correct Answer:

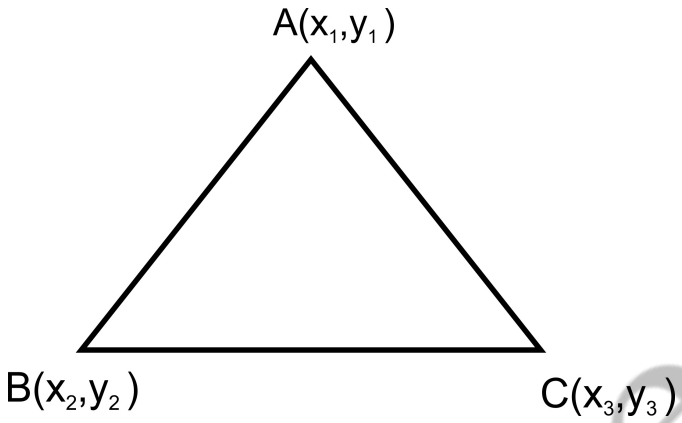
0.5

Solution:

Area of Triangle -

Area of Triangle

If vertices of a triangle ABC given as A (x_1, y_1), B (x_2, y_2) and C(x_3, y_3) then area of ΔABC is



$$\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Area of an n-sided polygon

The area of a polygon whose vertices are (x_1, y_1), (x_2, y_2), (x_3, y_3).....(x_n, y_n) taken in cyclic order is given by

$$= \frac{1}{2}\left|\{(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + \ldots\ldots + (x_ny_1 - x_1y_n)\}\right|$$

-

Tangents of Parabola in Parametric Form -

Tangents of Parabola in Parametric Form

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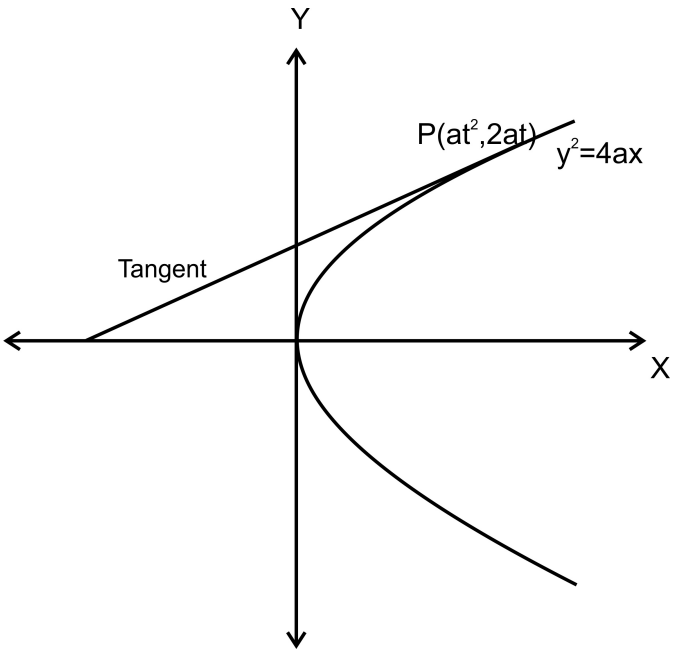
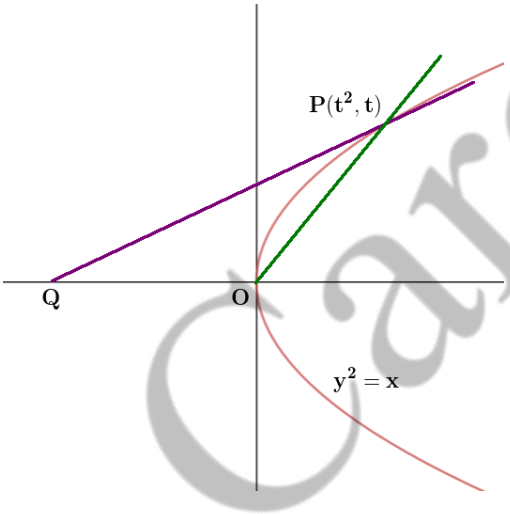


The equation of tangent to the parabola $y^2 = 4ax$ at the point $(at^2, 2at)$ is
$ty = x + at^2$

Equation of the tangent to the parabola $y^2 = 4ax$ at the point $P(x_1, y_1)$ is
$yy_1 = 2a(x + x_1)$
replace $x_1 \to at^2, y_1 \to 2at$
$y(2at) = 2a(x + at^2) \Rightarrow yt = x + at^2$

-



$2ty = x + t^2$
$Q(-t^2, 0)$

$$\frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ t^2 & t & 1 \\ -t^2 & 0 & 1 \end{vmatrix} = 4$$

$t = \pm 2$
$m = 1/2 = 0.5$

Correct Option (3)

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Q. 22 Let f(x) be a polynomial of degree 3 such that f(-1)=10, f(1)=-6,f(x) has a critical point at x=-1 and $f'(x)$ has a critical point at x=1, Then f(x) has a local minima at x= _____.

Correct Answer:
3

Solution:

Non-Monotonic Function and Critical Point -

Critical Point:

The critical point of a function is when its derivative does not exist or its derivative is equal to zero.

All the values of 'x' obtained by bellow conditions are said to be the critical points.

1. f(x) does not exists
2. f'(x) does not exists
3. f'(x) = 0.

Critical points are interior points of the intervals.

For the function f(x) = | x^2 - 4|, critical points are x = +2, -2 and x = 0 where its derivative is zero.

-

Let f (x) = $ax^3 + bx^2 + cx + d$

a = 1/4, d = 35/4, b = –3/4, c =9/4

f(x) = $a(x^3 – 3x^2 – 9x) + d$

$$f'(x) = \frac{3}{4}(x^2 - 2x - 3)$$

f ' (x) = 0 $\Rightarrow$ x = 3, –1

local minima exist at x = 3

OR

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$$f''(x) = \lambda(x-1)$$

$$\Rightarrow f'(x) = \lambda\frac{(x-1)^2}{2} - 2\lambda$$

$$\therefore f(x) = \frac{\lambda(x-1)^3}{6} - 2\lambda x + \mu$$

$$f(1) = -6 \Rightarrow -2\lambda + \mu = -6$$

$$f(-1) = 10 \Rightarrow -2\lambda + \mu = -6$$

$$\therefore \lambda = 6 \text{ and } \mu = 6$$

$$f'(x) = 0 \Rightarrow x = -1 \text{ or } x = 3$$

$$\therefore x = 3 \text{ has a local minima}$$

Correct Option (1)

Q. 23 If $\frac{\sqrt{2}\sin\alpha}{\sqrt{1+\cos 2\alpha}} = \frac{1}{7}$ and $\sqrt{\frac{1-\cos 2\beta}{2}} = \frac{1}{\sqrt{10}}, \alpha, \beta \in \left(0, \frac{\pi}{2}\right)$, then $\tan(\alpha + 2\beta)$ is equal to _____.

Option 1:

0

Option 2:

1

Option 3:

0.5

Option 4:

2

Correct Answer:

1

Solution:

Trigonometric Identities -

Trigonometric Identities-

These identities are the equations that hold true regardless of the angle being chosen.

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$$\sin^2 t + \cos^2 t = 1$$
$$1 + \tan^2 t = \sec^2 t$$
$$1 + \cot^2 t = \csc^2 t$$
$$\tan t = \frac{\sin t}{\cos t}, \quad \cot t = \frac{\cos t}{\sin t}$$

-

Double Angle Formula and Reduction Formula -

Double Angle Formula and Reduction Formula

$$\begin{aligned}
\sin(2\theta) &= 2\sin\theta\cos\theta \\
&= \frac{2\tan\theta}{1+\tan^2\theta} \\
\cos(2\theta) &= \cos^2\theta - \sin^2\theta \\
&= 1 - 2\sin^2\theta \\
&= 2\cos^2\theta - 1 \\
&= \frac{1-\tan^2\theta}{1+\tan^2\theta} \\
\tan(2\theta) &= \frac{2\tan\theta}{1-\tan^2\theta}
\end{aligned}$$

$$\tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$$
$$\tan(\theta+\theta) = \frac{\tan\theta + \tan\theta}{1 - \tan\theta\tan\theta}$$
$$\tan(2\theta) = \frac{2\tan\theta}{1-\tan^2\theta}$$

-

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$$\frac{\sqrt{2}\sin\alpha}{\sqrt{2}\cos\alpha} = \frac{1}{7}$$

$$\tan\alpha = \frac{1}{7}$$

$$\sin\beta = \frac{1}{\sqrt{10}}$$

$$\tan\beta = \frac{1}{\sqrt{3}}$$

$$\tan 2\beta = \frac{2\cdot\frac{1}{3}}{1-\frac{1}{9}} = \frac{\frac{2}{3}}{\frac{8}{9}} = \frac{3}{4}$$

$$\tan(\alpha+2\beta) = \frac{\tan\alpha+\tan 2\beta}{1-\tan\alpha\tan 2\beta} = 1$$

Correct Option (2)

Q. 24 The number of 4 letter words (with or without meaning) that can be formed from the eleven letters of the word 'EXAMINATION' is _____,

Option 1:

$8!$

Option 2:

$\frac{8!}{2!2!}$

Option 3:

3600

Option 4:

2454

Correct Answer:

2454

Solution:

EXAMINATION 2N, 2A, 2I, E, X, M, T, O

Case I All are different: so ${}^{8}C_{4}.4! = 1680$

Case II 2 same and 2 different: so ${}^{3}\mathrm{C}_{1}\cdot{}^{7}\mathrm{C}_{2}\cdot\frac{4!}{2!} = 756$

Case III 2 same of first kind and 2 same of second kind: so ${}^{3}\mathrm{C}_{2}\cdot\frac{4!}{2!\cdot 2!}$ = 18

Total = 2454

Correct Option (4)

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Q. 25 The sum, $\sum_{n=1}^{7} \frac{n(n+1)(2n+1)}{4}$ is equal to _____.

Option 1:
1024

Option 2:
810

Option 3:
504

Option 4:
1008

Correct Answer:
504

Solution:
Sum of Common Series

Sum of the square of first n-natural numbers

$$1^2 + 2^2 + 3^2 + 4^2 + \ldots\ldots\ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$$ Sum of the cube of first n-natural numbers

$$1^3 + 2^3 + 3^3 + 4^3 + \ldots\ldots\ldots + n^3 = \left\{\frac{n(n+1)}{2}\right\}^2$$

-

Now,

$$\frac{1}{4}\left[\sum_{n=1}^{7}\left(2n^3 + 3n^2 + n\right)\right]$$

$$\frac{1}{4}\left[2\left(\frac{7.8}{2}\right)^2 + 3\left(\frac{7.8.15}{6}\right) + \frac{7.8}{2}\right]$$

$$\frac{1}{4}[2 \times 49 \times 16 + 28 \times 15 + 28]$$

$$\frac{1}{4}[1568 + 420 + 28] = 504$$

Correct Option (3)

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8th January Shift-1

Physics

Q. 1 Effective capacitance of parallel combination of two capacitors C_1 and C_2 is $10\mu F$. When these capacitors are individually connected to a voltage source of $1V$, the energy stored in the capacitor C_2 is 4 times that of C_1. If these capacitors are connected in series, their effective capacitance will be ($in\ \mu F$):

Correct Answer:
1.6

Solution:
When capacitors are in parallel,

$$C_{eq} = C_1 + C_2$$

So, $C_1 + C_2 = 10\mu F$

Now when they are individually connected to 1V battery, then

The energy of C_2=4 x energy of C_1

$$\Rightarrow \frac{1}{2}C_2V^2 = 4 \times \frac{1}{2}C_1V^2$$

$$\Rightarrow C_2 = 4C_1$$

So, C_2=8 and C_1=2

So when both are in series, then $C_{eq} = \frac{C_1C_2}{C_1 + C_2} = \frac{8}{5} = 1.6\ \mu F$

So the correct option is 1.

Q. 2 A proton with the kinetic energy of $1MeV$ moves from south to north. It gets an acceleration of $10^{12}m/s^2$ by an applied magnetic field (west to east). The value of magnetic field (in mT):

(Rest mass of a proton is $1.6 \times 10^{-27}kg$)

Correct Answer:
0.71

Solution:

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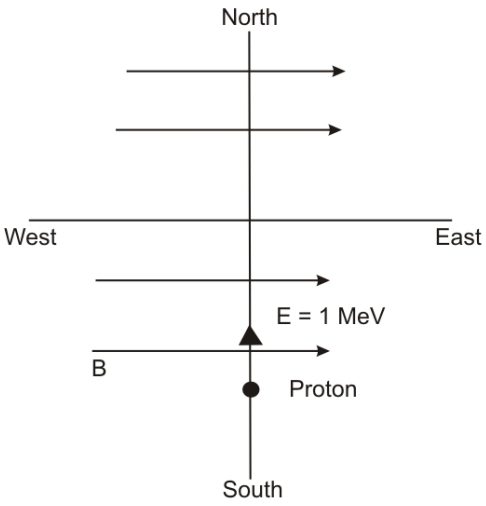


$\because$ K.E. $= 1.6 \times 10^{-13} = \frac{1}{2} \times 1.6 \times 10^{-27} \mathrm{V}^2$

$\mathrm{V} = \sqrt{2} \times 10^7$

$\therefore \mathrm{Bqv} = \mathrm{ma}$

$\mathrm{B} = \frac{1.6 \times 10^{-27} \times 10^{12}}{1.6 \times 10^{-19} \times \sqrt{2} \times 10^7}$

$= 0.71 \times 10^{-3} \mathrm{T}$

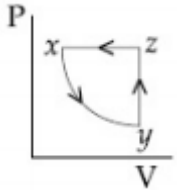
so $0.71 \mathrm{mT}$

So option (1) is correct

Q. 3 A thermodynamic cycle $xyzx$ is shown on a V - T diagram.

The P - V diagram that best describe this cycle is : (Diagrams are schematic and not to scale)

Option 1:



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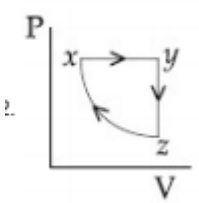
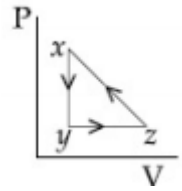
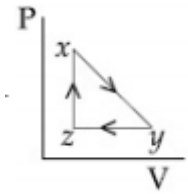
Option 2:



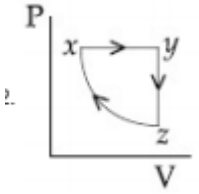
Option 3:



Option 4:



Correct Answer:



Solution:
Since xy is passing through the origin.

So,

$$V \propto T$$

$$\frac{V}{T} = constant$$

So for an ideal gas: **P= constant**

Now Y-Z is a constant volume graph (from the question). Since Z-X is a constant temperature graph it is hyperbolic in nature. So option 1 is correct

So the correct option is 2.

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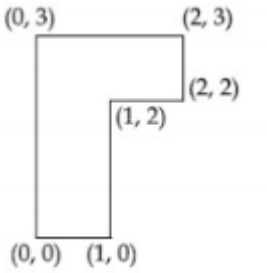
Q. 4 The coordinates of centre of mass of a uniform flag shaped lamina (thin flat plate) of mass $4kg$. (The coordinates of the same are shown in the figure) are :



Option 1:
$(1.25m, 1.50.m)$

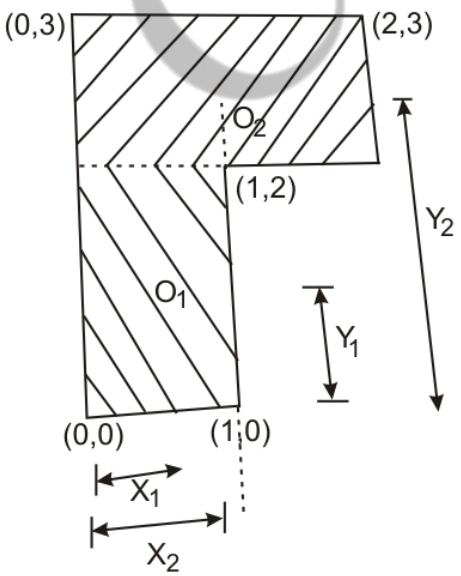
Option 2:
$(0.75m, 0.75m)$

Option 3:
$(0.75m, 1.75m)$

Option 4:
$(1m, 1.75m)$

Correct Answer:
$(0.75m, 1.75m)$

Solution:



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The co-ordinate of O_1 is (0.5,1), O_2 is (1, 2.5)

$$x_{cm} = \frac{m \times 0.5 + m \times 1}{2m} = 0.75$$

$$y_{cm} = \frac{m \times +m \times \frac{5}{2}}{2m} = 1.75$$

So the correct option is 3.

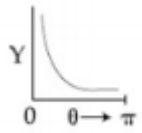
Q. 5 The graph which depicts the results of Rutherford gold foil experiment with α—particles is:

θ: Scattering angle

Y : Number of scattered α— particles detected

(Plots are schematic and not to scale)

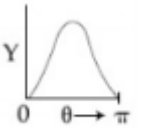
Option 1:



Option 2:

Option 3:

Option 4:



Correct Answer:

Solution:

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As $Y = N(\theta) \propto \frac{1}{\sin^4\left(\frac{\theta}{2}\right)}$

Here when $\theta = 0$ Sin will be zero then Y will tend to infinity.

and that is satisfied by the only option 1

So the correct option is 1.

Q. 6 Consider a solid sphere of Radius R and mass density $\rho(r) = \rho_0\left(1 - \frac{r^2}{R^2}\right), \ 0 < r \leq R.$ The minimum density of a liquid in which it will float is :

Option 1:
$\frac{2\rho_0}{5}$

Option 2:
$\frac{\rho_0}{5}$

Option 3:
$\frac{2\rho_0}{3}$

Option 4:
$\frac{\rho_0}{3}$

Correct Answer:
$\frac{2\rho_0}{5}$

Solution:

As

Buoyant force-

- The buoyant force is given as

$F_B = \rho V g$

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Where F_B=Buoyant force

ρ= density of the fluid

V= Volume of the solid body immersed in the liquid or Volume of the fluid displaced

So

Applying force balance

$$\rho_l \times \frac{4}{3}\pi R^3 = \rho_o(4\pi)\int r^2\left(1-\frac{r^2}{R^2}\right)dr$$

$$\Rightarrow \rho_l \times \frac{4}{3}\pi R^3 = \rho_o(4\pi)\left[\frac{r^3}{3}-\frac{r^5}{5R^2}\right]_0^R$$

$$\Rightarrow \rho_l \times \frac{4}{3}\pi R^3 = \rho_o(4\pi)\left[\frac{R^3}{3}-\frac{R^3}{5}\right]$$

$$\Rightarrow \rho_l \times \frac{4}{3}\pi R^3 = \rho_o(4\pi)\left[\frac{2R^3}{15}\right]$$

$$\Rightarrow \rho_l = \frac{2}{5}\rho_o$$

So the correct option is 1.

Q. 7 In finding the electric field using Gauss law the formula $\left|\vec{E}\right| = \frac{q_{enc}}{\epsilon_0 |A|}$ is applicable. In the formula ϵ_0 is permittivity of free space, A is the area of Gaussian surface and q_{enc} is charge enclosed by the Gaussian surface. This equation can be used in which of the following situation?

Option 1:
For any choise of Gaussian surface.

Option 2:
Only when the Gaussian is an equipotential surface.

Option 3:
Only when Gaussian surface is an equipotential surface and $\left|\vec{E}\right|$ is constant on the surface.

Option 4:
Only when $\left|\vec{E}\right|$ = constant on the surface.

Correct Answer:
Only when Gaussian surface is an equipotential surface and $\left|\vec{E}\right|$ is constant on the surface.

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Solution:
As Gauss's law is one of the fundamental laws of physics which states that the net flux of an electric field in a closed surface is directly proportional to the enclosed electric charge.

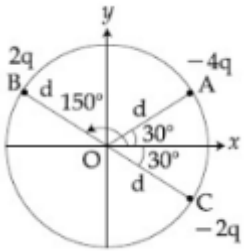
And

The relation of gauss's law is valid when

The magnitude of the electric field is constant & the surface is equipotential

So the correct option is 3.

Q. 8 Three charged particles A, B and C with charges $-4q, 2q \ and \ -2q$ are present on the circumference of a circle of radius d. The charged particles A, C and the centre O of the circle formed an equilateral triangle as shown in figure. Electric field at O along x-direction is:



Option 1:
$\frac{\sqrt{3}q}{4\pi\epsilon_0 d^2}$

Option 2:
$\frac{3\sqrt{3}q}{4\pi\epsilon_0 d^2}$

Option 3:
$\frac{\sqrt{3}q}{\pi\epsilon_0 d^2}$

Option 4:
$\frac{2\sqrt{3}q}{\pi\epsilon_0 d^2}$

Correct Answer:
$\frac{\sqrt{3}q}{\pi\epsilon_0 d^2}$

Solution:

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Let E_1 be the resultant electric field due to charge 2q and - 2q. The resultant E_1 is in the direction of -2q

$$E_1 = \frac{k2q}{d^2} + \frac{k2q}{d^2} = \frac{k4q}{d^2}$$

E_2 due to -4q is in the direction of -4q and magnitude is given by

$$E_2 = \frac{k4q}{d^2}$$

The net electric field in the x-direction is

$$E_x = E_1 cos30 + E_2 cos30 = \frac{\sqrt{3}}{2} \times 2 \times \frac{4q}{4\pi\epsilon_0 d^2}$$
$$E_x = \sqrt{3}\frac{q}{\pi\epsilon_0 d^2}$$

So the correct option is 3.

Q. 9 The length of a potentiometer wire is $1200cm$ and it carries a current of $60mA$. For a cell of emf $5V$ and internal resistance of 20Ω, the null point on it is found to be at $1000cm$. The resistance of whole wire is $(in\ \Omega)$:

Correct Answer:
100

Solution:

In the potentiometer, a battery of known emf E is connected in the secondary circuit. A constant current I is flowing through AB from the driver circuit. The jockey is a slide on potentiometer wire AB to obtain null deflection in the galvanometer. Let l be the length at which galvanometer shows null deflection.

Since the potential of wire AB (V) is proportional to the length AB(L).

Similarly $E \ \alpha \ l$

So we get

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$$\frac{V}{E}=\frac{L}{l}$$

$$V=E\frac{L}{l}$$

$$V=5\times\frac{1200}{1000}=6V$$

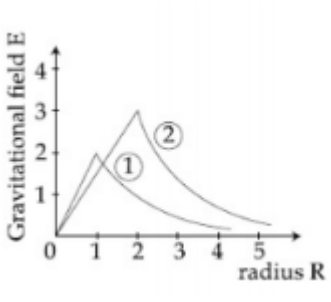
given current through potentiometer wire =60mA

V=iR

$$60\times10^{-3}R=6$$
$$\Rightarrow R=100\Omega$$

So the correct option is 4.

Q. 10 Consider two solid sphers of radii $R_1=1m,\ R_2=2m$ and masses M_1 and M_2, respectively. The gravitational field due to sphere ① and ② are shown. The value of $\frac{M_1}{M_2}$ is:



Option 1:
$\frac{2}{3}$

Option 2:
$\frac{1}{6}$

Option 3:
$\frac{1}{2}$

Option 4:
$\frac{1}{3}$

Correct Answer:
$\frac{1}{6}$

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Solution:
Considering Case 2 and case 1, one by one

and using $I = \frac{GM}{R^2}$ $\quad$ (Gravitational field on surface of sphere)

R-Radius of sphere

$$3 = \frac{\mathrm{Gm_2}}{2^2}$$

$$2 = \frac{\mathrm{Gm_1}}{1^2}$$

$$\therefore \frac{3}{2} = \frac{1}{4}\frac{\mathrm{m_2}}{\mathrm{m_1}}$$

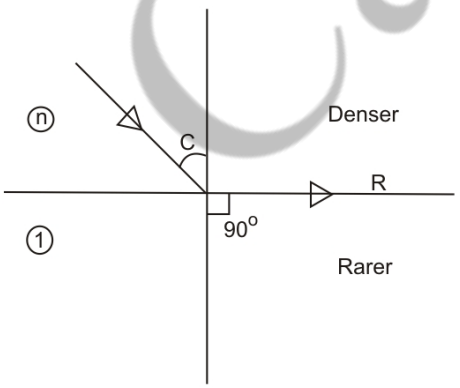
$$\frac{\mathrm{m_1}}{\mathrm{m_2}} = \frac{1}{6}$$

So the correct option is 2.

Q. 11 The critical angle (in degree)of a medium for a specific wavelength, if the medium has relative permitivity 3 and relative permeability $\frac{4}{3}$ for this wavelength, will be :

Correct Answer: 30

Solution:



$$C_{vacuum} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \; and \; C_{medium} = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}}$$

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and $n = \frac{C_{vacuum}}{C_{medium}} = \sqrt{\mu_r \epsilon_r} = \sqrt{\frac{4}{3} * 3} = 2$

By using snells law

$$n * sinC = 1 * sin(90^0)$$
$$\Rightarrow n * sinC = 1$$
$$\Rightarrow sinC = \frac{1}{2} \Rightarrow critical\ \ angle = C = 30^0$$

So the correct option is 2.

Q. 12 The dimension of stopping potential v_0 in photoelectric effect in units of Planck's constant 'h', speed of light 'c' and Gravitational constant 'G' and ampere A is :

Option 1:
$h^{-2/3} c^{-1/3} G^{4/3} A^{-1}$

Option 2:
$[h]^0 [G]^{-1} [c]^5 [A]^{-1}$

Option 3:
$h^2 G^{3/2} c^{1/3} A^{-1}$

Option 4:
$h^{1/3} G^{2/3} c^{1/3} A^{-1}$

Correct Answer:
$[h]^0 [G]^{-1} [c]^5 [A]^{-1}$

Solution:

Let $V_0 = [h]^p [G]^q [c]^r [A]^s$

Now, $K_{max} = eV_0 \Rightarrow [K_{max}] = [eV_0] \Rightarrow [V_0] = \frac{[K_{max}]}{[e]} \Rightarrow [V_0] = \frac{[ML^2T^{-2}]}{[AT]}$

$\therefore [V_0] = [ML^2T^{-3}A^{-1}]$

$E = hf \Rightarrow [h] = \frac{[E]}{[f]} \Rightarrow \frac{[ML^2T^{-2}]}{[T^{-1}]} \Rightarrow [h] = [ML^2T^{-1}]$

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$$F = \frac{Gm^2}{r^2} \Rightarrow [G] = \frac{[F][r^2]}{[m^2]} = \frac{[MLT^{-2}][L^2]}{[M^2]} = [M^{-1}L^3T^{-2}]$$

So, $[ML^2T^{-3}A^{-1}] = [ML^2T^{-1}]^p[M^{-1}L^3T^{-2}]^q[LT^{-1}]^r[A]^s$

From here we will get: p-q=1---------(1)

2p+3q+r=2----------(2)

-p-2q-r=-3----------(3)

s=-1----------(4)

From equation 1, 2,3 and 4 we will get: p=0,q=-1, r=5 and s=-1

So, $[V_0] = [h]^0[G]^{-1}[c]^5[A]^{-1}$

Q. 13 A leak proof cylinder of length $1m$, made of a metal which has very low coefficient of expansion is floating vertically in water at $0^\circ C$ such that its height above the water surface is $20cm$. When the temperature of water is increased to $4^\circ C$, the height of the cylinder above the water surface becomes $21cm$. The density of water at $T = 4^\circ C,$ relative to the density at $T = 0^\circ C,$ is close to :

Correct Answer: 1.01

Solution:

As

Buoyant force-

- The buoyant force is given as

$$F_B = \rho V g$$

Where F_B=Buoyant force

ρ= density of the fluid

V= Volume of the solid body immersed in the liquid or Volume of the fluid displaced

So

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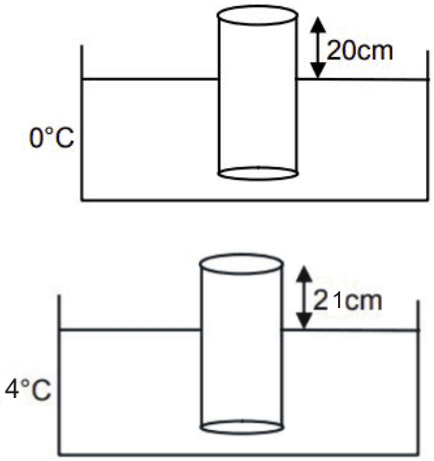


$$mg = V\rho g$$

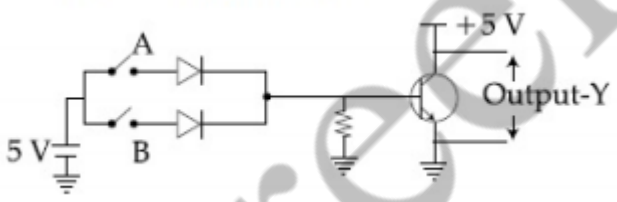
$$\mathrm{mg} = \mathrm{A}(80)\mathrm{p}_{0^\circ\mathrm{C}}\mathrm{g}$$

$$\mathrm{mg} = \mathrm{A}(79)\rho_{4^\circ\mathrm{C}}\mathrm{g}$$

$$\frac{\rho_{4^\circ\mathrm{C}}}{\rho_{0^\circ\mathrm{C}}} = \frac{80}{79} = 1.01$$

So the correct option is 2.

Q. 14 Boolean relation at the output stage - Y for the following circuit is :



Option 1:

$\bar{A} + \bar{B}$

Option 2:

$a + b$

Option 3:

$A \cdot B$

Option 4:

$\bar{A} \cdot \bar{B}$

Correct Answer:

$\bar{A} \cdot \bar{B}$

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Solution:

The first part of the given circuit represents or gate whose output is A+B

The second part of the circuit is NOT gate whose input is A+B and output is $\overline{A+B} = \bar{A}.\bar{B}$

So the correct option is 4.

Q. 15 When photon of energy $4.0eV$ strikes the surface of a metal A, the ejected photoelectrons have maximum kinetic energy $T_A\ eV$ and de-Broglie wavelength λ_A. The maximum kinetic energy of photoelectrons liberated from another metal B by photon of energy $4.50eV$ is $T_B = (T_A - 1.5)eV$. If the de-Broglie wavelength of these photoelectrons $\lambda_B = 2\lambda_A$, then the work function(in eV) of metal B is:

Correct Answer:
4

Solution:

As we know $\lambda = \dfrac{h}{\sqrt{2mT}}$

where T=kinetic energy

So $\dfrac{\lambda_A}{\lambda_B} = \sqrt{\dfrac{T_A}{T_B}} = \dfrac{1}{2} ...(1)$

Also $Given\ \ T_B = T_A - 1.5 ...(2)$

Using equation (1) and (2)

$T_A = 2\ eV,\ and\ T_B = 1.5\ eV$

And using the Einstein equation

$$E_B = T_B + W_B$$
$$\Rightarrow W_B = E_B - T_B = (4.5 - 0.5)eV = 4\ eV$$

So the correct option is 1.

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Q. 16 A particle of mass m is fixed to one end of a light spring having force constant k and unstretched length l. The other end is fixed. The system is given an angular speed ω about the fixed end of the spring such that it rotates in a circle in gravity free space. Then the stretch in the spring is :

Option 1:

$$\frac{ml\omega^2}{k-\omega m}$$

Option 2:

$$\frac{ml\omega^2}{k-m\omega^2}$$

Option 3:

$$\frac{ml\omega^2}{k+m\omega^2}$$

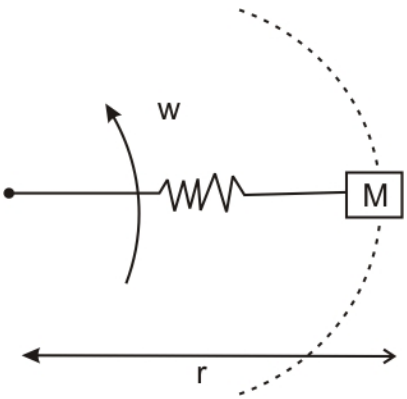
Option 4:

$$\frac{ml\omega^2}{k+m\omega}$$

Correct Answer:

$$\frac{ml\omega^2}{k-m\omega^2}$$

Solution:



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As natural lentgh=l

Let elongation=x

Mass m is moving with angular velocity ω in a radius r

where $r = l + x$

Due to elongation x spring force is given by $F_s = Kx$

And $F_C = m\omega^2 r = m\omega^2(l + x)$

as $F_C = F_s$

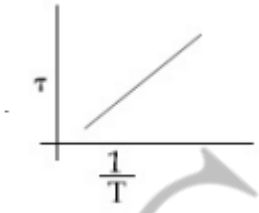
So

$$Kx = m\omega^2(l + x)$$
$$\Rightarrow x = \frac{m\omega^2 l}{K - m\omega^2}$$

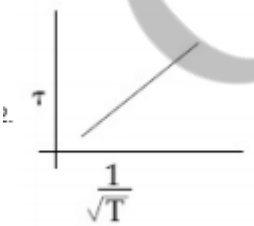
So the correct option is 2.

Q. 17 The plot that depicts the behaviour of the mean free time τ (time between two successive collisions) for the molecules of an ideal gas, as a function of temperature (T), qualitatively, is: (Graphs are schematic and not drawn to scale)

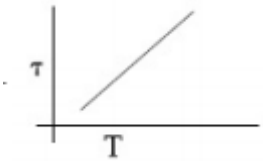
Option 1:



Option 2:



Option 3:



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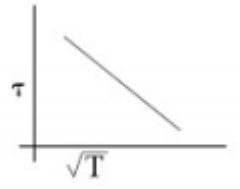
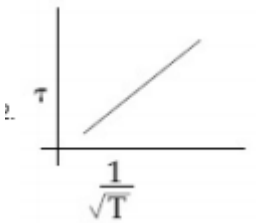
Option 4:



Correct Answer:



Solution:
As relaxation time is given as

$$\tau = \frac{\lambda}{V_{rms}}$$

where λ=mean free path

and $V_{rms} \propto \sqrt{T}$

while $\boldsymbol{\lambda = \frac{1}{\sqrt{2}\pi N d^2}}$

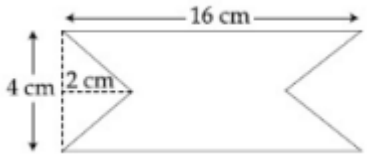
where N=number of molecules per unit volume

So keeping all the other quantities are constant.

we get $\tau \ \ \alpha \ \frac{1}{\sqrt{T}}$

So the correct graph is given in option 2

Q. 18 At time $t = 0$ magnetic field of 1000 Gauss is passing perpendicularly through the area defined by the closed loop shown in the figure. If the magnetic field reduces linearly to $500\ Gauss$, in the next $5s$, then induced EMF (in μV) in the loop is :



Correct Answer:
56

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Solution:

Area of the given loop

$A = Area\ of\ rectangle - Area\ of\ 2\ triangles\ with\ height\ 2cm$
$A = 16 \times 4 - 2 \times 4 = 56cm^2$

Magnitude of emf

$E = A\frac{dB}{dt} = 56 \times 10^{-4} \times (\frac{(1000 - 500) \times 10^{-4}}{5})$
$E = 5600 \times 10^{-8}V = 56\mu\ V$

So the correct graph is given in option 1.

Q. 19 The magnifying power of a telescope with tube length $60cm$ is 5. What is the focal length (in cm) of its eye piece?

Correct Answer:
10

Solution:
Let the focal length of the objective as f_0
and focal length of the eyepiece as f_e
Magnifying power, $M = \frac{f_0}{f_e}$
Tube length, $L = f_0 + f_e$

Given: Magnifying power $= 5$

$\therefore \frac{f_0}{f_e} = 5$
$\therefore f_0 = 5f_e$
$and\ \mathrm{L} = \mathrm{f_0} + \mathrm{f_e} = 60$
$5\mathrm{f_e} + \mathrm{f_e} = 60 \quad [\because \mathrm{f_0} = 5\mathrm{f_e}]$
$6\mathrm{f_e} = 60$
$\mathrm{f_e} = \frac{60}{6} = 10\mathrm{cm}$
So the focal length of eye piece is 10cm.

So the correct answer is given in option 3.

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Q. 20 Consider a uniform rod of mass $M = 4m$ and length l pivoted about its centre. A mass m moving with velocity v making angle $\theta = \frac{\pi}{4}$ to the rod's long axis collides with one end of the rod and sticks to it. The angular speed of the rod - mass system just after the collision is :

Option 1:
$\frac{4}{7}\frac{v}{l}$

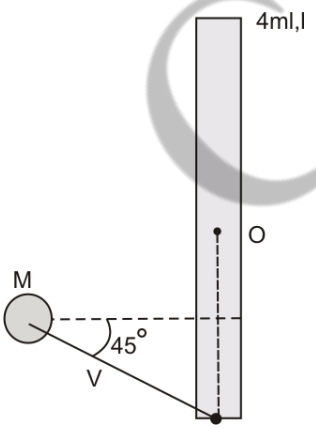
Option 2:
$\frac{3\sqrt{2}}{7}\frac{v}{l}$

Option 3:
$\frac{3}{7\sqrt{2}}\frac{v}{l}$

Option 4:
$\frac{3}{7}\frac{v}{l}$

Correct Answer:
$\frac{3\sqrt{2}}{7}\frac{v}{l}$

Solution:



Let us conserve angular momentum about O:-

So, $L_i = \left(\frac{mv}{\sqrt{2}}\right) \times \frac{l}{2}$, where $\left(\frac{mv}{\sqrt{2}}\right)$ is linear momentum and $\left(\frac{l}{2}\right)$ is the distance from centre O.

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Now, $L_f = I\omega$

Here, $I = \frac{4ml^2}{12} + m\left(\frac{l}{2}\right)^2 = \frac{7ml^2}{12}$

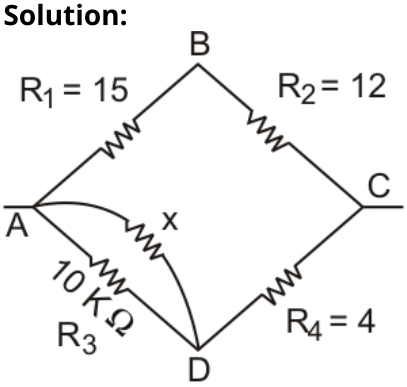
So, $L_i = L_f \Rightarrow \omega = \frac{6v}{7\sqrt{2}l} = \frac{3\sqrt{2}v}{7l}$

So the correct graph is given in option 2.

Q. 21 Four resistances of $15\Omega, 12\Omega, 4\Omega \ and \ 10\Omega$ respectively in cyclic order to form Wheatstone's network. The resistance that is to be connected in parallel with the resistance of 10Ω to balance the network is _____Ω.

Correct Answer:
10

Solution:



$$R_3 = \frac{10 \times x}{10 + x}$$

For balanced Wheatstone bridge $\frac{R_1}{R_2} = \frac{R_3}{R_4} \ \ or \ \ \frac{R_1}{R_3} = \frac{R_2}{R_4}$

$R_2 = 12\Omega \ \ and \ \ R_4 = 4\Omega$

As $\frac{R_2}{R_4} = \frac{12}{4} = 3$

So using $R_1 = 15\Omega$

We get $R_3 = R_{AD} = 5\Omega$

let we connected x-ohms in parallel to 10-ohm resistance

i.e $R_3 = 5\Omega = \frac{x * 10}{x + 10}$

we get $x = 10\Omega$

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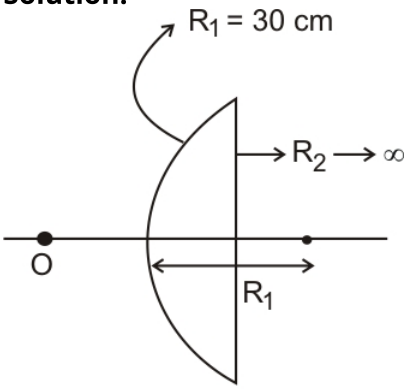
So the answer will be 10.

Q. 22 A point object in air is in front of the curved surface of a $plano-convex$ lens. The radius of curvature of the curved surface is $30cm$ and the refractive index of the lens material is $1.5,$ then the focal length of the lens (in cm) is _____.

Correct Answer: 60

Solution:



$$\frac{1}{f} = (\mu-1)\left\{\frac{1}{R_1} - \frac{1}{R_2}\right\}$$

So, $\frac{1}{f} = (1.5-1)\left(\frac{1}{30} - \frac{1}{\infty}\right) = \frac{0.5}{30} = \frac{1}{60}$

So f= 60 cm

So the answer will be 60.

Q. 23 A one metre long (both ends open) organ pipe is kept in a gas that has double the density of air at STP. Assuming the speed of sound in air at STP is $300m/s$, the frequency difference between the fundamental and second harmonic of this pipe is _____Hz.

Correct Answer: 106

Solution:

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$$V = \sqrt{\frac{B}{\rho}}$$

$$\frac{V_{pipe}}{V_{air}} = \frac{\sqrt{\frac{B}{2\rho}}}{\sqrt{\frac{B}{\rho}}} = \frac{1}{\sqrt{2}}$$

$$V_{pipe} = \frac{V_{air}}{\sqrt{2}}$$

$f_n = \frac{(n+1)V_{pipe}}{2\ell}$

$f_1 - f_0 = \frac{V_{pipe}}{2\ell} = \frac{300}{2\sqrt{2}} = 106.38 = 106 \text{ Hz } (\text{If } \sqrt{2} = 1.41)$

So the answer will be 106

Q. 24 A body A, of mass $m = 0.1\ kg$ has an initial velocity of $3\ \hat{i} ms^{-1}$. It collides elastically with another body, B of the same mass which has an initial velocity of $5\ \hat{j} ms^{-1}$. After collision, A moves with a velocity $\vec{v} = 4(\hat{i} + \hat{j})$. The energy of B after collision is written as $\frac{x}{10} J$. The value of x is _____.

Correct Answer: 1

Solution:

For elastic collision

$KE_i = KE_f$

$\frac{1}{2}m \times 25 + \frac{1}{2} \times m \times 9 = \frac{1}{2}m \times 32 + \frac{1}{2}mv^2$

$34 = 32 + v^2 \Rightarrow v^2 = 2$

$\text{KE} = \frac{1}{2} \times 0.1 \times 2 = 0.1\text{J} = \frac{1}{10}$

$x = 1$

So the answer will be 1

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Q. 25 A particle is moving along the x- axis with its coordinate with time 't' given by $x(t) = 10 + 8t - 3t^2$. Another particle is moving along the y-axis with its coordinate as a function of time given by $y(t) = 5 - 8t^3$. At $t = 1$ s, the speed of the second particle measured in the frame of the first particle is given by $\sqrt{v}$. Then v (in m/s) is _____.

Correct Answer:
580

Solution:
$x_A = -3t^2 + 8t + 10$
$\vec{v}_A = (-6t + 8)\hat{i}$
$= 2\hat{i}$
$Y_B = 5 - 8t^3$
$\vec{v}_B = -24t^2\hat{j} = -24\hat{j}$
$\sqrt{v} = |\vec{v}_B - \vec{v}_A| = |-24\hat{j} - 2\hat{i}|$
$\sqrt{v} = \sqrt{580}$

So the value of v is 580

Chemistry

Q. 1 For the Balmer series in the spectrum of H atom, $\bar{v} = \mathrm{R_H}\left\{\frac{1}{\mathrm{n_1^2}} - \frac{1}{\mathrm{n_2^2}}\right\}$ the correct statements among (I) to (IV) are:

(1) As wavelength decreases, the lines in the series converge
(II) The integer n_1 is equal to 2
(III) The lines of longest wavelength corresponds to n_2 = 3
(IV) The ionization energy of hydrogen can be calculated from wavenumber of these lines

Option 1:
$(II), (III), (IV)$

Option 2:
$(I), (III), (IV)$

Option 3:
$(I), (II), (III)$

Option 4:
$(I), (II), (IV)$

Correct Answer:
$(I), (II), (III)$

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Solution:
Line Spectrum of Hydrogen-like atoms

Where R is called Rydberg constant, R = 1.097 X 10^7, Z is the atomic number

n_1= 1, 2, 3....

n_2= n_1+1, n_1+2 .....

Lyman Series spectrum:

Where

n_1= 1 and n_2= 2, 3, 4....

This lies in the Ultraviolet region.

Balmer Series Spectrum:

Where n_1= 2 and n_2= 3, 4, 5, 6....

It lies in the visible region.

Paschen, Bracket and Pfund Series spectrums:

these lies in Infrared Region.

-

Since, $\Delta E \propto \frac{1}{\lambda}$ so for wavelength to be longest or maximum the energy gap should be minimum. For the Balmer series, the transition for the longest wavelength is from n = 3 to n = 2. Thus clearly, n_1 value for the Balmer series is 2. Further, as wavelength decreases, the lines in the series converge.

Therefore, Option(3) is correct.

Q. 2 The strength of an aqueous NaOH solution is most accurately determined by titrating:
(Note: consider that an appropriate indicator is used)

Option 1:
Aq.NaOH in a pipette and aqueous oxalic acid in a burette

Option 2:
Aq.NaOH in a volumetric flask and concentrated H_2SO_4 in a conical flask

Option 3:
Aq.NaOH in a burette and concentrated H_2SO_4 in a conical flask

Option 4:
Aq.NaOH in a burette and aqueous oxalic acid in a conical flask

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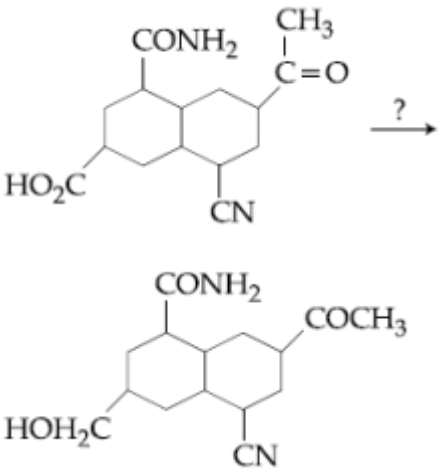
Correct Answer:
Aq.NaOH in a burette and aqueous oxalic acid in a conical flask

Solution:
As we have learnt,

To calculate the strength of NaOH, it is titrated against oxalic acid. For this purpose, NaOH is kept in a burette and oxalic acid in a conical flask.

Therefore, Option(4) is correct.

Q. 3 The most suitable reagent for the given conversion is:



Option 1:
B_2H_6

Option 2:
$LiAlH_4$

Option 3:
$NaBH_4$

Option 4:
H_2/Pd

Correct Answer:
B_2H_6

Solution:
As we have learnt,

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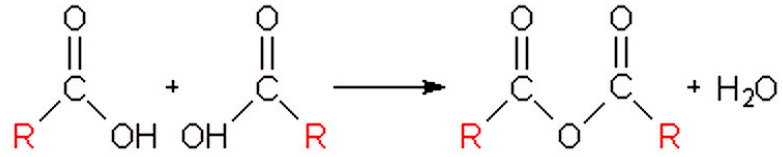
Chemical Properties of Carboxylic Acids -

Formation of Anhydride

Carboxylic acid on heating with H_2SO_4 or P_2O_5 gives corresponding anhydride. The reaction occurs as follows:

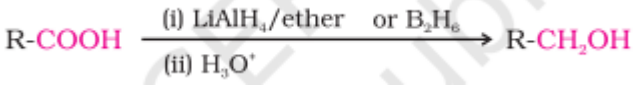


Reaction with Ammonia

On heating with NH_3, carboxylic acid gives ammonium salt which on further heating at high temperature gives amides. The reaction occurs as follows:

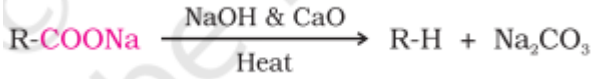
Reduction

Carboxylic acids are reduced to 1° alcohol by $LiALH_4$ or better with B_2H_6. B_2H_6 does not easily reduce esters, nitro, and halo groups, and $NaBH_4$ does not reduce (-COOH) group. The reaction occurs as follows:



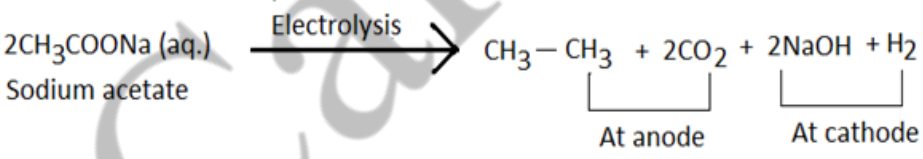
Decarboxylation

When sodium salts of carboxylic acids are heated with soda lime (NaOH and CaO in 3:1 ratio), they form hydrocarbons by losing CO_2. This is known as decarboxylation. The reaction occurs as follows:



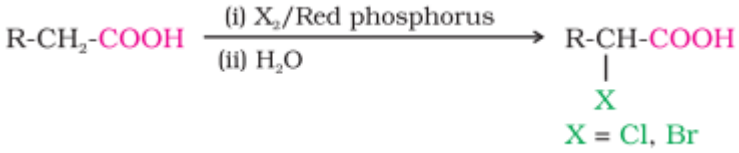
Kolbe's electrolysis

The electrolysis of aqueous solution of sodium or potassium salts of carboxylic acid makes the carboxylic acid to undergo decarboxylation to form alkane. The reaction occurs as follows:



Halogenation

The method of preparing α-chloro or α-bromo acid is by Hell-Volhard-Zelinsky reaction, which is carried out by treating the acid with Cl_2 or Br_2 in the presence of a small amount of red phosphorous. The reaction occurs as follows:



Ring substitution

Aromatic carboxylic acid undergoes substitution electrophilic reactions in which the (-COOH) group acts as a deactivating and *meta*-directing group. It does not undergo Friedel-Crafts reaction because the (-

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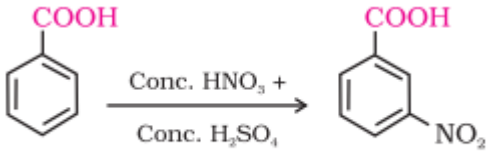
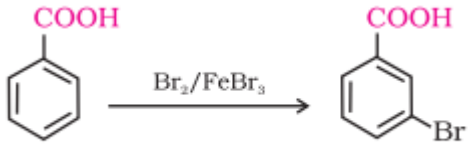
COOH) group is deactivating and the catalyst $AlCl_3$ gets bonded to the (COOH) group. The reactions occur as follows:



m-Nitrobenzoic acid



m-Bromobenzoic acid

-

B_2H_6 is very selective and usually used to reduce acid to alcohol.

Therefore, Option(1) is correct.

Q. 4 The number of bonds between sulphur and oxygen atoms in $S_2O_8^{2-}$ and the number of bonds between sulphur and sulphur atoms in rhombic sulphur, respectively, are:

Option 1:

$8\ and\ 6$

Option 2:

$4\ and\ 6$

Option 3:

$8\ and\ 8$

Option 4:

$4\ and\ 8$

Correct Answer:

$8\ and\ 8$

Solution:

As we have learnt,

Sulphur - Allotropic Forms -

Sulphur forms numerous allotropes of which the yellow rhombic (α-sulphur) and monoclinic (β -sulphur) forms are the most important. The stable form at room temperature is rhombic sulphur, which transforms to monoclinic sulphur when heated above 369 K.

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Rhombic sulphur (α-sulphur)

This allotrope is yellow in colour, m.p. 385.8 K and specific gravity 2.06. Rhombic sulphur crystals are formed on evaporating the solution of roll sulphur in CS_2. It is insoluble in water but dissolves to some extent in benzene, alcohol and ether. It is readily soluble in CS_2

Monoclinic sulphur (β-sulphur)

Its m.p. is 393 K and specific gravity 1.98. It is soluble in CS_2. This form of sulphur is prepared by melting rhombic sulphur in a dish and cooling, till crust is formed. Two holes are made in the crust and the remaining liquid poured out. On removing the crust, colourless needle-shaped crystals of β-sulphur are formed. It is stable above 369 K and transforms into α-sulphur below it. Conversely, α-sulphur is stable below 369 K and transforms into β-sulphur above this. At 369 K both the forms are stable. This temperature is called transition temperature. Both rhombic and monoclinic sulphur have S_8 molecules. These S_8 molecules are packed to give different crystal structures. The S_8 ring in both forms is puckered and has a crown shape. The molecular dimensions are given in the figure below.

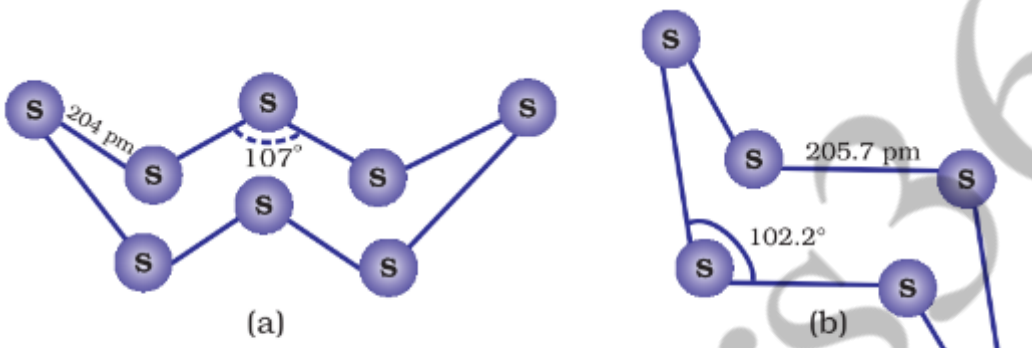
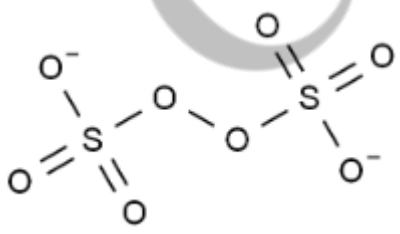


Fig. 7.5: The structures of (a) S_8 ring in rhombic sulphur and (b) S_6 form

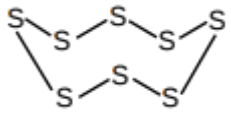
Several other modifications of sulphur-containing 6-20 sulphur atoms per ring have been synthesised in the last two decades. In cyclo-S_6, the ring adopts the chair form and the molecular dimensions are as shown in the figure. At elevated temperatures (~1000K), S_2 is the dominant species and is paramagnetic like O_2.

-

The structure of $S_2O_8^{2-}$ is given below:



Rhombic sulphur



Thus, the number of S-S bonds is 8 and the number of S-O bonds is also 8.

Therefore, Option(3) is correct.

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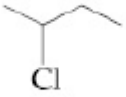
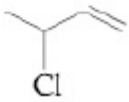
Q. 5 The decreasing order of reactivity towards dehydrohalogenation (E_1) reaction of the following compounds is:

(A)

(B)

(C) 

(D) 

Option 1:

$B > A > D > C$

Option 2:

$B > D > C > A$

Option 3:

$B > D > A > C$

Option 4:

$D > B > C > A$

Correct Answer:

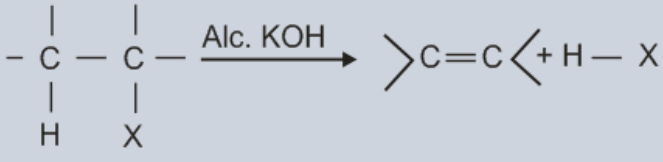
$D > B > C > A$

Solution:

As we have learnt,

Dehydrohalogenation of Alkyl Halides -

When alkyl halides are heated with reagents like $C_2H_5O^-$, alc. KOH, $NaNH_2$, KNH_2, $(Me)_3COK$ etc. then alkenes are formed as the product. The general reaction occurs as follows:



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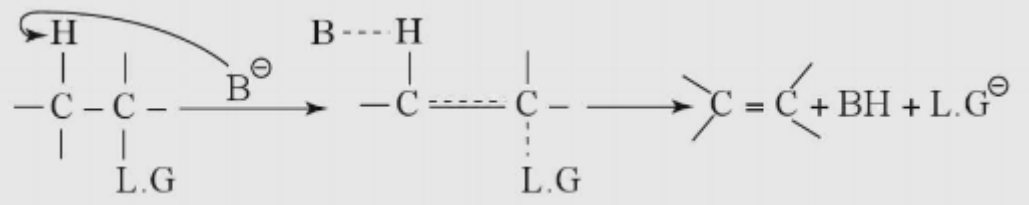
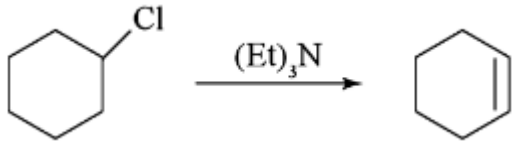
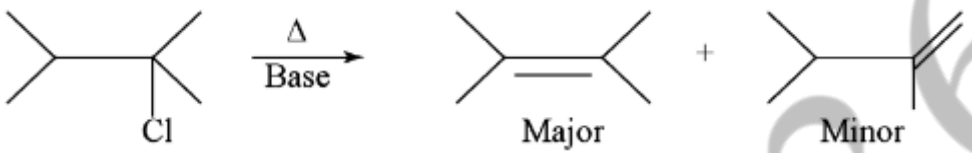
Mechanism



Some examples include:

(1) 

(2) 

-

E1 reaction proceeds via carbocation formation, therefore greater the stability of carbocation, faster the E1 reaction.

Therefore, Option(4) is correct.

Q. 6 A flask contains a mixture of isohexane and 3-methylpentane. One of the liquids boils at $63^{\circ}C$ while the other boils at $60^{\circ}C$. What is the best way to separate the two liquids and which one will be distilled out first?

Option 1:
Fractional distillation, isohexane

Option 2:
simple distillation,3-methylpentane

Option 3:
fractional distillation, 3 - methylpentane

Option 4:
simple distillation, isohexane

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Correct Answer:
Fractional distillation, isohexane

Solution:
As we have learnt,

Simple distillation is used for the separation of liquids having a significant difference in their boiling points. If the difference in the boiling points is not much, then Fractional distillation is used as the preferred method of separation.

Liquid having lower boiling point is distilled first in fractional distillation.

So, it is isohexane which is distilled first as it has a lower value of boiling point.

Therefore, Option(1) is correct.

Q. 7 The third ionization enthalpy is minimum for:

Option 1:
Mn

Option 2:
Co

Option 3:
Ni

Option 4:
Fe

Correct Answer:
Fe

Solution:
Importance of Ionisation Enthalpy

Ionisation enthalpy is an important factor for determining the nature of an element. The elements with low ionisation enthalpies are metals while the elements with higher ionisation enthalpies are non-metals.

The stability of oxidation states of an element can also be determined on the basis of the value of ionisation enthalpies.

(i) When the difference between two successive ionisation enthalpies for an atom is approximately 965 - 1450kJ/mol or less, then higher oxidation state will be more stable.

For example, Li has IE_1 = 520kJ/mol and IE_2 = 7298.1kJ/mol.

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Thus, $IE_2 - IE_1$ = (7298.1 - 520.3)
= 6777.8kJ/mol

Therefore, Li^+ or first oxidation state of Li is more stable.

(ii) When the difference between two successive ionisation enthalpies for an atom is higher than 1450kJ/mol, then lower oxidation state will be more stable.

For example, Ca has IE_1 = 589.8kJ/mol and IE_2 = 1145.4kJ/mol.

Thus, $IE_2 - IE_1$ = (1145.4 - 589.8)

= 555.6kJ/mol

Therefore, Ca^{2+} or second oxidation state of Ca is more stable.

-

The third ionisation potential data for the given element is given below:

Co^{3+} - 3237

Mn^{3+} - 3258

Fe^{3+} - 2967

Ni^{3+} - 3400

The electronic configuration of Fe^{2+} is [Ar]3d^6. Thus it is easier to remove one electron to get [Ar]3d^5 electronic configuration as it is half-filled and thus it is stable.

Therefore, Option(4) is correct.

Q. 8 As per Hardy-Schulze formulation, the flocculation values of the following for ferric hydroxide sol are in the order:

Option 1:
$AlCl_3 > K_3[Fe(CN)_6] > K_2CrO_4 > KBr = KNO_3$

Option 2:
$K_3[Fe(CN)_6] < K_2CrO_4 < AlCl_3 < KBr < KNO_3$

Option 3:
$K_3[Fe(CN)_6] < K_2CrO_4 < KBr = KNO_3 = AlCl_3$

Option 4:
$K_3[Fe(CN)_6] > AlCl_3 > K_2CrO_4 > KBr > KNO_3$

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Correct Answer:

$K_3[Fe(CN)_6] < K_2CrO_4 < KBr = KNO_3 = AlCl_3$

Solution:

As per Hardy Schulze rules coagulation power.

$$\text{Coagulation value or flocculation value} \propto \frac{1}{\text{Coagulation power}}$$

Since $Fe(OH)_3$ is positively charged sol, hence, the anionic charge will flocculate.

As per Hardy Schulze rules coagulation power of anion follows the order :

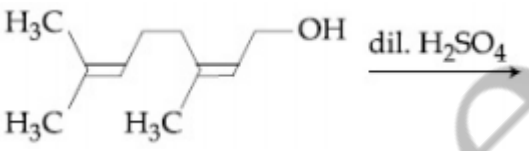
$[Fe(CN)_6]^{3-} > CrO_4^{2-} > Cl^- = Br^- = NO_3^-$

Higher the coagulation power lower will be its flocculation value therefore order will be :

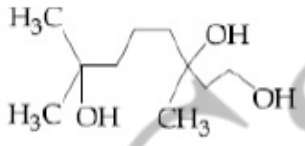
$[Fe(CN)_6]^{3-} < CrO_4^{2-} < Cl^- = Br^- = NO_3^-$

Therefore, Option(3) is correct.

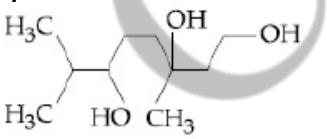
Q. 9 The major product of the following reaction is:



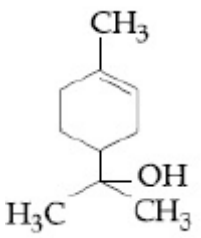
Option 1:



Option 2:



Option 3:



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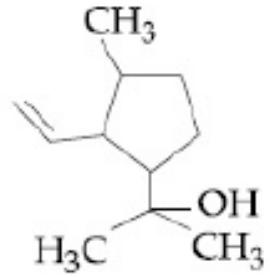
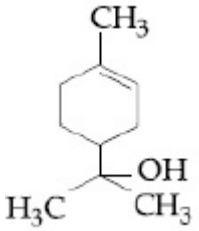
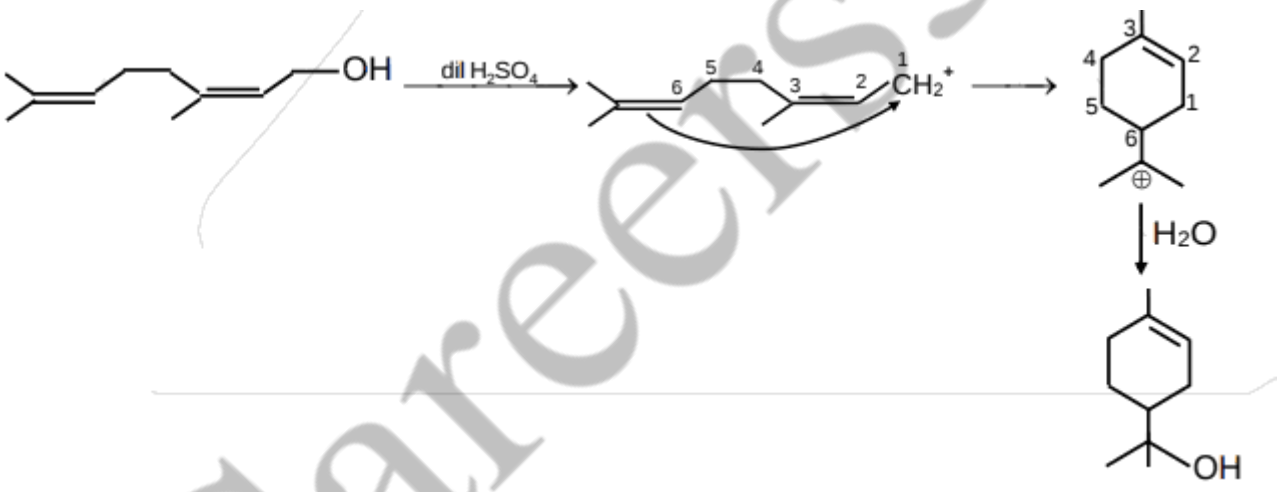
Option 4:



Correct Answer:



Solution:



Therefore, Option(3) is correct.

Q. 10 The first ionization energy (in kJ\mol) of Na, Mg, Al and Si respectively, are:

Option 1:
496,737,577,786

Option 2:
496,577,737,786

Option 3:
496,577,786,737

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Option 4:
786,737,577,496

Correct Answer:
496,737,577,786

Solution:
As we have learnt,

Ionisation Enthalpy or Ionisation Potential -

Ionisation Enthalpy

Ionisation enthalpy may be defined as the minimum energy required to remove the most loosely bound electron from an isolated gaseous atom to convert it into gaseous monovalent positive ion.

$$M(g) \overset{IE_1}{\rightarrow} M^+(g) + e$$

IE1 is ionisation enthalpy or also known as first ionisation enthalpy.

Ionisation Potential

Ionisation enthalpy is also known as ionisation potential. It is the minimum potential difference required to remove the outermost electron from a gaseous atom to form a cation.

Factors Affecting Ionisation Enthalpy

The ionisation enthalpy of any atom is affected by the following factors.

- **Size of the atom:** The larger the size of an atom, the lower is the ionisation enthalpy. As the atomic size increases, the distance between the outermost electrons and the nucleus increases due to which the force of attraction between the nucleus and these outermost electrons decreases, thus it becomes easy to remove an electron from the atom and hence the ionisation enthalpy decreases. Thus,

  Ionisation enthalpy ∝ 1/Atomic size

- **Screening effect:** The higher is the value of screening effect, the lower is the ionisation enthalpy. As the screening effect increases, the repulsion between the electrons increases and thus the removal of electron from the atom becomes easier. Thus,

  Ionisation enthalpy ∝ 1/Screening effect

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- **Nuclear charge:** As the nuclear charge increases, the force of attraction between the nucleus and electrons also increases and thus the removal of electron from the atom becomes difficult and hence the ionisation enthalpy increases. Thus,

  Ionisation enthalpy ∝ Nuclear charge

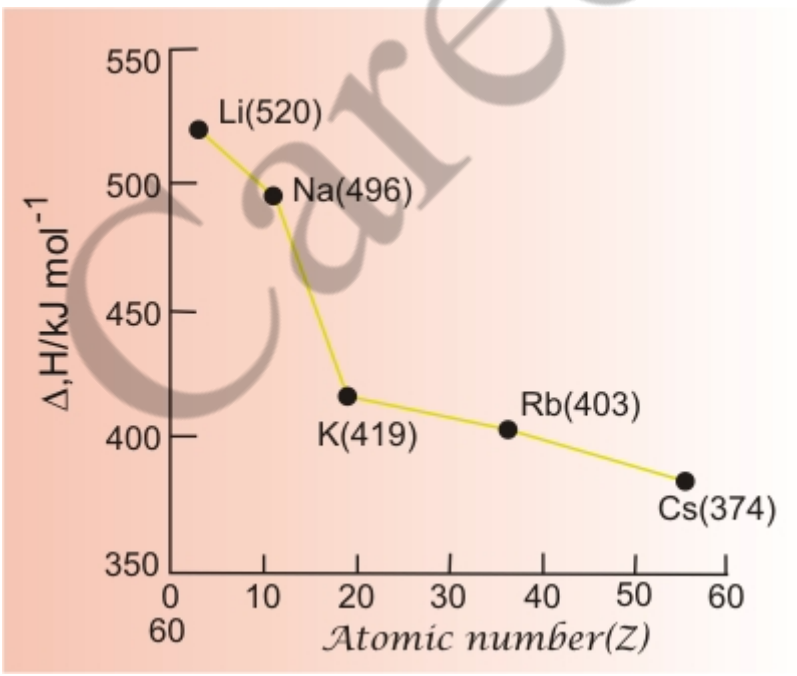
- **Half filled and fully filled orbitals:** The atoms with half filled and fully filled orbitals are more stable than other atoms. Thus removing an electron from these atoms requires a little more energy. Thus for these atoms with half filled and fully filled orbitals, the ionisation enthalpy is higher than others.

- **Shape of orbital:** The ionisation enthalpy also depends on the shape of orbital in which the last electron enters. The more the orbital is close to the nucleus, the more energy is required to remove the electron in the same orbit. Thus, the ionisation enthalpy for the orbitals from the same orbit follows the given order:

  s > p > d > f

Variation of Ionisation Enthalpy

- In moving down the group, the ionisation enthalpy decreases. As we move down in the group, the number of shells increases due to which the force of attraction between the nucleus and the outer electrons decreases, thus removing an electron from the atom becomes easy and hence ionisation enthalpy decreases. There are some exceptions after the element with atomic number 72. The elements with atomic number from 73 to 82 have higher ionisation enthalpy than the earlier elements in their respective group. This deviation of behaviour is because of the lanthanide contraction.



- In moving from left to right in a period, the ionisation enthalpy increases. In period, the nuclear charge increases but the number of shells remain the same, thus the force of attraction between the nucleus and the outer electrons increases and hence the ionisation enthalpy increases. In a period, some elements like Be, Mg, N and P have exceptionally higher ionisation enthalpies than expected. This is because of their half filled or fully filled outer orbitals.

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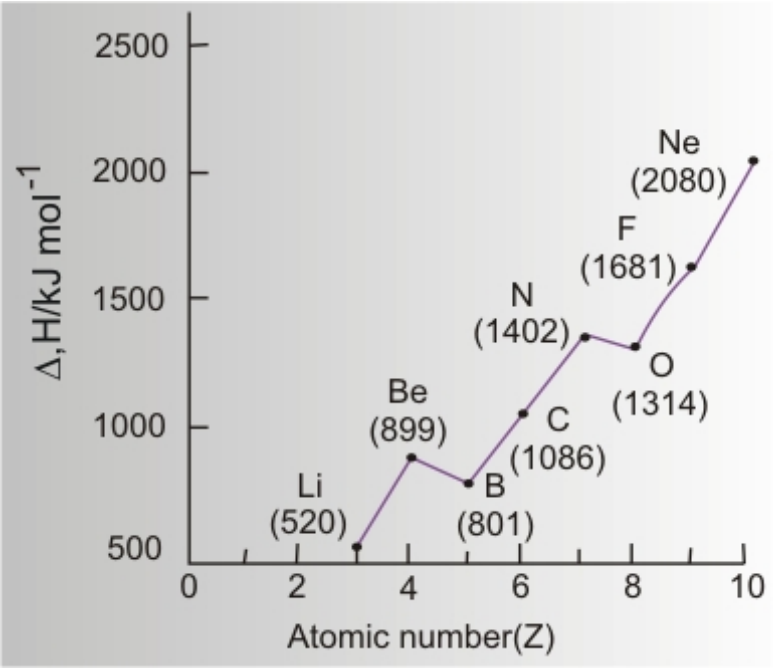

Importance of Ionisation Enthalpy

Ionisation enthalpy is an important factor for determining the nature of an element. The elements with low ionisation enthalpies are metals while the elements with higher ionisation enthalpies are non-metals.

The stability of oxidation states of an element can also be determined on the basis of the value of ionisation enthalpies.

(i) When the difference between two successive ionisation enthalpies for an atom is approximately 965 - 1450kJ/mol or less, then higher oxidation state will be more stable.

For example, Li has IE_1 = 520kJ/mol and IE_2 = 7298.1kJ/mol.

Thus, $IE_2 - IE_1$ = (7298.1 - 520.3)
= 6777.8kJ/mol

Therefore, Li^+ or first oxidation state of Li is more stable.

(ii) When the difference between two successive ionisation enthalpies for an atom is higher than 1450kJ/mol, then lower oxidation state will be more stable.

For example, Ca has IE_1 = 589.8kJ/mol and IE_2 = 1145.4kJ/mol.

Thus, $IE_2 - IE_1$ = (1145.4 - 589.8)

= 555.6kJ/mol

Therefore, Ca^{2+} or second oxidation state of Ca is more stable.

-

The correct ionization enthalpy values is given below:

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496,737,577,786

Therefore, Option(1) is correct.

Q. 11 Arrange the following compounds in increasing order of C-OH bond length:
methanol, phenol, *p*-ethoxyphenol

Option 1:
phenol < methanol < p-ethoxyphenol

Option 2:
methanol < p-ethoxyphenol < phenol

Option 3:
phenol <p-ethoxyphenol < methanol

Option 4:
methanol < phenol < p-ethoxyphenol

Correct Answer:
phenol <p-ethoxyphenol < methanol

Solution:
Phenol has one -OH group and the oxygen will resonate with benzene. Here the bond between O and C will double bond most of the time.

p-ethoxyphenol has one -OH group and one $-OCH_3$, from both groups oxygen will resonate with benzene. There are two oxygen is present, so they will resonate one by one. So, the Bond length between C and O will be larger here than phenol.

Methanol has one -OH group and there will be no resonating between C and O. So, the Length between C and O will be maximum.

The correct order of bond length is given below:
Phenol < *p*-ethoxyphenol < methanol

Therefore, Option(3) is correct.

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Q. 12 Among the gases $(a) - (e)$, the gases that cause greenhouse effect are:

$(a)\ CO_2$
$(b)\ H_2O$
$(c)\ CFCs$
$(d)\ O_2$
$(e)\ O_3$

Option 1:
(a), (b), (c) and (d)

Option 2:
(a), (c), (d) and (e)

Option 3:
(a) and (d)

Option 4:
(a), (b), (c) and (e)

Correct Answer:
(a), (b), (c) and (e)

Solution:
As we have learnt,

The gases that cause greenhouse effect are: $CO_2, O_3\ and\ H_2O.$
It is to be noted that some other examples of Greenhouse gases are Chlorofluorocarbons and Methane.

Therefore, Option(4) is correct

Q. 13 The predominant intermolecular forces present in ethyl acetate, a liquid, are:

Option 1:
London dispersion, dipole-dipole and hydrogen bonding

Option 2:
hydrogen bonding and London dispersion

Option 3:
Dipole-dipole and hydrogen bonding

Option 4:
London dispersion and dipole-dipole

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Correct Answer:
London dispersion and dipole-dipole

Solution:
Ethyl acetate is a polar molecule so London dispersion and dipole-dipole interaction will be present there.

Therefore, Option(4) is correct.

Q. 14 When gypsum is heated to $393\ K$, it forms:

Option 1:
$CaSO_4.\frac{1}{2}\ H_2O$

Option 2:
Dead burnt plaster

Option 3:
$CaSO_4.5\ H_2O$

Option 4:
Anhydrous $CaSO_4$

Correct Answer:
$CaSO_4.\frac{1}{2}\ H_2O$

Solution:
The reaction occurs as follows:

$$\underset{\text{Gypsum}}{\mathrm{CaSO_4.2H_2O}} \xrightarrow{393\mathrm{K}} \underset{\text{Plaster of Paris}}{\mathrm{CaSO_4.\frac{1}{2}H_2O}} + \frac{3}{2}\mathrm{H_2O}$$

Therefore, Option(1) is correct.

Q. 15 Which of the following statement is not true for glucose?

Option 1:
The penta acetate of glucose does not react with hydroxylamine to give oxime

Option 2:
Glucose reacts with hydroxylamine to form oxime

Option 3:
Glucose gives Schiff's test for aldehyde

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Option 4:
Glucose exists in two crystalline forms α and β

Correct Answer:
Glucose gives Schiff's test for aldehyde

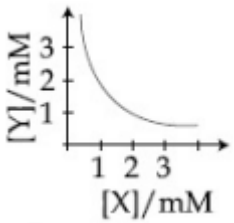
Solution:
As we have learnt,

Evidence for Ring Structure of Glucose -
1. Glucose does not react with sodium bisulphite. It confirms the absence of free -CHO group.
2. Glucose does not give Schiff's test and DNP test. It confirms the absence of free -CHO group.
3. Glucose pentaacetate does not react with hydroxylamine. It means the absence of free -CHO group.

Glucose does not react with Schiff's test because Schiff's reagent is used for aldehydes. Glucose exists in its cyclic form in which the aldehyde group is not free but is instead used for hemiacetal formation.

Therefore, Option(3) is correct.

Q. 16 The stoichiometry and solubility product of a salt with the solubility curve given below, respectively:



Option 1:
$X_2Y, 2 \times 10^{-9} M^3$

Option 2:
$XY_2, 1 \times 10^{-9} M^3$

Option 3:
$XY_2, 4 \times 10^{-9} M^3$

Option 4:
$XY, 2 \times 10^{-6} M^3$

Correct Answer:
$XY_2, 4 \times 10^{-9} M^3$

Solution:
1) $\mathrm{X_2Y(s) \rightarrow 2X^+(aq) + Y^-(aq)}$

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2×10^{-3} 10^{-3}

$K_{sp} = [X^+]^2 [Y^-] = (2 \times 10^{-3})^2 \times 10^{-3} = 4 \times 10^{-6}$

2) $XY_2(s) \rightarrow X^+(aq) + 2Y^-(aq)$

10^{-3} 2×10^{-3}

$K_{sp} = [X^+]^2 [Y^-] = 10^{-3} \times (2 \times 10^{-3})^2 = 4 \times 10^{-6}$

3) $XY(s) \rightarrow X^+(aq) + Y^-(aq)$

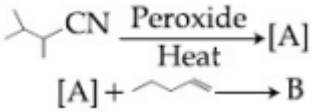
10^{-3} 10^{-3}

$K_{sp} = [X^+] [Y^-] = 10^{-3} \times 10^{-3} = 10^{-6}$

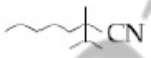
So, $XY_2, 4 \times 10^{-9} M^3$ will be correct.

Therefore, **Option(3) is correct.**

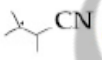
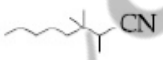
Q. 17 The major products A and B in the following reactions are:



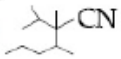
Option 1:

A = and B = 

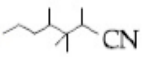
Option 2:

A =  and B = 

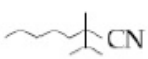
Option 3:

A = and B = 

Option 4:

A = and B = 

Correct Answer:

A = and B = 

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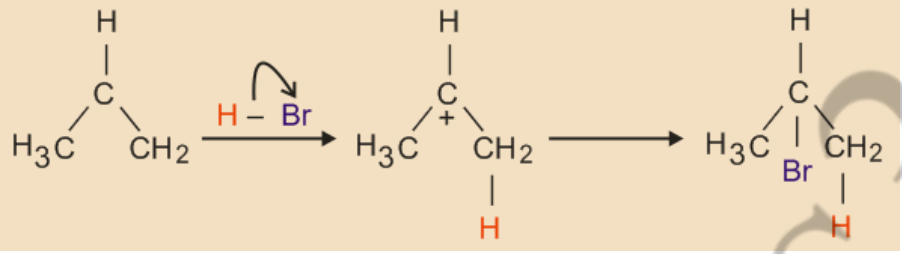
Solution:
As we have learnt,

Markonikov and Anti-markonikov Reaction -

Markovnikov's rule
This rule states that the acid hydrogen of the protic acid gets attached to the carbon with more hydrogen substituents and negative part adds to the atom with less number of hydrogen atoms.

Mechanism
The addition of halogens and halogen acids takes place by electrophilic addition(EA) reaction. +E mechanism is that when electrons of the π-bonds are transferred to that atom of the multiple bonds to which the reagent finally gets attached. First the electrophile(H^+) adds to the positive C atom and hence this step is slow and the rate-determining step. Afterwards, the negative part of the reagent (Br^-) adds to the positive C atom. Thus, it is known as (+E) reaction.



Rule 1: In alkene and alkyne, (+E) reaction takes place, first electrophile adds and then the negative part of the reagent is finally added.

Rule 2: In general, when inductively electron-withdrawing group(-I) is attached to (C=C) and has a lone pair of electrons then +R effect is operative than -I effect and Markovnikov's addition takes place.

Rule 3: If inductively electron-withdrawing group(-I) is not attached to (C=C), is one or more C atom away from (C=C), and has a lone pair of electrons, then -I effect is operative than +R effect and anti-Markovnikov's addition takes place.

Anti-Markovnikov's rule
In the presence of peroxide, such as benzoyl peroxide and light, the addition of HBr(not HCl and HI) to unsymmetrical alkenes occurs contrary to Markovnikov's rule.

Mechanism
The mechanism of this process occurs in three steps:

1. **Chain initiation:** Hydrogen Peroxide is an unstable molecule, if we heat it, or shine it with sunlight, two free radicals of OH will be formed. These OH radicals will go on and attack HBr, which will take the Hydrogen and create a Bromine radical. Hydrogen radical do not form as they tend to be extremely unstable with only one electron, thus bromine radical which is more stable will be readily formed.

2. **Chain propagation:** The Bromine Radical will go on and attack theless substituted carbon of the alkene. This is because after the bromine radical attacked the alkene a carbon radical will be formed. A carbon radical is more stable when it is at a more substituted carbon due to induction and hyperconjugation. Thus, the radical will be formed at the more substituted carbon, while the bromine is bonded to the less substituted carbon. After a carbon radical is formed, it will go on and attack the hydrogen of HBr, and

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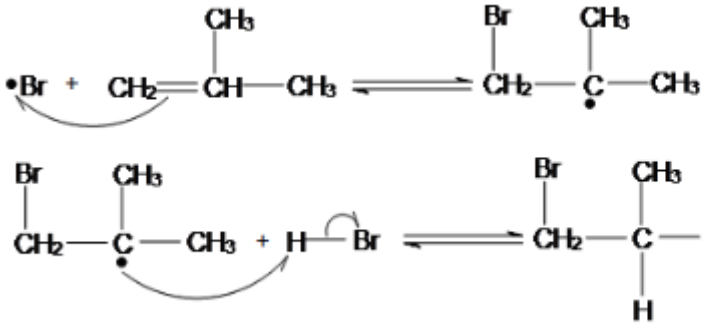
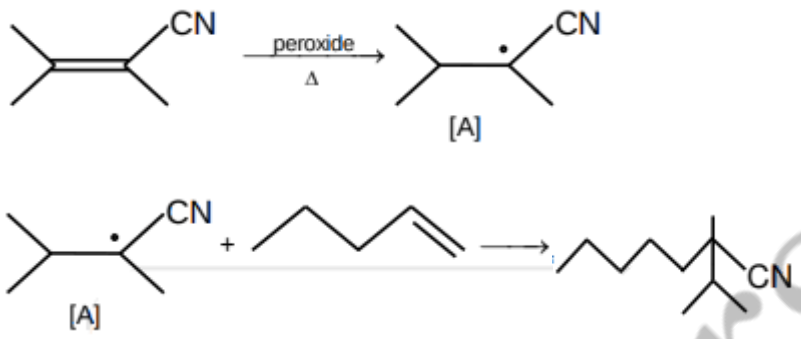
thus a bromine radical will be formed again.



3. **Chain termination:** In termination step, two bromine radicals combined to give bromine. This radical addition of bromine to alkene by radical addition reaction will go on until all the alkene turns into bromoalkane, and this process will take some time to finish.

-



Therefore, Option(1) is correct.

Q. 18 The rate of a certain biochemical reaction at physiological temperature (T) occurs 10^6 times faster with enzyme than without. The change in the activation energy upon adding enzyme is:

Option 1:

$-6RT$

Option 2:

$-6(2.303)RT$

Option 3:

$+6RT$

Option 4:

$+6(2.303)RT$

Correct Answer:

$-6(2.303)RT$

Solution:

The equation for activation energy is given as follows:

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$$\log k = \frac{-E_a}{2.303RT} + \log A_o$$

Now,

$$\ln\left(\frac{k_2}{k_1}\right) = -\frac{E_{a_2} - E_{a_1}}{RT}$$

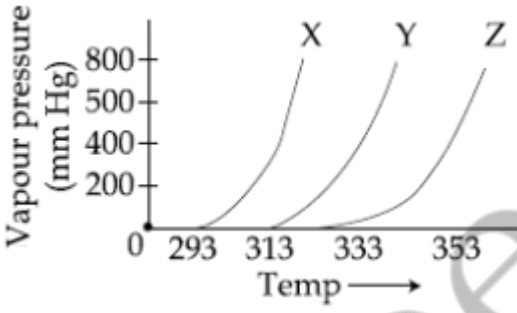
$$\log\frac{10^6}{1} = \frac{-1}{2.303RT}(E_{a_2} - E_{a_1})$$

$$(E_{a_2} - E_{a_1}) = -6(2.303)RT$$

Catalyst decrease the required activation energy for a reaction to proceed and thus increases the reaction rate.

Therefore, Option(2) is correct.

Q. 19 The graph of vapour pressure and temperature for three different liquids X, Y, and Z is shown below:



The following inferences are made:

(A) X has higher intermolecular interactions compared to Y.

(B) X has lower intermolecular interactions compared to Y

(C) Z has lower intermolecular interactions compared to Y

The correct inference(s) is /are :

Option 1:
(C)

Option 2:
(A)

Option 3:
(B)

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Option 4:
(A) and (C)

Correct Answer:
(B)

Solution:
Vaporizing starting means boiling started.

Given in the graph, at zero vapour pressure the boiling starting point.

So, the Order of B.P. is Z > Y > X

If the boiling point is low it will easily vaporize due to weak intermolecular force of attraction.

Order of Intermolecular Interaction : Z > Y > X.

the intermolecular force of attraction increases, the boiling point increases then vapour pressure decreases.

So, X has lower intermolecular interactions compared to Y and Z

Therefore, Option(3) is correct.

Q. 20 The complex that can show *fac*- and *mer*- isomers is:

Option 1:
$[Co(NH_3)_3(NO_2)_3]$

Option 2:
$[Pt(NH_3)_2Cl_2]$

Option 3:
$[Co(NH_3)_4Cl_2]^+$

Option 4:
$[COCl_2(en)_2]$

Correct Answer:
$[Co(NH_3)_3(NO_2)_3]$

Solution:
Complexes of the type $[Ma_3b_3]$ exist in tw ogeometrical forms which are named as facial (*fac*-) and meridional (*mer*-).

The complex $[Co(NH_3)_3(NO_2)_3]^+$ can be represented in fac- and mer- isomeric forms as follows:

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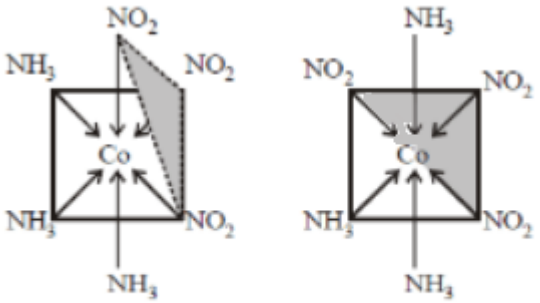
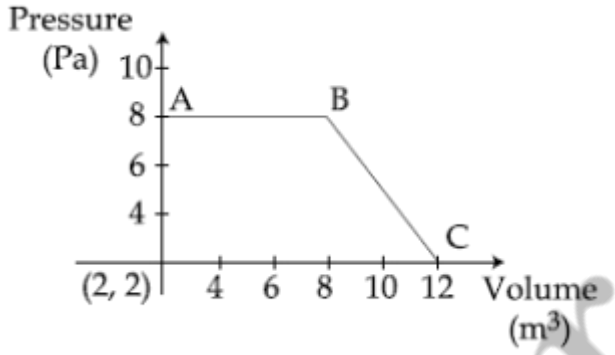


Therefore, Option(1) is correct.

Q. 21 The magnitude of work done by a gas that undergoes a reversible expansion along the path ABC shown in the figure is



Correct Answer:
48

Solution:

$$W = (8-2)^2 + \frac{1}{2}((12-8)\times(8-2))$$

$$W = 48$$

Q. 22 Ferrous sulphate heptahydrate is used to fortify foods with iron. The amount (in grams) of the salt required to achieve 10ppm of iron in 100kg of wheat is:_____
Atomic weight of Fe = 55.85; S = 32.00; O = 16.00;

Correct Answer:
4.96

Solution:

$$\text{ppm} = \frac{\text{Solute in gram}}{\text{Solution in gram}} \times 10^6$$

$$10 = \frac{\text{mass of Fe in gram}}{100 \text{ X } 1000} \times 10^6$$

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Fe = 1 gram

56 g of Fe in 1 mol of $FeSO_4.7H_2O$ (molar mass 278 gram /mol)

Therefore 1 gram of Fe is present in = (1/56) moles of $FeSO_4.7H_2O$

mass of $FeSO_4.7H_2O$ = $\frac{1}{56} \times 278 = 4.96 \text{ gram}$

Q. 23 The number of chiral centres in penicillin is:

Option 1:
3

Option 2:
4

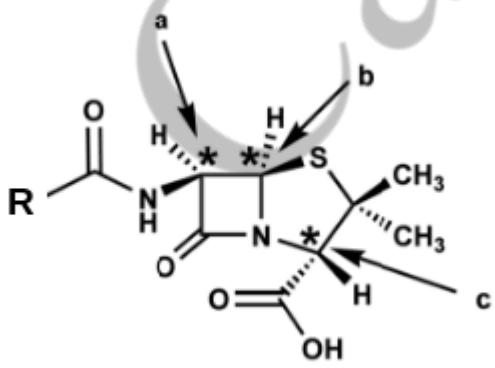
Option 3:
5

Option 4:
6

Correct Answer:
3

Solution:
An asymmetric carbon atom (chiral carbon) is **a carbon atom that is attached to four different types of atoms or groups of atoms**.

As shown in the figure below, there are 3 chiral carbons in penicillin.



Option 1 is correct.

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Q. 24 The volume (in ml) of 0.125 M $AgNO_3$ required to quantitatively precipitate chloride ions in 0.3g of $[Co(NH_3)_6]Cl_3$ is:

$$Molecular\ Mass\ of\ [Co(NH_3)_6]Cl_3 = 267.46\ g/mol$$

$$Molecular\ Mass\ of\ AgNO_3 = 169.87\ g/mol$$

Correct Answer:
27

Solution:
The reaction occurs as follows:

$$\mathrm{AgNO_3\ +\ [Co(NH_3)_6]Cl_3\ \rightarrow\ 3AgCl}$$

Molecular weight of $[Co(NH_3)_6]Cl_3$ = 267.48g/mol

Number of moles of $[Co(NH_3)_6]Cl_3$ = 0.3/267.48 = 1.125 x 10^{-3} moles.

1 mole of $[Co(NH_3)_6]Cl_3$ gives 3 moles of AgCl.

Thus, moles of AgCl = 3 x 1.125 x 10^{-3} moles

Now, 3 moles of $AgNO_3$ reacts with 3 moles of AgCl

Thus, moles of $AgNO_3$ = 3 x 1.125 x 10^{-3} moles

Now, volume(L) of $AgNO_3$ = (3 x 1.125 x 10^{-3})/0.125 = 27 x 10^{-3} L
volume(in mL) of $AgNO_3$ = 27ml

Q. 25 What would be the electrode potential for the given half cell reaction at pH = 5?

$$\mathrm{2H_2O \rightarrow O_2 + 4H^{\oplus} + 4e^-; E^0_{red}\ = 1.23\ V}$$

$$\mathrm{(R = 8.314\ J\ mol^{-1}\ K^{-1};\ Temp\ = 298\ K\ ;}$$

oxygen under std. atm. the pressure of 1 bar)

Correct Answer:
0.93

Solution:
Given half cell reaction

$$\mathrm{2H_2O \rightarrow O_2 + 4H^+;\quad E^o_{red} = 1.23\ V}$$

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$4H^+ + 4e^- \rightarrow 2H_2; \quad E^0_{red} = 0$ (for hydrogen)

$$E_{cell} = E^0 - \frac{0.0591}{n} \log \left[\frac{\text{Product}}{\text{Reactant}} \right]$$

$$E_{cell} = 1.23 - \frac{0.0591}{4} \log \frac{[P_{H_2}]^2}{[H^+]^4}; \quad [P_{H_2} = 1 \text{ bar given }]$$

$$E_{cell} = 1.23 - \frac{0.0591}{4} \times 4 \times \left[-\log \left(H^+ \right) \right]$$

But $pH = 5, H^+ = 10^{-5}M$

$$\therefore E_{cell} = 1.23 - 0.0591 \times \left[-\log \left(10^{-5} \right) \right]$$

$$E_{cell} = 1.23 - 0.0591 \times 5 \times 1$$

$$E_{cell} = 1.23 - 0.2955$$

$$E_{cell} = 0.9345 \approx 0.93$$

$$E_{cell} = 0.93 \text{ V}$$

Maths

Q. 1 For which of the following ordered pairs (μ, δ), the system of linear equations

$$x + 2y + 3z = 1$$

$$3x + 4y + 5z = \mu$$

$$4x + 4y + 4z = \varrho$$

is inconsistent ?

Option 1:
$(4, 6)$

Option 2:
$(3, 4)$

Option 3:
$(1, 0)$

Option 4:
$(4, 3)$

Correct Answer:
$(4, 3)$

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Solution:

Solution of System of Linear Equations Using Matrix Method -

let us consider n linear equations in n unknowns, given as below

$$
\begin{aligned}
&a_{11}x_1 + a_{12}x_2 + ... + a_{1n}x_n = b_1\\
&a_{21}x_1 + a_{22}x_2 + ... + a_{2n}x_n = b_2\\
&... \quad ... \quad ... \quad ... \qquad ... \qquad ...\\
&... \quad ... \quad ... \quad ... \qquad ... \qquad ...\\
&a_{n1}x_1 + a_{n2}x_2 + ... + a_{nn}x_n = b_n
\end{aligned}
$$

here $x_1, x_2, ...x_n$ are unknown variables
if $b_1 = b_2 = ... = b_n = 0$ then the system of equation is known as homogenous system of equation and if any of $b_1, b_2, ...b_n$ is not known then it is called non homogenous system of equation

The above system of equations can be written in matrix form as

$$
\begin{bmatrix}
a_{11} & a_{12} & ... & ... & a_{1n}\\
a_{21} & a_{22} & ... & ... & a_{2n}\\
... & ... & ... & ... & ...\\
... & ... & ... & ... & ...\\
a_{n1} & a_{n2} & ... & ... & a_{nn}
\end{bmatrix}
\begin{bmatrix} x_1\\ x_2\\ ...\\ ...\\ x_n \end{bmatrix}
=
\begin{bmatrix} b_1\\ b_2\\ ...\\ ...\\ b_n \end{bmatrix}
$$

$\Rightarrow AX = B$, where

$$
A = \begin{bmatrix}
a_{11} & a_{12} & ... & ... & a_{1n}\\
a_{21} & a_{22} & ... & ... & a_{2n}\\
... & ... & ... & ... & ...\\
... & ... & ... & ... & ...\\
a_{n1} & a_{n2} & ... & ... & a_{nn}
\end{bmatrix}, X = \begin{bmatrix} x_1\\ x_2\\ ...\\ ...\\ x_n \end{bmatrix}, B = \begin{bmatrix} b_1\\ b_2\\ ...\\ ...\\ b_n \end{bmatrix}
$$

Premultiplying equation AX=B by A^{-1}, we get

$A^{-1}(AX) = A^{-1}B \Rightarrow (A^{-1}A)X = A^{-1}B$

$\Rightarrow IX = A^{-1}B$

$\Rightarrow X = A^{-1}B$

$$\Rightarrow X = \frac{adjA}{|A|}B$$

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Types of equation :

1. **System of equations is non-homogenous:**

i. If $|A| \neq 0$, then the system of equations is consistent and has a unique solution $X = A^{-1}B$

ii. If $|A| = 0$ and $(\text{adj } A)\cdot B \neq 0$, then the system of equations is inconsistent and has no solution.

ii. If $|A| = 0$ and $(\text{adj } A)\cdot B = 0$, then the system of equations is consistent and has infinite number of solutions.

2. **System of equations is homogenous:**

i. If $|A| \neq 0$, then the system of equations has only trivial solution and it has one solution.

ii. If $|A| = 0$ then the system of equations has non-trivial solution and it has an infinite number of solution.

ii. If number of equation < number of unknown then it has non-trivial solution.

-

$$D = \begin{vmatrix} 3 & 4 & 5 \\ 1 & 2 & 3 \\ 4 & 4 & 4 \end{vmatrix}$$

$$= 2\begin{vmatrix} 3 & 4 & 5 \\ 1 & 2 & 3 \\ 2 & 2 & 2 \end{vmatrix}$$

$$R_1 \to R_1 - (R_2 + R_3)$$

$$D = \begin{vmatrix} 3 & 4 & 5 \\ 1 & 2 & 3 \\ 2 & 2 & 2 \end{vmatrix} = 0$$

Hence, it has infinitely many solutions

$$\mathrm{P}_3 \equiv \alpha \mathrm{P}_1 + \beta \mathrm{P}_2$$

$$3\alpha + \beta = 4 \ \ \& \ \ 4\alpha + 2\beta = 4 \Rightarrow \alpha = 2 \ \ \& \ \ \beta = -2$$

$$2\mu - 2 = \delta$$

For non infinite solution $2\mu - 2 \neq \delta$

Correct Option (4)

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Q. 2 Let $y = y(x)$ be a solution of the differential equation,
$\sqrt{1-x^2}\frac{dy}{dx} + \sqrt{1-y^2} = 0, |x| < 1.$

If $y\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$, then $y\left(\frac{-1}{\sqrt{2}}\right)$ is equal to :

Option 1:
$-\frac{1}{\sqrt{2}}$

Option 2:
$-\frac{\sqrt{3}}{2}$

Option 3:
$\frac{1}{\sqrt{2}}$

Option 4:
$\frac{\sqrt{3}}{2}$

Correct Answer:
$\frac{1}{\sqrt{2}}$

Solution:

Formation of Differential Equation and Solutions of a Differential Equation -

This is the **general solution** of the differential equation (2), which represents the family of the parabola (when a = 1) and one member of the family of parabola is given in Eq (1).

Also, Eq (1) is a **particular solution** of the differential equation (2).

The **solution** of the differential equation is a relation between the variables of the equation not containing the derivatives, but satisfying the given differential equation.

A **general solution** of a differential equation is a relation between the variables (not involving the derivatives) which contains the same number of the arbitrary constants as the order of the differential equation.

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Particular solution of the differential equation obtained from the general solution by assigning particular values to the arbitrary constant in the general solution.

-

$$\sqrt{1-x^2}\frac{dy}{dx} = -\sqrt{1-y^2}$$

$$\frac{dy}{\sqrt{1-y^2}} = -\frac{dx}{\sqrt{1-x^2}} \Rightarrow \sin^{-1} y = -\sin^{-1} x + c$$

$$x = \frac{1}{2}, y = \frac{\sqrt{3}}{2} \Rightarrow \frac{\pi}{2} = -\frac{\pi}{6} - c$$

$$\sin^{-1} y = \frac{\pi}{2} - \sin^{-1} x$$

$$= \cos^{-1} x \Rightarrow y = \sin(\cos^{-1} x)$$

$$y\left(\frac{1}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}}$$

Correct Option (3)

Q. 3 If a, b and c are the greatest values of ${}^{19}C_p, {}^{20}C_q, {}^{21}C_r$ respectively, then:

Option 1:

$$\frac{a}{11} = \frac{b}{22} = \frac{c}{42}$$

Option 2:

$$\frac{a}{10} = \frac{b}{11} = \frac{c}{42}$$

Option 3:

$$\frac{a}{11} = \frac{b}{22} = \frac{c}{21}$$

Option 4:

$$\frac{a}{10} = \frac{b}{11} = \frac{c}{21}$$

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Correct Answer:
$\frac{a}{11}=\frac{b}{22}=\frac{c}{42}$

Solution:
Binomial Coefficient of the middle term is greatest.

Now,

$^{n}C_{r}$ is max at middle term
$a={}^{19}C_{p}={}^{19}C_{10}={}^{19}C_{9}$
$b={}^{20}C_{q}={}^{20}C_{10}$
$c={}^{21}C_{r}={}^{21}C_{10}={}^{21}C_{11}$

$$\frac{a}{{}^{19}C_{9}}=\frac{b}{{}^{20}C_{10}}=\frac{c}{{}^{21}C_{11}}$$

$$\frac{a}{{}^{19}C_{9}}=\frac{b}{\frac{20}{10}\cdot{}^{19}C_{9}}=\frac{c}{\frac{21}{11}\cdot\frac{20}{10}{}^{19}C_{9}}$$

$$\frac{a}{1}=\frac{b}{2}=\frac{c}{42/11}$$
$$\frac{a}{11}=\frac{b}{22}=\frac{c}{42}$$

Correct Option (1)

Q. 4 Which one of the following is a tautology ?

Option 1:
$(P\wedge(P\rightarrow Q))\rightarrow Q$

Option 2:
$P\wedge(P\vee Q)$

Option 3:
$Q\rightarrow(P\wedge(P\rightarrow Q))$

Option 4:
$P\vee(P\wedge Q)$

Correct Answer:
$(P\wedge(P\rightarrow Q))\rightarrow Q$

Solution:

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Tautology And Contradiction -

Tautology

A compound statement is called **tautology** if it is always true for all possible truth values of its component statement.

For example, (p ⇒ q) ∨ (q ⇒ p)

Contradiction (fallacy)

A compound statement is called a **contradiction** if it is always false for all possible truth values of its component statement.

For example, ~(p ⇒ q) ∨ (q ⇒ p)

Truth Table

p	q	$p \to q$	$q \to p$	$(p \to q) \vee (q \to p)$	$\sim ((p \to q) \vee (q \to p))$
T	T	T	T	T	F
T	F	F	T	T	F
F	T	T	F	T	F
F	F	T	T	T	F

Quantifiers

Quantifiers are phrases like 'These exist' and "for every". We come across many mathematical statements containing these phrases.

For example –

p : For every prime number x, √x is an irrational number.

q : There exists a triangle whose all sides are equal.

-

Practise Session - 2 -

Q1. Write the truth table for the following statement pattern:

1. $\sim (\mathrm{p} \vee ((\sim \mathrm{q} \Rightarrow \mathrm{q}) \wedge \mathrm{q}))$
2. $(\mathrm{p} \wedge \mathrm{q}) \vee (\sim \mathrm{r})$

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$\mathbf{p}$	$\mathbf{q}$	$(\mathbf{p} \rightarrow \mathbf{q})$
F	F	$\mathbf{T}$
F	T	$\mathbf{T}$
T	F	F
T	T	$\mathbf{T}$

$\mathbf{p}$	$\mathbf{q}$	$(\mathbf{p} \wedge \mathbf{q})$
F	F	$\mathbf{F}$
F	T	F
T	F	F
T	T	T

$\mathbf{p}$	$\mathbf{q}$	$(p \wedge (p \rightarrow q)) \rightarrow q$
F	F	$\mathbf{T}$
F	T	$\mathbf{T}$
T	F	T
T	T	$\mathbf{T}$

Correct Option (1)

Q. 5 Let $f: R \rightarrow R$ be such that for all $x \epsilon R (2^{1+x} + 2^{1-x})$, $f(x)$ and $(3^x + 3^{-x})$ are in A.P., then the minimum value of $f(x)$ is :

Option 1:
0

Option 2:
4

Option 3:
3

Option 4:
2

Correct Answer:
3

Solution:
Important Properties of an AP

If a, b, c are in AP, then 2b = a + c

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and

AM ≥ GM

Applying this to given question,

$$2f(x) = \left(2\frac{1}{2^x} + 2.2^x\right) + \left(3^x + \frac{1}{3^x}\right)$$

$$Now,\ AM \geq GM$$

$$\frac{a+b}{2} \geq \sqrt{ab}$$
$$\frac{3^x + \frac{1}{3^x}}{2} \geq \sqrt{1}$$
$$\left(3^x + \frac{1}{3^x}\right) \geqslant 2$$

$$Value\ equals\ 2\ at\ \ 3^x = \frac{1}{3^x} \Rightarrow x = 0$$

$$Similarly, \frac{1}{2^x} + 2^x \geq 2, and\ this\ \ holds\ \ at\ \ x = 0$$

$$So, 2(minimum\ of\ \ f(x)) = 2(2) + (2)$$
$$\min\ \ f(x) = 3$$

Correct Option (3)

Q. 6 The locus of a point which divides the line segment joining the point $(0, -1)$ and a point on the parabola, $x^2 = 4y$, internally in the ratio $1 : 2$, is :

Option 1:
$9x^2 - 12y = 8$

Option 2:
$4x^2 - 3y = 2$

Option 3:
$x^2 - 3y = 2$

Option 4:
$9x^2 - 3y = 2$

Correct Answer:
$9x^2 - 12y = 8$

Solution:

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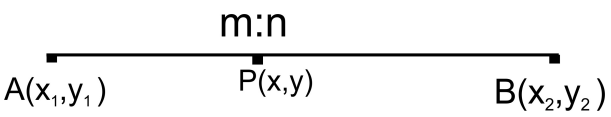
Section Formula -

Section Formula

1. **Internal division**

The coordinates of the point P (x, y) dividing the line segment joining the two points A (x_1, y_1) and B (x_2, y_2) internally in the ratio m : n is given by

$$\mathbf{x = \frac{mx_2 + nx_1}{m + n}}, \quad \mathbf{y = \frac{my_2 + ny_1}{m + n}}$$



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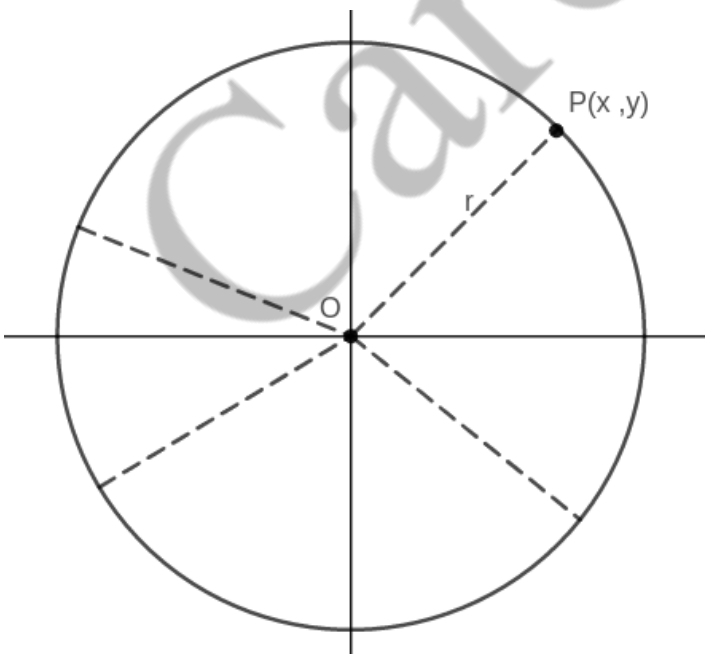
Locus and its Equation -

Locus and its Equation

When point move in a plane under certain geometrical conditions, then the point traces out a path, This path of the moving point is known as a locus.

For example, let a point O(0,0) is a fixed point (i.e. origin) and a variable point P (x, y) is in the same plane. If point P moves in such a way that the distance OP is constant r, then point P traces out a circle whose center is O(0, 0) and radius is r.



Steps to Finding the Equation of Locus

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1. Consider the point (h, k) whose locus is to be found.

2. Express the given condition as an equation in terms of the known quantities and unknown parameters.

3. Eliminate the parameters so that eliminant consists only locus coordinates h, k, and known quantities.

4. Now, replace the locus coordinate (h, k) with (x, y) in eliminant.

-

$\text{Let point } P \text{ be } \left(2t, t^2\right) \text{ and } Q \text{ be } (h, k)$

$h = \frac{2t}{3}, k = \frac{-2+t^2}{3}$

$\text{Hence locus is } 3k + 2 = \left(\frac{3h}{2}\right)^2 \Rightarrow 9x^2 = 12y + 8$

Correct Option (1)

Q. 7 For $a > 0$, let the curves $C_1 : y^2 = ax$ and $C_2 : x^2 = ay$ intersect at origin O and a point P. Let the line $x = b(0 < b < a)$ intersect the chord OP and the x-axis at points Q and R,respectively. If the line $x = b$ bisects the are bounded by the curves, C_1 and C_2, and the area of $\Delta OQR = \frac{1}{2}$, then 'a' satisfies the equation :

Option 1:
$x^6 - 12x^3 + 4 = 0$

Option 2:
$x^6 - 12x^3 - 4 = 0$

Option 3:
$x^6 + 6x^3 - 4 = 0$

Option 4:
$x^6 - 6x^3 + 4 = 0$

Correct Answer:
$x^6 - 12x^3 + 4 = 0$

Solution:

Parabola -

Parabola

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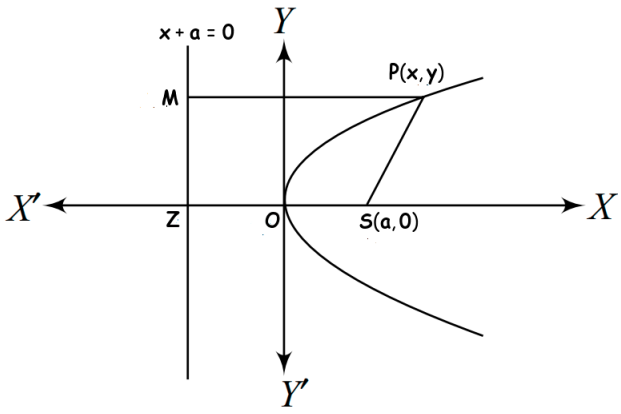
A parabola is the set of all points (x, y) in a plane that are the same distance from a fixed line, called the directrix, and a fixed point (the focus) not on the directrix in the plane.

Standard equation of a parabola



Let focus of parabola is S(a, 0) and directrix be x + a = 0, and axis as x-axis

P(x, y) is any point on the parabola.

Now, from the definition of the parabola,

$$\begin{aligned} & SP = PM \\ \Rightarrow \quad & SP^2 = PM^2 \\ \Rightarrow \quad & (x-a)^2 + (y-0)^2 = (x+a)^2 \\ \Rightarrow \quad & y^2 = 4ax \end{aligned}$$

which is the required equation of a standard parabola

-

Area Bounded by Curves When Intersects at More Than One Point -

Area bounded by the curves $y = f(x)$, $y = g(x)$ and intersect each other in the interval [a, b]

First find the point of intersection of these curves $y = f(x)$ and $y = g(x)$, let the point of intersection be x = c

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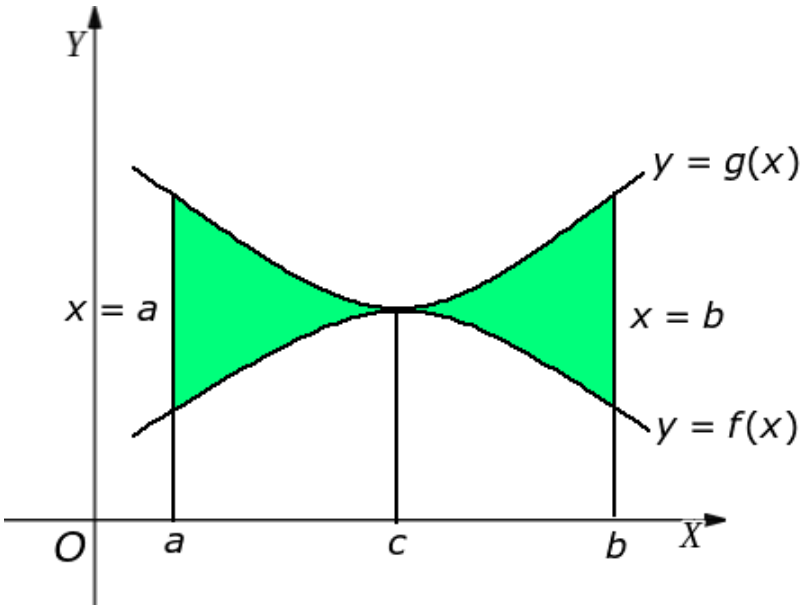


Area of the shaded region

$$= \int_a^c \{f(x) - g(x)\}dx + \int_c^b \{g(x) - f(x)\}dx$$

When two curves intersects more than one point

rea bounded by the curves *y=f(x), y=g(x)* and intersect each other at three points at *x = a, x = b amd x = c.*

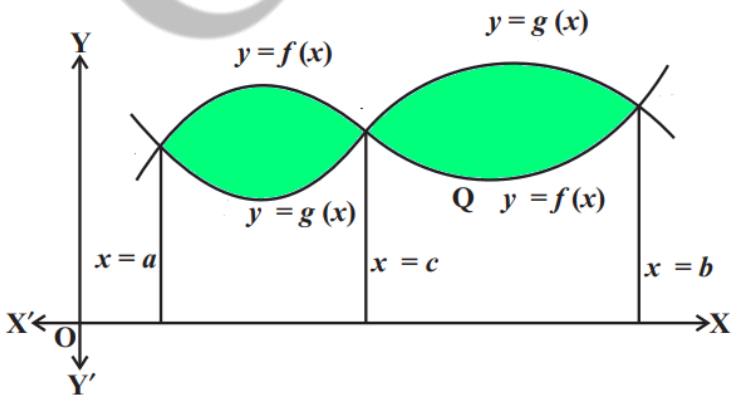
To find the point of intersection, solve *f(x) = g(x).*

For x ∈ (a, c), f(x) > g(x) and for x ∈ (c, b), g(x) > f(x).

Area bounded by curves,

$$\mathrm{A} = \int_a^b |f(x) - g(x)|\, dx$$

$$= \int_a^c (f(x) - g(x))\, dx + \int_c^b (g(x) - f(x))\, dx$$



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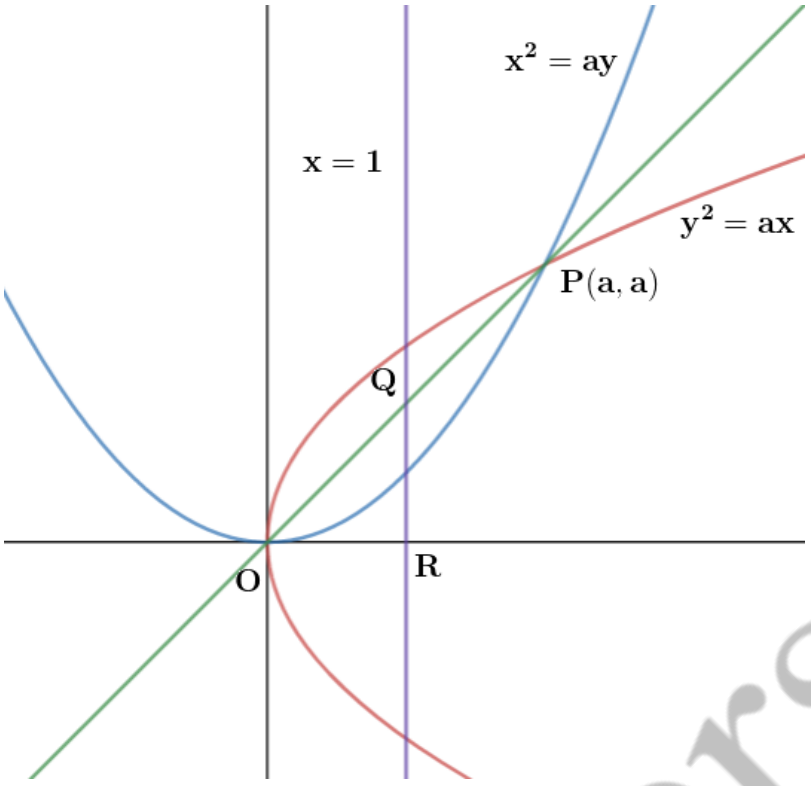
$\frac{1}{2}(b \times b) = \frac{1}{2}$
$b = 1$



$$\int_0^1 \left(\sqrt{a}\sqrt{x} - \frac{x^2}{a}\right) dx = \frac{a^2}{6} \text{ by property of parabola}$$

By solving above you will get

$$a^6 - 12a^3 + 4 = 0$$

Correct option (1)

Q. 8 The inverse function of

$$f(x) = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}}, x\epsilon(-1, 1), \text{is}$$

Option 1:

$$\frac{1}{4}(\log_8 e)\log_e\left(\frac{1-x}{1+x}\right)$$

Option 2:

$$\frac{1}{4}(\log_8 e)\log_e\left(\frac{1+x}{1-x}\right)$$

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Option 3:

$\frac{1}{4}\log_e\left(\frac{1+x}{1-x}\right)$

Option 4:

$\frac{1}{4}\log_e\left(\frac{1-x}{1+x}\right)$

Correct Answer:

$\frac{1}{4}(\log_8 e)\log_e\left(\frac{1+x}{1-x}\right)$

Solution:

Inverse of a function -

Function f: X → Y is invertible if and only if f is a bijective function.

The function g is called the inverse of f and is denoted by f^{-1} .

Let us consider a one-one function with domain A and range B. Where, A={1,2,3,4} and B={2,4,6,8} and f: A → B is given f(x)=2x, then write f and f^{-1} as a set of ordered pairs.

Here member y ? B arises from one and one member of x ? A.

So, f ={(1,2) (2,4) (3,6) (4,8)}

And f^{-1} = {(2,1) (4,2) (6,3) (8,4)}

In above definition domain of f = {1,2,3,4} = range of f^{-1}

Range of f = {2,4,6,8} = domain of f^{-1} .

Steps to find the inverse of a function:

i) First we write f(x) as y and equate y=f(x), where f(x) is a function in x

ii) Then we separate the variable x as the dependent variable and express it in terms of y by assuming y as the dependent variable

iii) Then we write f(y)=x where f(y) is a function in y

iv) And finally, we replace every y by x

Properties of an inverse function

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i) The inverse of a bijection is unique.

ii) if f: A → B is a bijection and g: B → A is the inverse of f, then $fog = I_B$ and $gof = I_A$, where I_A and I_B are identity functions on the sets A and B, respectively.

iii) inverse of a bijection is also a bijection.

iv) If f: A → B and g: B → C are two bijections, then $(\mathrm{got})^{-1} = \mathrm{f}^{-1}\mathrm{og}^{-1}$

-

$$f(x) = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}}$$
$$f(x) = \frac{8^{4x} - 1}{8^{4x} + 1} \text{put } 8^{4x} = t$$
$$y = \frac{t-1}{t+1}$$
$$yt + y = t - 1$$
$$\frac{y+1}{1-y} = t$$
$$\frac{y+1}{1-y} = 8^{4x}$$
$$\rightarrow ln\left(\frac{y+1}{1-y}\right) = 4x \ln 8$$
$$x = \frac{1}{4\ln 8} \ln\left(\frac{y+1}{1-y}\right)$$
$$f^{-1}(x) = \frac{1}{4\ln 8} \ln\left(\frac{x+1}{1-x}\right)$$

Correct Option (2)

Q. 9 $\lim\limits_{x\to 0}\left(\frac{3x^2+2}{7x^2+2}\right)^{1/x^2}$ is equal to :

Option 1:
E

Option 2:
$\frac{1}{e^2}$

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Option 3:

$\frac{1}{e}$

Option 4:

e^2

Correct Answer:

$\frac{1}{e^2}$

Solution:

Limits of the form (1 power infinity) -

Limits of the form 1^∞ (1 power infinity)

To find the limit of the exponential of form 1^∞ , we will use the following results

If $\lim_{x\to a}\ f(x) = \lim_{x\to a}\ g(x) = 0$

Then,

$\lim_{x\to a}\ [1+f(x)]^{\frac{1}{g(x)}} = e^{\lim_{x\to a}\frac{f(x)}{g(x)}}$.

Or

When, $\lim_{x\to a}\ f(x) = 1 and \lim_{x\to a}\ g(x) = \infty$

Then,

$$\lim_{x\to a}\ [f(x)]^{g(x)} = \lim_{x\to a}\ [1+f(x)-1]^{g(x)}$$
$$= e^{\lim_{x\to a}\,(f(x)-1)g(x)}$$

Some particular cases

$$(a)\quad \lim_{x\to 0}\ (1+x)^{\frac{1}{x}} = e$$

$$(b)\quad \lim_{x\to 0}\ \left(1+\frac{1}{x}\right)^{x} = e$$

$$(c)\quad \lim_{x\to 0}\ (1+cx)^{\frac{1}{x}} = e^{c}$$

$$(d)\quad \lim_{x\to 0}\ \left(1+\frac{c}{x}\right)^{x} = e^{c}$$

-

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Limits of the form (1 power infinity) -

Limits of the form 1^{∞} (1 power infinity)

To find the limit of the exponential of form 1^{∞} , we will use the following results

If $\lim_{x\to a}\ f(x) = \lim_{x\to a}\ g(x) = 0$

Then,

$$\lim_{x\to a}\ [1+f(x)]^{\frac{1}{g(x)}} = e^{\lim_{x\to a}\ \frac{f(x)}{g(x)}}.$$

Or

When, $\lim_{x\to a}\ f(x) = 1 and \lim_{x\to a}\ g(x) = \infty$

Then,

$$\lim_{x\to a}\ [f(x)]^{g(x)} = \lim_{x\to a}\ [1+f(x)-1]^{g(x)}$$

$$= e^{\lim_{x\to a}\ (f(x)-1)g(x)}$$

Some particular cases

$$(a)\quad \lim_{x\to 0}\ (1+x)^{\frac{1}{x}} = e$$

$$(b)\quad \lim_{x\to 0}\ \left(1+\frac{1}{x}\right)^{x} = e$$

$$(c)\quad \lim_{x\to 0}\ (1+cx)^{\frac{1}{x}} = e^{c}$$

$$(d)\quad \lim_{x\to 0}\ \left(1+\frac{c}{x}\right)^{x} = e^{c}$$

-

$$\lim_{x\to 0}\left(\frac{3x^2+2}{7x^2+2}\right)^{\frac{1}{x^2}} = \lim_{x\to 0}\left(1+\frac{3x^2+2}{7x^2+2}-1\right)^{\frac{1}{x^2}}$$

$$\lim_{x\to 0}\left(1+\frac{-4x^2}{7x^2+2}\right)^{1/x^2} = e^{\lim_{x\to 0}\left(-\frac{4x^2}{7x^2+2}\right)\left(\frac{1}{x^2}\right)}$$

$$= \frac{1}{e^2}$$

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Correct Option (2)

Q. 10 Let, $f(x) = \left(\sin(\tan^{-1} x) + \sin(\cot^{-1} x)\right)^2 - 1,$

$|x| > 1.$ $If \frac{dy}{dx} = \frac{1}{2}\frac{d}{dx}\left(\sin^{-1}(f(x))\right)$ and $y(\sqrt{3}) = \frac{\pi}{6}$, then $y(-\sqrt{3})$ is equal to :

Option 1:

$\frac{\pi}{3}$

Option 2:

$\frac{2\pi}{3}$

Option 3:

$-\frac{\pi}{6}$

Option 4:

$\frac{5\pi}{6}$

Correct Answer:

$\frac{5\pi}{6}$

Solution:

Integration as Reverse Process of Differentiation -

Integration is the reverse process of differentiation. In integration, we find the function whose differential coefficient is given.

For example,

$$\frac{\mathrm{d}}{\mathrm{dx}}(\sin \mathrm{x}) = \cos \mathrm{x}$$

$$\frac{\mathrm{d}}{\mathrm{dx}}\left(\mathrm{x}^2\right) = 2\mathrm{x}$$

$$\frac{\mathrm{d}}{\mathrm{dx}}\left(\mathrm{e}^{\mathrm{x}}\right) = \mathrm{e}^{\mathrm{x}}$$

In the above example, the function cos x is the derived function of sin x. We say that sin x is an anti derivative (or an integral) of cos x. Similarly, x^2 and e^x are the anti derivatives (or integrals) of 2x and e^x respectively.

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Also note that the derivative of a constant (C) is zero. So we can write the above examples as:

$$\frac{d}{dx}(\sin x + c) = \cos x$$
$$\frac{d}{dx}\left(x^2 + c\right) = 2x$$
$$\frac{d}{dx}\left(e^x + c\right) = e^x$$

Thus, anti derivatives (or integrals) of the above functions are not unique. Actually, there exist infinitely many anti derivatives of each of these functions which can be obtained by selecting C arbitrarily from the set of real numbers.

For this reason C is referred to as arbitrary constant. In fact, C is the parameter by varying which one gets different anti derivatives (or integrals) of the given function.

If the function F(x) is an antiderivative of f(x), then the expression F(x) + C is the indefinite integral of the function f(x) and is denoted by the symbol ∫ f(x) dx.

By definition,

$$\int f(x)dx = F(x) + c, \quad \text{where } F'(x) = f(x) \text{ and } 'c' \text{ is constant.}$$

-

Trigonometric Identities -

Trigonometric Identities-

These identities are the equations that hold true regardless of the angle being chosen.

$$\sin^2 t + \cos^2 t = 1$$
$$1 + \tan^2 t = \sec^2 t$$
$$1 + \cot^2 t = \csc^2 t$$
$$\tan t = \frac{\sin t}{\cos t}, \quad \cot t = \frac{\cos t}{\sin t}$$

-

Principal Value of function f-1 (f (x)) -

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Principal Value of function $f^{-1}(f(x))$

1. $\sin^{-1}(\sin(\theta)) = \theta$ for all $\theta \in [-\pi/2, \pi/2]$
2. $\cos^{-1}(\cos(\theta)) = \theta$ for all $\theta \in [0, \pi]$
3. $\tan^{-1}(\tan(\theta)) = \theta$ for all $\theta \in (-\pi/2, \pi/2)$
4. $\cot^{-1}(\cot(\theta)) = \theta$ for all $\theta \in (0, \pi)$
5. $\sec^{-1}(\sec(\theta)) = \theta$ for all $\theta \in [0, \pi] - \{\pi/2\}$
6. $\csc^{-1}(\csc(\theta)) = \theta$ for all $\theta \in [-\pi/2, \pi/2] - \{0\}$

-

$$y = \frac{1}{2}\sin^{-1}(f(x)) + c$$

$$\text{Let } \theta = \tan^{-1} x \Rightarrow \cot^{-1} x = \tfrac{\pi}{2} - \theta$$

$$f(x) = \left\{\sin\theta + \sin\left(\tfrac{\pi}{2} - \theta\right)\right\}^2 - 1$$

$$\sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta - 1 = \sin 2\theta$$

$$y = \frac{1}{2}\sin^{-1}\left(\sin\left(2\tan^{-1} x\right)\right) + C$$

$$\frac{\pi}{6} = \frac{1}{2}\sin^{-1}\left(\sin\left(2\frac{\pi}{3}\right)\right) + C$$

$$\frac{\pi}{6} = \frac{1}{2}\frac{\pi}{3} + C \Rightarrow C = 0$$

$$y = \frac{1}{2}\sin^{-1}\left(\sin\left(-2\frac{\pi}{3}\right)\right) = -\frac{\pi}{6}$$

Correct Option (4)

Q. 11 If the equation, $x^2 + bx + 45 = 0 (b \epsilon R)$ has conjugate complex roots and they satisfy $|z + 1| = 2\sqrt{10}$, then:

Option 1:
$b^2 + b = 12$

Option 2:
$b^2 - b = 42$

Option 3:
$b^2 - b = 30$

Option 4:
$b^2 + b = 72$

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Correct Answer:

$b^2 - b = 30$

Solution:

Nature of Roots -

Let the quadratic equation is $ax^2 + bx + c = 0$

D is the discriminant of the equation.

i) if D < 0, then root are in the form of complex number,

If a,b,c ∈ R (real number) then roots will be conjugate of each other, means if p + iq is one of the roots then other root will be p - iq

-

Let $z = \alpha \pm i\beta$ be roots of the equation

so $2\alpha = -b \text{ and } \alpha^2 + \beta^2 = 45, (\alpha + 1)^2 + \beta^2 = 40$

So, $(\alpha + 1)^2 - \alpha^2 = -5$

$\Rightarrow \quad 2\alpha + 1 = -5 \quad \Rightarrow \quad 2\alpha = -6$

$\text{so b} = 6$

$Hence, \quad b^2 - b = 30$

Correct Option 3

Q. 12 The mean and the standard deviation (s.d.) of 10 observations are 20 and 2 respectively. Each of these 10 observations is multiplied by p and then reduced by q, where $p \neq 0$ and $q \neq 0$. If the new mean and new s.d. become half of their original values, then q is equal to :

Option 1:

-20

Option 2:

-5

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Option 3:

10

Option 4:

-10

Correct Answer:

-20

Solution:

Some Important Point Regarding Statistics -

Some Important Point Regarding Statistics

1. The sum of the deviation of an observation from their mean is equal to zero. i.e. $\sum_{i=1}^{n}(x_i - \bar{x}) = 0.$
2. The sum of the square of the deviation from the mean is minimum, i.e. $\sum_{i=1}^{n}(x_i - \bar{x})^2 \text{ is least.}$
3. The mean is affected accordingly if the observations are given a mathematical treatment i.e. addition, subtraction multiplication by a constant term.
4. If set of n_1 observations has mean $\bar{x}_1$ and set of n_2 observations has mean $\bar{x}_2$, then their combined mean is $\frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$ and the .
5. If set of n_1 observations has mean $\bar{x}_1$ and set of n_2 observations has mean $\bar{x}_2$ then their combined variance is given by

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$$

where, $d_1 = \bar{x}_2 - \bar{x} \quad \text{and} \quad d_2 = \bar{x}_1 - \bar{x}$

$$\sigma_1^2 = \frac{1}{n_1}\sum_{i=1}^{n_1}(x_{1i} - \overline{x_1})^2$$

$$\sigma_2^2 = \frac{1}{n_2}\sum_{j=1}^{n_2}(x_{2j} - \overline{x_2})^2$$

$\bar{x}_1, \quad \bar{x}_2$are the means and $\sigma_1, \quad \sigma_2$ are the standard deviations of two series.

-

$$\text{old series} \Rightarrow x_{old} = x_1, x_2, x_3 \ldots\ldots x_n$$

Old Mean

$$\overline{x}_{old} = \frac{(x_1 + x_2 + x_3 \ldots\ldots x_n)}{n} = 20$$

Old Standard Deviation

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$$\sigma_{old}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

If each observation is multiplied with p & then q is subtracted

$$\text{new series} \Rightarrow x_{new} = px_1 - q,\ px_2 - q,\ px_3 - q \ldots\ldots\ px_n - q$$

New mean is the half of old mean

$$\overline{x}_{new} = \frac{p\left(x_1 + x_2 + x_3 \ldots\ldots x_n\right) - nq}{n} = 10$$

$$\overline{x}_{new} = p\,\overline{x}_{old} - q = 10$$

$$\overline{x}_{new} = 20p - q = 10 \qquad \ldots(i)$$

and new standard deviations is the half of old mean

$$\sigma_{new}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_{new} - \bar{x}_{new})^2$$

$$\sigma_{new}^2 = \frac{1}{n}\sum_{i=1}^{n}\left[(px_i - q) - (p\bar{x}_{old} - q)\right]^2$$

$$\sigma_{new}^2 = \frac{1}{n}\sum_{i=1}^{n}(px_i - p\bar{x}_{old})^2$$

$$\sigma_{new}^2 = \frac{1}{n}p^2\sum_{i=1}^{n}(x_i - \bar{x}_{old})^2$$

$$\sigma_{new} = |p|\sigma_{old}$$

$$1 = 2|p|$$

$$p = \pm 0.5$$

Case 1 $p = 0.5$

Using equation 1

we get

$$20p - q = 10$$
$$10 - q = 10$$
$$q = 0$$

Case 1 $p = -0.5$

Using equation 1

we get

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$20p - q = 10$
$-10 - q = 10$
$q = -20$

Correct Option (1)

Q. 13 If $\int \frac{\cos\ x\ dx}{\sin^3 x(1+\sin^6 x)^{2/3}} = f(x)(1+\sin^6 x)^{1/\lambda} + c$ where c is a constant of integration, then $\lambda f\left(\frac{\pi}{3}\right)$ is equal to :

Option 1:
$-\frac{9}{8}$

Option 2:
$\frac{9}{8}$

Option 3:
2

Option 4:
-2

Correct Answer:
-2

Solution:

Integration Using Substitution -

The method of substitution is one of the basic methods for calculating indefinite integrals.

Substitution - change of variable

To solve the integrate of the form

$$I = \int f\,(g(x)) \cdot g'(x)\,dx,$$

where $g(x)$ is continuously differentiable function.
put $g(x) = t,\ \ g'(x)\,dx = dt$
After substitution, we get $\int f(t)\,dt.$
Evalute this integration and substitute back the value of t.

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$\sin x = t \quad \Rightarrow \cos x dx = dt$

$$\int \frac{dt}{t^2\left(1+t^6\right)^{2/3}} = \int \frac{dt}{t^3 t^4\left(\frac{1}{t^6}+1\right)^{2/3}}$$

$$u = \frac{1}{t^6} + 1 \Rightarrow du = -\frac{6}{t^7}dt$$

$$\frac{du}{-6} = \frac{dt}{t^7}$$

$$= \int \frac{du}{-6u^{2/3}} = -\frac{1}{2}u^{1/3} + C$$

$$= -\frac{1}{2}\left(\frac{1}{t^6}+1\right)^{1/3} + C$$

$$= -\frac{1}{2}\left(\frac{(1+\sin^6 x)^{1/3}}{\sin^2 x}\right)$$

$$f(x) = -\frac{1}{2}\frac{1}{\sin^2 x} \quad \lambda = 3$$

$$\lambda f(\pi/3) = -2$$

Correct option (4)

Q. 14 Let A and B be two independent events such that $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{6}$. Then, which of the following is TRUE?

Option 1:

$P(B/A) = \frac{1}{2}$

Option 2:

$P\left(A/B'\right) = \frac{1}{3}$

Option 3:

$P\left(A/B\right) = \frac{2}{3}$

Option 4:

$P\left(A'/B'\right) = \frac{1}{3}$

Correct Answer:

$P\left(A/B'\right) = \frac{1}{3}$

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Solution:
A & B are independent events

So, A & B' are also independent events, and hence

$$P\left(\frac{A}{B'}\right) = P(A) = \frac{1}{3}$$

Correct Option (2)

Q. 15 Let the volume of a parallelopiped whose coterminous edges are given by $\vec{u} = \hat{i} + \hat{j} + \lambda\hat{k}, \vec{v} = \hat{i} + \hat{j} + 3\hat{k}$ and $\vec{w} = 2\hat{i} + \hat{j} + \hat{k}$ be 1 cu. unit. If θ be the angle between the edges $\vec{u}$ and $\vec{w}$, then $\cos\Theta$ can be :

Option 1:
$\frac{7}{6\sqrt{6}}$

Option 2:
$\frac{5}{7}$

Option 3:
$\frac{7}{6\sqrt{3}}$

Option 4:
$\frac{5}{3\sqrt{3}}$

Correct Answer:
$\frac{7}{6\sqrt{3}}$

Solution:

Dot (Scalar) Product in Terms of Components -

Angle between two vectors

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$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$$

$$\Rightarrow \quad \cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$$

$$\Rightarrow \quad \theta = \cos^{-1}\left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}\right)$$

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

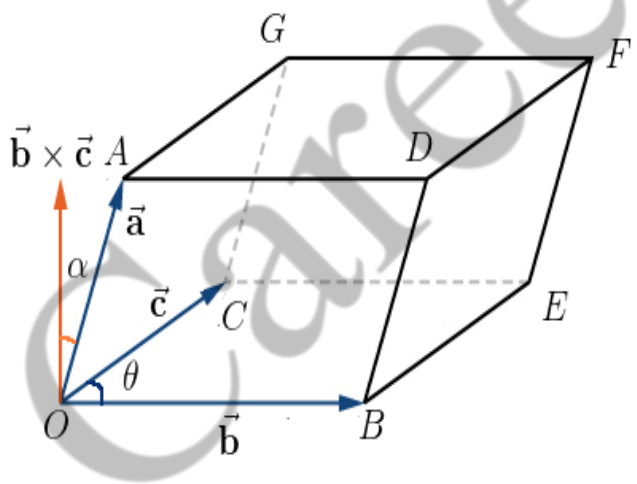
$$\Rightarrow \quad \theta = \cos^{-1}\left(\frac{a_1b_1 + a_2b_2 + a_3b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2}\sqrt{b_1^2 + b_2^2 + b_3^2}}\right)$$

-

Geometrical Interpretation of Scalar Triple Product -

Let vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ represented the sides of a parallelepiped OA, OB and OC respectively. Then, $\vec{b} \times \vec{c}$ is a vector perpendicular to the plane of $\vec{b}$ and $\vec{c}$. Let Θ be the angle between vectors $\vec{b}$ and $\vec{c}$ and α be the angle between $\vec{a}$ and $\vec{b} \times \vec{c}$.

If $\hat{n}$ is a unit vector along $\vec{b} \times \vec{c}$, then α is the angle between $\hat{n}$ and $\vec{a}$.



$$\begin{aligned}\left[\vec{a}\ \vec{b}\ \vec{c}\right] &= \vec{c} \cdot \left(\vec{b} \times \vec{c}\right) \\ &= \vec{a} \cdot (\mathbf{bc}\sin\theta\ \hat{\mathbf{n}}) \\ &= (\mathbf{bc}\sin\theta)(\vec{a} \cdot \hat{\mathbf{n}}) \\ &= (\mathbf{bc}\sin\theta)(\mathbf{a} \cdot \mathbf{1} \cdot \cos\alpha) \\ &= (\mathbf{a} \cdot \cos\alpha)(\mathbf{bc}\sin\theta) \\ &= (\text{Height}) \cdot (\text{Area of Base}) \\ &= \text{Volume of parallelepiped}\end{aligned}$$

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$$\pm 1 = \begin{vmatrix} 1 & 1 & \lambda \\ 1 & 1 & 3 \\ 2 & 1 & 1 \end{vmatrix} \Rightarrow = -\lambda + 3 = \pm 1 \Rightarrow \lambda = 2 \text{ or } \lambda = 4$$

$$\text{For } \lambda = 4 \frac{1}{90}$$

$$\cos\theta = \frac{2+1+4}{\sqrt{6}\sqrt{18}} = \frac{7}{6\sqrt{3}}$$

Correct Option (3)

Q. 16 Let two points be $A(1, -1)$ and $B(0, 2)$. If a point $P(x', y')$ be such that the area of $\Delta PAB = 5\ sq.$ units and it lies on the line, $3x + y - 4\lambda = 0$, then a value of λ is :

Option 1:

4

Option 2:

1

Option 3:

-3

Option 4:

3

Correct Answer:

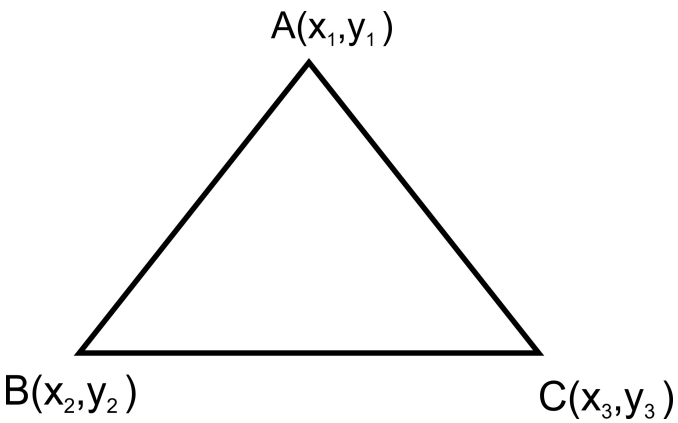
3

Solution:

Area of Triangle -

Area of Triangle

If vertices of a triangle ABC given as A (x_1, y_1), B (x_2, y_2) and C(x_3, y_3) then area of ΔABC is



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$$\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Area of an n-sided polygon

The area of a polygon whose vertices are (x_1, y_1), (x_2, y_2), (x_3, y_3).....(x_n, y_n) taken in cyclic order is given by

$$= \frac{1}{2}\left|\{(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + \ldots\ldots + (x_ny_1 - x_1y_n)\}\right|$$

-

$$D = \frac{1}{2}\begin{vmatrix} 0 & 2 & 1 \\ 1 & -1 & 1 \\ x' & y' & 1 \end{vmatrix}$$

$$-2\,(1 - x') + (y' + x') = \pm 10$$

$$-2 + 2x' + y' + x' = \pm 10$$

$$3x' + y' = 12 \text{ or } 3x' + y' = -8$$

$$\lambda = 3, -2$$

Correct Option (4)

Q. 17 The shortest distance between the lines

$$\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1} \text{ and } \frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4} \text{ is :}$$

Option 1:

$2\sqrt{30}$

Option 2:

$\frac{7}{2}\sqrt{30}$

Option 3:

3

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Option 4:

$3\sqrt{30}$

Correct Answer:

$3\sqrt{30}$

Solution:

Shortest Distance between Two Lines -

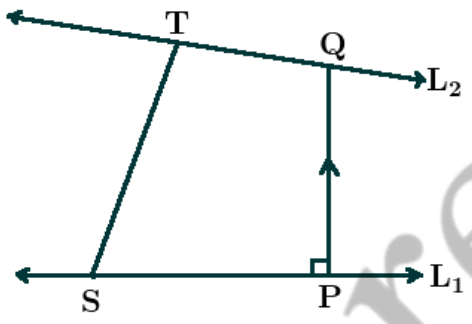
Distance between two skew lines

If L_1 and L_2 are two skew lines, then there is one and only one line perpendicular to each of lines L1 and L2 which is known as the line of shortest distance.

Vector form

$L_1: \quad \vec{\mathbf{r}} = \vec{\mathbf{r}}_0 + \lambda\vec{\mathbf{b}}$

$L_2: \quad \vec{\mathbf{r}} = \vec{\mathbf{r}'}_0 + \mu\vec{\mathbf{b}'}$



If $\overrightarrow{PQ}$ is the shortest distance vector between L_1 and L_2, then it being perpendicular to both $\vec{\mathbf{b}}$ and $\vec{\mathbf{b}'}$, therefore, the unit vector $\hat{\mathbf{n}}$ along $\overrightarrow{PQ}$ would be

$$\hat{\mathbf{n}} = \frac{\vec{\mathbf{b}} \times \vec{\mathbf{b}'}}{\left|\vec{\mathbf{b}} \times \vec{\mathbf{b}'}\right|}$$

Then, $\overrightarrow{PQ} = d\hat{\mathbf{n}}$

where "d" is the magnitude of the shortest distance vector. Let θ be the angle between $\overrightarrow{ST}$ and $\overrightarrow{PQ}$.

Then

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$$\mathrm{PQ} = \mathrm{ST}\,|\cos\theta|$$

but, $$\cos\theta = \left|\frac{\overrightarrow{\mathrm{PQ}}\cdot\overrightarrow{\mathrm{ST}}}{|\overrightarrow{\mathrm{PQ}}||\overrightarrow{\mathrm{ST}}|}\right|$$

$$= \left|\frac{d\hat{n}\cdot\left(\vec{r'}_0 - \vec{r}_0\right)}{d\,\mathrm{ST}}\right| \quad \left(\text{since } \overrightarrow{\mathrm{ST}} = \vec{r'}_0 - \vec{r}_0\right)$$

$$= \left|\frac{\left(\vec{b}\times\vec{b'}\right)\cdot\left(\vec{r'}_0 - \vec{r}_0\right)}{\mathrm{ST}\left|\vec{b}\times\vec{b'}\right|}\right|$$

Hence, the required shortest distance is

$$d = \mathrm{PQ} = \mathrm{ST}|\cos\theta|$$

or $$\mathbf{d} = \left|\frac{\left(\vec{\mathbf{b}}\times\vec{\mathbf{b'}}\right)\cdot\left(\vec{\mathbf{r}}'_0 - \vec{\mathbf{r}}_0\right)}{\left|\vec{\mathbf{b}}\times\vec{\mathbf{b'}}\right|}\right|$$

-

$\overrightarrow{\mathrm{AB}} = 6\hat{i} + 15\hat{j} + 3\hat{k}$
$\vec{p} = 3\hat{i} - \hat{j} + \hat{k}$
$\vec{q} = -3\hat{i} + 2\hat{j} + 4\hat{k}$

$$\vec{p}\times\vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 1 \\ -3 & 2 & 4 \end{vmatrix} = \hat{i}(-4-2) - \hat{j}(12+3) + \hat{k}(6-3)$$

$\Rightarrow -6\hat{i} - 15\hat{j} + 3\hat{k}$

$$\text{shortest dist} = \frac{|\overrightarrow{\mathrm{AB}}\cdot(\overrightarrow{\mathrm{p}}\times\overrightarrow{\mathrm{q}})}{|\overrightarrow{\mathrm{p}}\times\overrightarrow{\mathrm{q}}|}$$

$=3\sqrt{30}$

Correct Option (4)

Q. 18 Let the line $y = mx$ and the ellipse $2x^2 + y^2 = 1$ intersect at a point P in the first quadrant. If the normal to this ellipse at P meets the co-ordinate axes at $\left(-\frac{1}{3\sqrt{2}}, 0\right)$ and $(0, \beta)$, then β is equal to :

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Option 1:

$\frac{2}{\sqrt{3}}$

Option 2:

$\frac{2}{3}$

Option 3:

$\frac{2\sqrt{2}}{3}$

Option 4:

$\frac{\sqrt{2}}{3}$

Correct Answer:

$\frac{\sqrt{2}}{3}$

Solution:

Equation of Normal in Point Form and Parametric Form -

Equation of Normal in Point Form and Parametric Form

Point form:

The equation of normal at (x_1, y_1) to the ellipse, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2$.

-

Let P be (x_1, y_1)

Equation of normal at P is $\frac{x}{2x_1} - \frac{y}{y_1} = -\frac{1}{2}$

It passes through $\left(-\frac{1}{3\sqrt{2}}, 0\right) \Rightarrow \frac{-1}{6\sqrt{2}x_1} = -\frac{1}{2} \Rightarrow x_1 = \frac{1}{3\sqrt{2}}$

So $y_1 = \frac{2\sqrt{2}}{3}$ (as P lies in 1 'quadrant)

So $\beta = \frac{y_1}{2} = \frac{\sqrt{2}}{3}$

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Correct option (4)

Q. 19 If c is a point at which Roll's theorem holds for the function, $f(x) = \log_e\left(\frac{x^2+\alpha}{7x}\right)$ in the interval $[3, 4]$, where $\alpha \in R$ then $f''(c)$ is equal to :

Option 1:
$-\frac{1}{24}$

Option 2:
$-\frac{1}{12}$

Option 3:
$\frac{\sqrt{3}}{7}$

Option 4:
$\frac{1}{12}$

Correct Answer:
$\frac{1}{12}$

Solution:

Rolle's Theorem -

Rolle's Theorem

Informally, Rolle's theorem states that if the outputs of a differentiable function f are equal at the endpoints of an interval, then there must be an interior point c where f ′(c) = 0

Rolle's Theorem Statement:

Let f be a real-valued function defined on the closed interval [a, b] such that

1. it is continuous on the closed interval [a, b],
2. it is differentiable on the open interval (a, b), and
3. f (a) = f (b).

There then exists at least one c ∈ (a, b) such that f ′(c) = 0.

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$$f(3) = f(4)$$
$$\frac{9+\alpha}{3} = \frac{16+\alpha}{4}$$

$$\alpha = 12$$

$$f'(x) = \frac{x^2-12}{x\left(x^2+12\right)}$$
$$f'(c) = 0$$
$$\therefore \quad c = \sqrt{12}$$
$$\therefore \quad f''(c) = \tfrac{1}{12}$$

Correct Option (4)

Q. 20 Let $f(x) = x\ \cos^{-1}\left(-\sin|x|\right), x\,\epsilon\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, then which of the following is true ?

Option 1:
$f'(0) = -\frac{\pi}{2}$,

Option 2:
f' is decreasing in $\left(-\frac{\pi}{2}, 0\right)$ and increasing in $\left(0, \frac{\pi}{2}\right)$

Option 3:
f is not differentiable at $x = 0$

Option 4:
f' is increasing in $\left(-\frac{\pi}{2}, 0\right)$ and decreasing in $\left(0, \frac{\pi}{2}\right)$

Correct Answer:
f' is decreasing in $\left(-\frac{\pi}{2}, 0\right)$ and increasing in $\left(0, \frac{\pi}{2}\right)$

Solution:

Application of Monotonicity (Part 1) -

-

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$$f'(x) = x\left(\pi - \cos^{-1}(\sin|x|)\right)$$
$$= x\left(\pi - \left(\tfrac{\pi}{2} - \sin^{-1}(\sin|x|)\right)\right)$$
$$= x\left(\tfrac{\pi}{2} + |x|\right)$$

$$f(x) = \begin{cases} x\left(\frac{\pi}{2} + x\right) & x \geq 0 \\ x\left(\frac{\pi}{2} - x\right) & x < 0 \end{cases}$$
$$f'(x) = \begin{cases} \frac{\pi}{2} + 2x & x \geq 0 \\ \frac{\pi}{2} - 2x & x < 0 \end{cases}$$

$f'(x)$ is increasing in $\left(0, \frac{\pi}{2}\right)$ and decreasing in $\left(\frac{-\pi}{2}, 0\right)$

Correct Option (2)

Q. 21 An urn contains 5 red marbles, 4 black marbles, and 3 white marbles. Then the number of ways in which 4 marbles can be drawn so that at most three of them are red is:

Correct Answer:
490

Solution:
Number of red balls selected can be 0 or 1 or 2 or 3

So, the following cases are possible

$$\begin{array}{lcccc} \text{Red Balls}(5): & 0 & 1 & 2 & 3 \\ \text{Other Balls}(7): & 4 & 3 & 2 & 1 \\ & {}^5C_0 \times {}^7C_4 & {}^5C_1 \times {}^7C_3 & {}^5C_2 \times {}^7C_2 & {}^5C_3 \times {}^7C_1 \end{array}$$

$$= 1 \times 35 + 5 \times 35 + 10 \times 21 + 10 \times 7 = 490$$

Q. 22 Let the normal at a point P on the curve $y^2 - 3x^2 + y + 10 = 0$ intersect the y-axis at $\left(0, \frac{3}{2}\right)$. If m is the slope of the tangent at P to the curve, then $|m|$ is equal to

---

Correct Answer:
4

Solution:

Equation of Straight Line (Part 1) -

Equation of Straight Line

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(b) Point-Slope form

Let the equation of give line l with slope ‘m’ is

$y = mx + c$ .....(i)

(x_1,y_1) lies on the line i

$y_1 = mx_1 + c$ .....(ii)

From (i) and (ii) [(ii) - (i)]

$y - y_1 = m(x - x_1)$

The equation of a straight line whose slope is given as ‘m’ and passes through the point (x_1,y_1) is $\mathbf{y - y_1 = m(x - x_1)}$.

-

$$2yy' + y' - 6x = 0$$
$$y' = \frac{6x}{2y+1}$$
$$\frac{-1}{y'} = \frac{-(2y+1)}{6x} (\text{slope of the normal})$$

Equation of the normal $y - y_1 = \frac{-(2y_1+1)}{6x_1}(x - x_1)$

Normal intersect at (0,3/2)

$$\frac{3}{2} - y_1 = \frac{-(2y_1+1)}{6x_1}(0 - x_1)$$
$$8y_1 - 8 = 0$$
$$y_1 = 1$$
$$x_1 = \pm 2$$
$$|m| = 4$$

Q. 23 The least positive value of 'a' for which the equation, $2x^2 + (a-10)x + \frac{33}{2} = 2a$ has real roots is

Option 1:
8

Option 2:
6

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Option 3:
4

Option 4:
2

Correct Answer:
8

Solution:

Nature of Roots -

Let the quadratic equation is $ax^2 + bx + c = 0$

D is the discriminant of the equation.

ii) If D > 0, then roots will be real and distinct.

$$x_1 = \frac{-b + \sqrt{D}}{2a} \text{ and } x_2 = \frac{-b - \sqrt{D}}{2a}$$

$$\text{Then, } ax^2 + bx + c = a(x - x_1)(x - x_2)$$

iii) if roots D = 0, then roots will be real and equal, then

$$x_1 = x_2 = \frac{-b}{2a}$$

$$\text{Then, } ax^2 + bx + c = a(x - x_1)^2$$

-

$$D \geqslant 0$$

$$(a-10)^2 - 8\left(\frac{33}{2} - 2a\right) \geq 0$$

$$a^2 + 100 - 20a - 132 + 16a \geq 0$$

$$a^2 - 4a - 32 \geqslant 0$$

$$a^2 - 8a + 4a - 32 \geq 0$$

$$(a+4)(a-8) \geq 0$$

$$a \leq -4 \text{ or } a \geq 8$$

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least positive value is 8.

Q. 24 The sum $\sum_{k=1}^{20}(1+2+3+.....+k)$ is

Option 1:
1540

Option 2:
1680

Option 3:
1260

Option 4:
1450

Correct Answer:
1540

Solution:
Sum of Common Series

Sum of the square of first n-natural numbers

$$1^2+2^2+3^2+4^2+.....+n^2=\frac{n(n+1)(2n+1)}{6}$$

Now,

$$\sum_{k=1}^{20}\frac{k(k+1)}{2}=\frac{1}{2}\left(\sum_{k=1}^{20}k^2+\sum_{k=1}^{20}k\right)$$

$$=\frac{1}{2}\left(\frac{20(20+1)(2\times20+1)}{6}+\frac{20(20+1)}{2}\right)$$

$$=\frac{1}{2}((70\times41)+210)=1540$$

Q. 25 The number of all 3×3 matrices A, with enteries from the set $\{-1,0,1\}$ such that the sum of the diagonal elements of AA^T is 3, is

Option 1:
672

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Option 2:
512

Option 3:
1024

Option 4:
256

Correct Answer:
672

Solution:
Let matrix A be

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$A^T = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$$

$$\text{trace}(AA^T) = a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 = 3$$

So out of 9 elements, 3 elements must be equal to 1 or −1, and the rest elements must be 0.

Possible cases

$$\begin{aligned} 0,0,0,0,0,0,1,1,1 &\rightarrow \text{Total possibilities} = {}^9C_6 \\ 0,0,0,0,0,0,-1,-1,-1 &\rightarrow \text{Total possibilities} = {}^9C_6 \\ 0,0,0,0,0,0,1,1,-1 &\rightarrow \text{Total possibilities} = {}^9C_6 \times 3 \\ 0,0,0,0,0,0,-1,1,-1 &\rightarrow \text{Total possibilities} = {}^9C_6 \times 3 \end{aligned}$$

$$\text{Total number of cases} = {}^9C_6 \times 8 = 672$$

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9th January Shift-1

Physics

Q. 1 An electric dipole of moment $\vec{p} = (-\hat{i} - 3\hat{j} + 2\hat{k}) \times 10^{-29} Cm$ is at the origin (0,0,0). The electric field due to this dipole at $\vec{r} = +\hat{i} + 3\hat{j} + 5\hat{k}$ (note that $\vec{r} \cdot \vec{p} = 0$) is parallel to:

Option 1:
$+\hat{i} - 3\hat{j} - 2\hat{k}$

Option 2:
$-\hat{i} - 3\hat{j} + 2\hat{k}$

Option 3:
$+\hat{i} + 3\hat{j} - 2\hat{k}$

Option 4:
$-\hat{i} + 3\hat{j} - 2\hat{k}$

Correct Answer:
$+\hat{i} + 3\hat{j} - 2\hat{k}$

Solution:
$\vec{p} \cdot \vec{r} = 0$

E must be antiparallel to p

$$\vec{E} = -\lambda \vec{p}$$

$$\vec{A} = ai + bj + ck$$
$$\Rightarrow \frac{a}{\lambda} = \frac{b}{3\lambda} = \frac{-c}{2\lambda} = k$$
$$\Rightarrow A = \lambda(i + 3j - 2k)$$

Hence the correct option is (3).

Q. 2 Consider two ideal diatomic gases A and B at same temperature T. Molecules of the gas A are rigid and have a mass m. Molecules of the gas B have an additional vibrational mode and have a mass $\frac{m}{4}$. The ratio of the specific heats $\left(C_V^A \text{ and } C_V^B\right)$ of gas A and B respectively is:

Option 1:
5:9

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Option 2:
7:9

Option 3:
3:5

Option 4:
5:7

Correct Answer:
5:7

Solution:

Molar heat capacity of A at constant volume $= \frac{5R}{2}$

Molar heat capacity of B at constant volume $= \frac{7R}{2}$

Dividing both $\frac{(C_V)_A}{C_V)_B} = \frac{5}{7}$

Hence the correct option is (4).

Q. 3 Two particles of equal mass m have respective initial velocities $u\hat{i}$ and $u\left(\frac{\hat{i}+\hat{j}}{2}\right)$. They collide completely inelastically. The energy lost in the process is:

Option 1:
$\frac{3}{8}mu^2$

Option 2:
$\sqrt{\frac{2}{3}}mu^2$

Option 3:
$\frac{1}{3}mu^2$

Option 4:
$\frac{1}{8}mu^2$

Correct Answer:
$\frac{1}{8}mu^2$

Solution:

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Conserving Momentum-

$$m.u\hat{i} + m\left(\frac{u}{2}\hat{i} + \frac{u}{2}\hat{j}\right) = 2m(u_1\hat{i} + u_2\hat{j})$$

On solving

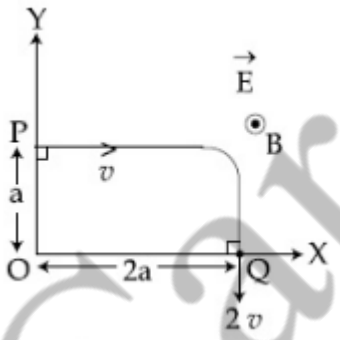
$u_1 = \frac{3u}{4}$ and $u_1 = \frac{u}{4}$

Change in K.E

$$\left[\frac{1}{2}mu^2 + \frac{1}{2}m\left(\frac{u}{2}\sqrt{2}\right)^2\right] - \left[\frac{1}{2}(2m)\left(\frac{9u^2}{16} + \frac{u^2}{16}\right)\right]$$
$$= \frac{3mu^2}{4} - \frac{5mu^2}{8} = \frac{mu^2}{8}$$

Hence the correct option is (4).

Q. 4 A charged particle of mass 'm' and charge 'q' moving under the influence of uniform electric field $E\hat{i}$ and a uniform magnetic field $B\hat{k}$ follows a trajectory from point P to Q as shown in figure. The velocities at P and Q are respectively, $v\hat{i}$ and $-2v\hat{j}$. Then which of the following statements (A, B, C, D) are the correct? (trajectory shown is schematic and not to scale)



(A) $E = \frac{3}{4}\left(\frac{mv^2}{qa}\right)$

(B) Rate of work done by the electric field at P is $\frac{3}{4}\left(\frac{mv^2}{a}\right)$

(C) Rate of work done by both the fields at Q is Zero.

(D) The difference between the magnitude of angular momentum of the particle at P and Q is $2mav$

Option 1:
(A), (C), (D)

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Option 2:
(A), (B), (C)

Option 3:
(A), (B), (C), (D)

Option 4:
(B), (C), (D)

Correct Answer:
(A), (B), (C)

Solution:
A) by work-energy theorem

$$W_{mag} + W_{ele} = \frac{1}{2}m(2v)^2 - \frac{1}{2}m(v)^2$$

$$0 + qE_0 2a = \frac{3}{2}mv^2$$

$$E_0 = \frac{3}{4}\frac{mv^2}{qa}$$

(B) Rate of change of work done at A = power of electric force

$$= qE_0V$$

$$= \frac{3}{4}\frac{mv^3}{a}$$

(C) at Q, $\frac{dw}{dt} = 0$ for both forces

(D) $\Delta\vec{L} = (-m(2v)(2a)\hat{k}) - (-mva\hat{k})$
$|\Delta\vec{L}| = 3mva$

Hence, the correct option is (2).

Q. 5 A particle moving with kinetic energy E has de Broglie wavelength λ. If energy ΔE is added to its energy, the wavelength become $\frac{\lambda}{2}$. Value of ΔE, is:

Option 1:
$2e$

Option 2:
$4e$

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Option 3:
$3E$

Option 4:
E

Correct Answer:
$3E$

Solution:

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$$
$$\Rightarrow \frac{\lambda_2}{\lambda_1} = \frac{\frac{\lambda}{2}}{\lambda} = \frac{1}{2}$$
$$\Rightarrow \frac{E_1}{E_2} = \frac{E}{E + \Delta E}$$
$$\Rightarrow \frac{\lambda_2}{\lambda_1} = \sqrt{\frac{E_1}{E_2}}$$
$$\Rightarrow \frac{1}{4} = \frac{E}{E + \Delta E}$$
$$\Rightarrow \Delta E = 3E$$

Hence the option correct option is (3).

Q. 6 A vessel of depth 2h is half filled with a liquid of refractive index $2\sqrt{2}$ and the upper half with another liquid of refractive index $\sqrt{2}$. The liquids are immersible. The apparent depth of the inner surface of the bottom of vessel will be:

Option 1:
$\frac{h}{\sqrt{2}}$

Option 2:
$\frac{3}{4}h\sqrt{2}$

Option 3:
$\frac{h}{3\sqrt{2}}$

Option 4:
$\frac{h}{2(\sqrt{2}+1)}$

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Correct Answer:

$\frac{3}{4}h\sqrt{2}$

Solution:

$$d = \frac{h}{\sqrt{2}} + \frac{h}{2\sqrt{2}} \Rightarrow d = \frac{h}{\sqrt{2}} \times \frac{3}{2} = \frac{3\sqrt{2}h}{4}$$

Hence the option correct option is (2).

Q. 7 Radiation with wavelength 6561 $\overset{o}{A}$ falls on a metal surface to produce photoelectrons. The electrons are made to enter a uniform magnetic field of $3 \times 10^{-4}T$. If the radius of the largest circular path followed by the electrons is 10mm, the work function (in eV) of the metal is close to:

Correct Answer:

1.1

Solution:

From photoelectric equation:-

$$KE_{max} = E - \phi$$
$$= \frac{12400}{\lambda(\ln A)} - \phi \quad (in\ eV)$$
$$\therefore r = \frac{\sqrt{2mKE}}{eB}$$
$$KE_{max} = \frac{r^2e^2B^2}{2m}$$
$$= \frac{r^2e2B^2}{2m} \quad (in\ eV)$$
$$\therefore \phi = \frac{12400}{6556} - \frac{r^2eB^2}{2m}$$
$$= 1.1eV$$

Hence the option correct option is (1) .

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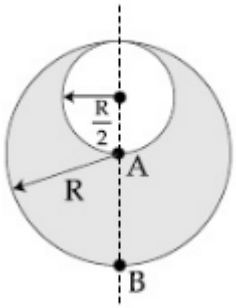
Q. 8 Consider a sphere of radius R which carries a uniform charge density ρ. If a sphere of radius $\frac{R}{2}$ is carved out of it, as shown, the ratio $\frac{\left|\hat{E_A}\right|}{\left|\hat{E_B}\right|}$ of magnitude of electric field $\hat{E_A}$ and $\hat{E_B}$, respectively, at points A and B due to the remeaning portion is:



Option 1:
$\frac{21}{34}$

Option 2:
$\frac{18}{54}$

Option 3:
$\frac{17}{54}$

Option 4:
$\frac{18}{34}$

Correct Answer:
$\frac{18}{34}$

Solution:

Condider a sphere of density $-\rho$ and radius R/2 is removed from a sphere of radius R

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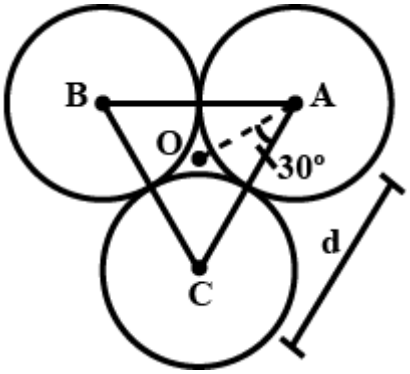
then, Electric field at A is $E_A = E_\rho + E_{-\rho} = 0 - \frac{\rho(\frac{R}{2})}{3\epsilon_0} = \frac{\rho R}{6\epsilon_0}$

electric field at B is $E_B = E_\rho + E_{-\rho} = \frac{\rho(R)}{3\epsilon_0} - \frac{k\rho\frac{4}{3}\pi\left(\frac{R}{2}\right)^3}{\left(\frac{3R}{2}\right)^2} = \frac{17\rho R}{54\epsilon_0}$

$$\left|\frac{E_A}{E_B}\right| = \frac{9}{17} \Rightarrow \frac{18}{34}$$

Hence the correct option is (4).

Q. 9



Three solid spheres each of mass m and diameter d are stuck together such that the lines connecting the centres form an equilateral triangle of the side of length d. The ratio $\frac{I_O}{I_A}$ of moment of inertia I_0 of the system about an axis passing the centroid and about centre of any of spheres I_A and perpendicular to the plane of the triangle is:

Option 1:
$\frac{15}{13}$

Option 2:
$\frac{13}{15}$

Option 3:
$\frac{13}{23}$

Option 4:
$\frac{23}{13}$

Correct Answer:
$\frac{13}{23}$

Solution:

As a moment of inertia of one sphere of radius R about its centre= $I = \frac{2}{5}mR^2$

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from the figure given it is clear that $OC = \frac{d}{\sqrt{3}}$

M.I about O $= 3\left[\left(\frac{2}{5}M\left(\frac{d}{2}\right)^2 + M\left(\frac{d}{\sqrt{3}}\right)^2\right)\right] = \frac{13}{10}Md^2$

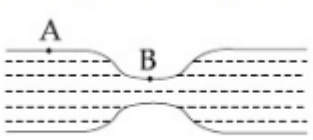
M.I about A $= 2\left[\left(\frac{2}{5}M\left(\frac{d}{2}\right)^2 + M\left(d\right)^2\right)\right] + \frac{2}{5}M\left(\frac{d}{2}\right)^2 = \frac{23}{10}Md^2$

New Ratio $= \frac{13}{23}$

Hence the option correct option is (3).

Q. 10 Water flows in a horizontal tube (see figure). The pressure of water changes by 700Nm^2 between A and B where the area of cross-section is 40cm^2 and 20 cm^2, respectively. Find the rate of flow of water through the tube.

(density of water = 1000kgm^{-3})



(Fig)

Option 1:
3020 cm^3/s

Option 2:
2420 cm^3/s

Option 3:
2720 cm^3/s

Option 4:
1810 cm^3/s

Correct Answer:
2720 cm^3/s

Solution:
Using the equation of continuity

$40V_A = 20V_B$
$\Rightarrow 2V_A = V_B$

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Using Bernoulli's Equation

$$P_A + \frac{1}{2}\rho V_A^2 = P_B + \frac{1}{2}\rho V_B^2$$

$$P_A - P_B = \frac{1}{2}\rho\left(V_B^2 - V_A^2\right)$$

$$\Delta P = \frac{1}{2} \times 1000\left(V_B^2 - \frac{V_B^2}{4}\right)$$

$$\Rightarrow \Delta P = 500 \times \frac{3V_B^2}{4}$$

$$\Rightarrow V_B = \sqrt{\frac{\Delta P \times 4}{1500}} = \sqrt{\frac{700 \times 4}{1500}} m/s$$

Volume flow rate = $20 \times 100 \times V_B = 2732 cm^3/s$

So, the closest option to this answer is option c.

Hence the correct option is (3).

Q. 11 Three harmonic waves having equal frequency v and same intensity I_0, have phase angles 0, $\frac{\pi}{4}$ and $-\frac{\pi}{4}$, respectively. When they are superimposed the intensity of the resultant wave is close to:

Option 1:
$0.2I_0$

Option 2:
I_0

Option 3:
$3I_0$

Option 4:
$5.8I_0$

Correct Answer:
$5.8I_0$

Solution:

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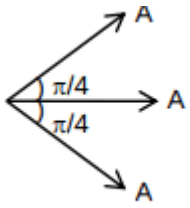

$$A_{res} = A + \frac{A}{\sqrt{2}} + \frac{A}{\sqrt{2}} = A(1+\sqrt{2})$$

$$I = (1+\sqrt{2})^2 I_0 = 5.8 I_0$$

Hence the correct option is (4).

Q. 12 The aperture diameter of a telescope is 5m. The separation between the moon and the earth is $4 \times 10^5 km$. With light of wavelength of 5000 $\overset{o}{A}$, the minimum separation between objects on the surface of moon, so that they are just resolved, is close to:

Option 1:
600 m

Option 2:
20 m

Option 3:
60 m

Option 4:
200 m

Correct Answer:
60 m

Solution:

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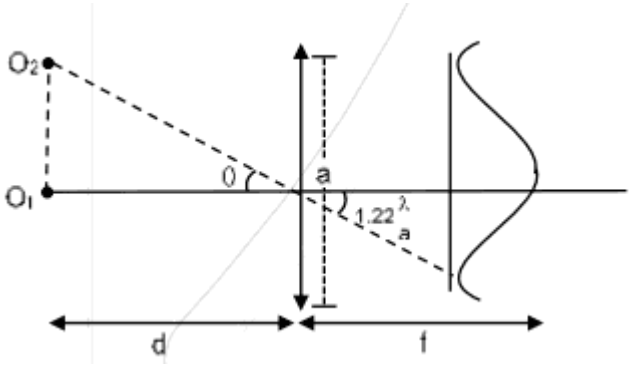


$$\theta = 1.22\frac{\lambda}{a}$$

Distance $= O_1O_2 = d\theta$

$$= O_1O_2 = 1.22\frac{\lambda}{a}d$$

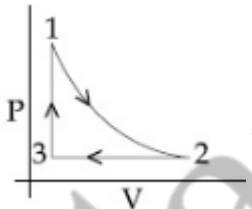
distance $= O_1O_2 = \dfrac{1.22 \times 5893 \times 10^{-10} \times 4 \times 10^8}{5} \approx 57.5m$

Answer from the options = 60m

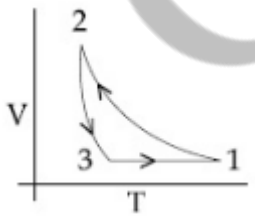
Hence the correct option is (3).

Q. 13 Which of the following is an equivalent cyclic process corresponding to the thermodynamic cycle given in the figure?

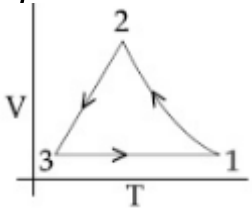
(Graphs are schematic and are not to scale)



Option 1:



Option 2:



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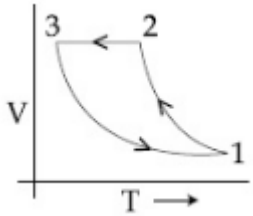
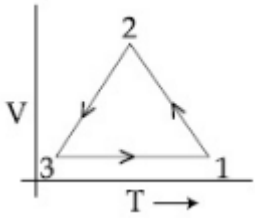
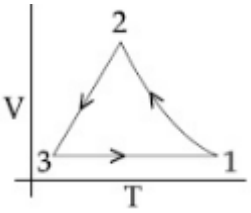
Option 3:



Option 4:



Correct Answer:



Solution:
Let 3=A, 1=B, 2=C

then

For process A - B; Volume is constant

$PV = nRT$ As P increases; T increases

For process BC

$PV^{\gamma} = constant$

$\Rightarrow TV^{\gamma-1} = constant$

For process C - A; pressure is constant

$V = kT$

Hence the correct option is (2).

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Q. 14 The electric fields of two plane electromagnetic plane waves in vacuum are given by

$\vec{E_1} = E_0\hat{j}\cos(\omega t - kx)$ and

$\vec{E_2} = E_0\hat{k}\cos(\omega t - ky)$

At t = 0, a particle of charge q is at origin with velocity $\vec{v} = 0.8c\hat{j}$ (c is the speed of light in vacuum

). The instantaneous force experienced by the particle is:

Option 1:
$E_0q\left(0.8\hat{i} - \hat{j} + 0.4\hat{k}\right)$

Option 2:
$E_0q\left(0.4\hat{i} - 3\hat{j} + 0.8\hat{k}\right)$

Option 3:
$E_0q\left(-0.8\hat{i} + \hat{j} + \hat{k}\right)$

Option 4:
$E_0q\left(0.8\hat{i} + \hat{j} + 0.2\hat{k}\right)$

Correct Answer:
$E_0q\left(0.8\hat{i} + \hat{j} + 0.2\hat{k}\right)$

Solution:

Magnetic field vectors associated with this electromagnetic wave are given by

$$\vec{B}_1 = \frac{E_0}{c}\hat{k}\cos(kx - \omega t) \quad \& \quad \vec{B}_2 = \frac{E_0}{c}\hat{i}\cos(ky - \omega t)$$

$$\vec{F} = q\vec{E} + q(\vec{V} \times \vec{B})$$

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$$= q(\vec{E}_1 + \vec{E}_2) + q\left(\vec{V} \times (\vec{B}_1 + \vec{B}_2)\right)$$

by putting the value of $\vec{E}_1, \vec{E}_2, \vec{B}_1$ & $\vec{B}_2$

The net Lorentz force on the charged particle is

$$\vec{F} = qE_0[0.8\cos(kx - \omega t)\hat{i} + \cos(kx - \omega t)\hat{j} + 0.2\cos(ky - \omega t)\hat{k}]$$

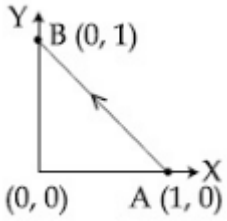
at t = 0 and at x = y = 0

$$\vec{F} = qE_0[0.8\hat{i} + \hat{j} + 0.2\hat{k}]$$

Hence the option correct option is (4).

Q. 15 Consider a force $\vec{F} = -x\hat{i} + y\hat{j}$. The work done by this force in moving a particle from point A(1,0) to B(0,1) along the line segment is :

(all quantities are in SI units)



Option 1:
$\frac{3}{2}$

Option 2:
2

Option 3:
1

Option 4:
$\frac{1}{2}$

Correct Answer:
1

Solution:

$$W = \int \vec{F} \cdot d\vec{s}$$

$$= (-x\hat{i} + y\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$$

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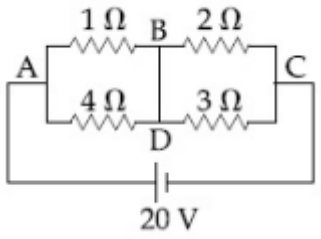
$$= \int_1^0 -xdx + \int_0^1 ydy$$

$$= -\frac{x^2}{2}\Big|_1^0 + \frac{y^2}{2}\Big|_0^1 = \left(0 + \frac{1}{2}\right) + \left(\frac{1}{2}\right) = 1J$$

Hence the correct option is (3).

Q. 16 In the given circuit diagram, a wire is joining points B and D. The current in this wire is: (current in amperes,A)



Correct Answer:
2

Solution:

$$\text{Reff} = \frac{4}{5} + \frac{6}{5} = 2\Omega$$

$$i = \frac{20}{2} = 10A$$

So after drawing the circuit

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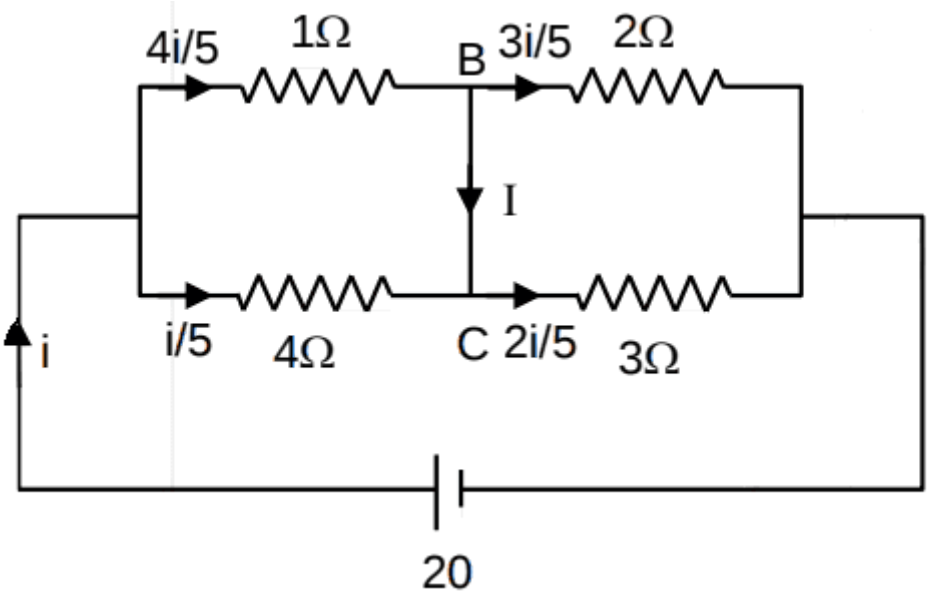

$$I = \frac{4i}{5} - \frac{3i}{5} = +\frac{i}{5} = 2A$$

Hence the option correct option is (1).

Q. 17 A long, straight wire of radius 'a' carries a current distributed uniformly over its cross-section. The ratio of the magnetic fields due to the wire at distance $\frac{a}{3}$ and 2a, respectively from the axis of the wire is:

Option 1:
$\frac{3}{2}$

Option 2:
2

Option 3:
$\frac{2}{3}$

Option 4:
$\frac{1}{2}$

Correct Answer:
$\frac{2}{3}$

Solution:

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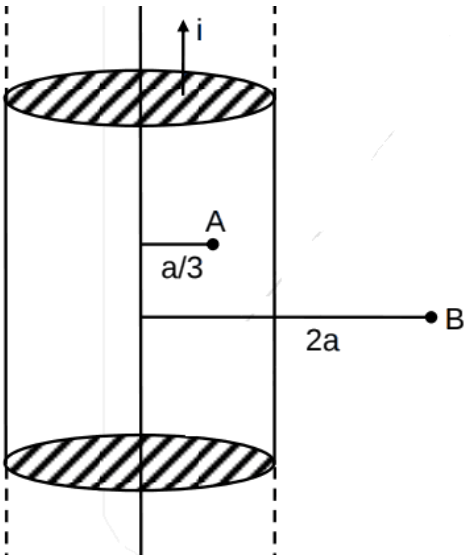

$$B_A = \frac{\mu_0 ir}{2\pi a^2} = \frac{\mu_0 ia}{\pi a^2 6} = \frac{\mu_0 i}{6\pi a}$$

$$B_B = \frac{\mu_0 i}{2\pi(2a)}$$

$$\frac{B_A}{B_B} = \frac{4}{6} = \frac{2}{3}$$

Hence the option correct option is (3).

Q. 18 A quantity *f* is given by $f = \sqrt{\frac{hc^5}{G}}$ where c is speed on light, G is universal gravitational constant and h is the Plank's constant. Dimension of *f* is that of:

Option 1:
area

Option 2:
volume

Option 3:
momentum

Option 4:
energy

Correct Answer:
energy

Solution:

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$$\sqrt{\frac{hc^5}{G}} = \sqrt{\frac{[ML^2T^{-1}]\,[LT^{-1}]^5}{[M^{-1}L^3T^{-2}]}}$$

$$\sqrt{\frac{hc^5}{G}} = \sqrt{\frac{[ML^7T^{-6}]}{[M^{-1}L^3T^{-2}]}}$$

$$\Rightarrow \sqrt{[M^2L^4T^{-4}]} = ML^2T^{-2}$$

ML^2T^{-2} **is also the dimension of energy.**

Hence the option correct option is (4).

Q. 19 If the screw on screw-gauge is given six rotations, it moves by 3 mm on the main scale. If there are 50 divisions on the circular scale the least count of the screw gauge is: (in cm)

Correct Answer:
0.001

Solution:

Pitch $= \frac{3}{6} = 0.5mm$

L.C $= \frac{0.5mm}{50} = \frac{1}{100}mm = 0.01mm = 0.001cm$

Hence the option correct option is (1).

Q. 20 A body A of mass m is moving in a circular orbit of radius R about a planet. Another body B of mass $\frac{m}{2}$ collides with A with a velocity which is half $\left(\frac{\vec{v}}{2}\right)$ the instantaneous velocity of $\vec{v}$ of A. The collision is completely inelastic. Then, the combined body:

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Option 1:
Escapes from the planet's gravitational field

Option 2:
starts moving in an elliptical orbit around the planet

Option 3:
falls vertically downwards towards the planet

Option 4:
continues to move in a circular orbit

Correct Answer:
starts moving in an elliptical orbit around the planet

Solution:
Conserving momentum:

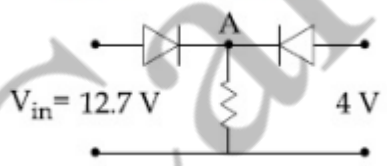
$$\frac{m}{2}\frac{v}{2} + mv = \left(m + \frac{m}{2}\right) v_t$$

$$v_t = \frac{5mv}{4 \times \frac{3m}{2}} = \frac{5v}{6}$$

$v_t < v_{orb} (= v)$ Thus the combined mass will go on to an elliptical path

Hence the correct option is (2).

Q. 21 Both the diodes used in the circuit shown are assumed to be ideal and have negligible resistance when these are forward biased. Built-in potential in each diode is 0.7V. For the input voltages shown in the figure, the voltage (in Volts) at a point A is _____



Option 1:
12

Option 2:
10

Option 3:
40

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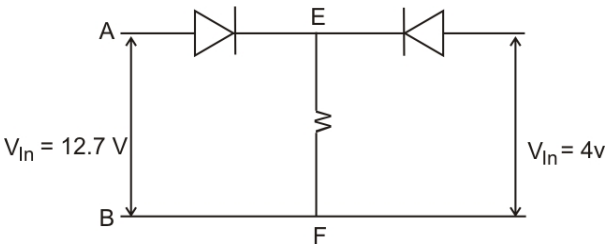
Option 4:
24

Correct Answer:
12

Solution:

We want to find $\mathbf{V_E}$

By applying kvl in AEFBA



12.7 - 0.7 - V_E = 0.

implies

V_E = 12V

Hence the correct option is (1).

Q. 22 The distance x covered by a particle in one dimensional motion varies with time t as $x^2 = at^2 + 2bt + c$. If the acceleration of the particle depends of x as x^{-n}, where n is an integer, the value of n is _____

Correct Answer:
3

Solution:

$$x^2 = at^2 + 2bt + c$$

$$\therefore 2x\frac{\mathrm{d}x}{\mathrm{d}t} = 2at + 2b$$

$$\Rightarrow v = \frac{at + b}{x}$$

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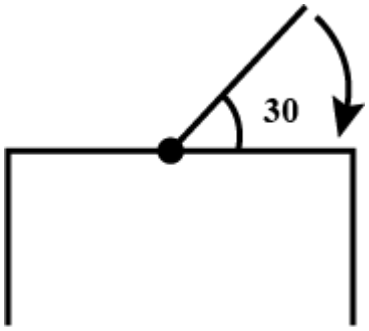
Again differentiating with respect to t

$$2x\frac{d^2x}{dt^2} + 2\left(\frac{dx}{dt}\right)^2 = 2a$$
$$\Rightarrow 2xa + 2v^2 = \text{constant (say } c)$$
$$\Rightarrow xa = c - \left(\frac{at+b}{x}\right)^2$$
$$x^3a = c(at^2+2bt+c)^2 - (at+b)^2$$
$$\Rightarrow a \propto x^{-3}$$

value of n is 3.

Hence the correct option is (1).

Q. 23 One end of a straight uniform 1m long bar is pivoted on a horizontal table. It is released from rest when it makes an angle 30° from the horizontal (see figure). Its angular speed when it hits the table is given as $\sqrt{n}s^{-1}$, where n is an integer. The value of n is _____.



Option 1:
15

Option 2:
12

Option 3:
24

Option 4:
30

Correct Answer:
15

Solution:

$$mg\frac{t}{2}\sin 30^\circ = \frac{1}{2}\frac{mt^2}{3}\omega^2$$

Solving

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$$\omega^2 = 15$$
$$\Rightarrow \omega = \sqrt{15}$$

Since, $\omega = \sqrt{n} = \sqrt{15}$, Therefore n= 15.

Hence the correct option is (1).

Q. 24 A body mass m = 10 kg is attached to one end of a wire of length 0.3m. The maximum angular speed (in rad s^{-1}) with which it can be rotated about its other end in space station is (Breaking stress of wire $= 4.8 \times 10^7 Nm^{-2}$ and area of cross-section of the wire $= 10^{-2} cm^2$) is:

Option 1:
4

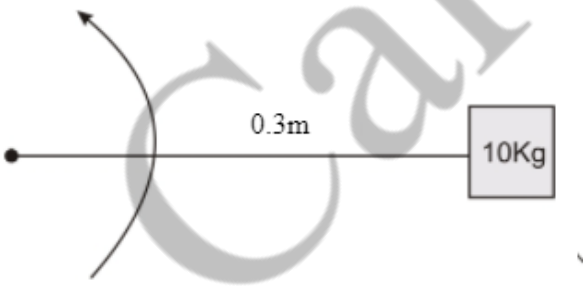
Option 2:
8

Option 3:
6

Option 4:
1

Correct Answer:
4

Solution:



$$\frac{F}{A} = \frac{mv^2}{lA}$$
$$\Rightarrow v = \sqrt{\frac{lF}{m}} = \sqrt{\frac{0.3}{10} \times 4.8 \times 10^7 \times 10^{-6}}$$
$$= \sqrt{3 \times 48 \times 10^4 \times 10^{-6}} = 1.2\text{m/s}$$
$$\omega = \frac{v}{l} = \frac{1.2}{0.3} = 4\text{rad/s}$$

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the correct option is (1).

Q. 25 In a fluorescent lamp choke (a small transformer) 100 V of reverse voltage is produced when the choke current changes uniformly from 0.25 A to 0 in a duration of 0.025 ms. The self-inductance of the choke (in mH) is estimated to be _____.

Correct Answer:
10

Solution:

$$V = L\frac{\mathrm{d}i}{\mathrm{d}t} \Rightarrow 100 = L\frac{0.25}{0.025}$$
$$\Rightarrow L = 10 \text{ mH}$$

Chemistry

Q. 1 If the magnetic moment of a dioxygen species is 1.73 B.M, it may be:

Option 1:
$O_2, O_2^- or O_2^+$

Option 2:
$O_2^- or O_2^+$

Option 3:
$O_2 or O_2^-$

Option 4:
$O_2 or O_2^+$

Correct Answer:
$O_2^- or O_2^+$

Solution:
As we have learnt,

Electronic Configuration for Molecules -

Electronic Configuration and Molecular Behaviour

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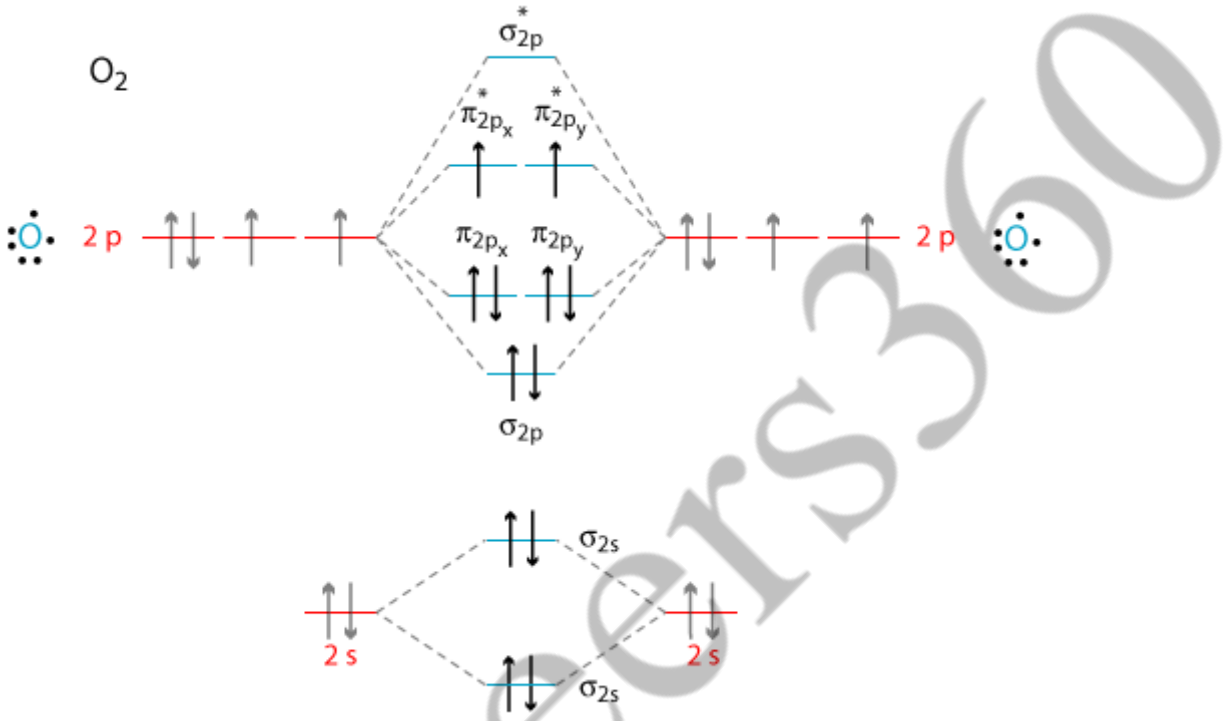
We predict the distribution of electrons in these molecular orbitals by filling the orbitals in the same way that we fill atomic orbitals, by using the Aufbau principle. Lower-energy orbitals fill first, electrons spread out among degenerate orbitals before pairing, and each orbital can hold a maximum of two electrons with opposite spins. Just as we write electron configurations for atoms, we can write the molecular electronic configuration by listing the orbitals with superscripts indicating the number of electrons present. For clarity, we place parentheses around molecular orbitals with the same energy. In this case, each orbital is at a different energy, so parentheses separate each orbital.

For example, the electronic configuration of O_2 molecule is given below:



Thus, the electronic configuration of O_2 molecule can be written as:

$\sigma_{1s}2\ \sigma^*_{1s}2\ \sigma_{2s}2\ \sigma^*_{2s}2\ \sigma_{2pz}2\ \pi_{2px}2\ \pi_{2py}2\ \pi^*_{2px}1\ \pi^*_{2py}1$

-

$$O_2 = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2pz^2 \pi 2px^2 = \pi 2py^2 \pi^* 2px^1 = \pi^* 2py^1$$

The magnetic moment of dioxygen species = 1.73 B.M means it has one unpaired electron. One unpaired electron is present in O_2^+, O_2^-.

Hence, the option number (2) is correct.

Q. 2 Which of these will produce the highest yield in Friedel Crafts reaction?

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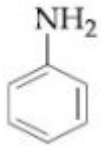
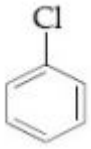
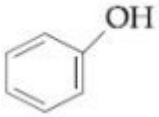
Option 1:



Option 2:

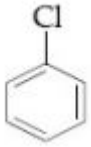


Option 3:



Option 4:

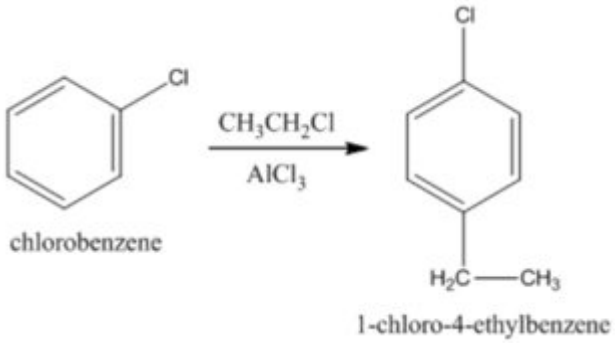
Correct Answer:



Solution:
As we have learnt,

Phenol does not undergo Friedel Craft Alkylation or Acylation. This is because oxygen's lone pair in phenol makes the coordinate bond with $AlCl_3$ (A Lewis acid) hence blocking it. Same is the case with Aniline. These are not preferred as substrates for Friedel Crafts Reactions.

Haloarenes undergo the usual electrophilic reactions of the benzene ring such as Friedel-Crafts reactions. Halogen atom besides being slightly deactivating is o, p-directing; therefore, further substitution occurs at the ortho- and para-positions with respect to the halogen atom.



chlorobenzene

1-chloro-4-ethylbenzene

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Hence, the option number (2) is correct.

Q. 3 'X' melts at low temperature and is a bad conductor of electricity in both liquid and solid state, X is:

Option 1:
Carbon tertachloride

Option 2:
Silicon carbide

Option 3:
Mercury

Option 4:
Zinc sulphide

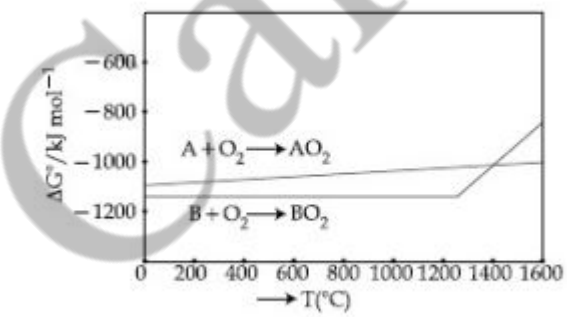
Correct Answer:
Carbon tertachloride

Solution:

$CCl_4 \rightarrow$ Non-conductor in solid and liquid phase.

Hence, the option number (1) is correct.

Q. 4 According to the following diagram, A reduces BO_2 when the temperature is:



Option 1:
$> 1200^{\circ}C but < 1400^{\circ}C$

Option 2:
$> 1400^{\circ}C$

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Option 3:

$< 1200^{\circ}C$

Option 4:

$< 1400^{\circ}C$

Correct Answer:

$> 1400^{\circ}C$

Solution:

As we have learnt,

Ellingham Diagram -

- Ellingham diagram normally consists of plots of $\Delta_f GV$ vs T for the formation of oxides of common metals and reducing agents i.e., for the reaction given below.

  $2xM(s) + O_2(g) \rightarrow 2M_xO(s)$

  In this reaction, gas is consumed in the formation of oxide hence, molecular randomness decreases in the formation of oxide which leads to a negative value of ΔS as a result sign of $T\Delta S$ term becomes positive. Subsequently, $\Delta_f G^o$ shifts towards higher side despite rising T. The result is a positive slope in the curve for most of the reactions for the formation of $M_xO(s)$.

- Each plot is a straight line and slopes upwards except when some change in phase ($s \rightarrow 1$ or $1 \rightarrow g$) takes place. The temperature at which such change occurs is indicated by an increase in the slope on the positive side (e.g., in the Zn, ZnO plot, the melting is indicated by an abrupt change in the curve).

- When the temperature is raised, a point is reached in the curve where it crosses $\Delta_r G^o=0$ line. Below this temperature, $\Delta_r G^o$ for the formation of oxide is negative so M_xO is stable. Above this point, free energy of formation of oxide is positive. The oxide, MxO will decompose on its own.

- Similar diagrams are constructed for sulfides and halides also. From them it becomes clear that why reduction of M_xS is difficult.

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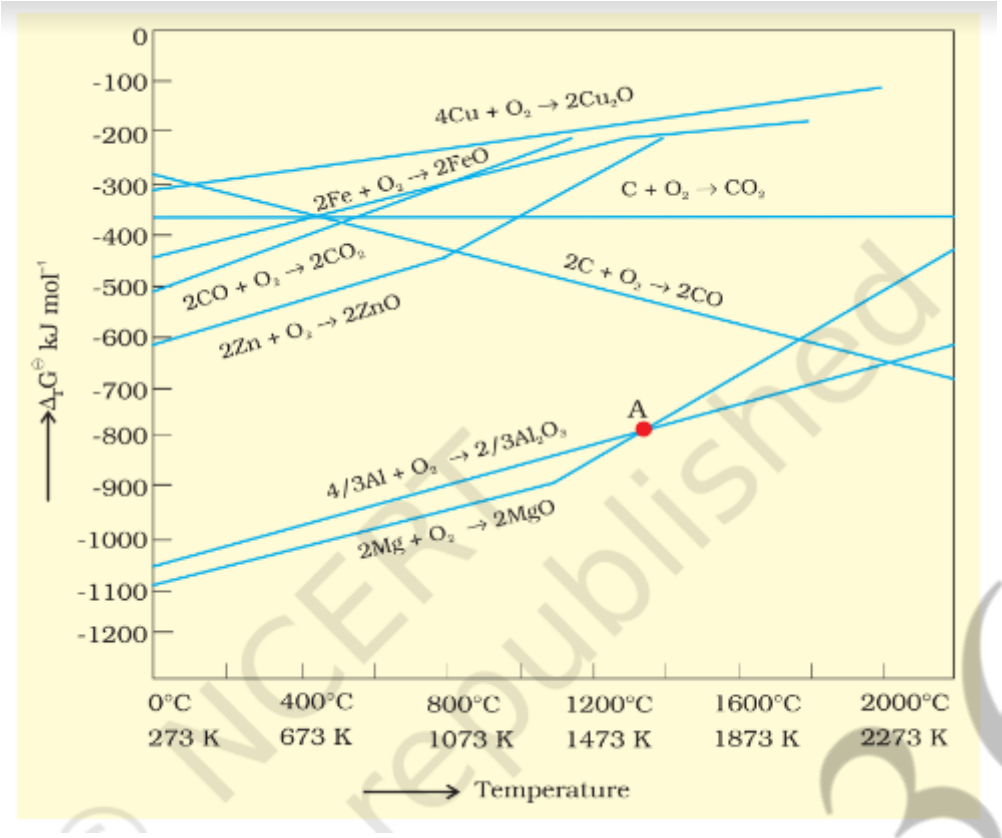

Limitations of Ellingham Diagram

- It only tells about the thermodynamic feasibility of the reduction of metal oxide by a reducing agent. It does not give any idea about the kinetics of the reaction.
- It always considers an equilibrium between the two reactions or consider an equilibrium between reactants and products. But if solids are there then no equilibrium exist.

-

$$A + BO_2 \longrightarrow B + AO_2$$
$$\Delta G = -ve$$
$$\text{Only above } 1400^\circ C$$

Hence, the option number (2) is correct.

Q. 5 The de Brogile wavelength of an electron in the 4th Bohr orbit is:

Option 1:

$6\pi a_0$

Option 2:

$2\pi a_0$

Option 3:

$8\pi a_0$

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Option 4:
$4\pi a_0$

Correct Answer:
$8\pi a_0$

Solution:
As we have learnt,

Debroglie wavelength -

Bohr model and de-broglie principle :

We have:

$n\lambda = 2\pi r_n$

n = 4

$$2\pi\frac{n^2}{z}a_o = n\lambda$$

$$2\pi\frac{4^2}{1}a_o = n\lambda$$

$$\lambda = 8\pi a_o$$

Therefore, **Option(3) is correct**

Q. 6 For the following reactions:

$$A \xrightarrow{700K} product$$

$$A \xrightarrow[catalyst]{500K} product$$

it was found that E_a is decreased by 30 kJ/mol in the presence of a catalyst. If the rate remains unchanged, the activation energy (in KJ/mol) for catalyzed reaction is (Assume pre-exponential factor is same):

Correct Answer:
75

Solution:
The formula of the rate of reaction

$$k = Ae^{-Ea/RT}$$

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E_a is decreased by 30 kJ/mol in the presence of a catalyst. If the rate remains unchanged

$$\mathrm{E_a(catalyst) - E_a = -30\ kJ/mol}$$

and

$$\mathrm{k_{catalyst} = k}$$

So,

$$\mathrm{Ae^{\frac{-Ea(catalyst)}{RT_{catalyst}}} = Ae^{-\frac{Ea}{RT}}}$$

$$\frac{\mathrm{E_a(catalyst)}}{\mathrm{T_{(catalyst)}}} = \frac{\mathrm{E_a}}{\mathrm{T}}$$

$$\frac{\mathrm{E_a(catalyst)}}{500} = \frac{\mathrm{E_a}}{700}$$

$$\frac{\mathrm{E_a(catalyst)}}{\mathrm{E_a}} = \frac{500}{700}$$

$$\frac{\mathrm{E_a(catalyst)}}{\mathrm{E_a(catalyst) - E_a}} = \frac{500}{500-700}$$

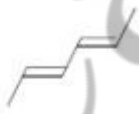
$$\frac{\mathrm{E_a(catalyst)}}{-30} = \frac{500}{-200}$$

$$\mathrm{E_a(catalyst)} = \frac{-30 \times 500}{-200}$$

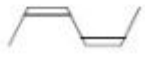
$$\mathrm{E_a(catalyst) = 75\ kJ/mol}$$

Hence, the option number (1) is correct.

Q. 7 The correct order of heat of combustion for the following alkadienes is:

(a) 

(b)

(c) 

Option 1:
b < c < a

Option 2:
a < b < c

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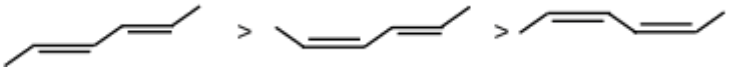
Option 3:
a < c < b

Option 4:
c < b < a

Correct Answer:
a < b < c

Solution:
In isomers of hydrocarbon, the heat of combustion depends upon their stabilities. As the stability increases the heat of combustion decreases.

Stability order 

Hence, the option number (2) is correct.

Q. 8 Complex X of composition $Cr(H_2O)_6Cl_n$ has a spin only magnetic moment of 3.83 BM. It reacts with $AgNO_3$ and shows geometrical isomerism. The IUPAC nomenclature of X is:

Option 1:
Tetraaquadichlorido chromium (IV) chloride dihydrate

Option 2:
Hexaaqua chromium (III) chloride

Option 3:
Dichloridotetraaqua chromium (IV) chloride dihydrate

Option 4:
Tetraaquadichlorido chromium (III) chloride dihydrate

Correct Answer:
Tetraaquadichlorido chromium (III) chloride dihydrate

Solution:
$\mathrm{Cr\,(H_2O)_6\,Cl_n;}\quad (\mu_{\mathrm{complex}})\,\mathrm{spin}\ = 3.8\ \mathrm{B.M}$

From data of magnetic moment oxidation number of Cr should be +3
Hence complex is $Cr(H_2O)_6Cl_3$.

Complex shows geometrical isomerism therefore formula of the complex is $[Cr(H_2O)_4Cl_2]Cl.2H_2O$.

Its IUPAC Name: Tetraaquadichloridochromium(III) chloride dihydrate

Hence, **the option number (4) is correct.**

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Q. 9 The electronic configurations of bivalent europium and trivalent cerium are :

(atomic number : Xe = 54, Ce = 58, Eu = 63)

Option 1:
$[Xe]4f^{7}\ and\ [Xe]4f^{1}$

Option 2:
$[Xe]4f^{7}6s^{2}\ and\ [Xe]4f^{2}6s^{2}$

Option 3:
$[Xe]4f^{2}\ and\ [Xe]4f^{7}$

Option 4:
$[Xe]4f^{4}\ and\ [Xe]4f^{9}$

Correct Answer:
$[Xe]4f^{7}\ and\ [Xe]4f^{1}$

Solution:
The electronic configurations are given as

Eu^{2+} : $[Xe]4f^{7}$
Ce^{3+} : $[Xe]4f^{1}$

Hence, the option number (1) is correct.

Q. 10 B has smaller first ionization enthalpy than Be. Consider the following statements :

(I) It is easier to remove 2p electron than 2s electron.

(II) 2p electron of B is more shielded from the nucleus by the inner core of electrons than the 2s electrons of Be.

(III) 2s electron has more penetration power than 2p electron

(IV) atomic radius of B is more than Be

(atomic number B = 5, Be = 4)

The correct statements are :

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Option 1:
I, II and III

Option 2:
II , III and IV

Option 3:
I, III and IV

Option 4:
I, II and IV

Correct Answer:
I, II and III

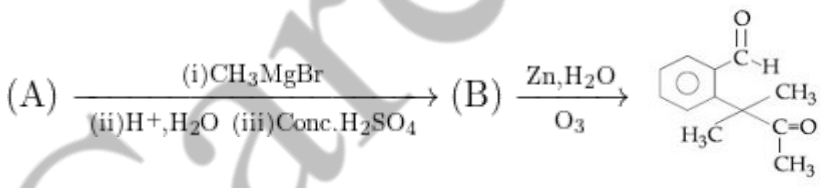
Solution:
Screening effect: The higher is the value of the screening effect, the lower is the ionisation enthalpy. As the screening effect increases, the repulsion between the electrons increases and thus the removal of an electron from the atom becomes easier. Thus,

Ionisation enthalpy ∝ 1/Screening effect

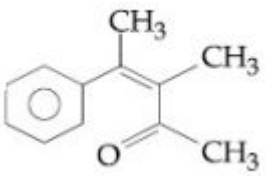
(i) It is easier to extract an electron from 2p orbital than 2s orbital
(ii) Penetration power of 2s orbital is greater than 2p orbital
(iii) There is Shielding of 2p electron by 2s electron

Hence, the option number (1) is correct.

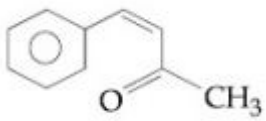
Q. 11 Identify (A) in the following reaction sequence:



Option 1:



Option 2:



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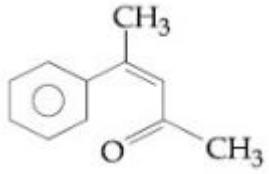
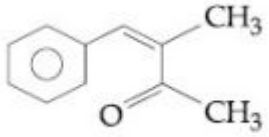
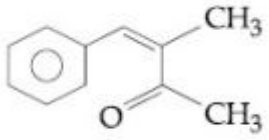
Option 3:



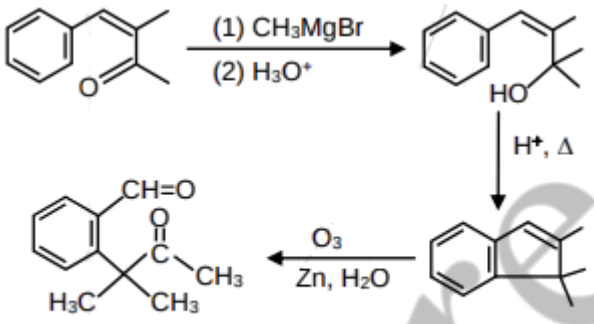
Option 4:



Correct Answer:



Solution:
The reaction will be this -



Hence, the option number (4) is correct.

Q. 12 The acidic , basic and amphoteric oxides, respectively, are:

Option 1:
Cl_2O, CaO, P_4O_{10}

Option 2:
N_2O_3, Li_2O, Al_2O_3

Option 3:
Na_2O, SO_3, Al_2O_3

Option 4:
MgO, Cl_2O, Al_2O_3

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Correct Answer:
N_2O_3, Li_2O, Al_2O_3

Solution:
As we have learnt,

Classification of Elements as Metals, Nonmetals and Metalloids -

The elements can be divided into three categories i.e, metals and non-metals and metalloids. Metals comprise more than 78% of all known elements and appear on the left side of the Periodic Table. Metals are usually solids at room temperature except mercury which is liquid at room temperature and even gallium and caesium also have very low melting points (303K and 302K. respectively). Metals usually have high melting and boiling points. They are good conductors of heat and electricity. They are malleable and ductile .

Non-metals are placed at the top right side of the Periodic Table. In a period, the property of elements change from metallic on the left to non-metallic on the right. Non-metals are usually solids or gases at room temperature with low melting and boiling points (boron and carbon are exceptions). They are poor conductors of heat and electricity. Most non- metallic solids are brittle and are neither malleable nor ductile. The elements become more metallic as we move from top to bottom in a group and the non-metallic character increases as we move from left to right in a period.

There are some elements whose properties are in between the metals and non-metals. They are known as metalloids. There are a total of 8 metalloids in the periodic table i.e, boron(B), silicon (Si), germanium (Ge), arsenic (As), antimony (Sb), tellurium (Te), polonium (Po) and astatine (At).

-

Non-metal oxides are acidic in nature
alkali metal oxides are basic in nature
Al_2O_3 is amphoteric.
Hence, the option number (2) is correct.

Q. 13 The K_{sp} for the following dissociation is 1.6×10^{-5}.

$$PbCl_2(s) = Pb^{2+}(aq) + 2Cl^{-}(aq)$$

Which of the following choices is correct for a mixture of 300mL, 0.134 M $Pb(NO_3)_2$ and 100 mL 0.4 M NaCl?

Option 1:
$Q < K_{sp}$

Option 2:
$Q > K_{sp}$

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Option 3:
$Q = K_{sp}$

Option 4:
Not enough data is provided

Correct Answer:
$Q > K_{sp}$

Solution:
The K_{sp} for the following dissociation is 1.6×10^{-5}.

$$PbCl_2(s) = Pb^{2+}(aq) + 2Cl^{-}(aq)$$

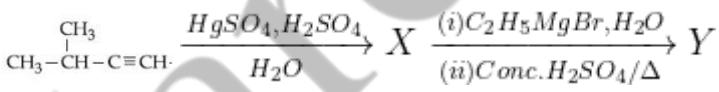
So,

$$Q = \left[Pb^{2+}\right]\left[Cl^{2}\right]^2$$
$$Q = \frac{300 \times 0.134}{400} \times \left[\frac{100 \times 0.4}{400}\right]^2$$
$$Q = \frac{3 \times 0.134}{4} \times (0.1)^2$$
$$Q = 0.105 \times 10^{-2}$$
$$Q = 1.005 \times 10^{-3}$$

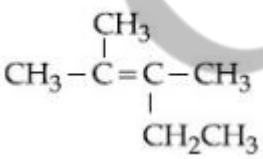
So, Q > Ksp

Hence, the option number (2) is correct.

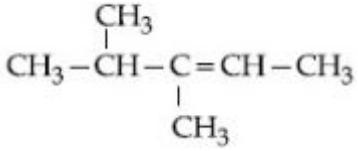
Q. 14 The major product (Y) in the following reaction is:



Option 1:



Option 2:



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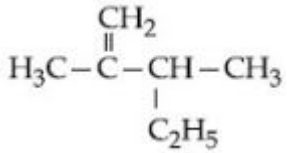
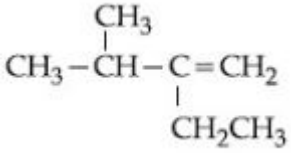
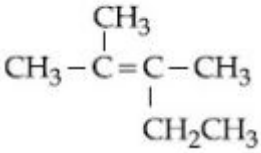
Option 3:



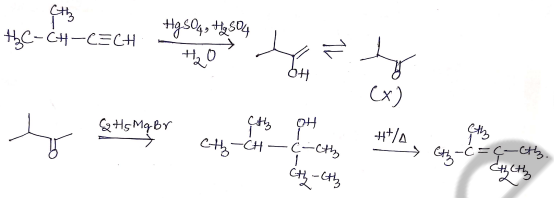
Option 4:



Correct Answer:



Solution:
The given reaction occurs as



Hence, the option number (1) is correct.

Q. 15 A chemist has 4 samples of artifical sweetener A, B, Cand D. To identify these samples, he performed certain experiments and noted the following observations:

(i) A and D both form blue - violet colour with ninhydrin.

(ii) Lassaigne extract of C gives positive $AgNO_3$ test and negative $Fe_4[Fe(CN)_6]_3$ test.

(iii) Lassaigne extract of B and D gives positive sodium nitroprusside test.

Based on these observations which option is correct?

Option 1:
A : Alitame
B : Saccharin
C : Aspartame
D : Sucralose

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Option 2:

A : Saccharin

B : Alitame

C : Sucralose

D : Aspartame

Option 3:

A : Aspartame

B : Alitame

C : Sucralose

D : Saccharin

Option 4:

A : Aspartame

B : Saccharin

C : Sucralose

D : Alitame

Correct Answer:

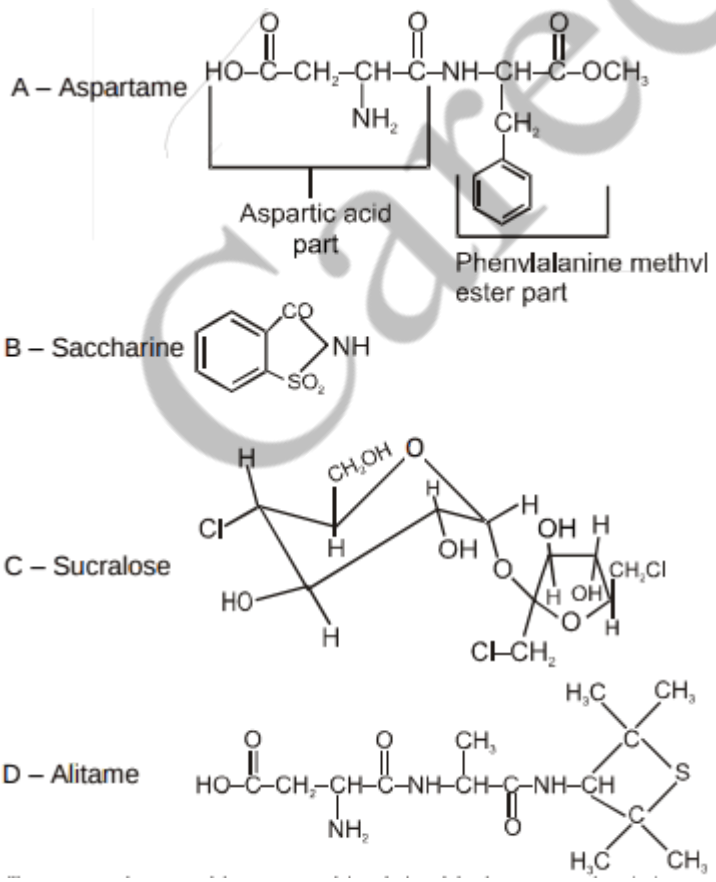
A : Aspartame

B : Saccharin

C : Sucralose

D : Alitame

Solution:



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(i) A & D give a positive test with ninhydrin because both have free carboxylic and amine groups.
(ii) C form precipitate with AgNO3 in the lassaigne extract of the sugar because it has chlorine atoms.
(iii) B & D give a positive test with sodium nitroprusside because both have sulphur atoms.

Hence, the option number (4) is correct.

Q. 16 The compound that cannot act both as oxidising and reducing agent is :

Option 1:
H_2SO_3

Option 2:
H_3PO_4

Option 3:
HNO_2

Option 4:
H_2O_2

Correct Answer:
H_3PO_4

Solution:
As we have learnt,

Oxoacids of Phosphorus -

Phosphorus forms a number of oxoacids. The important oxoacids of phosphorus with their formulas, methods of preparation and the presence of some characteristic bonds in their structures are given in Table given below.

Name	Formula	Oxidation state of phosphorus	Characteristic bonds and their number	Preparation
Hypophosphorous (Phosphinic)	H_3PO_2	+1	One P – OH Two P – H One P = O	white P_4 + alkali
Orthophosphorous (Phosphonic)	H_3PO_3	+3	Two P – OH One P – H One P = O	$P_2O_3 + H_2O$
Pyrophosphorous	$H_4P_2O_5$	+3	Two P – OH Two P – H Two P = O	$PCl_3 + H_3PO_3$
Hypophosphoric	$H_4P_2O_6$	+4	Four P – OH Two P = O One P – P	red P_4 + alkali
Orthophosphoric	H_3PO_4	+5	Three P – OH One P = O	$P_4O_{10}+H_2O$
Pyrophosphoric	$H_4P_2O_7$	+5	Four P – OH Two P = O One P – O – P	heat phosphoric acid
Metaphosphoric*	$(HPO_3)_n$	+5	Three P – OH Three P = O Three P – O – P	phosphorus acid + Br_2, heat in a sealed tube

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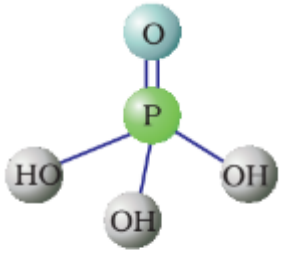
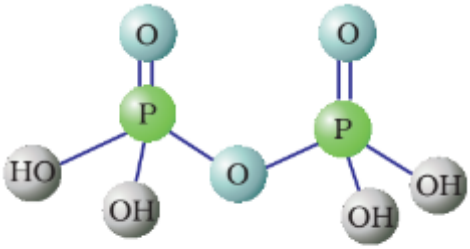
The compositions of the oxoacids are interrelated in terms of loss or gain of H_2O molecule or O-atom. The structures of some important oxoacids are given in later pictures. In oxoacids phosphorus is tetrahedrally surrounded by other atoms. All these acids contain at least one P=O bond and one P–OH bond. The oxoacids in which phosphorus has lower oxidation state (less than +5) contain, in addition to P=O and P–OH bonds, either P–P (e.g., in $H_4P_2O_6$) or P–H (e.g., in H_3PO_2) bonds but not both. These acids in +3 oxidation state of phosphorus tend to disproportionate to higher and lower oxidation states. For example, orthophophorous acid (or phosphorous acid) on heating disproportionates to give orthophosphoric acid (or phosphoric acid) and phosphine.

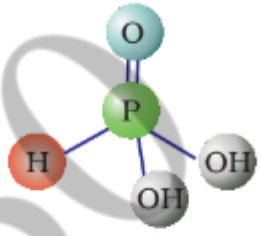
$$4H_3PO_3 \rightarrow 3H_3PO_4 + PH_3$$



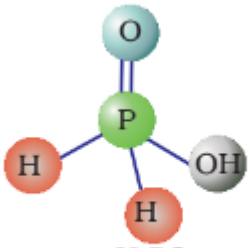
H_3PO_4
Orthophosphoric acid



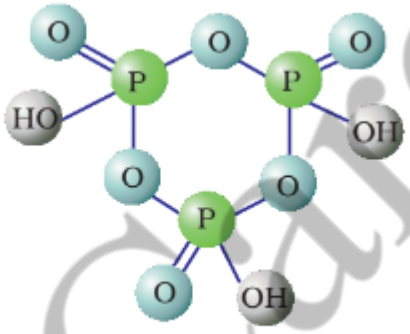
$H_4P_2O_7$
Pyrophosphoric acid



H_3PO_3
Orthophosphorous acid



H_3PO_2
Hypophosphorous acid



Cyclotrimetaphosphoric acid, $(HPO_3)_3$

Polymetaphosphoric acid, $(HPO_3)_n$

-

As in H_3PO_4 Phosphorous is present it's maximum oxidation number state hence it cannot act as a reducing agent.

Hence, the option number (2) is correct.

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Q. 17 If the enthalpy of atomisation for $Br_{2(l)}$ is xkJ/mol and bond enthalpy for Br_2 is ykJ/mol, the relation between them :

Option 1:
is x < y

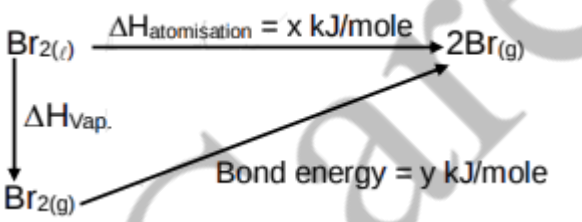
Option 2:
does not exist

Option 3:
is x >y

Option 4:
is x = y

Correct Answer:
is x >y

Solution:

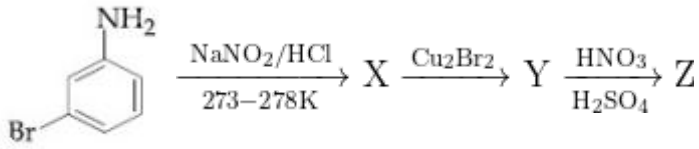


$$\Delta H_{atomisation} = \Delta H_{vap} + \text{Bond energy}$$

Hence x > y

Hence, the option number (3) is correct.

Q. 18 The major product Z is obtained in the following reaction scheme is :



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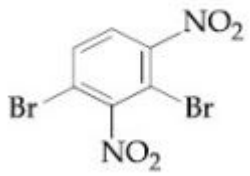
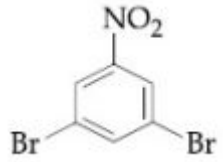
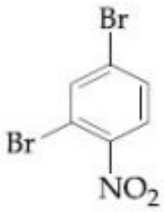
Option 1:



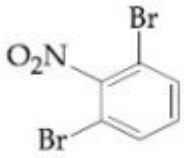
Option 2:



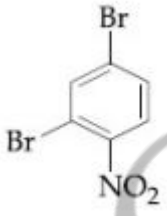
Option 3:



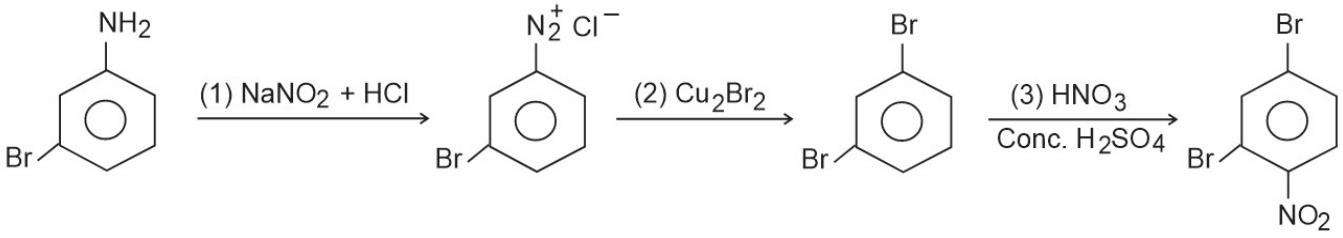
Option 4:



Correct Answer:



Solution:
The reaction will be -



Hence, option number (3) is correct.

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Q. 19 $[Pd(F)(Cl)(Br)(I)]^{2-}$ has n number of geometrical isomers. Then, the spin- only magnetic moment and crystal field stabilisation energy [CFSE] of $[Fe(CN)_6]^{n-6}$, respectively are ;

[Note : Ignore the pairing energy]

Option 1:
$1.73BM\ and -2.0\Delta_0$

Option 2:
$2.84\ BM\ and -1.6\Delta_0$

Option 3:
$0\ BM\ and -2.4\Delta_0$

Option 4:
$5.92\ BM\ and\ 0$

Correct Answer:
$1.73BM\ and -2.0\Delta_0$

Solution:
As we have learnt,

Square planar complexes of type $[Mabcd]$ have 3 geometrical isomers:

$$(\text{a trans b}); (\text{a trans c}); (\text{b trans c})$$

Therefore, number of Geometrical Isomers in square planar $[PdFClBrI]^{2-}$ is 3
Hence, n = 3

Now, the given complex is $[Fe(CN)_6]^{3-}$ which contains Fe^{3+} having a d^5 configuration. According to CFT, the configuration is $t_{2g}^5 e_g^0$ and the central ion has only 1 unpaired electron.

$$\mu = \sqrt{n(n+2)} = 1.73\text{B.M}$$

$$\text{CFSE} = -0.4\Delta_0 \times nt_{2g} + 0.6\Delta_0 \times n_{eg}$$

$$= -0.4\Delta_0 \times 5 = -2.0\Delta_0$$

Hence, **the option number (1) is correct.**

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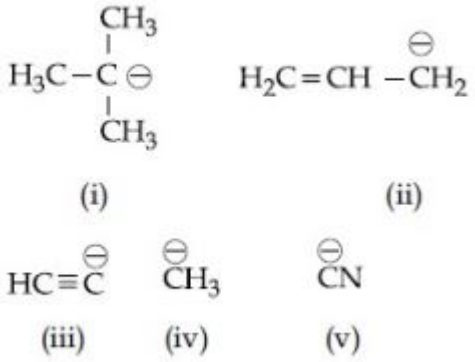
Q. 20 The increasing order of basicity for the following intermediates is (from weak to strong)



(i) (ii)

(iii) (iv) (v)

Option 1:
v < iii < ii < iv < i

Option 2:
iii < i < ii < iv < v

Option 3:
v < i < iv < ii < iii

Option 4:
iii < iv < ii < i < iv

Correct Answer:
v < iii < ii < iv < i

Solution:
1. Basicity is inversely proportional to electronegativity.

2. Basicity is inversely proportional to the stability of the negative charge.

i) is most unstable due to the three +I effects which will destabilise the negative charge, So it will be most basic in nature.

iv) will be second-most unstable, it will be second most basic among these.

the order which matches is: 1) v < iii < ii < iv < i

We do not compare ii, iii and v without checking the data of pka value.

Hence, option number (1) is correct.

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Q. 21 The hardness of a water sample containing $10^{-3} M\ MgSO_4$ expressed as $CaCO_3$ equivalents (in ppm) is_____.

(molar mass of $MgSO_4$ is 120.37 g/mol)

Option 1:
100ppm

Option 2:
200ppm

Option 3:
99.85ppm

Option 4:
111.5ppm

Correct Answer:
100ppm

Solution:
We know that,

The hardness of Water :
The hardness of water is due to the presence of bicarbonates, chlorides, and sulfates of calcium and magnesium bicarbonates causes temporary hardness while chlorides and sulphates cause permanent hardness. The extent of hardness is called degree of hardness. It is defined as the number of parts by weight of calcium carbonate present In one million parts by weight of water that is, in ppm (milligrams per litre) of $CaCO_3$

$$\text{Hardness of water } = \frac{\text{gm of calcium carbonate}}{10^6 \text{gm of water}}$$

$$\text{ppm}_{(\text{in terms of } CaCO_3)} = \frac{\text{weight of } CaCO_3 \text{ in gram}}{\text{weight of water in gram}} \text{ x } 10^6$$

$$1 \text{ L} = 1000 \text{ gram}$$

Now, Need to find the weight gram of $CaCO_3$
10^{-3} molar $CaSO_4$ = 10^{-3} moles of $CaSO_4$ present in 1 L solution.

We assume

$n_{CaCO3} = n_{CaSO4}$

weight of $CaCO_3$ / molar mass of $CaCO_3$ = mole of $CaSO_4$

weight of $CaCO_3$ /100 = 10^{-3}

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weight of $CaCO_3$ = 100 X 10^{-3}

And the volume of water = 1 L then

weight of water = 1000

$$\text{ppm}_{(\text{in terms of } CaCO_3)} = \frac{10^{-3} \text{ x } 100}{1000} \text{ x } 10^6$$

$$\text{ppm}_{(\text{in terms of } CaCO_3)} = 100\text{ppm}$$

Hence, the option number (1) is correct.

Q. 22 The molarity of HNO_3 in a sample which has density 1.4 g/mL and mass percentage of 63% is ____ (in **mol/L**) (Molecular weight of HNO_3 = 63)

Correct Answer:
14

Solution:

$$63\% \text{ w/w} \rightarrow \text{HNO}_3 \text{ solution}$$

Let, mass of solution be 100g

$\therefore$ Mass of HNO_3 is 63 g

$\therefore$ Moles of HNO_3 is 1

$$Now,\ volume\ of\ solution\ = \frac{100}{1.4} ml = \frac{1}{14}\ litres$$

$$Molarity = \frac{1}{\frac{1}{14}} = 14$$

Hence, the option number (1) is correct.

Q. 23 108 g of sliver (molar mass 108 g mol^{-1}) is deposited at cathode from $AgNO_3(aq)$ solution by a certain quantity of electricity. The volume (in L) of oxygen gas produced at 273K and 1 bar pressure from water by the same quantity of electricity is _____.

Option 1:
5.675 L

Option 2:
-

Option 3:
-

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Option 4:

-

Correct Answer:

5.675 L

Solution:

$Ag^+ + e^- \rightarrow Ag$

1F charge is required to deposit 1 mole of Ag

$H_2O \rightarrow \frac{1}{2}O_2 + 2H^+ 2e^-$

$2F \text{ charge deposit } \rightarrow \frac{1}{2} \text{ mole}$

$1F \text{ charge will deposit } \rightarrow \frac{1}{4} \text{ mole}$

$V_{O_2} = \frac{nRT}{P}$

$V_{O_2} = \frac{1}{4} x \frac{0.08341 \, x \, 273}{1}$

$V_{O_2} = \frac{1}{4} x \, 22.7$

$V_{O_2} = 5.675L$

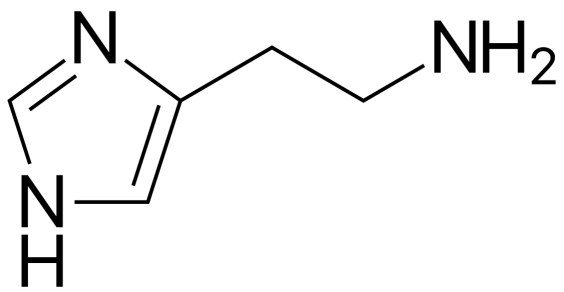
Q. 24 The mass percentage of nitrogen in histamine is ____.

Correct Answer:

37.78%

Solution:

The structure of histamine is given below:



The molecular formula of Histamine is $C_5H_9N_3$

The molecular mass of Histamine is 111.15

Now, mass % of nitrogen = (42/111.15) x 100 = 37.78%

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Q. 25 How much amount of NaCl (in grams) should be added to 600g of water to decrease the freezing point of water to $-0.2^{\circ}C$?

(Assume that the freezing point depression constant for water = 2K kg mol^{-1})

Correct Answer:
1.76

Solution:
$\Delta T_f = 0.2^{\circ}C$
$\Delta T_f = ik_f m$
$0.2 = 2 \times 2 \times \frac{w}{58.5} \times \frac{1000}{600}$

$w = \frac{0.2 \times 58.5 \times 600}{1000 \times 4} = \frac{1.2 \times 58.5}{40} = 1.76g$

Maths

Q. 1 Let C the centroid of the triangle with vertices $(3, -1), (1, 3)\ and\ (2, 4)$. Let P be the point of intersection of the lines $x + 3y - 1 = 0$ and $3x - y + 1 = 0$. Then the line passing through the points C and P also passes through the point:

Option 1:
$(-9, -7)$

Option 2:
$(-9, -6)$

Option 3:
$(7, 6)$

Option 4:
$(9, 7)$

Correct Answer:
$(-9, -6)$

Solution:

Centroid -

Centroid

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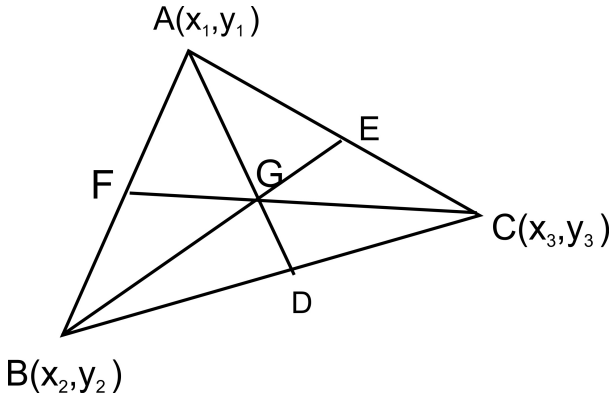
Centroid of a triangle is the point of intersection of the medians of the triangle. A centroid divides the median in the ratio 2:1.

Whereas, the median is the line joining the mid-points of the sides and the opposite vertices.

The coordinates of the centroid of a triangle (G) whose vertices are A (x_1, y_1), B (x_2, y_2) and C(x_3, y_3), is given by



$$\left(\frac{\mathbf{x_1+x_2+x_3}}{\mathbf{3}}, \frac{\mathbf{y_1+y_2+y_3}}{\mathbf{3}}\right)$$

If D (a_1, b_1), E (a_2, b_2) and F (a_3, b_3) are the mid point of ΔABC, then its centroid is given by

$$\left(\frac{\mathbf{a_1+a_2+a_3}}{\mathbf{3}}, \frac{\mathbf{b_1+b_2+b_3}}{\mathbf{3}}\right)$$

-

Point of intersection of two lines -

Point of intersection of two lines

Equation of two non-parallel line is

$$L_1 = a_1x + b_1y + c_1 = 0$$
$$L_2 = a_2x + b_2y + c_2 = 0$$

If P (x_1, y_1) is a point of intersection of L_1 and L_2 , then solving these two equations of the line by cross multiplication

$$\frac{x_1}{b_1c_2 - c_1b_2} = \frac{y_1}{c_1a_2 - a_1c_2} = \frac{1}{a_1b_2 - b_1a_2}$$

We get,

$$(\mathbf{x_1, y_1}) = \left(\frac{\mathbf{b_1c_2 - b_2c_1}}{\mathbf{a_1b_2 - a_2b_1}}, \frac{\mathbf{c_1a_2 - c_2a_1}}{\mathbf{a_1b_2 - a_2b_1}}\right)$$

-

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Equation of Straight Line (Part 2) -

Equation of Straight Line

(c) Two-point form

The equation of a straight line passing through the two given points (x_1,y_1) and $(x_1,y_1)$is given by

$$\mathbf{y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1)}.$$

-

The centroid of triangle ABC D(2,2)

Point of intersection P $\left(-\frac{1}{5}, \frac{2}{5}\right)$

equation of line DP is 8x – 11y + 6 = 0

Point (–9,–6) satisfies the equation

Q. 2 The product $2^{1/4} \cdot 4^{1/16} \cdot 8^{1/48} \cdot 16^{1/128} \ldots$ is equal to :

Option 1:
$2^{1/4}$

Option 2:
2

Option 3:
$2^{1/2}$

Option 4:
1

Correct Answer:
$2^{1/2}$

Solution:
Sum of an infinite GP

If a is the first term and r is the common ratio of a G.P. Then,

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$$S_\infty = \frac{a}{1-r}$$

S_∞ is the sum to infinite terms of the G.P.

Now,

$$2^{1/4} \cdot 4^{1/16} \cdot 8^{1/48} \ldots = 2^{\frac{1}{4}+\frac{1}{8}+\frac{1}{16}\cdots}$$
$$\Rightarrow 2^{\frac{\frac{1}{4}}{1-\frac{1}{2}}} = \sqrt{2}$$

Q. 3 A spherical iron ball of $10cm$ radius is coated with a layer of ice of uniform thickness that melts at a rate of $50cm^3/min$. When the thickness of ice is $5cm$, then the rate (in cm / min.) at which of the thickness of ice decreases, is :

Option 1:
$5/6\pi$

Option 2:
$1/54\pi$

Option 3:
$1/36\pi$

Option 4:
$1/18\pi$

Correct Answer:
$1/18\pi$

Solution:

Derivative as Rate Measure -

Derivative as Rate Measure

We have seen that for quantities that are changing over time, the rates at which these quantities change are given by derivatives. If two related quantities are changing over time, the rates at which the quantities change are related. For example, if a balloon is being filled with air, both the radius of the balloon and the volume of the balloon are increasing.

If a variable quantity y depends on and varies with a quantity x, then the rate of change of y with respect to x is $\frac{dy}{dx}$.

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A rate of change with respect to time is simply called the rate of change.

For example, the rate of change of displacement (s) of an object w.r.t. time is velocity (v). $v = \frac{ds}{dt}$

When limit $\Delta t \to 0$ is applied, the rate of change becomes instantaneous and we get the rate of change of displacement (s) w.r.t. time at an instant.

i.e. $\lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$

-

$$V = \frac{4}{3}\pi(10+a)^3$$
$$\frac{\partial V}{\partial t} = 4\pi(10+a)^2 \cdot \frac{\partial a}{\partial t} = 50\frac{cm^3}{\min}$$
$$4\pi(10+5)^2 \cdot \frac{\partial a}{\partial t} = 50\frac{cm^3}{\min}$$
$$\frac{\partial a}{\partial t} = \frac{1}{18\pi}\frac{cm}{\min}$$

Q. 4 Let f be any function continuous on $[a, b]$ and twice differentiable on (a, b). If for all $x\epsilon\,(a, b)\,, f'(x) > 0$ and $f''(x) < 0$, then for any $c\epsilon(a, b)$, $\frac{f(c)-f(a)}{f(b)-f(c)}$ is greater than :

Option 1:
$\frac{b-c}{c-a}$

Option 2:
1

Option 3:
$\frac{c-a}{b-c}$

Option 4:
$\frac{b+a}{b-a}$

Correct Answer:
$\frac{c-a}{b-c}$

Solution:

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Lagrange's Mean Value Theorem -

Lagrange's Mean Value Theorem

Rolle's theorem is a special case of the **Mean Value Theorem**. In Rolle's theorem, we consider differentiable functions f defined on a closed interval [a, b] with f(a) = f(b) . The Mean Value Theorem generalized Rolle's theorem by considering functions that do not necessarily have equal value at the endpoints.

Statement

Let f (x) be a function defined on [a, b] such that

1. it is continuous on [a, b],
2. it is differentiable on (a, b).

Then there exists a real number c (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Cauchy's mean value Theorem

Cauchy's mean value theorem, also known as the extended mean value theorem. It states that if both function f(x) and g(x) are continuous on the closed interval [a, b] and differentiable on the open interval (a, b) and g'(x) is not zero on that open interval, then there exists c in (a, b) such that

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

-

Use LMVT for x ∈[a,c]

$$\frac{\mathrm{f(c) - f(a)}}{\mathrm{c - a}} = \mathrm{f'}(\alpha), \alpha \in (\mathrm{a}, \mathrm{c})$$

also use LMVT for x ∈[c,b]

$$\frac{f(b) - f(c)}{b - c} = f'(\beta), \beta \in (c, b)$$

∵ f ''(x) < 0 $\rightarrow$ f '(x) is decreasing

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$f'(\alpha) > f'(\beta)$

$\frac{f(c)-f(a)}{c-a} > \frac{f(b)-f(c)}{b-c}$

$\frac{f(c)-f(a)}{f(b)-f(c)} > \frac{c-a}{b-c} (\because f(x) \text{ is increasing})$

Q. 5 The value of

$$\cos^3\left(\frac{\pi}{8}\right)\cdot\cos\left(\frac{3\pi}{8}\right)+\sin^3\left(\frac{\pi}{8}\right)\cdot\sin\left(\frac{3\pi}{8}\right) \text{ is :}$$

Option 1:

$\frac{1}{4}$

Option 2:

$\frac{1}{2\sqrt{2}}$

Option 3:

$\frac{1}{2}$

Option 4:

$\frac{1}{\sqrt{2}}$

Correct Answer:

$\frac{1}{2\sqrt{2}}$

Solution:

Trigonometric Identities -

Trigonometric Identities-

These identities are the equations that hold true regardless of the angle being chosen.

$\sin^2 t + \cos^2 t = 1$

$1 + \tan^2 t = \sec^2 t$

$1 + \cot^2 t = \csc^2 t$

$\tan t = \frac{\sin t}{\cos t}, \quad \cot t = \frac{\cos t}{\sin t}$

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Allied Angles (Part 1) -

Allied Angles (Part 1)

Two angles or numbers are called allied iff their sum or difference is a multiple of π/2

- $\sin(90^0 - \theta) = \cos(\theta)$
- $\cos(90^0 - \theta) = \sin(\theta)$
- $\tan(90^0 - \theta) = \cot(\theta)$
- $\csc(90^0 - \theta) = \sec(\theta)$
- $\sec(90^0 - \theta) = \csc(\theta)$
- $\cot(90^0 - \theta) = \tan(\theta)$

- $\sin(90^0 + \theta) = \cos(\theta)$
- $\cos(90^0 + \theta) = -\sin(\theta)$
- $\tan(90^0 + \theta) = -\cot(\theta)$
- $\csc(90^0 + \theta) = \sec(\theta)$
- $\sec(90^0 + \theta) = -\csc(\theta)$
- $\cot(90^0 + \theta) = -\tan(\theta)$

-

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$$\cos^3\frac{\pi}{8}\cos\frac{3\pi}{8}+\sin^3\frac{\pi}{8}\sin\frac{3\pi}{8}$$
$$\sin\left(\frac{\pi}{2}-\frac{3\pi}{8}\right)=\cos\left(\frac{3\pi}{8}\right)=\sin\left(\frac{\pi}{8}\right)$$

$$\cos^3\frac{\pi}{8}\sin\frac{\pi}{8}+\sin^3\frac{\pi}{8}\cos\frac{\pi}{8}$$
$$\sin\frac{\pi}{8}\cos\frac{\pi}{8}\left(\cos^2\frac{\pi}{8}+\sin^2\frac{\pi}{8}\right)$$
$$\frac{1}{2}\left(2\sin\frac{\pi}{8}\cos\frac{\pi}{8}\right)=\frac{1}{2}\sin\frac{2\pi}{8}=\frac{1}{2\sqrt{2}}$$

Q. 6 The number of real roots of the equation,

$e^{4x}+e^{3x}-4e^{2x}+e^{x}+1=0$ is :

Correct Answer:
1

Solution:

Transcendental function -

Transcendental functions: the functions which are not algebraic are called transcendental functions. Exponential, logarithmic, trigonometric and inverse trigonometric functions are transcendental functions.

Exponential Function: function f(x) such that $\mathrm{f(x)}=\mathrm{a^x}$ is known as an exponential function.

$$\text{base}: \quad a>0, a\neq 1$$
$$\text{domain}: x\in\mathbb{R}$$
$$\text{range}: f(x)>0$$

$$\text{base}: \quad a>0, a\neq 1$$
$$\text{domain}: x>0$$
$$\text{range}: f(x)\in\mathbb{R}$$

If a > 1 If a < 1

Properties of Logarithmic Function

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1. $\log_e(ab) = \log_e a + \log_e b$
2. $\log_e\left(\frac{a}{b}\right) = \log_e a - \log_e b$
3. $\log_e a^m = m \log_e a$
4. $\log_a a = 1$
5. $\log_{b^m} a = \frac{1}{m} \log_b a$
6. $\log_b a = \frac{1}{\log_a b}$
7. $\log_b a = \frac{\log_m a}{\log_m b}$
8. $a^{\log_a m} = m$
9. $a^{\log_c b} = b^{\log_c a}$
10. $\log_m a = b \Rightarrow a = m^b$

-

Quadratic Equation -

The root of the quadratic equation is given by the formula:

$$x = \frac{-b \pm \sqrt{D}}{2a}$$

or

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Where D is called the discriminant of the quadratic equation, given by $D = b^2 - 4ac$,

-

Let $e^x = t \in (0, \infty)$

Now the equation

$t^4 + t^3 - 4t^2 + t + 1 = 0$
$t^2 + t - 4 + \frac{1}{t} + \frac{1}{t^2} = 0$
$\left(t^2 + \frac{1}{t^2}\right) + \left(t + \frac{1}{t}\right) - 4 = 0$

Let $t + \frac{1}{t} = \alpha$

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$(\alpha^2-2)+\alpha-4=0$
$\alpha^2+\alpha-6=0$
$\alpha^2+\alpha-6=0$

$\alpha=-3,2$

Only positive value possible so $\alpha=2\Rightarrow \quad e^x+e^{-x}=2$

x=0 is the only solution.

Q. 7 The value of $\int_0^{2x}\frac{x\sin^8 x}{\sin^8 x+\cos^8 x}dx$ is equal to :

Option 1:
2π

Option 2:
4π

Option 3:
$2\pi^2$

Option 4:
π^2

Correct Answer:
π^2

Solution:

Properties of the Definite Integral (Part 2) - King's Property -

Property 4 (King's Property)

This is one of the most important properties of definite integration.

$$\int_a^b f(x)\,dx=\int_a^b f(a+b-x)\,dx$$

-

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Application of Periodic Properties in Definite Integration -

Property 9

If f(x) is a periodic function with period T, then the area under f(x) for n periods would be n times the area under f(x) for one period, i.e.

$$\int_0^{\mathbf{nT}} \mathbf{f(x)dx} = \mathbf{n}\int_0^{\mathbf{T}} \mathbf{f(x)dx}$$

-

$$I = \int_0^{2\pi} \frac{x\sin^8 x}{\sin^8 x + \cos^8 x}dx$$
$$I = \int_0^{2\pi} \frac{(2\pi - x)\sin^8(2\pi - x)}{\sin^8(2\pi - x) + \cos^8(2\pi - x)}dx$$
$$2I = \int_0^{2\pi} \frac{2\pi\sin^8 x}{\sin^8 x + \cos^8 x}dx$$
$$I = \int_0^{2\pi} \frac{\pi\sin^8 x}{\sin^8 x + \cos^8 x}dx$$
$$I = \int_0^{2\pi} \frac{\pi\sin^8 x}{\sin^8 x + \cos^8 x}dx$$
$$I = 4\int_0^{\pi/2} \frac{\pi\sin^8 x}{\sin^8 x + \cos^8 x}dx$$
$$I = 4\int_0^{\pi/2} \frac{\pi\sin^8(\pi/2 - x)}{\sin^8(\pi/2 - x) + \cos^8(\pi/2 - x)}dx$$
$$I = 2\int_0^{\pi/2} \pi dx$$
$$I = \pi^2$$

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Q. 8 If for some α and β in R, the intersection of the following three planes

$x + 4y - 2z = 1$

$x + 7y - 5z = \beta$

$x + 5y + \alpha z = 5$

is aline in R^3, then $\alpha + \beta$ is equal to :

Correct Answer:
10

Solution:

Cramer's law -

Cramer's law for the system of equations in two variables :

$$\text{Let } a_1x + b_1y = c_1 \text{ and } a_2x + b_2y = c_2, \text{where}$$
$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$
$$\text{on solving this equation by cross multiplication, we get}$$
$$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{a_1c_2 - a_2c_1} = \frac{1}{a_1b_2 - a_2b_1}$$
$$\text{or } \frac{x}{\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}} = \frac{y}{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}} = \frac{1}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$
$$\text{or } x = \frac{\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}, y = \frac{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

We can observe that first row in the numerator of x is of constants and 2nd row in the numerator is of constants, and the denominator is of the coefficient of variables.

We can follow this analogy for the system of equations of 3 variable where third in the numerator of the value of z will be of constant and denominator will be formed by the value of coefficients of the variables.

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so let us consider the system of equations

$a_1x + b_1y + c_1z = d_1$ ...(i)

$a_2x + b_2y + c_2z = d_2$ ...(ii)

$a_3x + b_3y + c_3z = d_3$ ...(iii)

then Δ, which will be determinant of coefficient of variables, will be

$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$, Δ_1 numerator of x is :

$\Delta_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$, similarly $\Delta_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$ and

$\Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$, if$\Delta \neq 0$ then $x = \frac{\Delta_1}{\Delta}$

$y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}$

i) If ? ≠ 0, then the system of equations has a unique finite solution and so equations are consistent, and solutions are $x = \frac{\Delta_1}{\Delta}, y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}$

ii) If ? = 0, and any of $\Delta_1 \neq 0$ *or* $\Delta_2 \neq 0$ *or* $\Delta_3 \neq 0$

Then the system of equations is inconsistent and hence no solution exists.

iii) If all $\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$ then

System of equations is consistent and dependent and it has an infinite number of solution.

-

$\Delta = 0 \Rightarrow \begin{vmatrix} 1 & 4 & -2 \\ 1 & 7 & -5 \\ 1 & 5 & \alpha \end{vmatrix} = 0$

$(7\alpha + 25) - (4\alpha + 10) + (-20 + 14) = 0$

$3\alpha + 9 = 0 \Rightarrow \alpha = -3$

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$$\text{Also} \quad D_3 = 0 \Rightarrow \begin{vmatrix} 1 & 4 & 1 \\ 1 & 7 & \beta \\ 1 & 5 & 5 \end{vmatrix} = 0$$

$1(35 - 5\beta) - (15) + 1(4\beta - 7) = 0$

$\beta = 13$

$\alpha + \beta = 10$

Q. 9 If $e_1 \ and \ e_2$ are the eccentricities of the ellipse, $\frac{x^2}{18} + \frac{y^2}{4} = 1$ and the hyperbola, $\frac{x^2}{9} - \frac{y^2}{4} = 1$ respectively and $(e_1, e_2)$is a point on the ellipse, $15x^2 + 3y^2 = k,$ then k is equal to :

Correct Answer:

16

Solution:

What is Ellipse? -

Ellipse

$$\mathbf{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1} \quad \text{where } , b^2 = a^2\left(1 - e^2\right)_{(a>b)}$$

1. a > b
2. the length of the major axis is 2a
3. the coordinates of the vertices are (±a, 0)
4. the length of the minor axis is 2b
5. the coordinates of the co-vertices are (0, ±b)

-

What is Hyperbola? -

Hyperbola:

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Eccentricity of Hyperbola:

Equation of the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

we have,

$$b^2 = a^2\left(e^2 - 1\right)$$

$$e^2 = \frac{b^2 + a^2}{a^2}$$

$$e = \sqrt{1 + \left(\frac{b^2}{a^2}\right)}$$

$$e = \sqrt{1 + \left(\frac{2b}{2a}\right)^2}$$

$$e = \sqrt{1 + \left(\frac{\text{conjugate axis}}{\text{transverse axis}}\right)^2}$$

-

$$e_1 = \sqrt{1 - \frac{4}{18}} = \sqrt{\frac{7}{9}} = \frac{\sqrt{7}}{3}$$

$$e_2 = \sqrt{1 + \frac{4}{9}} = \sqrt{\frac{13}{9}} = \frac{\sqrt{13}}{3}$$

$$15e_1^2 + 3e_2^2 = k \Rightarrow \quad k = 15\left(\frac{7}{9}\right) + 3\left(\frac{13}{9}\right)$$

So, k= 16

Q. 10 If $f(x) = \begin{cases} \frac{\sin(a+2)x + \sin x}{x}; & x < 0 \\ b; & x = 0 \\ \frac{(x+3x^2)^{\frac{1}{3}} - x^{\frac{1}{3}}}{x^{\frac{1}{3}}}; & x > 0 \end{cases}$ is continuous at $x = 0$, then $a + 2b$ is equal to :

Correct Answer:

0

Solution:

Continuity -

A function f(x) is continuous at a point a if and only if the following three conditions are satisfied:

1. f(a) is defined

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2. $\lim_{x \to a} f(x)$ exists

3. $\lim_{x \to a} f(x) = f(a)$

Or $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = f(a)$

$i.e.$ L.H.L. = R.H.L. value of a function at $x = a$

-

$$LHL = \frac{\sin(a+2)x}{x}\frac{a+2}{a+2} + \frac{\sin x}{x} = a + 3$$

$$RHL = \frac{(1+3x)^{1/3} - 1}{x} = 1$$

$a = -2, \quad b = 1$
$a + 2b = 0$

Q. 11

If the matrices $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{bmatrix}$, $B = adj\ A$ and $C = 3A$, then $\frac{|adj\ B|}{|C|}$ is equal to :

Correct Answer:
8

Solution:

Properties of adjoint of Matrix - Part 2 -

Properties of adjoint of matrix

5. If A is non-singular square matrix, then, $|\mathrm{adj}(\mathrm{adjA})| = |\mathrm{A}|^{(\mathrm{n}-1)^2}$

Proof: from the previous property, we know that

$\mathrm{adj}(\mathrm{adjA}) = |\mathrm{A}|^{(\mathrm{n}-2)}\mathrm{A}$
taking determinant on both sides, then
$|\mathrm{adj}(\mathrm{adjA})| = ||\mathrm{A}|^{(\mathrm{n}-2)}\mathrm{A}| = |\mathrm{A}|^{\mathrm{n}(\mathrm{n}-2)}|\mathrm{A}| = |\mathrm{A}|^{(\mathrm{n}-1)^2}$
(using $|kA| = k^n|A|$)

-

$$|A| = \begin{vmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{vmatrix} = 6$$

$$|\,\mathrm{adj\,B}| = |\,\mathrm{adj\ adj\,A}| = |\mathrm{A}|^{(n-1)^2} = |\mathrm{A}|^4 = (36)^2$$

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$$|C| = |BA| = 3^3 \times 6$$
$$\frac{|\text{adjB}|}{|C|} = \frac{36 \times 36}{3^3 \times 6} = 8$$

Q. 12 A circle touches the y-axis at the point $(0, 4)$ and passes through the point $(2, 0)$. Which of the following lines is not tangent to this circle ?

Option 1:
$4x - 3y + 17 = 0$

Option 2:
$3x + 4y - 6 = 0$

Option 3:
$4x + 3y - 8 = 0$

Option 4:
$3x - 4y - 24 = 0$

Correct Answer:
$4x + 3y - 8 = 0$

Solution:

Circle(Definition) -

Circle

Equation of circle

Central Form:

The equation of a circle with centre at C (h,k) and radius r is $(x-h)^2+(y-k)^2=r^2$.

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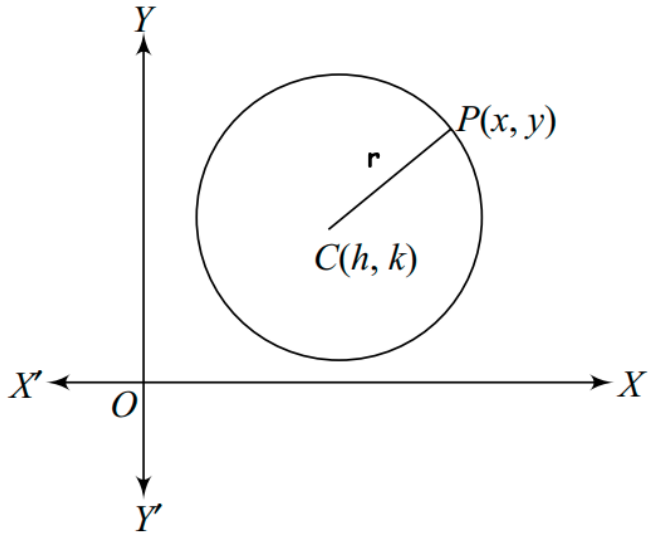

Let P(x, y) be any point on the circle. Then, by definition, | CP | = r.

Using the distance formula, we have

$$\sqrt{(x-h)^2+(y-k)^2} = r$$

i.e. $$(x-h)^2+(y-k)^2 = r^2$$

If the centre of the circle is the origin or (0,0) then equation of the circle becomes
$(x-0)^2+(y-0)^2 = r^2$
i.e. $x^2+y^2 = r^2$

General Form:

The equation of a circle with centre at (h,k) and radius r is

$\Rightarrow (x-h)^2+(y-k)^2 = r^2$
$\Rightarrow x^2+y^2-2hx-2ky+h^2+k^2-r^2 = 0 \quad \ldots(i)$
Which is of the form :
$\mathbf{x^2+y^2+2gx+2fy+c=0} \quad \ldots(ii)$

-

Family of Circles -

2. Equation of the family of circles passing through the point of intersection of a given circle S = 0 and a line L = 0is S + λL where, λ is a parameter.

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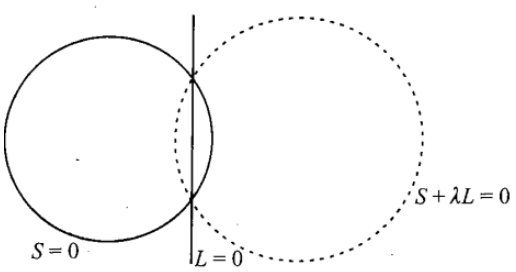

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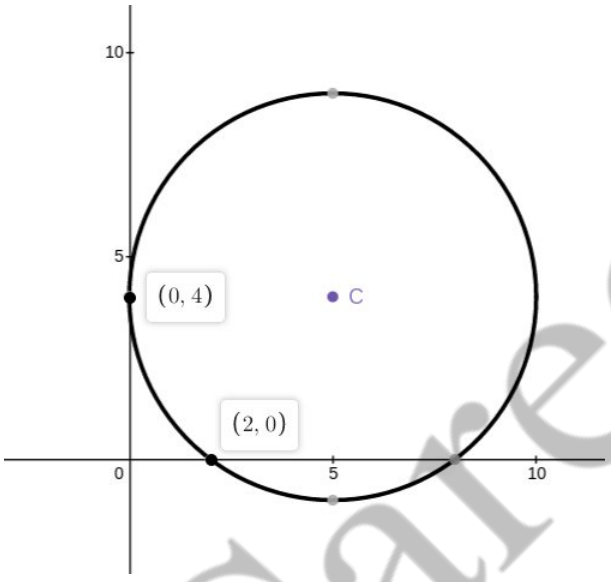

Equation of family of circle touches at (0,4) $(x-0)^2+(y-4)^2+\lambda x=0$

This family of circle passes through (2,0)

$4+16+2\lambda=0 \Rightarrow \lambda=-10$

$x^2+y^2-10x-8y+16=0$

Center (5, 4) and radius = 5

Now check the option

4x+3y-8=0

$$\left|\frac{4\times 5+3\times 4-8}{5}\right|=\frac{24}{5}\neq 5$$

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Q. 13 Let z be a complex number such that $\left|\frac{z-i}{z+2i}\right| = 1$ and $|z| = \frac{5}{2}$. Then the value of $|z+3i|$ is :

Option 1:

$\sqrt{10}$

Option 2:

$\frac{7}{2}$

Option 3:

$\frac{15}{4}$

Option 4:

$2\sqrt{3}$

Correct Answer:

$\frac{7}{2}$

Solution:

Distance formula, Equation of perpendicular bisector -

Distance between two points A(z_1) and B(z_2) can be found as

AB = $|z_2 - z_1|$ = | Affix of B - Affix of A |

$\text{Let } z_1 = x + iy \text{ and } z_2 = x + iy$
$\text{Then }, |z_1 - z_2| = |(x_1 - x_2) + i(y_1 - y_2)|$
$= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
$= \text{distance between points } (x_1, y_1) \text{ and}$
$(x_2, y_2) = \text{distance between between } z_1 \text{ and } z_2$
$\text{where } z_1 = x_1 + iy_1 \text{ and } z_2 = x_2 + iy_2$

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- The distance of a point from the origin is $|z - 0| = |z|$
- Three points $A(z_1)$, $B(z_2)$ and $C(z_3)$ are collinear, then AB + BC = AC

  i.e. $|z_2 - z_1| + |z_3 - z_2| = |z_3 - z_1|$

We can use the distance formula to find the equation of perpendicular bisector formula.

Let two fixed points $A(z_1)$ and $B(z_2)$ and a moving point C(z) which is always at the same

Distance from $A(z_1)$ and $B(z_2)$

AC = BC

Locus of a point equidistant from two given points.

$|z - z_1| = |z - z_2|$

z will lie on the perpendicular bisector of the line joining Z_1 and Z_2.

This is the equation of perpendicular of bisector as z always remains at the same distance from

$A(z_1)$ and $B(z_2)$

-

Area of triangle, circle (formula) -

Equation of Circle:

The equation of the circle whose center is at the point Z_0 and have radius r is given by

$|z - z_0| = r$

If the center is origin then, $z_0 = 0$, hence equation reduces to $|z| = r$

Interior of the circle is represented by $|z - z_0| < r$

The exterior is represented by $|z - z_0| > r$

Here z can be represented as x + iy and Z_0 is represented by $x_0 + iy_0$

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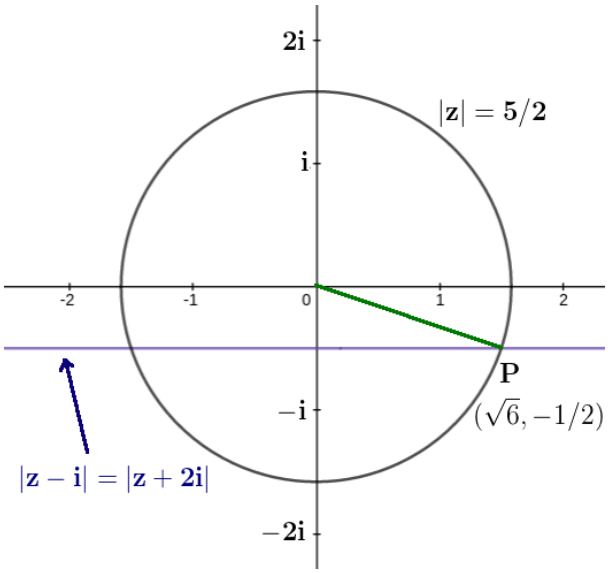
$|z-i| = |z+2i|$
this is the equation of perpendicular bisector



z = $(\sqrt{6}, -1, 2)$

$z = \sqrt{6} - 1/2i$

$|z+3i| = |\sqrt{6} + (3-1/2)i| = \sqrt{6+\frac{25}{4}} = \frac{7}{2}$

Q. 14 If $f'(x) = \tan^{-1}(\sec x + \tan x), -\frac{\pi}{2} < x < \frac{\pi}{2}$, and $f(0) = 0$, then $f(1)$ is equal to :

Option 1:
$\frac{\pi+1}{4}$

Option 2:
$\frac{\pi+2}{4}$

Option 3:
$\frac{1}{4}$

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Option 4:

$$\frac{\pi - 1}{4}$$

Correct Answer:

$$\frac{\pi + 1}{4}$$

Solution:

Principal Value of function f-1 (f (x)) -

Principal Value of function f^{-1} (f (x))

1. $\sin^{-1}(\sin(\theta)) = \theta$ for all $\theta \in [-\pi/2, \pi/2]$
2. $\cos^{-1}(\cos(\theta)) = \theta$ for all $\theta \in [0, \pi]$
3. $\tan^{-1}(\tan(\theta)) = \theta$ for all $\theta \in (-\pi/2, \pi/2)$
4. $\cot^{-1}(\cot(\theta)) = \theta$ for all $\theta \in (0, \pi)$
5. $\sec^{-1}(\sec(\theta)) = \theta$ for all $\theta \in [0, \pi] - \{\pi/2\}$
6. $\csc^{-1}(\csc(\theta)) = \theta$ for all $\theta \in [-\pi/2, \pi/2] - \{0\}$

-

Derivative of the Polynomial Function -

Derivative of the Polynomial Function

Finding derivatives of functions by using the definition of the derivative can be a lengthy and, for certain functions, a rather challenging process.

In this section, we will learn rules for finding derivatives that allow us to bypass this process. Let's begin with the basics.

1. $\frac{d}{dx}(\textbf{constant}) = \mathbf{0}$

$f(x) = c \;\; and \;\; f(x+h) = c$

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \to 0} \frac{c - c}{h} \\ &= \lim_{h \to 0} \frac{0}{h} \\ &= \lim_{h \to 0} 0 = 0 \end{aligned}$$

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The rule for differentiating constant functions is called the constant rule. It states that the derivative of a constant function is zero; that is, since a constant function is a horizontal line, the slope, or the rate of change, of a constant function, is 0.

2. $\frac{d}{dx}(\mathbf{x^n}) = \mathbf{nx^{n-1}}$

-

$$f'(x) = \tan^{-1}(\sec x + \tan x) = \tan^{-1}\left(\frac{1+\sin x}{\cos x}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{1-\cos\left(\frac{\pi}{2}+x\right)}{\sin\left(\frac{\pi}{2}+x\right)}\right) = \tan^{-1}\left(\frac{2\sin^2\left(\frac{\pi}{4}+\frac{x}{2}\right)}{2\sin\left(\frac{\pi}{4}+\frac{x}{2}\right)\cos\left(\frac{\pi}{4}+\frac{x}{2}\right)}\right)$$

$$= \tan^{-1}\left(\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right) = \frac{\pi}{4}+\frac{x}{2}$$

$$(f'(x)) = \frac{\pi}{4}+\frac{x}{2}$$

$$f(x) = \frac{\pi}{4}x + \frac{x^2}{4} + c$$

$$f(0) = c = 0 \Rightarrow \quad f(x) = \frac{\pi}{4}x + \frac{x^2}{4}$$

$$\text{So } f(1) = \frac{\pi+1}{4}$$

Q. 15 Negation of the statement :

'$\sqrt{5}$ is an integer or 5 is irational' is :

Option 1:
$\sqrt{5}$ is irrational or 5 ia an integer.

Option 2:
$\sqrt{5}$ is not an integer or 5 is not irrational.

Option 3:
$\sqrt{5}$ is an integer and 5 is irrational.

Option 4:
$\sqrt{5}$ is not an integer and 5 is not irrational.

Correct Answer:
$\sqrt{5}$ is not an integer and 5 is not irrational.

Solution:

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Logic Connectivity -

Negation of a Statement

The denial of a statement is called the negation of the statement. The negation of a statement is generally formed by introducing the word “not” at some proper place in the statement or by prefixing the statement with “It is not the case that” or “It is false that”.

The negation of a statement p in symbolic form is written as “~ p” and read as “not p”.

p : New Delhi is the capital of India.

The negation of this statement is

~p : New Delhi is not the capital of India.

Or, ~p : It is not the case that New Delhi is the capital of India.

Or, ~p : It is false that New Delhi is the capital of India.

The truth value of negation of a statement is always opposite to the truth value of the original statement.

p	$\sim p$
T	F
F	T

-

Practise Session - 1 -

Q1. Write the negations of the following:

1. Ram is a doctor or peon.
2. Room is clean and big.
3. $\sqrt{5}$ is a rational number.
4. If Ram is a doctor then he is smart

-

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$\sqrt{5}$ is not an integer and 5 is not an irrational Number

or $\sim (p \vee q) = \sim p \wedge \sim q$

Q. 16 If for all real triplets $(a, b, c),\ f(x) = a + bx + cx^2;\ \ then\ \ \int_0^1 f(x)dx$ is equal to:

Option 1:

$2\left\{3f\left(1\right) + 2f\left(\frac{1}{2}\right)\right\}$

Option 2:

$\frac{1}{3}\left\{f\left(0\right) + f\left(\frac{1}{2}\right)\right\}$

Option 3:

$\frac{1}{2}\left\{f\left(1\right) + 3f\left(\frac{1}{2}\right)\right\}$

Option 4:

$\frac{1}{6}\left\{f\left(0\right) + f(1) + 4f\left(\frac{1}{2}\right)\right\}$

Correct Answer:

$\frac{1}{6}\left\{f\left(0\right) + f(1) + 4f\left(\frac{1}{2}\right)\right\}$

Solution:

Integration as Reverse Process of Differentiation -

If the function F(x) is an antiderivative of f(x), then the expression F(x) + C is the indefinite integral of the function f(x) and is denoted by the symbol ∫ f(x) dx.

By definition,

$$\int \mathrm{f(x)dx} = \mathrm{F(x)} + \mathrm{c}, \quad \text{where } \mathrm{F'(x)} = \mathrm{f(x)} \text{ and } {}'\mathrm{c}' \text{ is constant.}$$

Rules of integration:

f(x) and g(x) are functions with antiderivatives ∫ f(x) and ∫ g(x) dx. Then,

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(a) $\int \mathrm{kf}(x)dx = k\int f(x)dx$ for any constant k.

(b) $\int (f(x)+g(x))dx = \int f(x)dx + \int g(x)dx$

(c) $\int (f(x)-g(x))dx = \int f(x)dx - \int g(x)dx$

Since, $\frac{\mathrm{d}}{\mathrm{dx}}(\mathrm{F(x)}) = \mathrm{f(x)} \quad \Leftrightarrow \quad \int \mathrm{f(x)dx} = \mathrm{F(x)} + \mathrm{c}$

Based on this definition and various standard formulas (which we studied in Limit, Continuity and Differentiability) we obtained the following **important integration formulae,**

1. $\frac{\mathrm{d}}{\mathrm{dx}}(\mathrm{kx}) = \mathrm{k} \Rightarrow \int \mathrm{kdx} = \mathrm{kx} + \mathrm{C},$ where k is a constant
2. $\frac{\mathrm{d}}{\mathrm{dx}}\left(\frac{\mathrm{x}^{\mathrm{n}+1}}{\mathrm{n}+1}\right) = \mathrm{x}^{\mathrm{n}}, \mathrm{n} \neq -1 \Rightarrow \int \mathrm{x}^{\mathrm{n}}\mathrm{dx} = \frac{\mathrm{x}^{\mathrm{n}+1}}{\mathrm{n}+1} + \mathrm{C}, \mathrm{n} \neq -1$
3. $\frac{\mathrm{d}}{\mathrm{dx}}(\log|\mathrm{x}|) = \frac{1}{\mathrm{x}} \Rightarrow \int \frac{1}{\mathrm{x}}\mathrm{dx} = \log|\mathrm{x}| + \mathrm{C},$ when $\mathrm{x} \neq 0$
4. $\frac{\mathrm{d}}{\mathrm{dx}}(\mathrm{e}^{\mathrm{x}}) = \mathrm{e}^{\mathrm{x}} \Rightarrow \int \mathrm{e}^{\mathrm{x}}\mathrm{dx} = \mathrm{e}^{\mathrm{x}} + \mathrm{C}$
5. $\frac{\mathrm{d}}{\mathrm{dx}}\left(\frac{\mathrm{a}^{\mathrm{x}}}{\log_{\mathrm{e}}\mathrm{a}}\right) = \mathrm{a}^{\mathrm{x}}, \mathrm{a} > 0, \mathrm{a} \neq 1 \quad \Rightarrow \int \mathrm{a}^{\mathrm{x}}\mathrm{dx} = \frac{\mathrm{a}^{\mathrm{x}}}{\log_{\mathrm{e}}\mathrm{a}} + \mathrm{C}$

-

Definite Integration -

Working Rule to evaluate definite Integral $\int_a^b f(x)dx$

1. First find the indefinite integration $\int \mathrm{f(x)}\ \mathrm{dx}$ and suppose the result be $\mathrm{F(x)}$
2. Next find the $\mathrm{F(b)}$ and $\mathrm{F(a)}$
3. And, finally value of definite integral is obtained by subtracting $\mathrm{F(a)}$ from $\mathrm{F(b)}$.
   Thus, $\int_{\mathrm{a}}^{\mathrm{b}} \mathrm{f(x)dx} = [\mathrm{F(x)}]_{\mathrm{a}}^{\mathrm{b}} = \mathrm{F(b)} - \mathrm{F(a)}.$

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$$\int_0^1 \left(a+bx+cx^2\right) dx = ax + \frac{bx^2}{2} + \frac{cx^3}{3}\Big|_0^1 = a + \frac{b}{2} + \frac{c}{3}$$

f(1) = a + b + c

f(0) = a

$$f\left(\frac{1}{2}\right) = a + \frac{b}{2} + \frac{c}{4}$$

Option (4) satisfies the given condition

Q. 17 If the number of five digit numbers with distinct digits and 2 at the 10th place is $336k$, then k is equal to :

Correct Answer:

8

Solution:

1st term can be$(1, 3, 4.....9)$: 8 options

and 2nd can be $(0, 1, 3...9)$ but one digit used to fill 1st term): 8 options

Similarly 3rd place: 7 options

4th place is Tens place: can be filled by only a single digit 2

5 th place: 6 options

$$336K = 8 \times 8 \times 7 \times 6 \times 1$$

so $k = 8$

Correct Option (1)

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Q. 18

Let the observations $x_i(1 \leq i \leq 10)$ satisfy the equations, $\sum_{i=1}^{10}(x_i - 5) = 10$ and $\sum_{i=1}^{10}(x_i - 5)^2 = 40$. If $\mu \; and \; \lambda$ are the mean and the variance of the observations, $x_1 - 3, x_2 - 3, \cdots, x_{10} - 3$, then the ordered pair (μ, λ) is equal to :

Option 1:
$(6, 3)$

Option 2:
$(3, 6)$

Option 3:
$(3, 3)$

Option 4:
$(6, 6)$

Correct Answer:
$(3, 3)$

Solution:

Mean -

MEAN (Arithmetic Mean)

The mean is the sum of the value of each observation in a dataset divided by the number of observations. For example, to calculate the mean weight of 50 people, add the 50 weights together and divide by 50. Technically this is the arithmetic mean.

Mean of the Ungrouped Data

If n observations in data are x_1, x_2, x_3, ..., x_n, then arithmetic mean $\bar{x}$ is given by

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \ldots \cdots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Shortcut Method

$$\text{Arithmetic mean } \; \bar{x} = A + \frac{\sum_{i=1}^{n} d_i}{n}$$
$$\text{where, } A = \text{assumed mean,}$$
$$d_i = \text{deviation from assumed mean} = x_i - A$$

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Mean of Discrete Frequency Distribution and Continuous Frequency Distribution

If observations in data are x_1, x_2, x_3, ..., x_n with respective frequencies f_1, f_2, f_3, ..., f_n ; then

Sum of the value of the observation = $f_1x_1 + f_2x_2 + f_3x_3 + ... + f_nx_n$

and Number of observations = $f_1 + f_2 + f_3 + ...+ f_n$

The mean in this case is given by

$$\bar{x} = \frac{f_1x_1 + f_2x_2 + f_3x_3 + ... + f_nx_n}{f_1 + f_2 + f_3 + ... + f_n} = \frac{\sum_{i=1}^{n} f_ix_i}{\sum_{i=1}^{n} f_i}$$

Shortcut Method

$$\text{Arithmetic mean } \bar{x} = A + \frac{\sum_{i=1}^{n} f_i(x_i - A)}{n}$$
$$\text{where, } A = \text{assumed mean.}$$

-

Dispersion (Variance and Standard Deviation) -

Variance and Standard Deviation

The mean of the squares of the deviations from the mean is called the variance and is denoted by σ^2 (read as sigma square).

Variance is a quantity which leads to a proper measure of dispersion.

The variance of n observations x_1 , x_2 ,..., x_n is given by

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Variance and Standard Deviation of a Discrete Frequency Distribution

The given discrete frequency distribution be

$$x \quad : \quad x_1, \quad x_2, \quad x_3, \quad \ldots \quad x_n$$

$$f \quad : \quad f_1, \quad f_2, \quad f_3, \quad \ldots \quad f_n$$

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$$\text{In this case, Variance } (\sigma^2) \quad = \frac{1}{N}\sum_{i=1}^{n} f_i\,(x_i - \bar{x})^2$$

$$\text{and, Standard Deviation}(\sigma) \; = \sqrt{\frac{1}{N}\sum_{i=1}^{n} f_i\,(x_i - \bar{x})^2}$$

$$\text{where, N} = \sum_{i=1}^{n} f_i$$

Variance and Standard deviation of a continuous frequency distribution

The formula for variance and standard deviation are the same as in the case of discrete frequency distribution. Here, x_i is the mid point of each class.

Another formula for Standard Deviation

$$\begin{aligned}
\text{Variance } (\sigma^2) &= \frac{1}{\text{N}}\sum_{i=1}^{n} f_i\,(x_i - \bar{x})^2 = \frac{1}{\text{N}}\sum_{i=1}^{n} f_i\left(x_i^2 + \bar{x}^2 - 2\bar{x}x_i\right) \\
&= \frac{1}{N}\left[\sum_{i=1}^{n} f_i x_i^2 + \sum_{i=1}^{n} \bar{x}^2 f_i - \sum_{i=1}^{n} 2\bar{x} f_i x_i\right] \\
&= \frac{1}{N}\left[\sum_{i=1}^{n} f_i x_i^2 + \bar{x}^2 \sum_{i=1}^{n} f_i - 2\bar{x}\sum_{i=1}^{n} x_i f_i\right] \\
&= \frac{1}{N}\left[\sum_{i=1}^{n} f_i x_i^2 + \bar{x}^2 N - 2\bar{x}\cdot N\bar{x}\right] \\
&= \frac{1}{N}\left[\sum_{i=1}^{n} f_i x_i^2 + \bar{x}^2 N - 2\bar{x}\cdot N\bar{x}\right] \\
&\left[\because \frac{1}{N}\sum_{i=1}^{n} x_i f_i = \bar{x} \text{ or } \sum_{i=1}^{n} x_i f_i = \text{N}\bar{x}\right] \\
&= \frac{1}{\text{N}}\sum_{i=1}^{n} f_i x_i^2 + \bar{x}^2 - 2\bar{x}^2 = \frac{1}{\text{N}}\sum_{i=1}^{n} f_i x_i^2 - \bar{x}^2
\end{aligned}$$

$$\text{or } \; \sigma^2 = \frac{1}{N}\sum_{i=1}^{n} f_i x_i^2 - \left(\frac{\sum_{i=1}^{n} f_i x_i}{N}\right)^2 = \frac{1}{N^2}\left[N\sum_{i=1}^{n} f_i x_i^2 - \left(\sum_{i=1}^{n} f_i x_i\right)^2\right]$$

$$\text{Thus, standard deviation } (\sigma) = \frac{1}{N}\sqrt{N\sum_{i=1}^{n} f_i x_i^2 - \left(\sum_{i=1}^{n} f_i x_i\right)^2}$$

-

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$$\text{Mean } (x_i - 5) = \frac{\sum (x_i - 5)}{10} = 1$$

$$\therefore \text{Mean}, \quad \lambda = (x_i - 5 + 2) + 2 = 3$$

$$\mu = \text{var} = \left[(x_1 - 5) = \frac{\sum (x_i - 5)^2}{10} - \frac{\Sigma (x_i - 5)}{10} \right] = 3$$

Correct Option (3)

Q. 19 The integral $\int \frac{dx}{(x+4)^{8/7}(x-3)^{6/7}}$ is equal to :

(where C ia a constant of integration)

Option 1:

$$-\left(\frac{x-3}{x+4}\right)^{-1/7} + C$$

Option 2:

$$\frac{1}{2}\left(\frac{x-3}{x+4}\right)^{3/7} + C$$

Option 3:

$$\left(\frac{x-3}{x+4}\right)^{1/7} + C$$

Option 4:

$$-\frac{1}{13}\left(\frac{x-3}{x+4}\right)^{-13/7} + C$$

Correct Answer:

$$\left(\frac{x-3}{x+4}\right)^{1/7} + C$$

Solution:

Integration Using Substitution -

The method of substitution is one of the basic methods for calculating indefinite integrals.

Substitution - change of variable

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To solve the integrate of the form

$$I = \int f(g(x)) \cdot g'(x)\, dx,$$

where $g(x)$ is continuously differentiable function.
put $g(x) = t$, $g'(x)\, dx = dt$
After substitution, we get $\int f(t)\, dt$.
Evalute this integration and substitute back the value of t.

-

$$\int \left(\frac{x-3}{x+4}\right)^{\frac{-6}{7}} \frac{1}{(x+4)^2} dx$$

Let $\frac{x-3}{x+4} = t^7$

$$\frac{7}{(x+4)^2} dx = 7t^6 dt$$

$$\int t^{-6} t^6 dt = t + c$$

$$\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}} + c$$

Correct Option (3)

Q. 20 In a box, there are 20 cards, out of which 10 are labelled as A and the remaining 10 are labelled as B. Cards are drawn at random, one after the other and with replacement, till a second A-card is obtained. The probability that the second A-card appears before the third B-card is:

Option 1:

$\frac{15}{16}$

Option 2:

$\frac{9}{16}$

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Option 3:

$\frac{13}{16}$

Option 4:

$\frac{11}{16}$

Correct Answer:

$\frac{11}{16}$

Solution:

-

AA + ABA + BAA + ABBA + BBAA + BABA

$$\frac{1}{4}+\frac{1}{8}+\frac{1}{8}+\frac{1}{16}+\frac{1}{16}+\frac{1}{16}=\frac{11}{16}$$

Correct Option (4)

Q. 21 If the vectors, $\vec{p}=(a+1)\hat{i}+a\hat{j}+a\hat{k}, q=\vec{a}\hat{i}+(a+1)\hat{j}+a\hat{k} and \vec{r}=a\hat{i}+a\hat{j}+(a+1)\hat{k}\,(a\epsilon R)$ are coplanar and $3\left(\vec{p}\cdot\vec{q}\right)^2-\lambda\left|\vec{r}\times\vec{q}\right|^2=0$, then the value of λ is _____.

Correct Answer:

1

Solution:

Linear Dependent Vectors -

If vectors $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k},\ \ \vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ and $\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$ are coplanar, then

$$\begin{vmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{vmatrix}=0$$

-

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Dot (Scalar) Product in Terms of Components -

If $\vec{\mathbf{a}} = a_1\hat{\mathbf{i}} + a_2\hat{\mathbf{j}} + a_3\hat{\mathbf{k}}$ and $\vec{\mathbf{b}} = b_1\hat{\mathbf{i}} + b_2\hat{\mathbf{j}} + b_3\hat{\mathbf{k}}$, then
$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$

-

Vector Product in Terms of Components -

If $\vec{\mathbf{a}} = a_1\hat{\mathbf{i}}+a_2\hat{\mathbf{j}}+a_3\hat{\mathbf{k}}$ and $\vec{\mathbf{b}} = b_1\hat{\mathbf{i}}+b_2\hat{\mathbf{j}}+b_3\hat{\mathbf{k}}$, then their cross product given by

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

-

$\vec{P}, \vec{Q}, \vec{R}$ are coplanar vectors so $\begin{vmatrix} a+1 & a & a \\ a & a+1 & a \\ a & a & a+1 \end{vmatrix} = 0$

By solving this a=-1/3 so now

$\vec{P} = \frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{1}{3}\hat{k}$
$\vec{Q} = \frac{1}{3}(-\hat{i} + 2\hat{j} - \hat{k})$
$\vec{R} = \frac{1}{3}(-\hat{i} - \hat{j} + 2\hat{k})$

$$\vec{P} \cdot \vec{Q} = \frac{1}{9}(-2 - 2 + 1) = -\frac{1}{3}$$

$$\vec{R} \times \vec{Q} = \frac{1}{9}\begin{vmatrix} i & j & k \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{vmatrix}$$

By solving you will get $\vec{R} \times \vec{Q} = \frac{i+j+k}{3}$

$$|\vec{R} \times \vec{Q}|^2 = \frac{1}{3}$$

$$3(\vec{P} \cdot \vec{Q})^2 - \lambda|\vec{R} \times \vec{Q}|^2 = 0$$

λ=1

Correct Option (1)

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Q. 22 The projection of the line segment joining the points $(1, -1, 3)$ *and* $(2, -4, 11)$ on the line joining the points $(-1, 2, 3)$ *and* $(3, -2, 10)$ is _____.

Correct Answer:

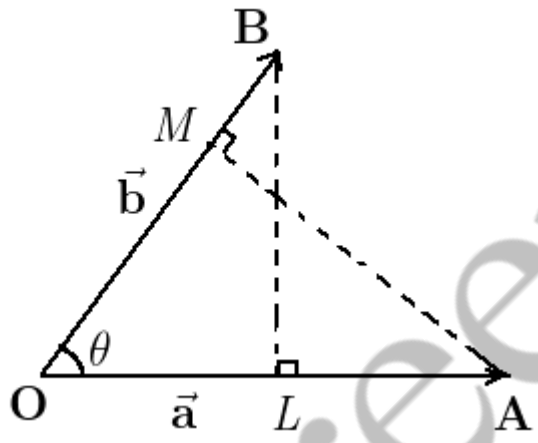
8

Solution:

Dot (Scalar) Product in Terms of Components -

If $\vec{\mathbf{a}} = a_1\hat{\mathbf{i}} + a_2\hat{\mathbf{j}} + a_3\hat{\mathbf{k}}$ and $\vec{\mathbf{b}} = b_1\hat{\mathbf{i}} + b_2\hat{\mathbf{j}} + b_3\hat{\mathbf{k}}$, then
$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$

Geometrical Interpretation of Scalar Product



$$\text{Projection of } \vec{\mathbf{a}} \text{ on } \vec{\mathbf{b}} = \frac{\vec{\mathbf{a}} \cdot \vec{\mathbf{b}}}{|\vec{\mathbf{b}}|} = \vec{\mathbf{a}} \cdot \frac{\vec{\mathbf{b}}}{|\vec{\mathbf{b}}|} = \vec{\mathbf{a}} \cdot \hat{\mathbf{b}}$$

$$\text{Projection of } \vec{\mathbf{b}} \text{ on } \vec{\mathbf{a}} = \frac{\vec{\mathbf{a}} \cdot \vec{\mathbf{b}}}{|\vec{\mathbf{a}}|} = \vec{\mathbf{b}} \cdot \frac{\vec{\mathbf{a}}}{|\vec{\mathbf{a}}|} = \vec{\mathbf{b}} \cdot \hat{\mathbf{a}}$$

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Let $A(1, -1, 3), B(2, -4, 11)$

$\overrightarrow{AB} = \hat{i} - 3\hat{j} + 8\hat{k}$

$C(-1, 2, 3), D(3, -2, 10)$

$\overrightarrow{CD} = 4\hat{i} - 4\hat{j} + 7\hat{k}$

$$\text{projection of } \overrightarrow{AB} \text{ on } \overrightarrow{CD} = \frac{\overrightarrow{AB} \cdot \overrightarrow{CD}}{|\overrightarrow{CD}|}$$

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$$= \left(\frac{4+12+56}{\sqrt{16+16+49}} \right) = \frac{72}{9} = 8$$

Correct Option (3)

Q. 23 The number of distinct solutions of the equation, $\log_{\frac{1}{2}} |\sin x| = 2 - \log_{\frac{1}{2}} |\cos x|$ in the interval $[0, 2\pi]$, is _____.

Correct Answer:
8

Solution:

Logarithmic Inequalities -

Logarithmic inequalities:

$$\log_a x > \log_a y = \begin{cases} x > y, & \text{if } a > 1 \\ x < y, & \text{if } 0 < a < 1 \end{cases}$$

$$\log_a x > y = \begin{cases} x > a^y, & \text{if } a > 1 \\ x < a^y, & \text{if } 0 < a < 1 \end{cases}$$

$$\log_a x < y = \begin{cases} x < a^y, & \text{if } a > 1 \\ x > a^y, & \text{if } 0 < a < 1 \end{cases}$$

Generally, from these inequalities, we can conclude that logarithmic functions are monotonically increasing for a >1 and decreasing for 0 < a < 1. Basic property which we must remember about log is that argument (means x or y) must be positive.

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Double Angle Formula and Reduction Formula -

Double Angle Formula and Reduction Formula

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$$\begin{aligned}
\sin(2\theta) &= 2\sin\theta\cos\theta \\
&= \frac{2\tan\theta}{1+\tan^2\theta} \\
\cos(2\theta) &= \cos^2\theta - \sin^2\theta \\
&= 1 - 2\sin^2\theta \\
&= 2\cos^2\theta - 1 \\
&= \frac{1-\tan^2\theta}{1+\tan^2\theta} \\
\tan(2\theta) &= \frac{2\tan\theta}{1-\tan^2\theta}
\end{aligned}$$

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Trigonometric Equations -

Trigonometric Equations

Trigonometric equations are, as the name implies, equations that involve trigonometric functions. Similar in many ways to solving polynomial equations or rational equations, only specific values of the variable will be solutions, if there are any solutions at all.

Solution of Trigonometric Equation

The value of an unknown angle which satisfies the given trigonometric equation is called a solution or root of the equation. For example, 2sinΘ = 1, clearly Θ = 30^0 and Θ = 150^0 satisfies the equation; therefore, 30^0 and 150^0 are solutions of the equation 2sinΘ = 1 between 0^0 and 360^0.

Principal Solution

The solutions of a trigonometric equation lie in the interval [0, 2π]. For example, 2sinΘ = 1 , then the two values of sinΘ between 0 and 2π are π/6 and 5π/6. Thus, π/6 and 5π/6 are the principal solutions of equation 2sinΘ = 1.

General Solution

As trigonometric functions are periodic, solutions are repeated within each period, so, trigonometric equations may have an infinite number of solutions. The solution consisting of all possible solutions of a trigonometric equation is called its general solution.

Some Important General Solutions of Equations

Equations	Solution
$\sin\theta = 0$	$\theta = n\pi, \quad n \in \mathbb{I}$

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$\cos\theta = 0 \quad \theta = (2n+1)\frac{\pi}{2}, \quad n \in \mathbb{I}$

$\tan\theta = 0 \quad \theta = n\pi, \quad n \in \mathbb{I}$

$\sin\theta = 1 \quad \theta = (4n+1)\frac{\pi}{2}, \quad n \in \mathbb{I}$

$\cos\theta = 1 \quad \theta = 2n\pi, \quad n \in \mathbb{I}$

$\sin\theta = -1 \quad \theta = (4n-1)\frac{\pi}{2}, \quad n \in \mathbb{I}$

$\cos\theta = -1 \quad \theta = (2n+1)\pi, \quad n \in \mathbb{I}$

$\cot\theta = 0 \quad \theta = (2n+1)\frac{\pi}{2}, \quad n \in \mathbb{I}$

-

$\log_{\frac{1}{2}}(|\sin(x)|) + \log_{\frac{1}{2}}(|\cos(x)|) = 2$

$\log_{\frac{1}{2}}(|\sin(x)|\,|\cos(x)|) = 2$

$\log_{\frac{1}{2}}(|\sin(x)|\,|\cos(x)|) = \log_{\frac{1}{2}}\left(\frac{1}{4}\right)$

$|\sin x||\cos x| = \frac{1}{4}$

$\sin 2\theta = \frac{1}{2}$

$x = \frac{\pi}{12} + \pi n,\ x = \frac{5\pi}{12} + \pi n$

Total number of solution is 8

Correct Option (4)

Q. 24 If for $x \geqslant 0, y = y(x)$ is the solution of the differential equation, $(x+1)dy = \left((x+1)^2 + y - 3\right)dx, y(2) = 0$, then $y(3)$ is equal to _____.

Correct Answer:

3

Solution:

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Linear Differential Equation -

The linear differential equations are those in which the variable and its derivative occur only in the first degree.

An equation of the form

$$\frac{dy}{dx} + P(x) \cdot y = Q(x) \qquad \ldots (\mathrm{i})$$

Where P(x) and Q(x) are functions of *x* only or constant is called a linear equation of the first order.

To solve the differential equation (i)

multiply both sides of Eq (i) by $\int e^{P(x)\,dx}$, we get

$$e^{\int P(x)dx}\left(\frac{dy}{dx} + P(x)y\right) = e^{\int P(x)dx} \cdot Q(x)$$

i.e. $$e^{\int P(x)dx} \cdot \frac{dy}{dx} + y\frac{d}{dx}\left(e^{\int P(x)dx}\right) = Qe^{\int P(x)dx}$$

or $$\frac{d}{dx}\left(ye^{\int P(x)dx}\right) = e^{\int P(x)dx} \cdot Q(x)$$

Integrating both sides, we get

or $$\int d\left(ye^{\int P(x)dx}\right) = \int \left(e^{\int P(x)dx} \cdot Q(x)\right)dx$$

$\Rightarrow$ $$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx}dx + C$$

Which is the required solution of the given differential equation.

The term $\int e^{P(x)\,dx}$ which convert the left hand expression of the equation into a perfect differential is called an Integrating factor (IF).

Thus, we remember the solution of the above equation as

$$y(\mathrm{IF}) = \int Q(\mathrm{IF})dx + C$$

NOTE :

Sometimes a given differential equation becomes linear if we take x as the dependent variable and y as the independent variable, i.e. it can be written in the form

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Thus, we remember the solution of the above equation as

$$y(\text{IF}) = \int Q(\text{IF})dx + C$$

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$$(1+x)\frac{dy}{dx} = \left((1+x)^2 + (y-3)\right)$$

$$(1+x)\frac{dy}{dx} - y = (1+x)^2 - 3$$

$$\frac{dy}{dx} - \frac{1}{(1+x)}y = (1+x) - \frac{3}{(1+x)}$$

$$IF = e^{-\int \frac{1}{1+x}dx} = \frac{1}{1+x}$$

$$\frac{d}{dx}\left(\frac{y}{1+x}\right) = 1 - \frac{3}{(1+x)^2}$$

$$x = 2, y = 0, c = -3$$

So at x=3 y=3

Correct Option (1)

Q. 25 The coefficient of x^4 in the expansion of $\left(1+x+x^2\right)^{10}$ is

Correct Answer:

615

Solution:

$\left(1+x+x^2\right)^{10}$ has general term $\frac{10!}{a!\ b!\ c!}x^b x^{2c}$

With a + b + c = 10 .....(i)

For coefficient of x^4, b + 2c = 4 .....(ii)

Possible combinations of a, b and c satisfying (i) and (ii) both are

1. When c = 0, b = 4 (from (ii)) , and a = 6 (from (i))

2. When c = 1, b = 2 (from (ii)) , and a = 7 (from (i))

3. When c = 2, b = 0 (from (ii)) , and a = 8 (from (i))

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So, (8,0,2), (7,2,1), (6, 4, 0)

So, coefficients are $\frac{10!}{8!\,0!\,2!}+\frac{10!}{7!\;2!\;1!}+\frac{10!}{6!\;4!\;0!}$ = 615

Correct Option (1)

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9th January Shift-2

Physics

Q. 1 For the four sets of three measured physical quantities as given below. Which of the following options is correct?

(i) $A_1 = 24.36, B_1 = 0.0724, C_1 = 256.2$

(ii) $A_2 = 24.44, B_2 = 16.082, C_2 = 240.2$

(iii) $A_3 = 25.2, B_3 = 19.2812, C_3 = 236.183$

(iv) $A_4 = 25, B_4 = 236.191, C_4 = 19.5$

Option 1:
$A_4 + B_4 + C_4 \ < \ A_1 + B_1 + C_1 \ = \ A_2 + B_2 + C_2 \ = \ A_3 + B_3 + C_3$

Option 2:
$A_1 + B_1 + C_1 \ = \ A_2 + B_2 + C_2 \ = \ A_3 + B_3 + C_3 \ = \ A_4 + B_4 + C_4$

Option 3:
$A_1 + B_1 + C_1 \ < \ A_3 + B_3 + C_3 \ < \ A_2 + B_2 + C_2 \ < \ A_4 + B_4 + C_4$

Option 4:
None of these

Correct Answer:
None of these

Solution:

$A_1 + B_1 + C_1 = 24.36 + 0.0724 + 256.2 = 280.6324 = 280.6$
$A_2 + B_2 + C_2 = 24.44 + 16.082 + 240.2 = 280.722 = 280.7$
$A_3 + B_3 + C_3 = 25.2 + 19.2812 + 236.183 = 280.6642 = 280.7$
$A_4 + B_4 + C_4 = 25 + 236.191 + 19.5 = 280.691 = 281$

Answer should be $A_1 + B_1 + C_1 < A_2 + B_2 + C_2 = A_3 + B_3 + C_3 < A_4 + B_4 + C_4$

So option (4) is correct.

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Q. 2 Two gases-argon (atomic radius 0.07 nm, atomic weight 40) and xenon (atomic radius 0.1 nm, atomic weight 140) have the same number density and are at the same temperature. The ratio of their respective mean free times is closest to :

Option 1:
4.67

Option 2:
2.3

Option 3:
1.83

Option 4:
1.07

Correct Answer:
1.07

Solution:

Mean free time $= \dfrac{1}{\sqrt{2}n_n d^2 v_{ms}}$

$\frac{t_{Ar}}{t_{xe}} = \frac{d_{xe}^2}{d_{Ar}^2} \times \sqrt{\frac{m_1}{m_2}}$

$= \left(\frac{0.1}{0.07}\right)^2 \times \sqrt{\frac{40}{140}}$

$= \left(\frac{10}{7}\right)^2 \times \sqrt{\frac{2}{7}} = 1.07$

Q. 3 A small spherical droplet of density d is floating exactly half immersed in a liquid of density p and surface tension T. The radius of the droplet is (take note that the surface tension applies an upward force on the droplet) :

Option 1:
$r = \sqrt{\dfrac{2T}{3(d+p)g}}$

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Option 2:

$$r = \sqrt{\frac{T}{(d+p)g}}$$

Option 3:

$$r = \sqrt{\frac{T}{(d-p)g}}$$

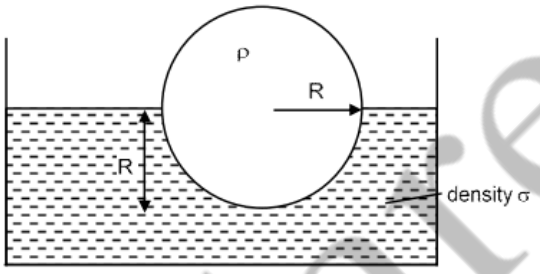
Option 4:

$$r = \sqrt{\frac{3T}{(2d-p)g}}$$

Correct Answer:

$$r = \sqrt{\frac{3T}{(2d-p)g}}$$

Solution:



$$dVg = \rho\left(\frac{V}{2}\right)g + T(2\pi r) \Rightarrow d \cdot \frac{4}{3}\pi r^3 g = \rho \cdot \frac{2}{3}\pi r^3 g + 2\pi r T$$

$$\Rightarrow \frac{2}{3}r^2 g(2d-\rho) = 2T$$

$$\Rightarrow r = \sqrt{\frac{3T}{(2d-\rho)g}}$$

Q. 4 In LC circuit the inductance $L = 40\ mH$ and capacitance $C = 100\ \mu F$. If a voltage $V(t) = 10\sin(314)t)$ is applied to the circuit , the current in the circuit is given as :

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Option 1:

$10 \cos 314\ t$

Option 2:

$0.52 \cos 314\ t$

Option 3:

$5.2 \cos 314\ t$

Option 4:

$0.52\ \sin 314\ t$

Correct Answer:

$0.52 \cos 314\ t$

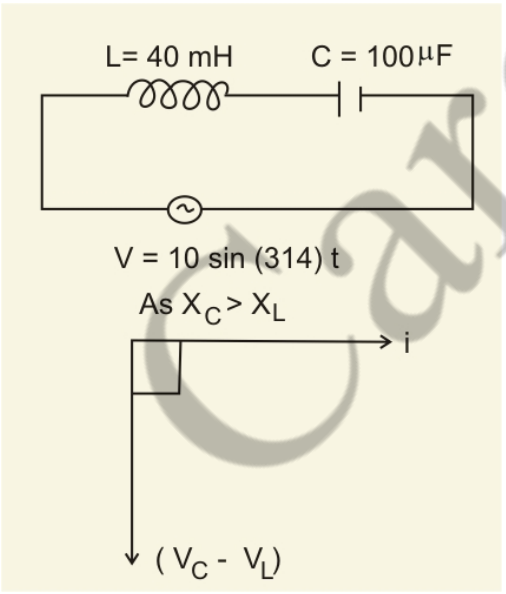
Solution:

As $Z = \sqrt{R^2 + (X_C - X_L)^2}$

But R=0

$$Z = (X_C - X_L) = \frac{1}{C_w} - Lw$$

$$\Rightarrow Z = \frac{1}{100*10^{-4}*314} - (314 * 40 * 10^{-3}) = 31.84 - 12.56 = 19.28\Omega$$



$$i = \frac{V_0}{Z} \sin\left(314t + \frac{\pi}{2}\right) = \frac{10}{19.28} \cos(314t) = 0.52 \cos(314t)$$

Hence the correct option is (2).

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Q. 5 A wire of length L and mass per unit length $6.0 \times 10^{-3} kgm^{-1}$ is put under tension of 540 N. Two consecutive frequencies at which it resonates are : 420 Hz and 490 Hz . Then L in meters is :

Correct Answer:
2.1

Solution:

Fundamental frequency = 490 – 420 = 70 Hz

$$70 = \frac{1}{2l}\sqrt{\frac{T}{\mu}}$$

$$\Rightarrow 70 = \frac{1}{2l}\sqrt{\frac{540}{6\times10^{-3}}}$$

$$\Rightarrow l = \frac{1}{2\times70}\sqrt{90 \times 10^3} = \frac{300}{140}$$

$$\Rightarrow l \approx 2.14m$$

Hence the correct option is (1).

Q. 6 Two steel wires having same length are suspended from a ceiling under the same load. If the ratio of their energy stored per unit volume is 1 : 4, the ratio of their diameters is :

Option 1:
$1 : 2$

Option 2:
$\sqrt{2} : 1$

Option 3:
$2 : 1$

Option 4:
$1 : \sqrt{2}$

Correct Answer:
$\sqrt{2} : 1$

Solution:

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$$\frac{du}{dv} = \frac{1}{2} stress \times \frac{stress}{Y}$$

$$= \frac{1}{2}\frac{F^2}{A^2 Y}$$

$$\frac{du}{dv} \propto \frac{1}{d^4}$$

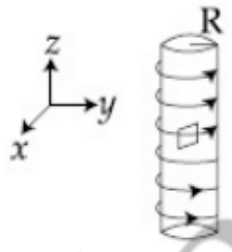
$$\frac{\left(\frac{du}{dv}\right)_1}{\left(\frac{du}{dv}\right)_2} = \frac{d_2^4}{d_1^4} = \frac{1}{4}$$

$$\frac{d_1}{d_2} = (4)^{1/4}$$

$$\frac{d_1}{d_2} = \sqrt{2} : 1$$

Hence the correct option is (2).

Q. 7 An electron gun is placed inside a long solenoid of radius R on its axis. The solenoid has n turns / length and carries a current I. The electron gun shoots an electron along the radius of the solenoid with speed ν . If the electron does not hit the surface of the solenoid, maximum possible value of ν is (all symbols have their standard meaning) :



Option 1:

$$\frac{2e\mu_0 nIR}{m}$$

Option 2:

$$\frac{e\mu_0 nIR}{2m}$$

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Option 3:

$$\frac{e\mu_0 nIR}{4m}$$

Option 4:

$$\frac{e\mu_0 nIR}{m}$$

Correct Answer:

$$\frac{e\mu_0 nIR}{2m}$$

Solution:

Charge = e

$$R_{max} = \frac{R}{2} = \frac{mv_{max}}{e\mu_0 In}$$

Therefore, $v_{max} = \frac{e\mu_0 nIR}{2m}$

$$R_{max} = \frac{R}{2} = \frac{mv_{max}}{e\mu_0 In}$$

Hence the correct option is (2).

Q. 8

A plane electromagnetic wave is propagating along the direction $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$, with its polarization along the direction $\hat{k}$. The correct form the magnetic field of the wave would be (here B_0 is an appropriate constant) :

Option 1:

$$B_0\frac{\hat{i}-\hat{j}}{\sqrt{2}}\cos\left(\omega t - k\frac{\hat{i+j}}{\sqrt{2}}\right)$$

Option 2:

$$B_0\frac{\hat{j}-\hat{i}}{\sqrt{2}}\cos\left(\omega t + k\frac{\hat{i+j}}{\sqrt{2}}\right)$$

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Option 3:

$$B_0\frac{\hat{i}+\hat{j}}{\sqrt{2}}\cos\left(\omega t-k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$$

Option 4:

$$B_0\,\hat{k}\,\cos\left(\omega t-k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$$

Correct Answer:

$$B_0\frac{\hat{i}-\hat{j}}{\sqrt{2}}\cos\left(\omega t-k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$$

Solution:

EM wave is in direction - $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

As we know that the axis of polarisation of the Em wave is same as Electric field direction that is - $\hat{k}$

$\vec{E}\times\vec{B}$ **direction of propagation of EM wave =** $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

$$\Rightarrow \vec{k}\times\vec{B}=\frac{\hat{i}+\hat{j}}{\sqrt{2}}$$

So B is along $\Rightarrow\frac{\hat{i}-\hat{j}}{\sqrt{2}}$

And the equation of the electromagnetic waves will be in terms of the $\cos(\omega t-\vec{K}\cdot\vec{r})$

So by concluding the above result we can deduce that the option (1) is correct.

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Q. 9 Planet A has mass M and radius R. Planet B has half the mass and half the radius of Planet A. If the escape velocities from the Planets A and B are ν_A and ν_B, respectively, then $\frac{\nu_A}{\nu_B} = \frac{n}{4}$. The value of n is :

Correct Answer:
4

Solution:

$v_e = \sqrt{\frac{2GM}{R}}$

$\therefore \frac{v_1}{v_2} = \frac{\sqrt{\frac{2GM}{R}}}{\sqrt{\frac{2GM/2}{R/2}}} = 1 = \frac{n}{4}$

$\Rightarrow n = 4$

Hence the correct option is (1).

Q. 10 A rod 'l' has non-uniform linear mass density given by $\rho(x) = a + b\left(\frac{x}{l}\right)^2$, where a and b are constants and $0 \leq x \leq l$. The value of x for the centre of mass of the rod is at :

Option 1:
$\frac{3}{2}\left(\frac{2a+b}{3a+b}\right)l$

Option 2:
$\frac{3}{2}\left(\frac{a+b}{2a+b}\right)l$

Option 3:
$\frac{3}{4}\left(\frac{2a+b}{3a+b}\right)l$

Option 4:
$\frac{4}{3}\left(\frac{a+b}{2a+3b}\right)l$

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Correct Answer:

$\frac{3}{4}\left(\frac{2a+b}{3a+b}\right)l$

Solution:

$$\text{Given, } \lambda = \left(\text{a} + \text{b}\left(\frac{\text{x}}{\text{l}}\right)^2\right)$$

$$\frac{\text{dN}}{\text{dx}} = \lambda$$

$$\text{dN} = \lambda\text{dx} = \left(\text{a} + \text{b}\left(\frac{\text{x}}{\text{l}}\right)^2\right) \cdot \text{dx}$$

$$= \frac{\int \text{x} \cdot \text{dm}}{\int \text{dm}} = \frac{\int \text{x}\lambda\text{dx}}{\int \lambda\text{dx}}$$

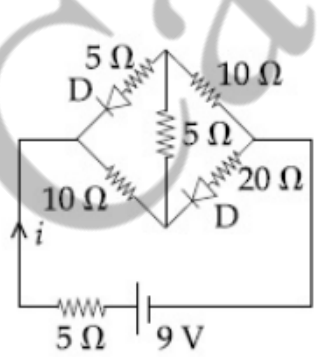
$$= \frac{\int_0^1 \text{x}\left(\text{a} + \left(\frac{\text{bx}^2}{1^2}\right)\right)\text{dx}}{\int_0^1 \left(\text{a} + \frac{\text{bx}^2}{1^2}\right)\text{dx}} = \frac{\int_0^1 \left(\text{ax} + \frac{\text{bx}^3}{1^2}\right)\text{dx}}{\int_0^1 \left(\text{a} + \frac{\text{bx}^2}{1^2}\right)\text{dx}}$$

$$= \frac{\left[\frac{ax^2}{2}\right]_0^1 + \frac{b}{1^2}\left[\frac{x^4}{4}\right]_0^1}{a[x]_0^1 + \frac{b}{1^2}\left[\frac{x^3}{3}\right]_0^1} = \frac{\frac{a1^2}{2} + \frac{b1^2}{4}}{al + \frac{bl}{3}}$$

$$= \frac{(2a+b)1}{(3a+b)4} \times 3$$

$$= \frac{31}{4}\left(\frac{2a+b}{3a+b}\right)$$

Q. 11 The current i in the network is :



Option 1:

$0.3A$

Option 2:

$0\ A$

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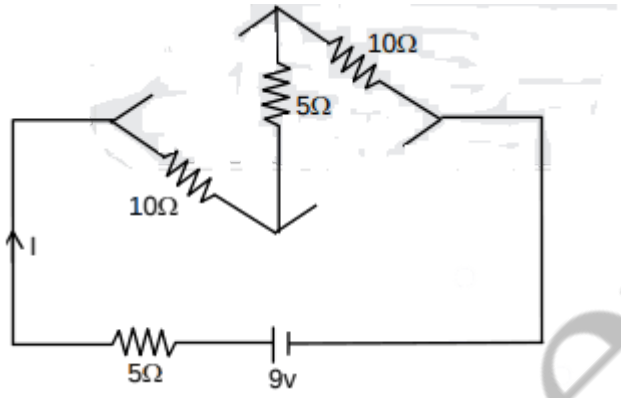
Option 3:
$0.2\ A$

Option 4:
$0.6\ A$

Correct Answer:
$0.3A$

Solution:
Both diodes are in reverse biased

So new circuits can be drawn as



So $I = \frac{9}{30} = \frac{3}{10} = 0.3\ A$

Hence the correct option is (1).

Q. 12 There is a small source of light at some depth below the surface of water (refractive index $=\frac{4}{3}$) in a tank of large cross sectional surface area. Neglecting any reflection from the bottom and absorption by water, percentage of light that emerges out of surface is (nearly) :

[Use the fact that surface area of a spherical cap of height h and radius of curvature r is $2\pi rh$]

Option 1:
$34\ ^{o}/_{o}$

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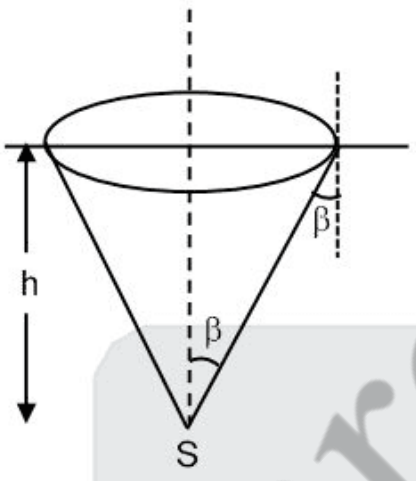
Option 2:
$17\ ^{o}/_{o}$

Option 3:
$50\ ^{o}/_{o}$

Option 4:
$21\ ^{o}/_{o}$

Correct Answer:
$17\ ^{o}/_{o}$

Solution:



$\sin\beta = \frac{3}{4}, \cos\beta = \frac{\sqrt{7}}{4}$

$Solid\ angle\ d\Omega = 2\pi R^2(1-\cos\beta)$

Percentage of light $= \frac{2\pi R^2(1-\cos\beta)}{4\pi R^2} \times 100 = \frac{1-\cos\beta}{2} \times 100 = \left(\frac{4-\sqrt{7}}{8}\right) \times 100 \approx 17\ ^{o}/_{o}$

Hence the correct option is (2).

Q. 13 The energy required to ionised a hydrogen like ion in its ground state is 9 Rydbergs. What is the wavelength (in nm) of the radiation emitted when the electron in this ion jumps from the second excited state to the ground state ?

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Correct Answer:
11.4

Solution:

$$\frac{hc}{\lambda} = (13.6eV)z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$n_1 = 1$$

$$n_2 = 3$$

$$\frac{hc}{\lambda} = (13.6eV)(3^2)\left[\frac{1}{1^2} - \frac{1}{3^2}\right]$$

$$\Rightarrow \frac{hc}{\lambda} = (13.6eV)(9)\left(\frac{8}{9}\right)$$

Wavelength$= \frac{1240}{8 \times 13.6}nm$

$$\lambda = 11.40nm$$

Hence the correct option is (1).

Q. 14 A small circular loop of conducting wire has radius a and carries current I. It is placed in a uniform magnetic field B perpendicular to its plane such that when it rotated slightly about its diameter and released, it starts performing simple harmonic motion of time period T. If the mass of the loop is m then :

Option 1:
$$T = \sqrt{\frac{\pi m}{IB}}$$

Option 2:
$$T = \sqrt{\frac{\pi m}{2IB}}$$

Option 3:
$$T = \sqrt{\frac{2m}{IB}}$$

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Option 4:

$T = \sqrt{\frac{2\pi m}{IB}}$

Correct Answer:

$T = \sqrt{\frac{2\pi m}{IB}}$

Solution:

$\tau = MB\sin\theta = I\alpha (using\ M = IA)$

$\Rightarrow \pi R^2 IB\theta = \frac{mR^2}{2}\alpha\ (Using\ \alpha = -\omega^2\theta)$

$\Rightarrow \omega = \sqrt{\frac{2\pi IB}{m}} = \frac{2\pi}{t}$

$\Rightarrow T = \sqrt{\frac{2\pi m}{IB}}$

Hence the correct option is (4).

Q. 15 A particle of mass m is projected with a speed u from the ground at an angle $\theta = \frac{\pi}{3}$ w.r.t. horizontal (x-axis). When it has reached its maximum height, it collides completely inelastically with another particle of the same mass and velocity $u\hat{i}$. The horizontal distance covered by the combined mass before reaching the ground is :

Option 1:

$\frac{3\sqrt{3}}{8}\ \frac{u^2}{g}$

Option 2:

$2\sqrt{2}\ \frac{u^2}{g}$

Option 3:

$\frac{3\sqrt{2}}{4}\ \frac{u^2}{g}$

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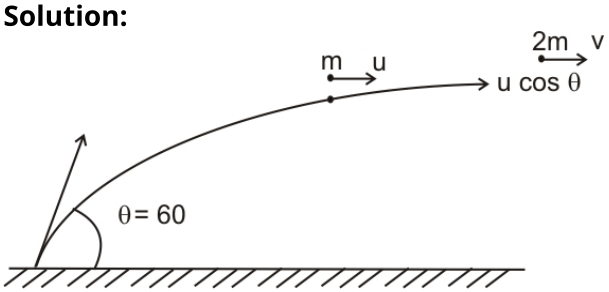
Option 4:

$\frac{5}{8}\ \frac{u^2}{g}$

Correct Answer:

$\frac{3\sqrt{3}}{8}\ \frac{u^2}{g}$

Solution:



$p_i = p_f$

$mu + mu\cos\theta = 2mv$

$\Rightarrow v = \frac{u(1+\cos 60^\circ)}{2} = \frac{3}{4}u$

So horizontal range after collision $= vt = v\sqrt{\frac{2H_{max}}{g}}$

$= \frac{3}{4}u\sqrt{\frac{2u^2\sin^2(60^\circ)}{2g^2}}$

$= \frac{3}{4}u^2\frac{\sqrt{\frac{3}{4}}}{g} = \frac{3\sqrt{3}u^2}{8g}$

Hence the correct option is (1).

Q. 16 A particle starts from the origin at t=0 with an initial velocity of $3.0\hat{i}\ m/s$ and moves in the x-y plane with a constant acceleration $\left(6.0\hat{i}+4.0\hat{j}\right)\ m/s^2$. The x-coordinate of the particle at the instant when its y-coordinate is 32 m is D meters. The value of D is :

Correct Answer:

60

Solution:

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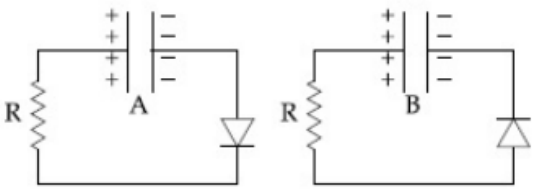
$$S_y = U_y t + \frac{1}{2} a_y t^2$$

$$\Rightarrow 32 = 0 + \frac{1}{2} * 4t^2 \Rightarrow t = 4\,\text{sec}$$

$$S_x = U_x t + \frac{1}{2} a_x t^2 = (3 * 4) + \frac{1}{2} * 6 * 16 = 60\ m$$

Hence the correct option is (1).

Q. 17 Two identical capacitors A and B, charged to the same potential 5V are connected in two different circuits as shown below at time t=0. If the charge on capacitors A and B at time t=CR is q_A and q_B respectively , then (Here e is the base of natural logarithm)



Option 1:
$Q_A = VC,\ Q_B = CV$

Option 2:
$Q_A = \frac{CV}{2},\ Q_B = \frac{VC}{e}$

Option 3:
$Q_A = \frac{VC}{e},\ Q_B = \frac{CV}{2}$

Option 4:
$Q_A = VC,\ Q_B = \frac{VC}{e}$

Correct Answer:
$Q_A = VC,\ Q_B = \frac{VC}{e}$

Solution:

Maximum charge on the capacitor$= 5CV$

(a) is reverse biased and (b) is forward biased

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(a) **(b)**

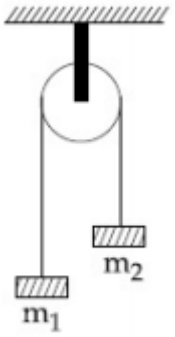
$Q_A = CV$

So, $q = q_{max}\left[e^{-t/RC}\right]$

$Q_B = \frac{CV}{e}$

Hence the correct option is (4).

Q. 18 A uniformly thick wheel with moment of inertia I and radius R is free to rotate about its centre of mass (see fig.) A massless string is wrapped over its rim and two blocks of masses m_1 and $m_2 (m_1 > m_2)$ are attached to the ends of the string. The system is released from rest. The angular speed of the wheel when m_1 descents by a distance h is :



Option 1:

$$\left[\frac{2(m_1+m_2)gh}{(m_1+m_2)R^2+I}\right]^{\frac{1}{2}}$$

Option 2:

$$\left[\frac{m_1+m_2}{(m_1+m_2)R^2+I}\right]^{\frac{1}{2}}gh$$

Option 3:

$$\left[\frac{(m_1-m_2)}{(m_1+m_2)R^2+I}\right]^{\frac{1}{2}}gh$$

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Option 4:

$$\left[\frac{2\left(m_1-m_2\right)gh}{\left(m_1+m_2\right)R^2+I}\right]^{\frac{1}{2}}$$

Correct Answer:

$$\left[\frac{2\left(m_1-m_2\right)gh}{\left(m_1+m_2\right)R^2+I}\right]^{\frac{1}{2}}$$

Solution:

$$k_1+U_1=k_1+k_2$$

$$0+0=\frac{1}{2}m_2v^2+\frac{1}{2}m_1v^2+\frac{1}{2}I\omega^2-m_1gh+m_2gh$$

$$(m_1-m_2)gh=\frac{1}{2}m_2(wR)^2+\frac{1}{2}m_1(wR^2)+\frac{1}{2}I\omega^2$$

$$w=\left[\frac{2(m_1-m_2)gh}{(m_1+m_2)R^2+I}\right]^{\frac{1}{2}}$$

Hence the correct option is (4).

Q. 19 An electron of mass m and magnitude of charge $|e|$ initially at rest gets accelerated by a constant electric field E.The rate of change of de-broglie wavelength of this electron at time t is ? (ignoring relativistic effects is) :

Option 1:

$$\frac{-h}{|e|\,Et^2}$$

Option 2:

$$\frac{|e|\,Et}{h}$$

Option 3:

$$-\frac{h}{|e|\,Et}$$

Option 4:

$$-\frac{h}{|e|\,E\sqrt{t}}$$

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Correct Answer:

$$\frac{-h}{|e|\,Et^2}$$

Solution:

$$\lambda_D = \frac{h}{mv}$$

$$\therefore \quad v = at$$

$$v = \frac{eE}{m}t \qquad \left(a = \frac{eE}{m}\right)$$

$$\lambda_D = \frac{h}{m\left(\frac{eE}{m}t\right)}$$

$$\lambda_D = \frac{h}{eEt}$$

$$\frac{d\lambda_d}{dt} = -\frac{h}{|e|\,Et^2}$$

Hence the correct option is (1).

Q. 20 A spring mass system (mass m, spring constant k and natural length l) rests in equilibrium on a horizontal disc.The free end of the spring is fixed at the centre of the disc. If the disc together with spring mass system, rotates about its axis with an angular velocity ω, $\left(k >> m\omega^2\right)$ the relative change in the length of the spring is best given by the option :

Option 1:

$$\frac{m\omega^2}{3k}$$

Option 2:

$$\frac{m\omega^2}{k}$$

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Option 3:

$\sqrt{\frac{2}{3}}\left(\frac{m\omega^2}{k}\right)$

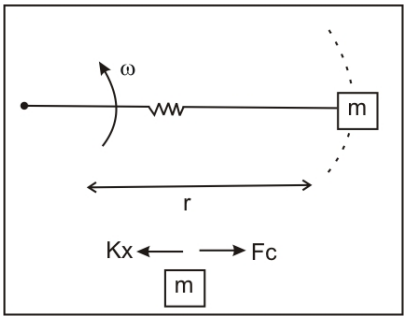
Option 4:

$\frac{2m\omega^2}{k}$

Correct Answer:

$\frac{m\omega^2}{k}$

Solution:



As natural lentgh=l_0

Let elongation=x

Mass m is moving with angular velocity ω in a radius r

where $r = l_0 + x$

Due to elongation x spring force is given by $F_s = Kx$

And $F_C = m\omega^2 r = m\omega^2(l_0 + x)$

as $F_C = F_s$

$Kx = m\omega^2(l_0 + x)$

$\Rightarrow x = \frac{m\omega^2 l_0}{K - m\omega^2}$

Using $k >> m\omega^2$

So, $\frac{x}{l_0}$ is equal to $\frac{m\omega^2}{k}$

Hence the correct option is (2).

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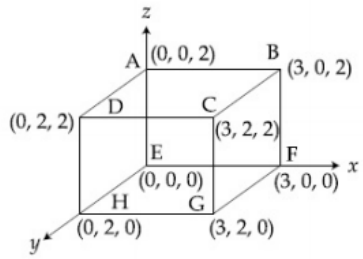
Q. 21 An electric field $\hat{E} = 4\,x\hat{i} - (y^2+1)\hat{j}\ N/C$ passes through the box shown in figure. The flux of the electric field through surfaces ABCD and BCGF are marked as $\mathrm{F_I}$ and $\mathrm{F_{II}}$ respectively. The difference between ($\mathrm{F_I}$-$\mathrm{F_{II}}$) is (in Nm^2/C)_____.



Correct Answer:
48

Solution:

Flux via ABCD

$$\phi_1 = \int \vec{E} \cdot d\vec{A} = 0$$

Flux via BCEF

$$\phi_2 = \int \bar{E} \cdot d\bar{A}$$

$$\phi_2 = \bar{E} \cdot \bar{A}$$

$$= (4x\hat{i} - (y^2+1)\hat{j}) \cdot 4\hat{i}$$

$$= 16x, x = 3$$

$$\phi_2 = 48\frac{N - m^2}{C}$$

$$|\phi_1 - \phi_2| = 48\frac{N - m^2}{C}$$

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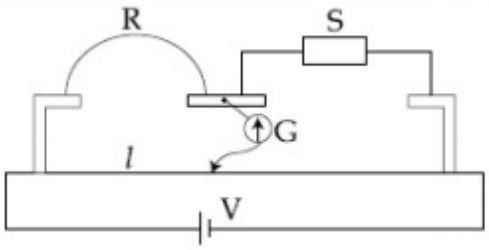
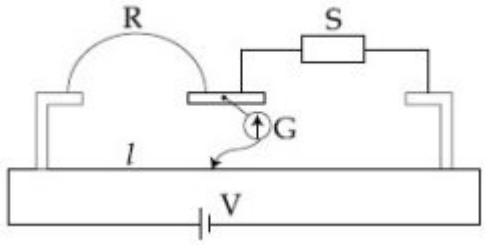
Q. 22 In a meter bridge experimental S is a standard resistance, R is resistance wire. It is found that balancing length is l=25 cm. If R is replaced by a wire of half length and half diameter that of R of same material, then the balancing distance l' (in cm) will now be _____.



Correct Answer: 40

Solution:



$$\frac{X}{R} = \frac{75}{25} = 3$$

$$R = \frac{P^l}{A} = \frac{4P^l}{\pi d^2}$$

$$R' = \frac{4\rho\left(\frac{l}{2}\right)}{\pi\left(\frac{d}{2}\right)^2} = 2R$$

then, $\frac{x}{R'} = \frac{X}{2R} = \frac{3}{2}$

$l = 40.00\ cm$

Q. 23 Starting at temperature 300 K, one mole of an ideal diatomic gas $(\gamma = 1.4)$ is first compressed adiabatically from volume V_1 tO $V_2 = \frac{V_1}{16}$. It is then allowed to expand isobarically to volume $2V_2$, If all the processes are quasi-static then the final temperature of the gas (in K) is (to the nearest integer) _____.

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Option 1:
1818

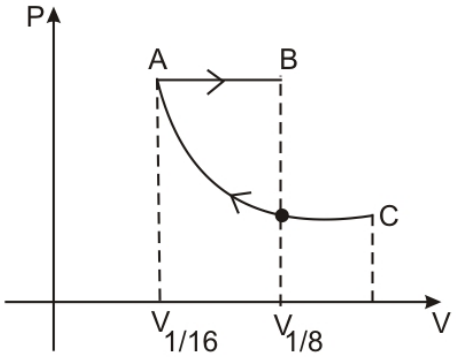
Option 2:
2073

Option 3:
1618

Option 4:
1563

Correct Answer:
1818

Solution:



For Process CA

$$PV^{\gamma} = constant$$

$$TV^{\gamma-1} = constant$$

$$300(V_1)^{1.4-1} = T_A\left(\frac{V_1}{16}\right)^{2/5}$$

$$T_A = 300 \times 2^{8/5}$$

Now for process AB

$$V_B/T_B = V_A/T_A$$

$$T_B = \frac{V_B T_A}{V_A} = 2 \times 300 \times 2^{8/5}$$

$$T_B = 1818K$$

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Q. 24 In a young's double slit experiment 15 fringes are observed on a small portion of the screen when light of wavelength 500 nm is used. Ten fringes are observed on the same section of the screen when another light source of wavelength λ is used. Then the value of λ is (in nm) ____.

Correct Answer: 750

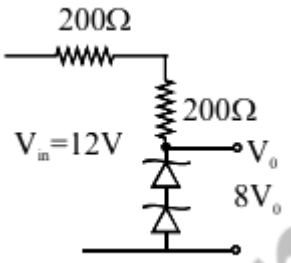
Solution:

$$15 \times 500 \times \frac{D}{d} = 10 \times \lambda_2 \times \frac{D}{d}$$

$$\lambda_2 = 15 \times 50nm$$

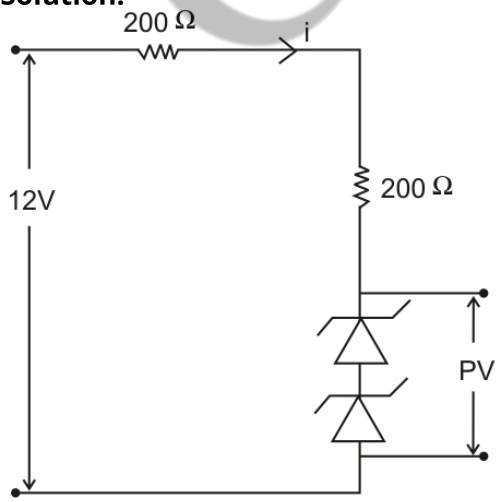
$$\lambda_2 = 750nm$$

Q. 25 The circuit shown below is working as a 8 V dc regulated voltage source . When 12 V is used as input, the power dissipated (in mW) in each diode is ; (considering both zener diodes are identical) ____.



Correct Answer:
40

Solution:



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$$i = \frac{(12-8)}{(200+200)}A = \frac{4}{400} = 10^{-2}A$$

$$Power\ loss\ in\ each\ diaode = (4)(10^{-2})W = 40mW$$

Chemistry

Q. 1 The reaction of $H_3N_3B_3Cl_3$ (A) with $LiBH_4$ in tetrahydrofuran gives inorganic benzene (B). Further, the reaction of (A) with (C) leads to $H_3N_3B_3(Me)_3$. Compounds (B) and (C) respectively, are :

Option 1:
Diborane and $MeMgBr$

Option 2:
Borazine and $MeBr$

Option 3:
Borazine and $MeMgBr$

Option 4:
Boron nitride and $MeBr$

Correct Answer:
Borazine and $MeMgBr$

Solution:
The reactions are given as:

$$B_3N_3H_3Cl_3 + LiBH_4 \rightarrow B_3N_3H_6 + LiCl + BCl_3$$

(A) (B)

$$B_3N_3H_3Cl_3 + 3MeMgBr \rightarrow B_3N_3H_3(CH_3)_3 + 3MgBrCl$$

(A) (C)

Hence, the option number (3) is correct.

Q. 2 Amongst the following, the form of water with the lowest ionic conductance at $298\ K$ is :

Option 1:
Saline water used for intravenous injection

Option 2:
water from a well

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Option 3:
distilled water

Option 4:
sea water

Correct Answer:
distilled water

Solution:
Pure water is non conducting. Conductance occurs through water due to the presence of ions. Usually, the water from different sources contain some amount of dissolved salts due to which it is conducting.

Distilled water is de-ionised water and hence it does not conduct electricity

Hence, the option number (3) is correct.

Q. 3 The first and second ionisation enthalpies of a metal are 496 and $4560\ \mathrm{kJ\ mol^{-1}}$, respectively. How many moles of HCl and $\mathrm{H_2SO_4}$, respectively, will be needed to react completely with 1 mole of the metal hydroxide ?

Option 1:
1 and 2

Option 2:
1 and 0.5

Option 3:
1 and 1

Option 4:
2 and 0.5

Correct Answer:
1 and 0.5

Solution:
According to the given data of I.E, this element must belong to group 1, and thus it is monovalent and forms hydroxide of the type M(OH).

$$\mathrm{MOH + HCl \longrightarrow MCl + H_2O}$$

$$\mathrm{2MOH + H_2SO_4 \longrightarrow M_2SO_4 + 2H_2O}$$

Thus, from the reaction stoichiometry it is clear that 1 mole of the metal hydroxide requires 1 mole and 0.5 moles of HCl and H_2SO_4 respectively for complete neutralisation.

Hence, **the option number (2) is correct.**

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Q. 4 The isomer (s) of $[Co\,(NH_3)_4\,Cl_2]$ that has/have a $Cl-Co-Cl$ angle of 90^o, is /are :

Option 1:
meridional and trans

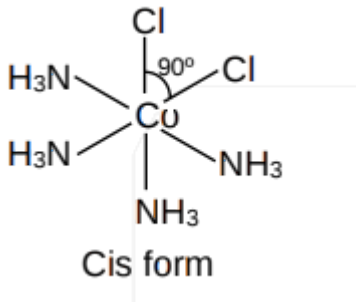
Option 2:
trans only

Option 3:
cis and trans

Option 4:
cis only

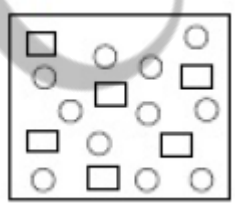
Correct Answer:
cis only

Solution:
The $Cl-Co-Cl$ bond angle of 90° can be found only in the cis isomer



Hence, **option number (4) is correct.**

Q. 5 In the figure shown below reactant A (represented by square) is in equilibrium with product B (represented by circle). The equilibrium constant is :



Option 1:
1

Option 2:
4

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Option 3:
8

Option 4:
2

Correct Answer:
2

Solution:
Now,

$$A \leftrightharpoons B$$

$$K_c = \frac{[B]}{[A]}$$

The concentration of [B] as represented by circle = 11

and the concentration of [A] as represented by square = 6

Therefore, the value of the equilibrium constant is near 2.

Hence, **Option number (4) is correct.**

Q. 6 Which of the following has the shortest $C - Cl$ bond ?

Option 1:
$Cl - CH = CH_2$

Option 2:
$Cl - CH = CH - CH_3$

Option 3:
$Cl - CH = CH - OCH_3$

Option 4:
$Cl - CH = CH - NO_2$

Correct Answer:
$Cl - CH = CH - NO_2$

Solution:
The resonance form of Cl–CH=CH–NO_2 is more stable than the resonance form of any other given compounds. Hence, the double bond character in the carbon-chlorine bond is maximum and the bond length is the shortest.

Hence, option number (4) is correct.

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Q. 7 Which polymer has 'chiral' monomer(s)?

Option 1:
Buna -N

Option 2:
Neoprene

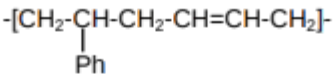
Option 3:
PHBV

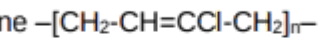
Option 4:
Nylon 6,6

Correct Answer:
PHBV

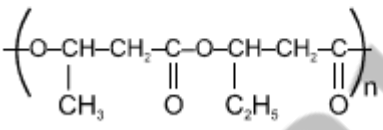
Solution:

Polymers

(1) Buna-S 

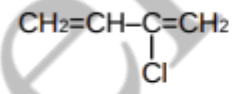
(2) Neoprene 

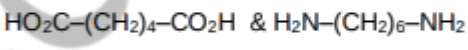
(3) Nylon-6,6

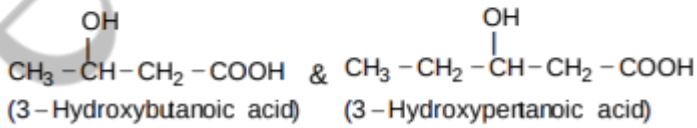
(4) 

PHBV

Monomers







In PHBV, both monomers have a chiral centre.

Hence, **the option number (3) is correct.**

Q. 8 5 g of zinc is treated separately with an excess of

(a) dilute hydrochloric acid and

(b) aqueous sodium hydroxide

The ratio of the volumes of H_2 evolved in these two reactions is :

Option 1:
$1:2$

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Option 2:
$1:4$

Option 3:
$1:1$

Option 4:
$2:1$

Correct Answer:
$1:1$

Solution:
$(\mathrm{Zn} + 2\mathrm{NaOH} \longrightarrow \mathrm{Na_2ZnO_2} + \mathrm{H_2})$
$(\mathrm{Zn} + 2\mathrm{HCl} \longrightarrow \mathrm{ZnCl_2} + \mathrm{H_2})$

According to stoichiometry in both the reactions, an equal number of moles of H_2 are evolved.

Hence, the Option number (3) is correct.

Q. 9 The solubility product of $Cr(OH)_3$ at 298 K is 6.0×10^{-31}. The concentration of hydroxide ions in a saturated solution of $Cr(OH)_3$ will be :

Option 1:
$(18 \times 10^{-31})^{1/2}$

Option 2:
$(2.22 \times 10^{-31})^{1/4}$

Option 3:
$(18 \times 10^{-31})^{1/4}$

Option 4:
$(4.86 \times 10^{-29})^{1/4}$

Correct Answer:
$(18 \times 10^{-31})^{1/4}$

Solution:
$\mathrm{Cr(OH)_3} \longrightarrow \mathrm{Cr^{+3}} + 3\mathrm{OH^-}$

s 3s

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$$K_{sp} = s \cdot (3s)^3$$

$$6 \times 10^{-31} = 27s^4$$

$$s = \left(\frac{6}{27} \times 10^{-31}\right)^{1/4}$$

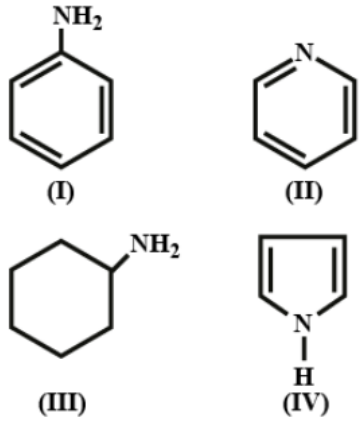
$$[OH^-] = 3s$$

$$[OH^-] = 3 \times \left(\frac{6}{27} \times 10^{-31}\right)^{1/4}$$

$$[OH^-] = \left(18 \times 10^{-31}\right)^{1/4} M$$

Hence, option number (3) is correct.

Q. 10 The decreasing order of basicity of the following amines is :



Option 1:
$(I) > (III) > (IV) > (II)$

Option 2:
$(II) > (III) > (IV) > (I)$

Option 3:
$(III) > (I) > (II) > (IV)$

Option 4:
$(III) > (II) > (I) > (IV)$

Correct Answer:
$(III) > (II) > (I) > (IV)$

Solution:
Basic strength depends upon the availability of lone pairs. Greater the resonance of lone pairs lesser the basic strength.

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In (III) and (II), the lone pairs are available, But nitrogen in III is sp^3 and in II is sp^2. So, III will have more basic strengths than II due to s-character.

In IV lone pair is not available due to aromatic resonance in the ring. it will be the least basic.

In I, the lone pair is in resonance with the benzene group. It will be less basic than II and III.

The decreasing order of basicity of the following amines is :

III > II > I > IV

Hence, option number (4) is correct.

Q. 11 Biochemical oxygen Demand(BOD) is the amount of oxygen required (in ppm):

Option 1:
For the photochemical breakdown of waste present in $1\ m^3$ volume of a water body

Option 2:
by anaerobic bacteria to breakdown inorganic waste present in a water body.

Option 3:
by bacteria to break-down organic waste in a certain volume of a water sample

Option 4:
for sustaining life in a water body

Correct Answer:
by bacteria to break-down organic waste in a certain volume of a water sample

Solution:
The amount of oxygen required by bacteria to break down the organic matter present in a certain volume of a sample of water is called Biochemical Oxygen Demand (BOD).

Hence, the option number (3) is correct.

Q. 12 The correct order of the spin-only magnetic moments of the following complexes is :

$(I)\ [Cr(H_2O)_6]\,Br_2$

$(II)\ Na_4\,[Fe(CN)_6]$

$(III)\ Na_3\,[Fe(C_2O_4)_3]\,(\Delta_0 > P)$

$(IV)\ (Et_4N)_2\,[CoCl_4]$

Option 1:
$(III) > (I) > (II) > (IV)$

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Option 2:
$(III) > (I) > (IV) > (II)$

Option 3:
$(I) > (IV) > (III) > (II)$

Option 4:
$(II) \approx (I) > (IV) > (III)$

Correct Answer:
$(I) > (IV) > (III) > (II)$

Solution:
As, $\mu_s = \sqrt{n(n+2)}$

Complex $(II) \rightarrow Fe^{+2} \Rightarrow$ S.F. L $\Rightarrow t_{2g}^{2,2} e_g^{0,0} \Rightarrow \mu_s = 0$

Complex (I) $\rightarrow Cr^{+2} \Rightarrow$ W.F.L $\Rightarrow t_{2g}^{1,1,1} e_g^{1,0} \Rightarrow \mu_s = \sqrt{24}$B.M

Complex (IV) $\rightarrow Co^{+2} \Rightarrow$ W.F.L $\Rightarrow e^{2,2} t_{2g}^{1,1,1} \Rightarrow \mu_s = \sqrt{15}$BM

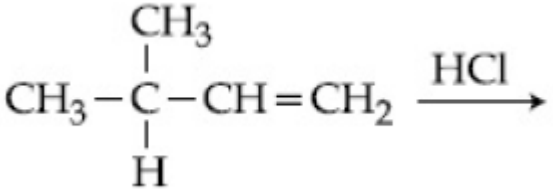
Complex (III) $\rightarrow Fe^{+3} \Rightarrow$ S.F.L $\Rightarrow t_{2g}^{2.1} e_g^{0.0} \Rightarrow \mu_s = \sqrt{3}$B.M

Hence, **Option number (3) is correct.**

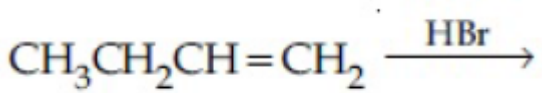
Q. 13 Which of the following reactions will not produce a racemic product ?

Option 1:

Option 2:



Option 3:



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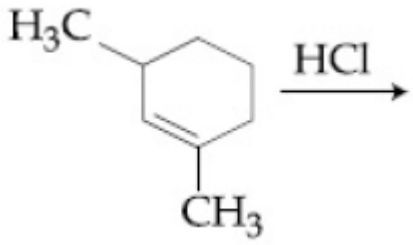
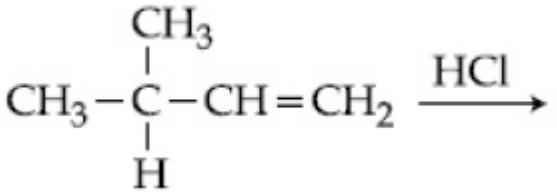
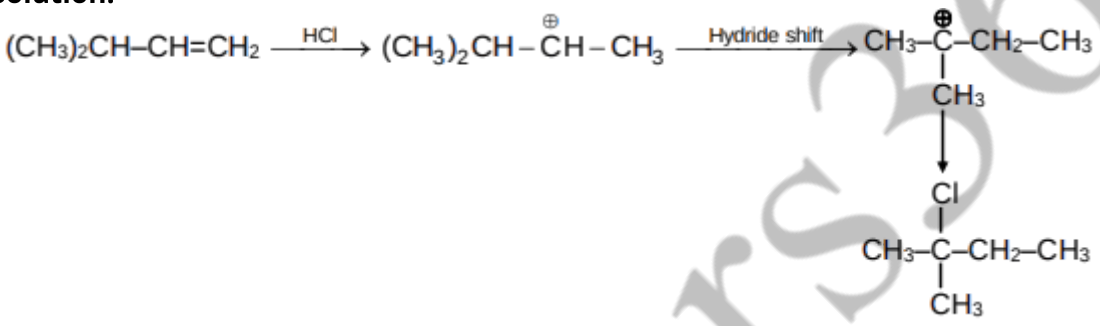
Option 4:



Correct Answer:



Solution:



This reaction will not form a racemic mixture as a product because the product formed in the reaction is not chiral.

Hence, the option number (2) is correct.

Q. 14 The true statement amongst the following is :

Option 1:
S is a function of temperature but δS is not a function of temperature.

Option 2:
Both δS and S are functions of temperature

Option 3:
Both S and δS are not functions of temperature.

Option 4:
S is not a function of temperature but δS is a function of temperature.

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Correct Answer:
Both δS and S are functions of temperature

Solution:

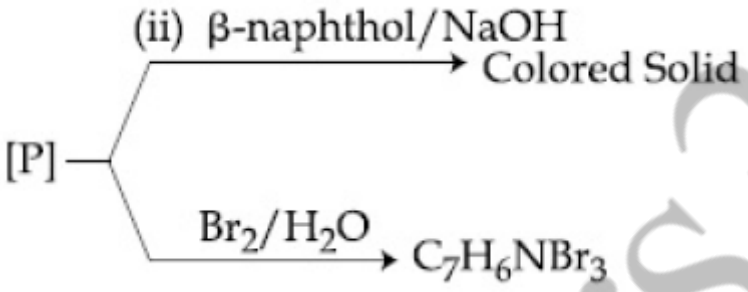
$$\Delta S = \int \frac{dq}{T}$$

$$\mathrm{S_T} = \int_0^{\mathrm{T}} \frac{\mathrm{ncdT}}{\mathrm{T}}$$

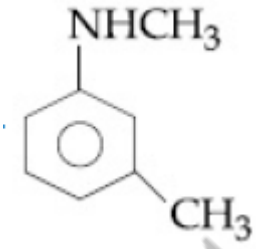
Hence, option number (2) is correct.

Q. 15 Consider the following reactions,

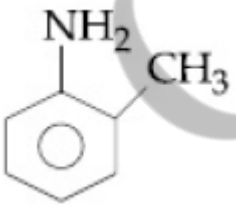
$(i)\ NaNO_2/HCl, 0-5^oC$



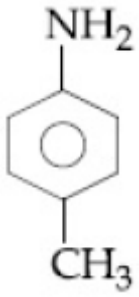
Option 1:



Option 2:



Option 3:



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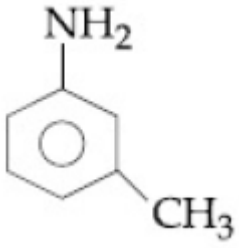
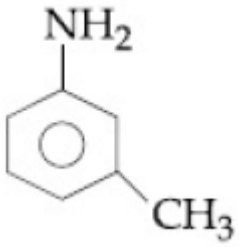
Option 4:



Correct Answer:



Solution:

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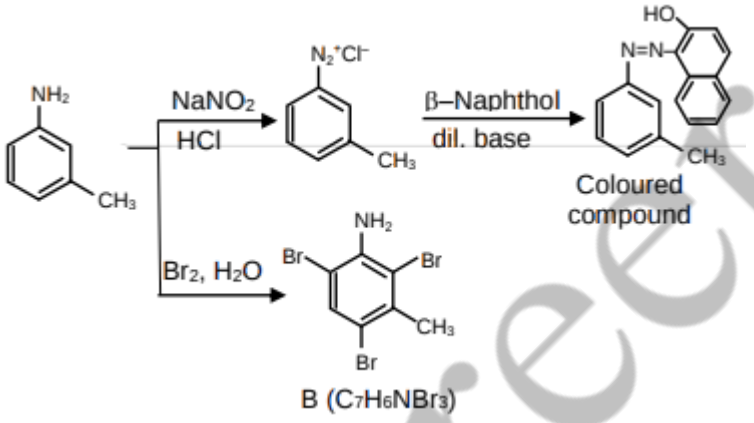

Hence, the option number (4) is correct.

Q. 16 A,B and C are three biomolecules. The results of the tests performed on them are given below :

	Molisch's Test	Barfoed Test	Biuret Test
A	Positive	Negative	Negative
B	Positive	Positive	Negative
C	Negative	Negative	Positive

A, B and C are respectively:

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Option 1:
A = Lactose, B = Glucose, C = Albumin

Option 2:
A = Lactose, B = Fructose, C = Alanine

Option 3:
A = Lactose, B = Glucose, C = Alanine

Option 4:
A = Glucose, B = Fructose, C = Albumin

Correct Answer:
A = Lactose, B = Glucose, C = Albumin

Solution:

Compound	Molisch's test	Barfoed test	Biuret test
Yes	Lactose, Glucose, Fructose	Glucose	Albumin
No		Sucrose	Alanine

Hence, the option number (1) is correct.

Q. 17 Among the statements $(a) - (d)$, the correct ones are :

(a) Lithium has the highest hydration enthalpy among the alkali metals.

(b) Lithium chloride is insoluble in pyridine.

(c) Lithium cannot form ethynide upon its reaction with ethyne.

(d) Both lithium and magnesium react slowly with H_2O.

Option 1:
(a), (b) and (d) only

Option 2:
(b) and (c) only

Option 3:
(a), (c) and (d) only

Option 4:
(a) and (d) only

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Correct Answer:
(a), (c) and (d) only

Solution:
(a) Among alkali metals, Li has the highest hydration tendency.

(b) Lithium chloride LiCl is soluble in Pyridine because it has significant covalent character due to high polarising power of lithium.

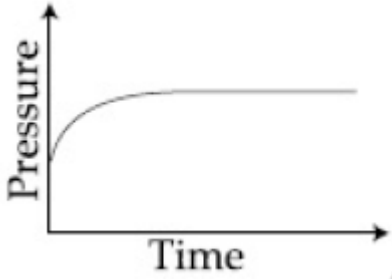
(c) Li doesn't form ethynide on reaction with ethyne because it has high ionisation enthalpy due to its extremely small size and high polarising power (charge/radius ratio). Its outermost electron is so close to the nucleus, so it is very difficult to remove the electron against the force of attraction applied by its nucleus.

(d) Li and Mg react slowly with water.

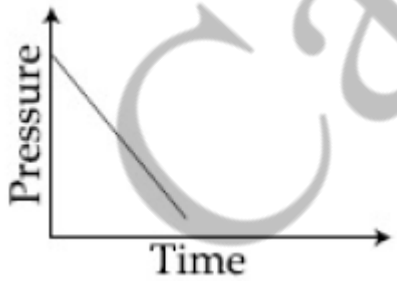
Hence, the option number (3) is correct.

Q. 18 A mixture of gases O_2, H_2 and CO are taken in a closed vessel containing charcoal. The graph that represents the correct behaviour of pressure with time is :

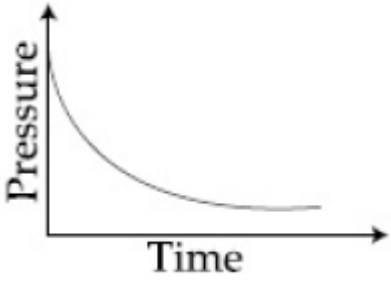
Option 1:



Option 2:



Option 3:



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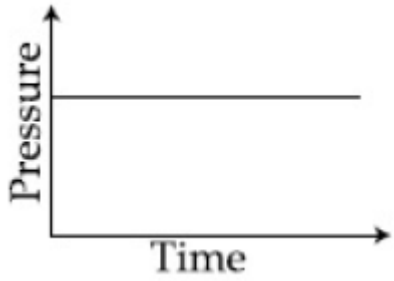
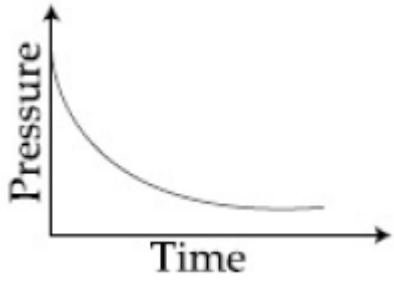
Option 4:



Correct Answer:



Solution:
We know that,

Charcoal is an adsorbent and the gases H_2, O_2 and CO are adsorbates. These gases get adsorbed on charcoal and with time pressure decreases.

Hence, the option number (3) is correct.

Q. 19 The number of sp^2 hybrid orbitals in a molecule of benzene is :

Option 1:
12

Option 2:
24

Option 3:
6

Option 4:
18

Correct Answer:
18

Solution:
The structure of Benzene-

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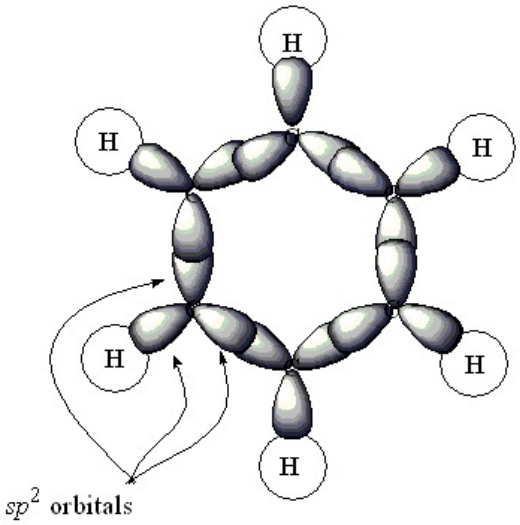
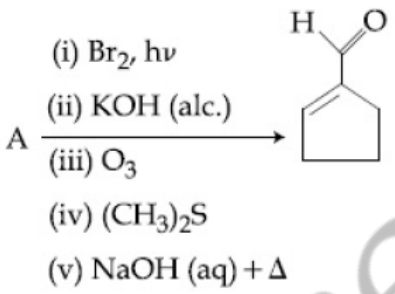


After observation of the image -

18 sp^2 hybrid orbitals are present in a molecule of benzene.

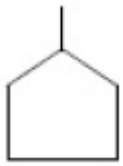
Hence, option number (4) is correct.

Q. 20 In the following reaction A is :

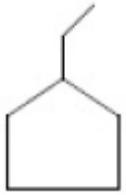


Option 1:

Option 2:



Option 3:



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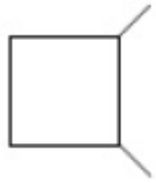
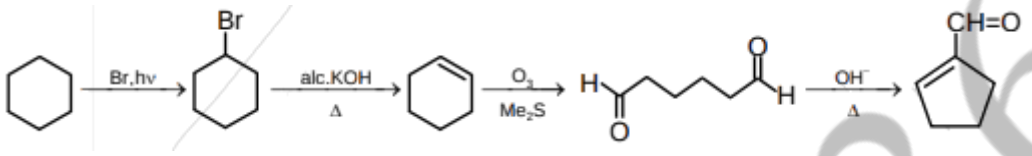
Option 4:



Correct Answer:

Solution:

@



Hence, the option number (1) is correct.

Q. 21 The sum of the total number of bonds between chromium and oxygen atoms in chromate and dichromate ions is

Option 1:

12

Option 2:

10

Option 3:

8

Option 4:

5

Correct Answer:

12

Solution:

The total number of Cr and O bonds is 12.

Therefore,option(1) is correct

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Q. 22 A sample of milk splits after 60 min. at 300 K and after 40 min. at 400 K when the population of lactobacillus acidophilus in it doubles. The activation energy (in kJ/mol) for this process is closest to_____.

$$(\text{Given, R} = 8.3 \text{ J mol}^{-1}\text{K}^{-1}, \ln(\frac{2}{3}) = 0.4, \text{e}^{-3} = 4.0)$$

Correct Answer:
3.984

Solution:
The formula of activation energy -

$$\ln\frac{\text{K}_2}{\text{K}_1} = \frac{\text{E}_\text{a}}{\text{R}}\left[\frac{1}{\text{T}_1} - \frac{1}{\text{T}_2}\right]$$

$$\ln\left(\frac{60}{40}\right) = \frac{\text{E}_\text{a}}{8.3}\,\text{x}\,\frac{100}{400\,\text{x}\,300}$$

$$\ln\left(\frac{3}{2}\right) \times 8.3 \times 1200 = \text{E}_\text{a}$$

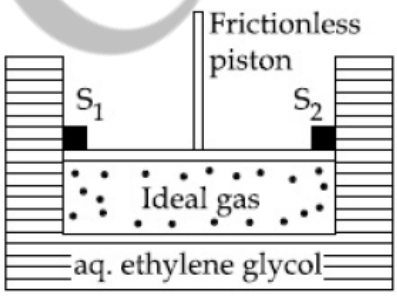
$$\text{E}_\text{a} = 0.4 \times 8.3 \times 1200$$

$$\text{E}_\text{a} = 3984 \text{ J/mol}$$

$$\text{E}_\text{a} = 3.984 \text{ kJ/mol}$$

Q. 23 A cylinder containing an ideal gas $(0.1\ mol\ of\ 1.0\ dm^3)$ is in thermal equilibrium with a large volume of 0.5 molal aqueous solution of enthylene glycol at its freezing point. If the stoppers S_1 and S_2 (as shown in the figure) are suddenly withdrawn, the volume of the gas in litres after equilibrium is achieved will be

$$(Given, K_f(water) = 2.0\ K\ kg\ mol^{-1}, R = 0.08\ dm^3 atm\ K^{-1} mol^{-1})$$



Correct Answer:
2.176

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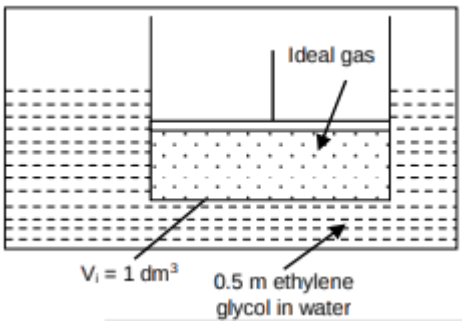
Solution:



$$K_f = 2.0K$$
$$m = 0.5m$$
$$\Delta T_f = K_f m$$
$$\Delta T_f = 0.5 \times 2$$
$$T_{initial} = 272K$$
$$n = 0.1\,mol$$
$$V = 1\,dm^3$$
$$P_{gas} = \frac{nRT}{V} = \frac{0.1 \times 0.08 \times 272}{1}$$
$$P_{gas} = 2.176\,atm$$

After releasing piston $P_1V_1 = P_2V_2$

$2.176 \times 1 = 1 \times V_2$

$V_2 = 2.176\ dm^3$

Q. 24 10.30 mg of O_2 is dissolved into a liter of sea water of density 1.03 g/mL. The concentration of O_2 in ppm is

Option 1:
10

Option 2:
20

Option 3:
25

Option 4:
40

Correct Answer:
10

Solution:

$$1030\ g \text{ of sea water contains } = 10.3 \times 10^{-3}\ g$$

$$10^6 gm \text{ of sea water contains } = \left(10.3 \times 10^{-3}\right) / 1030 \times 10^6$$

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$$\text{ppm} = \frac{10.3 \text{ x } 10^{-3}}{1030} \text{ x } 10^{6}$$

Thus, the correct answer is 10.

Therefore, the correct option is (1).

Q. 25 Consider the following reactions

$$A \xrightarrow[(ii)\ H_3O^+]{(i)\ CH_3MgBr} B \xrightarrow[573\ K]{Cu} 2-methyl-2-butene$$

The mass percentage of carbon in A is

Option 1:

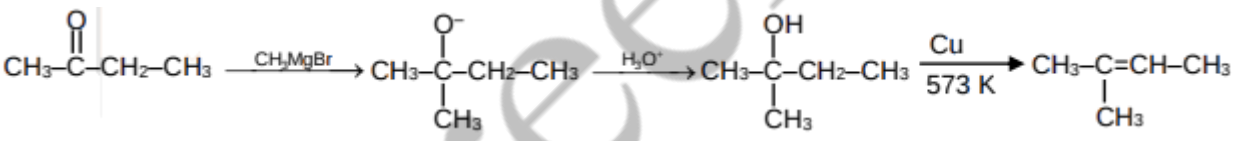
Option 2:

Option 3:

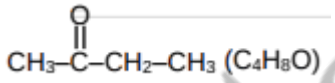
Option 4:

Correct Answer:

Solution:



Compounds is:



$$\text{Percentage carbon in compound A} = \left(\frac{12 \text{ x } 4}{48 + 16 + 8} \text{ x } 100\right) = 66.67$$

Maths

Q. 1 If $A = (x \epsilon \mathbf{R} : |x| < 2)$ and $B = (x \epsilon \mathbf{R} : |x - 2| \geqslant 3)$; then :

Option 1:

$A - B = [-1, 2)$

Option 2:

$B - A = \mathbf{R} - (-2, 5)$

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Option 3:
$A \cup B = \mathbf{R} - (2, 5)$

Option 4:
$A \cap B = (-2, -1)$

Correct Answer:
$B - A = \mathbf{R} - (-2, 5)$

Solution:

Union of sets, Properties of union -

Union of Set

Let A and B be any two sets. The union of A and B is the set which consists of all the elements of A and all the elements of B, the common elements being taken only once. The symbol 'ᴜ' is used to denote the union.

Symbolically, we write A U B = {x: x ∈ A or x ∈ B}.

-

Intersection of Set, Properties of Intersection -

The intersection of sets A and B is the set of all elements which are common to both A and B. The symbol '∩ 'is used to denote the intersection.

Symbolically, we write A ∩ B = {x: x ∈ A and x ∈ B}.

If A and B are two sets such that A ∩ B = φ, then A and B are called disjoint sets.

For example, let A = { 2, 4, 6, 8 } and B = { 1, 3, 5, 7 }. Then A and B are disjoint sets because there are no elements which are common to A and B.

-

Difference of set -

The difference of the sets A and B in this order is the set of elements which belong to A but not to B.

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Symbolically, we write A – B and read as “A *minus* B”.

I.e., If A = {1, 2, 3, 4} and B = {4, 5, 6, 8},

Then, A - B = {1, 2, 3} and B - A = {5, 6, 8}

-

$$A = \{x : x \in (-2, 2)\}$$
$$B = \{x : x \in (-\infty, -1] \cup [5, \infty)\}$$
$$A \cap B = \{x : x \in (-2, -1]\}$$
$$A \cup B = \{x : x \in (-\infty, 2) \cup [5, \infty)\}$$
$$A - B = \{x : x \in (-1, 2)\}$$
$$B - A = \{x : x \in (-\infty, -2] \cup [5, \infty)\}$$

Correct Option 2

Q. 2 If 10 distinct balls are kept in 4 distinct boxes randomly, then what is the probability that two boxes have 2 and 3 balls?

Option 1:
$\frac{965}{2^{10}}$

Option 2:
$\frac{945}{2^{10}}$

Option 3:
$\frac{945}{2^{11}}$

Option 4:
$\frac{965}{2^{11}}$

Correct Answer:
$\frac{945}{2^{10}}$

Solution:
110114

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DISTRIBUTION OF DISTINCT OBJECTS INTO DISTINCT PLACES -

suppose we have to arrange ‘n’ different objects into ‘r’ different groups, such that each group contains at least one or more than one object. To do this suppose we arrange ‘n’ different objects in a row first, which can be done in n n! ways, now after arranging them we need to add r-1 partition between them to make r group and we can choose those r-1 partitions (spaces) from n-1 partitions (spaces) available, so the total number of ways of arranging n different things into r different objects = $\mathrm{n!} \cdot^{\mathrm{n-1}} \mathrm{C_{r-1}}$.

-

Multiplication rule of probability for more than two events

If A, B and C are three events associated with sample space, then we have

P(A ∩ B ∩ C) = P(A) P(B|A) P(C|A ∩ B)

= P(A) P(B|A) P(C|AB)

Similarly, the multiplication rule of probability can be extended for four or more events.

-

Two boxes pick from 4 boxes is ${}^4C_2$ and can be arranged in 2! ways
$^4\mathrm{C}_2 \times 2 \times {}^{10}\mathrm{C}_2 \times {}^8\mathrm{C}_3 \times 2^5$

2^5 is the total number of possible cases in which remaining two boxes contains balls

4^{10} is total number of possible cases in which remaining four boxes contains balls

Required Probability=$\frac{^4\mathrm{C}_2 \times 2 \times {}^{10}\mathrm{C}_2 \times {}^8\mathrm{C}_3 \times 2^5}{4^{10}} = \frac{4 \times 3 \times 5 \times 9 \times 8 \times 7 \times 2^5}{2^{20}} = \frac{945}{2^{20}}$

Correct Option 2

Q. 3 If $x = 2\sin\theta - \sin 2\theta$ and $y = 2\cos\theta - \cos 2\theta,\ \theta \equiv [0, 2\pi]$, then $\frac{\mathrm{d}^2\mathrm{y}}{\mathrm{d}x^2}\ at\ \theta = \pi$ is :

Option 1:
$\frac{3}{8}$

Option 2:
$\frac{3}{4}$

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Option 3:

$\frac{3}{2}$

Option 4:

$-\frac{3}{4}$

Correct Answer:

$\frac{3}{8}$

Solution:

Differentiation of Function in Parametric Form -

Differentiation of Function in Parametric Form

Sometimes, x and y are given as functions of a single variable, i.e., x = f(t) and y = g(t) are two functions and t is a variable. In such cases x and y are called parametric functions or parametric equations and t is called the parameter.

To find $\frac{dy}{dx}$ in such cases, first find the relationship between x and y by eliminating the parameter t and then differentiate with respect to t.

But sometimes it is not possible to eliminate t, then in that case use

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{f'(t)}{g'(t)}$$

-

$$\frac{dx}{d\theta} = 2\cos\theta - 2\cos 2\theta$$
$$\frac{dy}{d\theta} = -2\sin\theta + 2\sin 2\theta$$
$$\therefore \frac{dy}{dx} = \frac{\sin 2\theta - \sin\theta}{\cos\theta - \cos 2\theta}$$
$$= \frac{2\sin\frac{\theta}{2}\cdot\cos\frac{3\theta}{2}}{2\sin\frac{\theta}{2}\sin\frac{3\theta}{2}} = \cot\frac{3\theta}{2}$$

$$\frac{d^2y}{dx^2} = \frac{-3}{2}\csc^2\frac{3\theta}{2}\cdot\frac{d\theta}{dx}$$
$$\frac{d^2y}{dx^2} = \frac{-\frac{3}{2}\csc^2\frac{3\theta}{2}}{2(\cos\theta - \cos 2\theta)}$$
$$\frac{d^2y}{dx^2}|_{\theta=\pi} = -\frac{3}{4(-1-1)} = \frac{3}{8}$$

Correct Option (1)

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Q. 4 Let f and g be differentiable functions on r such that fog is the identity function. If for some $a, b\epsilon\mathbf{R}, g'(a) = 5 \ and \ g(a) = b$ then $f'(b)$ is equal to :

Correct Answer:
$\frac{1}{5}$

Solution:

Rules of Differentiation (Chain Rule) -

Rules of Differentiation (Chain Rule)

Chain Rule or Derivation of Composite Function:

The chain rule, which states that the derivative of a composite function is the derivative of the outer function evaluated at the inner function times the derivative of the inner function.

Let f and g be functions. For all x in the domain of g for which g is differentiable at x and f is differentiable at g(x), the derivative of the composite function

h(x) = (f∘g)(x) = f (g(x)) Is given by

h′(x) = f′(g(x)) • g′(x)

Composites of Three or More Functions

For all values of x for which the function is differentiable, if k(x) = h(f(g(x)))Then,

$$k'(x) = h'\left(f(g(x))\right) \cdot f'\left(g(x)\right) \cdot g'(x)$$

-

$$f(g(x)) = x$$
$$\Rightarrow f'(g(x)) \cdot g'(x) = 1$$
$$Put \ \ x = a$$
$$\Rightarrow f'(g(a))g'(a) = 1$$
$$\Rightarrow f'(b) \times 5 = 1 \Rightarrow f'(b) = \frac{1}{5}$$

Correct Option 4

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Q. 5 In the expansion of $\left(\frac{x}{cos\theta}+\frac{1}{x\sin\theta}\right)^{16}$, if l_1 is the least value of the term independent of x when $\frac{\pi}{8}\leq\theta\leq\frac{\pi}{4}$ and l_2 is the least value of the term independent of x when $\frac{\pi}{16}\leq\theta\leq\frac{\pi}{8}$, then the ratio $l_2:l_1$ is equal to :

Option 1:
$16:1$

Option 2:
$8:1$

Option 3:
$1:8$

Option 4:
$1:16$

Correct Answer:
$16:1$

Solution:
General Term of Binomial Expansion
$(T_{r+1})^{\text{th}}$ term is called as general term in $(x+y)^n$ and general term is given by

$$\mathrm{T_{r+1}} = {}^{\mathrm{n}}\mathrm{C_r}\,\mathrm{x}^{\mathrm{n-r}}\cdot\mathrm{y}^{\mathrm{r}}$$

Term independent of x: It means term containing x^0,

Now,

$$\mathrm{T_{r+1}} = {}^{16}\mathrm{C_r}\left(\frac{\mathrm{x}}{\cos\theta}\right)^{16-\mathrm{r}}\left(\frac{1}{\mathrm{x}\sin\theta}\right)^{\mathrm{r}}$$

for $\mathrm{r}=8$ term is free from 'x'

$$\mathrm{T_9} = {}^{16}\mathrm{C_8}\frac{1}{\sin^8\theta\cos^8\theta}$$

$$\mathrm{T_9} = {}^{16}\mathrm{C_8}\frac{2^8}{(\sin 2\theta)^8}$$

$$\text{in } \theta\in\left[\frac{\pi}{8},\frac{\pi}{4}\right], L_1 = {}^{16}\mathrm{C_8}2^8$$

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$\because$ Min value of L_1 at $\theta = \pi/4$

$\text{in } \theta \in \left[\frac{\pi}{16}, \frac{\pi}{8}\right], L_2 = 16C_8 \frac{2^8}{\left(\frac{1}{\sqrt{2}}\right)^8} = {}^{16}C_8 \cdot 2^8 \cdot 2^4$

$\because$ Min value of L_2 at $\theta = \pi/8$

$\frac{L_2}{L_1} = \frac{16C_8 \cdot 2^8 2^4}{{}^{16}C_8 \cdot 2^8} = 16$

Correct option 1

Q. 6 Let $a, b \epsilon \mathbf{R}, a \neq 0$ be such that the equation, $ax^2 - 2bx + 5 = 0$ has a repeated root α, which is also a root of the equation, $x^2 - 2bx - 10 = 0$. If β is the other root of this equation, then $\alpha^2 + \beta^2$ is equal to:

Correct Answer:

25

Solution:

Nature of Roots -

Let the quadratic equation is $ax^2 + bx + c = 0$

D is the discriminant of the equation.

iii) if roots D = 0, then roots will be real and equal, then

$x_1 = x_2 = \frac{-b}{2a}$

Then, $ax^2 + bx + c = a(x - x_1)^2$

-

$ax^2 - 2bx + 5 = 0$ having equal roots or $D = 0$ and $\alpha = \frac{b}{a}$

$(2b)^2 = 4 \times 5 \times a \quad \Rightarrow \quad b^2 = 5a$

Put $\alpha = \frac{b}{a}$ in the second equation

$x^2 - 2bx - 10 = 0$

$\Rightarrow b^2 - 2ab^2 - 10a^2 = 0$

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$\Rightarrow 5a - 10a^2 - 10a^2 = 0$
$\Rightarrow 20a^2 = 5a$
$\Rightarrow a = \frac{1}{4}$ and $\mathrm{b}^2 = \frac{5}{4}$
$\alpha^2 = 20$ and $\beta^2 = 5$
$\alpha^2 + \beta^2$
$= 5 + 20$
$= 25$

Correct Option 2

Q. 7 Let a function $f : [0, 5] \rightarrow \mathbf{R}$ be continuous, $f(1) = 3$ and F be defined as :

$F(x) = \int_1^x t^2 g(t)\, dt$, where $g(t) = \int_1^t f(u)\, du$. Then for the function F, the point $x = 1$ is :

Option 1:
a point of infection.

Option 2:
a point of local maxima.

Option 3:
a point of local minima.

Option 4:
not a critical point.

Correct Answer:
a point of infection.

Solution:

Integration as Reverse Process of Differentiation -

Integration is the reverse process of differentiation. In integration, we find the function whose differential coefficient is given.

For example,

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$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}\left(x^2\right) = 2x$$

$$\frac{d}{dx}\left(e^x\right) = e^x$$

In the above example, the function cos x is the derived function of sin x. We say that sin x is an anti derivative (or an integral) of cos x. Similarly, x^2 and e^x are the anti derivatives (or integrals) of 2x and e^x respectively.

Also note that the derivative of a constant (C) is zero. So we can write the above examples as:

$$\frac{d}{dx}(\sin x + c) = \cos x$$

$$\frac{d}{dx}\left(x^2 + c\right) = 2x$$

$$\frac{d}{dx}\left(e^x + c\right) = e^x$$

Thus, anti derivatives (or integrals) of the above functions are not unique. Actually, there exist infinitely many anti derivatives of each of these functions which can be obtained by selecting C arbitrarily from the set of real numbers.

For this reason C is referred to as arbitrary constant. In fact, C is the parameter by varying which one gets different anti derivatives (or integrals) of the given function.

If the function F(x) is an antiderivative of f(x), then the expression F(x) + C is the indefinite integral of the function f(x) and is denoted by the symbol ∫ f(x) dx.

By definition,

$$\int f(x)dx = F(x) + c, \quad \text{where } F'(x) = f(x) \text{ and } 'c' \text{ is constant.}$$

-

Maxima and Minima of a Function -

Maxima and Minima of a Function

Let y = f(x) be a real function defined at x = a. Then the function f(x) is said to have a maximum value at x = a, if f(x) ≤ f(a) ∀ a ≥∈ R.

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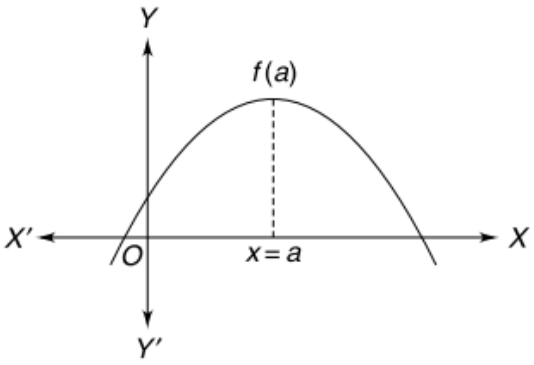
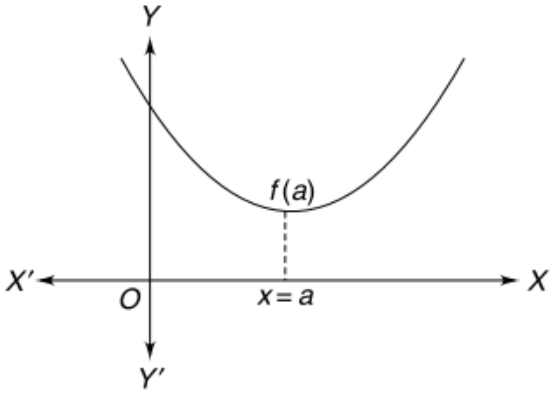
And also the function f(x) is said to have a minimum value at x = a, if f(x) ≥ f(a) ∀ a ∈ R

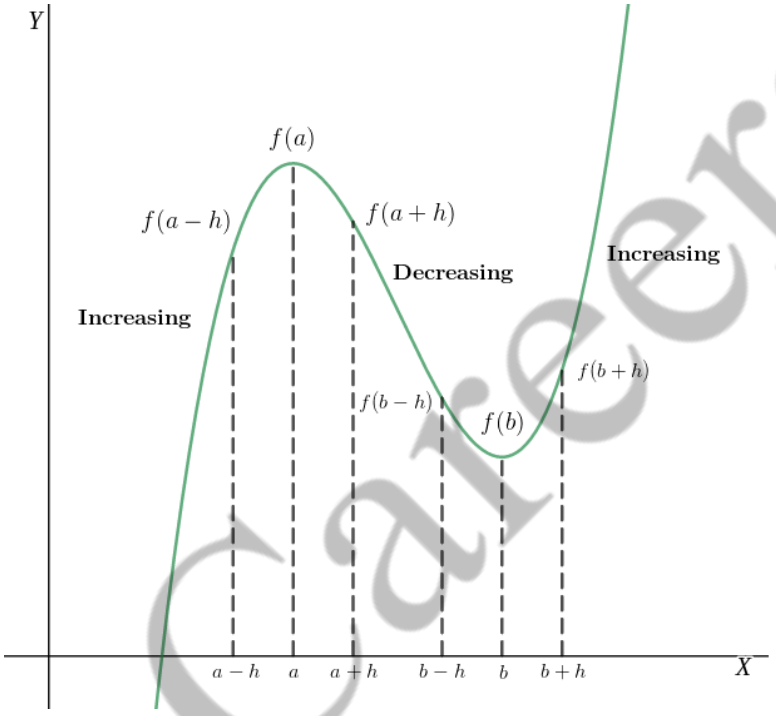




Concept of Local Maxima and Local Minima

The function f(x) is said to have a maximum (or we say that f(x) attains a maximum) at a point 'a' if the value of f(x) at 'a' is greater than its values for all x in a small neighborhood of 'a' .

In other words, f(x) has a maximum at x = 'a', if f(a + h) ≤ f(a) and f(a - h) ≤ f(a), where h ≥ 0 (very small quantity).



The function f(x) is said to have a minimum (or we say that f(x) attains a minimum) at a point 'b' if the value of f(x) at 'b' is less than its values for all x in a small neighborhood of 'b' .

In other words, f(x) has a maximum at x = 'b', if f(a + h) ≥ f(a) and f(a - h) ≥ f(a), where h ≥ 0 (very small quantity).

-

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$F'(x) = x^2 g(x)$

$\Rightarrow F'(1) = 1.\ g(1) = 0 \quad \ldots(1) \qquad (\because\ g(1) = 0)$

Now $(x) = 2xg(x) + x^2 g'(x)$

$\Rightarrow F''(x) = 2xg(x) + x^2 f(x) \quad (\because\ g'(x) = f(x))$

$\Rightarrow F''(1) = 0 + 1 \times 3$

$\Rightarrow F''(1) = 3 \qquad \ldots(2)$

From (1) and (2) F(x) has local minimum at x = 1

Correct Option (1)

Q. 8 Let $[t]$ denote the greatest integer $\leq t$ and $\lim\limits_{x \to 0} x \left[\frac{4}{x}\right] = A$. Then the function, $f(x) = \left[x^2\right] \sin(\pi x)$ is discontinuous, when x is equal to:

Option 1:
$\sqrt{A+1}$

Option 2:
$\sqrt{A}$

Option 3:
$\sqrt{A+5}$

Option 4:
$\sqrt{A+21}$

Correct Answer:
$\sqrt{A+1}$

Solution:

Piecewise function -

Greatest integer function

[1.75] = 1

[2.34] = 2

[-0.9] = -1

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[-4.8] = -5

From the definition of [x], we

can see that

Properties of greatest integer function:

i) [x] ≤ x < [x] + 1

ii) x - 1 < [x] < x

iii) I ≤ x < I+1 ⇒ [x] = I where I belongs to integer.

iv) [[x]]=[x]v)

v) $[x] + [-x] = \begin{cases} 0, & if \quad x \quad belongs\ to \quad integer \\ -1, & if \quad x \quad doesn't\ belongs\ to \quad integer \end{cases}$

vi) [x] + [-x] = 2x if x belongs to integer

2[x] + 1 if x doesn't belongs to integer

Fractional part function:

Properties of fractional part of x

i) {x}=x if 0 ≤ x < 1

ii) {x} = 0, if x belongs to integer

iii) {-x} =1-{x}, if x doesn't belongs integer

iv) {x ± integer } = {x}.

-

Algebra of Limits -

Algebra of Limits

Let f(x) and g(x) be defined for all x ≠ a over some open interval containing a. Assume that L and M are real numbers such that $\lim_{x \to a} f(x) = L$ and $\lim_{x \to a} g(x) = M$. Let c be a constant. Then, each of the following statements hold:

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Sum law for limits : $\lim_{x \to a}(f(x) + g(x)) = \lim_{x \to a} f(x) + \lim_{x \to a} g(x) = L + M$

Difference law for limits : $\lim_{x \to a}(f(x) - g(x)) = \lim_{x \to a} f(x) - \lim_{x \to a} g(x) = L - M$

Constant multiple law for limits : $\lim_{x \to a} cf(x) = c \cdot \lim_{x \to a} f(x) = cL$

Product law for limits : $\lim_{x \to a}(f(x) \cdot g(x)) = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x) = L \cdot M$

Quotient law for limits : $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)} = \frac{L}{M}$ for $M \neq 0$

Power law for limits : $\lim_{x \to a}(f(x))^n = \left(\lim_{x \to a} f(x)\right)^n = L^n$ for every positive integer n

Composition law of limit : $\lim_{x \to a}(fog)(x) = f\left(\lim_{x \to a} g(x)\right) = f(M)$, only if f(x) is continuous at g(x)=M.

If $\lim_{x \to a} f(x) = +\infty$ **or** $-\infty$, **then** $\lim_{x \to a} \frac{1}{f(x)} = 0$

-

Graph of Trigonometric Function (Part 1) -

Graph of Trigonometric Function (Part 1)

Sine Function

y = f(x) = sin(x)

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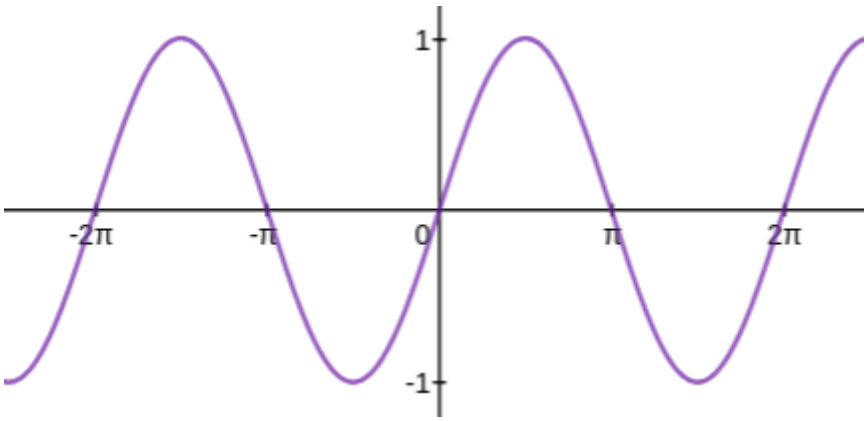

Domain is R

Range is [-1, 1]

-

$$\lim_{x\to 0} x\left(\frac{4}{x}-\left\{\frac{4}{x}\right\}\right)=A \quad \Rightarrow \lim_{x\to 0} 4-x\left\{\frac{4}{x}\right\}=A$$
$$\Rightarrow 4-0=A$$

$(1)\ x=\sqrt{A+1}\Rightarrow x=\sqrt{5}\Rightarrow$ discontinuous
$(2)\ x=\sqrt{A+21}\Rightarrow x=5\Rightarrow$ continuous
$(3)\ x=\sqrt{A}\Rightarrow x=2\Rightarrow$ continuous
$(4)\ x=\sqrt{A+5}\Rightarrow x=3\Rightarrow$ continuous

Correct Option (1)

Q. 9 Let $a-2b+c=1$.

If $f(x)=\begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$, then :

Option 1:
$f(-50)=501$

Option 2:
$f(-50)=-1$

Option 3:
$f(50)=1$

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Option 4:
$f(50) = -501$

Correct Answer:
$f(50) = 1$

Solution:

Elementary row operations -

Elementary row operations

Row transformation: Following three types of operation (Transformation) on the rows of a given matrix are known as elementary row operation (transformation).

i) Interchange of i^{th} row with j^{th} row, this operation is denoted by

$$\mathrm{R_i} \leftrightarrow \mathrm{R_j} \ \text{ or } \mathrm{R_{ij}}$$

ii) The multiplication of ith row by a constant k (k≠0) is denoted by

$$\mathrm{R_i} \leftrightarrow \mathrm{kR_i} \ \text{ or } \mathrm{k} \cdot \mathrm{R_i}$$

iii) The addition of ith row to the elements of jth row multiplied by constant k (k≠0) is denoted by

$$\mathrm{R_i} \leftrightarrow \mathrm{R_i} + \mathrm{kR_j} \ \text{ or } \mathrm{k} \cdot \mathrm{R_{ij}}$$

In the same way, three-column operations can also be defined too.

-

$$\text{Apply } \mathrm{R_1} = \mathrm{R_1} + \mathrm{R_3} - 2\mathrm{R_2}$$
$$\Rightarrow f(x) = \begin{vmatrix} 1 & 0 & 0 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix} \quad \Rightarrow f(x) = 1$$
$$\Rightarrow f(50) = 1$$

Correct Option (3)

Q. 10 Given : $f(x) = \begin{cases} x, & 0 \le x < \frac{1}{2} \\ \frac{1}{2}, & x = \frac{1}{2} \\ 1-x, & \frac{1}{2} < x \le 1 \end{cases}$ and $g(x) = \left(x - \frac{1}{2}\right)^2$, $x \epsilon \mathbf{R}$. Then the area (in sq. units) of the region bounded b the curves, $y = f(x)$ and $y = g(x)$ between the lines, $2x = 1 \ \ and \ \ 2x = \sqrt{3}$, is :

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Option 1:

$\frac{\sqrt{3}}{4}-\frac{1}{3}$

Option 2:

$\frac{1}{3}+\frac{\sqrt{3}}{4}$

Option 3:

$\frac{1}{2}+\frac{\sqrt{3}}{4}$

Option 4:

$\frac{1}{2}-\frac{\sqrt{3}}{4}$

Correct Answer:

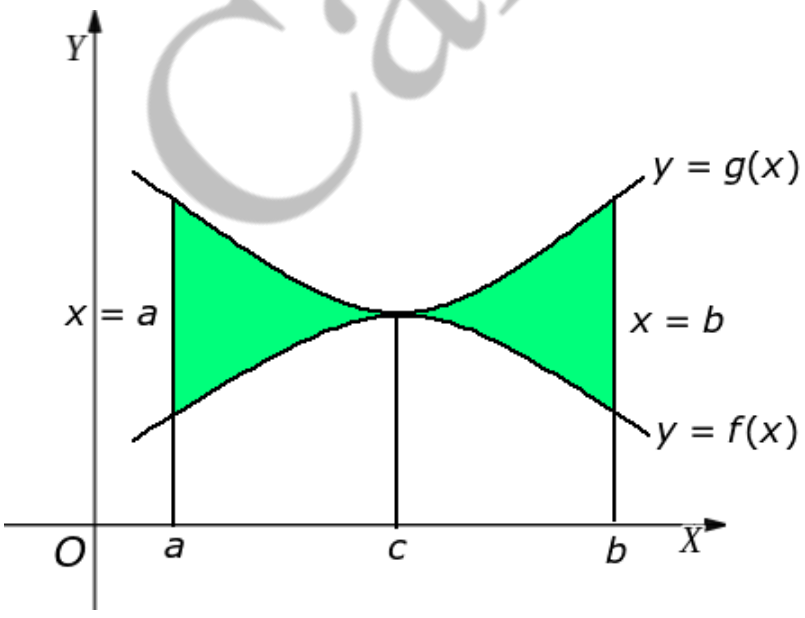
$\frac{\sqrt{3}}{4}-\frac{1}{3}$

Solution:

Area Bounded by Curves When Intersects at More Than One Point -

Area bounded by the curves $y = f(x)$, $y = g(x)$ and intersect each other in the interval [a, b]

First find the point of intersection of these curves $y = f(x)$ and $y = g(x)$, let the point of intersection be x = c



Area of the shaded region

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$$= \int_a^c \{f(x) - g(x)\}dx + \int_c^b \{g(x) - f(x)\}dx$$

When two curves intersects more than one point

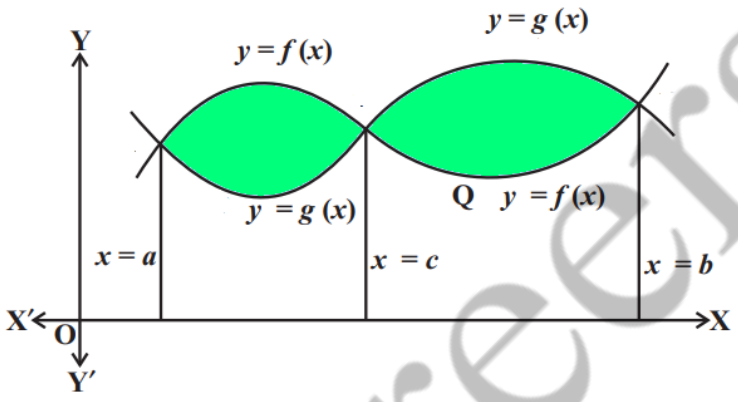
rea bounded by the curves *y=f(x), y=g(x)* and intersect each other at three points at *x = a, x = b amd x = c.*

To find the point of intersection, solve *f(x) = g(x).*

For x ∈ (a, c), f(x) > g(x) and for x ∈ (c, b), g(x) > f(x).

Area bounded by curves,

$$\mathrm{A} = \int_a^b |f(x) - g(x)|\, dx$$
$$= \int_a^c (f(x) - g(x))\, dx + \int_c^b (g(x) - f(x))\, dx$$



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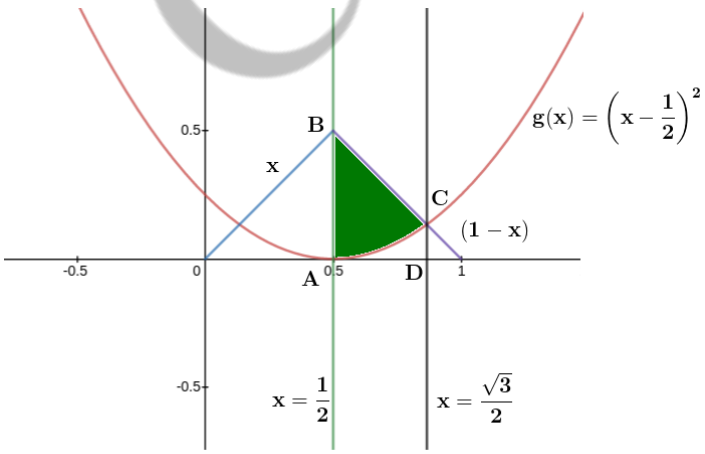

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Required area = Area of trapezium ABCD - $\int_{1/2}^{\sqrt{3}/2}\left(x-\frac{1}{2}\right)^2 dx$

$$=\frac{1}{2}\left(\frac{\sqrt{3}-1}{2}\right)\left(\frac{1}{2}+1-\frac{\sqrt{3}}{2}\right)-\frac{1}{3}\left(\left(x-\frac{1}{2}\right)^3\right)_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}}$$

$$=\frac{\sqrt{3}}{4}-\frac{1}{3}$$

Correct Option (1)

Q. 11 The following system of linear equations

$7x+6y-2z=0$

$3x+4+2z=0$

$x-2y-6z=0$, has

Option 1:
infinitely many solutions, (x,y,z) satisfying $y=2z$.

Option 2:
infinitely many solutions, (x,y,z) satisfying $x=2z$.

Option 3:
no solution

Option 4:
only the trivial solution.

Correct Answer:
infinitely many solutions, (x,y,z) satisfying $x=2z$.

Solution:

System of Homogeneous linear equations -

Let,
$a_1x+b_1y+c_1z=0 \quad ...(i)$
$a_2x+b_2y+c_2z=0 \quad ...(ii)$
$a_3x+b_3y+c_3z=0 \quad ...(iii)$
be three homogeneous equation
and let $\Delta=\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

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If ? ≠ 0, then x= 0, y = 0, z = 0 is the only solution of the above system. This solution is also known as a trivial solution.

If ? = 0, at least one of x, y and z are non-zero. This solution is called a non-trivial solution.

Explanation: using equation (ii) and (iii), we have

$$\frac{x}{b_2c_3 - b_3c_2} = \frac{y}{c_2a_3 - c_3a_2} = \frac{z}{a_2b_3 - a_3b_2}$$

$$\text{or } \frac{x}{\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}} = \frac{y}{\begin{vmatrix} c_2 & a_2 \\ c_3 & a_3 \end{vmatrix}} = \frac{z}{\begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}} = k(\text{say} \neq 0)$$

$$\therefore x = k\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}, y = k\begin{vmatrix} c_2 & a_2 \\ c_3 & a_3 \end{vmatrix} \text{ and } z = k\begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

$$\text{putting these value in equation (i), we have}$$

$$a_1\left\{k\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}\right\} + b_1\left\{k\begin{vmatrix} c_2 & a_2 \\ c_3 & a_3 \end{vmatrix}\right\} + c_1\left\{\begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}\right\} = 0$$

$$\Rightarrow a_1\begin{vmatrix} b_2 & c_2 \\ b_3 & c_2 \end{vmatrix} - b_1\begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1\begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix} = 0 \quad [\because k \neq 0]$$

$$\text{or } \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0 \text{ or } \Delta = 0$$

This is the condition for a system have Non-trivial solution.

-

$(1)\ 7x + 6y - 2z = 0$

$(2)\ 3x + 4y + 2z = 0$

$(3)\ x - 2y - 6z = 0$

$$\begin{vmatrix} 7 & 6 & -2 \\ 3 & 4 & 2 \\ 1 & -2 & -6 \end{vmatrix} = 7(-20) - 6(-20) - 2(-10) = -140 + 120 + 20 = 0$$

so infinite non-trivial solution exist

now equation (1) + 3 equation (3)

10x - 20z = 0

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x = 2z

Correct Option 2

Q. 12 If $p \rightarrow (p \wedge \sim q)$ is false, then the truth values of p and q are respectively :

Option 1:
F, T

Option 2:
T, F

Option 3:
F, F

Option 4:
T, T

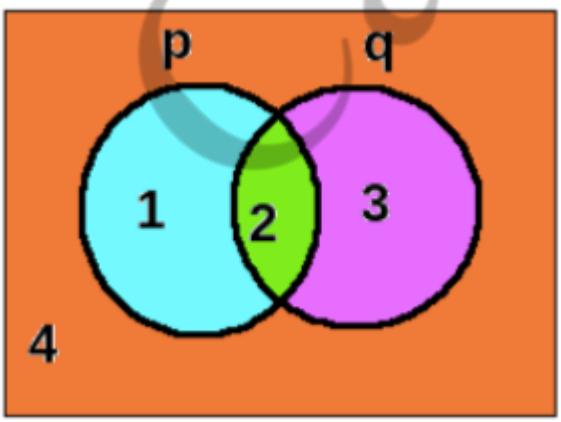
Correct Answer:
T, T

Solution:

Relation Between Set Notation and Truth Table -

Sets can be used to identify basic logical structures of statements. Statements have two fundamental roles either it is true or false.

Let us understand with an example of two sets p{1,2} and q{2,3}.



$p \vee q$	$p \cup q$	$1, 2, 3$
$p \wedge q$	$p \cap q$	2
p^c	$\sim p$	$3, 4$
q^c	$\sim q$	$1, 4$

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Using this relation we get

Element	p	q	$\sim p$	$\sim q$	$p \wedge q$	$p \vee q$	$\sim (p \wedge q)$	$\sim p \wedge \sim q$
1	T	F	F	T	F	T	T	F
2	T	T	F	F	T	T	F	T
3	F	T	T	F	F	T	T	F
4	F	F	T	T	F	F	T	F

-

Practise Session - 2 -

Q1. Write the truth table for the following statement pattern:

1. $\sim (\mathrm{p} \vee ((\sim \mathrm{q} \Rightarrow \mathrm{q}) \wedge \mathrm{q}))$
2. $(\mathrm{p} \wedge \mathrm{q}) \vee (\sim \mathrm{r})$

-

$\mathbf{p}$	$\mathbf{q}$	$(\mathbf{p} \rightarrow (\mathbf{p} \mathbf{\Lambda} - \mathbf{q}))$
F	F	$\mathbf{T}$
F	T	$\mathbf{T}$
$\mathbf{T}$	F	$\mathbf{T}$
$\mathbf{T}$	T	F

Correct Option (4)

Q. 13 The length of the minor axis (along y-axis) of an ellipse in the standard form is $\frac{4}{\sqrt{3}}$. If this ellipse touches the line, $x + 6y = 8;$ then its eccentricity is :

Correct Answer:

$\frac{1}{2}\sqrt{\frac{11}{3}}$

Solution:

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What is Ellipse? -

Ellipse

Standard Equation of Ellipse:

The standard form of the equation of an ellipse with center (0, 0) and major axis on the x-axis is

$$\frac{\mathbf{x^2}}{\mathbf{a^2}}+\frac{\mathbf{y^2}}{\mathbf{b^2}}=\mathbf{1} \quad \text{where }, b^2=a^2\left(1-e^2\right)_{(a>b)}$$

1. a > b
2. the length of the major axis is 2a
3. the coordinates of the vertices are (±a, 0)
4. the length of the minor axis is 2b
5. the coordinates of the co-vertices are (0, ±b)

-

Equation of Tangent of Ellipse in Parametric Form and Slope Form -

Slope Form:

The equation of tangent of slope m to the ellipse, $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ are $y=mx\pm\sqrt{a^2m^2+b^2}$ and coordinate of point of contact is $\left(\mp\frac{a^2m}{\sqrt{a^2m^2+b^2}},\pm\frac{b^2}{\sqrt{a^2m^2+b^2}}\right)$

-

$2b=\frac{4}{\sqrt{3}} \quad \Rightarrow \quad b=\frac{2}{\sqrt{3}}$

Equation of tangent $\equiv y=mx\pm\sqrt{a^2m^2+b^2}$

comparing with $\equiv y=\frac{-x}{6}+\frac{4}{3}$

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$$m = \frac{-1}{6} \text{ and } a^2m^2 + b^2 = \frac{16}{9}$$
$$\Rightarrow \quad \frac{a^2}{36} + \frac{4}{3} = \frac{16}{9}$$
$$\Rightarrow \quad \frac{a^2}{36} = \frac{16}{9} - \frac{4}{3} = \frac{4}{9}$$
$$\Rightarrow \quad a^2 = 16$$
$$e = \sqrt{1 - \frac{b^2}{a^2}}$$
$$e = \sqrt{\frac{11}{12}}$$

Correct Option (2)

Q. 14 If z be a complex number satisfying $|Re(z)| + |Im(z)| = 4$, then $|z|$ cannot be :

Option 1:
$\sqrt{7}$

Option 2:
$\sqrt{\frac{17}{2}}$

Option 3:
$\sqrt{10}$

Option 4:
$\sqrt{8}$

Correct Answer:
$\sqrt{7}$

Solution:

Complex number -

A complex number is the sum of a real number and an imaginary number. A complex number is expressed in standard form as a + bi where a is the real part and b is the imaginary part. For example, 5 + 2i is a complex number. So, too, is 3 + 4i√3.

We write the complex number by C or z = a + ib, *a* and *b* are real number (a, b ∈ R).

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- *a* is real part of the complex number and denoted by Re(z),
- *b* is the imaginary part of the complex number and denoted by Im(z),

E.g : z = 2 + 3i is a complex number.

With Re(z) = 2 and Im(z) = 3

-

Area of triangle, circle (formula) -

Equation of Circle:

The equation of the circle whose center is at the point Z_0 and have radius r is given by

$|z - z_0| = r$

If the center is origin then, $z_0 = 0$, hence equation reduces to |z| = r

Interior of the circle is represented by $|z - z_0| < r$

The exterior is represented by $|z - z_0| > r$

Here z can be represented as x + iy and Z_0 is represented by $x_0 + iy_0$

-

z = x + iy

|x| + |y| = 4

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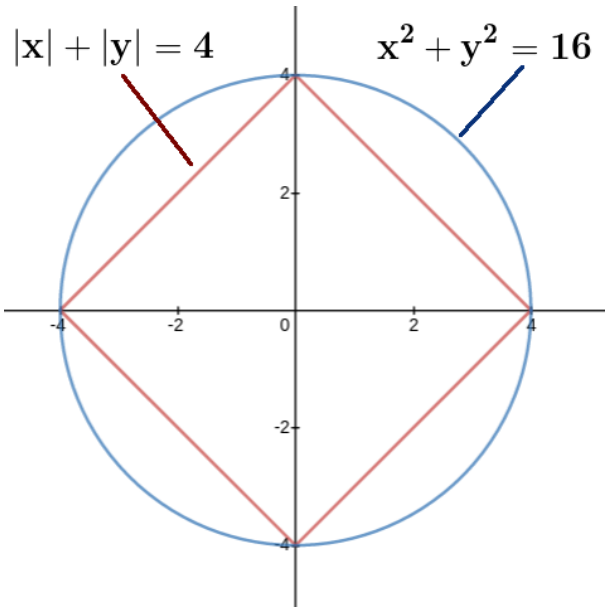

Minimum value of |z| = $2\sqrt{2}$

Maximum value of |z| = 4

$z \in [\sqrt{8}, \sqrt{16}]$

So |z| can't be $\sqrt{7}$

Correct Option (1)

Q. 15 If $x = \sum_{n=0}^{\infty}(-1)^n \tan^{2n}\theta \ \ and \ \ y = \sum_{n=0}^{\infty}\cos^{2n}\theta$, for $0 < \theta < \frac{\pi}{4}$, then :

Option 1:
$y(1+x) = 1$

Option 2:
$x(1-y) = 1$

Option 3:
$y(1-x) = 1$

Option 4:
$x(1+y) = 1$

Correct Answer:
$y(1-x) = 1$

Solution:
Sum of an infinite GP

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If a is the first term and r is the common ratio of a G.P. Then,

$$S_\infty = \frac{a}{1-r}$$

S_∞ is the sum to infinite terms of the G.P.

Trigonometric Identities

These identities are the equations that hold true regardless of the angle being chosen.

$$\sin^2 t + \cos^2 t = 1$$
$$1 + \tan^2 t = \sec^2 t$$
$$1 + \cot^2 t = \csc^2 t$$
$$\tan t = \frac{\sin t}{\cos t}, \quad \cot t = \frac{\cos t}{\sin t}$$

Now,

$$x = \sum_{n=0}^{\infty} (-1)^n \tan^{2n}\theta = 1 - \tan^2\theta + \tan^4\theta.....$$

$$y = \sum_{n=0}^{\infty} \cos^{2n}\theta = 1 + \cos^2\theta + \cos^4\theta.....$$

Use $S_\infty = \frac{1}{1-r}$

$$x = \frac{1}{1+\tan^2\alpha} = \cos^2\theta$$

$$y = \frac{1}{1-\cos^2\theta} = \frac{1}{\sin^2\theta}$$

$$\Rightarrow (1-x) = \sin^2\theta$$

$$\Rightarrow y(1-x) = 1$$

Correct Option (3)

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Q. 16 If $\frac{dy}{dx} = \frac{xy}{x^2+y^2}; y(1) = 1;$ then a value of x satisfying $y(x) = e$ is :

Option 1:
$\sqrt{3}\,e$

Option 2:
$\frac{1}{2}\sqrt{3}\,e$

Option 3:
$\sqrt{2}\,e$

Option 4:
$\frac{e}{\sqrt{2}}$

Correct Answer:
$\sqrt{3}\,e$

Solution:

Homogeneous Differential Equation -

This equation can be solved by the substitution y = vx.

$$\mathrm{y = vx}$$

$$\Rightarrow \quad \frac{\mathrm{dy}}{\mathrm{dx}} = \mathrm{v} + \mathrm{x}\frac{\mathrm{dv}}{\mathrm{dx}}$$

Thus, $\frac{dy}{dx} = \phi\left(\frac{y}{x}\right)$ transforms to

$$\mathrm{v} + \mathrm{x}\frac{\mathrm{dv}}{\mathrm{dx}} = \phi(\mathrm{v})$$

$$\Rightarrow \quad \frac{dv}{\phi(v) - v} = \frac{dx}{x}$$

The variables have now been separated and the solution is

$$\int \frac{\mathrm{dv}}{\phi(\mathrm{v}) - \mathrm{v}} = \ln \mathrm{x} + \mathrm{c}$$

After the integration v should be replaced by y/x
to get the required solution.

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$$\text{put } y = vx, \quad \frac{dy}{dx} = v + x\frac{dv}{dx}$$

$$v + x\frac{dv}{dx} = \frac{v}{1+v^2}$$

$$x\frac{dv}{dx} = \frac{-v^3}{1+v^2}$$

$$\frac{1+v^2}{v^3}dv = -\frac{dx}{x}$$

$$-\frac{x^2}{2y^2} + \ln y = c \Rightarrow c = -1/2$$

$$-\frac{x^2}{2y^2} + \ln y = c \Rightarrow c = -1/2$$

$$\text{Now } y(x) = e \quad \Rightarrow \quad 1 = \frac{x^2}{2e^2} - \frac{1}{2}$$

$$x = \pm\sqrt{3}e$$

Correct Option 1

Q. 17 If one end of a focal chord AB of the parabola $y^2 = 8x$ is at $A\left(\frac{1}{2}, -2\right)$, then the equation of the tangent to it at B is :

Option 1:
$x + 2y + 8 = 0$

Option 2:
$2x - y - 24 = 0$

Option 3:
$x - 2y + 8 = 0$

Option 4:
$2x + y - 24 = 0$

Correct Answer:
$x - 2y + 8 = 0$

Solution:

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Length of the Latus rectum and parametric form -

Parametric Equation:

From the equation of the parabola, we can write

$\frac{y}{2a} = \frac{2x}{y} = t$ here, t is a parametewr

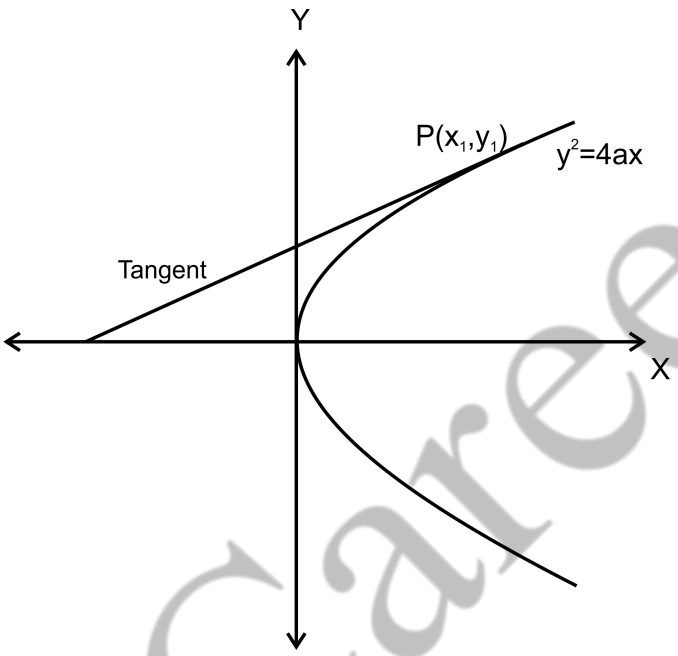
Then, $x = at^2$ and $y = 2at$ are called the parametric equations and the point $(at^2, 2at)$ lies on the parabola.

-

Tangents of Parabola in Point Form -

Tangents of Parabola in Point Form

Equation of the tangent to the parabola $y^2 = 4ax$ at the point $P(x_1, y_1)$ is $yy_1 = 2a(x + x_1)$



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$y^2 = 8x$ then $A(2t_1^2, 4t_1)$

given $A\left(\frac{1}{2}, -2\right) \Rightarrow t_1 = -1/2$

$t_1 \cdot t_2 = -1$

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$$\text{then } t_2 = 2 \Rightarrow B(8,8)$$
$$\text{Equation of tangent } 8y = 4(x+8)$$
$$2y = x + 8$$

Correct Option 3

Q. 18 Let a_n be the n[th] term of a G.P. of positive terms. If $\sum_{n=1}^{100} a_{2n+1} = 200 \text{ and } \sum_{n=1}^{100} a_{2n} = 100, \text{ then } \sum_{n=1}^{200} a_n$ is equal to :

Correct Answer:
150

Solution:
Sum of n-term of a GP

Let S_n be the sum of n terms of the G.P. with the first term 'a' and common ratio 'r'. Then

$$S_n = a(\frac{1-r^n}{1-r})$$

The above formula does not hold for r = 1, For r = 1 the sum of n terms of the G.P. is $S_n = na$.

Now,

$$\sum_{n=1}^{100} a_{2n+1} = a_3 + a_5 + \ldots\ldots a_{201} = 200 \Rightarrow \frac{ar^2\left(r^{200}-1\right)}{r^2-1} = 200$$
$$\sum_{n=1}^{100} a_{2n} = a_2 + a_4 + \ldots\ldots a_{200} = 100 = \frac{ar\left(r^{200}-1\right)}{r^2-1} = 100$$

On dividing above equations, r = 2

Adding both equations,

$$\Rightarrow a_2 + a_3 + \ldots\ldots\ldots\ldots a_{200} + a_{201} = 300$$
$$\Rightarrow r\left(a_1 + \ldots\ldots\ldots\ldots a_{200}\right) = 300$$
$$\sum_{n=1}^{200} a_n = \frac{300}{r} = 150$$

Correct Option 4

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Q. 19 A random variable X has the following probability distribution:

$$X: \quad 1 \quad 2 \quad 3 \quad 4 \quad 5$$

$$P(X): \quad K^2 \quad 2K \quad K \quad 2K \quad 5K^2$$

Then $P(X > 2)$ is equal to:

Option 1:

$\frac{7}{12}$

Option 2:

$\frac{23}{36}$

Option 3:

$\frac{1}{36}$

Option 4:

$\frac{1}{6}$

Correct Answer:

$\frac{23}{36}$

Solution:

$\sum P_i = 1 \Rightarrow 6k^2 + 5k = 1$

$\Rightarrow 6k^2 + 5k - 1 = 0$

$\Rightarrow k = \frac{1}{6},\ k = -1 \ (\text{invalid})$

$Now,\ P(X > 2) = P(3) + P(4) + P(5) = k + 2k + 5k^2$

$= \frac{1}{6} + \frac{2}{6} + \frac{5}{36} = \frac{6 + 12 + 5}{36} = \frac{23}{36}$

Correct Option 2

Q. 20 If $\int \frac{d\theta}{\cos^2\theta\,(\tan 2\theta + \sec 2\theta)} = \lambda \tan\theta + 2\log_e |f(\theta)| + C$ where C is a constant of integration, then the ordered pair $(\lambda, f(\theta))$ is equal to:

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Option 1:
$(-1, 1 - \tan\theta)$

Option 2:
$(-1, 1 + \tan\theta)$

Option 3:
$(1, 1 + \tan\theta)$

Option 4:
$(1, 1 - \tan\theta)$

Correct Answer:
$(-1, 1 + \tan\theta)$

Solution:

Integration as Reverse Process of Differentiation -

Rules of integration:

f(x) and g(x) are functions with antiderivatives ∫ f(x) and ∫ g(x) dx. Then,

$$\text{(a)} \int \mathrm{kf}(x)dx = k\int f(x)dx \text{ for any constant } k.$$

$$\text{(b)} \int (f(x) + g(x))dx = \int f(x)dx + \int g(x)dx$$

$$\text{(c)} \int (f(x) - g(x))dx = \int f(x)dx - \int g(x)dx$$

$$\text{Since, } \frac{\mathrm{d}}{\mathrm{dx}}(\mathrm{F(x)}) = \mathrm{f(x)} \Leftrightarrow \int \mathrm{f(x)dx} = \mathrm{F(x)} + \mathrm{c}$$

Based on this definition and various standard formulas (which we studied in Limit, Continuity and Differentiability) we obtained the following **important integration formulae,**

1. $\frac{\mathrm{d}}{\mathrm{dx}}(\mathrm{kx}) = \mathrm{k} \Rightarrow \int \mathrm{kdx} = \mathrm{kx} + \mathrm{C}, \text{ where k is a constant}$
2. $\frac{\mathrm{d}}{\mathrm{dx}}\left(\frac{\mathrm{x^{n+1}}}{\mathrm{n+1}}\right) = \mathrm{x^n}, \mathrm{n} \neq -1 \Rightarrow \int \mathrm{x^n dx} = \frac{\mathrm{x^{n+1}}}{\mathrm{n+1}} + \mathrm{C}, \mathrm{n} \neq -1$
3. $\frac{\mathrm{d}}{\mathrm{dx}}(\log|\mathrm{x}|) = \frac{1}{\mathrm{x}} \Rightarrow \int \frac{1}{\mathrm{x}}\mathrm{dx} = \log|\mathrm{x}| + \mathrm{C}, \text{ when } \mathrm{x} \neq 0$
4. $\frac{\mathrm{d}}{\mathrm{dx}}(\mathrm{e^x}) = \mathrm{e^x} \Rightarrow \int \mathrm{e^x dx} = \mathrm{e^x} + \mathrm{C}$
5. $\frac{\mathrm{d}}{\mathrm{dx}}\left(\frac{\mathrm{a^x}}{\log_{\mathrm{e}}\mathrm{a}}\right) = \mathrm{a^x}, \mathrm{a} > 0, \mathrm{a} \neq 1 \Rightarrow \int \mathrm{a^x dx} = \frac{\mathrm{a^x}}{\log_{\mathrm{e}}\mathrm{a}} + \mathrm{C}$

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$$\int \frac{\sec^2\theta}{\frac{1+\tan^2\theta}{1-\tan^2\theta}+\frac{2\tan\theta}{1-\tan^2\theta}}d\theta$$
$$=\int \frac{\sec^2\theta\,(1-\tan^2\theta)}{(1+\tan\theta)^2}d\theta$$
$$=\int \frac{\sec^2\theta(1-\tan\theta)}{1+\tan\theta}d\theta$$
$$\text{put}\ \ \tan\theta=t\Rightarrow \sec^2\theta d\theta=dt$$
$$=\int\left(\frac{1-t}{1+t}\right)dt=\int\left(-1+\frac{2}{1+t}\right)dt$$
$$=-t+2\log(1+t)+C$$
$$=-\tan\theta+2\log(1+\tan\theta)+C$$
$$\Rightarrow\quad \lambda=-1\ \text{and}\ f(x)=1+\tan\theta$$

Correct option 2

Q. 21 Let $\vec{a},\vec{b}\ and\ \vec{c}$ be three vertices such that $|\vec{a}|=\sqrt{3},\ |\vec{b}|=5,\ \ \vec{b}\cdot\vec{c}=10$ and the angle between $\vec{b}\ and\ \vec{c}\ is\ \frac{\pi}{3}$. If $\rightarrow\vec{A}$ is perpendicular to the vector $\vec{b}\times\vec{c}$, then $\left|\vec{a}\times\left(b\times\vec{c}\right)\right|$ is equal to _____.

Correct Answer:
30

Solution:

Dot (Scalar) Product of Two Vectors -

Dot (scalar) Product

If $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ are two non-zero vectors, then their scalar product (or dot product) is denoted by $\vec{\mathbf{a}}\cdot\vec{\mathbf{b}}$ and is defined as

$\vec{\mathbf{a}}\cdot\vec{\mathbf{b}}=|\vec{\mathbf{a}}||\vec{\mathbf{b}}|\cos\theta.\qquad(0\le\theta\le\pi)$ where, θ is the angle between $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$

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Vector (or Cross) Product of Two Vectors -

The vector product of two nonzero vectors $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$, is denoted by $\vec{\mathbf{a}} \times \vec{\mathbf{b}}$ and defined as,

$$\vec{\mathbf{a}} \times \vec{\mathbf{b}} = |\vec{\mathbf{a}}| \left|\vec{\mathbf{b}}\right| \sin\theta \hat{\mathbf{n}}$$

where Θ is the angle between $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$, 0 ≤ θ ≤ π and $\hat{\mathbf{n}}$ is a unit vector perpendicular to both $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$, such that $\vec{\mathbf{a}}$, $\vec{\mathbf{b}}$ and $\hat{\mathbf{n}}$ form a right hand system.

-

$$\vec{b}.\vec{c} = 10 \Rightarrow |\vec{b}||\vec{c}| \cos\left(\frac{\pi}{3}\right) = 10$$

$$\Rightarrow \quad 5 \cdot |\vec{c}| \cdot \frac{1}{2} = 10 \quad \Rightarrow |\vec{c}| = 4$$

$$\vec{a}(\vec{b} \times \vec{c}) = 0$$

$$\vec{a} \times (\vec{b} \times \vec{c}) = |\vec{a}||\vec{b} \times \vec{c}| \sin\left(\frac{\pi}{2}\right)$$

$$\sqrt{3} \times |\vec{b}||\vec{c}| \sin\frac{\pi}{3} \times 1 = \sqrt{3} \times 5 \times 4 \times \frac{\sqrt{3}}{2} = 30$$

Correct Option (2)

Q. 22 If $C_r = {}^{25}C_r$ and $C_0 + 5 \cdot C_1 + 9 \cdot C_2 + \cdots\cdots + (101) \cdot C_{25} = 2^{25} \cdot k$, then k is equal to _____.

Correct Answer:
51

Solution:
Properties of Binomial Theorem and Binomial Coefficient

$$\sum_{r=1}^{n} r \times {}^{n}C_r = n.2^{n-1}$$

Now,

General term for given series is: $(4r + 1).{}^{25}C_r$

Applying summation

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$$= 4\sum_{r=1}^{25} r \times \ {}^{25}C_r + 2^{25}$$
$$= 4 \times 25 \cdot 2^{24} + 2^{25} = 50 \cdot 2^{25} + 2^{25} = (50+1)2^{25} = 51 \cdot 2^{25}$$

k = 51

Correct option 3

Q. 23 If the curves, $x^2 - 6x + y^2 + 8 = 0 \; and \; x^2 - 8y + y^2 + 16 - k = 0, (k > 0)$ touch each other at a point, then the largest value of k is_____.

Option 1:
15

Option 2:
30

Option 3:
51

Option 4:
36

Correct Answer:
36

Solution:

Intersection of Two Circle -

CASE 2

When two circles touch each other externally
$|\mathbf{C_1C_2}| = \mathbf{r_1} + \mathbf{r_2},$

i.e, the distance between the centres is equal to the sum of radii, then two circles touch externally.

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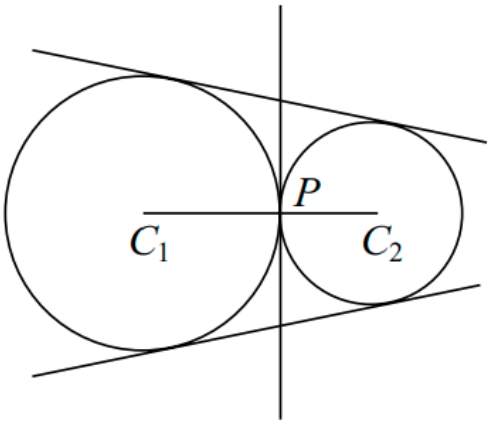

In this case, two direct common tangents are real and distinct while the transverse tangents are coincident.

In this case, point of contact P divides C_1 and C_2 internally in the ratio r_1:r_2.

$$\frac{C_1P}{C_2P} = \frac{r_1}{r_2}$$

$$\text{Coordinate of point P is } \left(\frac{r_1x_2 + r_2x_1}{r_1 + r_2}, \frac{r_1y_2 + r_2y_1}{r_1 + r_2}\right)$$

$$\text{The equation of tangent at point } P \text{ is } S_1 - S_2 = 0, \text{ where } S_1 = 0 \text{ and } S_2 = 0 \text{ are equations of circles.}$$

-

Two circles touch each other if $C_1C_2 = |r_1 \pm r_2|$

Distance between C2(3, 0) and C1(0, 4) is either $\sqrt{k} + 1$ or $|\sqrt{k} - 1|$

Also $C_1C_2 = \sqrt{4^2 + 3^2} = 5$

$\Rightarrow \sqrt{k} + 1 = 5$ or $|\sqrt{k} - 1| = 5 \quad \Rightarrow k = 16$ or $k = 36$

Maximum value of K is 36

Correct Option 4

Q. 24 The number of terms common to the two A.P's $3, 7, 11, \cdots\cdots\cdots, 407$ and $2, 9, 16, \cdots\cdots, 709$ is_____.

Correct Answer:

14

Solution:

General Term of an AP

The n^{th} term (general term) of the A.P. is $a_n = a + (n - 1)d$.

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In this Question,

$$\begin{aligned}
&\text{First common term} \quad = 23\\
&\text{common difference (this will equal the LCM of common differences of the APs involved)} \quad = 7 \times 4 = 28\\
&\text{Last term} \ \leq 407\\
\Rightarrow \quad & 23 + (n-1) \times 28 \leq 407\\
\Rightarrow \quad & (n-1) \times 28 \leq 384\\
\Rightarrow \quad & n \leq 13.71 + 1\\
\Rightarrow \quad & n \leq 14.71\\
\text{So,} \quad & n = 14
\end{aligned}$$

Correct Option 1

Q. 25 If the distance between the plane, $23x - 10y - 2z + 48 = 0$ and the plane containing the lines $\frac{x+1}{2} = \frac{y-3}{4} = \frac{z+1}{3}$ and $\frac{x+3}{2} = \frac{y+2}{6} = \frac{z-1}{\lambda}$ $(\lambda \epsilon \mathbf{R})$ is equal to $\frac{k}{\sqrt{633}}$, then k is equal to _____.

Correct Answer:

3

Solution:

Distance of a Point From a Plane -

Cartesian Form

Let P(x_1, y_1, z_1) be the given point with position vector $\vec{\mathbf{a}}$ and ax + by + cz + d = 0 be the Cartesian equation of the given plane. Then

$$\vec{a} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$$
$$\vec{n} = a\hat{i} + b\hat{j} + c\hat{k}$$

Hence, from Vector form of the perpendicular from P to the plane is

$$\left|\frac{\left(x_1\hat{i} + y_1\hat{j} + z_1\hat{k}\right) \cdot (a\hat{i} + b\hat{j} + c\hat{k}) + d}{\sqrt{a^2 + b^2 + c^2}}\right| = \left|\frac{ax_1 + by_1 + cz_1 + d}{\sqrt{a^2 + b^2 + c^2}}\right|$$

-

Lines must be intersecting

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$\Rightarrow (2a-1, 4a+3, 3a-1) = (2b-3, 6b-2, \lambda b+1)$

$\Rightarrow 2a-1 = 2b-3, 4a+3 = 6b-2, 3a-1 = \lambda b+1$

$\Rightarrow \mathrm{b} = \frac{1}{2}, \mathrm{a} = -\frac{1}{2}, \lambda = -7$

distance of plane contains given lines from given plane is same as distance between point $(-3, -2, 1)$ from given plane.

Required distance equal to

$$\frac{|-69+20-2+48|}{\sqrt{529+100+4}} = \frac{3}{\sqrt{633}} = \frac{\mathrm{k}}{\sqrt{633}} \Rightarrow \mathrm{k} = 3$$

Correct Option 4

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